A Wilson Loop after the First Renormalisation Step

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1 The Insertion at a Fixed Cutoff

This paper follows a Wilson loop through the first renormalisation-group step of Bałaban's construction of four-dimensional lattice Yang–Mills theory [3, 4]. That step is the block-spin transformation T of (1.3): the bond variables of a periodic fine lattice of spacing ε are averaged over blocks of side L, and the bare measure is mapped to the induced measure of the bond variables of the coarse lattice of spacing $L\varepsilon$. The loop is inserted in the bare measure and carried through T; the result (Section 1.2) writes its bare expectation as a perimeter factor times a corner factor times an average, over the coarse ensemble, of the loop at the background field reconstructed from the coarse field, plus two error terms that it bounds.

The gauge group G is a compact connected simple Lie group, g_0 is the bare coupling, and $\chi\colon G \to \mathbb{C}$ is a normalised irreducible character, so that $\chi(\mathbf{1}) = 1$. Let $\varepsilon > 0$ be the fine lattice spacing and $L > 1$ the blocking factor. The fine and coarse bond fields are U and V, respectively. Choose an oriented square contour C on the fine lattice and an oriented square $\bar{C}$ on the lattice of spacing $L\varepsilon$, with each side of $\bar{C}$ represented by the corresponding straight fine path in C. This assumption fixes the path correspondence; the two contours describe the same square in the underlying torus.

For a field U and an ordered contour $C = (b_1, \dots, b_n)$, holonomy and the Wilson loop are

$$\mathrm{Hol}_U C = U(b_1) \cdots U(b_n), \qquad W_\chi(C; U) = \chi(\mathrm{Hol}_U C)\,. \tag{1.1}$$

The bond field satisfies $U(b^{-1}) = U(b)^{-1}$ when a contour traverses a bond backwards. The character makes W_χ gauge invariant.

Let E and E_1 be the positive fine and coarse bonds of the four-dimensional periodic lattices, and let $\mathcal{P}_T$ be the unoriented fine plaquettes. Fix the action's unitary realisation $G \subset U(N)$ and $\mathrm{tr}_N = N^{-1}\,\mathrm{Tr}$. With ∂p denoting the oriented boundary of p, the bare action and density are

$$S(U) = \sum_{p \in \mathcal{P}_T} (1 - \mathrm{Re}\,\mathrm{tr}_N\, U(\partial p)), \qquad \rho_0(U) = \exp(-g_0^{-2} S(U) - E_\varepsilon)\,. \tag{1.2}$$

Here E_ε is a field-independent vacuum scalar, retained in both numerator and denominator. Product normalised Haar measures on G^E and G^{E_1} are denoted by $\mathrm{d}U$ and $\mathrm{d}V$. The full first-step operator, before the small-field/large-field decomposition, is [3, (0.1)]:

$$(T\rho)(V) = \int \left(\prod_{c \in E_1} \delta_G(M(U, c) V(c)^{-1}) \right) \rho(U)\, \mathrm{d}U\,. \tag{1.3}$$

Here δ_G is the group Dirac measure normalised against Haar measure, $\int \delta_G(WV^{-1})\,\mathrm{d}V = 1$ for every $W \in G$, and M is the averaging operation, defined as follows. For a positive coarse bond c let $P_c(U)$ be the transport along the straight fine path from c_- to c_+, let $\beta_c(U)$ be the logarithmic average of [4, (0.4)], displayed in (4.1) below, and let $\mathcal{O}_c$ be the set of fine configurations U at which every logarithm in $\beta_c(U)$ has its argument in the chart domain (the chart is the inverse of $A \mapsto e^{iA}$ on the Hilbert ball $h(A) < \varrho_G$, fixed in the conventions paragraph of Section 3). Put

$$M(U,c) = \begin{cases} e^{i\beta_c(U)} P_c(U), & U \in \mathcal{O}_c, \\ P_c(U), & U \notin \mathcal{O}_c. \end{cases} \tag{1.4}$$

[3, (0.1)] integrates over every fine configuration, so it needs an averaging operation on all of G^E; Bałaban's own formula is [4, (0.4)], defined where its logarithms are, and Bałaban states in [3, p. 243] that any averaging operation satisfying general properties analogous to [4, (0.5)–(0.9)] may be taken. Equation (1.4) is the choice made here; the properties used in this paper are the following, (ii) being the analogue of the gauge covariance [4, (0.6)], proved directly from (1.4). The properties (0.5)–(0.9) of the group mean of [4], which the licence of [3, p. 243] presupposes for its averaging operation, are not used for M off $\mathcal{O}_c$: on the branches the averaging operation is evaluated on $\mathcal{O}_c$ only, where it is [4, (0.4)] ((B1), Section 3.1), and its value off $\mathcal{O}_c$ enters only through the constraint on the rows of Ω_1^c.

Lemma 1.1 (The Averaging Operation)**.** *(i) $\mathcal{O}_c$ is open, and $M(\bullet, c)\colon G^E \to G$ is Borel, analytic on $\mathcal{O}_c$, where it is [4, (0.4)], and a function of the bonds with both endpoints in the two blocks of c alone. (ii) For every fine gauge transformation $g\colon T \to G$, $U^g(b) = g(b_-)U(b)g(b_+)^{-1}$, one has $U^g \in \mathcal{O}_c$ if and only if $U \in \mathcal{O}_c$, and $M(U^g, c) = g(c_-)M(U,c)g(c_+)^{-1}$; in particular $M(U^g) = M(U)$ when g is the identity at the block centres.*

Proof. (i) The chart domain is open, being the image of the open ball under $A \mapsto e^{iA}$, which is injective with invertible differential there (the conventions paragraph of Section 3). Each loop $U(\gamma)U([x,x'])U(\gamma')^{-1}P_c(U)^{-1}$ of (4.1) is a product of bond variables and their inverses, continuous in U, so $\mathcal{O}_c$, the set where finitely many such loops lie in the chart domain, is open; the chart is analytic on the chart domain (inverse of an analytic map with invertible differential), so β_c and $e^{i\beta_c}P_c$ are analytic on $\mathcal{O}_c$, and P_c is analytic on G^E. A map that is continuous on an open set and continuous on its complement is Borel. The paths γ, $[x,x']$, γ' and the straight path of c have all their bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$: $\gamma \subset \mathcal{B}_{c_-}$, $\gamma' \subset \mathcal{B}_{c_+}$, and the straight path from $x \in \mathcal{B}_{c_-}$ to $x + Le_\mu \in \mathcal{B}_{c_+}$ leaves $\mathcal{B}_{c_-}$ at the bond at which it enters $\mathcal{B}_{c_+}$. (ii) Every loop is based at c_-, so under g it is conjugated by $g(c_-)$. The Hilbert ball is Ad-invariant, h being Ad-invariant, and $e^{i\,gAg^{-1}} = ge^{iA}g^{-1}$; so the chart domain is invariant under conjugation, and for W in it $\frac{1}{i}\log(gWg^{-1}) = g(\frac{1}{i}\log W)g^{-1}$, by the injectivity of $A \mapsto e^{iA}$ on the ball. Hence $U^g \in \mathcal{O}_c$ if and only if $U \in \mathcal{O}_c$, $\beta_c(U^g) = g(c_-)\beta_c(U)g(c_-)^{-1}$ and $e^{i\beta_c(U^g)} = g(c_-)e^{i\beta_c(U)}g(c_-)^{-1}$ there, while $P_c(U^g) = g(c_-)P_c(U)g(c_+)^{-1}$ everywhere. The product gives the covariance on $\mathcal{O}_c$, and off $\mathcal{O}_c$ it is the covariance of P_c. *q.e.d.*

The right side of (1.3) is read as follows. For bounded Borel h on G^{E_1} and $W \in G^{E_1}$, $\int \mathrm{d}V\, h(V) \prod_{c \in E_1} \delta_G(W(c)V(c)^{-1}) = h(W)$; so for a nonnegative or integrable Borel function ρ on G^E, $T\rho$ is the image measure of $\rho\,\mathrm{d}U$ under the Borel map $M\colon G^E \to G^{E_1}$ of Lemma 1.1, $(T\rho)(A) = \int_{M^{-1}(A)} \rho\,\mathrm{d}U$ for Borel $A \subseteq G^{E_1}$, that is $\int h\,\mathrm{d}(T\rho) = \int h(M(U))\rho(U)\,\mathrm{d}U$. This is the meaning of (1.3) and of every group δ-function of a coarse variable in this paper: it is

integrated against that variable first. Bałaban writes [3, (0.1)] as a Haar density; for integrable ρ the finite measure $T\rho$ has a Haar density exactly when it is absolutely continuous with respect to $\mathrm{d}V$, and then $(T\rho)(V)$ denotes a version of that density. No absolute continuity is asserted or needed: every identity below in which $\rho_1(V)$ or $N_1(V)$ appears is an identity of measures, in which $\rho_1(V)\,\mathrm{d}V$ and $N_1(V)\,\mathrm{d}V$ stand for the measures ρ_1 and N_1 of (1.5), and Bałaban's transformations of (0.1) into (1.11)–(1.15) [3] are identities for the integrals of bounded Borel functions of V against these measures under the branch hypotheses (B1)–(B4) of Section 3.1, which record the facts about Bałaban's construction that these identities use. Bałaban's additional operation **R** [3, p. 243], applied after each transformation T and changing the densities on the large-field regions, is not applied here: the first-step objects are those of T alone. The bare expectation is $\mathbb{E}_\varepsilon[F] \coloneqq \int F\rho_0\,\mathrm{d}U / \int \rho_0\,\mathrm{d}U$ for bounded Borel F on G^E. Bałaban's cutoffs, gauge insertions and changes of variables give alternative representations of this same integral; their full history sum is retained [3]: the partitions of unity [3, (1.1), (1.4)] sum to one pointwise, so the sum over histories of the branch integrals is the full integral $\int \rho_0\,\mathrm{d}U$ whatever the value of M off $\bigcup_c \mathcal{O}_c$, the branch integrals using M on $\mathcal{O}_c$ only (Section 3.1). For a finite or complex measure μ on G^E, $T\mu \coloneqq M_*\mu$ is the same pushforward, so that T acts on the measures defined next. The outgoing measure and the measure with the insertion are

$$\rho_1 = T[\rho_0], \qquad N_1 = T[\rho_0 W_\chi(C;\bullet)]\ . \tag{1.5}$$

Here ρ_1 is a finite positive measure on G^{E_1} with $\rho_1(G^{E_1}) = \int \rho_0\,\mathrm{d}U$, and N_1 is a complex measure with $|N_1(A)| \leqslant \int_{M^{-1}(A)} \rho_0 |W_\chi|\,\mathrm{d}U \leqslant \rho_1(A)$ for every Borel A, so that $|N_1| \leqslant \rho_1$ for the total variation and $N_1 \ll \rho_1$. The outgoing Wilson function is the Radon–Nikodym derivative

$$W_1(C;V) = \frac{\mathrm{d}N_1}{\mathrm{d}\rho_1}(V)\ , \tag{1.6}$$

defined for ρ_1-almost every V, with $|W_1| \leqslant 1$ ρ_1-almost everywhere; a Borel version with $|W_1| \leqslant 1$ everywhere is fixed. Where ρ_1 and N_1 have Haar densities, $W_1(C;V) = N_1(V)/\rho_1(V)$ for ρ_1-almost every V, the density-ratio form in which [3] writes such quantities. This definition carries the bare insertion through the existing finite-cutoff transformation. It does not introduce a Wilson-loop counterterm.

The intermediate integration over regular fluctuations does not yet produce the measures (1.5) on G^{E_1}. It also retains the exterior fine field and the unintegrated fluctuations. Those variables are integrated by the outer operation in [3, (1.29)]; the measure ρ_1 is obtained from the same operation without the insertion. The vacuum construction is given in [3, (1.11)–(1.30)].

1.1 Normalisation

The bound $|N_1| \leqslant \rho_1$ is the positivity of the blocking kernel together with $|W_\chi| \leqslant 1$. The physical partition function and the expectation after the transformation are

$$\mathcal{Z}_0 = \int \rho_0(U)\,\mathrm{d}U = \rho_1(G^{E_1}) > 0\ , \tag{1.7}$$

$$\mathbb{E}_\varepsilon[W_\chi(C)] = \frac{1}{\mathcal{Z}_0} N_1(G^{E_1}) = \frac{1}{\mathcal{Z}_0} \int W_1(C;V)\,\mathrm{d}\rho_1(V)\ . \tag{1.8}$$

Both integrals retain the sum over small-field and large-field regions. Conditional fluctuation integrals used below have their own normalising factors; those factors remain inside the construction of N_1 and ρ_1.

1.2 The Main Result

The main statement of the paper concerns $\mathbb{E}_\varepsilon[W_\chi(C)]$ for a square contour: after the first step the expectation is a perimeter factor times a corner factor, times the expectation, in the coarse ensemble enriched by the large-field branch data, of $\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n; U_1)$: the fine loop at the background U_1 reconstructed from the coarse field and the exterior fine field, with a finite-geometry exponent and the indicator of the covered event, plus two error terms. The objects are defined in Sections 3–8 and the appendices; they are listed here so that the statement can be read on its own, and nothing in the list is assumed. Theorem 1.8 gives the same form for the product of the Wilson functions of two vertex-disjoint squares in planes parallel to that of e_1, e_2, with the product of the two coefficients, the two finite-geometry exponents and a cross exponent bounded in terms of the distance between the squares; its objects specific to the pair are defined in Section 10 and referenced from its statement.

Notation. C_n is a coarse translate of the counterclockwise square with vertices 0, nLe_1, $nL(e_1+e_2)$, nLe_2 in fine-lattice units (Section 7.4), $n \geqslant 1$; it has $n_C = 4nL$ contour bonds, $4n$ coarse segments indexed cyclically by $j \in \{0, \dots, 4n-1\}$, and $P(C_n)$ is its perimeter as it enters (8.1) (Section F.1). $W_\chi(C; U)$ is the normalised character χ of the holonomy of the fine field U around C, ε the fine lattice spacing, and σ names the cutoff norm of (4.12), the operator or the normalised Hilbert–Schmidt norm (conventions of Section 3). T is the operator (1.3) with the averaging operation (1.4); ρ_1, N_1, W_1 and $\mathcal{Z}_0$ are (1.5), (1.6) and (1.7), and $\mathbb{E}_\varepsilon$ is the bare expectation defined after (1.3). $\xi = (\Omega_1, R_1, V, V_0, A_{\mathrm{e}})$ is the branch data of Section 3.1 (the small-field region Ω_1, the large-fluctuation region R_1, which with Ω_1 determines the filled region Λ_1, the coarse field, the exterior fine field and the exterior free fluctuation; at the original carrier there are no labels), $U_1 = U_1(V, V_0)$ the background of Section 3.1, $\mathbb{E}_{\mathrm{h}}$ the history expectation (D.20) (weight $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$, normalisation $\mathcal{Z}_0$; positive, with $\mathbb{E}_{\mathrm{h}}[1] = 1$), and A_{cov} the covered event of Lemma 3.28. $\kappa_{C_n} = e^{-\delta m^\sigma(g_0,\varepsilon)P(C_n)}(z_{\mathrm{c}}^\sigma(g_0, \pi/2))^4$ is the scalar (8.1), with the perimeter coefficient δm^σ of (5.39) and the corner factor z_{c}^σ of (6.12), in the cutoff prescription σ fixed throughout. E_{def} is the signed-defect term (E.1), E_{bdry} the exterior-response and uncovered term (D.21), and $E^\circ_\xi = E^\circ_{I,C_n,T}$ the finite-geometry exponent (F.4) of Proposition F.2, a function of the history through the carrier $I = \Lambda_1^*$ alone. $d_j = \max(0, d_T(K_j, I^c) - R_0)$ and $\iota_T = \max(0, N_{\min}/4 - R_0)$ are the distances (F.1): K_j the union of the free-input stencils of the fine traversals of segment j, I^c the complement of the carrier in the torus free carrier, d_T the periodic fine-bond metric, $N_{\min}$ the shortest fine period, and $R_0 = 4(1+R_*+R_H+L)$ the fixed range of Section 7 (R_* the perimeter range, not the walk range n_*R of Section B.8; this R_0 is not the scale R_0 of [3]). The scales of the statement are τ, the marked-forest parameter of (5.18), $\tau = C[g_0(1+r)^3 + e^{-cr^2}]$ with the cutoff radius $r = A_1(\log g_0^{-2})^{p_0}$ of (4.12); $\pi_* = \min(p_0(g_0), p_0(g_1))$ with $p_0(g) = A_0(\log g^{-2})^{p_0}$ and $g_1 = g_1(g_0)$ the first-step coupling of [3, (1.27)]; $\varepsilon_1 = g_1 p_0(g_1)$; ρ of (E.11), $\rho = O(g_0^2(1+r)^4)$; and $b = \lfloor LMR_1/16 \rfloor$ with the scale $R_1 = R_1(g_0)$ of [3] (Section B.8). The constants C, c of Proposition F.2, C'_L, c_* of (E.3) and Proposition D.15, C_χ, c of (E.24), $C_{\mathrm{resp},\chi}$ of (D.41) (the constant after the replacement made there, bearing the name of the constant of Corollary D.8 it is built from), and c_q of (9.6) ($c_q = c_*$) are fixed before the coupling and depend on L, the group, the character, M_1, M_2, M, the fixed expansion and chart choices and the constants A_0, A_1, p_0, B_3, β_0 of [3] only (the last through c_θ of (D.26), hence $c_* = c_q$, which involve $\delta_0 = (A_1/A_0)\varepsilon_0$, B_3 and the bound [3, (2.6)] on g_1/g_0), not on g_0, n, the torus or the history; the constants C, c of different statements are distinct and are named by their statement; $K = 256$ is the fixed constant bounding, in units of $\ell = LMR_1$, the distance of a witness from its tested bond (Sections D.7 and 9.3, Lemma 9.45; distinct from the periods K_μ and the stencils K_j), and $C_K = 4(4K\ell + 9)^4$ depends on g_0 through the $R_1 = R_1(g_0)$ of [3], a power of L with $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$; δ_q^I of (9.53) depends on g_0 (through τ, C_K and the

one-point rate q of Section D.7), not on n, the torus or the history, and is smaller than every power of g_0 (Remark 1.3(1)).

Hypotheses. (H1) G is a compact connected simple Lie group in its fixed unitary realisation $G \subset U(N)$ and χ a normalised irreducible character (the opening of this section and the conventions of Section 3). (H2) The parameters satisfy the standing conditions (3.1), the constants are fixed in the order of Section B.8, and g_0 lies below the threshold of Section B.8: its items 1–6 and the further conditions listed there, among them the threshold g_* of Proposition D.15 (which contains that of Proposition D.17), the thresholds of Proposition E.22 (those of Lemmas E.1, E.9 and E.10, $R_\Upsilon \geqslant 2$, $u \leqslant u_*$, $v \leqslant 1$ and the small-coupling conditions of Sections E.9.1 and E.10.5), the common small-coupling thresholds of Proposition 7.3 and (9.39)(ii), the thresholds of Lemmas D.12 and D.14 (required already for (ii), through g_*) and the further threshold of (9.53); Corollaries 1.4 and 1.5 use in addition the further threshold stated in Corollary 1.4. (H3) The torus is one of those of [4] (p.251, where the period is $2L^m$ in physical units with the lattice spacing $\varepsilon = L^{-K}$, so that the exponent of [4] is $m - K$ in terms of the m here; the family of tori is the same): the fine periods are $N_\mu = 2L^m$ fine-lattice units in every direction, with the integer m so large that N_μ is a multiple of $\ell = LMR_1$ (a power of L: M by (3.1), R_1 by [3, p. 246]; the periods are then compatible and even, Section D.7) exceeding 200ℓ (Section 9.4.1), and every coarse period $K_\mu = N_\mu/L = 2L^{m-1}$ satisfies $K_\mu \geqslant 3(n+2)$ and exceeds $\max(2, 2R_{\mathrm{p}} + 8)$ ((3.1)); for (iii) in addition $N_{\min} \geqslant 4R_0 + (4/c)\log(4n)$ with the c of Proposition F.2 ((9.39)(iii)). These are lower bounds on m in terms of the fixed loop and the coupling; nothing is let tend to infinity. The collar conditions (7.28) concern M, M_2 and R_1, not m, and are among the thresholds of (H2) (item 1 of Section B.8 and (9.39)(i)). The results of Section D.7 and of the later sections are stated for general compatible even periods and are used here for these. (H4) Bałaban's results are used as stated and cited, in the readings recorded where [3] fixes no convention: the branch hypotheses (B1)–(B4) of Section 3.1 with Bałaban's derivation of [3, (1.11)–(1.25)] as recorded there (the constants that Bałaban does not display are Bałaban's and are independent of the coupling and of the branch: B_3 depends on d and L only [7, Theorem 1, p. 279]; the $O(L^2)$ of p.247 behind the domain statement of (B1) and item 6 of Section B.8 is that of the reasoning of [9, Lemma 1, p. 79], which [3] invokes there and whose bound $|V' - 1| < 4d^2\alpha_0 + \alpha_1$ has absolute constants; the $O(1)$ of (B3)(iii) is obtained in the general step of [3, p. 268, (3.17)–(3.19)], from $44d^2B_3$ and [8, (106)–(108), (159)–(163)], whose constants are displayed there; the bound on $\mathbf{H}_{1,Ax}$ of p.251, behind the condition "MR_1 large" of item 5 of Section B.8, is derived in Lemma D.18 from [7, Proposition 9, (172), and (190)] and the block-local restoration of the axial gauge, with constants fixed before g_0; the $O(1)$ of (B1)(iv) is displayed in [8, p. 25], $d(2d+1)$; and the decay rate δ_{c} of the bound of (B3)(iii) is that of the exponential decay property [7, (190)], which Bałaban invokes in [3, p. 266] for the functions $\mathbf{H}_{k,\square}$ and Lemma D.18 uses for $\mathbf{H}_{1,Ax}$: in the Landau gauge it is the rate $\delta_{\mathrm{P}}/8$ of [7, (190)] in the distance $d(y, y')$ of [6, (2.46)], whose scale weights are fixed powers of L at the first step (Section D.8.3), with δ_{P} the common decay rate defined in [10, p. 423] after Theorem 3.12, "a positive, absolute constant depending on d and L only", for every background satisfying [10, (3.35)–(3.36)], a class that contains the minimal configurations by [7, Theorem 1, (9)–(10)]; the $\widetilde{Q}$ of [3] is LQ_{U_1}, the linear part of the constraint function [4, (2.4), p. 266] (decomposed as $L\widetilde{Q}B' + \widetilde{C}(B')$ at p.267), and the σ in [3, (1.14)–(1.15)] is the normalised Haar density σ/σ_0 of [11, (18), p. 260] ($\mathrm{d}U' = \sigma(A')\,\mathrm{d}A'$, $\sigma_0 = \sigma(0)$), the transformation to which Bałaban refers at the foot of [3, p. 248] (the step list of Section 3.1); the terms that [3] does not define are taken in the sense Bałaban gives them: the "domain of integration in (1.11)", p.248, is the set to which the characteristic factors of (1.11) restrict the integral, the support of every one of them (as [4, p. 265] defines its characteristic function "in such a way that the domain of integration in (2.1) is restricted to" a set of configurations),

under which $U_1 \in D^{(11)}(V, V_0)$, (B1)(iii), and Remark D.16; in (B1), $U_1(\Omega_1 \cap \Lambda_1^c, V)$ is the minimal configuration of (2.12) and (2.15) for the determining data named at p.251, as [3] writes the operation it calls similar to (1.21)–(1.23) at the general step, p.269, with data from both scales, so that (1.21) depends on (V, V_0); the rows of Ω_1 and of Ω_1^c (Section 3) are those of [11, Sect. A], which [3, p. 248] follows: a row with one block off Ω_1 is a row of Ω_1, and its average involves V_0 on the outer block as a constant ([11, (12), p. 258]: $U = V_0$ on Ω_1^c, $\bar{U} = V$ on $\Omega_1^{(1)}$; the δ-functions of Ω_1^c are those of the rows $b' \subset \Omega_1^c$, p.259), which is the form of (B2) on such rows; the $X^\sim$ of [3] adds every cube meeting X, at a face, an edge, or a vertex ([4, p. 257]: $\tilde{\square}^n$ is the cube of side $(1 + 2n)M$ with the centre of $\square$); the M_1 of [3] is a lower bound on the cube side ([10, Theorem 3.1, p. 397] holds "for $M \geqslant M_1$", and M_1 "was fixed in" that paper, [9, p. 77]); the "compatible" partitions of [3] (p.245, where the partitions introduced there "have to be compatible", also with π_0 and with the partitions introduced later, and p.257, "compatible with all other partitions") are nested, the cubes of the coarser of two compatible partitions being unions of cubes of the finer, the sense of [9, p. 98], "compatible with the block structure of the lattice T, i.e. for every j the cube $\square_j$ is a sum of the big blocks" (Section B.2); "regular" in (2.12) is membership in the class [7, (6)], to which Bałaban refers in [3, p. 256] for the precise definitions, with any radius allowed by [7, Theorem 1], which gives the same minimal orbit for each (Lemma D.11); and the $\widetilde{D}$ in [3, (1.13)–(1.14)] is the solution of the row equations [11, (17), pp. 259–260], the transformation to which Bałaban refers in [3, p. 248], solved "separately for each function $Q(A', c)$", with "a unique solution ... for A sufficiently small", analytic with a Taylor expansion beginning with second order terms, which is the small solution φ of (B4)(i) and the only chart-class family whose fine field lies in $D^{(11)}(V, V_0)$ (existence, uniqueness, analyticity, and the positivity of the Jacobian (B4)(iii) are proved in Lemma B.8, from (B1) (with (B3)(i), through (B1)(i)) and (B2) with the readings listed here, the fine periods being at least $8M$, and threshold items 5 and 6); the readings that are this paper's are marked as such where they are made, and they are the following: the bonds crossing from Ω_1 to Ω_1^c taken as variables of the integral over Ω_1 (Section 3), the rule of [11, (8), p. 258] for plaquettes, "plaquettes with at least one corner belonging to Ω_1", applied to bonds; this paper's convention $\chi_1(\square') = 0$, $\chi_1^c(\square') = 1$, $\chi_0'(\square') = 0$, $\chi_0'^c(\square') = 1$ where the configuration $U_{1,\square'}(V)$ does not exist, which keeps (1.4) and (1.8) partitions of unity; three corrections of the printed text of [3]: v read from (1.14) where [3, p. 249] points to "(1.15)" (before (8.5)), the "$z \neq y$" in [3, (1.5)] read as $x \neq y$, as in (1.7) ((8.7)), and the first member of the chain of [3] $\delta_0 = g_0 A_1 p_0(g_0) = (A_1/A_0)\varepsilon_0$, p.247, read as $g_0(A_1/A_0)p_0(g_0)$, the value of the last member and of $\delta_k = g_k A_1/A_0\, p_0(g_k)$ at p.265, which fixes the radius $r = \delta_0/g_0$ at which κ_{C_n} is identified (Section B.8); the domain of the integral [3, (1.11)] is otherwise Bałaban's: its averages and means are the principal logarithms of [8, p. 21, (22)–(25)]; on the whole domain of integration, the designated bonds included, the fluctuation lies in the chart ball of the conventions paragraph of Section 3 by [3, pp. 247–248] (Bałaban's statements, justified there by the reasoning of Lemma 1 of [9]), and the arguments of the averages and means lie there by [8, pp. 24–25, (44)–(46)] with (B3)(i), p.248, both derived in (B1)), [4, (0.4) and (0.10)–(0.11)] for the averaging operation and the group means, Bałaban's transfer statements [3, p. 256] (all the results of [7] used for the averaging and determining sets of [3]), [4] after (0.10) and after (0.12), through which Section D.8 takes over the constraint [7, (3)] and [8, Proposition 1, (51)] for the averaging [4, (0.4)], the results of [7] for the determining sets $\mathbf{B}_1(Q_w^{[4]})$ and $\mathbf{B}_1(\Lambda_1^c)$ of [3] in place of the spacing condition [7, (1)] for their fine layers, and the restoring bound (D.48), for the local minima and for the function of (1.21) ([3, p. 269]), [3, (1.5)–(1.7)] for the gauge-fixing functional, [3, (1.27)] for g_1, p.246 for R_1 and ε_1, and [3, (2.4)–(2.9)] for the relations between the scales R_j, g_j and ε_j (Section B.8), the conditions of [3] on the parameters as taken over in item 5 of Section B.8, and the validity of Bałaban's construction [3] for the values

$M_1 < M_2 < M$ fixed there following Bałaban's prescription (the conventions after (3.1)). No property of this paper's own objects is a hypothesis: every lemma and proposition cited below is proved in the paper under (H1)–(H4).

Theorem 1.2 (A Wilson Loop after the First Renormalisation Step)**.** *Under (H1)–(H4):*

(i) (Form.)

$$\begin{aligned} \mathbb{E}_\varepsilon[W_\chi(C_n)] &= \kappa_{C_n}\,\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)\big] + E_{\mathrm{def}} + E_{\mathrm{bdry}}\ , \\ \kappa_{C_n} &= e^{-\delta m^\sigma(g_0,\varepsilon)P(C_n)}\big(z^\sigma_{\mathrm{c}}(g_0,\pi/2)\big)^4 \neq 0\ , \end{aligned} \tag{1.9}$$

where the perimeter coefficient δm^σ and the corner factor z^σ_{c} depend on g_0, ε, L, the group, the character and σ only, the fixed averaging and mean prescriptions of [4, (0.4) and (0.10)–(0.11)], the gauge-fixing functional of [3, (1.5)–(1.7)], the designated-bond convention (Section F.1) and the cutoff test (1.18) at the radius $r = A_1(\log g_0^{-2})^{p_0}$ (the constants A_1, p_0 of [3], with its δ_0 as corrected in Section B.8 and (H4)) being part of the setting, so that κ_{C_n} depends on n through the perimeter $P(C_n)$ alone and not on the torus or the history, and every term is integrable against the history weight.

(ii) (Bounds.)

$$\begin{aligned} |E^\circ_\xi| &\leqslant C\tau^2\Big[\sum_{j=0}^{4n-1} e^{-cd_j} + 4ne^{-c\iota_T} + (n+1)e^{-cn}\Big] \leqslant C\tau^2(9n+1) \quad \text{on } A_{\mathrm{cov}}\ , \\ |E_{\mathrm{def}}| &\leqslant C'_L n_C e^{-(c_*/2)\pi_*^2} + e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1\ , \\ |E_{\mathrm{bdry}}| &\leqslant C_{\mathrm{resp},\chi}\, n_C(1+g_0 r)\, e^{-c_*\pi_*^2}\ . \end{aligned} \tag{1.10}$$

(iii) (Size of the leading term.) $|\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)]| \leqslant e^{C\tau^2(9n+1)}$, and under the additional hypotheses of (H2), (H3) for (iii),

$$\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}} e^{|E^\circ_\xi|}\big] \leqslant \exp\big[C\tau^2(2+1/c) + n_C\delta_q^I\big]\ , \qquad \delta_q^I \leqslant \exp\big[-c_q\pi_*^2/(2C_K)\big]\ , \tag{1.11}$$

so that the leading term $\kappa_{C_n}\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)]$ of (1.9) has modulus at most $|\kappa_{C_n}|\exp[C\tau^2(2+1/c)+n_C\delta_q^I]$, and $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^\circ_\xi}-1|] \leqslant \exp[C\tau^2(2+1/c)+n_C\delta_q^I]-1$.

Proof. (i) is Corollary F.3, whose hypotheses are the conventions of Section F.1 and those of Proposition 7.3 (the square geometry and collar conditions of (H3), the thresholds of (H2)); $\kappa_{C_n} \neq 0$ is an exponential by (5.39) and (6.12), as stated in Corollary 9.43(a) and after (8.18), δm^σ is the absolutely convergent sum (5.39), characterised there as a limit, and z^σ_{c} the exponential (6.12) of a convergent corner sum, from both of which n and the torus are absent. In (ii) the first bound is Proposition F.2 with the elementary bounds ($e^{-cd_j} \leqslant 1$, $e^{-c\iota_T} \leqslant 1$, $(n+1)e^{-cn} \leqslant n+1$); the second is (E.79) of Proposition E.22, whose hypotheses are those stated at the start of Section E.11 (Section 3, Appendix B, the conventions of Section E.8.2) and the thresholds named in (H2); the third is (D.41) of Proposition D.15, under compatible even periods and (D.2) (a further condition of Section B.8). In (iii) the first bound is the positivity of $\mathbb{E}_{\mathrm{h}}$ (D.20), $\mathbb{E}_{\mathrm{h}}[1] = 1$, $|W_\chi| \leqslant 1$ and the crude bound of (ii); (1.11) and the two consequences are (9.53) of Corollary 9.47(c) and the sentence following it, from Corollary 9.43(c) applied to E°_ξ (whose bound has the form (9.40)) under $4ne^{-c\iota_T} \leqslant 1$, and from (9.48) of Lemma 9.46 at $\lambda = C\tau^2$, whose joint witness bound (9.44) (Lemma 9.45) rests on Section D.7. The preamble of Corollary 9.47 ((9.39) and the hypotheses of Proposition 9.34, that is those of Proposition 9.28 and (9.29), and of Lemma 9.44, the setting

of Proposition 9.39) consists of the square geometry and period conditions of (H3) and of thresholds of Sections 9.3–9.6 listed among the further conditions of Section B.8, so it is implied by (H2) and (H3); its item (c) at $D = I$ states the conditions it uses. *q.e.d.*

Remark 1.3 (Regime of the Statement). (1) The scales: as $g_0 \to 0$, $\tau = C[g_0(1+r)^3 + e^{-cr^2}] \to 0$ (with the single radius $r = \delta_0/g_0 = A_1(\log g_0^{-2})^{p_0}$ of the notation paragraph and Section B.8, the radius of the cutoff test Θ^σ as well), while $e^{-c_*\pi_*^2}$, $e^{-(c_*/2)\pi_*^2}$ and δ_q^I are smaller than every power of g_0 ($\pi_*^2 \geqslant A_0^2 2^{-2p_0}(\log g_0^{-2})^{2p_0}$ by the bounds (9.9) as used in Corollary 9.47(c), with $p_0 \geqslant 5\mathfrak{r} \geqslant 10$, and for δ_q^I the last member of (1.11) with the arithmetic preceding (9.52), $C_K \leqslant C(L^2M)^4 2^{4\mathfrak{r}}(\log g_0^{-2})^{4\mathfrak{r}}$ and $2p_0 - 4\mathfrak{r} \geqslant 12$), and $\rho\varepsilon_1 \to 0$. The n-dependence of the bounds is explicit: pointwise on A_{cov}, E_ξ° is at most linear in n; under the torus condition $4ne^{-c\iota_T} \leqslant 1$ the growth is carried by the $4n$ boundary terms e^{-cd_j}, each at most one and exponentially small when the carrier complement is far from segment j, while the remaining terms $4ne^{-c\iota_T} + (n+1)e^{-cn}$ of (ii) are at most $2 + 1/c$ (Corollary 9.43(c), applied to E_ξ° as in the proof above); in history average the exponential of $|E_\xi^\circ|$ carries the perimeter-rate factor $e^{n_C\delta_q^I}$ of (iii), where δ_q^I is governed by the C_K-th root q^{1/C_K} of the one-point rate q of Section D.7, while Corollary 1.4 below separates the deterministic torus and corner part of E_ξ° and pays its carrier part at the rate $q^{1/16}$, which gives (iii) with $\delta_q^\sharp$ in place of δ_q^I; E_{bdry} and the first term of the bound (1.10) on E_{def} are perimeter rates of the same kind; and the second term of that bound is small only when $n_C\rho\varepsilon_1 \ll 1$ ($e^x - 1 \geqslant x$). The bounds (1.10) on E_{def} and E_{bdry} are absolute, not relative to $\kappa_{C_n} = e^{-\delta m^\sigma P(C_n)}(z_{\mathrm{c}}^\sigma)^4$; whether they lie below $|\kappa_{C_n}| = e^{-\delta m^\sigma P(C_n)}|z_{\mathrm{c}}^\sigma|^4$ (δm^σ is real, by the argument following (5.39)) depends on the sizes of δm^σ and z_{c}^σ, for which Sections 5 and 6 give two-sided bounds: $|\kappa_v^\sigma| \leqslant C\tau^2$ (the sentence following (5.33), by (5.31), with (5.39)) and $|\zeta_{v,w}^\sigma| \leqslant C\tau^2$ ((6.5), Section 6) give $|\delta m^\sigma|P(C_n) = n|\sum_v \kappa_v^\sigma| \leqslant 4Cn\tau^2$ and $e^{-4C\tau^2} \leqslant |z_{\mathrm{c}}^\sigma|^4 \leqslant e^{4C\tau^2}$, hence $e^{-4C\tau^2(n+1)} \leqslant |\kappa_{C_n}| \leqslant e^{4C\tau^2(n+1)}$, with C independent of g_0, n and the torus (an upper bound is also at hand from the expectation: at any configuration of A_{cov}, $|\kappa_{C_n}| = |S_{\xi,C_n}^\circ|e^{-\Re E_\xi^\circ} \leqslant e^{C\tau^2(9n+1)}$ by (ii) and $|S_{\xi,C_n}^\circ| \leqslant 1$, the proof of Proposition E.22): E_{bdry} and the first term of the bound on E_{def} are smaller than every power of g_0 at fixed n, and the ratios of their bounds to $|\kappa_{C_n}|$ are, respectively, $C_{\mathrm{resp},\chi}n_C(1+g_0r)e^{-c_*\pi_*^2}/|\kappa_{C_n}|$ and $C_L'n_Ce^{-(c_*/2)\pi_*^2}/|\kappa_{C_n}|$, below one exactly when $\log(C_{\mathrm{resp},\chi}n_C(1+g_0r)) + \delta m^\sigma P(C_n) + 4\log|z_{\mathrm{c}}^\sigma|^{-1} < c_*\pi_*^2$, respectively $\log(C_L'n_C) + \delta m^\sigma P(C_n) + 4\log|z_{\mathrm{c}}^\sigma|^{-1} < (c_*/2)\pi_*^2$, conditions that involve the length of the loop through n_C and $P(C_n)$; by the lower bound on $|\kappa_{C_n}|$ they hold whenever $\log(C_{\mathrm{resp},\chi}n_C(1+g_0r)) + 4C\tau^2(n+1) < c_*\pi_*^2$, respectively $\log(C_L'n_C) + 4C\tau^2(n+1) < (c_*/2)\pi_*^2$, that is for n up to a fixed multiple of π_*^2/τ^2 at small coupling, and for every fixed n once g_0 is small ($\tau \to 0$, $\pi_* \to \infty$); when $\delta m^\sigma > 0$ the exact conditions fail at fixed g_0 for all large n, which the hypotheses allow on correspondingly large tori. The second term of the bound on E_{def}, at most $2x$ with $x = C_\chi n_C\rho(\varepsilon_1 + e^{-cb/(LM)})$ when $x \leqslant 1$ ($e^x - 1 \leqslant 2x$ for $0 \leqslant x \leqslant 1$), is bounded only by a quantity polynomially small in g_0 at fixed n ($\rho = O(g_0^2(1+r)^4)$); its comparison with $|\kappa_{C_n}|$ at the original carrier is made by the relative form of (E.78): the far-history term there retains $|S_{\xi,C}^\circ| = |\kappa_{C_n}||e^{E_\xi^\circ}|$ on A_{cov} (F.3), so by (1.11) (the far-history term is the $\mathbb{E}_{\mathrm{h}}$-average, with the weight w_ξ and the normaliser $\mathcal{Z}_0$ of (iii), of $\mathbf{1}_{A_{\mathrm{cov}}\cap\mathsf{N}_b^{\mathrm{c}}}|S_{\xi,C}^\circ|(e^x - 1) \leqslant \mathbf{1}_{A_{\mathrm{cov}}}|\kappa_{C_n}|e^{|E_\xi^\circ|}(e^x - 1)$) it is at most $|\kappa_{C_n}|e^{C\tau^2(2+1/c)+n_C\delta_q^I}(e^x - 1)$, a relative bound, also with $\delta_q^\sharp$ in place of δ_q^I (Corollary 1.4); only the near-history term $C_L'n_Ce^{-(c_*/2)\pi_*^2}$ is absolute. (2) At the retreated carrier of the labelled history terms, Section 9 gives the same identification $S_\Lambda^\circ = \kappa_C e^{E_\Lambda^\circ}$ on $A_{\mathrm{cov}}^{\mathrm{fin}}$ (Corollary 9.43(a)), the bound on the source-containing boundary terms relative to the leading coefficient (Corollary 9.47(a); its factor $e^{C\tau^2(9n+1)}$ is exponential in n at fixed g_0, so that bound is small at fixed n_C only) and the bound (9.36) on the labelled defect (Proposition 9.40; its near-history part $C_L'n_Ce^{-(c_*/2)\pi_*^2}$ carries no factor $|\kappa_C|$ and is absolute,

while its far-history part retains $|S^\circ_\Lambda| = |\kappa_C||e^{E^\circ_\Lambda}|$ on $A^{\mathrm{fin}}_{\mathrm{cov}}$ and is relative by Corollary 9.43(a) and the exponential moment (9.50) of Corollary 9.47(c), as at the original carrier in (1)), and the first and exponential moments of E°_Λ (Corollary 9.47(b),(c)). (3) Not proved here: a bound on the history average of $e^{E^\circ_\xi}$ literally independent of n (the proved form is the perimeter rate of (iii)); a lower bound on $|\kappa_{C_n}|$ uniform in n ($|\kappa_{C_n}| = e^{-\delta m^\sigma P(C_n)}|z^\sigma_{\mathrm{c}}|^4$ is bounded below uniformly in n if and only if $\delta m^\sigma \leqslant 0$, the sign of δm^σ is not determined here, and the lower bound $e^{-4C\tau^2(n+1)}$ of (1) decays in n); a lower bound on the modulus of the leading expectation $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)]$ uniform in n (Corollary 1.5 below gives one at a fixed loop, with an error $\eta_n = a_n + e^{n_C\delta^\sharp_q} - 1$ whose leading part for $n \leqslant \varepsilon_1^{-1}$ and nontrivial χ is the area term $2\mathrm{Lip}_\chi c_{\mathrm{h}} n^2\varepsilon_1$, and when $\eta_n < 1$ "below $|\kappa_{C_n}|$" in (1) compares with the leading term up to the factor $(1-\eta_n)^{-1}e^{C\tau^2(2+1/c)}$); and the second renormalisation step: neither the coupling nor the scale is iterated, and the output of the step is the pair of the complex measure $\kappa_{C_n}^{-1}N_1$ and the measure ρ_1 as described in Section 8, not a statement about a coarse-lattice Wilson loop against a local effective density (Conjecture 11.1 of Section 11 states the form expected after $k+1$ steps for $n = L^k$).

Corollary 1.4 (Factorisation of the Leading Term)**.** *Under (H1)–(H4), including the additional hypotheses of (H2) and (H3) for (iii) of Theorem 1.2, write the exponent* (F.4) *on* A_{cov} *as* $E^\circ_\xi = E^\partial_\xi + E^\circ_{T,n}$*, with the carrier part and the torus and corner part*

$$E^\partial_\xi = F_{I,C,T} - F_{C,T}\,, \qquad E^\circ_{T,n} = (F_{C,T} - F_{C,\infty}(n)) + E^\sigma_n \tag{1.12}$$

in the notation of Proposition F.2. Then $E^\circ_{T,n}$ *does not depend on the history,* $\kappa_{C_n}e^{E^\circ_{T,n}} = \mathcal{W}^{\circ,\sigma}_T(C_n)$ *is the flat reference loop* (5.5) *of* C_n *on the torus, and*

$$\begin{aligned} \kappa_{C_n}\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_\xi}W_\chi(C_n;U_1)\big] &= \mathcal{W}^{\circ,\sigma}_T(C_n)\big(\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}}W_\chi(C_n;U_1)\big] + R_n\big)\,, \\ |E^\circ_{T,n}| &\leqslant C\tau^2\big[4ne^{-c\iota_T} + (n+1)e^{-cn}\big] \leqslant C\tau^2(2+1/c)\,, \\ |R_n| \leqslant \mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^\partial_\xi}-1|\big] &\leqslant e^{n_C\delta^\sharp_q}-1\,, \qquad \delta^\sharp_q \coloneqq C\tau^2 N'_{\mathrm{lat}}(c)\,e^{C\tau^2L^4C_K\Phi_c}\,q^{1/16}\,, \end{aligned} \tag{1.13}$$

with C*,* c *of Proposition F.2,* $N'_{\mathrm{lat}}(c)$ *and* Φ_c *of Lemma 9.46 (they depend on* L *and* c *only,* (9.45)*),* C_K *of Lemma 9.45 and the one-point rate* $q \leqslant e^{-c_q\pi_*^2}$ *of Section D.7 (*$g_0 < g_*$*, as in* (9.53)*). For* g_0 *below a further threshold, depending on* L*,* M*,* M_1*,* M_2*, the group, the character, the source constants and* c*,* $C\tau^2L^4C_K\Phi_c \leqslant 1$ *and* $\delta^\sharp_q \leqslant eC\tau^2N'_{\mathrm{lat}}(c)\,e^{-c_q\pi_*^2/16}$*: the carrier part is paid at a fixed fraction of the exponent of the one-point rate, with no loss growing as* $g_0 \to 0$*. Moreover* $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}e^{|E^\circ_\xi|}] \leqslant \exp[C\tau^2(2+1/c) + n_C\delta^\sharp_q]$*, the bound* (1.11) *of (iii) with* $\delta^\sharp_q$ *in place of* δ^I_q*. Consequently* $\mathbb{E}_\varepsilon[W_\chi(C_n)] = \mathcal{W}^{\circ,\sigma}_T(C_n)(\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}W_\chi(C_n;U_1)] + R_n) + E_{\mathrm{def}} + E_{\mathrm{bdry}}$ *by* (1.9)*.*

Proof. The decomposition is (F.4). In the proof of Proposition 7.3 at $D = I$, from which Proposition F.2 is obtained, the three parts are bounded separately: $|E^\partial_\xi| \leqslant C\tau^2\sum_{j=0}^{4n-1}e^{-cd_j}$ by (7.44), $|F_{C,T} - F_{C,\infty}(n)| \leqslant C\tau^2(4n)e^{-c\iota_T}$ by (7.46), and $|E^\sigma_n| \leqslant C\tau^2(n+1)e^{-cn}$ by (6.11); the bound of Proposition F.2 is their sum, with the largest of the three constants and the smallest of the three rates; these are the C, c of that proposition fixed in the notation paragraph, the c of (H3) and (9.39)(iii) is this one, and each of the three bounds holds with them. $F_{C,T}$ is the logarithm, on the connected source branch, of $\mathcal{W}^{\circ,\sigma}_T(C_n) = N^\sigma_{C_n,T}/D^\sigma_T$ (5.5), $F_{C,\infty}(n)$ is the bulk logarithm (6.9) and E^σ_n the corner-extraction error; none of them involves the carrier I or any other datum of the history, so $E^\circ_{T,n}$ is a number fixed by g_0, n, the torus and the setting. By the identity $F_{I,C,T} = \log\kappa_{C_n} + E^\circ_\xi$ of Proposition F.2, $\log\kappa_{C_n} + E^\circ_{T,n} = F_{I,C,T} - E^\partial_\xi = F_{C,T}$, whence $\kappa_{C_n}e^{E^\circ_{T,n}} = \mathcal{W}^{\circ,\sigma}_T(C_n)$. The second form of

the bound on $E^\circ_{T,n}$ uses $4ne^{-c\iota_T} \leqslant 1$, which is (9.39)(iii) in (H3), and $(n+1)e^{-cn} \leqslant 1 + 1/c$. Since $E^\circ_{T,n}$ is a constant and the integrands are bounded ($|E^\partial_\xi| \leqslant 4C\tau^2 n$ on A_{cov}, $|W_\chi| \leqslant 1$), $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)] = e^{E^\circ_{T,n}} \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\partial_\xi} W_\chi(C_n;U_1)]$, and the identity holds with $R_n = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}(e^{E^\partial_\xi} - 1) W_\chi(C_n;U_1)]$. By the positivity of $\mathbb{E}_{\mathrm{h}}$ (D.20), $|W_\chi| \leqslant 1$, $e^{|x|} - 1 \geqslant |e^x - 1|$ and the pointwise bound of (9.48), with $\lambda = C\tau^2$ and the marks $m_b = \mathbf{1}\{b \in E(B_\xi)\}$, $|R_n| \leqslant \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} |e^{E^\partial_\xi} - 1|] \leqslant \mathbb{E}_{\mathrm{h}}[\exp(\lambda \sum_b \varphi_c(b) m_b)] - 1$, using $\mathbb{E}_{\mathrm{h}}[1] = 1$. Let L^p be the smallest power of L exceeding $\sigma_K = 2K\ell + 4$ (Lemma 9.45), so $L^p \leqslant L\sigma_K$, and put $s_\mu = \min(L^p, N_\mu/2)$; N_μ/s_μ is even, as $N_\mu = 2L^m$ by (H3). Partition the sites of the torus into the boxes $Q = \prod_\mu [k_\mu s_\mu, (k_\mu + 1)s_\mu)$, $k_\mu \in \mathbb{Z}/(N_\mu/s_\mu)$, and colour Q by $(k_\mu \bmod 2)_\mu$, sixteen colours. Two distinct boxes of one colour differ in some direction μ by a nonzero even index difference modulo N_μ/s_μ, which forces $N_\mu/s_\mu \geqslant 4$ and $s_\mu = L^p$, so their sites are at periodic ℓ^∞ distance at least $L^p > \sigma_K$. For a finite set S of bonds let $N(\mathsf{S})$ be the number of boxes containing the initial site of a bond of S; one bond of S in each such box of the most frequent colour gives a subset S' at pairwise ℓ^∞ distance $> \sigma_K$ with $|\mathsf{S}'| \geqslant N(\mathsf{S})/16$, so the first inequality of (9.44) and $q \leqslant 1$ give $\mathbb{E}_{\mathrm{h}}[\prod_{b\in\mathsf{S}} m_b] \leqslant q^{N(\mathsf{S})/16}$. Expand $\exp(\lambda \sum_b \varphi_c(b) m_b) = \prod_b (1 + y_b m_b)$, $y_b = e^{\lambda\varphi_c(b)} - 1 \geqslant 0$, into the sum over S of $\prod_{b\in\mathsf{S}} y_b m_b$, as in the proof of Lemma 9.46 (finitely many nonnegative terms); since $N(\mathsf{S}) = \sum_Q \mathbf{1}\{\mathsf{S} \text{ has a bond with initial site in } Q\}$, the sum factorises over the boxes:

$$\begin{aligned}\mathbb{E}_{\mathrm{h}}\Big[\prod_b (1 + y_b m_b)\Big] &\leqslant \sum_{\mathsf{S}} \prod_{b\in\mathsf{S}} y_b\, q^{N(\mathsf{S})/16} = \prod_Q \Big(1 + q^{1/16}\Big(\prod_{b\in Q}(1+y_b) - 1\Big)\Big) \\ &\leqslant \exp\Big(q^{1/16} \sum_Q \big(e^{\lambda \sum_{b\in Q}\varphi_c(b)} - 1\big)\Big)\,,\end{aligned}$$

the products and sums over $b \in Q$ running over the bonds with initial site in Q. A box has at most $L^{4p} \leqslant L^4\sigma_K^4$ sites, hence at most $4L^4\sigma_K^4 \leqslant L^4 C_K$ bonds, so $\lambda \sum_{b\in Q} \varphi_c(b) \leqslant \lambda L^4 C_K \Phi_c$ by (9.45); with $e^x - 1 \leqslant x e^x$ and $\sum_b \varphi_c(b) \leqslant n_C N'_{\mathrm{lat}}(c)$ (proof of Lemma 9.46) the exponent is at most $n_C \lambda N'_{\mathrm{lat}}(c) e^{\lambda L^4 C_K \Phi_c} q^{1/16} = n_C \delta_q^\sharp$. The same pointwise bound gives $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{|E^\partial_\xi|}] \leqslant \mathbb{E}_{\mathrm{h}}[\exp(\lambda \sum_b \varphi_c(b) m_b)] \leqslant e^{n_C \delta_q^\sharp}$, and with $|E^\circ_\xi| \leqslant |E^\circ_{T,n}| + |E^\partial_\xi|$ the moment bound follows. For the further threshold: $C_K \leqslant 4(4K+9)^4 (L^2 M)^4 2^{4\mathfrak{r}} (\log g_0^{-2})^{4\mathfrak{r}}$ and $\tau \leqslant C[g_0(1+r)^3 + e^{-cr^2}]$ with $r = A_1(\log g_0^{-2})^{p_0}$ (the comparison in Corollary 9.47(c)), so $\tau^2 C_K \to 0$ as $g_0 \to 0$, and $q^{1/16} \leqslant e^{-c_q \pi_*^2/16}$. The last statement is (1.9) with the identity of (1.13) substituted. *q.e.d.*

Corollary 1.5 (The Leading Term at a Fixed Loop)**.** *Under the hypotheses of Corollary 1.4, let π_χ be the irreducible unitary representation of G with $\chi = (\dim \pi_\chi)^{-1} \operatorname{Tr} \pi_\chi$, Lip_χ a constant with $\|\pi_\chi(g) - \pi_\chi(h)\|_{\mathrm{op}} \leqslant \mathrm{Lip}_\chi \|g - h\|_{\mathrm{op}}$ for $g, h \in G$ (it exists because π_χ is smooth and $G \subset U(N)$ is a compact submanifold, and it depends on the group and the character only), C_L the contour constant with $|\Gamma_C| \leqslant C_L n_C$ (Section D.7, the paragraph before Lemma D.9), depending on L only, and q the one-point rate of Lemma D.14. Then*

$$\begin{aligned}&\big|\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}} W_\chi(C_n;U_1)\big] - 1\big| \leqslant a_n \coloneqq 2\mathrm{Lip}_\chi c_{\mathrm{h}} n^2 \varepsilon_1 + \big(2(nL+1)^2 + C_L n_C\big) q\,, \\ &\kappa_{C_n} \mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)\big] = \mathcal{W}_T^{\circ,\sigma}(C_n)\big(1 + R_n^\sharp\big)\,, \qquad |R_n^\sharp| \leqslant \eta_n \coloneqq a_n + e^{n_C\delta_q^\sharp} - 1\,,\end{aligned} \tag{1.14}$$

with $c_{\mathrm{h}} = \sqrt{N}$, $\varepsilon_1 = g_1 p_0(g_1)$ and $\delta_q^\sharp$ of (1.13). Hence, when $\eta_n < 1$, the leading term of (1.9) is nonzero, $(1 - \eta_n)|\mathcal{W}_T^{\circ,\sigma}(C_n)| \leqslant |\kappa_{C_n} \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)]| \leqslant (1 + \eta_n)|\mathcal{W}_T^{\circ,\sigma}(C_n)|$, and $e^{-C\tau^2(2+1/c)}|\kappa_{C_n}| \leqslant |\mathcal{W}_T^{\circ,\sigma}(C_n)| \leqslant e^{C\tau^2(2+1/c)}|\kappa_{C_n}|$ with the C, c of Corollary 1.4. For every fixed n, $\eta_n \to 0$ as $g_0 \to 0$; at small coupling η_n is small for n up to a small multiple of $\varepsilon_1^{-1/2}$, and the bound is not uniform in n.

Proof. Let S_n be the set of the $(nL+1)^2$ fine sites of the closed square bounded by C_n in its plane, and $A_\square = \{S_n \subseteq \Lambda_1\}$, an event of the regions. If $A_\square$ fails, some $x \in S_n$ lies in $B_\xi = \Lambda_1^c$ and the positive bond $z_x = (x, x+e_1)$ lies in $E(B_\xi)$ (notation of Section D.7); so $\mathbf{1}_{A_\square^c} \leqslant \sum_{x\in S_n} \mathbf{1}\{z_x \in E(B_\xi)\}$, and $\mathbb{P}_\mathrm{h}(A_\square^c) \leqslant (nL+1)^2 q$ by $p_z \leqslant q$ (Lemma D.14) and the positivity of $\mathbb{E}_\mathrm{h}$ (D.20). On a branch in $A_\square$ every fine plaquette p of the square has its vertices in Λ_1, a union of blocks: the block of a vertex x lies in Λ_1, a bond with both endpoints in that block is free (a designated bond crosses between two blocks) and belongs to I, and its initial site is within ℓ^∞ distance $L-1$ of x, so every bond of p is within distance $L \leqslant 3R_\mathrm{c}$ of I, and Lemma B.4 gives $\|U_1(\partial p) - \mathbf{1}\|_\mathrm{op} \leqslant 2c_\mathrm{h}\varepsilon_1 L^{-2}$. Lattice Stokes: order the $(nL)^2$ plaquettes of the square by rows, starting at the base corner x_0 of C_n, the image of the vertex 0 of the square of the notation paragraph under the coarse translation that gives C_n (bottom row first, left to right in each row), and let Q_k be the union of the first k of them, whose boundary, oriented counterclockwise and based at x_0, is a closed lattice path; $Q_1 = p_1$ is the plaquette at x_0, for $k \geqslant 1$ the shared path s below does not contain x_0, and $\partial Q_{(nL)^2}$ is C_n (another base point or orientation replaces the holonomy h below by a conjugate or by h^{-1}, which changes neither $\|h-\mathbf{1}\|_\mathrm{op}$ nor $|\chi(h)-1|$, since $\chi(h^{-1}) = \overline{\chi(h)}$ and $\chi(\mathbf{1}) = 1$); adding the next plaquette p replaces the one or two bonds s it shares with ∂Q_k (its lower and left sides that lie in Q_k) by the complementary path s' of ∂p, and if the based boundary path of Q_k is $\alpha s \beta$, then $\alpha s' \beta$ is that of Q_{k+1} and its holonomy is $vU(\partial p)^{\pm 1}v^{-1}$ times the holonomy of $\alpha s \beta$, with v the holonomy of αs and ∂p based at the endpoint of s. For unitaries, $\|vwv^{-1}-\mathbf{1}\| = \|w-\mathbf{1}\| = \|w^{-1}-\mathbf{1}\|$ and $\|w_1w_2-\mathbf{1}\| \leqslant \|w_1-\mathbf{1}\| + \|w_2-\mathbf{1}\|$, so on $A_\square$ the holonomy h of U_1 around C_n satisfies $\|h-\mathbf{1}\|_\mathrm{op} \leqslant (nL)^2 \cdot 2c_\mathrm{h}\varepsilon_1 L^{-2} = 2c_\mathrm{h} n^2 \varepsilon_1$, and, χ being a class function with $|\operatorname{Tr} X| \leqslant \dim \pi_\chi \, \|X\|_\mathrm{op}$, $|W_\chi(C_n;U_1) - 1| = |\chi(h) - \chi(\mathbf{1})| \leqslant \|\pi_\chi(h) - \mathbf{1}\|_\mathrm{op} \leqslant 2\mathrm{Lip}_\chi c_\mathrm{h} n^2 \varepsilon_1$. Since $\mathbb{E}_\mathrm{h}$ is positive with $\mathbb{E}_\mathrm{h}[1] = 1$,

$$\mathbb{E}_\mathrm{h}\big[\mathbf{1}_{A_\mathrm{cov}} W_\chi(C_n;U_1)\big] - 1 = \mathbb{E}_\mathrm{h}\big[\mathbf{1}_{A_\mathrm{cov}\cap A_\square}(W_\chi - 1)\big] + \mathbb{E}_\mathrm{h}\big[\mathbf{1}_{A_\mathrm{cov}\cap A_\square^c}(W_\chi - 1)\big] - \mathbb{P}_\mathrm{h}(A_\mathrm{cov}^c) \,,$$

and the three terms are at most $2\mathrm{Lip}_\chi c_\mathrm{h} n^2\varepsilon_1$, $2\mathbb{P}_\mathrm{h}(A_\square^c) \leqslant 2(nL+1)^2 q$ ($|W_\chi| \leqslant 1$) and $C_L n_C q$ (D.25) in modulus, which is the first bound. By Corollary 1.4 the leading term is $\mathcal{W}_T^{\circ,\sigma}(C_n)(\mathbb{E}_\mathrm{h}[\mathbf{1}_{A_\mathrm{cov}} W_\chi(C_n;U_1)] + R_n) = \mathcal{W}_T^{\circ,\sigma}(C_n)(1+R_n^\sharp)$ with $R_n^\sharp = (\mathbb{E}_\mathrm{h}[\mathbf{1}_{A_\mathrm{cov}} W_\chi(C_n;U_1)] - 1) + R_n$ and $|R_n| \leqslant e^{n_C \delta_q^\sharp} - 1$, which is the second; the modulus bounds follow from $1-\eta_n \leqslant |1+R_n^\sharp| \leqslant 1+\eta_n$ and $\mathcal{W}_T^{\circ,\sigma}(C_n) = \kappa_{C_n} e^{E_{T,n}^\circ}$ with $|E_{T,n}^\circ| \leqslant C\tau^2(2+1/c)$. As $g_0 \to 0$, $\varepsilon_1 \to 0$ ($g_1 \leqslant (1+\beta_0)g_0$ and $g_0 \leqslant (1+\beta_0)g_1$, [3, (2.6)] as quoted in Section B.8; the second inequality keeps ε_1^{-1} polynomial in g_0^{-1}), $q \leqslant e^{-c_* \pi_*^2}$ (Proposition D.15) and, below the further threshold of Corollary 1.4, $\delta_q^\sharp \leqslant eC\tau^2 N'_\mathrm{lat}(c) e^{-c_q \pi_*^2/16}$, with $\pi_* \to \infty$; so $\eta_n \to 0$ at fixed n, and the terms other than $2\mathrm{Lip}_\chi c_\mathrm{h} n^2 \varepsilon_1$ are smaller than every power of g_0 for $n \leqslant \varepsilon_1^{-1}$. *q.e.d.*

Proposition 1.6 (The Leading Term and the Coarse Loop)**.** *Assume (H1)–(H4). Write the coarse bonds of C_n in the order of traversal from the vertex 0 as $c_0, \dots, c_{4n-1}$, and for a coarse bond c traversed in the negative direction put $V(c) = V(-c)^{-1}$ and $P_c(U) = P_{-c}(U)^{-1}$. Let*

$$\mathrm{Hol}_V C_n = V(c_0)V(c_1)\cdots V(c_{4n-1}) \,, \qquad W_\chi(C_n;V) = \chi(\mathrm{Hol}_V C_n)$$

be the holonomy and the Wilson function of the coarse field along C_n, so that $\mathrm{Hol}_{U_1} C_n = P_{c_0}(U_1)\cdots P_{c_{4n-1}}(U_1)$ in the convention of (1.1)*. Put*

$$s \coloneqq 6c_\mathrm{h}\varepsilon_1 \,, \qquad t \coloneqq e^s - 1 \,,$$

with c_h and $\varepsilon_1 = g_1 p_0(g_1)$ as in Lemma B.4, and assume $s \leqslant 1/10$ and $t \leqslant r_G$, with r_G the group constant of Appendix B recalled in Section 3 (conditions on g_0). Then on A_cov, at every branch:

(i) for every j, with c the positive coarse bond such that $c_j = \pm c$: $U_1 \in \mathcal{O}_c$, $V(c) = e^{i\beta_c(U_1)}P_c(U_1)$, and

$$\big\|V(c_j)P_{c_j}(U_1)^{-1} - \mathbf{1}\big\|_{\mathrm{op}} \leqslant t\,, \qquad \|\beta_c(U_1)\|_{\mathrm{op}} \leqslant 2\arcsin(t/2) \leqslant 1.06\,s\,; \tag{1.15}$$

(ii) with $H_0 = \mathbf{1}$, $H_j = P_{c_0}(U_1)\cdots P_{c_{j-1}}(U_1)$ and $\widetilde{R}_j = H_j V(c_j) P_{c_j}(U_1)^{-1} H_j^{-1}$,

$$\mathrm{Hol}_V C_n = \widetilde{R}_0\widetilde{R}_1\cdots\widetilde{R}_{4n-1}\ \mathrm{Hol}_{U_1} C_n\,, \qquad \big\|\widetilde{R}_0\cdots\widetilde{R}_{4n-1} - \mathbf{1}\big\|_{\mathrm{op}} \leqslant e^{4ns} - 1\,, \tag{1.16}$$

where $V(c_j)P_{c_j}(U_1)^{-1}$ depends on U_1 only through the bonds with both endpoints in the two blocks of c_j;

(iii) with π_χ and Lip_χ as in Corollary 1.5 (a Lipschitz constant of π_χ on G for $\|\bullet\|_{\mathrm{op}}$),

$$\big|W_\chi(C_n;V) - W_\chi(C_n;U_1)\big| \leqslant \mathrm{Lip}_\chi\big(e^{6c_{\mathrm{h}}n_C\varepsilon_1/L} - 1\big)\,. \tag{1.17}$$

Consequently, under (H1)–(H4) and the two conditions above,

$$\mathbb{E}_\varepsilon[W_\chi(C_n)] = \kappa_{C_n}\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_\xi}W_\chi(C_n;V)\big] + E_{\mathrm{coarse}} + E_{\mathrm{def}} + E_{\mathrm{bdry}}\,,$$
$$|E_{\mathrm{coarse}}| \leqslant |\kappa_{C_n}|\,\mathrm{Lip}_\chi\big(e^{6c_{\mathrm{h}}n_C\varepsilon_1/L} - 1\big)e^{C\tau^2(9n+1)}\,,$$

and, under the additional hypotheses of (H2) and (H3) for Theorem 1.2(iii),

$$|E_{\mathrm{coarse}}| \leqslant |\kappa_{C_n}|\,\mathrm{Lip}_\chi\big(e^{6c_{\mathrm{h}}n_C\varepsilon_1/L} - 1\big)\exp\big[C\tau^2(2+1/c) + n_C\delta_q^I\big]\,,$$

with E_{def}, E_{bdry} and the constants of Theorem 1.2.

Proof. Location. Fix j and let c be the positive coarse bond with $c_j = \pm c$, and $\mathcal{E}_c$ the set of bonds with both endpoints in the two blocks $\mathcal{B}_{c_-}$, $\mathcal{B}_{c_+}$ of c. Every loop in β_c, $U(\gamma)U([x,x'])U(\gamma')^{-1}P_c(U)^{-1}$ with $\gamma \in \mathcal{G}(c_-,x)$, $\gamma' \in \mathcal{G}(c_+,x')$, uses only bonds of $\mathcal{E}_c$: γ and γ' are monotone paths inside the boxes $\mathcal{B}_{c_-}$, $\mathcal{B}_{c_+}$, and $[x,x']$ and the straight path of c run from $\mathcal{B}_{c_-}$ to $\mathcal{B}_{c_+}$. The plaquettes used below lie in the box $\mathcal{B}_{c_-}\cup\mathcal{B}_{c_+}$, of ℓ^∞ diameter $2L-1$. On A_{cov}, $\Lambda_b \subseteq I = \Lambda_1^*$ for every fine contour bond b of c (Section D.6); Λ_b is $\{b\}$ or the set of free bonds of $\mathcal{E}_c$. Each block of c contains $(L-1)/2$ fine bonds of the straight path of c with both endpoints in that block; they are free (a designated bond crosses between two blocks), so they lie in I, and each has an endpoint in Λ_1. Since Λ_1 is a union of blocks (the proof of Corollary 1.5), both blocks of c lie in Λ_1. By Lemma 3.11(i), every site within ℓ^∞ distance $LMR_1 - 1 > 2L$ of Λ_1 lies in $\Omega_1^{\sim -1}$, so both blocks of c lie in Ω_1, and the constraint $\bar{U}_1 = V$ of [3, (1.12)] on $\Omega_1^{(1)}$ (Section D.8.1) gives $V(c) = M(U_1,c)$. Every plaquette of the box has a bond within distance $8L+1 < 3R_{\mathrm{c}}$ of I ($R_{\mathrm{c}} \geqslant 2R_{\mathrm{loc}}+1 = 4L+5$), so Lemma B.4 gives $\|U_1(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant 2c_{\mathrm{h}}\varepsilon_1 L^{-2}$ for each of them.

Area. Let ℓ be one of the loops, with $x = c_- + \sum_\nu n_\nu e_\nu$, $|n_\nu| \leqslant (L-1)/2$ ($\mathcal{B}_{c_-}$ is the block of the odd L of (3.1) centred at c_-), and let γ_+ be the translate of γ by Le_μ, $c = (c_-, c_- + Le_\mu)$; $\gamma_+ \in \mathcal{G}(c_+,x')$. Passing from the order of directions of γ' to that of γ_+ by adjacent transpositions inverts each pair of directions at most once, and exchanging two consecutive straight segments of lengths $|n_\alpha|$, $|n_\beta|$ moves the path across $|n_\alpha||n_\beta|$ plaquettes. Moving $[x,x']$ back along γ one bond at a time crosses L plaquettes for a bond of direction $\pm e_\nu$, $\nu \neq \mu$, and none for a bond of direction $\pm e_\mu$, where the two paths coincide after cancellation of backtracks. So ℓ is reduced to the empty loop by at most

$$A_L = \sum_{\alpha<\beta}|n_\alpha||n_\beta| + L\sum_{\nu\neq\mu}|n_\nu| \leqslant \tfrac{3}{2}(L-1)^2 + \tfrac{3}{2}L(L-1) \leqslant 3L^2$$

plaquette moves, by $\sum_{\alpha<\beta} a_\alpha a_\beta \leqslant \frac{3}{8}(\sum_\alpha a_\alpha)^2$ for four nonnegative numbers and $\sum_\nu |n_\nu| \leqslant 2(L-1)$ for the first term, and by $|n_\nu| \leqslant (L-1)/2$ for each of the three $\nu \neq \mu$ for the second.

One loop. A move across a plaquette p multiplies the transport of the loop at its base point by $gU_1(\partial p)^{\pm 1}g^{-1}$, with $g \in G$ a transport along the current loop. So $U_1(\ell)$ is a product of at most A_L such conjugates. For unitary W and g, $\|gWg^{-1}-\mathbf{1}\|_{\mathrm{op}} = \|W-\mathbf{1}\|_{\mathrm{op}} = \|W^{-1}-\mathbf{1}\|_{\mathrm{op}}$, and $\|\prod_m(\mathbf{1}+a_m)-\mathbf{1}\|_{\mathrm{op}} \leqslant \prod_m(1+\|a_m\|_{\mathrm{op}})-1$; hence

$$\|U_1(\ell)-\mathbf{1}\|_{\mathrm{op}} \leqslant (1+2c_{\mathrm{h}}\varepsilon_1 L^{-2})^{3L^2}-1 \leqslant t\ .$$

The eigenvalues $e^{i\varphi}$ of $U_1(\ell)$, $\varphi \in (-\pi,\pi]$, satisfy $2|\sin(\varphi/2)| = |e^{i\varphi}-1| \leqslant t$, so the Hermitian matrix A with $e^{iA} = U_1(\ell)$ whose eigenvalues are these φ has $\|A\|_{\mathrm{op}} \leqslant 2\arcsin(t/2)$. The function $\phi(s) = 2\arcsin((e^s-1)/2)$ is convex on $[0,1/10]$, its derivative $e^s(1-(e^s-1)^2/4)^{-1/2}$ being increasing, so $\phi(s) \leqslant 10\phi(1/10)s \leqslant 1.06s$. Since $U_1(\ell) \in G$ and $\|U_1(\ell)-\mathbf{1}\|_{\mathrm{op}} \leqslant t \leqslant r_G$, $U_1(\ell)$ lies in the chart domain (Section 3): $U_1(\ell) = e^{iB}$ with $B \in \mathfrak{g}$ and $h(B) < \varrho_G$, so $\|B\|_{\mathrm{op}} \leqslant \sqrt{N}h(B) < \pi$. A Hermitian B with $e^{iB} = U_1(\ell)$ and spectrum in $(-\pi,\pi)$ is the function of $U_1(\ell)$ that takes the value φ on the eigenspace of $e^{i\varphi}$, $\varphi \in (-\pi,\pi)$, since $\lambda \mapsto e^{i\lambda}$ is injective on $(-\pi,\pi)$; so $B = A$. Hence $A \in \mathfrak{g}$, $h(A) < \varrho_G$, and A is the chart value $\frac{1}{i}\log U_1(\ell)$, the chart being the inverse of $A \mapsto e^{iA}$ on the chart ball. Hence $U_1 \in \mathcal{O}_c$.

One coarse bond. $\beta_c(U_1)$ is a convex combination of the matrices $\frac{1}{i}\log U_1(\ell)$ (the weights of (4.1) are nonnegative and sum to one, $|\mathcal{B}_{c_-}| = L^4$), so $\|\beta_c(U_1)\|_{\mathrm{op}} \leqslant 2\arcsin(t/2)$, and $V(c)P_c(U_1)^{-1} = e^{i\beta_c(U_1)}$ has $\|e^{i\beta_c(U_1)}-\mathbf{1}\|_{\mathrm{op}} \leqslant 2\sin(\arcsin(t/2)) = t$. For $c_j = -c$, $V(c_j)P_{c_j}(U_1)^{-1} = P_c(U_1)^{-1}e^{-i\beta_c(U_1)}P_c(U_1)$, a conjugate of $e^{-i\beta_c(U_1)}$, with the same bound. This is (i), and the dependence stated in (ii) is Lemma 1.1(i).

The loop. By induction on m, $V(c_0)\cdots V(c_{m-1}) = \widetilde{R}_0\cdots\widetilde{R}_{m-1}H_m$, the step being

$$\widetilde{R}_0\cdots\widetilde{R}_{m-1}H_mV(c_m) = \widetilde{R}_0\cdots\widetilde{R}_{m-1}\big(H_mV(c_m)P_{c_m}(U_1)^{-1}H_m^{-1}\big)H_mP_{c_m}(U_1)\ .$$

At $m = 4n$ this is the identity in (1.16). Each $\widetilde{R}_j$ is a conjugate of $V(c_j)P_{c_j}(U_1)^{-1}$, so $\|\widetilde{R}_j-\mathbf{1}\|_{\mathrm{op}} \leqslant t$ and $\|\widetilde{R}_0\cdots\widetilde{R}_{4n-1}-\mathbf{1}\|_{\mathrm{op}} \leqslant (1+t)^{4n}-1 = e^{4ns}-1$. For (iii), χ is $(\dim\pi_\chi)^{-1}\operatorname{Tr}\pi_\chi$ and $|(\dim\pi_\chi)^{-1}\operatorname{Tr}X| \leqslant \|X\|_{\mathrm{op}}$, so

$$\begin{aligned}|W_\chi(C_n;V)-W_\chi(C_n;U_1)| &\leqslant \|\pi_\chi(\mathrm{Hol}_V C_n)-\pi_\chi(\mathrm{Hol}_{U_1}C_n)\|_{\mathrm{op}}\\ &\leqslant \mathrm{Lip}_\chi\|\mathrm{Hol}_V C_n-\mathrm{Hol}_{U_1}C_n\|_{\mathrm{op}}\ ,\end{aligned}$$

and $\|\mathrm{Hol}_V C_n-\mathrm{Hol}_{U_1}C_n\|_{\mathrm{op}} = \|(\widetilde{R}_0\cdots\widetilde{R}_{4n-1}-\mathbf{1})\,\mathrm{Hol}_{U_1}C_n\|_{\mathrm{op}} \leqslant e^{4ns}-1$, with $4ns = 24c_{\mathrm{h}}n\varepsilon_1 = 6c_{\mathrm{h}}n_C\varepsilon_1/L$.

The expectation. On A_{cov}, $W_\chi(C_n;U_1) = W_\chi(C_n;V)+(W_\chi(C_n;U_1)-W_\chi(C_n;V))$. Insert this into (1.9) and put $E_{\mathrm{coarse}} = \kappa_{C_n}\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_\xi}(W_\chi(C_n;U_1)-W_\chi(C_n;V))]$. Since $|e^{E^\circ_\xi}| \leqslant e^{|E^\circ_\xi|}$, (1.17) with $|E^\circ_\xi| \leqslant C\tau^2(9n+1)$ on A_{cov} (1.10) and $|\mathbb{E}_{\mathrm{h}}[X]| \leqslant \sup|X|$ (Section D.6) gives the first bound on E_{coarse}, and (1.17) with (1.11) the second. *q.e.d.*

Remark 1.7 (What the Proposition Does and Does Not Compare). (1) (1.16) is exact: after the first step the fine loop at the background is the coarse holonomy with one insertion per coarse bond, $\mathrm{Hol}_{U_1}C_n = V(c_0)Q_0V(c_1)Q_1\cdots V(c_{4n-1})Q_{4n-1}$ with $Q_j = V(c_j)^{-1}P_{c_j}(U_1)$. For $c_j = \pm c$, Q_j is $P_c(U_1)^{-1}e^{-i\beta_c(U_1)}P_c(U_1)$ if $c_j = c$ and $e^{i\beta_c(U_1)}$ if $c_j = -c$, a function of U_1 on the bonds of $\mathcal{E}_{c_j}$, with $\|Q_j-\mathbf{1}\|_{\mathrm{op}} \leqslant t$. The $\widetilde{R}_j$ of (ii) are conjugates of the $V(c_j)P_{c_j}(U_1)^{-1}$ by the transports H_j along the earlier bonds, and are not local. The insertions are of first order in the plaquette deviations of U_1, of size $O(\varepsilon_1)$ per coarse bond.

(2) Per fine contour bond the exponent in (1.17) is $6c_{\mathrm{h}}\varepsilon_1/L$, of order $L^{-1}g_0(\log g_0^{-2})^{p_0}$, while that of E_{def} is $C_\chi\rho(\varepsilon_1+e^{-cb/(LM)})$, of order g_0^3 times a power of $\log g_0^{-2}$. For a square of

fixed physical side ℓ_C, n_C is of order ℓ_C/ε, and E_{coarse} is controlled by the proposition only while $n_C\varepsilon_1/L$ is small.

(3) Both $W_\chi(C_n;U_1)$ and $W_\chi(C_n;V)$ are functions of the coarse field and the branch data: the first step leaves no integral over the young variable a over which their difference is averaged ($\mathbb{E}_{\mathrm{h}}$ integrates over the branch data, the retained fluctuation A_{e} included). Every average of the difference is taken in the coarse ensemble $\mathbb{E}_{\mathrm{h}}$, which Theorem 1.2 leaves unevaluated. A bound on E_{coarse} that uses cancellations of the first-order part of the insertions is therefore a statement about the coarse measure, and it is not given here.

Theorem 1.8 (Two Disjoint Squares after the First Renormalisation Step)**.** *Let $C=C_1\sqcup C_2$, χ_1, χ_2, $n_C=n_{C_1}+n_{C_2}$ and $W_{\chi_1,\chi_2}(C;U)=\chi_1(\mathrm{Hol}_U C_1)\chi_2(\mathrm{Hol}_U C_2)$ be as in Section 10.1: C_i is a coarse translate of the counterclockwise square of side n_iL in a plane parallel to that of e_1,e_2, χ_i a normalised irreducible character, and $V(C_1)\cap V(C_2)=\varnothing$. Assume (H1)–(H4) for (C_1,χ_1) and for (C_2,χ_2) in place of (C_n,χ), and let g_0 lie below the thresholds of Section 10: those of Lemmas 10.4, 10.6 and 10.7, and for (iii) that of* (10.10)*. Let A_{cov} be the covered event of the pair, E_{def} and E_{bdry} the terms* (E.1) *and* (D.21) *formed with the observable and the groups of the pair (Section 10), κ_{C_i} and $E^\circ_{\xi,i}$ the coefficient and the exponent of Theorem 1.2 for (C_i,χ_i), $E^\times_\xi$ the cross exponent of Lemma 10.7(b), and $\Delta_\times$, d as in* (10.2)*. Then:*

(i) (Form.)

$$\mathbb{E}_\varepsilon[W_{\chi_1,\chi_2}(C)]=\kappa_{C_1}\kappa_{C_2}\,\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_{\xi,1}+E^\circ_{\xi,2}+E^\times_\xi}W_{\chi_1,\chi_2}(C;U_1)\big]+E_{\mathrm{def}}+E_{\mathrm{bdry}}\ , \tag{1.18}$$

every term being integrable against the history weight; $\kappa_{C_i}\neq 0$ depends on n_i through $P(C_i)$ alone, as in Theorem 1.2(i).

(ii) (Bounds.) On A_{cov}, $|E^\circ_{\xi,i}|\leqslant C\tau^2\big[\sum_{j\in J_{C_i}}e^{-cd_j}+4n_ie^{-c\iota_T}+(n_i+1)e^{-cn_i}\big]\leqslant C\tau^2(9n_i+1)$ for $i=1,2$, with J_{C_i} the $4n_i$ coarse segments of C_i, and

$$\begin{aligned}|E^\times_\xi|&\leqslant C_\times\rho\Delta_\times\leqslant C_\times\rho\, n_Ce^{-c_\times d/(LM)}\ ,\\ |E_{\mathrm{def}}|&\leqslant e^{C_\times\rho\Delta_\times}\Big[C'_Ln_Ce^{-(c_*/2)\pi_*^2}+e^{C_\chi n_C\rho(\varepsilon_1+e^{-cb/(LM)})+C_\times\rho\Delta_\times}-1\Big]\ ,\\ |E_{\mathrm{bdry}}|&\leqslant C_{\mathrm{resp},\chi_1\chi_2}\,n_C(1+g_0r)\,e^{-c_*\pi_*^2}\ .\end{aligned} \tag{1.19}$$

(iii) (Size of the leading term.)

$$|\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}e^{E^\circ_{\xi,1}+E^\circ_{\xi,2}+E^\times_\xi}W_{\chi_1,\chi_2}(C;U_1)]|\leqslant e^{C\tau^2(9n_1+9n_2+2)+C_\times\rho\Delta_\times}\ ,$$

and under the condition of (H3) for (iii) for each square,

$$\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}}e^{|E^\circ_{\xi,1}|+|E^\circ_{\xi,2}|}\big]\leqslant\exp\big[2C\tau^2(2+1/c)+n_C\delta_q^{I,2}\big]\ ,\qquad \delta_q^{I,2}\leqslant\exp\big[-c_q\pi_*^2/(2C_K)\big]\ , \tag{1.20}$$

with $\delta_q^{I,2}$ of (10.10)*, so that the leading term of* (1.18) *has modulus at most $|\kappa_{C_1}\kappa_{C_2}|\exp[2C\tau^2(2+1/c)+n_C\delta_q^{I,2}+C_\times\rho\Delta_\times]$, and*

$$\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^\circ_{\xi,1}+E^\circ_{\xi,2}+E^\times_\xi}-1|]\leqslant\exp[2C\tau^2(2+1/c)+n_C\delta_q^{I,2}+C_\times\rho\Delta_\times]-1\ .$$

Here C and the c of the bounds on $E^\circ_{\xi,i}$ and of (iii) are the larger, respectively the smaller, of the constants of Proposition F.2 for the two characters, and d_j is the distance (F.1) *of segment j of C_i at the carrier $I=\Lambda_1^*$ (not the distance d of* (10.2)*); $C_\chi=\max_i C_{\chi_i}$ and the c of the*

bound on E_{def} are those of Lemma 10.7, formed from the constants of (E.24)*, and $C_\times$, $c_\times$ the constants of that lemma; ρ is* (E.11) *with $u = C^{\mathrm{src}}_{\chi_1\chi_2} g_0(1+r)$, the enlarged source constant of Lemma 10.4, which is distinct from this C_χ; $C_{\mathrm{resp},\chi_1\chi_2} = C_L \max((C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2})S_c, 1)$ with the constants C_{resp,χ_i} of Corollary D.8 for χ_i (those of* (D.8) *as enlarged in Lemma D.6, before the replacement made in* (D.41)*) and C_L, S_c of Proposition D.15; C'_L, c_*, c_q, C_K, τ, π_*, ε_1, b are those of Theorem 1.2, a constant that depends on the character being taken at its larger value for χ_1, χ_2 when it is a prefactor and at its smaller value when it is a decay rate. The constants C, c, C_χ, $C^{\mathrm{src}}_{\chi_1\chi_2}$, $C_\times$, $c_\times$, $C_{\mathrm{resp},\chi_1\chi_2}$, C'_L, c_*, c_q are fixed before the coupling; C_K and the scales τ, π_*, ε_1, ρ, b depend on g_0 as in Theorem 1.2; none of them depends on n_1, n_2, the positions of the squares, the torus or the history.*

Proof. Section 10.8. *q.e.d.*

Remark 1.9 (The Cross Terms). No condition on the distance of the squares is imposed beyond $V(C_1) \cap V(C_2) = \varnothing$. The cross factor $e^{E^\times_\xi}$ and the factors $e^{C_\times \rho \Delta_\times}$ in (1.19) and in (iii) are close to one when $C_\times \rho \Delta_\times$ is small. Since $\Delta_\times \leqslant n_C$ and $\rho = O(g_0^2(1+r)^4)$, this holds at fixed n_1, n_2 for all positions of the squares once g_0 is small; at fixed coupling, $C_\times \rho \Delta_\times \leqslant \epsilon$ whenever $d/(LM) \geqslant c_\times^{-1}(\log(C_\times \rho n_C) + \log \epsilon^{-1})$, which the hypotheses allow on tori with correspondingly large periods. As $C_\times \rho \Delta_\times \to 0$ the bound on $E^\times_\xi$ tends to zero, and those on E_{def} and E_{bdry} take the form of (1.10) with $n_C = n_{C_1} + n_{C_2}$ and the constants C_χ, c, ρ, $C_{\mathrm{resp},\chi_1\chi_2}$ of Theorem 1.8.

2 Extraction from the Effective Interaction

Bałaban first constructs the effective interaction as a function of the remaining field. He then extracts its relevant coefficients at the identity and subtracts the corresponding local terms from that field-dependent interaction. We use the convention in [4, (1.6)–(1.7)] and following [3, (1.30)], in which the Gaussian normalisation is included in the effective interaction. The intermediate definition in [4, (2.13)] keeps that contribution separate.

In the small-field construction of [4], with no large-field regions, let B be a small coarse Lie-algebra field, and let $U_1(e^{iB})$ be the constrained critical configuration with coarse field e^{iB}. Write $\mathbf{E}^{(1)}$ for the first-step effective interaction, including the Gaussian normalisation, and put

$$\mathcal{E}_1(B) = \mathbf{E}^{(1)}(g_0, U_1(e^{iB})) \,. \tag{2.1}$$

Choose an orthonormal basis of the Lie algebra for the action's invariant inner product. The index a denotes a basis element, μ a bond direction and x its initial coarse site. The second derivative is

$$\Pi^{ab}_{1,\mu\nu}(x,y) = \left. \frac{\partial^2 \mathcal{E}_1(B)}{\partial B^a_\mu(x)\, \partial B^b_\nu(y)} \right|_{B=0} \,. \tag{2.2}$$

After the colour reduction and the translation-invariant infinite-lattice limit, let $\Pi_{1,\mu\nu}(x)$ denote the resulting scalar kernel with its second site fixed at the origin. In Bałaban's conventions the coupling counterterm is

$$\beta_1(g_0) = \sum_{x \in \mathbb{Z}^4} x_1 x_2 \Pi_{1,12}(g_0, x) \,. \tag{2.3}$$

The definition and the symmetry argument are given in [4, (1.20)–(1.22), (5.37)–(5.43)].

In a proper regular region Λ_1, the subtraction has the form

$$\mathbf{E}^{(1)}(\Lambda_1, g_0, U_1) - \mathbf{E}^{(1)}(\Lambda_1, g_0, \mathbf{1}) - \beta_1(g_0) S_{\phi_1}(U_1) \; . \tag{2.4}$$

Here ϕ_1 is the localisation cutoff in [3, (1.27)], supported in Λ_1 and equal to one on its specified interior. If $x(p)$ is the initial site of an oriented representative of the plaquette p, then

$$S_{\phi_1}(U) = \sum_p \phi_1(x(p)) \big(1 - \operatorname{Re} \operatorname{tr}_N U(\partial p)\big) \; . \tag{2.5}$$

The boundary interaction and the exterior integrations remain in [3, (1.29)–(1.30)]. Bałaban states after (1.30) that the regular interaction is the interaction of [4] restricted to Λ_1. Within the cube-localised construction of [4], the changes of reconstruction and summation domain are controlled in [4, (4.35)–(4.37)].

For the Wilson loop, the quantities to extract are the scalar density along a straight side and the scalar contribution of a corner. We denote them by δm and z_{c}. Their construction below uses connected terms of the normalised Wilson integral at the identity background. Section 7 identifies the straight-side coefficient with the flat projection of the conditional Wilson kernel in the first RG step, and estimates the error caused by its integration boundary. Section 8 gives the outgoing insertion with the reconstructed closed flux factors and the remaining conditional source terms retained. Section 3 writes the conditional first-step integral with the insertion, at the actual background and exterior fields, as an exact sum of local closed insertions paired with signed local measures; the estimates it states as lemmas are proved in Appendix B.

3 The Sourced Effective Interaction at the First Step

This section carries out the first step of the procedure in Section 2: the conditional first-step integral with the Wilson insertion is written as an exact sum of local closed insertions paired with signed local measures, at the actual background and exterior fields. Every displayed identity is exact at fixed volume. What the section does not prove, and states instead as lemmas with proofs in Appendix B, is the existence of the objects in those identities and one cluster identity: the two-sided bounds on the young block $H_{\xi,II}$ of the regional Hessian (Lemma 3.4), the analyticity of the reconstruction on the cutoff box (Lemma 3.6), the finiteness of the selected integrals (Lemma 3.16) and the absolute convergence of the cluster sums (Lemmas 3.19 and 3.20, the latter including the cluster identity (3.74), whose proof is Appendix B, Section B.6). Each identity below holds under those lemmas, under the standing conditions (3.1), and under the branch hypotheses (B1)–(B4), which record what is taken from Bałaban's construction of the branch density. The other inputs are Bałaban's derivation of [3, (1.11)–(1.25)], as recorded after (3.7), [4, (0.4) and (0.10)–(0.11)] for the averaging operation and the axial transports, Appendix A for the flat values, and the Ad-invariance (5.4) of the bond norm. The bounds on the terms of the final identity are Appendix D and Section 9.

Conventions. G is the compact connected simple gauge group fixed at the start of the paper, in its fixed unitary realisation, so that group elements near the identity are $U = e^{iA}$ with A in the real vector space $\mathfrak{g}$ of Hermitian matrices $i^{-1}\times$(Lie algebra); $\langle X, Y\rangle = \operatorname{tr}_N(XY)$ is the fixed Ad-invariant real inner product on $\mathfrak{g}$ of the flat calculation (the opening paragraph of Section 4; it is the inner product for which every constant of Appendix A, in particular γ_L, is derived), $h(X) = \langle X, X\rangle^{1/2}$, $d_G = \dim \mathfrak{g}$. Every quadratic form of this section is defined

without reference to this choice; the operators representing the forms, the Hilbert norms, the Lebesgue measures $\mathrm{d}a$, the δ-functions of [3, (1.14)], the Haar density σ and the determinants all refer to this one inner product, and a constant said to depend on L and the group only is a constant for it. The identity [3, (1.15)] is an identity between the fine integral, which involves no inner product, and an expression whose measures $\mathrm{d}A$, δ-functions and densities σ, constants σ_0, $z^{(0)}$ are taken for one and the same inner product on $\mathfrak{g}$, whichever it is; it therefore holds for the fixed inner product with the function σ and the constants σ_0, $z^{(0)}$ belonging to it, and these are the values meant below. log is the group chart: the inverse of $A \mapsto e^{iA}$ on the Ad-invariant Hilbert ball $h(A) < \varrho_G := \pi/\sqrt{N}$, whose range in G is the chart domain. On this ball $\|A\|_{\mathrm{op}} \leqslant \sqrt{N}\, h(A) < \pi$ (for h the normalised Hilbert–Schmidt norm, $h(A)^2 = N^{-1}\sum_j \lambda_j^2 \geqslant N^{-1}\|A\|_{\mathrm{op}}^2$), so the eigenvalues of A lie in $(-\pi,\pi)$: $A \mapsto e^{iA}$ is injective there (an eigenvalue λ of A is recovered from the eigenvalue $e^{i\lambda}$ of e^{iA} on the same eigenspace), its differential is invertible (the differential of exp at iA is singular only when two eigenvalues of iA differ by a nonzero multiple of $2\pi i$), and its inverse is the restriction to the chart domain of the principal logarithm $\frac{1}{i}\log$ used in the flat calculation ("the chart domain" of $\mathcal{O}_c$, the sentence preceding (4.1) and (1.4); "the regular chart containing the identity", the sentence following (4.5)); every logarithm below is taken in this chart. The chart is Bałaban's: [8, p. 21, (22)–(25)] defines $\log U = iA$ for a unitary $U = \sum_j e^{i\lambda_j} P_j$ (spectral projections P_j, distinct $\lambda_j \in]-\pi,\pi]$) by $A = \sum_j \lambda_j P_j$, Hermitian with $|A| \leqslant \pi$, with the bounds $|U-1| \leqslant |\log U| \leqslant \frac{\pi}{2}|U-1|$; the chart logarithm is the restriction of this principal logarithm to the chart domain, on which it is analytic, and the bound (25) holds for the normalised Hilbert–Schmidt norm h as for the operator norm, since $|e^{i\lambda}-1| = 2|\sin(\lambda/2)| \geqslant \frac{2}{\pi}|\lambda|$ for $\lambda \in]-\pi,\pi]$: $h(\frac{1}{i}\log W) \leqslant \frac{\pi}{2} h(W-\mathbf{1})$, and a unitary $W \in G$ with $\|W-\mathbf{1}\|_{\mathrm{op}} \leqslant r_G$, in particular one with $h(W-\mathbf{1}) \leqslant r_G/\sqrt{N}$, lies in the chart domain, r_G being the group constant of Appendix B (the eigenvalue bound alone does not place W in the chart domain for a general compact $G \subset U(N)$: for $G = SU(N)$ with $N \geqslant 10$, the element $W = e^{2\pi i/N}\mathbf{1}$ has $h(W-\mathbf{1}) = 2\sin(\pi/N) < 2/\sqrt{N}$, while its principal logarithm $(2\pi/N)\mathbf{1}$ does not lie in $\mathfrak{su}(N)$). In particular "$\mathsf{M}(U)(c)$ is defined" means throughout that every logarithm in the averaging formula [4, (0.4)] for the row c has its argument in the chart domain, that is $U \in \mathcal{O}_c$, where the averaging operation (1.4) is [4, (0.4)] (it is defined for every U; off $\mathcal{O}_c$ it is the straight transport, a value that enters only through the constraint $\bar{V}_0 = V$ on the rows of Ω_1^c, Section 3.1), and "the local group means $\mathcal{U}_U(y,x)$ are defined" that the transports along the path family from y to x lie in the small-diameter domain of the fixed group mean selected after [4, (0.9)], with the relative logarithms of (0.10) in the chart domain. We use that specified analytic mean, with the left-right covariance asserted in [4, (0.6) and (0.10)], and define $\mathcal{U}_U(y,x)$ by (0.11). No uniqueness among all solutions having logarithms in the chart is asserted. The implicit function theorem identifies the specified mean with its near-diagonal solution on a sufficiently small neighbourhood. The same convention defines the domain $D^{(11)}$ of Section 3.1, the set $\mathcal{X}$ of the measurability paragraph and the objects of Section 3.2. The chart being Bałaban's, the domain statement that on the domain of integration in [3, (1.11)] the arguments of these logarithms and the fluctuation lie in the chart domain is the smallness of the fluctuation in [3, pp. 247–248] (by Lemma 1 of [9]) and of the contour variables ([8, pp. 24–25]), derived in (B1) for ε_0 small; its constants $O(L^2)$ and $O(1)$ are those of [9, Lemma 1, p. 79] and [8, p. 25], independent of the coupling, as recorded in (B1) and in the hypothesis (H4) of the main theorem. Appendix B uses besides ϱ_G an operator-norm radius r_G and the comparison constants $c_{\mathrm{h}} = \sqrt{N}$, $c_{\mathrm{n}} = 1/N$ between h, the matrix norm $|\bullet|$ of the cutoff and $\|\bullet\|_{\mathrm{op}}$ ($h \leqslant |\bullet| \leqslant \|\bullet\|_{\mathrm{op}} \leqslant \sqrt{N}h$, where $h(Y) = (N^{-1}\operatorname{tr} Y^*Y)^{1/2}$ for an arbitrary matrix Y, so that $\|Y\|_{\mathrm{op}} \leqslant c_{\mathrm{h}}|Y|$ holds for every matrix, not only for Hermitian ones, and for either cutoff norm); they are recalled where used. The right inverse h_U and the force J_U of Section 3.2 always carry their background subscript, which distinguishes them from the norm

h and the carrier J. $|X|$ is the bond norm of [3] used in (4.12), (5.3) (the operator or the normalised Hilbert–Schmidt norm, fixed once; both are Ad-invariant by (5.4)). For $B \in \mathfrak{g}^S$ the Hilbert norm is $\|B\|^2 = \sum_{b\in S} h(B_b)^2$ and the phase norm is $\|B\|_\infty = \max_b h(B_b)$. E_* is the set of free bonds (all positive fine bonds except the designated bonds $b_0(c)$, [4, p. 267]: $b_0(c)$ is the fine bond contained in the coarse bond c whose endpoints lie in the two blocks of c, the sentence following (4.2)); $\mathcal{B}_y$ is the block with centre y and $b_\pm$ the endpoints of a bond b; distances of bonds are periodic ℓ^∞ distances of initial vertices. Bonds and rows of a set X of sites follow [3, (1.8)–(1.9)]: the bonds of X are the positive fine bonds with at least one endpoint in X, $X^{(1)}$ is the set of block centres in X, the rows of X are the coarse bonds with at least one endpoint in $X^{(1)}$, written $X^{[1]}$, and X^* is the set of bonds of X other than the designated bonds $b_0(c)$, $c \in X^{[1]}$ (so a bond of X^* may have one endpoint outside X, and the variables of a row of X include its designated bond). The complement is treated differently, following this paper's reading of [3, (1.11)–(1.12)], which say only that $V_0 = U|_{\Omega_1^c}$ are the integration variables localised in Ω_1^c (p.248), read with the bonds crossing from Ω_1 to Ω_1^c as variables of the integral over Ω_1 (the rule of [11, (8), p. 258] for plaquettes, "plaquettes with at least one corner belonging to Ω_1", applied to bonds; the rows are those of [11, (12), p. 258], which [3, p. 248] follows, (H4)): the bonds of Ω_1^c, written $b \in \Omega_1^c$, are the positive fine bonds with both endpoints off Ω_1, the restriction $V_0 = U|_{\Omega_1^c}$ is the restriction to these bonds, and the rows of Ω_1^c are the coarse bonds with both block centres off Ω_1; so every positive fine bond is a bond of Ω_1 or a bond of Ω_1^c and not both, and every coarse bond is a row of Ω_1 or a row of Ω_1^c and not both. $X^\sim$ is Bałaban's enlargement [3] of X by all cubes touching X (p.245; including the cubes meeting X only at an edge or a vertex, as in [4, p. 257], where $\tilde{\Box}^n$ is the cube of side $(1+2n)M$ with the centre of $\Box$, (H4); used for the distance statement at the end of this sentence and in Lemma 3.11(i)) of the partition in terms of which X is defined (p.245, where the partition into cubes $\Box$ of side LM_2R_0 is introduced, $\Box^\sim$ is $\Box$ enlarged by one layer of M_2R_0-cubes, and "usually" $X^\sim$ enlarges by a layer of the cubes of the partition defining X; here LMR_1-cubes for Ω_1, R_1' and Λ_1, $L^2M_2R_1$-cubes for $\Box'$, Q_1 and R_1, p.246), $X^{\sim-1} = ((X^c)^\sim)^c$ is the shrinking and $X^{\sim k}$ the k-fold iterate; a site of $X^{\sim-1}$ is at ℓ^∞ distance at least the cube side from X^c. The exponent of [3] in the definition of the scales, R_j the smallest power of L with $R_j \geqslant (\log g_j^{-2})^{\mathfrak{r}}$, $\mathfrak{r} \geqslant 2$ and $p_0 \geqslant 5\mathfrak{r}$ (p.245, p.246, [3, (2.5)]), is written $\mathfrak{r}$ here, because r is the cutoff radius of (4.12). The scales of [3] satisfy $R_0 \leqslant (L+1)R_1$ by [3, (2.9), p. 256], hence $R_0 \leqslant LR_1$ as both are powers of L; and M_1 is the M_1 of [3, p. 245], the size of cubes for which the theorems of the previous papers, in particular those on the variational problem, are valid (a lower bound on the cube side: [10, Theorem 3.1, p. 397] holds "for $M \geqslant M_1$", and M_1 "was fixed in" that paper, [9, p. 77]; [3] takes $M_2 = L^2M_1$ as an example, which is consistent with the conditions below). The identity of a matrix space is written $\mathbf{1}$; the action is S (the $A(U)$ of [3]) and the scalar normaliser E of [3] is written E: it is the vacuum scalar E_ε of (1.2), a constant independent of the configuration and of the branch, which cancels in every ratio of this paper and enters only the absolute identities for $\mathcal{Z}_0$ (Section D.7). The standing conditions on the parameters are

$$\begin{gathered} L > 11 \text{ odd ([4, p. 251])}; \quad M_1 < M_2 < M \text{ odd powers of } L,\ M \geqslant 8LM_2; \\ \text{(3.59)}; \quad \delta_M \leqslant 1,\ h_+^2\delta_M^2/\gamma \leqslant \tfrac{1}{8},\ 4C_{\mathrm{d}}\delta_M^2(LM+1)^4 \leqslant \tfrac{1}{2}; \\ \text{every fine period a multiple of } LMR_1 \text{ and at least } 8M; \\ \text{every coarse period} > \max(2, 2R_{\mathrm{p}}+8); \\ g_0 \text{ below the threshold of Appendix B, Section B.8,} \\ \text{which includes } LMR_1 > 4R_{\mathrm{c}} + 4L^2M_2R_1 + 8 \\ \text{and } \widehat{\mu} = C_{L,R_{\mathrm{c}}}\varepsilon_1 < \min\bigl(\tfrac{1}{2}\tau_L,\ \gamma_L/(4\max(1,c_L))\bigr) \\ \text{(the hypotheses of Lemma 3.4),} \end{gathered} \tag{3.1}$$

where R_{p} and R_{c} are the constants of that appendix ((B.7) and Lemma 3.11), γ, h_+ those of (3.56), δ_M, C_{d} those of Lemma 3.16, τ_L, c_L those of Lemma 3.3 and γ_L, $C_{L,R_{\mathrm{c}}}$ those of Lemma 3.4, all depending on L and the group only (δ_M also on M), and $\varepsilon_1 = g_1 p_0(g_1)$ the quantity of [3, p. 246], which depends on the coupling; the threshold depends on L, the group, the character χ (through the collar constants C'_χ, K_χ of item 4 of Section B.8), M_1, M_2, M, on the constants A_0, A_1, the exponents p_0 and $\mathfrak{r}$ of [3], on the constants B_3, δ_{c} and the $O(1)$ of the comparison after [3, (1.19)] recorded in (B3)(iii), and on the map $g_0 \mapsto g_1$ of [3, (1.27)], which are fixed as in [3] with A_0/A_1 large enough for (B3)(i); the order of choice of every constant and the list of conditions are those of Section B.8, whose closing paragraph also lists the further conditions imposed where the statements about the square and the appendices state them (Sections D.1, D.7, 9.3–9.9). The threshold also takes over the conditions of [3] on the parameters (item 5 of Section B.8). Those on the coupling, ε_0 small enough for Theorem 1 of [7] (reference [15] of [3], whose bracketed numbers are those of the bibliography of [4]; [3, p. 248]), MR_1 large (p.251, Lemma D.18), and M_2R_1 large enough for (B3)(ii) (p.250), are conditions that g_0 lies below a value depending on L, the group, M_1, M_2, M, and the constants of [3], met for small g_0 at fixed M_1, M_2, M since $R_1 \to \infty$. Those on M_1, M_2, M are not listed in [3]: M_1 is the cube size for which the theorems of the previous papers hold (p.245), M_2 is "not too much larger" than M_1, e.g. $M_2 = L^2 M_1$, and M is "much larger than M_2", its choice "dictated by many conditions involving this constant" (p.245). They are taken over as the hypothesis that Bałaban's construction [3] is valid for the values $M_1 < M_2 < M$ used here, which follow Bałaban's prescription (Section B.8); this paper adds to them only the conditions displayed above. The scale integer R_1 is the power of L of [3, p. 246] (Section 3.1; the same letter denotes the large-fluctuation region of [3], and in every product LMR_1, $L^2M_2R_1$ it is the scale), so the first-step cubes of sides LMR_1 and $L^2M_2R_1$, the LM-, M- and M_1-cubes of Section 3.5 and the blocks form nested partitions of the fine torus (all sides are powers of L, and the LM-cubes nest in the $L^2M_2R_1$-cubes once $LM_2R_1 \geqslant M$, threshold item 1 of Section B.8, which holds for small g_0 since $R_1 \to \infty$); the collar inequality holds once the scale satisfies $R_1 \geqslant R_{\mathrm{c}} + 3$, since $LMR_1 \geqslant 8L^2M_2R_1$. The condition "every coarse period > 2" is the operative hypothesis of Lemma 3.2 (it makes $b_0(c)$ the only designated bond in $\mathcal{E}_c$: with a coarse period equal to two, two coarse bonds would join the same pair of blocks). Every statement of this section is made under (3.1). When $I = \varnothing$ below, $\mathfrak{g}^I$ is a point, $\mathrm{d}a$ the unit mass, every sum reduces to its empty term and every identity holds trivially.

Some letters carry more than one meaning, each fixed by its context, as follows. R_* is the walk range n_*R in Section 3.5 and in the order of choice of Section B.8; from Section 5 on it is the fixed range introduced there, bounding the label-support diameters and the source-to-stencil distances (fixed independently of the coupling), and it is this R_* that enters $R_0 = 4(1 + R_* + R_H + L)$. R_0 is the scale of [3] where its scales and partitions are quoted (the sentence on $R_0 \leqslant (L+1)R_1$ above, p.245, and Lemma D.11); from Section 7 on it is this paper's range constant $R_0 = 4(1+R_*+R_H+L)$, as stated there. R_1 is the scale integer in every product LMR_1, $L^2M_2R_1$ and wherever it is called the scale, and the large-fluctuation region of [3] where it is called a region. M is the cube side; the group mean is the sans-serif $\mathsf{M}(U)(c)$; $M(W)$ is in Section 8 the level-wise collection $M^j(W)$ of [3, (2.11)] at the first step, and from Section 9.6.1 on the positive group majorant of a group W, with $M_v(\mathfrak{c})$ the tree majorant of a tuple (Section E.5.2). ℓ is the macro side LMR_1 in Section 9.4.1 and in Appendix D, where it is so declared, and a selection label under summation signs ($\sum_\ell$, Sections 3.7 and 9.6.1); $\ell_i(Y) = d_i(Y) + 5$ are the shifted tree lengths of a domain (defined with (B.1)). m_L is the Hessian constant of Lemma 3.3; m alone is the tuple length in $\mathfrak{c} = (D_1, \dots, D_m)$ (Section 3.6) and the segment length in coarse units in (7.30); $m(D) = \varepsilon_* e^{-\lambda_* \ell_1(D)}$ is the majorant of Lemma 3.19; δm^σ is the mass coefficient (5.39). $\theta \geqslant 0$ is the Mayer constant of Section B.5

(Section 3.6 on); in Appendix D, from (D.26) on, θ is the witness threshold defined there, θ_ϑ its ϑ-dependent form (Section 9.3), and θ_{h}, θ_{x} are the constants of Section E.7.2; ϑ and ϑ_* are different letters throughout. C_L is the coercivity constant (A.24) of Appendix A in Lemma 3.4 ($\gamma_L = C_L^{-1}$) and that appendix; it is the contour constant $|\Gamma_C| \leqslant C_L n_C$ in Section D.7, at the opening of Appendix E, in Section E.2 (with $C'_L = 8C_L$), in Corollary 1.5 and in Remark 11.2; elsewhere it is a constant depending on L (in Section D.8.1 also on N) as stated where it appears. $K = 256$ is the fixed integer of σ_K, C_K (Lemma 9.45); in (H3) K is the exponent of [4] in $\varepsilon = L^{-K}$; the number of steps in Section 11 is $k+1$. The italic $B_\xi = \Lambda_1^c$ is the filled bad region (with $B_\xi^{\mathrm{fin}} = (\Lambda_1^{\mathrm{fin}})^c$), the calligraphic $\mathcal{B}_\xi$, $\mathcal{B}_\xi^\Lambda$ the exterior response functional, and the fraktur $\mathfrak{B}$ a compatible family of the expansion; B_3 is the constant of [3]; the fraktur $\mathfrak{B}_0 = (\Lambda_1^{(0)})^c$ is a set of sites and $\mathfrak{B}_Q$ a determining set (Section D.8.1), the calligraphic $\mathcal{B}_y$ the block with centre y, and B without the subscript ξ a generic vector of $\mathfrak{g}^S$, the ball B_δ, the bound B_C or the host constants B_h, B_{h}; in Section 3.1 the letter F denotes integrands, and the logarithms $F_{I,C,T}$, $F_{C,T}$, $F_{C,\infty}(n)$ and the integrals $F_\xi(t)$ are named with their subscripts; elsewhere F_γ is a set of edges of a polymer and $F(\Gamma)$ a weight in a cluster sum.

3.1 Branch Data, the Outer Operation and the Conditional Integral

Fix one term of the history sum of [3, (1.11)–(1.30)]: the regions Ω_1, R_1 ([3] uses R_1 both for the large-fluctuation region of (1.18) and for the scale integer of p.246, the smallest power of L with $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$, which grows as $g_0 \to 0$; both uses are kept, the scale appearing in the products LMR_1, $L^2M_2R_1$, MR_1, M_2R_1 and in its defining inequality $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$; here $g_1 = g_1(g_0)$ is the first-step coupling of [3], the constant value on $\Lambda_1^{\sim -1}$ of the solution of (1.27), $g_0^{-2} = g_1^{-2} + \beta_1(g_0)$ with the β_1 of [3], a function of g_0 alone, and $\varepsilon_1 = g_1 p_0(g_1)$ as on p.246), the derived Λ_1, S_1 of (1.20) with Λ_1 the filled domain of the modification following (1.20), the new field V, the integration variable V_0 on Ω_1^c, and the background $U_1 = U_1(V, V_0)$ of (1.12) in its axial mean gauge, equal to V_0 on Ω_1^c. The young carrier is $I = \Lambda_1^*$, the retained free carrier is $J = \Omega_1^* \setminus I$, and $A_{\mathrm{e}} \in \mathfrak{g}^J$ is the retained fluctuation. The young variable is $a \in \mathfrak{g}^I$; both are extended by zero to Ω_1^* where needed. The branch data are $\xi = (\Omega_1, R_1, V, V_0, A_{\mathrm{e}})$. The constraint $\delta(\bar{U}V^{-1})$ of [3, (1.11)] is a product over the coarse bonds, and it is split by rows: on the rows of Ω_1 (coarse bonds with a block centre in Ω_1) the constraint $\bar{U} = V$ is the critical-point constraint of (1.12), imposed on the integral over the bonds of Ω_1 (which include the bonds crossing from Ω_1 to Ω_1^c; the rows with one block centre off Ω_1 average also over bonds of Ω_1^c, which enter as the constant V_0); read, like every group δ-function here, by integrating against the coarse variables it constrains, it is the fibre measure of the pushforward of the fine measure under the map $\Phi(U) = (\bar{U}|_{\Omega_1^{[1]}}, U|_{\Omega_1^c})$, a submersion on the open set $\mathcal{O}^\Phi$ of fine configurations at which the averages and means on the rows of Ω_1 are defined and $D\Phi$ is onto, which contains the set $\mathcal{X}$ of the measurability paragraph below and, on every branch, $D(V, V_0)$ (shown there) (the fibre measures of a C^1 submersion f of an open set $\mathcal{Y}$ of a manifold with a smooth positive density, here G^E with Haar measure or $\mathfrak{g}^E$ with Lebesgue measure, into such a manifold of dimension k, are defined at *every* point y of $\mathsf{f}(\mathcal{Y})$, not only almost everywhere: by the submersion theorem $\mathcal{Y}$ is covered by open sets $\mathcal{Y}_j$ with C^1 diffeomorphisms ψ_j from open sets of $\mathbb{R}^k \times \mathbb{R}^{n-k}$ onto $\mathcal{Y}_j$ and $\mathsf{f}(\psi_j(y,z)) = y$; with a continuous partition of unity (θ_j) subordinate to $(\mathcal{Y}_j)$ put $\mu_y^{\mathsf{f}}(g) \coloneqq \sum_j \int (\theta_j g)(\psi_j(y,z)) \,|\det D\psi_j(y,z)|\, \mathrm{d}z$ for Borel $g \geqslant 0$ on $\mathcal{Y}$, the determinants taken with respect to the densities, that of the manifold on $\mathcal{Y}_j$ and, on $\mathbb{R}^k \times \mathbb{R}^{n-k}$, the product of the target density in the variable y and Lebesgue measure $\mathrm{d}z$, so that $\mathrm{d}y$ in (i) is the target density. Then (i) $\int_{\mathcal{Y}} \varphi(\mathsf{f}(x)) g(x)\, \mathrm{d}x = \int \varphi(y)\, \mu_y^{\mathsf{f}}(g)\, \mathrm{d}y$ for Borel $\varphi \geqslant 0$ (the change of

variables formula and Fubini's theorem); (ii) $y \mapsto \mu^{\mathsf{f}}_y(g)$ is continuous for every continuous g with compact support in $\mathcal{Y}$ (dominated convergence); (iii) the family does not depend on the choices made, two families with (i) and (ii) agreeing for almost every y by (i) and hence for every y by (ii); (iv) for every $y \in \mathsf{f}(\mathcal{Y})$ and every such g, $\mu^{\mathsf{f}}_y(g) = \lim_{\delta\to 0} |E_\delta|^{-1} \int_{\mathsf{f}^{-1}(E_\delta)} g \,\mathrm{d}x$ for every family of Borel sets E_δ with $B_{c\delta}(y) \subseteq E_\delta \subseteq B_{C\delta}(y)$ for constants $0 < c \leqslant C$, by (i) with $\varphi = \mathbf{1}_{E_\delta}$ and (ii) (the mean of a continuous function over E_δ tends to its value at y); so μ^{f}_y is carried by $\mathsf{f}^{-1}(y)$, and the fibre measures of the restriction of f to an open subset of $\mathcal{Y}$ are the restrictions of the μ^{f}_y. From (iv) follows the transformation rule, at every level: if Ξ is a C^1 diffeomorphism of an open set $\mathcal{N}_0$ onto an open subset of $\mathcal{Y}$, ψ a C^1 diffeomorphism of an open neighbourhood of $\mathsf{f}(\Xi(\mathcal{N}_0))$ onto its image, and $\mathsf{f}' \coloneqq \psi \circ \mathsf{f} \circ \Xi$ on $\mathcal{N}_0$ (a submersion), then $\mu^{\mathsf{f}'}_{\psi(y)}(g \circ \Xi \,|\det D\Xi|) = |\det D\psi(y)|^{-1} \mu^{\mathsf{f}}_y(g)$ for every $y \in \mathsf{f}(\Xi(\mathcal{N}_0))$ and every Borel $g \geqslant 0$ on $\Xi(\mathcal{N}_0)$: $\int_{\mathcal{N}_0} \mathbf{1}_{\mathsf{f}'(x')\in B_\delta(\psi(y))}\, g(\Xi(x'))\,|\det D\Xi(x')|\,\mathrm{d}x' = \int_{\Xi(\mathcal{N}_0)} \mathbf{1}_{\mathsf{f}(x)\in\psi^{-1}(B_\delta(\psi(y)))}\, g(x)\,\mathrm{d}x$ exactly, for every δ (the change of variables formula for Ξ); the sets $E_\delta = \psi^{-1}(B_\delta(\psi(y)))$ satisfy the condition of (iv) with $|E_\delta|/|B_\delta| \to |\det D\psi(y)|^{-1}$; dividing by $|B_\delta|$ and letting $\delta \to 0$ gives the identity for continuous g of compact support, and two Radon measures on the fibre, a C^1 submanifold, agreeing on these agree. For Φ on $\mathcal{O}^\Phi$: for Borel $F \geqslant 0$ supported in $\mathcal{O}^\Phi$ and Borel $\varphi \geqslant 0$ (hence for bounded Borel φ when F is integrable), $\int \varphi(\Phi(U))F(U)\,\mathrm{d}U = \int \varphi(V,V_0) \int_{\Phi^{-1}(V,V_0)} F \,\mathrm{d}\mu_{V,V_0}\,\mathrm{d}V|_{\Omega_1^{[1]}}\mathrm{d}V_0$ with $\mu_{V,V_0} \coloneqq \mu^\Phi_{(V,V_0)}$, defined for every $(V,V_0) \in \Phi(\mathcal{O}^\Phi)$, in particular for every admissible pair, and since Φ preserves the coordinate $U|_{\Omega_1^c}$ its straightening maps may be taken to preserve it, so that for every (V,V_0) the measure $\mu^\Phi_{(V,V_0)}$ is the fibre measure over V of the submersion $U|_{\Omega_1} \mapsto \bar{U}|_{\Omega_1^{[1]}}$ on the slice $U|_{\Omega_1^c} = V_0$, the Jacobians agreeing because the differentials of the straightening maps are block triangular with the identity in the V_0 block), and the chart, linearisation, and elimination steps of [3] listed after (3.7) compute μ_{V,V_0} in the variables of (1.14)–(1.15), in which the constraint is linear, by the transformation rule applied at the level of the pair itself and two facts. First, if Θ is a C^1 diffeomorphism of an open set $\mathcal{N}_\Theta$ onto its image and $\mathcal{Q} \circ \Theta = \Lambda$ on $\mathcal{N}_\Theta$ with Λ linear and onto, then $\mathcal{Q}$ is a submersion on $\Theta(\mathcal{N}_\Theta)$ and its fibre measures there are $\mu^{\mathcal{Q}}_y = \Theta_*(|\det D\Theta|\,\mu^\Lambda_y)$ for every $y \in \Lambda(\mathcal{N}_\Theta)$, the transformation rule with $\Xi = \Theta$ and $\psi = \mathbf{1}$ (the change of variables formula $\int_{\Theta(\mathcal{N}_\Theta)} \varphi(\mathcal{Q}(x))F(x)\,\mathrm{d}x = \int_{\mathcal{N}_\Theta} \varphi(\Lambda(x'))F(\Theta(x'))\,|\det D\Theta(x')|\,\mathrm{d}x'$ alone gives the identity for almost every y with an exceptional set depending on $\mathcal{Q}$, here on $U_1(V,V_0)$, which would not cover the level 0 needed at every pair; the rule at every level does). Second, the fibre measure over every y, in particular over 0, of a linear surjection Λ whose restriction Λ_{d} to the designated variables is invertible is Lebesgue measure on the free variables x_{f} times $|\det \Lambda_{\mathrm{d}}|^{-1}$, since $(y, x_{\mathrm{f}}) \mapsto (x_{\mathrm{f}}, \Lambda_{\mathrm{d}}^{-1}(y - \Lambda_{\mathrm{f}} x_{\mathrm{f}}))$ is a single straightening map of Λ with Jacobian $|\det \Lambda_{\mathrm{d}}|^{-1}$ (the sifting of $\delta(\Lambda x)$ over the designated variables); here that restriction is $\bigoplus_c \alpha_c(U_1)$ up to the normalisation of the δ-functions, which gives $z^{(0)}(U_1)$. The transformation rule is applied first on the slice $U|_{\Omega_1^c} = V_0$ with Ξ the composition of the chart $A' \mapsto e^{iA'}$ and the right translation by U_1 on the bonds of Ω_1 ($|\det D\Xi| = \prod_b \sigma(A'(b))$, the Haar density, with respect to Lebesgue and Haar measure) and ψ the row chart $W \mapsto \frac{1}{i}\log(WV^{-1})$ on a neighbourhood of V in $G^{\Omega_1^{[1]}}$ ($|\det D\psi(V)| = \sigma_0^{-|\Omega_1^{[1]}|}$ with respect to Haar and Lebesgue measure), under which the slice submersion becomes $\mathcal{Q}_{U_1}$ of (3.9) and the level V becomes the level 0: $\mu_{V,V_0} = \sigma_0^{-|\Omega_1^{[1]}|}\Xi_*(\prod_b \sigma(A'(b))\,\mu_0^{\mathcal{Q}_{U_1}})$ on Ξ of the chart domain (the rule is applied on $\mathcal{N}_0 \coloneqq \Xi^{-1}(\mathsf{f}^{-1}(\mathrm{dom}\,\psi))$, f the slice submersion, the part of the chart domain whose row averages lie in the domain of the row chart; every point of the fibre over V in Ξ of the chart domain lies in $\Xi(\mathcal{N}_0)$, since there $\mathsf{f} = V \in \mathrm{dom}\,\psi$, and the fibre measure of a restriction is the restriction of the fibre measure, so nothing is lost), which is the conversion of the row δ-functions in the step list, at the level of the pair itself. The two facts are then applied to $\mathcal{Q} = \mathcal{Q}_{U_1}$ at fixed (V,V_0), at the level 0, in the chart variable A' of (1.13), with

$\Theta(A) = A - h_{U_1}\widetilde{D}(A)$ on the open set $\mathcal{N}_\Theta$ of Lemma B.8(iv), on which $\widetilde{D} = \varphi$ is analytic, solves the row equations and has $\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}) > 0$, on which Θ takes chart-valued phases (so that $\Theta(\mathcal{N}_\Theta) \subseteq \mathcal{N}_0$, the set on which the transformation rule was applied above: the phases of $\Theta(A)$ are chart-valued, and at every row c the row phase $\Theta(A)|_{\mathcal{E}_c}$ has norm below $\vartheta_* < 2\varrho_*$, where $\mathcal{N}_{U_1,c}$, and hence $\mathcal{Q}_{U_1,c}$ of (3.9), is defined and analytic (Lemma B.7(i), which applies at every row by Lemma B.8(i)), so that the row averages lie in the domain of the row chart), and which contains the argument of every $U \in D^{(11)}(V, V_0)$ (Lemma B.8(ii)) and does not depend on the region R_1 (this step precedes the insertion of (1.18), which introduces R_1): $\mathcal{Q}_{U_1} \circ \Theta = LQ_{U_1} = \Lambda$ on $\mathcal{N}_\Theta$ (the row equations and $LQ_{U_1}h_{U_1} = \mathbf{1}$), and Θ is injective on $\mathcal{N}_\Theta$, since $\Theta(A) = \Theta(A'') = q$ gives $\widetilde{D}(A) = \mathcal{N}_{U_1}(q) = \widetilde{D}(A'')$ by the row equations $d_c = \mathcal{N}_{U_1,c}(A - h_{U_1}d)$, hence $A = q + h_{U_1}\widetilde{D}(A) = A''$; so Θ is a diffeomorphism of $\mathcal{N}_\Theta$ onto the open set $\Theta(\mathcal{N}_\Theta)$ by the inverse function theorem; on the rows of Ω_1^c (both block centres off Ω_1) it is the constraint $\bar{V}_0 = V$ on $V_0 = U|_{\Omega_1^c}$, written $\delta(\bar{V}_0 V^{-1})$ below, and the constrained integral $\int \mathrm{d}V_0|_{\Omega_1^c}\, \delta(\bar{V}_0 V^{-1}) \cdots$ in the form in which Bałaban writes it in [3, (1.14)–(1.15)] (in (1.11) it is part of $\int \mathrm{d}U\, \delta(\bar{U}V^{-1})$) is taken as it stands there and read, like every group δ-function of a coarse variable in this paper (the paragraph following Lemma 1.1), by integrating it against the coarse variables it constrains: for bounded Borel h on G^{E_1} and nonnegative Borel F, $\int \mathrm{d}V\, h(V) \int \mathrm{d}V_0\, \delta(\bar{V}_0 V^{-1}) F(V, V_0) = \int \mathrm{d}V|_{\Omega_1^{[1]}} \int \mathrm{d}V_0\, h(V) F(V, V_0)$ with $V(c) = M(V_0, c)$ of (1.4) on the rows c of Ω_1^c (a function of the bonds of Ω_1^c alone, Lemma 1.1(i)); so $\int \mathcal{O}[F](V)\, \mathrm{d}V$ below is the integral of F against the measure in V that the outer operation defines, and $\mathcal{O}[F](V)$ is a Haar density only where that measure has one. Let $D^{(11)} = D^{(11)}(\Omega_1, V)$ be the domain of integration in [3, (1.11)]: the fine configurations U with $\chi_1(\Omega_1)\zeta(\Omega_1^c)\chi_0'(\Omega_1) \neq 0$ on which the block averages $\mathsf{M}(U)$ are defined on every row of Ω_1 and the local group means $\mathcal{U}_U(y, x)$, $x \in \mathcal{B}_y$, $y \in \Omega_1^{(1)}$, of the gauge-fixing functional $\mathbf{G}(\Omega_1^{(1)}, U)$ of (1.11) are defined (in the chart of the preamble, Bałaban's principal logarithm; by [3, pp. 247 and 248] with Lemma 1 of [9] and [8, pp. 24–25], derived in (B1), these two conditions are implied by the nonvanishing of the cutoffs and the constraint for ε_0 small, so that $D^{(11)}$ is the domain of integration in [3, (1.11)], taken as it stands; they are kept explicit so that $D^{(11)} \subseteq \mathcal{O}_c$ on every row of Ω_1 is visible), subject to the constraint $\bar{U} = V$ on the rows of Ω_1 and $\bar{V}_0 = V$ on the rows of Ω_1^c. Its fibre over the restriction is $D^{(11)}(V, V_0) = \{U \in D^{(11)} : U|_{\Omega_1^c} = V_0\}$. A pair (V, V_0), written (Ω_1, V, V_0) where the region is to be displayed and called a pair throughout (Ω_1 is fixed on the branch), is *admissible* when $D^{(11)}(V, V_0) \neq \varnothing$; (B1) provides the background $U_1 \in D^{(11)}(V, V_0)$ for every admissible pair. The domain of integration of (1.15) with the cutoff (1.25) is in general smaller than $D^{(11)}(V, V_0)$ (the small- and large-fluctuation tests of (1.18) in (1.25) are further conditions; on R_1 the large-fluctuation tests exclude the configurations near U_1); it is described in the variables of (1.15) as follows. For an admissible pair, a free vector $B \in \mathfrak{g}^{\Omega_1^*}$ and a family $d = (d_c)_{c \in \Omega_1^{[1]}}$ of solutions of the row equations (3.24) at the argument $A = g_0 \mathsf{C}_{U_1} B$ (that is, $d_c = \mathcal{N}_{U_1,c}(A - h_{U_1}d)$ for every row c), put $q_d(B) \coloneqq A - h_{U_1}d \in \mathfrak{g}^{\Omega_1}$, the phase (3.25) formed with this family ([3, (1.13)–(1.14)] at the background U_1; on the free bonds $q_d(B)(b) = g_0 B(b)$), and let $U_d(B)$ be the fine field equal to $e^{iq_d(B)}U_1$ on the bonds of Ω_1 and to V_0 on the bonds of Ω_1^c. The family d is *of chart class* when $h(q_d(B)(b)) < \varrho_G$ for every bond b of Ω_1, that is, when the whole phase $q_d(B)$ is chart-valued, the class of the variable A' of [3, (1.13)]. Let

$$\mathcal{A}_\xi \coloneqq \{B \in \mathfrak{g}^{\Omega_1^*} : \text{there is a family } d \text{ of chart class at } g_0 \mathsf{C}_{U_1} B \\ \text{with } U_d(B) \in D^{(11)}(V, V_0) \text{ and the cutoff (1.25) nonzero at } U_d(B)\}\,, \tag{3.2}$$

the cutoff (1.25) being a function of the fine field and the branch data (its tests run over the free bonds $(\square'^{\sim 2})^*$, at which the scaled variable is $g_0^{-1}\frac{1}{i}\log(UU_1^{-1})$, by (1.18)–(1.19), p.250, and (1.25), p.252, in the variables of (1.15), and through $\chi_0'(S_1)$ over the fine field; no test

involves a designated value of the linearised variable, so the cutoff does not involve $\widetilde{D}$). For $B \in \mathcal{A}_\xi$ the family d of chart class with $U_d(B) \in D^{(11)}(V, V_0)$ is unique and equals $\varphi(g_0 \mathsf{C}_{U_1} B)$ (Lemma B.8(iii)), and $\widetilde{D}_{U_1}(g_0 \mathsf{C}_{U_1} B) \coloneqq d$, $q_{U_1}(B) \coloneqq q_d(B)$ and $U(B) \coloneqq U_d(B)$ are defined on $\mathcal{A}_\xi$ by this uniqueness; the cutoff (1.25) is then a function of B, U_1 and the branch data. Since $q_{U_1}(B)$ is chart-valued, it is the chart logarithm of the fluctuation $U(B)U_1^{-1}$, so that $B \mapsto U(B)$ is injective on $\mathcal{A}_\xi$ ($U(B)$ determines $U(B)U_1^{-1}$, hence its chart logarithm $q_{U_1}(B)$, hence B on the free bonds, where $q_{U_1}(B)(b) = g_0 B(b)$). Conversely, every $U \in D^{(11)}(V, V_0)$ at which the cutoff (1.25) is nonzero is of the form $U(B)$ with $B \in \mathcal{A}_\xi$: by (B1) the fluctuation UU_1^{-1} lies in the chart domain bond by bond on Ω_1, so $A' \coloneqq \frac{1}{i}\log(UU_1^{-1})$ of (1.13) is defined and chart-valued; the constraint $\mathsf{M}(U)(c) = V(c) = \mathsf{M}(U_1)(c)$ on the rows of Ω_1 gives $\mathcal{Q}_{U_1,c}(A') = 0$ in (3.9), that is $LQ_{U_1}A' = -\mathcal{N}_{U_1}(A')$; with $d \coloneqq \mathcal{N}_{U_1}(A')$ the vector $A' + h_{U_1}d$ lies in the kernel of Q_{U_1} ($LQ_{U_1}h_{U_1} = \mathbf{1}$) and agrees with A' on the free bonds (h_{U_1} has its outputs at the designated bonds), so it equals $g_0 \mathsf{C}_{U_1} B$ with $B \coloneqq g_0^{-1} A'|_{\Omega_1^*}$, the kernel of Q_{U_1} being the graph of C_{U_1} over the free bonds by (3.12) ($\alpha_c(U_1)$ invertible, (B2) and Lemma 3.2); hence $A' = g_0 \mathsf{C}_{U_1} B - h_{U_1} d$, $d_c = \mathcal{N}_{U_1,c}(A') = \mathcal{N}_{U_1,c}(A - h_{U_1}d)$ solves every row equation, d is of chart class because A' is chart-valued, and $U_d(B) = e^{iA'}U_1 = U$ on Ω_1 while $U|_{\Omega_1^c} = V_0$; so $B \in \mathcal{A}_\xi$ and $U(B) = U$. Thus $\mathcal{A}_\xi$ is the domain of integration of (1.15) with the cutoff (1.25) in the free variable B of (1.15), and $B \mapsto U(B)$ is a bijection of $\mathcal{A}_\xi$ onto $D(V, V_0) \coloneqq \{U \in D^{(11)}(V, V_0) : \text{the cutoff (1.25) is nonzero at } U\} = U(\mathcal{A}_\xi)$, the same domain in the fine variable; put $D(\Omega_1, R_1, V) \coloneqq \bigcup_{V_0} D(V, V_0)$. By (B4)(ii) the differential of $B \mapsto U(B)$ is injective at every point of $\mathcal{A}_\xi$, with image the tangent space of the fibre $\Phi^{-1}(V, V_0)$ at $U(B)$. For an admissible pair and $A_\mathrm{e} \in \mathfrak{g}^J$, $a \in \mathfrak{g}^I$ let $U_\xi(a) = U(\Psi_\xi(a + A_\mathrm{e}))$ be the reconstructed fine field (3.26) (the inverse of the changes of variables [3, (1.13), (1.14), (1.22)] at the background U_1, defined where those are: where $(a + A_\mathrm{e})(b) \in \mathcal{W}_b$ for every $b \in S_1^*$, Section 3.4, and every row solution is defined at $g_0 \mathsf{C}_{U_1} \Psi_\xi(a + A_\mathrm{e})$), and let

$$\mathcal{D}_\xi \coloneqq \{(A_\mathrm{e}, a) \in \mathfrak{g}^J \times \mathfrak{g}^I : \ \Psi_\xi(a + A_\mathrm{e}) \text{ is defined and } \Psi_\xi(a + A_\mathrm{e}) \in \mathcal{A}_\xi\} \tag{3.3}$$

be the domain of integration of (1.15) with the cutoff (1.25) in the variables of (1.22); by (B4)(ii) (Lemma 3.8) Ψ_ξ maps $\mathcal{D}_\xi$ bijectively onto $\mathcal{A}_\xi$, so the reconstruction $(A_\mathrm{e}, a) \mapsto U_\xi(a)$ is a bijection of $\mathcal{D}_\xi$ onto $D(V, V_0)$. In particular $\mathcal{D}_\xi \neq \varnothing$ exactly when $\mathcal{A}_\xi \neq \varnothing$, that is when $D(V, V_0) \neq \varnothing$: no admissible pair with nonempty domain of integration is lost in passing to the variables of (1.22). A *branch* is a value of the data ξ such that (Ω_1, V, V_0) is admissible and $(A_\mathrm{e}, a) \in \mathcal{D}_\xi$ for some a. On a branch the set $\{a : (A_\mathrm{e}, a) \in \mathcal{D}_\xi\}$ is the open box $\{\Theta_\xi = 1\}$ of (3.42) (Lemma 3.11(iv)), and the large-field factor $\zeta_\xi(A_\mathrm{e})$ of (3.6) and the retained cutoff $\chi_{\xi,\mathrm{e}}(A_\mathrm{e})$ of (3.43) are defined and nonzero. The branch indicator $\iota_\xi(A_\mathrm{e})$ is one when ξ is a branch and zero otherwise; as the fibre over a branch value is the open box, which contains $a = 0$, $\iota_\xi(A_\mathrm{e}) = 1$ exactly when $(A_\mathrm{e}, 0) \in \mathcal{D}_\xi$, a Borel condition (a section of the Borel set $\mathcal{D}_\xi$, measurability paragraph). It is a factor of the integrand of (3.4), which therefore vanishes off the branches, and every object below is defined on branches only. The sets $\mathcal{A}_\xi$, $D(V, V_0)$ and $\mathcal{D}_\xi$ depend on ξ only through (Ω_1, R_1, V, V_0): the cutoff (1.25) in the variable B of (1.15) and the map Ψ_ξ do not involve A_e, which is a variable of these sets.

For a branch function $\mathfrak{M}_\xi$, the outer operation is

$$\mathcal{O}[\mathfrak{M}](V) = \sum_{\Omega_1, R_1} \chi_1(\Omega_1) \int \delta(\bar{V}_0 V^{-1}) \left[\int \iota_\xi(A_\mathrm{e}) \zeta_\xi(A_\mathrm{e}) \chi_{\xi,\mathrm{e}}(A_\mathrm{e}) \kappa_\xi \mathfrak{M}_\xi \, \mathrm{d}A_\mathrm{e}|_J \right] \mathrm{d}V_0|_{\Omega_1^c} \ , \tag{3.4}$$

with the prefactor

$$\kappa_\xi = z^{(0)}(U_1) \, g_0^{d_G |\Omega_1^*|} \sigma_0^{|\Omega_1^*|} z^{-(L^4-1)|\Omega_1^{(1)}|} \exp\left(-g_0^{-2} S(U_1) - \mathsf{E}\right) \ . \tag{3.5}$$

Here $\chi_1(\Omega_1)$ is the background test of [3, (2.16)–(2.17)] at $k=1$, a function of V; $\delta(\bar V_0 V^{-1})$ is the part of the constraint $\delta(\bar U V^{-1})$ of (1.11) on the rows of Ω_1^c (both blocks off Ω_1), as in [3, (1.14)–(1.15)], where it stands in front of the V_0 integral. The large-field factor $\zeta(\Omega_1^c)$ of (1.11) is defined by the characteristic functions and gauge-fixing exponentials of (1.6) and (1.8) whose cubes or blocks lie in Ω_1^c, resummed over the admissible P_0, P_1, Q_1 (p.248). It is a sum over the admissible (P_0, P_1, Q_1) compatible with Ω_1 of products of nonnegative functions of the fine field U on the bonds near their cubes or blocks and of constants; the factors of (1.4), (1.6) and (1.8) not displayed in (1.11), and the bonds involved, are the following. The large-field factors $\chi_0^c(\square)$ of (1.1) for the LM_2R_0-cubes $\square \subseteq P_0$, and the factors $\chi_0(\square)$, $\square \subseteq P_0^c$, that survive the removal of (B3)(i), those of the cubes $\square \not\subseteq \Omega_1^\sim$: every such cube is at ℓ^∞ distance at least LMR_1 from Ω_1 ($\Omega_1^\sim$ adds a layer of LMR_1-cubes, the partitions are nested, and the large-field regions are at distance at least $2LMR_1$ from Ω_1, p.248), and the factor tests the plaquettes of $\square^\sim$, whose bonds are within $M_2R_0 \leqslant LM_2R_1 \leqslant MR_1/8 < LMR_1$ of $\square$ (conventions: $R_0 \leqslant LR_1$ by [3, (2.9)]; $M \geqslant 8LM_2$), so it is a function of V_0. The factors $\chi_1^c(\square')$, $\square' \subseteq P_1$, and $\chi_1(\square')$ of (1.4) for the cubes $\square' \subseteq P_1^c$ not in Ω_1 (those in Ω_1 form $\chi_1(\Omega_1)$) are functions of the block field V (through the local minima $U_{1,\square'}(V)$ of (1.2)). The gauge-fixing factors χ_{Ax} of the blocks $\mathcal{B}_y$, $y \in P_1^1$, not in $\Omega_1^\sim$ (those in $\Omega_1^\sim$ are removed by (B3)(i)), the exponentials $e^{-g_0^{-2}\mathbf{G}(P_1^1 \setminus \Omega_1^{(1)}, U)}$ of (1.6)–(1.7) of the blocks of P_1^1 off Ω_1, and the constants $z^{-(L^4-1)|P_1^1 \setminus \Omega_1^{(1)}|}$ of the normaliser of (1.5)–(1.6) for these blocks (the exponent of (1.11) counts the blocks of $\Omega_1^{(1)} \subseteq P_1^1$ only): a block off Ω_1 lies in Ω_1^c, since Ω_1 is a union of blocks, and the axial transports $\mathcal{U}_U(y,x)$ use the bonds inside the block, which are bonds of Ω_1^c, so these are functions of V_0. The factors $\chi_0'^c(\square')$, $\square' \subseteq Q_1$, and $\chi_0'(\square')$ of (1.8) for the $L^2M_2R_1$-cubes $\square' \subseteq \Omega_1^c$ not in Q_1 (those in Ω_1 form $\chi_0'(\Omega_1)$) involve the fine field on $(\square'^{\sim 2})^*$, the bonds within $2L^2M_2R_1+1$ of $\square'$, together with the local minima $U_{1,\square'}(V)$ of (1.2): for $\square' \subseteq Q_1$ these are bonds of Ω_1^c (Q_1 is at distance at least $2LMR_1$ from Ω_1 by (1.10)), and for the other cubes they are bonds of Ω_1^c and bonds of Ω_1 within $2L^2M_2R_1+1$ of Ω_1^c, which lie in the layer $\Omega_1 \setminus \Omega_1^{\sim -1} \subseteq S_1$ ([3, (1.20)]), since $LMR_1 \geqslant 8L^2M_2R_1$. So $\zeta(\Omega_1^c)$ is a nonnegative function of V, V_0 and of the fine field on the bonds of Ω_1 within $2L^2M_2R_1+1$ of Ω_1^c, the last dependence coming from the factors $\chi_0'(\square')$, $\square' \subseteq \Omega_1^c \setminus Q_1$, only. The phrase of [3] "localized in Ω_1^c" refers to the cubes and blocks, and $\zeta(\Omega_1^c)$ is not a function of V, V_0 alone. In the variables of (1.22) it is

$$\zeta_\xi(A_{\mathrm{e}}) = \zeta(\Omega_1^c)\big(U_\xi(a)\big)\ , \tag{3.6}$$

with $U_\xi(a)$ the reconstructed fine field (3.26); the right side does not depend on a, because V, V_0 are branch data and the bonds of Ω_1 on which $\zeta(\Omega_1^c)$ depends are at distance at least $LMR_1 - 2L^2M_2R_1 - 2 > 2R_{\mathrm{c}} + 2L$ from Λ_1, so their links are the links of far rows, functions of A_{e} and the branch data only (Lemma 3.5, (B4)). It is therefore a factor of the A_{e} integrand, as written in (3.4); where exactly it stands is immaterial for the identities of this section, which concern the a-integral at fixed branch data. The constants: σ_0 is the Haar constant of (1.14), z the gauge-fixing normaliser of (1.5) and E the E of [3], all independent of the branch; $z^{(0)} = z^{(0)}(U_1)$ is the factor of the δ-function integration over the designated variables (p.249); it is $\prod_{c \in \Omega_1^{[1]}} |\det_{\mathfrak{g}} \alpha_c(U_1)|^{-1}$ times a constant fixed by the normalisation of the δ-functions in [3] and by $|\Omega_1^{[1]}|$, with α_c of (3.11), so it depends on the branch through U_1 (the linearised constraint $(\widetilde{Q}A')(c) = 0$ of p.249 is the row c of $Q_{U_1}A' = 0$, whose coefficient at $A'(b_0(c))$ is $\alpha_c(U_1)$, Lemma 3.2 and [4, p. 267]). None of these is used except as a factor of κ_ξ, which is the prefactor of [3] verbatim. The retained cutoff $\chi_{\xi,\mathrm{e}}$ is defined in (3.43): it is the product of the retained factors of [3, (1.25)], namely the good free-bond tests of (1.18) at the bonds of J, the large-fluctuation tests on R_1 and $\chi_0'(S_1)$, all composed with the change of variables (1.22), and it depends on A_{e} and the branch data only (Lemma 3.11(ii)).

The conditional integrand $\mathcal{K}_\xi(a)$, $a \in \mathfrak{g}^I$, is the integrand of [3, (1.15)] in the variables of (1.22) at the actual branch data, divided by the factors already in κ_ξ, ζ_ξ and $\chi_{\xi,\mathrm{e}}$ and by the factors $\chi_1(\Omega_1)$, $\delta(\bar{V}_0 V^{-1})$ standing outside the A_e integral; it is written out in (3.44), and $\mathcal{J}^{\mathrm{a}}_\xi[f] = \int f \mathcal{K}_\xi \, \mathrm{d}a$ is the conditional functional.

Measurability and integrability. The integrand of [3, (1.15)] with the cutoff (1.25), in the variables $(V, V_0, A_\mathrm{e}, a)$ of (1.22), is a finite composition of indicators of sets defined by inequalities between continuous functions (the tests), of the background $U_1(V, V_0)$, of the local minima $U_{1,\square'}(V)$ of (1.2) entering χ'_0 and of the field $\mathbf{H}_{1,Ax}$ of (1.21) entering the bondwise maps (1.22), of the analytic objects of [3] in (B4) and of the bondwise maps (1.22). The background is a Borel function of the admissible pairs, by the following argument, which uses (B1), (B2) and the smoothness of the constraint. Let $\mathcal{X}$ be the open set of fine configurations U on which $\mathsf{M}(U)$ is defined on every row of Ω_1, the local group means $\mathcal{U}_U(y, x)$, $x \in \mathcal{B}_y$, $y \in \Omega_1^{(1)}$, are defined (chart convention of the preamble; they are analytic there) and every $\alpha_c(U)$, $c \in \Omega_1^{[1]}$, is invertible (Section 3.2), and let $\Phi(U) = (\bar{U}|_{\Omega_1^{[1]}}, U|_{\Omega_1^c})$ on $\mathcal{X}$; $D\Phi(U)$ is onto, since $LQ_U h_U = \mathbf{1}$ with h_U supported on the designated bonds of Ω_1, so each fibre $\Phi^{-1}(V, V_0) \cap \mathcal{X}$ is a C^∞ submanifold and the tangent space of the fibre is the kernel of Q_U, the range of C_U. Let $\mathrm{Crit} = \{U \in \mathcal{X} : D(S + \mathbf{G}(\Omega_1^{(1)}, \bullet))(U) \circ \mathsf{C}_U = 0\}$ be the set of critical points of the constrained functional of [3, (1.12)] on the fibres; it is relatively closed in $\mathcal{X}$, being defined by the vanishing of a continuous function; the critical points of this functional, the points of the fibre at which the differential of the functional vanishes on the tangent space of the fibre, are the points of Crit (on $\mathcal{X}$ the fibres are manifolds with tangent space the range of C_U, and $U_1 \in \mathcal{X}$ by (B2) and its axial mean gauge). The set $\{(V, U) : U \in D^{(11)}(\Omega_1, V)\}$ is Borel in (V, U): it is a finite union of sets defined by non-strict and strict inequalities between functions of (V, U) that are continuous in U and Borel in V (such functions are jointly Borel: G^E is separable metric, and a function continuous in U and Borel in V is a pointwise limit of jointly Borel functions, the standard argument for Carathéodory functions), and by the equalities $\bar{U}(c) = V(c)$ on the rows c of Ω_1 (continuous in U on the open set where the averages are defined) and $\mathsf{M}(U|_{\Omega_1^c}, c) = V(c)$ on the rows of Ω_1^c (the global average (1.4), defined for every U and Borel in U: analytic on the open set $\mathcal{O}_c$ and, off it, the straight transport of the preamble, a continuous function) (the tests involve V, U and the Borel functions $U_{1,\square'}(V)$ of (B1); each factor $\chi'_0(\square')$ is zero for V outside the Borel set $\mathcal{V}_{\mathrm{ax}}(\square')$ of Lemma 3.1 for the determining set of $\square'$, where the configuration it compares with does not exist, by the convention of (B1), so that $\{(V, U) : \chi'_0(\square') \neq 0\} = \{(V, U) : V \in \mathcal{V}_{\mathrm{ax}}(\square'),\ \sup_{b \in (\square'^{\sim 2})^*} |U(b) U_{1,\square'}(V)(b)^{-1} - \mathbf{1}| < 2\delta_0\}$ is Borel, and $U_{1,\square'}(V)$ is never evaluated off $\mathcal{V}_{\mathrm{ax}}(\square')$; likewise the factors of (1.4), functions of V alone, are Borel: by the convention of (B1), $\chi_1(\square') = \mathbf{1}\{V \in \mathcal{V}_{\mathrm{ax}}(\square')\}\, \mathbf{1}\{\sup_{p \subset \square'^{\sim}} |U_{1,\square'}(V)(\partial p) - \mathbf{1}| < \varepsilon_1 L^{-2}\}$ and $\chi_1^c(\square') = 1 - \chi_1(\square')$), intersected with the open set where the averages and the means are defined; and by (B1) for every admissible pair the fibre $D^{(11)}(V, V_0)$ contains exactly one point of Crit, namely $U_1(V, V_0)$. So the graph $\{(V, V_0, U) : U \in D^{(11)}(\Omega_1, V) \cap \mathrm{Crit},\ \Phi(U) = (V|_{\Omega_1^{[1]}}, V_0)\}$ of $(V, V_0) \mapsto U_1(V, V_0)$ is a Borel set (the intersection of a Borel set with the graph of the continuous map Φ on the open set $\mathcal{X}$, a Borel set, relatively closed over $\mathcal{X}$), and a function whose graph is a Borel subset of a product of Polish spaces is Borel on a Borel domain: the domain is the projection of a Borel set with singleton sections, Borel by the Lusin–Novikov theorem [14, Theorem 18.10], and a function on a Borel domain whose graph is Borel is Borel, [14, Theorem 14.12]. Hence the set of admissible pairs is Borel and U_1 is Borel on it. The set $\mathcal{A}_\xi$ of (3.2) is Borel, and q_{U_1} is Borel on it, without circularity between the existence and the uniqueness of the chart-class family: let P be the set of pairs $(B, d) \in \mathfrak{g}^{\Omega_1^*} \times \mathfrak{g}^{\Omega_1^{[1]}}$ such that d solves every row equation at $g_0 \mathsf{C}_{U_1} B$, is of chart class, and $U_d(B) \in D^{(11)}(V, V_0)$ with the cutoff (1.25) nonzero at $U_d(B)$. It is Borel, jointly with

(V, V_0) through the Borel background U_1: the row equations are equalities between functions continuous on the open set where the relative averages are defined, the chart-class condition is open, $U_d(B)$ is continuous in (B, d, U_1), $D^{(11)}(V, V_0)$ is a section of the Borel set above, and the cutoff is a finite product of indicators of inequalities between Borel functions. By definition $\mathcal{A}_\xi$ is the projection of P to the first factor, and every section $\mathsf{P}(B)$ has at most one point, by the uniqueness of the chart-class family (Lemma B.8(iii)); so $\mathcal{A}_\xi$ is Borel by the Lusin–Novikov theorem [14, Theorem 18.10] (the projection of a Borel set with countable sections), and $B \mapsto d$ is Borel on $\mathcal{A}_\xi$ by [14, Theorem 14.12] (its graph is P); hence q_{U_1}, $U(B)$ and the row solutions $\widetilde{D}_c$ of (B4)(i) are Borel on $\mathcal{A}_\xi$, jointly with the branch data; so is $\frac{\delta}{\delta A}\widetilde{D}$ at the arguments $A = g_0 \mathsf{C}_{U_1} B$, $B \in \mathcal{A}_\xi$, by the following formula. With $A' = q_{U_1}(B)$, the identity $\mathcal{Q}_{U_1}(A - h_{U_1}\widetilde{D}(A)) = LQ_{U_1}A$ holds on the open set $\mathcal{N}_\Theta \ni A$ of Lemma B.8(iv), on which $\widetilde{D} = \varphi$ solves the row equations and is analytic (by (3.9) and $LQ_{U_1}h_{U_1} = \mathbf{1}$), so by the chain rule $D\mathcal{Q}_{U_1}(A') \circ (\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}(A)) = LQ_{U_1}$. Applied to $h_{U_1}w$, $w \in \mathfrak{g}^{\Omega_1^{[1]}}$, this gives $D\mathcal{Q}_{U_1}(A')h_{U_1} \circ (\mathbf{1} - \frac{\delta}{\delta A}\widetilde{D}(A)h_{U_1}) = LQ_{U_1}h_{U_1} = \mathbf{1}$ on $\mathfrak{g}^{\Omega_1^{[1]}}$, so the square map $D\mathcal{Q}_{U_1}(A')h_{U_1}$ is invertible, and then $\frac{\delta}{\delta A}\widetilde{D}(A) = (D\mathcal{Q}_{U_1}(A')h_{U_1})^{-1}\big(D\mathcal{Q}_{U_1}(A') - LQ_{U_1}\big)$, a continuous function of (U_1, A') on the open set where the averages are defined, composed with the Borel maps U_1 and q_{U_1}; hence it is Borel. The same identity shows that $\mathbf{1} - \frac{\delta}{\delta A}\widetilde{D}(A)h_{U_1}$ is invertible, hence $\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}(A)) = \det(\mathbf{1} - \frac{\delta}{\delta A}\widetilde{D}(A)h_{U_1}) \neq 0$ (Sylvester's determinant identity), which is the nonvanishing part of (B4)(iii) at every argument of the domain of integration. Moreover $D(V, V_0) \subseteq \mathcal{O}^\Phi$, the set on which Φ is a submersion (the paragraph defining $D^{(11)}$): for $B \in \mathcal{A}_\xi$, $A = g_0\mathsf{C}_{U_1}B$ and $A' = q_{U_1}(B)$, the identity $D\mathcal{Q}_{U_1}(A') \circ (\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D}(A)) = LQ_{U_1}$ just derived has right side onto by (B2) and second factor invertible; hence $D\mathcal{Q}_{U_1}(A')$ is onto. Since $\mathcal{Q}_{U_1,c}(A') = \frac{1}{i}\log(\mathsf{M}(e^{iA'}U_1)(c)V(c)^{-1})$ with $\mathsf{M}(e^{iA'}U_1)(c) = V(c)$, where the differential of the chart is the identity, and $A' \mapsto e^{iA'}U_1$ has invertible differential (A' is chart-valued), the differential of $U \mapsto \bar{U}|_{\Omega_1^{[1]}}$ at $U(B) = e^{iA'}U_1$ is onto, that is, $D\Phi(U(B))$ is onto. Borel as well is $\mathcal{D}_\xi$, the preimage of $\mathcal{A}_\xi$ under Ψ_ξ, which is continuous in its argument on its open domain and depends on the pair through the Borel field $\mathbf{H}_{1,Ax}$ below, hence is jointly Borel by the argument just given, and $D(V, V_0) = U(\mathcal{A}_\xi)$ is Borel as the image of a Borel set under the injective Borel map $B \mapsto U(B)$ (injective, as shown after (3.2)), by the Lusin–Souslin theorem for Borel injections [14, Corollary 15.2]; the Jacobians and the bondwise maps are continuous in their arguments; the local minima $U_{1,\square'}(V)$ of (1.2) are Borel functions of V, and the configuration $U_1(\Omega_1 \cap \Lambda_1^c, V)$ of (1.21) is a Borel function of (V, V_0) ([3] states that it depends on $V|_{\Omega_1 \cap \Lambda_1^c}$; its defining conditions as recorded in (B1) involve V_0 where the rows of $(\Lambda_1^\sim)^c$ cross to Ω_1^c, and the Borel choice is made in the pair): they are the Borel choices of Lemma 3.1 fixed in (B1); hence $\mathbf{H}_{1,Ax} = L\frac{1}{i}\log\big(U_1U_1(\Omega_1 \cap \Lambda_1^c, V)^{-1}\big)$ of (1.21) is Borel in (V, V_0) on the bonds of S_1^*, where alone it is used (the chart logarithm is continuous on the chart domain, which contains the argument there by Lemma D.18 and (3.40)). So the integrand is jointly Borel, and nonnegative. By Tonelli's theorem the section integral $\mathcal{J}_\xi^{\mathrm{a}}[1] = \int \mathcal{K}_\xi \,\mathrm{d}a$ is a $[0, \infty]$-valued Borel function of the branch data, and $\int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]](V)\,\mathrm{d}V$ is the integral of the nonnegative integrand over all variables, equal to the fine integral $\mathcal{Z}_0$ of (3.7) below, which is finite (a bounded function on a compact group); hence the outer integrals converge, $\mathcal{J}_\xi^{\mathrm{a}}[1] < \infty$ for almost every value of the branch data, and on every branch $\mathcal{J}_\xi^{\mathrm{a}}[1] < \infty$, since the integrand (3.44) is bounded on the bounded box $\{\Theta_\xi = 1\}$ (its exponent is the sum of an analytic function bounded on the closed box, Lemma 3.6, and of a constant of the branch, Lemma 3.9). The Wilson function $f_\xi(a) = W_\chi(C; U_\xi(a))$ is jointly Borel for the same reason and $|f_\xi| \leqslant 1$, so $\mathcal{J}_\xi^{\mathrm{a}}[f_\xi]$ is a Borel function of the branch data with $|\mathcal{J}_\xi^{\mathrm{a}}[f_\xi]| \leqslant \mathcal{J}_\xi^{\mathrm{a}}[1]$, and Fubini's theorem applies to it. No bound on $\mathcal{J}_\xi^{\mathrm{a}}[1]$ uniform in the

branch is claimed or needed here. Then

$$\mathcal{Z}_0 = \int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]](V)\,\mathrm{d}V, \qquad \mathbb{E}_\varepsilon[W_\chi(C)] = \frac{1}{\mathcal{Z}_0}\int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[f_\xi]](V)\,\mathrm{d}V\ , \tag{3.7}$$

with $f_\xi(a) = W_\chi(C; U_\xi(a))$ the Wilson function of the reconstructed fine field $U_\xi(a)$ of (3.26). The first identity is obtained by Bałaban's steps from the fine integral to [3, (1.15)] with the cutoff (1.25), performed with $\zeta(\Omega_1^c)$ kept inside the V_0 integral and the cutoff in its exact composed form (the estimate-based removals of [3] are recorded in (B3)); the second is the same derivation with the bounded function $W_\chi(C;\bullet)$ of the fine field, invariant under all fine gauge transformations, inserted under the fine integral. The steps, in the following order, which is that of [3] except at two places (p.248 drops χ_0 and χ_{Ax} before the resummation (1.11), and p.249–250 integrates the δ-functions over the designated variables before inserting (1.18); the operations concerned are multiplications of the integrand by functions of the fine field equal to one, regroupings of a finite sum and the evaluation of δ-functions, and the resulting integrand, a function of the fine field, is the same in either order): the insertion of the block-average δ-functions and of the partitions of unity (1.1), (1.4) (multiplications by functions equal to one, invariant under the fine gauge transformations of (1.5): the tests of (1.1) are plaquette tests and those of (1.4) are functions of V), with the group Dirac measure normalised against Haar measure, $\int \delta(\bar{U}V^{-1})\,\mathrm{d}V = 1$ for every U (the sentence following (1.3); [3, (0.1)] is the blocking operator whose δ-functions these are; this is why $\mathcal{Z}_0 = \int \varrho_1\,\mathrm{d}V$ in the notation of [3]); the Faddeev–Popov gauge fixing (1.5)–(1.6), an identity for the integral of any function invariant under the gauge transformations u of (1.5), which introduces the normaliser $z^{-(L^4-1)|P_1^1|}$ of (1.5) for the blocks of the gauge-fixed region; the insertion of the partition of unity (1.8) after the gauge is fixed (its tests compare U with the axial-gauge minima $U_{1,\square'}(V)$ and are not gauge invariant); the resummation (1.11) over P_0, P_1, Q_1 at fixed Ω_1 (p.248, a regrouping of a finite sum; R_1 is introduced afterwards, in (1.18)), after which the gauge-fixed blocks displayed are those of $\Omega_1^{(1)}$ and the displayed normaliser is $z^{-(L^4-1)|\Omega_1^{(1)}|}$, the gauge-fixing factors and normalisers of the other blocks of P_1^1 being part of $\zeta(\Omega_1^c)$; the removal of (B3)(i), which replaces factors by one where the retained factors are nonzero; the change of variables $U = U'U_1$ on the bonds of Ω_1 at fixed (V, V_0) (a translation of the Haar measure) and $A' = \frac{1}{i}\log U'$ of (1.13), the chart being a bijection of the domain onto its image with the Haar density as Jacobian, $\mathrm{d}U' = \sigma(A')\,\mathrm{d}A'$ bond by bond with $\sigma(0) = \sigma_0$ (the domain lies in the chart domain by (B1)); the conversion of the group δ-functions on the rows of Ω_1: by (3.9) and $\mathsf{M}(U_1)(c) = V(c)$, $\mathsf{M}(e^{iA'}U_1)(c)V(c)^{-1} = e^{i\mathcal{Q}_{U_1,c}(A')}$, and $\delta_G(e^{iX}) = \sigma(X)^{-1}\delta(X) = \sigma_0^{-1}\delta(X)$ on the support, so the row δ-functions become $\sigma_0^{-|\Omega_1^{[1]}|}\prod_c \delta(\mathcal{Q}_{U_1,c}(A'))$ (the transformation rule of the paragraph preceding (3.2), applied at the level of the pair (V, V_0) itself, as is every later change of variables of this list: the fibre measures are transported at that level, not for almost every level); the linearising substitution $A' = B - h_{U_1}\widetilde{D}(B)$ of (1.14) ([3] writes both variables as A; in this step list B is the unscaled variable of (1.14) and A the scaled one of p.249, whereas Section 3.1 writes B for the scaled free vector and $A = g_0\mathsf{C}_{U_1}B$ for the unscaled kernel vector, and the lemmas follow Section 3.1), the change of variables by the diffeomorphism Θ of the paragraph preceding (3.2), defined there on the set $\mathcal{N}_\Theta$ of Lemma B.8(iv), which does not depend on R_1, since this step precedes the insertion of (1.18), which introduces R_1 ((B4)(i), (iii) and Lemma B.8(iv); Θ is injective with $\mathcal{Q}_{U_1}\circ\Theta = LQ_{U_1}$), with the Jacobian $|\det(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D})|$ (the first fact of that paragraph, at the level 0), after which $\mathcal{Q}_{U_1}(A') = LQ_{U_1}B$ (the equation of [4, p. 267], (3.24)) and the constraint is the linear constraint $\delta(\widetilde{Q}B)$ of (1.14), the $\widetilde{Q}$ of [3] being LQ_{U_1}, the linear part of the constraint function $\widetilde{Q}$ of [4, (2.4), p. 266] (the $\widetilde{Q}$ of [4] in the equation quoted at (3.24) is Q_U, so that its $L\widetilde{Q}$ is this operator; $\delta(LQ_{U_1}B) = L^{-d_G|\Omega_1^{[1]}|}\delta(Q_{U_1}B)$, and the constant $L^{-d_G|\Omega_1^{[1]}|}$ of this rescaling of the δ-functions is part of $z^{(0)}$, p.249), while the Haar factor

$\prod_{b\in\Omega_1}\sigma(A'(b))$ is written as $\sigma_0^{\#\{\text{bonds of }\Omega_1\}}$ times the normalised density $\prod_b\sigma(A'(b))/\sigma_0$, the $\sigma(A-h\widetilde{D}(A))$ in [3, (1.14)–(1.15)] (the σ there is the normalised density, as in [11, (18), p. 260], to which Bałaban refers at the foot of [3, p. 248] and where σ/σ_0 appears with the exponent $\log\sigma_0|\Omega_1^*|$ of (1.14)): the $\sigma_0^{|\Omega_1^{[1]}|}$ of this product cancels the $\sigma_0^{-|\Omega_1^{[1]}|}$ of the conversion, and $\sigma_0^{\#\{\text{bonds of }\Omega_1\}-|\Omega_1^{[1]}|}=\sigma_0^{|\Omega_1^*|}$ is the factor in [3, (1.14) and (1.15)]; the scaling $B=g_0A$ of p.249 (in [3]: $A=g_0A'$), with Jacobian $g_0^{d_G\times(\text{number of bonds of }\Omega_1)}$ and $\delta(\widetilde{Q}g_0A)=g_0^{-d_G|\Omega_1^{[1]}|}\delta(\widetilde{Q}A)$, so that the power of g_0 is $g_0^{d_G|\Omega_1^*|}$; the insertion of the partition of unity (1.18) in the scaled variable (p.250) and the removal of (B3)(ii), which together produce the cutoff (1.25); the integration of the δ-functions $\delta(\widetilde{Q}A)$ over the designated variables $A(b_0(c))$, $c\in\Omega_1^{[1]}$, which is the elimination of (B2) and produces the factor $z^{(0)}(U_1)$, after which the remaining Lebesgue measure is $\mathrm{d}A|_{\Omega_1^*}$ and the normalised Haar density remains at every bond of Ω_1, at the designated bonds evaluated at the eliminated values (it is the Haar primitive (3.30), part of the v of [3]); and the bondwise change of variables (1.22) on S_1^*, a change of variables by (B4)(ii) with the Jacobian $|\det D\Psi_\xi|$. Each step is an identity for the integral of any bounded measurable function of the fine field, and the fine field is expressed through the integration variables by the reconstruction (3.26). No factor is moved out of the history sum. The outer operation is Bałaban's expansion at the stage of [3, (1.15)] with the cutoff (1.25); the conditional integration (1.26) of [3], its cluster expansion and the renormalisation (1.27)–(1.28) are replaced by the construction of this section and are not used, except that the first-step coupling $g_1(g_0)$ defined by (1.27) on $\Lambda_1^{\sim-1}$ enters, as a parameter depending on g_0 alone, the scale R_1 and the quantity ε_1 of Lemma 3.4 (Section 3.1). The following facts about the branch density are taken from [3] as stated there, except where an item records a reading, a derivation or a proof (among them the Borel choices, the convention and the derivations in (B1), the Remark under (B2), the parts of (B4)(i) and (iii) proved in Lemma B.8, and the proofs in (B4)(ii)). Together with the lemmas proved in Appendix B, the standing conditions, and the further statements of [3] named in this section (the definitions (1.1)–(1.10) and (1.18)–(1.22), the local minima $U_{1,\square'}$ of (1.2) and the field $\mathbf{H}_{1,Ax}$ of (1.21), the axial mean gauge of U_1 of p.248, the scale relations (2.6) and (2.9) of p.255–256, the coupling map (1.27) with its β_1, (2.16)–(2.17) for $\chi_1(\Omega_1)$, (2.4)–(2.5) and p.246 for ε_j and R_j, (1.17) for the gauge action on phases, and, through (B1), Theorem 1 of [7], reference [15] of [3]; (26) of [7] is cited in [3, p. 248] for the expansion of the action), they are the inputs of this section:

(B1) [3, (1.11)] is taken as it stands, with its domain of integration $D^{(11)}$ (Section 3.1); the local minima $U_{1,\square'}(V)$ of (1.2) are minimal configurations of the variational problem [3, (2.12)] for the determining set of (1.2), elements of the unique minimal orbit (p.256) taken in the axial gauge (p.247); the configuration $U_1(\Omega_1\cap\Lambda_1^c,V)$ of (1.21) is determined, p.251, by the conditions of (1.12) on the domain $(\Lambda_1^{\sim})^c$ (the constraint $\bar{U}=V$, the rows crossing to Ω_1^c involving V_0 as in (B2), and the minimality of the constrained functional, whose critical point is its minimum, p.248) and on $\Lambda_1^{\sim}\setminus\Lambda_1$ by the determining set $\mathbf{B}_1(\Lambda_1^c)$ and the configuration $Q_1^{s*}V$, in the axial gauge; [3] names the determining data only at p.251, and the configuration is the minimal configuration in the sense of (2.12) (p.256: the functional A^η minimised over the regular configurations satisfying the constraints) for the constraints named, as [3] writes the operation it calls similar to (1.21)–(1.23) at the general step, p.269, where the function is $U(\mathbf{B}(\Lambda_{k+1}^c\cap\Lambda_k)\cup\mathbf{B}_{k+1},\bullet)$ of (2.15) with data from both scales; the phrase of [3] "depends on V restricted to $\Omega_1\cap\Lambda_1^c$" is accordingly its statement about the dependence on V, the crossing rows involving V_0 as for U_1, so that the Borel choice below is made in the pair (V,V_0). The integrals over V in [3] presuppose that these configurations depend

measurably on the data. By Lemma 3.1, which covers conditions that are equalities between continuous functions and the minimality of a continuous functional over a set so defined, a Borel choice exists for each of the two problems, in V for (1.2) and in (V, V_0) for (1.21); one such choice is fixed for each. Where the data are regular the chosen configuration is Bałaban's: the minimal configurations for given data form one orbit, in the regularity class [7, (6)] ([3, p. 256], "the minimal orbit, i.e., the set of minima"; [7, Theorem 1, p. 279]; the class and radius of Section D.7.1, and of Section D.8.1 for the configuration of (1.21)), of the gauge transformations equal to the identity on the set $\mathfrak{B}_k$ of determining points ([7, (4), p. 278]: every fine site of the scale-zero determining set and the block centres of the scale-one set), and the axial gauge determines a unique element in each such orbit: the axial condition $U(y, x) = \mathbf{1}$ of [3] is posed on the mean contour variables of [4, (0.11)] ([3, p. 247]), which a transformation w of the orbit changes to $w(y)U(y, x)w(x)^{-1}$ by [4, (0.6)], so that $w(y) = \mathbf{1}$ and the axial condition for both configurations force $w(x) = \mathbf{1}$ ([9, p. 78] for the single-contour axial gauge); so the axial representative is unique, and the statements of [3] about these configurations that are used here (the tests (1.4) and (1.8) and (B3)(iii)) hold for the chosen configurations; (3.40) is proved for them in Lemma D.18. The domain of (1.11) is described here. The averages and means of (1.11) are the principal logarithms of [8, p. 21, (22)–(25)], Bałaban's own convention, restricted in the preamble to the chart ball. That on the domain of integration every argument of a logarithm of [4, (0.4)] on a row of Ω_1 lies in the chart domain, every family of transports entering a mean (0.10)–(0.11) at a block centre of $\Omega_1^{(1)}$ lies in the small-diameter domain of the mean, and the fluctuation $U' = UU_1^{-1}$ lies in the chart domain on every bond of Ω_1, the designated bonds $b_0(c)$ included, so that $A' = \frac{1}{i}\log U'$ of (1.13) is defined, is the smallness statement of [3, pp. 247–248], which Bałaban justifies by the reasoning of the proof of Lemma 1 of [9]; it is derived here from Bałaban's statements with that proof as the model of the reasoning, not from the lemma of [9] applied as stated (its hypotheses, the exact axial condition, and [9, (1.7)] on the blocks concerned, are not literally those of the domain of [3]), as follows. (i) On the domain of integration the cutoffs χ_0 and χ_{Ax} of (1.5)–(1.6) localised in $\Omega_1^\sim$ equal one ((B3)(i), p.248): $|U(\partial p) - 1| < \varepsilon_0$ for every plaquette $p \subset \Omega_1^\sim$ and $|U(y, x) - 1| < \varepsilon_0$ for every mean (0.11) at $y \in \Omega_1^{\sim(1)}$, $x \in B(y)$. (ii) For such U [3] states $|UU_{1,\square'}^{-1} - 1| < O(L^2)\varepsilon_0$ on every bond of $\square'^{\sim 2}$ (p.247, "on $\square'^{\sim 2}$", the designated bonds not excluded, whereas the test (1.8) runs over $(\square'^{\sim 2})^*$), "by the same reasoning as in the proof of Lemma 1 [14]", that is Lemma 1 of [9] (p. 79, (1.24)–(1.25)): if two configurations V_0 and $V'V_0$ have plaquette variables within $\alpha_0 L^{-2}$ of one on Ω_1 ((1.7) there), the same variables on the axial contours $\Gamma_{y,x}$, $x \in B(y)$, and block averages within α_1 of each other on $\Omega_1^{(1)}$, then $|V' - 1| < 4d^2\alpha_0 + \alpha_1$ on every bond of Ω_1; its proof (pp. 79–80) bounds the bonds inside a block from the plaquettes and the axial conditions, the bonds joining two blocks from these, and the designated bond b_0 of each coarse bond c from the block-average condition (pp.79–80), and the bound is local (p.80); [3] applies the reasoning with $V_0 = U_{1,\square'}$ in the axial gauge, the block averages of both configurations equal to V on the blocks of $\square'^{\sim 4}$ ((1.2)–(1.3), p.246, and the constraint of (1.11)) and the axial condition on U holding within ε_0 by (i). (iii) U_1 lies in the domain of integration (p.248; the "domain of integration in (1.11)" of [3] is the support of every characteristic factor of (1.11), the set to which they restrict the integral, as in [4, p. 265], (H4) and Remark D.16), so (ii) applies to U_1 too, and $U' = (UU_{1,\square'}^{-1})(U_{1,\square'}U_1^{-1})$ is within $2\,O(L^2)\varepsilon_0$ of one on every bond of $\square'^{\sim 2}$, for every cube $\square' \subset \Omega_1$ (p.248: "both fields in the parenthesis ... are small"). (iv) The arguments of [4, (0.4)] on a row of Ω_1 are closed contours in two adjacent blocks of $\Omega_1^\sim$, and two transports of a family $\mathbf{G}(y, x)$ differ by such a contour; by the reasoning of [8, pp. 24–25]

((44)–(46): in a local axial gauge every bond of the four blocks $\Delta(p')$ is within $dL\alpha_0$ of one and every contour Γ in them within $|\Gamma|dL\alpha_0 = O(1)L^2\alpha_0$, "for $L^2\alpha_0$ sufficiently small"), with $\alpha_0 = \sqrt{N}\varepsilon_0$ from (i) (the $|\bullet|$ of [8] is the operator norm, which the norm of (i) bounds up to the factor $\sqrt{N}$), every such closed contour variable, a conjugation-invariant quantity, is within $O(1)L^2\varepsilon_0$ of one, and so is the diameter of each family of transports. The constant $O(1)$ of (iv) is displayed by [8], $|\Gamma| \leqslant (2d+1)L$ and the factor d, up to the elementary count of the contours of [4, (0.4) and (0.10)–(0.11)] in the four blocks and the factor $\sqrt{N}$; the constant $O(L^2)$ of (ii) is not displayed in [3]; it is that of the reasoning of [9, Lemma 1, p. 79], whose bound $4d^2\alpha_0 + \alpha_1$ has absolute constants, and, like the constants of (B3)(iii), it is independent of the coupling and of the branch ((H4)), so that for ε_0 small at fixed L the quantities of (iii) and (iv) lie below ϱ_G, $r_G/\sqrt{N}$, and the small-diameter radius of the mean ("ε_0 sufficiently small", [3, p. 248]; item 5 of Section B.8). This is the domain $D^{(11)}$ of Section 3.1. A convention of this paper: every factor of (1.4) and (1.8) defined through the configuration $U_{1,\square'}(V)$ of (1.2) is assigned a value for the V for which that configuration does not exist ($V \notin \mathcal{V}_{\mathrm{ax}}(\square')$, Lemma 3.1), namely $\chi_1(\square') = 0$, $\chi_1^c(\square') = 1$, $\chi_0'(\square') = 0$ and $\chi_0'^c(\square') = 1$, so that (1.4) and (1.8) remain partitions of unity ([3, (1.2)–(1.4) and (1.8), pp. 246–247] define these factors through $U_{1,\square'}(V)$ without an existence clause; the convention is this paper's), and $U_{1,\square'}(V)$ enters only where it exists. Under it, where $\chi_1(\Omega_1) \neq 0$ the configuration $U_{1,\square'}(V)$ exists for every cube $\square' \subset \Omega_1$, and where a factor $\chi_0'(\square')$ is nonzero it exists for that cube. For every admissible (Ω_1, V, V_0) the background $U_1 = U_1(V, V_0)$ of (1.12) exists, is the unique critical point of the constrained functional of [3] in the domain of integration in (1.11), that is in the fibre $D^{(11)}(V, V_0)$, and is a minimum with strictly positive second differential ([3, p. 248]: "there exists exactly one critical point" of (1.12) in the domain of integration in (1.11), by Theorem 1 of [7] (reference [15] of [3], the variational problem of the averaging operation) for ε_0 small; "it is a minimum of (1.12), with a strictly positive second order differential", stated there without further citation); the fluctuation $U' = UU_1^{-1}$ lies in the chart domain for every $U \in D^{(11)}(V, V_0)$ (p.248, derived above), so that $A' = \frac{1}{i}\log U'$ of (1.13) is defined.

(B2) For every admissible pair (Ω_1, V, V_0) the constraint $\delta(\widetilde{Q}A')$ of (1.14) is solved on every row $c \in \Omega_1^{[1]}$ for the designated variable $A'(b_0(c))$ as a linear function of the free variables of the two blocks of c ([3, p. 249], stated at the critical point U_1, which exists for every admissible pair by (B1); [3] states this for the rows of Ω_1 without distinguishing the rows with one block off Ω_1; on such a row the average involves V_0 on the outer block as a constant, as in the first-step problem [11, (12), p. 258] ($U = V_0$ on Ω_1^c, $\bar{U} = V$ on $\Omega_1^{(1)}$), which [3, p. 248] follows, (H4), and $b_0(c)$, a bond of Ω_1, is the eliminated variable, following the choice of $A'(b_0(c))$ in [3, p. 249] as the variable solved for on every row). In the notation of Section 3.2 this is the invertibility of $\alpha_c(U_1)$ for every $c \in \Omega_1^{[1]}$, so that U_1 lies in the open set $\mathcal{X}$ of the measurability paragraph for every admissible pair. *Remark.* The invertibility is also proved here, at every row $c \in \Omega_1^{[1]}$, near or far: $\alpha_c(U)$ depends on $U|_{\mathcal{E}_c}$ only, Lemma 3.2 gives its invertibility for backgrounds e^{iA} with $\|A\|_\infty \leqslant \tau_L$ on $\mathcal{E}_c$ and shows that a site gauge transformation multiplies it on both sides by invertible maps, and the first assertion of Lemma B.8(i) of Appendix B, proved from (B1)(i), (iii), (iv), the first condition of threshold item 6 (with (B.6)), the fine periods being at least $8M$, the inclusion $\mathcal{E}_c \subseteq \mathfrak{P}_c$ given by (3.10) and Lemma B.5, without (B2), gives a site gauge on the two blocks of c in which $U_1 = e^{i\phi}$ on $\mathcal{E}_c$ with $\|\phi\|_\infty \leqslant \tau_L/2$. So of (B2) only its form on the rows with one block off Ω_1 (the outer block entering as the constant V_0, and $A'(b_0(c))$ the eliminated variable) is an input.

(B3) The estimate-based simplifications in Bałaban's derivation of [3, (1.25)] hold under Bałaban's parameter hierarchy, in the following form: the dropped factors are equal to one wherever the retained factors are nonzero. (i) The removal of the factors $\chi_0(\square)$ and χ_{Ax} of the cubes and blocks localised in $\Omega_1^\sim$ after (1.10) (A_0 sufficiently larger than A_1, p.248). (ii) The removal of the second factor of (1.19) on the good cubes (M_2R_1 large, p.250), which leaves χ_0' on the cubes of S_1 only (p.250); the condition "M_2R_1 large enough" is a threshold on the coupling (it makes $O(1)B_3e^{-\delta_c LM_2R_1}\varepsilon_1$ much smaller than δ_0, both sides depending on g_0 through ε_1, δ_0 and R_1), taken over in item 5 of Section B.8. (iii) The comparison on which (ii) rests, p.250: for every $L^2M_2R_1$-cube $\square' \subseteq \Omega_1^{\sim -1} \setminus R_1$ (the $\square' \subset R_1^c$ of [3]; then $\square'^{\sim 4} \subseteq \Omega_1$, p.250), (by the convention of (B1), $U_{1,\square'}(V)$ exists for every cube $\square' \subset \Omega_1$ where $\chi_1(\Omega_1) \neq 0$, so the comparison is between two existing configurations), $|U_1(b)U_{1,\square'}(b)^{-1} - \mathbf{1}| < O(1)B_3e^{-\delta_c LM_2R_1}\varepsilon_1$ for the bonds b of $\square'^{\sim 2}$, with the constant B_3, the decay rate δ_c (the δ of [3]) and the $O(1)$ independent of the coupling and of the branch (Bałaban's constants, (H4): B_3 is that of [7, Theorem 1, p. 279], δ_c the rate of the exponential decay property [7, (190)], which Bałaban invokes in [3, p. 266] for the functions $\mathbf{H}_{k,\square}$, and the $O(1)$ that of Bałaban's derivation of the same comparison [3, (3.19), p. 268] from these facts and [8, (106)–(108), (159)–(163)]; only their independence from the coupling is used, in Appendix B, Lemma B.4). Item (iii) is used in Appendix B, Lemma B.4; item (i) is used only through its conclusion, the form (1.11) of the integrand and hence the form (1.25) of the cutoff, and item (ii) through (1.25) and directly in the proof of Lemma 3.11(iv). The rewriting of the retained tests in the new variables in [3, p. 252] is not used: the retained cutoff is defined in (3.43) as the exact composition.

(B4) For every admissible pair (Ω_1, V, V_0) and every region R_1 of the history sum, the expressions in [3, (1.14)–(1.15) and (1.22)–(1.24)] are defined on the domain of integration: the linearising correction $\widetilde{D}$ of (1.14), p.249 (at the foot of [3, p. 248] Bałaban refers to the transformations analysed between (16)–(18) of [16], or between (I.1.8)–(I.1.9) of [I], where [I] is [4] in the reference list of [3, p. 285]; [4, (1.8)–(1.9), p. 261] concern the functions J_j, and the linearisation of the unit-lattice step of [4] is the one following [4, (2.10), p. 267], where Bałaban states for its correction that its defining equation has exactly one solution, analytic in its argument, with an expansion beginning with quadratic terms; here $\widetilde{D}$ is the small solution φ of (i)); the phase $A - h_{U_1}\widetilde{D}(A)$; the Haar density $\sigma(A - h_{U_1}\widetilde{D}(A))$; the logarithm $\mathrm{Tr}\log(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\widetilde{D})$ of the Jacobian of (1.14); and the bondwise maps of (1.22) with their Jacobian. Three further properties of these objects are recorded as part of (B4); they are Bałaban's statements and facts proved in Lemma B.8. (i) The defining equation of $\widetilde{D}$ is the system (3.24) over the rows $c \in \Omega_1^{[1]}$, which decouples row by row (Lemma 3.5), as in [11, (17), pp. 259–260], the transformation to which Bałaban refers in [3, p. 248], where the equation is solved "separately for each function $Q(A', c)$"; the row component $\widetilde{D}_c(A)$ depends on $A|_{\mathcal{E}_c}$ only. [4, p. 267] states for the equation $L\widetilde{Q}B' + \widetilde{C}(B') = L\widetilde{Q}B$, $B' = B - h\widetilde{D}(B)$, that "there exists exactly one solution of this equation, and that it is an analytic function of B", in the unit-lattice step of [4], without naming the class in which the solution is unique, and [11, p. 260] states for (17) that "there exists a unique solution $\tilde{D}$ of this equation for A sufficiently small, and it is an analytic function of A with a Taylor expansion beginning with second order terms"; these statements are quoted as Bałaban's description of $\widetilde{D}$ and are used only for the identification at the end of (i). The row equation (3.24) involves the eliminated phase only through e^{iq}, so its solutions are determined only up to the periods of the exponential; the class used here is the chart class of Section 3.1 (the whole phase in the chart ball $h(q) < \varrho_G$, the class of the chart-valued variable A' of [3,

(1.13)]) with the fine field in $D^{(11)}(V, V_0)$. In this class the row solution is determined by Lemma B.8 of Appendix B, proved from (B1) (with (B3)(i), through (B1)(i)), (B2), the compatible partitions of [3] being nested ((H4)), the fine periods being at least $8M$, Lemmas B.5, B.7 and 3.2, and threshold items 5 and 6: at every row $c \in \Omega_1^{[1]}$, near or far, Lemma B.7 applies to U_1 and gives the small solution φ_c, analytic on the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$, real on real phases, with $\varphi_c(0) = 0$ and $D\varphi_c(0) = 0$, unique among the solutions d_c with $h(d_c) \leqslant \rho_{\mathrm{d}}$ (Lemma B.8(i)); the argument A of every $U \in D^{(11)}(V, V_0)$ lies in the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*/2$ at every row (Lemma B.8(ii)); and every chart-class family d at $g_0 \mathsf{C}_{U_1} B$ with $U_d(B) \in D^{(11)}(V, V_0)$ is $\varphi(g_0 \mathsf{C}_{U_1} B)$, $\varphi = (\varphi_c)_c$ (Lemma B.8(iii)). The row solution of this paper is $\widetilde{D}_c = \varphi_c$ at every row, and $\widetilde{D} = \varphi$ is analytic and solves the row equations on the open set $\mathcal{N}_\Theta$ of Lemma B.8(iv), which contains all arguments of the domain of integration. Every solution of the row system with chart-valued phase and fine field in $D^{(11)}(V, V_0)$ is φ, by Lemma B.8(iii); and the $\widetilde{D}$ in [3, (1.13)–(1.14), pp. 248–249] is φ: for A sufficiently small both are small solutions of the row equations ($\varphi(0) = 0$), so they agree by the uniqueness stated in [11], and the $\widetilde{D}$ of [3], the analytic function so described by [11] and [4], therefore agrees with φ on the connected ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ on which φ is analytic, a ball containing the arguments of the domain of integration (Lemma B.8(ii)). (ii) The substitutions (1.14) and (1.22) are changes of variables: the map $B \mapsto U(B)$ of Section 3.1, the composition of (1.13) and (1.14), is a bijection of $\mathcal{A}_\xi$ onto $D(V, V_0)$ (shown there from (B1), (B2) and Lemma B.8), and its differential, that of $B \mapsto g_0 \mathsf{C}_{U_1} B$ followed by the substitution $A' = A - h_{U_1} \widetilde{D}(A)$ and the exponential, is injective at every point of $\mathcal{A}_\xi$ with image the tangent space of the fibre $\Phi^{-1}(V, V_0)$ at $U(B)$ (the Jacobian $\det(\mathbf{1} - h_{U_1} \frac{\delta}{\delta A} \widetilde{D})$ of (1.14) is nonzero there, (iii), C_{U_1} is injective, and the image of $(\mathbf{1} - h_{U_1} \frac{\delta}{\delta A} \widetilde{D}) \mathsf{C}_{U_1}$ is the kernel of $D\mathcal{Q}_{U_1}(A')$ by the identity $D\mathcal{Q}_{U_1}(A') \circ (\mathbf{1} - h_{U_1} \frac{\delta}{\delta A} \widetilde{D}) = LQ_{U_1}$ of the measurability paragraph, $LQ_{U_1} \mathsf{C}_{U_1} = 0$, and a count of dimensions, $D\mathcal{Q}_{U_1}(A')$ being onto; the map $B \mapsto U(B)$ goes from $\mathfrak{g}^{\Omega_1^*}$ into the fibre, so its differential is not square); and Ψ_ξ of (3.38), acting bond by bond on S_1^*, is a C^∞ map which maps $\mathcal{D}_\xi$ bijectively onto $\mathcal{A}_\xi$ with invertible differential at every point of $\mathcal{D}_\xi$: this is Lemma 3.8 of Section 3.4, proved from the bound on $\mathbf{H}_{1,Ax}$ of [3, p. 251], derived in Lemma D.18 from [7, Theorem 1, (172), Proposition 9, and (190)], the application of [7] to the determining set $\mathbf{B}_1(\Lambda_1^c)$ in [3, p. 256], and the restoring bound (D.48), from the membership $U_1 \in D^{(11)}(V, V_0)$ of (B1), that is the cutoff $\chi_0'(\Omega_1)$ at U_1 ([3, p. 248]), and from the choice of the cutoff g of (1.22) made there (pp.249 and 252 of [3] are where Bałaban performs the substitutions with the Jacobians written there; the invertibility of $D\Psi_\xi$ is used in Lemma 3.9(ii)). The Jacobian of (1.22) is used here only as $|\det D\Psi_\xi|$, the exact Jacobian of (3.38), not through the expression [3, (1.24)]. (iii) The determinant $\det(\mathbf{1} - h_{U_1} \frac{\delta}{\delta A} \widetilde{D})$ is nonzero on $\mathcal{A}_\xi$, by (i) and Sylvester's identity (the measurability paragraph of Section 3.1); [3] performs (1.14) as a change of variables and writes its Jacobian as $\exp \mathrm{Tr} \log(I - h(\frac{\delta}{\delta A} \widetilde{D})(A))$ (p.249). The determinant is positive there, so that the logarithm of [3] is the real logarithm and equals $\sum_c \log \det_{\mathfrak{g}}(\mathbf{1} - h_{U_1}(c) \partial_{A(b_0(c))} \widetilde{D}_c)$ (Lemma 3.7): this is Lemma B.8(iv), (B.17), at every row; the identities of this paper use the absolute Jacobian $|\det(\mathbf{1} - h_{U_1} \frac{\delta}{\delta A} \widetilde{D})|$, for which the nonvanishing suffices. For the rows within $2R_{\mathrm{c}} + 2L$ of I and the inputs of Lemma 3.6, the expressions of (B4) coincide with the analytic objects of that lemma, by the uniqueness of the small solution stated there; elsewhere they enter only through a-independent factors (Lemma 3.9, (3.6)).

The Borel choice fixed in (B1) exists:

Lemma 3.1 (Borel Choice of Minimal Configurations)**.** *Let E be a finite set of fine bonds,*

$\mathbf{B}$ *a set of blocks, and* $\mathcal{U} \subseteq G^E$ *a finite intersection of open and closed subsets of the compact metric space* G^E*, contained in the open set on which the group means* $U(y,x)$ *of [4, (0.11)] at the block centres* y *of* $\mathbf{B}$ *are defined (chart convention of the preamble). For the determining sets* $\mathbf{B}$ *of [3], that of (1.2) and that of (1.21),* $E = E(\mathbf{B})$ *is the set of fine bonds of the region determined and* $\mathcal{U} = \mathcal{U}_{\mathbf{B}}$ *the set of regular configurations, on which the averaging operation* $M_{\mathbf{B}}$ *of [3, (2.11)] is defined: the configurations of the class (2) of [7, p. 278] (reference [15] of [3]): finitely many strict inequalities between continuous functions of* U *(the plaquette variables* $|U(\partial p) - 1|$ *and the covariant divergences* $|(D_U^{\eta *}\partial U)(b)|$*, polynomials in the matrix entries of the bond variables and their inverses), with the averages (the logarithmic averages of [4, (0.4)]) and the means* $U(y,x)$ *defined in the chart convention, an open set; the space [7, (6)] is its intersection with the constraint set (3), which is the constraint* $\mathfrak{c} = 0$ *below;* $M_{\mathbf{B}}$ *is continuous on* $\mathcal{U}$*. Let* P *be a Polish space of parameters,* $\mathcal{W}$ *a relatively open subset of* $\mathcal{U} \times P$ *(the set on which the averages entering the conditions are defined;* $\mathcal{W} = \mathcal{U} \times P$ *is allowed),* $\mathfrak{c}$ *and* $\mathfrak{d}$ *continuous maps on* $\mathcal{W}$ *into metric spaces, and* S *a continuous real function on* $\mathcal{W}$ *(the action, which may involve the parameter through boundary plaquettes). For* $p \in P$ *let* $\mathcal{W}(p) = \{U : (U,p) \in \mathcal{W}\}$*, an open subset of* $\mathcal{U}$*, let* $\mathcal{K}(p) = \{U \in \mathcal{W}(p) : \mathfrak{c}(U,p) = 0\}$*, let* $\mathcal{V}$ *be the set of* p *with* $\mathcal{K}(p) \neq \varnothing$ *on which* $S(\bullet, p)$ *attains its infimum* $m(p)$ *over* $\mathcal{K}(p)$*, and let*

$$\begin{aligned}\mathcal{M}_{\mathrm{ax}} = \{(p,U) : p \in \mathcal{V},\ U \in \mathcal{K}(p),\ S(U,p) = m(p),\ \mathfrak{d}(U,p) = 0,\\ U(y,x) = \mathbf{1} \text{ for every centre } y \text{ of a block of } \mathbf{B} \text{ whose bonds lie in } E\\ \text{and } x \in \mathcal{B}_y \setminus \{y\}\}\end{aligned} \tag{3.8}$$

be the set of minimal configurations satisfying the further equalities $\mathfrak{d} = 0$*, in the axial gauge (p.248), with projection* $\mathcal{V}_{\mathrm{ax}}$ *to the first factor. Then* $\mathcal{V}$*,* m*,* $\mathcal{M}_{\mathrm{ax}}$ *and* $\mathcal{V}_{\mathrm{ax}}$ *are Borel, and there is a Borel map* $\mathcal{V}_{\mathrm{ax}} \ni p \mapsto U_{\mathbf{B}}(p)$ *with* $(p, U_{\mathbf{B}}(p)) \in \mathcal{M}_{\mathrm{ax}}$ *for every* $p \in \mathcal{V}_{\mathrm{ax}}$*. For the problem (1.2),* $U_{1,\square'}(V) = U(\mathbf{B}_1(\square'^{\sim 4}), M^\bullet(Q_1^{s*}V))$*, the parameter is* $p = V|_{\square'^{\sim 4}}$ *(p.246),* $\mathcal{W} = \mathcal{U} \times P$*,* $\mathfrak{c}(U,V) = M_{\mathbf{B}}(U)\,(M^\bullet(Q_1^{s*}V))^{-1} - \mathbf{1}$*, so that* $\mathcal{K}(V) = \{U \in \mathcal{U}_{\mathbf{B}} : M_{\mathbf{B}}(U) = M^\bullet(Q_1^{s*}V)\}$ *is the constraint set of the variational problem [3, (2.12)] for the data of (1.2), and* $\mathfrak{d} \equiv 0$*; for a regular* V*, [3, p. 256] gives the unique minimal orbit of (2.12), so* $V \in \mathcal{V}$*, and this orbit has a regular representative in the axial gauge, so that* $V \in \mathcal{V}_{\mathrm{ax}}$ *(the proof below, from the gauge invariance stated at [7, p. 278]), which is the representative chosen in [3, p. 247], "taken in the axial gauge". For the configuration of (1.21),* $p = (V, V_0)$*,* E *the bonds of* $\Omega_1 \cap \Lambda_1^c$ *(the "function* $U_1(\Omega_1 \cap \Lambda_1^c, V)$*" of [3, p. 251]),* $\mathcal{W}$ *the relatively open set of the* (U,p) *at which the averages of the rows crossing to* Ω_1^c*, which involve* V_0 *on the outer block, are defined,* $\mathfrak{c}$ *collects the constraints* $\bar{U} = V$ *on the rows of* $(\Lambda_1^\sim)^c$ *(with* V_0 *on the blocks off* Ω_1*) and* $M_{\mathbf{B}_1(\Lambda_1^c)}(U) = M^\bullet(Q_1^{s*}V)$ *on the bonds of the determining set in* $\Lambda_1^\sim \setminus \Lambda_1$ *(its fine layer fixed to* $Q_1^{s*}V$ *and its coarse bonds to the averages of* $Q_1^{s*}V$*; the form of the constraint of (1.2); [3] names the determining data, "the set* $\mathbf{B}_1(\Lambda_1^c)$ *and the configuration* $Q_1^{s*}V$*", p.251),* $\mathfrak{d} \equiv 0$*, and* $S(U,p)$ *is the functional* A^η *of (2.12) for the determining set* $\mathbf{B}_1(\Lambda_1^c)$*, in which the plaquettes meeting* Ω_1^c *involve* V_0*, as recorded in (B1) from pp.251 and 269: the minimality* $S(U) = m(p)$ *over* $\mathcal{K}(p)$ *is the condition of (1.12) on* $(\Lambda_1^\sim)^c$*, the critical point of (1.12) being its minimum (p.248), and no critical-point equation is imposed separately (the constrained critical-point equation written with* C_U *presupposes the invertibility of the* $\alpha_c(U)$ *and is a consequence of the minimality where it is defined, not a condition of the problem); only the form of these conditions is used, each being an equality between continuous functions of* (U,p) *or the minimality of a continuous functional over a set defined by such equalities, and the existence of the configuration described in [3], for every admissible pair* (Ω_1, V, V_0) *(Section 3.1), is Bałaban's application of Theorem 1 of [7] to the determining set* $\mathbf{B}_1(\Lambda_1^c)$*: [3, p. 251] introduces the configuration for the terms of its history sum, whose domain of integration is* $D^{(11)}(V, V_0)$*, and uses it wherever (1.15) is taken, and p.251 states its determining conditions and no existence theorem for each pair; its existence is the existence*

of the minimal orbit of [7, Theorem 1] for these determining data, the application Bałaban makes of [7] to all the determining sets of [3, p. 256] (as recorded in Section D.8.1), taken over as Bałaban's. Theorem 1 of [7] (reference [15] of [3]) is cited in [3, p. 248], for the critical point of (1.12).

Proof. The set $\mathcal{K} = \{(p,U) : (U,p) \in \mathcal{W},\ \mathfrak{c}(U,p) = 0\}$ is Borel in $P \times G^E$, as the zero set of a continuous function on the Borel set $\mathcal{W}$ (relatively open in the Borel set $\mathcal{U} \times P$), and so is $\mathcal{K}_t = \{(p,U) \in \mathcal{K} : S(U,p) < t\}$ for every rational t. Every section $\mathcal{K}_t(p) = \{U \in \mathcal{W}(p) : \mathfrak{c}(U,p) = 0,\ S(U,p) < t\}$ is the intersection of $\mathcal{U}$, a finite intersection of open and closed subsets of the compact metric space G^E, with the open set $\mathcal{W}(p)$, a closed set (the zero set of $\mathfrak{c}(\bullet,p)$ is relatively closed in $\mathcal{W}(p)$, the trace of a closed set) and an open set (the trace of $\{S(\bullet,p) < t\}$, relatively open in $\mathcal{W}(p)$), hence K_σ: an open subset of a compact metric space is a countable union of compact sets, and the intersection of a K_σ set with a closed set, or with an open set, is K_σ. By the Arsenin–Kunugui theorem (a Borel subset of a product of Polish spaces whose sections are K_σ has a Borel projection and a Borel uniformisation, [14, Theorem 35.46(ii)]), $\operatorname{proj}\mathcal{K}_t = \{p : \inf_{\mathcal{K}(p)} S < t\}$ is Borel for every rational t, and so is $\operatorname{proj}\mathcal{K} = \{p : \mathcal{K}(p) \neq \varnothing\}$; hence $\widetilde{m}(p) = \inf_{\mathcal{K}(p)} S$ is a Borel function on $\operatorname{proj}\mathcal{K}$. The set $\mathcal{M} = \{(p,U) \in \mathcal{K} : S(U,p) = \widetilde{m}(p)\}$ is Borel, its sections $\mathcal{K}(p) \cap \{S = \widetilde{m}(p)\}$ are K_σ, so $\mathcal{V} = \operatorname{proj}\mathcal{M}$ is Borel by the same theorem, and $m = \widetilde{m}$ on $\mathcal{V}$. The axial gauge conditions $U(y,x) = \mathbf{1}$ are relatively closed conditions on the open set on which the means $U(y,x)$ of [4, (0.11)] are defined ([4, p. 253] defines the mean for sets of sufficiently small diameter, and it is continuous there in the chart convention), an open set containing $\mathcal{U}$ by hypothesis; and $\mathfrak{d}(U,p) = 0$ is a relatively closed condition on $\mathcal{W}$. So $\mathcal{M}_{\mathrm{ax}} = \mathcal{M} \cap \{\mathfrak{d} = 0\} \cap \{U \text{ axial}\}$ is Borel, and its sections are K_σ, being intersections of K_σ subsets of $\mathcal{U}$ with closed sets (the trace on $\mathcal{U}$ of a relatively closed subset of an open set containing $\mathcal{U}$ is the trace of a closed set); the theorem gives that $\mathcal{V}_{\mathrm{ax}}$ is Borel and that $\mathcal{M}_{\mathrm{ax}}$ has a Borel uniformisation, a Borel map $U_{\mathbf{B}}$ on $\mathcal{V}_{\mathrm{ax}}$ with graph in $\mathcal{M}_{\mathrm{ax}}$. The statements about Bałaban's problems in the lemma are those of [3]: a regular V determines the unique minimal orbit of (2.12), whose elements are the minimal configurations (p.256), and the configurations $U_{1,\square'}$ are taken in the axial gauge (p.247); a gauge transformation equal to $\mathbf{1}$ at the block centres and to the mean $U(y,x)$ elsewhere brings a minimal configuration to the axial gauge, leaving the block averages of [4, (0.4)] at the block centres unchanged ([4, (0.4)] transforms at the block centres, proof of Lemma 3.2) and $S(U)$ unchanged when S is gauge invariant, as the action is; the configuration so obtained is regular, satisfies the constraint and is minimal: [7] states that its class (2) is gauge invariant and its constraint set (3) invariant under the gauge transformations equal to one on its set $\mathfrak{B}_k$ of determining points ((4), p.278), which the transformation used is: it is the identity at the block centres, and on the blocks of the fine determining layer, where the constraint fixes the configuration to the straight extension $Q_1^{s*}V$, whose intrablock contours are the identity (Section D.8.2), the mean $U(y,x)$ is the identity; and that the space (6) and the action (5) are invariant under these (p.278); so $\mathcal{M}_{\mathrm{ax}}(V) \neq \varnothing$; the description of the configuration of (1.21) is p.251, taken as stated. *q.e.d.*

3.2 The Constrained Hessian at a Regular Background and Its Stencils

Let M be Bałaban's averaging operation (4.1) (written M there; here M is the cube side), and let Ω be a region, a union of blocks, and U a configuration of the fine torus for which $\mathsf{M}(U)(c)$ is defined on every row $c \in \Omega^{[1]}$ of that region and the local group means $\mathcal{U}_U(y,x)$, $x \in \mathcal{B}_y$, are defined for every centre $y \in \Omega^{(1)}$ (chart convention of the preamble; for $U = U_1$ on Ω_1 both hold, because $U_1 \in D^{(11)}$ and U_1 is in its axial mean gauge, $\mathcal{U}_{U_1}(y,x) = \mathbf{1}$);

the objects below are then defined for phases q in a neighbourhood of zero depending on U, which Lemma 3.2 and Appendix B quantify for the backgrounds that occur. The objects of this subsection are defined relative to the region Ω: the rows are the coarse bonds $c \in \Omega^{[1]}$ (at least one block in Ω), the block sums run over the centres $y \in \Omega^{(1)}$, the free carrier is Ω^*, and the links of U on bonds with both endpoints off Ω enter as fixed constants, with the phase variable equal to zero there. The torus itself ($\Omega = T$) is written without a superscript; the branch uses $\Omega = \Omega_1$ and $U = U_1$. Define, for $c \in \Omega^{[1]}$,

$$\mathcal{Q}_{U,c}(q) = \frac{1}{i} \log \left(\mathsf{M}(e^{iq}U)(c)\, \mathsf{M}(U)(c)^{-1} \right), \quad Q_U = \frac{1}{L} D\mathcal{Q}_U(0), \quad \mathcal{N}_U(q) = \mathcal{Q}_U(q) - LQ_U q \,, \tag{3.9}$$

so that $\mathcal{N}_U$ is the remainder denoted $\widetilde{C}$ in the linearisation following [4, (2.10)]; the logarithm is the group chart of the conventions above, defined when the relative average lies in the chart domain. For a coarse bond c with designated fine bond $b_0(c)$, and a coarse centre y, put

$$\mathcal{E}_c = \{r \,|\, \{r_-, r_+\} \subset \mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}\}, \qquad \mathcal{E}_y = \{r \,|\, \{r_-, r_+\} \subset \mathcal{B}_y\} \,, \tag{3.10}$$

r ranging over the positive fine bonds.

Lemma 3.2 (Two-Block Structure)**.** *Let $L \geqslant 3$ be odd and let every coarse period exceed two. The row components $\mathcal{Q}_{U,c}$ and $\mathcal{N}_{U,c}$ depend only on the phases and background bonds in $\mathcal{E}_c$, and $b_0(c)$ is the only designated bond in $\mathcal{E}_c$. Hence*

$$Q_{0,U}(c, c') = \delta_{cc'} \alpha_c(U), \qquad \alpha_c(U) = Q_U(c, b_0(c)) \,, \tag{3.11}$$

where $Q_{0,U}$ is the restriction of Q_U to the designated bonds, with $\alpha_c(\mathbf{1}) = L^{-4}\mathbf{1}$. There is a phase radius $\tau_L > 0$, depending on L and the group only, such that for $U = e^{iA}$ with $\|A\|_\infty \leqslant \tau_L$ every $\alpha_c(U)$ is analytic in A with smallest singular value at least $L^{-4}/2$. Under a site gauge transformation u, $\alpha_c(U^u) = \mathrm{Ad}_{u(c_-)}\, \alpha_c(U)\, \mathrm{Ad}^{-1}_{u(b_0(c)_-)}$.

Proof. Every path in the row of [4, (0.4)] for c lies in the two endpoint blocks, since the row averages parallel transports from the centre of $\mathcal{B}_{c_-}$ to the centre of $\mathcal{B}_{c_+}$ along paths inside the two blocks; a designated bond crosses its own coarse interface, and two adjacent blocks share exactly the interface labelled by c, so $b_0(c)$ is the only designated bond with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$ (for $L \geqslant 3$ odd the other designated bonds cross other interfaces). At the flat field, differentiating [4, (0.4)] gives (A.2) of Appendix A: the straight-path term $(Q_1 a)(c) = L^{-5} \sum_{x \in \mathcal{B}_{c_-}} a([x, x'])$ contains $b_0(c)$ in exactly L of its L^4 translated straight paths of L bonds each (the L starting points on the line through $b_0(c)$), each with coefficient L^{-5}, and the mean-path term involves the paths Φ inside the blocks, which contain no designated bond; this gives $\alpha_c(\mathbf{1}) = L^{-4}\mathbf{1}$, the designated coefficient $Q_{0,\infty} = L^{-4} I$ computed in Appendix A (paragraph "Construction on the infinite lattice", after (A.26); the computation is local and holds on every torus). For $U = e^{iA}$, $\alpha_c(U)$ is an analytic function of $A|_{\mathcal{E}_c}$ near zero (a composition of exponentials, the averaging and the chart logarithm) with $\alpha_c(\mathbf{1}) = L^{-4}\mathbf{1}$; the smallest singular value is continuous, so for each of the finitely many row types modulo $L\mathbb{Z}^4$ (translations commute with M) there is a radius on which α_c is analytic with smallest singular value at least $L^{-4}/2$, and τ_L is the smallest of these radii; it depends on L and the group only. The gauge covariance is $\mathsf{M}(U^u)(c) = u(c_-)\mathsf{M}(U)(c)u(c_+)^{-1}$ ([4, (0.4)] transforms at the block centres) together with $(R_u q)(b) = \mathrm{Ad}_{u(b_-)}\, q(b)$ for the phase ([3, (1.17)]): $\mathcal{Q}_{U^u,c}(R_u q) = \mathrm{Ad}_{u(c_-)}\, \mathcal{Q}_{U,c}(q)$, hence $Q_{U^u} R_u = \mathrm{Ad}_{u(c_-)}\, Q_U$ row by row, and the $b_0(c)$ column gives the formula. *q.e.d.*

By (B2) and Lemma 3.2, $\alpha_c(U)$ is invertible on every row of the regions used below. With $LQ_U h_U = \mathbf{1}$ for the supported right inverse h_U, whose only nonzero outputs are at the

designated bonds, the elimination maps are

$$\begin{aligned}(\mathsf{C}_U B)(b_0(c)) &= -\alpha_c(U)^{-1}\sum_{b\in\mathcal{E}_c\cap E_*} Q_U(c,b)B(b),\\ (h_U v)(b_0(c)) &= h_U(c)v(c), \qquad h_U(c)=\frac{1}{L}\alpha_c(U)^{-1}\ ,\end{aligned} \tag{3.12}$$

and $(\mathsf{C}_U B)(r) = B(r)$ on free bonds $r \in E_*$; thus $Q_U\mathsf{C}_U = 0$. The free-input and background stencils of a fine bond are

$$(\mathcal{F}_r,\mathcal{A}_r) = \begin{cases}(\{r\},\varnothing), & r\in E_*\ ,\\ (\mathcal{E}_c\cap E_*,\mathcal{E}_c), & r = b_0(c)\ .\end{cases} \tag{3.13}$$

The constrained quadratic form is [3, (1.14)–(1.16)] on the region Ω. With J_U and Δ_U the first derivative and Hessian at $q=0$ of $q\mapsto S(e^{iq}U)$, $q\in\mathfrak{g}^\Omega$ (the plaquettes with a bond in Ω contribute; links off Ω are the fixed constants), and the relative gauge functional

$$\mathscr{G}_U^\Omega(q) = \sum_{y\in\Omega}\sum_{x\in\mathcal{B}_y\setminus\{y\}}\big(1-\operatorname{Re}\operatorname{tr}_N(\mathcal{U}_{e^{iq}U}(y,x)\mathcal{U}_U(y,x)^{-1})\big)\ ,\qquad G_U^{(2),\Omega}(a) = D^2\mathscr{G}_U^\Omega(0)[a,a]\ , \tag{3.14}$$

where $\mathcal{U}_U(y,x)$ is the axial transport from the block centre y to x of the gauge-fixing functional $\mathbf{G}$ in [3, (1.11)] and the sum is over the block centres of Ω, the regional Hessian H_U^Ω on $\mathfrak{g}^{\Omega^*}$ is the symmetric operator of the quadratic form (B is extended by zero to the free bonds outside Ω^*, so that $a=\mathsf{C}_U B$ is defined on every row of $\Omega^{[1]}$)

$$\langle B,\mathsf{H}_U^\Omega B\rangle = \langle a,\Delta_U a\rangle + G_U^{(2),\Omega}(a) - \langle J_U, h_U D^2\mathcal{N}_U(0)[a,a]\rangle,\qquad a=\mathsf{C}_U B\ , \tag{3.15}$$

with the rows of $\mathcal{N}_U$ and h_U those of $\Omega^{[1]}$. The right side equals $D^2(S+\mathscr{G}_U^\Omega)(0)[a,a] - DS(0)[h_U D^2\mathcal{N}_U(0)[a,a]]$, which does not involve the inner product; H_U^Ω is its operator for the fixed one, and its entries $\mathsf{H}_U^\Omega(b,b')\in\operatorname{End}(\mathfrak{g})$ are those of the polarised bilinear form. This is [3, (1.16)], $\langle A,\Delta_1 A\rangle = \langle A,\Delta A\rangle - 2\langle hC^{(2)}(A),J_1\rangle + \mathbf{G}^{(2)}(A)$, with, as in [3], $C^{(2)}(A) = \frac{1}{2}D^2\mathcal{N}_U(0)[A,A]$ (the quadratic term of the linearising transformation, [4, p. 267], $\widetilde{D}^{(2)} = \widetilde{C}^{(2)}$) and $\mathbf{G}^{(2)}(A) = D^2\mathscr{G}(0)[A,A]$; at $\Omega=\Omega_1$, $U=U_1$ the relative gauge functional is the absolute one of [3], $\mathscr{G}_{U_1}^{\Omega_1}(q) = \mathbf{G}(\Omega_1^{(1)},e^{iq}U_1)$ with $\mathbf{G}$ of (1.7), (1.11), because U_1 is in its axial mean gauge, $\mathcal{U}_{U_1}(y,x)=\mathbf{1}$ for $x\in\mathcal{B}_y$, $y\in\Omega_1^{(1)}$ (Section 3.1; [3, p. 248]), so that $\mathbf{G}(\Omega_1^{(1)},U_1)=0$; and $\mathsf{H}_{U_1}^{\Omega_1}$ is the $C^*\Delta_1 C$ of [3, (1.15)], the same quadratic form written as an operator for the fixed inner product. The precision of the branch is

$$H_\xi = \mathsf{H}_{U_1}^{\Omega_1}\quad\text{on }\mathfrak{g}^{\Omega_1^*},\qquad H_{\xi,II} = \iota_I^* H_\xi \iota_I,\qquad H_{\xi,IJ} = \iota_I^* H_\xi \iota_J\ , \tag{3.16}$$

with ι_I,ι_J the extensions by zero. With $s_{p,U}(q) = 1-\operatorname{Re}\operatorname{tr}_N((e^{iq}U)(\partial p))$, $\mathscr{G}_{y,U}$ the y-summand of (3.14), and $a=\mathsf{C}_U B$, the form decomposes into local terms,

$$\begin{aligned}\langle B,\mathsf{H}_U^\Omega B\rangle &= \sum_{p\cap\Omega\neq\varnothing}\mathfrak{a}_p(B;U) + \sum_{y\in\Omega}\mathfrak{g}_y(B;U) - \sum_{c\in\Omega^{[1]}}\mathfrak{t}_c(B;U),\\ \mathfrak{a}_p &= D^2 s_{p,U}(0)[a,a],\\ \mathfrak{g}_y &= D^2\mathscr{G}_{y,U}(0)[a,a],\\ \mathfrak{t}_c &= \tfrac{1}{L}\langle J_U(b_0(c)),\alpha_c(U)^{-1}D^2\mathcal{N}_{U,c}(0)[a,a]\rangle\ .\end{aligned} \tag{3.17}$$

Their free-input and background sets are read off from Lemma 3.2 and (3.12): $a(r)$ depends on $B|_{\mathcal{F}_r}$ and on the background on $\mathcal{A}_r$, so $\mathfrak{a}_p$ has free-input set $\bigcup_{r\subset\partial p}\mathcal{F}_r$ and background

set $\bigcup_{r\subset\partial p}(\{r\}\cup\mathcal{A}_r)$; $\mathfrak{g}_y$ involves the phases on the axial paths inside $\mathcal{B}_y$, which contain no designated bond (designated bonds cross interfaces), so its free-input and background sets are $\mathcal{E}_y\cap E_*$ and $\mathcal{E}_y$; $\mathfrak{t}_c$ involves $\mathcal{N}_{U,c}$, α_c and $J_U(b_0(c))$, the latter depending on the plaquettes through $b_0(c)$, which lie in $\mathcal{B}_{c_-}\cup\mathcal{B}_{c_+}$ for $L\geqslant 3$, so its sets are $\mathcal{E}_c\cap E_*$ and $\mathcal{E}_c$. The entries of H^Ω_U are the sums of the polarised entries of these terms. For a pair of positive bonds (b,b') let $\mathcal{T}_{bb'}$ be the set of terms of (3.17) (plaquettes, centres, rows of the torus) whose free-input set contains both b and b', and let the *entry stencil* $\mathcal{S}_{bb'}$ be the union of the background sets of these terms (one of finitely many types modulo $L\mathbb{Z}^4$; it is empty unless $d(b,b')\leqslant 4L$, Lemma 3.3). The entry function is

$$f_{bb'}(U)=\sum_{\mathfrak{T}\in\mathcal{T}_{bb'}}(\text{polarised entry of }\mathfrak{T}\text{ at }(b,b'))\in\mathrm{End}(\mathfrak{g}),\qquad f_{bb'}(A)\coloneqq f_{bb'}(e^{iA})\,,\quad(3.18)$$

defined for every configuration U on $\mathcal{S}_{bb'}$ for which $\mathsf{M}(U)$ is defined and $\alpha_c(U)$ is invertible on the rows of the terms in $\mathcal{T}_{bb'}$ and the local group means $\mathcal{U}_U(y,x)$ are defined for the centres y of the terms in $\mathcal{T}_{bb'}$ (the terms involve C_U on those rows and the means of those centres), whatever the configuration elsewhere; the torus form itself is not needed for its definition. The regional form is the compression of the torus form: for a background U on which H^Ω_U is defined (Section 3.2: $\mathsf{M}(U)$ defined and $\alpha_c(U)$ invertible on the rows of $\Omega^{[1]}$, the means $\mathcal{U}_U(y,x)$ defined for $y\in\Omega^{(1)}$) and every pair $(b,b')\in\Omega^*\times\Omega^*$,

$$\mathsf{H}^\Omega_U(b,b')=f_{bb'}(U|_{\mathcal{S}_{bb'}})\,,\qquad(3.19)$$

because a term whose free-input set contains the bond b of Ω^* has its index in Ω (a plaquette containing b, or containing a bond r with $b\in\mathcal{F}_r$, meets Ω; a centre y with $b\in\mathcal{E}_y$ lies in Ω; a row c with $b\in\mathcal{E}_c\cap E_*$ has an endpoint block containing an endpoint of b, so $c\in\Omega^{[1]}$), hence every term of $\mathcal{T}_{bb'}$ is a term of the regional sum, and the regional form is defined under the weaker requirement on U stated above (Lemma B.6 records this in the appendix).

Lemma 3.3 (Gauge Law and Locality of the Hessian)**.** *Let u be a site gauge transformation of the torus, acting on the background by $U\mapsto U^u$ and on phases by $(R_uq)(b)=\mathrm{Ad}_{u(b_-)}\,q(b)$ ([3, (1.17)]), and let $R_{u,*}$ be the restriction of R_u to the free carrier Ω^*; it is orthogonal for the invariant inner product. Then, for every region Ω,*

$$\mathsf{H}^\Omega_{U^u}=R_{u,*}\mathsf{H}^\Omega_U R^{-1}_{u,*}\,.\qquad(3.20)$$

Every entry $\mathsf{H}^\Omega_U(b,b')$ is a finite composition of local analytic operations in the background bonds of the entry stencil $\mathcal{S}_{bb'}$, which lies within the range $\rho_L=4L$ of b, including the fibre inverses α_c^{-1}, and it vanishes for $d(b,b')>\rho_L$ (the stencils are listed in Appendix B, Lemma B.3). The entry function $f_{bb'}$ of (3.18) *is entrywise gauge covariant: for a site gauge transformation u,*

$$f_{bb'}(U^u)=\mathrm{Ad}_{u(b_-)}\,f_{bb'}(U)\,\mathrm{Ad}^{-1}_{u(b'_-)}\,.\qquad(3.21)$$

The norms are the operator norm for the Hilbert norms induced by h and, for $Df_{bb'}(A)$, the norm $\sup_{\|X\|_\infty\leqslant1}\|Df_{bb'}(A)[X]\|$. Let τ_L denote the radius of Lemma 3.2, reduced so that these operations are analytic on $\|A\|_\infty\leqslant\tau_L$ and reduced once more in (B.6) *of Appendix B; every statement is made for this final value. Define*

$$m_L=\max_b\sum_{b'}\sup_{\|A\|_\infty\leqslant\tau_L}\|f_{bb'}(A)\|,\qquad c_L=\max_b\sum_{b'}\sup_{\|A\|_\infty\leqslant\tau_L}\|Df_{bb'}(A)\|\,.\qquad(3.22)$$

These depend on L, the group and the invariant inner product only.

Proof. The intertwinings are $\mathsf{M}(U^u)(c) = u(c_-)\mathsf{M}(U)(c)u(c_+)^{-1}$, hence $\mathcal{Q}_{U^u} \circ R_u = R_u^{(1)}\mathcal{Q}_U$ with $(R_u^{(1)}v)(c) = \mathrm{Ad}_{u(c_-)}\, v(c)$ (proof of Lemma 3.2), and therefore $Q_{U^u}R_u = R_u^{(1)}Q_U$, $\mathcal{N}_{U^u} \circ R_u = R_u^{(1)}\mathcal{N}_U$, $h_{U^u} = R_u h_U (R_u^{(1)})^{-1}$ (from the covariance of α_c in Lemma 3.2) and $\mathsf{C}_{U^u} = R_u \mathsf{C}_U R_{u,*}^{-1}$. The three terms of (3.15) are then each preserved: $S((e^{iR_uq})U^u) = S((e^{iq}U)^u) = S(e^{iq}U)$ gives $\Delta_{U^u} = R_u\Delta_U R_u^{-1}$ and $J_{U^u} = R_u J_U$; each relative mean $\mathcal{U}_{e^{iqU}}(y,x)\mathcal{U}_U(y,x)^{-1}$ is conjugated by $u(y)$, which preserves the trace; and $\langle J_{U^u}, h_{U^u}D^2\mathcal{N}_{U^u}(0)[R_ua, R_ua]\rangle = \langle R_uJ_U, R_uh_UD^2\mathcal{N}_U(0)[a,a]\rangle$ equals the untransformed force term because R_u is orthogonal for the invariant inner product. The same three identities hold term by term in (3.17) (each term is a function of U and $a = \mathsf{C}_U B$ on its own stencil, and $\mathsf{C}_{U^u}R_{u,*}B = R_u\mathsf{C}_U B$), so the polarised entry of each term transforms as $\mathrm{Ad}_{u(b_-)}(\bullet)\,\mathrm{Ad}^{-1}_{u(b'_-)}$, which is (3.21). Locality: the index of a term is the lower vertex $x(p)$ of its plaquette, the centre y or the initial vertex of $b_0(c)$, and the distance of a bond to the index is that of its initial vertex. By the two-block structure of Lemma 3.2 substituted into (3.17), the free-input and background bonds of a block or row term have initial vertices in the one or two blocks of the index, within ℓ^∞ distance $2L-1$ of it (the sites of two adjacent blocks are within $2L-1$ of each other); a bond r of ∂p has initial vertex within 1 of $x(p)$, and the bonds of $\mathcal{F}_r \cup \mathcal{A}_r$ are within $2L-1$ of r, so the bonds of a plaquette term are within $2L$ of its index. Hence a term whose free-input set contains b and b' has $d(b,b') \leqslant 4L = \rho_L$, and its background bonds are within $4L$ of b. The stencils are listed in Lemma B.3. *q.e.d.*

Lemma 3.4 (Regular-Background Bounds)**.** *Let $\gamma_L = C_L^{-1}$ be the flat lower bound* (A.25) *of Appendix A, τ_L the final radius of Lemma 3.3, and let R_c (the collar range, Lemma 3.11) and C_{L,R_c} be the constants fixed in Appendix B, Section B.2; they depend on L and the group only. Whenever* (3.1) *holds and $\widehat{\mu} = C_{L,R_c}\varepsilon_1$, $\varepsilon_1 = g_1 p_0(g_1)$ ([3, p. 246], with g_1 the first-step coupling), satisfies $\widehat{\mu} < \tau_L/2$ and $\max(1, c_L)\widehat{\mu} < \gamma_L/4$, on every branch (where H_ξ is defined, by (B1) and (B2), as the operator of a quadratic form at fixed volume)*

$$\frac{\gamma_L}{2}\,\mathbf{1} \leqslant H_{\xi,II} \leqslant m_L\,\mathbf{1}\,, \tag{3.23}$$

and $H_{\xi,II}(b,b') = 0$ for $d(b,b') > \rho_L$ (the entry stencil $\mathcal{S}_{bb'}$ is empty beyond that distance, Lemma B.3). The same bounds hold for every principal restriction of $H_{\xi,II}$. Moreover, for every row $b \in I$ the row sum over all free bonds, those outside I included, obeys $\sum_{b' \in \Omega_1^*} \|(H_\xi)_{bb'}\| \leqslant m_L$, with $\|\bullet\|$ the operator norm for h and m_L of (3.22).

The proof is Appendix B, Section B.2: for $b, b' \in I$ the entries $(H_\xi)_{bb'}$ are $f_{bb'}(U_1|_{\mathcal{S}_{bb'}})$ by (3.19) (Lemma B.6); a partition of unity on the coarse grid, a local gauge on each patch bringing U_1 to e^{iA} with $\|A\|_\infty \leqslant \widehat{\mu}$ (Lemmas B.4 and B.5), and the entrywise fundamental theorem of calculus compare the form with the flat one.

3.3 The Source Phase and the Local Primitives

On the branch, put $U = U_1$ and $\Omega = \Omega_1$. The definitions and Lemma 3.5 of this subsection use only (B2), (B4), Lemma 3.6 (for the row solutions on the near rows), Lemmas B.7 and B.8 of Appendix B (for the row solutions on the far rows and the agreement of the definitions), Section 3.2, the map Ψ_ξ of (3.38) with (3.27) and (3.39), and Lemma 3.11(i),(iii). They are stated for the background $U = U_1$ of the branch. For the reference background $U = \mathbf{1}$ of Section 3.8 only the near rows and the germs occur (the reference inputs vanish off I), the row solutions are those of Lemma 3.6 in its reference form (Lemma B.13), (B4) is not invoked, and v, V below are defined by the same formulas. Lemma 3.7 uses in addition that U is a

critical point of S under the block-average constraint and is in the axial mean gauge, which holds for U_1 by (B1) and Section 3.1 and for $U = \mathbf{1}$ because the flat configuration minimises S and $\mathscr{G}_{\mathbf{1}}^{\Omega_1}$ outright and has $\mathcal{U}_{\mathbf{1}}(y,x) = \mathbf{1}$. For a free vector $B \in \mathfrak{g}^{\Omega_1^*}$ put $A = g_0 \mathsf{C}_U B \in \mathfrak{g}^{\Omega_1}$. The first-step correction $\widetilde{D}_U$ (for $U = U_1$, the correction of [3, (1.13)–(1.14)], as identified in (B4)(i)) is the solution, selected below, of the system

$$\widetilde{D}_{U,c}(A) = \mathcal{N}_{U,c}\big(A - h_U \widetilde{D}_U(A)\big), \qquad c \in \Omega_1^{[1]} , \tag{3.24}$$

the form of the equation of [4, p. 267] for Bałaban's own step ($L\widetilde{Q}B' + \widetilde{C}(B') = L\widetilde{Q}B$ with $B' = B - h\widetilde{D}(B)$, $\widetilde{C} = \mathcal{N}_U$, $\widetilde{Q} = Q_U$) written row by row. Let $\rho_{\mathrm{d}} > 0$ be the radius $2K_*\vartheta_*^2$ of Appendix B, Lemma B.7(ii); it depends on L and the group only. A solution d_c of the row equation of c with $h(d_c) < \rho_{\mathrm{d}}$ is called *small*. The row solution $\widetilde{D}_{U,c}(A)$ is defined as follows: for $U = U_1$ on the rows not within $2R_{\mathrm{c}} + 2L$ of I (the far rows; a row is within ρ of a bond set when every bond of its two-block set is, see below) it is the row solution $\widetilde{D}_c(A) = \varphi_c(A|_{\mathcal{E}_c})$ of (B4)(i), defined on the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ (Lemma B.8(i)); the arguments $A = g_0 \mathsf{C}_{U_1} B$, $B \in \mathcal{A}_\xi$, of the domain of integration (3.2) lie in the ball of radius $\vartheta_*/2$, $\|A|_{\mathcal{E}_c}\| < \vartheta_*/2$ (Lemma B.8(ii)), and there its value is the component at c of the chart-class family of Section 3.1 (Lemma B.8(iii)); on the rows within $2R_{\mathrm{c}} + 2L$ of I (the near rows), for $U = U_1$ and for the reference background, it is the unique small solution of the row equation of c at the argument $A|_{\mathcal{E}_c}$ given by Lemma 3.6 on the row polydisc $\mathcal{U}_c$; and on a fixed neighbourhood $\mathcal{N}_c$ of $A|_{\mathcal{E}_c} = 0$, for every row, it is the germ of Lemma 3.5, where $\mathcal{N}_c$ is the neighbourhood given by that lemma, reduced so that it lies in the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ and the germ takes values with $h < \rho_{\mathrm{d}}$ (both hold near zero, where the germ vanishes, and both conditions are open; the phase $A - h_U(c)d_c$ then has norm below $\vartheta_* + H_*\rho_{\mathrm{d}} < 2\vartheta_* \leqslant \varrho_G$ and is chart-valued). Where two of these definitions apply they agree: each produces a solution d_c of the row equation with $h(d_c) \leqslant \rho_{\mathrm{d}}$ at an argument in the ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ (the far-row values by Lemma B.8(i), the near-row values by Lemma 3.6, the germ by the reduction of $\mathcal{N}_c$), and Lemma B.7(ii), which applies at every row for $U = U_1$ (Lemma B.8(i)) and for the reference background $U = \mathbf{1}$ (zero phases), gives exactly one such solution. The reconstructed phase is the function of the free vector

$$q_U(B) = A - h_U \widetilde{D}_U(A) \in \mathfrak{g}^{\Omega_1}, \qquad A = g_0 \mathsf{C}_U B , \tag{3.25}$$

defined on free and designated bonds of Ω_1 wherever every row solution is defined. The integration variable of the branch is $a + A_{\mathrm{e}} \in \mathfrak{g}^{\Omega_1^*}$ in the variables of [3, (1.22)]; with Ψ_ξ the bondwise map (3.38) of Section 3.4 from these variables to those of (1.15), the reconstructed fine field is

$$U_\xi(a)(b) = e^{iq_U(\Psi_\xi(a+A_{\mathrm{e}}))(b)} U_1(b) \quad (b \in \Omega_1), \qquad U_\xi(a)(b) = V_0(b) \quad (b \in \Omega_1^c) , \tag{3.26}$$

the fine field $U = U'U_1$ of [3, (1.13)] at the actual branch data, $U_\xi(a) = U(\Psi_\xi(a + A_{\mathrm{e}}))$ in the notation of Section 3.1; it is defined for $(A_{\mathrm{e}}, a) \in \mathcal{D}_\xi$ by the definition of $\mathcal{D}_\xi$ and $\mathcal{A}_\xi$, and by Lemma 3.6 and Lemma 3.5 its links on the rows within $2R_{\mathrm{c}} + 2L$ of I are defined for every a in the enlarged box defined there, the other links not depending on a. Write $U_\xi^0 = U_\xi(0)$. Since $I \cap S_1^* = \varnothing$ ((3.39) below) and Ψ_ξ acts bond by bond and is the identity off S_1^*,

$$\Psi_\xi(a + A_{\mathrm{e}}) = a + \Psi_\xi(A_{\mathrm{e}}) \qquad (a \in \mathfrak{g}^I,\ A_{\mathrm{e}} \in \mathfrak{g}^J) . \tag{3.27}$$

A row c is said to lie within ρ of a bond set S when every bond of $\mathcal{E}_c$ is within distance ρ of S; as the bonds of $\mathcal{E}_c$ have their initial vertices in two adjacent blocks, this holds as soon as one bond of $\mathcal{E}_c$ is within $\rho - (2L - 1)$ of S.

Lemma 3.5 (Rowwise Locality of the Reconstruction)**.** *The equation of row c in* (3.24) *involves $\widetilde{D}_{U,c}$ alone and depends only on $A|_{\mathcal{E}_c}$ and $U|_{\mathcal{E}_c}$; so the system decouples row by row, and a solution d_c of the row equation at the argument $A|_{\mathcal{E}_c}$ is a function of $A|_{\mathcal{E}_c}$ and $U|_{\mathcal{E}_c}$. The row equation has a unique analytic germ of solutions at $(A|_{\mathcal{E}_c}, d_c) = (0,0)$, with $d_c(0) = 0$ and $Dd_c(0) = 0$; it is the germ at zero of every solution continuous at zero with value zero, in particular, for $U = U_1$, of the row solution $\widetilde{D}_c = \varphi_c$ of (B4)(i). Consequently, wherever the row solutions are defined (Section 3.3): on a free bond r, $q_U(B)(r) = g_0 B(r)$; on $b_0(c)$, $q_U(B)(b_0(c))$ depends only on $B|_{\mathcal{E}_c \cap E_*}$ and on $U|_{\mathcal{E}_c}$; and $q_U(B)$ vanishes on all of $\mathcal{E}_c$ whenever $B = 0$ on $\mathcal{E}_c \cap E_*$ and the row solution is the germ value there. In particular, by* (3.27)*, the link $U_\xi(a)(r)$ depends on a only through $a|_{\mathcal{F}_r \cap I}$ for $r \in \Omega_1$, and not at all for $r \in \Omega_1^c$.*

Proof. $\mathcal{E}_c$ contains no designated bond other than $b_0(c)$ (Lemma 3.2), so $h_U \widetilde{D}_U(A)$ restricted to $\mathcal{E}_c$ involves only $\widetilde{D}_{U,c}$, and $\mathcal{N}_{U,c}$ depends on its argument restricted to $\mathcal{E}_c$ (Lemma 3.2); by (3.12), $A|_{\mathcal{E}_c}$ depends on $B|_{\mathcal{E}_c \cap E_*}$ only. At $(0,0)$ the derivative in d_c of $d_c - \mathcal{N}_{U,c}(A - h_U(c) d_c)$ is the identity, since $D\mathcal{N}_U(0) = D\mathcal{Q}_U(0) - LQ_U = 0$, and $\mathcal{N}_{U,c}(0) = 0$ (the zero phase reproduces the averages of U). The implicit function theorem gives the germ, unique among continuous solutions near $(0,0)$, with $d_c(0) = 0$ and $Dd_c(0) = D\mathcal{N}_{U,c}(0)(\dots) = 0$. For $U = U_1$ the row solution φ_c of (B4)(i) is analytic with $\varphi_c(0) = 0$ and solves the row equation (Lemma B.8(i)), so its germ at zero is this one. If $B = 0$ on $\mathcal{E}_c \cap E_*$ then $A = 0$ on $\mathcal{E}_c$, so the germ value is $\widetilde{D}_{U,c}(0) = 0$ and $q_U(B) = 0$ on $\mathcal{E}_c$. The last claim follows from (3.26), (3.27) and $\mathcal{F}_{b_0(c)} = \mathcal{E}_c \cap E_*$: the links of $U_\xi(a)$ on the bonds of $\mathcal{E}_c$ involve the row solution of c at the argument $g_0 \mathsf{C}_U \Psi_\xi(a + A_{\mathrm{e}})|_{\mathcal{E}_c}$, which involves a only through $a|_{\mathcal{E}_c \cap E_* \cap I}$. *q.e.d.*

The analyticity of the reconstruction on the cutoff box is the following lemma, proved in Appendix B (Lemmas B.7 and B.9); the enlarged box is $\{a \in \mathfrak{g}^I \,|\, |a(b)| \leqslant 2r \ \forall b \in I\}$ with r the cutoff radius of (3.42), R_{c} is the collar range of Lemma 3.11, the row polydisc of a row c is $\mathcal{U}_c = \{B \in \mathfrak{g}^{\Omega_1^*} \,|\, |B(b)| < 2 + 4r \ \forall b \in \mathcal{E}_c \cap E_*\}$, and the block polydisc of a centre y is $\mathcal{U}_y = \{B \,|\, |B(b)| < 2 + 4r \ \forall b \in \mathcal{E}_y\}$ (every bond of $\mathcal{E}_y$ is free). A centre y lies within ρ of a bond set S when every bond of $\mathcal{E}_y$ does.

Lemma 3.6 (Charts)**.** *Let $\vartheta_* > 0$ be the phase radius of Appendix B, Lemma B.7 ($\vartheta_* \leqslant \tau_L/2$; it depends on L and the group only). On every branch, and for the reference background $U = \mathbf{1}$ of Section 3.8, for every row $c \in \Omega_1^{[1]}$ within $2R_{\mathrm{c}} + 2L$ of I and every real phase A with $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$ (Hilbert norm of $\mathfrak{g}^{\mathcal{E}_c}$), the row equation of c at A has exactly one small solution, $\widetilde{D}_{U,c}(A) = \varphi_c(A|_{\mathcal{E}_c})$, where φ_c is analytic on the complex ball $\{\|A|_{\mathcal{E}_c}\| < \vartheta_*\}$, real on real phases, with $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$; under the threshold of Section B.8 (item 3) $g_0 \mathsf{C}_U \mathcal{U}_c$ lies in the ball $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$, so the row solution of Section 3.3 is defined at $A = g_0 \mathsf{C}_U B$ for every $B \in \mathcal{U}_c$; the determinant $\det(\mathbf{1} - h_U(c) \partial_{A(b_0(c))} \widetilde{D}_{U,c}(A))$ is positive there, and the following are analytic functions of $B|_{\mathcal{E}_c \cap E_*}$ on $\mathcal{U}_c$: $\widetilde{D}_{U,c}$, the phase q_U on $\mathcal{E}_c$, the normalised Haar logarithm $\log(\sigma(q_U(\bullet)(b))/\sigma(0))$ for $b \in \mathcal{E}_c$, the real logarithm of that determinant, and, when c is within $2R_{\mathrm{c}}$ of I, the plaquette terms $s_{p,U}(q_U(\bullet))$ of the plaquettes p with a bond in $\mathcal{E}_c$ (as functions of B on the intersection of the polydiscs of the rows of p, which are rows within $2R_{\mathrm{c}} + 2L$ of I); the first four vanish at $B = 0$, $\widetilde{D}_{U,c}$ and $q_U(\bullet)(b_0(c)) - g_0(\mathsf{C}_U B)(b_0(c))$ together with their first derivatives. For every centre $y \in \Omega_1^{(1)}$ within $2R_{\mathrm{c}} + 2L$ of I and every $B \in \mathcal{U}_y$, the relative means $\mathcal{U}_{e^{iqU}}(y,x)\mathcal{U}_U(y,x)^{-1}$, $x \in \mathcal{B}_y$, at $q = q_U(B) = g_0 B$ on $\mathcal{E}_y$ exist and lie within r_G of $\mathbf{1}$ in operator norm, in the domain of the chart logarithm $\log_G$ of Appendix B, and the gauge term $\mathscr{G}_{y,U}(q_U(\bullet))$ is an analytic function of $B|_{\mathcal{E}_y}$ on $\mathcal{U}_y$ vanishing at $B = 0$. All these functions are the compositions of analytic functions φ of the real phase $A|_{\mathcal{E}_c}$ (or $q|_{\mathcal{E}_y}$) with the linear map $B \mapsto g_0 \mathsf{C}_U B$, where φ is analytic on a complex neighbourhood of the closed phase ball $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$ (which contains $g_0 \mathsf{C}_U \mathcal{U}_c$; for the row solution and the*

phase, on the complex ball of radius ϑ_*) and has derivatives up to order five bounded on that closed ball by constants depending on L and the group only; hence their derivatives in B up to order five are bounded on the polydiscs by such constants, and the primitives of this subsection formed from them satisfy, on the enlarged box at the inputs $a+t\Psi_\xi(A_{\mathrm{e}})$, $t\in[0,1]$, $|\mathfrak{p}(a+t\Psi_\xi(A_{\mathrm{e}}))|\leqslant C_{\mathrm{p}}g_0(1+2r)^3$ with C_{p} depending on L and the group only (the prefactor g_0^{-2} of $E_{\xi,\alpha}$ is absorbed by the third-order Taylor remainder, Lemma B.9, (B.18)*). Where $(A_{\mathrm{e}},a)\in\mathcal{D}_\xi$, these objects coincide with the objects of [3, (1.14)–(1.15)], by (B4)(i) and the uniqueness of the small solution (the phase $g_0\mathsf{C}_U\Psi_\xi(a+A_{\mathrm{e}})$ lies in the ball of radius $\vartheta_*/2$ on every such row, by the inclusion stated next with $t=1$, a being in the open box by Lemma 3.11(iv)). Every $a+t\Psi_\xi(A_{\mathrm{e}})$ with a in the enlarged box and $t\in[0,1]$ lies in $\mathcal{U}_c$ and $\mathcal{U}_y$ for every such row and centre (Lemma 3.11(i),(iii): $\Psi_\xi(A_{\mathrm{e}})=A_{\mathrm{e}}$ and $|A_{\mathrm{e}}(b)|<r$ on the bonds of $\mathcal{E}_c$, $\mathcal{E}_y$, which are within $2R_{\mathrm{c}}+4L-1\leqslant 3R_{\mathrm{c}}$ of I, the range of Lemma 3.11(i),(iii)), so every near primitive of this subsection is an analytic function of (a,t) there, with its value and its derivatives in a up to order four bounded as in* (B.18)–(B.19) *of Lemma B.9 (under threshold item 4 of Section B.8, $g_0s_\xi^2\leqslant 4g_0(1+r)^3\leqslant 4\omega_*/C_0$, so these are bounds by constants depending on L and the group only).*

The centred exponent and its primitives are those of [3, (1.14)–(1.16)], written as functions of the free vector $B\in\mathfrak{g}^{\Omega_1^*}$. With $s_{p,U}(q)=1-\operatorname{Re}\operatorname{tr}_N((e^{iq}U)(\partial p))$ for a plaquette p meeting Ω_1 (links off Ω_1 fixed at V_0),

$$E_{\xi,p}(B)=-\frac{1}{g_0^2}\big[s_{p,U}(q_U(B))-s_{p,U}(0)\big]\,, \tag{3.28}$$

and $E_{\xi,y}(B)=-g_0^{-2}\mathscr{G}_{y,U}(q_U(B))$ from the y-summand of (3.14), $y\in\Omega_1$. For either label α,

$$p_{\xi,\alpha}(B)=E_{\xi,\alpha}(B)-DE_{\xi,\alpha}(0)[B]-\frac{1}{2}D^2E_{\xi,\alpha}(0)[B,B]\,. \tag{3.29}$$

With $\sigma(q)\,\mathrm{d}q$ the pullback under $q\mapsto e^{iq}$ of the Haar measure $\mathrm{d}U$ of [3, (1.11)] (the density σ of (1.14), for the fixed inner product as explained in the conventions; its normalisation enters only the constants κ_ξ and c_ξ), for every bond b of Ω_1,

$$p^{\mathrm{H}}_{\xi,b}(B)=\log\sigma(q_U(B)(b))-\log\sigma(0)\,, \tag{3.30}$$

and for every row $c\in\Omega_1^{[1]}$

$$p^{\mathrm{J}}_{\xi,c}(B)=\log\Big|\det_{\mathfrak{g}}\Big(\mathbf{1}-h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c}(g_0\mathsf{C}_UB)\Big)\Big|\,, \tag{3.31}$$

where $\partial_{A(b_0(c))}\widetilde{D}_{U,c}$ is the partial derivative of the row solution, regarded as a function of the $|\mathcal{E}_c|$ independent variables $A(b)$, $b\in\mathcal{E}_c$, with respect to the variable at $b_0(c)$, evaluated at $A=g_0\mathsf{C}_UB$ (this is the designated block of the $\frac{\delta}{\delta A}\widetilde{D}$ of [3], proof of Lemma 3.7). A primitive is defined at B when the row solutions of its rows are defined at $g_0\mathsf{C}_UB$ (Section 3.3: on the domain of integration, on the polydiscs for the rows near I, and near $B|_{\text{inputs}}=0$); for a primitive whose row solutions are not defined at $B=0$ on the domain of integration (which happens when $R_1\neq\varnothing$, since the large-fluctuation tests exclude $A=0$ on R_1), the Taylor polynomial at $B=0$ in (3.29) is formed from the germs of Lemma 3.5; the V of [3] is the sum of the terms of order higher than two of the expansion at $A=0$ (p.249, [4, p. 267]), and Lemma 3.7 identifies the two. The letter $\mathfrak{p}$ denotes a primitive of any of the three kinds. By Lemma 3.5 the free-input set of $E_{\xi,p}$ and $p_{\xi,p}$ is $\bigcup_{r\subset\partial p}\mathcal{F}_r$, that of $E_{\xi,y}$ and $p_{\xi,y}$ is $\mathcal{E}_y\cap E_*=\mathcal{E}_y$, that of $p^{\mathrm{H}}_{\xi,b}$ is $\mathcal{F}_b$ and that of $p^{\mathrm{J}}_{\xi,c}$ is $\mathcal{E}_c\cap E_*$; the background set of each is the set of bonds

whose links enter (bonds of Ω_1 and, for plaquettes and rows at the boundary of Ω_1, bonds of Ω_1^c carrying V_0), as in (3.17); the rows of a primitive are the rows c with $b_0(c)$ in its background set (a gauge primitive $E_{\xi,y}$ has no rows: $\mathcal{E}_y$ contains no designated bond, and $q_U(B) = g_0 B$ on it). A primitive is *near* if its free-input set meets I and *far* otherwise; a far primitive is a function of A_{e} and the branch data alone (Lemma 3.5, on the domain of (B4)). A near primitive has a bond $b_1 \in I$ in its free-input set, and every bond of $\mathcal{E}_c$ for a row c of the primitive, and every bond of $\mathcal{E}_y$ for a gauge primitive, is within $4L+3 \leqslant 2R_{\mathrm{c}}$ of b_1 (two bonds of a two-block set are within $2L-1$ of each other, the bonds of a plaquette within 1); so the rows of a near primitive, and the centre of a near gauge primitive, lie within $2R_{\mathrm{c}}$ of I, and by Lemma 3.6 every near primitive is defined and analytic in (a,t) at the inputs $a + t\Psi_\xi(A_{\mathrm{e}})$, a in the enlarged box, $t \in [0,1]$. An *arc* γ of the contour C is a set of consecutive bonds of C (with their orientations), and γ_+ is the set of their positive representatives; the free-input and background stencils of an arc are

$$F_\gamma = \bigcup_{r \in \gamma_+} \mathcal{F}_r, \qquad D_\gamma = \gamma_+ \cup \bigcup_{c \,|\, b_0(c) \in \gamma_+} \mathcal{E}_c \,. \tag{3.32}$$

Lemma 3.7 (Exact Primitive Sum)**.** *Let v and V be the functions of [3, (1.15), p. 249]: $v(A)$ is the sum of the logarithms of the Haar densities $\sigma(A - h\widetilde{D}(A))$ over the bonds of Ω_1 and of $\log|\det(\mathbf{1} - h\frac{\delta}{\delta A}\widetilde{D})|$, the logarithm of the absolute Jacobian of (1.14), which is, in the notation of [3], $\mathrm{Tr}\log(\mathbf{1} - h\frac{\delta}{\delta A}\widetilde{D})$ where the determinant is positive (on the near rows, factor by factor, by Lemma 3.6; on all of $\mathcal{A}_\xi$ by (B4)(iii), Lemma B.8(iv)), the identities below using the absolute value throughout, and $V(A)$ is the remainder after the Taylor polynomial of degree two at $A = 0$ of the action and the gauge-fixing functional at the phase $A - h\widetilde{D}(A)$ (the "terms of order higher than two" of [3]; no series expansion is used, and the Taylor coefficients at $A = 0$ are those of the germs of Lemma 3.5). For $B \in \mathcal{A}_\xi$, and more generally for every $B \in \mathfrak{g}^{\Omega_1^*}$ at which every row solution is defined with the row solutions of the near rows those of Lemma 3.6 and those of the far rows the germs at zero argument ($B = 0$ on their two-block sets), with $A = g_0 \mathsf{C}_U B$,*

$$v(A) - v(0) - g_0^{-2} V(A) = \sum_\alpha p_{\xi,\alpha}(B) + \sum_b p^{\mathrm{H}}_{\xi,b}(B) + \sum_c p^{\mathrm{J}}_{\xi,c}(B) \,, \tag{3.33}$$

the first sum over the plaquettes p meeting Ω_1 and the centres $y \in \Omega_1^{(1)}$, the second over the bonds b of Ω_1, the third over the rows $c \in \Omega_1^{[1]}$ (the same ranges in every sum $\sum_\alpha$, $\sum_b$, $\sum_c$ below), so that the exponent of the second factor of [3, (1.15)] at the free vector B, with the quadratic form written through the regional Hessian (3.16)*,*

$$\mathscr{E}_\xi(B) = -\frac{1}{2}\langle B, H_\xi B\rangle + v(g_0 \mathsf{C}_U B) - g_0^{-2} V(g_0 \mathsf{C}_U B) \,, \tag{3.34}$$

satisfies $\mathscr{E}_\xi(0) = v(0)$ with $v(0) = \sum_b \log(\sigma(0)/\sigma_0) = 0$ (sum over the bonds of Ω_1; the Haar densities in the v of [3] are divided by their value σ_0 at the identity, as fixed in the step list after (3.7)*) and*

$$\begin{aligned} \mathscr{E}_\xi(B) - \mathscr{E}_\xi(0) &= -\frac{1}{2}\langle B, H_\xi B\rangle + \sum_\alpha p_{\xi,\alpha}(B) + \sum_b p^{\mathrm{H}}_{\xi,b}(B) + \sum_c p^{\mathrm{J}}_{\xi,c}(B) \\ &= \sum_\alpha E_{\xi,\alpha}(B) + \sum_b p^{\mathrm{H}}_{\xi,b}(B) + \sum_c p^{\mathrm{J}}_{\xi,c}(B) \,. \end{aligned} \tag{3.35}$$

Proof. The sum $\sum_\alpha E_{\xi,\alpha}(B)$ is $-g_0^{-2}$ times the difference of the action plus gauge-fixing functional at the phase $q_U(B)$ and at $q = 0$: here the gauge-fixing functional of [3, (1.11),

(1.14)] is $\mathbf{G}(\Omega_1^{(1)}, e^{iq}U_1) = \mathscr{G}_{U_1}^{\Omega_1}(q)$ because U_1 is in the axial mean gauge (Section 3.2, after (3.15)). Since $D\widetilde{D}_U(0) = 0$, $Dq_U(0)[B] = g_0\mathsf{C}_U B$, and the linear term of the Taylor expansion in B at zero is $\sum_\alpha DE_{\xi,\alpha}(0)[B] = -g_0^{-1}\big(DS(U_1)[\mathsf{C}_U B] + D\mathscr{G}_U^{\Omega_1}(0)[\mathsf{C}_U B]\big) = 0$: the relative gauge functional is a sum of nonnegative terms vanishing at $q = 0$, so its first variation vanishes; by [3, (1.12)] U_1 is a critical point of $S + \mathbf{G}(\Omega_1^{(1)}, \bullet)$ on the configurations with $\mathsf{M}(U) = V$ on $\Omega_1^{(1)}$ and $U = V_0$ on Ω_1^c, and $\mathbf{G}(\Omega_1^{(1)}, \bullet) \geqslant 0$ attains its minimum 0 at U_1, so $D\mathbf{G}(\Omega_1^{(1)}, U_1) = 0$ and $DS(U_1)[\delta] = 0$ for every phase δ supported on the bonds of Ω_1 with $D\mathcal{Q}_U(0)\delta = LQ_U\delta = 0$ (the tangent space of the constraint $\mathsf{M}(e^{i\delta}U_1) = V$); and $\mathsf{C}_U B$ is such a phase by $Q_U\mathsf{C}_U = 0$ (3.12). The quadratic term is $-\frac{1}{2}\langle B, H_\xi B\rangle$ by (3.15), with $A = g_0\mathsf{C}_U B$: the second variation of the action is $\langle A, \Delta_U A\rangle$ plus the force term $-2\langle J_U, h_U\frac{1}{2}D^2\mathcal{N}_U(0)[A, A]\rangle$ from the second-order part $-h_U\widetilde{D}_U^{(2)}(A)$ of the phase, and that of the gauge functional is $G_U^{(2),\Omega_1}(A)$. Subtracting both from $\sum_\alpha E_{\xi,\alpha}$ gives $\sum_\alpha p_{\xi,\alpha}$, which is therefore $-g_0^{-2}V(A)$. The Haar part of v is $\sum_b \log\sigma(q_U(B)(b))$ bond by bond, so v minus its Haar part at $B = 0$ gives $\sum_b p^{\mathrm{H}}_{\xi,b}$. For the Jacobian, $\frac{\delta}{\delta A}\widetilde{D}$ has rows indexed by $\Omega_1^{[1]}$ and columns by the bonds of Ω_1, and h_U maps row c to the bond $b_0(c)$; in the ordering (free bonds, designated bonds) the matrix $\mathbf{1} - h_U\frac{\delta}{\delta A}\widetilde{D}_U$ is block triangular with identity block on the free bonds and designated block $\mathrm{diag}_c(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c})$ by Lemma 3.2 (row c depends on $A|_{\mathcal{E}_c}$, which contains the single designated bond $b_0(c)$), so its determinant is $\prod_c \det_{\mathfrak{g}}(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c})$, positive on $\mathcal{A}_\xi$ by (B4)(iii), and at the other points admitted in the statement because each near-row factor is positive by Lemma 3.6 and each far-row factor equals one at zero argument ($D\widetilde{D}_{U,c}(0) = 0$); so Tr log of the matrix, the term in the v of [3], is the real logarithm of the positive determinant, and since each factor is a nonzero real, the logarithm of the absolute value of the product is the sum of the logarithms of the absolute values of the factors, $\sum_c \log|\det_{\mathfrak{g}}(\cdots)| = \sum_c p^{\mathrm{J}}_{\xi,c}(B)$; this vanishes at $B = 0$ since $D\widetilde{D}_U(0) = 0$; hence $v(0)$ is the Haar part at $B = 0$. The quadratic form of (1.15) in the free variable, $\frac{1}{2}\langle B, C^*\Delta_1 CB\rangle$, is $\frac{1}{2}\langle B, H_\xi B\rangle$ by the identification $\mathsf{H}_{U_1}^{\Omega_1} = C^*\Delta_1 C$ of Section 3.2 (Δ_1 of (1.16) already contains $\mathbf{G}^{(2)}$ and the force term; the g_0^{-2} of the action cancels the g_0^2 of $A = g_0\mathsf{C}_U B$). Subtracting $\mathscr{E}_\xi(0) = v(0)$ from (3.34) and inserting (3.33) gives the first form of (3.35); the second form restores the quadratic term to $\sum_\alpha p_{\xi,\alpha}$. *q.e.d.*

Each primitive is assigned a host Y_α in the native localisation family $\mathbf{D}_0$: lift its free-input and background stencil $\widetilde{\mathcal{S}}_\alpha$ to $\mathbb{Z}^4$ about its base (plaquette lower vertex, block centre, initial vertex of its bond, or initial vertex of $b_0(c)$ for $p^{\mathrm{J}}_{\xi,c}$; the stencil has ℓ^∞ diameter at most $4L < M$, below every period, so the lift about the base is well defined), let Z_α be the M-cubes $Q_z = Mz + [-M/2, M/2]^4$ meeting it, and put

$$Y_\alpha = \pi_T\Big(\bigcup_{z\in\prod_\mu([k^-_{\alpha,\mu}-1,k^+_{\alpha,\mu}+1]\cap\mathbb{Z})} Q_z\Big), \qquad k^-_{\alpha,\mu} = \min_{z\in Z_\alpha} z_\mu, \quad k^+_{\alpha,\mu} = \max_{z\in Z_\alpha} z_\mu\ , \tag{3.36}$$

with π_T the reduction modulo the periods; since the stencil has diameter at most $4L < M$, it meets at most two consecutive M-cubes in each direction, so the rectangle has side at most 4 cubes, and every period being at least $8M$, π_T is injective on it. Y_α is connected and contains the stencil in its interior. The potential is, for $Y \in \mathbf{D}_0$ and $a \in \mathfrak{g}^I$,

$$\mathbf{V}_\xi(Y, a) = \sum_{\mathfrak{p}\ \text{near},\ Y_{\mathfrak{p}}=Y} \mathfrak{p}(a) \quad \text{if } |a(b)| < r\ \forall b \in E(Y), \qquad \mathbf{V}_\xi(Y, a) = 0 \quad \text{otherwise,} \tag{3.37}$$

the sum over the near primitives $\mathfrak{p}$ of all three kinds assigned to Y, with $E(Y) = \{b \in I \,|\, b \cap Y \neq \varnothing\}$ and the primitives at zero retained input ($t = 0$). It depends on $a|_{E(Y)}$ only,

since the free-input set of a primitive assigned to Y lies in Y, hence in $E(Y)$ (Lemma 3.5), and it is bounded on the local box by Lemma 3.6 (each summand by $C_{\mathrm{p}} g_0 (1+2r)^3$). On the open box $\{|a(b)| < r \ \forall b \in I\}$, $\sum_Y \mathbf{V}_\xi(Y,a) = \sum_{\mathfrak{p} \text{ near } \mathfrak{p}} \mathfrak{p}(a)$.

3.4 The Boundary Law, the Exterior Interpolation and the Collar

The change of variables. Let $S_1 = (\Omega_1 \setminus \Omega_1^{\sim -1}) \cup R_1$ ([3, (1.20)]) and let S_1^* be its free bonds. [3, (1.22)] replaces the variable A of (1.15) on each bond $b \in S_1^*$ by a new variable $A'(b)$ and leaves the other bonds unchanged: $A = \Psi_\xi A'$ with

$$\begin{aligned} (\Psi_\xi A')(b) &= g_0^{-1} \log \left(e^{i g_0 A'(b)} e^{-i \omega_b(A'(b)) \mathbf{H}_{1,Ax}(b)} \right) \quad (b \in S_1^*), \\ (\Psi_\xi A')(b) &= A'(b) \quad (b \in \Omega_1^* \setminus S_1^*) \,, \end{aligned} \tag{3.38}$$

where ω_b is the C^∞ cutoff g of [3, p. 252] ($0 \leqslant g \leqslant 1$, $g = 0$ on $|A'| \leqslant \frac{4}{3} g_0^{-1} \delta_0$, $g = 1$ where $|e^{i g_0 A'} - 1| \geqslant \frac{5}{3}\delta_0$) for b in the set $(R_1^{\sim 2} \cap S_1)^*$ named in the definition preceding (1.22) (p.251); [3] specifies g by these properties only, and one such g with $|Dg(A')| \leqslant 6 g_0/\delta_0$ for every A' (operator norm for the cutoff norm $|\bullet|$) is fixed once: the function $\ell(A') = \min(1, \max(0, (|g_0 A'| - \frac{17}{12}\delta_0)/(\frac{1}{6}\delta_0)))$ is Lipschitz with constant $6 g_0/\delta_0$, vanishes for $|g_0 A'| \leqslant \frac{17}{12}\delta_0$ and equals one for $|g_0 A'| \geqslant \frac{19}{12}\delta_0$, and its convolution with a C^∞ probability density supported in $\{|A'| \leqslant \delta_0/(12 g_0)\}$ is C^∞ with values in $[0,1]$, has the same Lipschitz constant, vanishes for $|g_0 A'| \leqslant \frac{4}{3}\delta_0$ and equals one for $|g_0 A'| \geqslant \frac{5}{3}\delta_0$, hence wherever $|e^{i g_0 A'} - \mathbf{1}| \geqslant \frac{5}{3}\delta_0$, as $|e^{iX} - \mathbf{1}| \leqslant |X|$; and $\omega_b = 1$ for the other bonds of S_1^* (p.252 describes the second set as $S_1^* \setminus (R_1^{\sim 2})^*$, which differs from the complement of the first set in S_1^* on the bonds joining $R_1^{\sim 2} \setminus S_1$ to $S_1 \setminus R_1^{\sim 2}$; the definition of p.251 is used, and (B4)(ii) refers to the map so defined; on the bonds of the difference the map is the second form of [3], g replaced by one, and Bałaban's remark that these changes of variables "do not change the functions $\chi^{(0)}$" (p.252), its rewriting (1.23) of the cutoff and its rewriting of the retained tests in the new variables, p.252, with the localisation property stated there ((B3)), are not used, the cutoff being composed exactly in (3.43)), $\mathbf{H}_{1,Ax}(b) \in \mathfrak{g}$ is the field $\mathbf{H}_{1,Ax}(\Omega_1 \cap \Lambda_1^c, \bullet)$ of [3, (1.21)] at the actual branch data, as it appears in (1.22), without the prefactor L^{-1} of (1.21) (a function of the pair (V, V_0) through $U_1 = U_1(V, V_0)$ and $U_1(\Omega_1 \cap \Lambda_1^c, V)$, the latter stated by [3, p. 251] to depend on $V|_{\Omega_1 \cap \Lambda_1^c}$; Borel in (V, V_0) by the measurability paragraph of Section 3.1; with (1.21) and $\omega_b = 1$, $e^{i g_0 (\Psi_\xi A')(b)} U_1(b) = e^{i g_0 A'(b)} e^{-i(1-L^{-1}) \mathbf{H}_{1,Ax}(b)} U_1(\Omega_1 \cap \Lambda_1^c, b)$, which is not the second factor of [3, (1.23)]: as printed, (1.22) carries $\mathbf{H}_{1,Ax}$ where (1.21) carries $L^{-1}\mathbf{H}_{1,Ax}$, and (1.24) has $+F$ against the $-F$ of (1.22); the identities of this paper use (1.22) as printed, with the exact Jacobian $|\det D\Psi_\xi|$ and the cutoff composed in the variable of (1.15), never (1.23)–(1.24), so they do not depend on the resolution of this discrepancy), and log is the group chart, the inverse of $A \mapsto e^{iA}$ on the chart ball (so that log is the chart logarithm of the fluctuation, written $\frac{1}{i}\log$ in the notation of [3] for the matrix logarithm, and $g_0^{-1}\log(\cdots)$ is the variable of (1.15)). Ψ_ξ acts bond by bond, and ψ_b denotes its component at $b \in S_1^*$, a C^∞ map (g is C^∞, not analytic) on the open set $\mathcal{W}_b \subseteq \mathfrak{g}$ of those $A'(b)$ for which $g_0 A'(b)$ lies in the chart ball and the argument of the chart logarithm lies in the chart domain (on the bonds where (1.22) is the identity, the variable A' of (1.22) is the variable A of (1.15), which on the free bonds is g_0^{-1} times the fluctuation variable of (1.13); on S_1^* the new variable A' is restricted to the values with $g_0 A'$ in the chart ball as well, which loses no preimage, every preimage under ψ_b of a point of $\mathcal{A}_\xi$ lying in $g_0^{-1}K$ with K the ball of the proof of Lemma 3.8(b),(c); the change of variables is made on this domain). By Lemma 3.8 below, which is the (1.22) part of (B4)(ii), Ψ_ξ maps $\mathcal{D}_\xi$ bijectively onto $\mathcal{A}_\xi$, the domain of integration of (1.15) with the cutoff (1.25) (Section 3.1), and $D\Psi_\xi$ is invertible there, so that the change of

variables formula applies with the Jacobian $|\det D\Psi_\xi| = \prod_{b\in S_1^*} |\det D\psi_b|$; $\chi_{\xi,\mathrm{e}}(A_\mathrm{e})$ below is set to zero when $A_\mathrm{e}(b) \notin \mathcal{W}_b$ for some $b \in S_1^*$. Thus $\log|\det D\Psi_\xi(A')|$ depends on $A'|_{S_1^*}$ only, and it is C^∞; what Lemma 3.9 uses of it is that it is finite at $(A_\mathrm{e}, 0) \in \mathcal{D}_\xi$ and independent of a (a constant on the branch, by $I \cap S_1^* = \varnothing$ and (3.27)). The integration variable of the branch is $A' = a + A_\mathrm{e}$ with $a \in \mathfrak{g}^I$ and $A_\mathrm{e} \in \mathfrak{g}^J$. A bond of I has an endpoint in $\Lambda_1 \subseteq \Omega_1^{\sim -2} \cap (R_1'^{\sim})^c$, a bond of S_1^* has an endpoint in S_1, and $\Omega_1^{\sim -2}$ is separated from $\Omega_1 \setminus \Omega_1^{\sim -1}$, as $(R_1'^{\sim})^c$ is from $R_1 \subseteq R_1'$, by a full layer of LMR_1-cubes (the enlargement adjoins every touching cube, Section 3); hence, with distances between initial vertices,

$$I \cap S_1^* = \varnothing, \qquad d(I, S_1^*) \geqslant LMR_1 - 2\,. \tag{3.39}$$

The filling modification following [3, (1.20)] only removes cubes from Λ_1, so (3.39) holds for the final Λ_1. This gives (3.27).

Lemma 3.8 (The Boundary Change Is a Bijection)**.** *Let $r_G \in (0,1]$ be the group constant of Appendix B (defined there with the chart logarithm $\log_G$): every $U \in G$ with $\|U - \mathbf{1}\|_\mathrm{op} \leqslant r_G$ lies in the chart domain N_G, the image of the chart ball $\{h(A) < \varrho_G\}$ under $A \mapsto e^{iA}$. Let $\delta_0 \leqslant \delta_G := \min\bigl(\log(3/2)/(12\sqrt{N}),\, r_G/(10\sqrt{N})\bigr)$, a condition on the coupling ($\delta_0 = (A_1/A_0)\varepsilon_0 \to 0$). Let (Ω_1, V, V_0) be an admissible pair and R_1 a large-fluctuation region of the history sum (Section 3.1), so that U_1 and the configuration $U_1(\Omega_1 \cap \Lambda_1^c, V)$ of (1.21) exist ((B1) and Lemma 3.1) and $\mathbf{H}_{1,Ax}(b) := L\log\bigl(U_1(b)\,U_1(\Omega_1 \cap \Lambda_1^c, V)(b)^{-1}\bigr)$, formed with the representatives chosen in Lemma 3.1, is defined wherever this chart logarithm is; under* (3.40) *it is the analytic $\mathbf{H}_{1,Ax}$ of [3, (1.21)], since $L^{-1}\mathbf{H}_{1,Ax}(b)$ then lies in the chart ball and (1.21) determines it there as this logarithm. For $b \in S_1^*$ let $\square'(b) \subseteq S_1$ be an $L^2M_2R_1$-cube with $b \in (\square'(b)^{\sim 2})^*$ (one exists: S_1 is a union of such cubes, $\Omega_1 \setminus \Omega_1^{\sim -1}$ being a union of LMR_1-cubes, each a union of $L^2M_2R_1$-cubes (the cube partitions of [3, pp. 245–246] are nested: each cube of a coarser partition is a union of cubes of each finer one, the L-blocks included, the side ratios $M/(LM_2)$, a power of L by the standing choice, and LM_2R_1 being integers), and R_1 a union of them, p.250; b has an endpoint in S_1, hence in such a cube, and is not designated for a row outside $S_1^{[1]}$: if $b = b_0(c)$, the block containing its endpoint in S_1 lies in S_1 by the nesting, so $c \in S_1^{[1]}$, which $b \in S_1^*$ excludes), put $Z_b := U_1(b)U_{1,\square'(b)}(b)^{-1}$, and recall the bound on $\mathbf{H}_{1,Ax}$ of [3, p. 251], proved in Lemma D.18 at the tolerance*

$$|\mathbf{H}_{1,Ax}(b)| \leqslant \frac{\delta_0}{40} \qquad (b \in S_1^*)\,, \tag{3.40}$$

the input discussed after the proof; the inequality $|Z_b - \mathbf{1}| < 2\delta_0$ ($b \in S_1^*$) used with it is (B1)'s, derived in the proof. Let $\mathcal{T}_b := \{A \in \mathfrak{g} : h(g_0A) < \varrho_G,\ |e^{ig_0A}Z_b - \mathbf{1}| < 2\delta_0\}$. Then for every $b \in S_1^*$: (a) $B(b) \in \mathcal{T}_b$ for every $B \in \mathcal{A}_\xi$; (b) $\mathcal{T}_b \subseteq \psi_b(\mathcal{W}_b)$; (c) ψ_b is injective on $\psi_b^{-1}(\mathcal{T}_b)$, and $D\psi_b$ is invertible at every point of $\psi_b^{-1}(\mathcal{T}_b)$. Consequently Ψ_ξ maps $\mathcal{D}_\xi$ bijectively onto $\mathcal{A}_\xi$, and $D\Psi_\xi$ is invertible at every point of $\mathcal{D}_\xi$.

Proof. Norm facts, for the cutoff norm $|\bullet|$ (operator or normalised Hilbert–Schmidt) and Hermitian X with $\|X\|_\mathrm{op} \leqslant \pi$: $|e^{iX} - \mathbf{1}| \leqslant |X| \leqslant \frac{\pi}{2}|e^{iX} - \mathbf{1}|$ (eigenvalue by eigenvalue, $2|\sin(\theta/2)| \leqslant |\theta| \leqslant \pi|\sin(\theta/2)|$ for $|\theta| \leqslant \pi$, in both norms); $|UY| = |YU| = |Y|$ for unitary U, so $|UU' - \mathbf{1}| \leqslant |U - \mathbf{1}| + |U' - \mathbf{1}|$ for unitary U, U'; a unitary $U \in G$ with $|U - \mathbf{1}| < r_G/\sqrt{N}$ lies in the chart domain, since $\|U - \mathbf{1}\|_\mathrm{op} \leqslant \sqrt{N}|U - \mathbf{1}|$ for either cutoff norm and by the definition of r_G (the eigenvalue argument alone does not give this for a general compact $G \subset U(N)$: the Hermitian logarithm Y of U with eigenvalues in $(-\pi, \pi]$ need not lie in the Lie algebra $\mathfrak{g}$, so r_G is a constant of the group); and on $\{|X| \leqslant 12\delta_0\}$, writing $e^{iX} = \mathbf{1} + iX + R(X)$, one has $\|DR(X)\| \leqslant e^{\|X\|_\mathrm{op}} - 1 \leqslant e^{12\sqrt{N}\delta_0} - 1 \leqslant \frac{1}{2}$ (the derivative of X^k is a sum of k

terms X^jVX^{k-1-j}, each of norm at most $\|X\|_{\mathrm{op}}^{k-1}|V|$), so $|e^{iX}-e^{iX'}| \geqslant \frac{1}{2}|X-X'|$ for $|X|,|X'| \leqslant 12\delta_0$.

Fix $b \in S_1^*$, write $H = \mathbf{H}_{1,Ax}(b)$ and $\omega = \omega_b$, and pass to the group-scale variable $x = g_0A'$: $\psi_b(A') = g_0^{-1}F(g_0A')$ with $F(x) = \log(e^{ix}W(x))$, $W(x) = e^{-i\widetilde{\omega}(x)H}$, $\widetilde{\omega}(x) = \omega(x/g_0)$, on $\mathfrak{B}_b = g_0\mathcal{W}_b = \{x : h(x) < \varrho_G,\ e^{ix}W(x)$ in the chart domain$\}$; $|D\widetilde{\omega}| \leqslant 6/\delta_0$, and $|W(x)-\mathbf{1}| \leqslant \widetilde{\omega}(x)|H| \leqslant |H| \leqslant \delta_0/40$. Moreover $|Z_b - \mathbf{1}| < 2\delta_0$: by (B1) the background U_1 lies in $D^{(11)}(V,V_0)$, the domain of integration in (1.11) ([3, p. 248]), on which the factor $\chi_0'(\square'(b))$ of $\chi_0'(\Omega_1)$ is nonzero, that is $|U_1(b')U_{1,\square'(b)}(b')^{-1} - \mathbf{1}| < 2\delta_0$ for every $b' \in (\square'(b)^{\sim 2})^*$ ([3, (1.8)]), and b is one of these bonds.

(a) For $B \in \mathcal{A}_\xi$, $U(B)(b) = e^{ig_0B(b)}U_1(b)$ (b is free, so $q_{U_1}(B)(b) = g_0B(b)$, Lemma 3.5), and $g_0B(b)$ is the component at b of the chart logarithm $\frac{1}{i}\log U'$ of (1.13) (the variable written A' there, not the new variable A' of (1.22)), so $h(g_0B(b)) < \varrho_G$ by the chart-class condition in the definition (3.2) of $\mathcal{A}_\xi$; the cutoff (1.25) is nonzero at $U(B)$, in particular its factor $\chi_0'(\square'(b))$, the test $|U(B)(b')U_{1,\square'(b)}(b')^{-1} - \mathbf{1}| < 2\delta_0$ for $b' \in (\square'(b)^{\sim 2})^*$ ([3, (1.8) and p. 250]; a function of the fine field, composed with the changes of variables (1.14) and $A = g_0A'$ of p.249, Section 3.1), and at $b' = b$ this reads $|e^{ig_0B(b)}Z_b - \mathbf{1}| < 2\delta_0$.

(b) Let $y = g_0A$ with $A \in \mathcal{T}_b$. Then $|e^{iy} - \mathbf{1}| \leqslant |e^{iy}Z_b - \mathbf{1}| + |Z_b - \mathbf{1}| < 4\delta_0$, and $\|y\|_{\mathrm{op}} < \pi$ since $h(y) < \varrho_G$, so $|y| \leqslant \frac{\pi}{2}\cdot 4\delta_0 < 6.3\,\delta_0$. Let $K = \{x \in \mathfrak{g} : |x| \leqslant 7\delta_0\}$, compact and convex, with $K \subseteq \mathfrak{B}_b$: $h(x) \leqslant |x| \leqslant 7\delta_0 < \varrho_G$ (as $7\delta_0 \leqslant 7r_G/(10\sqrt{N}) < \pi/\sqrt{N} = \varrho_G$, $r_G \leqslant 1$) and $|e^{ix}W(x) - \mathbf{1}| \leqslant |x| + |H| \leqslant 7\delta_0 + \delta_0/40 < 8\delta_0 \leqslant r_G/\sqrt{N}$, so $e^{ix}W(x)$, an element of G, lies in the chart domain. For $x \in K$, $e^{iF(x)} = e^{ix}W(x)$ gives $|e^{iF(x)} - e^{ix}| = |W(x) - \mathbf{1}| \leqslant |H|$ and $|e^{iF(x)} - \mathbf{1}| \leqslant 7\delta_0 + |H|$, so $|F(x)| \leqslant \frac{\pi}{2}(7\delta_0 + \delta_0/40) < 11.1\,\delta_0 < 12\delta_0$ ($\|F(x)\|_{\mathrm{op}} < \pi$, $F(x)$ lying in the chart ball) and $|F(x) - x| \leqslant 2|e^{iF(x)} - e^{ix}| \leqslant 2|H| \leqslant \delta_0/20$, both $F(x)$ and x lying in $\{|X| \leqslant 12\delta_0\}$. The continuous map $T(x) = x - F(x) + y$ maps K into itself, $|T(x)| \leqslant \delta_0/20 + 6.3\,\delta_0 < 7\delta_0$, so by Brouwer's fixed point theorem it has a fixed point $x \in K$, that is $F(x) = y$; then $A' = x/g_0 \in \mathcal{W}_b$ and $\psi_b(A') = A$.

(c) Let $x, x' \in \mathfrak{B}_b$ with $F(x) = F(x') = y$ and $y/g_0 \in \mathcal{T}_b$. Then $e^{ix} = e^{iy}W(x)^{-1}$ gives $|e^{ix} - \mathbf{1}| \leqslant |e^{iy} - \mathbf{1}| + |H| < 4\delta_0 + \frac{1}{40}\delta_0$, so $|x| \leqslant \frac{\pi}{2}\cdot\frac{161}{40}\delta_0 < 6.4\,\delta_0$, and likewise $|x'| < 6.4\,\delta_0$; both lie in K. From $e^{ix}W(x) = e^{ix'}W(x')$, $e^{ix} - e^{ix'} = e^{ix'}\big(W(x')W(x)^{-1} - \mathbf{1}\big)$ with $W(x')W(x)^{-1} = e^{i(\widetilde{\omega}(x)-\widetilde{\omega}(x'))H}$, so $|e^{ix} - e^{ix'}| \leqslant |\widetilde{\omega}(x) - \widetilde{\omega}(x')|\,|H| \leqslant \frac{6}{\delta_0}|x - x'|\cdot\frac{\delta_0}{40} = \frac{3}{20}|x - x'|$, while $|e^{ix} - e^{ix'}| \geqslant \frac{1}{2}|x - x'|$; hence $x = x'$. For the differential at such an $x \in K$: with $P(x) = e^{ix}W(x)$, $F = \log\circ P$, and $D\log$ is invertible on the chart domain, so it suffices that $DP(x)$ is injective; $DP(x)[v] = D(e^{i\bullet})(x)[v]\,W(x) + e^{ix}DW(x)[v]$ with $|e^{ix}DW(x)[v]| = |D\widetilde{\omega}(x)[v]|\,|H| \leqslant \frac{3}{20}|v|$ and $|D(e^{i\bullet})(x)[v]\,W(x)| = |iv + DR(x)[v]| \geqslant \frac{1}{2}|v|$, so $DP(x)[v] = 0$ forces $v = 0$.

The consequences: Ψ_ξ is bondwise and the identity off S_1^*. For $B \in \mathcal{A}_\xi$, (a) and (b) give A' with $A'(b) \in \mathcal{W}_b$ for $b \in S_1^*$ and $\Psi_\xi(A') = B$, and $(A_{\mathrm{e}}, a) = (A'|_J, A'|_I) \in \mathcal{D}_\xi$ by (3.3); so $\Psi_\xi(\mathcal{D}_\xi) = \mathcal{A}_\xi$, the inclusion $\subseteq$ being the definition. If $\Psi_\xi(A') = \Psi_\xi(A'') = B \in \mathcal{A}_\xi$, then bond by bond $\psi_b(A'(b)) = \psi_b(A''(b)) = B(b) \in \mathcal{T}_b$ by (a), so $A'(b) = A''(b)$ by (c); and $D\Psi_\xi = \bigoplus_{b\in S_1^*} D\psi_b \oplus \mathbf{1}$ is invertible at every point of $\mathcal{D}_\xi$ by (c). Moreover, by (b), (c) and the inverse function theorem, Ψ_ξ restricted to the open set $\prod_{b\in S_1^*}\psi_b^{-1}(\mathcal{T}_b) \times \mathfrak{g}^{\Omega_1^*\setminus S_1^*}$ ($\mathcal{T}_b$ is open, being defined by strict inequalities between continuous functions) is a C^∞ diffeomorphism onto the open set $\prod_{b\in S_1^*}\mathcal{T}_b \times \mathfrak{g}^{\Omega_1^*\setminus S_1^*}$, which contains $\mathcal{A}_\xi$ by (a); so $\mathcal{D}_\xi$ lies in the former open set, on which $\log|\det D\Psi_\xi|$ is C^∞. *q.e.d.*

The inputs. The bound (3.40) is Lemma D.18 (Section D.8.6). Bałaban states in [3, p. 251] the bound $|\mathbf{H}_{1,Ax}| < O(1)B_3\exp(-\delta MR_1)\varepsilon_1$, "much smaller than δ_0, for MR_1 large enough",

for $\mathbf{H}_{1,Ax}$ "on $\square'$" for the cubes $\square' \subset S_1$, from the distance between $\square'$ and the support of the argument of $\mathbf{H}_{1,Ax}$ (one layer of M_1-cubes touching $\partial\Lambda_1$) and the exponential decay property of [7, (190)], which is stated there for the Landau gauge and which [3] transfers to the axial gauge ("on the unit lattice it holds for any gauge, for which propagators are exponentially decaying, hence for the axial gauge too"). The lemma derives a bound of this form on every bond of S_1^* from [7, Proposition 9, (172), and (190)], integrated in the Landau gauge, and from the block-local restoration of the axial gauge, with constants C_{H}, c_{H} fixed before g_0; with $\varepsilon_1/\delta_0 \leqslant (1+\beta_0)A_0/A_1$ ([3, (2.8)] and $\delta_0 = (A_1/A_0)\varepsilon_0$), the tolerance $\delta_0/40$ is a condition "MR_1 large", met for small g_0 at fixed M since $R_1 \to \infty$ (threshold item 5 of Section B.8). The second inequality of the proof, $|Z_b - \mathbf{1}| < 2\delta_0$ on every bond of S_1^*, is the statement of [3] that the critical point U_1 of (1.12) lies in the domain of integration in (1.11) (p.248, by Theorem 1 of [7]; recorded in (B1) as $U_1 \in D^{(11)}(V, V_0)$): that domain carries the factor $\chi_0'(\Omega_1) = \prod_{\square' \subset \Omega_1} \chi_0'(\square')$ of (1.8) and (1.11), and $\chi_0'(\square'(b)) \neq 0$ at U_1 is the inequality $|U_1(b')U_{1,\square'(b)}(b')^{-1} - \mathbf{1}| < 2\delta_0$ for every $b' \in (\square'(b)^{\sim 2})^*$, b among them. The tolerance is the cutoff's $2\delta_0$. Bałaban states the comparison of [3, p. 250] between U_1 and $U_{1,\square'}$ for the cubes $\square' \subset R_1^c$ with $\square'^{\sim 4} \subseteq \Omega_1$ (p.250; recorded in (B3)(iii) for the cubes of $\Omega_1^{\sim -1} \setminus R_1$, for all of which this holds), and its derivation of (1.25) keeps χ_0' on the cubes of S_1 (p.250); so for Z_b formed with a cube $\square'(b)$ of the layer $\Omega_1 \setminus \Omega_1^{\sim -1}$ no tolerance smaller than the cutoff's is stated (for the layer bonds of $(R_1^{\sim 2} \cap S_1)^*$, the "very small" of [3, p. 252], discussed below, concerns the comparison with the cubes of R_1 through (1.21), not Z_b), and none is derived here: for a cube $\square'$ of the layer within four $L^2M_2R_1$-layers of Ω_1^c, $\square'^{\sim 4} \not\subseteq \Omega_1$, and U_1 is tied by (1.12) to the exterior field V_0, which $\zeta(\Omega_1^c)$ constrains near Ω_1 through small-field tests only: on $\Omega_1^{\sim}$ only small-field characteristic functions remain (p.248), the χ_0 and χ_{Ax} localised there being equal to one and dropped (p.248, (B3)(i)), and the factors $\chi_0'(\square')$ of (1.8) of the cubes $\square' \subseteq \Omega_1^c$ outside Q_1 that carry one test the bonds of $(\square'^{\sim 2})^*$, among them bonds of Ω_1 for the cubes adjacent to it, with the tolerance $2\delta_0$. The "very small" of [3, p. 252] concerns $U_1(\Omega_1 \cap \Lambda_1^c)U_{1,\square'}^{-1}$ on $\square'^{\sim 2}$, in Bałaban's discussion of the bonds of $(R_1^{\sim 2} \cap S_1)^*$, not Z_b, and serves Bałaban to place $g(A') = 1$ where the second factor of (1.23) vanishes; neither it nor (1.23) is used here, the cutoff being composed exactly in (3.43), and the proof above uses the inequality $|Z_b - \mathbf{1}| < 2\delta_0$ in the same way on the bonds of both forms. The proof depends on that inequality only through the numerical constants $4\delta_0$, $7\delta_0$ and $12\delta_0$ of (b) and (c) (and δ_G), and goes through with $2\delta_0$ replaced by any tolerance $\eta \leqslant C\delta_0$ after the corresponding change of these constants; so a regularity bound for $U_1U_{1,\square'}^{-1}$ of the kind of p.247 with a coupling-independent constant would serve in place of the cutoff membership of (B1). The condition $\delta_0 \leqslant \delta_G$ and the choice of g above are this paper's; the condition on MR_1 at the tolerance $\delta_0/40$ (with ε_1/δ_0 bounded by [3, (2.8)] and the constants of Lemma D.18 fixed before g_0) and the condition of [3] on M_2R_1 are listed under item 5 of Section B.8, and $\delta_0 \leqslant \delta_G$ among the further conditions listed after its enumeration.

The cutoff. The small-field test of [3, (1.18)] for a cube $\square'$ of side $L^2M_2R_1$ is the indicator of $\{\sup_{b \in (\square'^{\sim 2})^*} |A(b)| < r\}$, $r = \delta_0/g_0$, in the bond norm of [3] with the normalisation (4.13), and it is the product of one indicator per bond of $(\square'^{\sim 2})^*$. Let

$$\mathcal{G}_\xi = \bigcup_{\square' \subset R_1^c} (\square'^{\sim 2})^*, \qquad R_1^c = \Omega_1^{\sim -1} \setminus R_1 \,, \tag{3.41}$$

the union over the good cubes of (1.18). Since $\Lambda_1 \subseteq R_1^c$ and both are unions of such cubes, the first two factors of [3, (1.25)] are the product of the tests over all good cubes, which by idempotence of indicators is $\prod_{b \in \mathcal{G}_\xi} \mathbf{1}_{\{|A(b)|<r\}}$. The young cutoff is

$$\Theta_\xi(a) = \prod_{b \in I} \mathbf{1}_{\{|a(b)|<r\}} \,, \tag{3.42}$$

and the retained cutoff is defined as the remaining factors of (1.25), composed with the change of variables:

$$\chi_{\xi,\mathrm{e}}(A_{\mathrm{e}}) = \prod_{b\in\mathcal{G}_\xi\cap J} \mathbf{1}_{\{|(\Psi_\xi A_{\mathrm{e}})(b)|<r\}} \cdot \big[\chi^{(0)c}(R_1)\chi_0'(S_1)\big](\Psi_\xi A_{\mathrm{e}}) \,, \tag{3.43}$$

where $\chi^{(0)c}(R_1)$ is the product of the large-fluctuation tests of (1.18) over the cubes of R_1 and $\chi_0'(S_1)$ the product of the tests of (1.8) over the cubes of S_1, both as functions of the variable A of (1.15) on Ω_1^* (through the free links $e^{ig_0A(b)}U_1(b)$ in the case of (1.8)) and of the branch data (V_0 on Ω_1^c, U_1 and the local minima $U_{1,\square'}$). The rewriting of these factors in the new variables in [3, p. 252, (1.23)] is not used; the exact compositions are carried as they stand. Lemma 3.11(ii) shows that (3.43) is indeed the complement of Θ_ξ in the exactly transformed cutoff.

The conditional integrand. With $\mathscr{E}_\xi$ the exponent of the second exponential of [3, (1.15)], written out in (3.34), the conditional integrand of Section 3 is

$$\mathcal{K}_\xi(a) = \begin{cases} \exp\Big(\mathscr{E}_\xi\big(\Psi_\xi(a+A_{\mathrm{e}})\big) + \log|\det D\Psi_\xi(a+A_{\mathrm{e}})|\Big), & \Theta_\xi(a)=1,\\ 0, & \Theta_\xi(a)=0, \end{cases} \tag{3.44}$$

$$\mathcal{J}_\xi^{\mathrm{a}}[f] = \int f(a)\mathcal{K}_\xi(a)\,\mathrm{d}a \,,$$

the exponent being defined on the box by (B4) and Lemma 3.6: by Lemma 3.11(ii) below, the integrand of [3, (1.15)] in the variables of (1.22), with the cutoff (1.25) in its exact composed form, is $\kappa_\xi\,\chi_{\xi,\mathrm{e}}(A_{\mathrm{e}})\,\mathcal{K}_\xi(a)$, as used in (3.7) ($\mathscr{E}_\xi$ and κ_ξ are the exponent and prefactor of [3] as printed, with σ, σ_0 and $z^{(0)}$ for the fixed inner product as explained in the conventions; only the a-dependence of $\mathcal{K}_\xi$ is used below). The change-of-variables term, which Bałaban expands as $V^{(0)}(S_1)$ ([3, (1.24) and p. 252]), is kept exact inside $\mathcal{K}_\xi$; Lemma 3.9 shows that it is a constant on the branch.

The exterior interpolation. Write $H = H_{\xi,II}$. For $t\in[0,1]$ put

$$\begin{aligned} P_{\xi,t}(a) &= \sum_{\mathfrak{p}\ \text{near}} \mathfrak{p}(a+tA_{\mathrm{e}}) \,, \qquad j_\xi(a) = \langle a, H_{\xi,IJ}A_{\mathrm{e}}\rangle \,,\\ K_{\xi;t}(a) &= \mathbf{1}_{\mathcal{D}_I}(a)\exp\left(-\frac{1}{2}\langle a,Ha\rangle - tj_\xi(a) + P_{\xi,t}(a)\right)\\ &\qquad (\mathcal{D}_I=\{a\,|\,|a(b)|<r\ \forall b\in I\}\ \text{the open box},\ \mathbf{1}_{\mathcal{D}_I}=\Theta_\xi),\\ I_{\xi;t} &= \int K_{\xi;t}(a)\,\mathrm{d}a, \qquad \mathrm{d}\nu_{\xi;t} = \frac{K_{\xi;t}}{I_{\xi;t}}\,\mathrm{d}a \,, \end{aligned} \tag{3.45}$$

the sum running over the near primitives of Section 3.3 (all three kinds). By Lemma 3.6 and Lemma 3.9 (i), each near primitive is defined and analytic in (a,t) on the enlarged box at the inputs $a+tA_{\mathrm{e}}$, $t\in[0,1]$; $K_{\xi;t}$ (zero off the open box) is therefore bounded and Borel on $\mathfrak{g}^I$, positive on the open box, and $0<I_{\xi;t}<\infty$. On the open box $P_{\xi,0}(a)=\sum_Y \mathbf{V}_\xi(Y,a)$ is the centred potential (3.37), and $K_{\xi;0}$ is the centred kernel of Section 3.5. This is the interpolation of [3, (3.27)–(3.29)] in the exterior field, restricted to the terms that depend on a. The Wilson function and the centred coefficient are

$$f_\xi(a) = W_\chi(C;U_\xi(a)), \qquad F_\xi(t) = \int f_\xi\,\mathrm{d}\nu_{\xi;t} \,, \tag{3.46}$$

with $U_\xi(a)$ the reconstructed fine field (3.26) at the actual retained data, including the change of variables; it is not interpolated. By Lemmas 3.6 and 3.5, f_ξ is defined and analytic on the

enlarged box: each contour link $U_\xi(a)(r)$ depends on a only through $a|_{\mathcal{F}_r\cap I}$, through a row within $2R_{\mathrm{c}}$ of I when $\mathcal{F}_r$ meets I, and is constant in a otherwise. Off the enlarged box f_ξ is extended by zero, which affects no integral against $\nu_{\xi;t}$.

Lemma 3.9 (The Collar)**.** *On every branch:*

1. $\Psi_\xi(A_{\mathrm{e}})(b) = A_{\mathrm{e}}(b)$ *for every* $b \in J$ *with* $d(b, I) \leqslant 2R_{\mathrm{c}}$*; every near primitive and every entry of* $H_{\xi,IJ}$ *involves* A_{e} *only on such bonds, so* $\mathfrak{p}(a + t\Psi_\xi(A_{\mathrm{e}})) = \mathfrak{p}(a + tA_{\mathrm{e}})$ *for near* $\mathfrak{p}$ *and* $\langle a, H_{\xi,IJ}\Psi_\xi(A_{\mathrm{e}})\rangle = j_\xi(a)$*. For every row* c *within* $2R_{\mathrm{c}}$ *of* I *the source phase on the bonds of* $\mathcal{E}_c$ *is, componentwise,*

$$\begin{aligned} q_U\big(\Psi_\xi(a + A_{\mathrm{e}})\big)(b) &= q_U(a + A_{\mathrm{e}})(b) \\ &= \big[g_0\mathsf{C}_U(a + A_{\mathrm{e}}) - h_U\widetilde{D}_U\big(g_0\mathsf{C}_U(a + A_{\mathrm{e}})\big)\big](b) \qquad (b \in \mathcal{E}_c)\ . \end{aligned} \tag{3.47}$$

2. *With the branch constant*

$$c_\xi = \mathscr{E}_\xi(0) - \frac{1}{2}\langle\Psi_\xi(A_{\mathrm{e}}), H_\xi\Psi_\xi(A_{\mathrm{e}})\rangle + \sum_{\mathfrak{p}\ \text{far}} \mathfrak{p}\big(\Psi_\xi(A_{\mathrm{e}})\big) + \log|\det D\Psi_\xi(A_{\mathrm{e}})|\ , \tag{3.48}$$

where $\mathscr{E}_\xi(0) = v(0)$ *and the far primitives are evaluated at the retained field alone,*

$$\mathcal{K}_\xi(a) = e^{c_\xi}K_{\xi;1}(a) \quad (a \in \mathfrak{g}^I), \qquad \mathcal{J}^{\mathrm{a}}_\xi[f] = e^{c_\xi}I_{\xi;1}\int f\,\mathrm{d}\nu_{\xi;1}, \qquad \frac{\mathcal{J}^{\mathrm{a}}_\xi[f_\xi]}{\mathcal{J}^{\mathrm{a}}_\xi[1]} = F_\xi(1)\ . \tag{3.49}$$

Proof. (i) By Lemma 3.11(i), a bond within $2R_{\mathrm{c}}$ of I is not in S_1^*, where Ψ_ξ is the identity. A near primitive has all its inputs within $4L+3 \leqslant 2R_{\mathrm{c}}$ of I (Section 3.3); an entry $(H_{\xi,IJ})_{bb'}$ with $b \in I$ vanishes unless $d(b, b') \leqslant \rho_L = 4L < 2R_{\mathrm{c}}$ (Lemma 3.3), since $R_{\mathrm{c}} \geqslant 2R_{\mathrm{loc}} + 1 = 4L + 5$. The phase on a row c within $2R_{\mathrm{c}}$ of I depends on the free vector on $\mathcal{E}_c \cap E_*$ (Lemma 3.5), and every bond of $\mathcal{E}_c$ is within $2R_{\mathrm{c}}$ of I by the definition of the row distance (Section 3.3), so Ψ_ξ is the identity on these bonds; on $\mathcal{E}_c$ only the row c contributes to $h_U\widetilde{D}_U$ (Lemma 3.5), which gives (3.47) as an identity between the $\mathcal{E}_c$-components.

(ii) Put $A^\Psi_{\mathrm{e}} = \Psi_\xi(A_{\mathrm{e}}) \in \mathfrak{g}^J$, so that $\Psi_\xi(a + A_{\mathrm{e}}) = a + A^\Psi_{\mathrm{e}}$ by (3.27). By Lemma 3.7,

$$\mathscr{E}_\xi(a + A^\Psi_{\mathrm{e}}) = \mathscr{E}_\xi(0) - \frac{1}{2}\langle a, Ha\rangle - \langle a, H_{\xi,IJ}A^\Psi_{\mathrm{e}}\rangle - \frac{1}{2}\langle A^\Psi_{\mathrm{e}}, H_\xi A^\Psi_{\mathrm{e}}\rangle + \sum_{\mathfrak{p}\ \text{near}} \mathfrak{p}(a + A^\Psi_{\mathrm{e}}) + \sum_{\mathfrak{p}\ \text{far}} \mathfrak{p}(A^\Psi_{\mathrm{e}})\ ,$$

where the far primitives do not depend on a because their free-input sets miss I. By (i) the third term is $-j_\xi(a)$ and the near sum is $P_{\xi,1}(a)$. The Jacobian factor depends on $(a + A_{\mathrm{e}})|_{S_1^*} = A_{\mathrm{e}}|_{S_1^*}$ only. The constant c_ξ is finite: $\det D\Psi_\xi(A_{\mathrm{e}}) \neq 0$ by (B4)(ii) (the differential of (1.22) is invertible on $\mathcal{D}_\xi$, and $D\Psi_\xi$ at $a + A_{\mathrm{e}}$ depends on $A_{\mathrm{e}}|_{S_1^*}$ only), and the far primitives are finite: their free-input sets miss I, so their values at A^Ψ_{e} are their values at $a_0 + A^\Psi_{\mathrm{e}} = \Psi_\xi(a_0 + A_{\mathrm{e}}) \in \mathcal{A}_\xi$ for any a_0 with $(A_{\mathrm{e}}, a_0) \in \mathcal{D}_\xi$ (a branch has one), where every row solution is defined and the primitives are finite by (B4). Collecting the a-independent terms gives $\mathcal{K}_\xi = e^{c_\xi}K_{\xi;1}$, and the two remaining identities follow by integration, dividing by $\mathcal{J}^{\mathrm{a}}_\xi[1] = e^{c_\xi}I_{\xi;1} > 0$. *q.e.d.*

The constant c_ξ depends on the branch data (through A_{e}, U_1, V_0 and Ω_1) and is not a universal constant: it enters the outer weight $\mathcal{J}^{\mathrm{a}}_\xi[1] = e^{c_\xi}I_{\xi;1}$ of the branch in (3.7). It is never estimated separately: Section 9 compares $\mathcal{O}[\mathcal{J}^{\mathrm{a}}_\xi[1]\,F_\xi(1)]$ with $\mathcal{O}[\mathcal{J}^{\mathrm{a}}_\xi[1]]$, the same nonnegative weight on both sides.

Lemma 3.10 (Boundary Response)**.** *For bounded Borel f on $\mathfrak{g}^I$, with $\mathrm{Cov}_\nu(f,l) = \int fl\,\mathrm{d}\nu - \int f\,\mathrm{d}\nu \int l\,\mathrm{d}\nu$,*

$$\int f\,\mathrm{d}\nu_{\xi;1} = \int f\,\mathrm{d}\nu_{\xi;0} + \mathcal{B}_\xi[f], \qquad \mathcal{B}_\xi[f] = \int_0^1 \mathrm{Cov}_{\nu_{\xi;t}}(f, -j_\xi + \partial_t P_{\xi,t})\,\mathrm{d}t\ , \tag{3.50}$$

where $\partial_t P_{\xi,t}(a) = \sum_{\mathfrak{p}\ \mathrm{near}} D\mathfrak{p}(a+tA_\mathrm{e})[A_\mathrm{e}]$.

Proof. For a in the box the exponent $E_t(a) = -\frac{1}{2}\langle a, Ha\rangle - tj_\xi(a) + P_{\xi,t}(a)$ is jointly analytic in (a,t) on a neighbourhood of $\overline{\mathcal{D}_I} \times [0,1]$, since each near primitive is analytic in the free vector on its row polydiscs (Lemma 3.6) and the input $a + tA_\mathrm{e}$ is affine in (a,t) with values in those polydiscs; its t-derivative $-j_\xi(a) + \partial_t P_{\xi,t}(a)$, with $\partial_t P_{\xi,t}(a) = \sum_\mathfrak{p} D\mathfrak{p}(a+tA_\mathrm{e})[A_\mathrm{e}]$ the derivative in the direction A_e, is continuous and hence bounded on $\overline{\mathcal{D}_I} \times [0,1]$. Hence $t \mapsto \int f K_{\xi;t}\,\mathrm{d}a$ and $t \mapsto I_{\xi;t}$ are C^1 with derivatives $\int f\,\partial_t E_t\,K_{\xi;t}\,\mathrm{d}a$ and $\int \partial_t E_t\,K_{\xi;t}\,\mathrm{d}a$ (differentiation under the integral over the bounded box, dominated by a constant times $K_{\xi;t} \leqslant e^{\sup|E_t|}$), so $F(t) = \int f\,\mathrm{d}\nu_{\xi;t}$ has $F'(t) = \mathrm{Cov}_{\nu_{\xi;t}}(f, \partial_t E_t)$. Integrate over $[0,1]$. *q.e.d.*

By Lemmas 3.9 and 3.10, on every branch

$$\frac{\mathcal{J}^{\mathrm{a}}_\xi[f_\xi]}{\mathcal{J}^{\mathrm{a}}_\xi[1]} = F_\xi(1) = F_\xi(0) + \mathcal{B}_\xi[f_\xi]\ , \tag{3.51}$$

the first term being the centred expectation of Section 3.7 and the second the exterior response, estimated in Appendix D.

Lemma 3.11 (Collar Coverage)**.** *Let $R_{\mathrm{loc}} = 2L+2$, which exceeds the ℓ^∞ diameter $2L-1$ of every two-block set $\mathcal{E}_c$, hence of the sets $\mathcal{E}_y$, $\mathcal{F}_r$, $\mathcal{A}_r$ and of the free-input and background sets of the block, row, Haar and Jacobian primitives (those of a plaquette primitive have diameter at most $4L < 2R_{\mathrm{loc}}$, proof of Lemma 3.3), and let $R_\mathrm{c} \geqslant 2R_{\mathrm{loc}} + 1$ be the collar range fixed in Appendix B after* (B.7)*; it depends on L and the group only. Under* (3.1)*, which include $M \geqslant 8LM_2$ and $LMR_1 > 4R_\mathrm{c} + 4L^2M_2R_1 + 8$, on every branch:*

1. *Every site within ℓ^∞ distance $LMR_1 - 1$ of Λ_1 lies in $\Omega_1^{\sim -1} \setminus R_1' \subseteq R_1^c$; $I \subseteq \mathcal{G}_\xi \subseteq \Omega_1^*$; and every free bond b of Ω_1 with $d(b,I) \leqslant 3R_\mathrm{c}$ lies in $\mathcal{G}_\xi \setminus S_1^*$.*
2. *The cutoff (1.25) composed with the change of variables, as a function of $A' = a + A_\mathrm{e}$, equals $\Theta_\xi(a)\chi_{\xi,\mathrm{e}}(A_\mathrm{e})$ with $\chi_{\xi,\mathrm{e}}$ of* (3.43)*, and $\chi_{\xi,\mathrm{e}}$ depends on A_e and the branch data only.*
3. *$|A_\mathrm{e}(b)| < r$ for every $b \in J$ with $d(b,I) \leqslant 3R_\mathrm{c}$; that is,*

$$r_{\mathrm{e},\xi} = \max\big(\{|A_\mathrm{e}(b)|\,|\,b \in J,\ d(b,I) \leqslant 3R_\mathrm{c}\} \cup \{0\}\big) < r\ . \tag{3.52}$$

4. *For an admissible pair and a branch value of A_e, $\{a \in \mathfrak{g}^I : (A_\mathrm{e}, a) \in \mathcal{D}_\xi\} = \{a : \Theta_\xi(a) = 1\}$, the open box of* (3.42)*.*

Proof. (i) Ω_1 and R_1' are unions of LMR_1-cubes ([3, (1.10), p. 251]) and $\Lambda_1 \subseteq \Omega_1^{\sim -2} \cap (R_1'^{\sim})^c$, so every site within ℓ^∞ distance $LMR_1 - 1$ of Λ_1 lies in $\Omega_1^{\sim -1} \setminus R_1' \subseteq R_1^c$. Let b be a free bond of Ω_1 with $d(b,I) \leqslant 3R_\mathrm{c}$: its initial vertex is within $3R_\mathrm{c}+1$ of a site of Λ_1, so the cube $\square'$ of side $L^2M_2R_1$ containing it lies within $3R_\mathrm{c} + 1 + L^2M_2R_1 < LMR_1 - 1$ of Λ_1 ((3.1)), hence $\square' \subseteq R_1^c$ is a good cube; b has an endpoint in $\square' \subseteq \square'^{\sim 2}$ and is not designated, so $b \in (\square'^{\sim 2})^* \subseteq \mathcal{G}_\xi$. By (3.39), $d(I, S_1^*) \geqslant LMR_1 - 2 > 3R_\mathrm{c}$, so $b \notin S_1^*$. Taking $d(b,I) = 0$

gives $I \subseteq \mathcal{G}_\xi$ (a bond of Λ_1 is a bond of Ω_1, and a designated bond with an endpoint in Λ_1 belongs to a row of Λ_1 because Λ_1 is a union of blocks, so $I \subseteq \Omega_1^*$). For $\mathcal{G}_\xi \subseteq \Omega_1^*$: a good cube $\square' \subseteq \Omega_1^{\sim -1}$ has $\square'^{\sim 2} \subseteq \Omega_1$ since $2L^2 M_2 R_1 < LMR_1$ ($M \geqslant 8LM_2$); a bond of $(\square'^{\sim 2})^*$ has an endpoint in Ω_1, and a designated bond with an endpoint in $\square'^{\sim 2}$, a union of blocks, belongs to a row with an endpoint in $(\square'^{\sim 2})^{(1)}$ and is excluded by the definition of * in (1.8).

(ii) The first two factors of (1.25) are $\prod_{b \in \mathcal{G}_\xi} \mathbf{1}_{\{|A(b)|<r\}}$ at $A = \Psi_\xi(a + A_\mathrm{e})$. For $b \in I \subseteq \mathcal{G}_\xi$, $A(b) = a(b)$ by (3.27), giving $\Theta_\xi(a)$; for $b \in \mathcal{G}_\xi \setminus I = \mathcal{G}_\xi \cap J$ (by (i)), $A(b) = (\Psi_\xi A_\mathrm{e})(b)$, giving the first factor of (3.43). The factor $\chi^{(0)c}(R_1)$ involves A on $(\square'^{\sim 2})^*$ for cubes $\square' \subseteq R_1$, and $\chi'_0(S_1)$ involves the free links $e^{ig_0 A(b)} U_1(b)$ on bonds $b \in (\square'^{\sim 2})^* \cap \Omega_1^*$ for cubes $\square' \subseteq S_1$, together with V_0 on the bonds of $(\square'^{\sim 2})^*$ with both endpoints off Ω_1, U_1 and $U_{1,\square'}$. All these bonds have both endpoints within $2L^2 M_2 R_1 + 1$ of $R_1 \cup S_1$, while every site of Λ_1 is at distance at least $LMR_1 > 2L^2 M_2 R_1 + 1$ from $R_1 \cup S_1$; so none of them has an endpoint in Λ_1, none lies in I, and on them $A = \Psi_\xi(A_\mathrm{e})$. Hence the last two factors of (1.25), composed with Ψ_ξ, are functions of $\Psi_\xi(A_\mathrm{e})$ and the branch data.

(iii) For $b \in J$ with $d(b, I) \leqslant 3R_\mathrm{c}$, (i) gives $b \in \mathcal{G}_\xi \cap J$ and $b \notin S_1^*$, so $\mathbf{1}_{\{|(\Psi_\xi A_\mathrm{e})(b)|<r\}} = \mathbf{1}_{\{|A_\mathrm{e}(b)|<r\}}$ is a factor of $\chi_{\xi,\mathrm{e}}(A_\mathrm{e})$, which is nonzero on a branch (the maximum over the empty set is read as 0).

(iv) Let a_0 be such that $(A_\mathrm{e}, a_0) \in \mathcal{D}_\xi$ (a branch), so $B_0 = \Psi_\xi(a_0 + A_\mathrm{e}) \in \mathcal{A}_\xi$ and, by (ii), $\Theta_\xi(a_0)\chi_{\xi,\mathrm{e}}(A_\mathrm{e}) \neq 0$. Let a be any point of the open box and $B = \Psi_\xi(a + A_\mathrm{e}) = a + \Psi_\xi(A_\mathrm{e})$ ((3.27); $\Psi_\xi(A_\mathrm{e})$ is defined, since B_0 is). It is to be shown that $B \in \mathcal{A}_\xi$, by checking the three conditions of (3.2). The cutoff (1.25) at B is $\Theta_\xi(a)\chi_{\xi,\mathrm{e}}(A_\mathrm{e}) = \chi_{\xi,\mathrm{e}}(A_\mathrm{e}) \neq 0$, by (ii). Every row equation has a solution at $g_0 \mathsf{C}_U B$: on the near rows by Lemma 3.6, since $B = a + A_\mathrm{e}$ on their two-block sets by (i) and lies in the row polydiscs by (iii); on the far rows the argument $(g_0 \mathsf{C}_U B)|_{\mathcal{E}_c}$ equals $(g_0 \mathsf{C}_U B_0)|_{\mathcal{E}_c}$ (Lemma 3.5: it involves B only through $B|_{\mathcal{E}_c \cap E_*}$, and $\mathcal{E}_c$ misses I for a far row, every bond of which is at distance more than $2R_\mathrm{c} + 2L - (2L-1) > 0$ from I), where the solution exists because $B_0 \in \mathcal{A}_\xi$. So $U(B)$ is defined, and on the far rows and the bonds of Ω_1^c its links are those of $U(B_0)$. It remains to show $U(B) \in D^{(11)}(V, V_0)$. The restriction to Ω_1^c is V_0. The block averages are defined on every row of Ω_1 and satisfy the constraint $\bar{U} = V$ there: $U(B) = e^{iq_U(B)} U_1$ with $q_U(B) = A - h_U \widetilde{D}_U(A)$, $A = g_0 \mathsf{C}_U B$, and on every row c the row equation gives $\mathcal{Q}_{U,c}(q_U(B)) = LQ_U q_U(B) + \mathcal{N}_{U,c}(q_U(B)) = LQ_U(A - h_U \widetilde{D}_U(A)) + \widetilde{D}_{U,c}(A) = -\widetilde{D}_{U,c}(A) + \widetilde{D}_{U,c}(A) = 0$, using $Q_U \mathsf{C}_U = 0$ and $LQ_U h_U = \mathbf{1}$; so $\mathsf{M}(U(B))(c) = \mathsf{M}(U_1)(c) = V(c)$ by (3.9) and (B1). The constraint $\bar{V}_0 = V$ on the rows of Ω_1^c involves V_0 and V only and holds because the pair is admissible. The tests of (1.11): $\chi_1(\Omega_1)$ is a function of V, $\zeta(\Omega_1^c)$ does not depend on a ((3.6)), and both are nonzero at $U(B_0) \in D^{(11)}(V, V_0)$. The factor $\chi'_0(\Omega_1)$ is the product of $\chi'_0(\square')$ over the $L^2 M_2 R_1$-cubes $\square' \subseteq \Omega_1$ ([3, (1.8)]); the cubes of $S_1 = (\Omega_1 \setminus \Omega_1^{\sim -1}) \cup R_1$ ([3, (1.20)]) give the factor $\chi'_0(S_1)$ of (1.25), nonzero at B by the first condition, and every other cube $\square' \subseteq \Omega_1$ lies in $\Omega_1^{\sim -1} \setminus R_1$, so it is a good cube with $\square'^{\sim 4} \subseteq \Omega_1$ (as $4L^2 M_2 R_1 < LMR_1$), and its factor $\chi'_0(\square')$ is the second factor of (1.19), equal to one at B by (B3)(ii), the retained factors being nonzero there. Hence $U(B) \in D^{(11)}(V, V_0)$, so $B \in \mathcal{A}_\xi$ and $(A_\mathrm{e}, a) \in \mathcal{D}_\xi$. Conversely $(A_\mathrm{e}, a) \in \mathcal{D}_\xi$ gives $\Psi_\xi(a + A_\mathrm{e}) \in \mathcal{A}_\xi$, so the cutoff (1.25) is nonzero there, and $\Theta_\xi(a) = 1$ by (ii). *q.e.d.*

The exterior response has Bałaban's integration-by-parts form ([3, (3.28)]). Put

$$\ell_\xi = H_{\xi,IJ} A_\mathrm{e}, \qquad m_\xi = H_{\xi,II}^{-1} \ell_\xi, \qquad D_{m_\xi} f(a) = \frac{\mathrm{d}}{\mathrm{d}s} f(a + s m_\xi)\Big|_{s=0}, \tag{3.53}$$

let $\mathcal{D}_I$ be the open box of (3.45), with outward unit normal $n(a)$ and surface measure $\mathrm{d}S$ for the Euclidean structure of $\langle \bullet, \bullet \rangle$ on $\mathfrak{g}^I$ (for which $\mathrm{d}a$ is Lebesgue measure), defined almost

everywhere on its Lipschitz boundary (a product of balls of the norm $|\bullet|$), and

$$\mathsf{Q}_{\xi,t}(a) = -\frac{1}{2}\langle a, H_{\xi,II}a\rangle - t\langle \ell_\xi, a\rangle + P_{\xi,t}(a), \quad \mathcal{T}_{\xi,t}[f] = \frac{1}{I_{\xi;t}}\int_{\partial\mathcal{D}_I} f(a)e^{\mathsf{Q}_{\xi,t}(a)}\langle m_\xi, n(a)\rangle \,\mathrm{d}S(a)\ , \tag{3.54}$$

so that $K_{\xi;t} = \mathbf{1}_{\mathcal{D}_I}e^{\mathsf{Q}_{\xi,t}}$ and $j_\xi(a) = \langle \ell_\xi, a\rangle$.

Lemma 3.12 (Integration by Parts with the Hard Cutoff)**.** *Assume Lemma 3.4, so that $H_{\xi,II}$ is invertible and m_ξ is defined. If f is C^1 on a neighbourhood of $\overline{\mathcal{D}_I}$, then for every $t \in [0,1]$, with expectations under $\nu_{\xi;t}$,*

$$\mathrm{Cov}(f, -j_\xi + \partial_t P_{\xi,t}) = -\mathbb{E}[D_{m_\xi}f] + \mathrm{Cov}(f, \partial_t P_{\xi,t} - D_{m_\xi}P_{\xi,t}) + \mathcal{T}_{\xi,t}[f] - \mathbb{E}[f]\mathcal{T}_{\xi,t}[1]\ . \tag{3.55}$$

Proof. $e^{\mathsf{Q}_{\xi,t}}$ is C^1 on the enlarged box, a neighbourhood of $\overline{\mathcal{D}_I}$, by Lemma 3.6. The divergence theorem on the bounded Lipschitz domain $\mathcal{D}_I$ gives $\int_{\mathcal{D}_I} D_{m_\xi}(fe^{\mathsf{Q}_{\xi,t}})\,\mathrm{d}a = I_{\xi;t}\mathcal{T}_{\xi,t}[f]$, and $D_{m_\xi}\mathsf{Q}_{\xi,t} = -j_\xi - t\langle m_\xi, \ell_\xi\rangle + D_{m_\xi}P_{\xi,t}$ because $\langle m_\xi, H_{\xi,II}a\rangle = \langle \ell_\xi, a\rangle = j_\xi(a)$. Dividing by $I_{\xi;t}$, $\mathbb{E}[D_{m_\xi}f] + \mathbb{E}[f(-j_\xi - t\langle m_\xi, \ell_\xi\rangle + D_{m_\xi}P_{\xi,t})] = \mathcal{T}_{\xi,t}[f]$. Subtract the same identity for $f = 1$ multiplied by $\mathbb{E}[f]$; the constant $-t\langle m_\xi, \ell_\xi\rangle$ cancels, and the two covariances combine as stated. *q.e.d.*

Since f_ξ is analytic on the enlarged box (after (3.46)), Lemma 3.12 applies to f_ξ. The estimate of the right-hand side is Lemma D.6 in Appendix D.

3.5 A Fixed Walk Expansion of the Centred Kernel

Write $H = H_{\xi,II}$ and $\mathsf{G} = H^{-1}$. Notation of Sections 3.5–3.8: $E_0 = E(Z_0)$ denotes the seed bonds (the designated bonds E_0 of (4.3) do not occur here); $N_i(Y)$ is a cube count (the measure N_1 of (1.5) does not occur); σ_Δ are weakening variables (the Haar density σ does not occur); B is the conditioned variable on $\mathfrak{g}^{E_0}$ (the free vector B of Sections 3.2–3.4 does not occur); D without argument is a component; J_D, J_Z are sets of young bonds (the retained carrier J does not occur); δ_M is a decay constant (the Mayer constant δ of Appendix B does not occur); $\gamma = \gamma_L/2$ is a spectral bound (arcs γ of the contour do not occur); a connected final domain is any nonempty wall-connected union of LM-cubes; and a_j, a_*, a_W are coefficients and weights, distinct from the young variable a. By Lemma 3.4, on every branch

$$\gamma\mathbf{1} \leqslant H \leqslant h_+\mathbf{1}, \qquad \gamma = \frac{\gamma_L}{2}, \quad h_+ = m_L\ , \tag{3.56}$$

and by Lemma 3.3, whose entries $\mathsf{H}^{\Omega_1}_{U_1}(b,c)$ at $b,c \in I$ are those of H, $H(b,c) = 0$ when $d_{\infty,T}(b_-, c_-) > R$, where $R \geqslant \max(\rho_L, 2)$ is fixed so that $H(b,c)$ is determined by the background, by the region Ω_1 and by membership in I within distance R of b_-. Norms on $\mathfrak{g}^I$ are those of the invariant real inner product, with Hermitian extension on the complexification. The constants γ, h_+, R depend on L and the group only (with the invariant inner product fixed in the conventions), and every choice below is made before the branch fields and the coupling, in the order γ, h_+, R; then κ_1; then n_*, R_*, C_{w}; then M_1, $\rho_{\mathrm{w}} = R_* + R + 1 + 2M_1$ and M; then g_0. None of $\gamma, h_+, R, \kappa_1, C_{\mathrm{d}}$ depends on M_1, M or the coupling (C_{d} of Lemma 3.16 is a function of γ, h_+, d_G).

Fix an aligned auxiliary cubical partition of side M_1 in fine units, with cubes $M_1 z + [-M_1/2, M_1/2]^4$; the M-cube partition π_0 and the LM-cube partition π_1 are nested over it, M/M_1 and L are odd integers, every fine period is a multiple of LMR_1 and at least $8M$, and M_1 and M are odd ((3.1): M_1, M_2, M are powers of the odd number L),

so that cube faces lie at half-integer coordinates and no lattice site lies on a face. Cubes and neighbourhoods are periodic images; all radii used below are smaller than M, so the periods $\geqslant 8M$ make every neighbourhood a lifted copy of its $\mathbb{Z}^4$ model. Set $s_H = 2h_+$, $\mathsf{A}_H = \mathbf{1} - H/s_H$, $q_H = 1 - \gamma/s_H < 1$ and $V_n = 4(2nR+3)^4$. A block of A_H^n vanishes beyond distance nR, has operator norm at most q_H^n and depends only on coefficients within distance $(n+1)R$ of its first bond, so its block-norm row and column sums are at most $V_n q_H^n$. Let $\kappa_1 \geqslant 1$ be the constant fixed in Appendix B, Section B.5 ((B.27); it depends on L and the group only), the spatial Cauchy radius being e^{κ_1}; put $a_* = e^{81\kappa_1}$ and choose n_* with

$$\beta = V_{n_*} q_H^{n_*}, \qquad a_* \beta \leqslant \frac{1}{4} \, . \tag{3.57}$$

Put $R_* = n_* R$, $\mathsf{P}_H = \mathsf{A}_H^{n_*}$, $V_* = V_{n_*}$ and $a_j = 4^{-j}\binom{2j}{j} \in (0,1]$. The binomial series gives

$$\mathsf{G}^{1/2} = s_H^{-1/2} \sum_{r=0}^{n_*-1} \sum_{m=0}^{\infty} a_{n_* m + r} \mathsf{A}_H^r \mathsf{P}_H^m \, . \tag{3.58}$$

For a free bond b let D_b be the union of the M_1-cubes meeting the closed ball of radius $R_* + R + 1$ about b_-, with one further M_1-layer; it is connected, lies in the ball of radius $\rho_{\mathrm{w}} = R_* + R + 1 + 2M_1$ about b_-, and contains the bonds and coefficient stencils of every block starting at b in (3.58) with an interior margin. Choose

$$M \geqslant 4(\rho_{\mathrm{w}} + R + R_* + 2) \, . \tag{3.59}$$

Then $2\rho_{\mathrm{w}} < M \leqslant LM$, so D_b meets at most $2^4 = 16 \leqslant 81$ cubes of π_1. A walk is a tuple $\omega = (r; b_0, \dots, b_{m+1})$ with nonzero block product

$$(\mathsf{A}_H^r)_{b_0 b_1} \prod_{j=1}^{m} (\mathsf{P}_H)_{b_j b_{j+1}} \, , \tag{3.60}$$

and localisation domain $D(\omega) = \bigcup_{j \leqslant m} D_{b_j}$, which is connected and contains both endpoint bonds in its interior.

For an M-cube Q let $Q^{\sim}$ be the union of the 3^4 cubes of π_0 meeting Q, so that $\operatorname{int} Q^{\sim}$ is the open ℓ^∞ neighbourhood of radius M of Q. For a union Z_0 of M-cubes (the seed, selected in Section 3.6) let $\widetilde{Z}_0 = \bigcup_{Q \subseteq Z_0} Q^{\sim}$ be its one-M-layer enlargement and Z_0' the union of the LM-cubes meeting $\operatorname{int} Q^{\sim}$ for some cube Q of Z_0. Both are unions over the cubes of Z_0, so both commute with unions of seeds, and $\operatorname{int} Q^{\sim} \subseteq \operatorname{int} Z_0'$ for every cube Q of Z_0, since every LM-cube incident to a point of the open set $\operatorname{int} Q^{\sim}$ meets it. Assign $\sigma_\Delta = 1$ to the cubes $\Delta \in \pi_1$ inside Z_0' and complex variables σ_Δ to the others, and weight each walk by

$$w_\omega(\sigma) = \prod_{\Delta \in \pi_1, \ \Delta \not\subseteq Z_0', \ \Delta \cap D(\omega) \neq \varnothing} \sigma_\Delta \, . \tag{3.61}$$

The weighted series is

$$S_{Z_0}(\sigma)_{b_0 b_{m+1}} = s_H^{-1/2} \sum_{r=0}^{n_*-1} \sum_{m=0}^{\infty} a_{n_* m + r} \sum_{b_1, \dots, b_m} w_\omega(\sigma) \, (\mathsf{A}_H^r)_{b_0 b_1} \prod_{j=1}^{m} (\mathsf{P}_H)_{b_j b_{j+1}}, \qquad \omega = (r; b_0, \dots, b_{m+1}) \, , \tag{3.62}$$

the inner sum running over the intermediate bonds; $S_{Z_0}(1) = \mathsf{G}^{1/2}$ by (3.58).

Lemma 3.13 (The Walk Series)**.** *On $|\sigma_\Delta| \leqslant e^{\kappa_1}$, $|w_\omega(\sigma)| \leqslant a_*^{m+1}$, and the series $S_{Z_0}(\sigma)$ converges absolutely in both Schur norms, uniformly on the polydisc, with operator norm and absolute block row and column sums at most*

$$C_{\mathrm{w}} = s_H^{-1/2} \frac{n_* V_* a_*}{1 - a_*\beta} \,. \tag{3.63}$$

It is holomorphic on the polydisc of radii $e^{\kappa_1 + (\log 2)/81}$. The bound is independent of Z_0, the volume and the number of its components. Under a site gauge transformation u of the background, $S_{Z_0}(\sigma)$ formed from $\iota_I^* \mathsf{H}_{U_1^u}^{\Omega_1} \iota_I$ in place of H equals $R_{u,*} S_{Z_0}(\sigma) R_{u,*}^{-1}$, with $R_{u,*}$ restricted to I and the same domains and weights.

Proof. $D(\omega)$ meets at most $81(m+1)$ cubes of π_1, which gives the weight bound. The unweakened grouped products have absolute block row and column sums at most $V_*\beta^m$ after summing the intermediate bonds, so the weighted series is dominated by $s_H^{-1/2} n_* V_* \sum_m (a_*\beta)^m a_*$. Increasing each radius by the factor $e^{(\log 2)/81}$ multiplies a_* by 2 and keeps $a_*\beta \leqslant 1/2$. The gauge law is (3.20) restricted to I (so A_H is conjugated by $R_{u,*}$) applied to each block; the weights and domains do not involve the background. *q.e.d.*

Put $E_0 = E(Z_0) = \{b \in I \,|\, b \cap Z_0 \neq \varnothing\}$, let ι_{E_0} be extension by zero and $\Pi_{I \setminus E_0}$ the complementary orthogonal projection, and set

$$C_{E_0} = (\iota_{E_0}^* H \iota_{E_0})^{-1}, \qquad \Gamma_{E_0}(\sigma) = \iota_{E_0}^* H \Pi_{I \setminus E_0} S_{Z_0}(\sigma) \,. \tag{3.64}$$

Pairings $\langle \bullet, \bullet \rangle$ involving complex vectors are the complex-bilinear extensions of the real pairing throughout.

Lemma 3.14 (Gaussian Conditioning)**.** *Let μ_0 be the standard Gaussian measure on $\mathfrak{g}^I$ and $\mu_{C_{E_0}}$ the centred Gaussian measure on $\mathfrak{g}^{E_0}$ of covariance C_{E_0}. For bounded Borel F on $\mathfrak{g}^{E_0}$,*

$$\int F(a|_{E_0}) \, \mathrm{d}\mu_{\mathsf{G}}(a) = \int \left[\int F(B) \, e^{-\frac{1}{2}\langle \Gamma_{E_0}(1)X, C_{E_0}\Gamma_{E_0}(1)X\rangle} e^{-\langle B, \Gamma_{E_0}(1)X\rangle} \, \mathrm{d}\mu_{C_{E_0}}(B) \right] \mathrm{d}\mu_0(X) \,. \tag{3.65}$$

Proof. Write $a = \mathsf{G}^{1/2} X$, so that a has law μ_{G} when X has law μ_0. Given $a|_{I \setminus E_0}$, the conditional law of $a|_{E_0}$ under μ_{G} is Gaussian with covariance C_{E_0} and mean $-C_{E_0} \iota_{E_0}^* H \Pi_{I \setminus E_0} a = -C_{E_0}\Gamma_{E_0}(1)X$ (complete the square in $\frac{1}{2}\langle a, Ha\rangle$), and its density with respect to $\mu_{C_{E_0}}$ is the displayed factor. This is [5, (2.5)–(2.6)]. *q.e.d.*

The weakened density is

$$e^{-\frac{1}{2}\langle \Gamma_{E_0}(\sigma)X, C_{E_0}\Gamma_{E_0}(\sigma)X\rangle} e^{-\langle B, \Gamma_{E_0}(\sigma)X\rangle} \, \mathrm{d}\mu_{C_{E_0}}(B) \, \mathrm{d}\mu_0(X) \,. \tag{3.66}$$

C_{E_0} is real, positive and independent of σ; its determinant is one fixed positive number on the whole polydisc.

Lemma 3.15 (Component Blocks)**.** *Let $Z \supseteq Z_0'$ be a union of LM-cubes and set $\sigma_\Delta = 0$ for $\Delta \not\subseteq Z$. For each wall-component D of Z put $J_D = \{b \in I \,|\, b \subset \operatorname{int} D\}$ and $E_D = E_0 \cap J_D$. Then $E_0 \subseteq \bigsqcup_D J_D$, $E_D = E(Z_0 \cap D)$, initial vertices in different E_D have distance at least $2M - 2 > R$, and*

$$C_{E_0} = \bigoplus_D (\iota_{E_D}^* H \iota_{E_D})^{-1}, \qquad \Gamma_{E_0}(\sigma) = \bigoplus_D \Gamma_{E_D}(\sigma|_D) \,, \tag{3.67}$$

where $\Gamma_{E_D}(\sigma|_D)$ is constructed from the seed $Z_0 \cap D$ alone, with final domain D (so $\sigma_\Delta = 0$ for every cube Δ of π_1 not in D) and the same I, H and background. Columns of $\Gamma_{E_0}(\sigma)$ outside $\bigsqcup_D J_D$ vanish, and $\Gamma_{E_0}(\sigma)$ does not depend on σ_Δ for cubes Δ in a component containing no seed cube.

Proof. A point is interior to a union of closed grid cubes exactly when all its incident cubes belong to it, and the incident family is wall connected (any two cubes containing a point are joined by a chain of cubes containing that point, consecutive ones sharing a wall); hence $\operatorname{int} Z = \bigsqcup_D \operatorname{int} D$, a disjoint union of open sets. For an M-cube Q of Z_0 the open neighbourhood $\operatorname{int} Q^\sim$ is connected and lies in $\operatorname{int} Z_0' \subseteq \operatorname{int} Z$, so it lies in one component interior; neighbourhoods in different components are disjoint, so their seed cubes have distance at least $2M$, and a bond meeting a seed cube lies in the same component interior. This gives $E_0 \subseteq \bigsqcup J_D$, $E_D = E(Z_0 \cap D)$ and the separation, which by finite range makes $\iota_{E_0}^* H \iota_{E_0}$ block diagonal. Consider an entry $(\Gamma_{E_0}(\sigma))_{eb} = \sum_{b'} H_{eb'} (\Pi_{I\setminus E_0})_{b'b'} S_{Z_0}(\sigma)_{b'b}$ with $e \in E_D$. A nonzero term has $d(e_-, b'_-) \leqslant R$; e meets a seed cube Q, so e_- is within 1 of Q and the open ball of radius $R+2$ about e_- lies in the open M-neighbourhood $\operatorname{int} Q^\sim$ of Q, because $R+3 \leqslant M$ by (3.59), hence in $\operatorname{int} D$; so $b' \subset \operatorname{int} D$. A walk from b' to b with $w_\omega \neq 0$ has every cube of π_1 meeting $D(\omega)$ inside Z, so $D(\omega) \subset \operatorname{int} Z$; $D(\omega)$ is connected and contains b', so $D(\omega) \subset \operatorname{int} D$ and $b \in J_D$. Its coefficients, domains and weakening variables are independent of the seed; the cubes with $\sigma_\Delta = 1$ inside D are those of $Z_0' \cap D = (Z_0 \cap D)'$ (Z_0' is the union over the seed cubes Q of the LM-cubes meeting $\operatorname{int} Q^\sim$; such a cube meets $\operatorname{int} Z$ at a point whose incident LM-cubes all lie in Z and form a wall-connected family, so it lies in the component containing Q); and $\Pi_{I\setminus E_0}$ agrees with $\Pi_{I\setminus E_D}$ on J_D. This gives the block decomposition and the vanishing of the columns outside $\bigsqcup J_D$. Finally, a walk contributing to a row $e \in E_D$ has $D(\omega) \subset \operatorname{int} D$, so its weight involves only σ_Δ with $\Delta \subseteq D$; a component without seed cube has no row at all. *q.e.d.*

The estimates on this kernel are collected in one statement.

Lemma 3.16 (Gaussian Payment)**.** *Let $N_M = \lfloor M/(2R_*) \rfloor$, $\delta_M = 2C_{\mathrm{w}} 4^{-N_M}$ and $A_* = h_+(\gamma^{-1/2}+1)$; by* (3.1)*, $\delta_M \leqslant 1$ and $h_+^2 \delta_M^2/\gamma \leqslant 1/8$. Write $\pi_i(X)$ for the set of π_i-cubes of a union X of such cubes. On every branch, for every seed Z_0, every union $Z \supseteq Z_0'$ of LM-cubes with $\sigma_\Delta = 0$ for $\Delta \not\subseteq Z$, and every σ in the closed polydisc $\{|\sigma_\Delta| \leqslant e^{\kappa_1}\ \forall \Delta \in \pi_1(Z) \setminus \pi_1(Z_0')\}$:*

1. *$\|\Gamma_{E_0}(\sigma) - \Gamma_{E_0}(1)\| \leqslant h_+ \delta_M$ and $\|C_{E_0}^{1/2}(\Gamma_{E_0}(\sigma) - \Gamma_{E_0}(1))\| \leqslant h_+ \gamma^{-1/2} \delta_M$, in the operator norms for the Hilbert norms on $\mathfrak{g}_{\mathbb{C}}^I$ and $\mathfrak{g}_{\mathbb{C}}^{E_0}$.*
2. *For $0 \leqslant \alpha_G \leqslant \min(\gamma/2, \gamma^2/(16A_*^2))$ and $J_Z = \bigsqcup_D J_D$, the integral of the absolute value of* (3.66) *against $e^{\alpha_G \|B\|^2/2}$ is at most $\exp(C_{\mathrm{d}}(\alpha_G + \delta_M^2)|J_Z|)$, $C_{\mathrm{d}} = C_{\mathrm{d}}(\gamma, h_+, d_G)$; with $4C_{\mathrm{d}}\delta_M^2 (LM+1)^4 \leqslant 1/2$ (*(3.1)*) and $\alpha_G \leqslant (8C_{\mathrm{d}}(LM+1)^4)^{-1}$ this is at most $e^{N_1(Z)}$, and the integral of the absolute value factors over the components D of Z (the signed density factors by Lemma 3.15).*
3. *For $0 < \gamma_2$ and the constant $c_{\mathrm{n}} > 0$ of the group with $h(X)^2 \geqslant c_{\mathrm{n}} |X|^2$ on $\mathfrak{g}$, $\prod_{b \in \mathsf{P}} \mathbf{1}_{\{|B_b| \geqslant r\}} \leqslant e^{-a_{\mathrm{bad}}|\mathsf{P}|} \exp(\frac{\gamma_2}{2} \sum_{b \in \mathsf{P}} h(B_b)^2)$ with $a_{\mathrm{bad}} = c_{\mathrm{n}} \gamma_2 r^2/2$, for every $\mathsf{P} \subseteq E_0$ (a pointwise inequality, used in the appendix with $\gamma_2 \leqslant \alpha_G$).*

The proof is Appendix B, Section B.4.

3.6 The Selected Integrals and the Polymer Representation

Connectivity of cubes is wall connectivity ([4], before (0.24)); two domains are compatible when they share neither a cube nor a wall ([5, (2.11)]); $N_i(Y)$ is the number of π_i-cubes of Y, $d_i(Y)$ its tree length ([4, p. 257], written out in (B.1)) and $\ell_i(Y) = d_i(Y) + 5$. The localisation family $\mathbf{D}_0$ consists of the potential hosts (3.36) and the probe hosts (3.75) below; all are connected unions of M-cubes. A potential family indexed by $\mathbf{D}_0$ is a collection $\{\mathbf{W}(Y,\bullet)\}_{Y\in\mathbf{D}_0}$ of Borel functions $\mathfrak{g}^I \to \mathbb{C}$, each depending on a only through $a|_{E(Y)}$, $E(Y) = \{b \in I \,|\, b \cap Y \neq \varnothing\}$, and bounded on the local box $\{|a_b| < r \ \forall b \in E(Y)\}$; only the values on the local boxes enter below, because every selected host Y comes with the factors χ_b, $b \in E(Y) \subseteq E(Y_0)$. The potential $\mathbf{V}_\xi$ of (3.37) is such a family; $\mathbf{V}_\xi(Y,\bullet) = 0$ on hosts carrying no near primitive.

Definition 3.17 (Selections and Selected Integrals)**.** *A selection is a triple $\mathfrak{s} = (\mathcal{D}, \mathsf{P}, Z)$: a finite $\mathcal{D} \subseteq \mathbf{D}_0$, a finite $\mathsf{P} \subseteq I \setminus E(Y_0)$ where $Y_0 = \bigcup_{Y\in\mathcal{D}} Y$, and a final domain Z. Its seed is $Z_0 = Y_0 \cup \bigcup\{Q \in \pi_0 \,|\, Q \cap b \neq \varnothing$ for some $b \in \mathsf{P}\}$, with $E_0 = E(Z_0)$ and Z_0' as in Section 3.5, and a final domain is a union $Z \supseteq Z_0'$ of LM-cubes. The selected integral of a bounded Borel F on $\mathfrak{g}^{E_0}$ is*

$$\begin{aligned}\mathcal{A}^{\mathrm{f}}_{\mathfrak{s}}[F] = \prod_{\Delta\in\pi_1(Z)\setminus\pi_1(Z_0')} \int_0^1 \mathrm{d}s_\Delta \oint_{|\sigma_\Delta|=e^{\kappa_1}} \frac{\mathrm{d}\sigma_\Delta}{2\pi i(\sigma_\Delta - s_\Delta)^2} \\ \cdot \prod_{Y\in\mathcal{D}} \int_0^1 \mathrm{d}t_Y \int \mathrm{d}\mu_0(X) \int \mathrm{d}\mu_{C_{E_0}}(B) \\ \cdot\, e^{-\frac12\langle \Gamma_{E_0}(\sigma)X, C_{E_0}\Gamma_{E_0}(\sigma)X\rangle} e^{-\langle B, \Gamma_{E_0}(\sigma)X\rangle} (-1)^{|\mathsf{P}|} \\ \cdot \prod_{b\in E(Y_0)} \chi_b(B) \prod_{b\in\mathsf{P}} (1-\chi_b(B)) \prod_{Y\in\mathcal{D}} \mathbf{W}(Y,B) e^{t_Y \mathbf{W}(Y,B)} F(B)\,,\end{aligned} \tag{3.68}$$

with $\chi_b(B) = \mathbf{1}_{\{|B_b|<r\}}$ in the bond norm of [3] (the indicator of (3.42)) and $\sigma_\Delta = 0$ for $\Delta \not\subseteq Z$. The vacuum activity of a connected final domain Z is

$$k^{\mathrm{f}}_\xi(Z) = \sum_{(\mathcal{D},\mathsf{P})\,|\,Z_0\neq\varnothing,\ Z_0'\subseteq Z} \mathcal{A}^{\mathrm{f}}_{(\mathcal{D},\mathsf{P},Z)}[1]\,, \tag{3.69}$$

and $\Xi^{\mathrm{f}}_\xi = \sum_{\mathcal{P}\ \text{compatible}} \prod_{Z\in\mathcal{P}} k^{\mathrm{f}}_\xi(Z)$ (empty family included), $(\Xi^{\mathrm{f}}_\xi)^{[Y]}$ the same sum over families compatible with Y.

The selected integral is finite: the integrand is holomorphic in σ on the polydisc of Lemma 3.13 for fixed (X, B, t), its absolute value is integrable in (X, B) uniformly on the closed polydisc by Lemma 3.16(ii) together with the boundedness of F and of each $\mathbf{W}(Y,\bullet)$ on its local box (enforced by the factors χ_b, $b \in E(Y_0)$), and the contours have positive distance from $[0, 1]$. Every selected datum depends only on B, since $E(Y) \subseteq E_0$ for $Y \in \mathcal{D}$ and $\mathsf{P} \subseteq E_0$. A fine bond meets at most two M-cubes (no site lies on a face), so at most $2|\mathsf{P}|$ cubes of the seed are not cubes of Y_0. The seed of a union of selections (disjoint host families and bad-bond sets) is the union of the seeds, and $\widetilde{Z}_0$, Z_0' are unions over the seed cubes (Section 3.5), so all three commute with unions.

Lemma 3.18 (Polymer Representation)**.** *Let $\mathbf{W}$ be a potential family indexed by $\mathbf{D}_0$ and $\mathfrak{n}_\xi = (2\pi)^{d_G|I|/2} \det(H)^{-1/2}$. Then*

$$\int \Theta_\xi(a) \exp\Big(-\frac12\langle a, Ha\rangle + \sum_{Y\in\mathbf{D}_0} \mathbf{W}(Y,a)\Big)\mathrm{d}a = \mathfrak{n}_\xi\, \Xi^{\mathrm{f}}_\xi[\mathbf{W}]\,, \tag{3.70}$$

with the activities of Definition 3.17 formed from $\mathbf{W}$*, and the sum* $\Xi^{\mathrm{f}}_{\xi}[\mathbf{W}]$ *is finite (finitely many selections at fixed volume). In particular, for* $\mathbf{W} = \mathbf{V}_{\xi}$*,*

$$I_{\xi;0} = \mathfrak{n}_{\xi}\, \Xi^{\mathrm{f}}_{\xi}\,, \qquad \Xi^{\mathrm{f}}_{\xi} = \frac{I_{\xi;0}}{\mathfrak{n}_{\xi}} > 0\,. \tag{3.71}$$

Proof. The left side is $\mathfrak{n}_{\xi} \int \Theta_{\xi} e^{\sum_Y \mathbf{W}(Y,\bullet)} \mathrm{d}\mu_{\mathsf{G}}$. Expand each $e^{\mathbf{W}(Y)} = 1 + \int_0^1 \mathbf{W}(Y) e^{t_Y \mathbf{W}(Y)} \mathrm{d}t_Y$; this gives the sum over $\mathcal{D}$. Write $\Theta_{\xi} = \prod_{E(Y_0)} \chi_b \prod_{I \setminus E(Y_0)} (1 - (1 - \chi_b))$ and expand the second product; this gives the sum over P with the sign $(-1)^{|\mathsf{P}|}$. The remaining integrand depends on $a|_{E_0}$ only (each $\mathbf{W}(Y,\bullet)$, $Y \in \mathcal{D}$, through $a|_{E(Y)}$ with $E(Y) \subseteq E_0$, and it is evaluated on the local box of Y because of the factors χ_b, $b \in E(Y_0)$), so Lemma 3.14 applies and produces (3.66) at $\sigma = 1$. For fixed (X, B, t) the density (3.66) is holomorphic in σ on the polydisc (Lemma 3.13); for each $\Delta \in \pi_1$ not in Z_0' write its value at $\sigma_\Delta = 1$ as the value at $\sigma_\Delta = 0$ plus $\int_0^1 \partial_{s_\Delta} \mathrm{d}s_\Delta$, expand the product over these cubes, label each term by $Z = Z_0' \cup \{\text{differentiated cubes}\}$, and represent each derivative by its Cauchy integral along $|\sigma_\Delta| = e^{\kappa_1}$; these are pointwise identities in (X, B, t), and the resulting parameter integrals may be taken outside the Gaussian integrals by Fubini, since the absolute value of the density is integrable uniformly on the polydisc (Lemma 3.16(ii)) and the other factors are bounded. This is (3.68); every step is an exact identity at fixed volume. The infinite objects are the walk series $S_{Z_0}(\sigma)$, which converges absolutely by Lemma 3.13, and the Gaussian integrals over (X, B) at complex σ, whose absolute convergence is Lemma 3.16(ii) (the walk-series bound alone does not control the real part of the quadratic exponent). Every cube of Z_0' is wall-connected inside Z_0' to a cube containing a seed cube: a cube Δ of Z_0' meets $\operatorname{int} Q^{\sim}$ for a seed cube Q at a point x; since $\operatorname{int} Q^{\sim}$ is open, x may be taken in the interior of Δ, so x lies in an M-cube $Q' \subseteq Q^{\sim} \cap \Delta$, and Q' is joined to Q by a chain of face-adjacent M-cubes of $Q^{\sim}$, whose LM-cubes start with Δ, are consecutively equal or wall adjacent, and all belong to Z_0' (each contains an M-cube of $Q^{\sim}$, whose interior lies in $\operatorname{int} Q^{\sim}$). Hence a wall-component of Z either contains a seed cube or consists of differentiated cubes only; in the second case $\Gamma_{E_0}(\sigma)$ does not depend on its σ-variables (Lemma 3.15), the Cauchy integral of a function constant in σ_Δ vanishes, and the term is zero. So only final domains all of whose components contain a seed cube contribute. For these, Lemma 3.15 factors the density and the Gaussian measures over the components D of Z ($\mu_{C_{E_0}} = \bigotimes_D \mu_{C_{E_D}}$ and the density is a product by the block decomposition; the X-integral factors because each block of Γ_{E_0} reads X on J_D only, and the remaining Gaussian factor of μ_0 integrates to one), and the parameter integrals are indexed by cubes and hosts, each inside one component. The cutoff, potential and sign factors factor because each selected host is a connected union of M-cubes inside Z_0, hence inside one component, and each bad bond lies in the component containing its (at most two, face-adjacent) incident cubes. Given Z with components $D_1, \dots, D_m$, the map $(\mathcal{D}, \mathsf{P}) \mapsto ((\mathcal{D}_j, \mathsf{P}_j))_j$, with $\mathcal{D}_j$ the hosts inside D_j and P_j the bad bonds inside D_j, is a bijection from the selections with final domain Z (all $Z_0' \subseteq Z$, every component seeded) onto the tuples of selections $(\mathcal{D}_j, \mathsf{P}_j, D_j)$ with nonempty seeds and final domains D_j: the inverse is the union, because the seed rule, the enlargement and the coarse cover commute with unions and $(Z_0 \cap D_j)' \subseteq D_j$ (it is wall-connected to its seed inside $Z_0' \subseteq Z$). Summing over Z gives the compatible-family sum, in which the empty selection contributes $\int \mathrm{d}\mu_{\mathsf{G}} = 1$. Equation (3.71) is the case $\mathbf{W} = \mathbf{V}_{\xi}$: $K_{\xi;0} = \Theta_{\xi} \exp(-\frac{1}{2}\langle a, Ha\rangle + \sum_Y \mathbf{V}_{\xi}(Y, a))$ by (3.45) and (3.37) (on the open box $\sum_Y \mathbf{V}_{\xi}(Y, a) = P_{\xi,0}(a)$), and positivity holds because $K_{\xi;0} > 0$ on the open box. *q.e.d.*

Lemma 3.19 (Activity Bounds)**.** *With the constants* $\lambda_*, \varepsilon_* > 0$ *and* $\theta \geqslant 0$ *of Appendix B, Section B.5 (fixed there before the coupling together with its auxiliary constants* $\kappa, \delta, \alpha_6, \omega_*$*,*

independently of M_1 and M), and g_0 below its threshold, for $\mathbf{W} = \mathbf{V}_\xi$ on every branch and every connected final domain Z:

$$|k^{\mathrm{f}}_\xi(Z)| \leqslant \varepsilon_* e^{-\lambda_* \ell_1(Z)}, \qquad \sum_{Z' \not\sim Z} |k^{\mathrm{f}}_\xi(Z')| e^{(1+6\theta)N_1(Z')} \leqslant \frac{1}{8} N_1(Z) \,, \tag{3.72}$$

where Z' ranges over the connected final domains and $Z' \not\sim Z$ means incompatible (a domain is incompatible with itself). The same holds for the reference data of Section 3.8 (Lemma B.13).

Lemma 3.20 (Vacuum Clusters)**.** *Under Lemma 3.19, for an ordered tuple $\mathfrak{c} = (D_1, \dots, D_m)$, $m \geqslant 1$, of connected final domains with Ursell coefficient $\varphi^T(\mathfrak{c}) = \sum_{\mathcal{G}} (-1)^{|\mathcal{G}|}$, the sum over the connected spanning subgraphs $\mathcal{G}$ of the incompatibility graph on $\{1, \dots, m\}$ ($j \neq j'$ joined when $D_j \not\sim D_{j'}$; a domain is incompatible with itself, so repeated domains are joined; $\varphi^T = 1$ for $m = 1$), put*

$$\Phi^{\mathrm{v}}_\xi(\mathfrak{c}) = \frac{\varphi^T(\mathfrak{c})}{m!} \prod_{j=1}^{m} k^{\mathrm{f}}_\xi(D_j) \,. \tag{3.73}$$

For every finite set $\mathcal{A}$ of connected final domains,

$$\sum_{\mathfrak{c} \ \mathit{hits}\ \mathcal{A}} |\Phi^{\mathrm{v}}_\xi(\mathfrak{c})| e^{2\theta \sum_j N_1(D_j)} \leqslant \frac{1}{8} \sum_{D \in \mathcal{A}} N_1(D) \,,$$

where $\mathfrak{c}$ hits $\mathcal{A}$ when some D_j is incompatible with a member of $\mathcal{A}$, and

$$\frac{(\Xi^{\mathrm{f}}_\xi)^{[\cup \mathcal{A}]}}{\Xi^{\mathrm{f}}_\xi} = \exp\Big(- \sum_{\mathfrak{c} \ \mathit{hits}\ \mathcal{A}} \Phi^{\mathrm{v}}_\xi(\mathfrak{c}) \Big) \,. \tag{3.74}$$

The proof, including the identity (3.74), is Appendix B, Section B.6: the Kotecký–Preiss criterion with the pinned-tree bound.

3.7 Probe Kernels and the Centred Expectation

For each $b \in I$ let $\mathcal{R}_b = \{r \text{ a bond of } \Omega_1 \,|\, b \in \mathcal{F}_r\}$ be its forward influence set (the bonds whose reconstructed link depends on $a(b)$, Lemma 3.5); it lies within ℓ^∞ distance $2L+1$ of b. Lift $\{b\} \cup \mathcal{R}_b$ to $\mathbb{Z}^4$ about b_- (its diameter is below every period), and let

$$Y_b = \pi_T\big(\text{the smallest rectangle of } M\text{-cubes containing the one-fine-layer neighbourhood of the lift, with one } M\text{-layer adjoined}\big) \,; \tag{3.75}$$

the lift together with its one-fine-layer neighbourhood has ℓ^∞ extent at most $2(2L+1)+3 = 4L+5 < M$ in every direction ($M \geqslant 8LM_2 \geqslant 8L^2$), so it meets at most two M-cubes per direction and the rectangle has side at most 4 cubes, of length $4M < 8M$, so π_T is injective on it. Y_b is a connected member of $\mathbf{D}_0$, contains b and $\mathcal{R}_b$ in its interior, and $N_0(Y_b) \leqslant 4^4 \leqslant s_{\mathrm{p}} = 5^4$. For a union R of LM-cubes its eligible labels are

$$\mathsf{L}_R = \{b \in I \,|\, Y_b \subseteq R\} \,, \tag{3.76}$$

and $\overline{\mathcal{D}}_{\mathsf{L}}$ is the closed box $|a_b| \leqslant r$, $b \in \mathsf{L}$, in $\mathfrak{g}^{\mathsf{L}}$. Each Y_b is a nonempty wall-connected union of M-cubes, so $Y_b \subseteq R$ places it inside exactly one wall-component of R; hence L_R is the

disjoint union of the L_D over the components D of R. For $\lambda \in \mathfrak{g}_{\mathbb{C}}^I$, with the complex bilinear extension of the real pairing, the probe family is

$$L_Y(\lambda, a) = \sum_{b \,|\, Y_b = Y} \langle \lambda_b, a_b \rangle, \qquad \widetilde{\mathbf{V}}_{\xi,\lambda}(Y, a) = \mathbf{V}_\xi(Y, a) + L_Y(\lambda, a) \,, \tag{3.77}$$

a potential family indexed by $\mathbf{D}_0$ for each λ (both terms depend on $a|_{E(Y)}$ only, since $Y_b = Y$ gives $b \in E(Y)$, and both are bounded on the local box of Y), entire in λ. Its partition function is

$$\mathscr{Z}^{\mathrm{c}}_\xi(\lambda) = \int e^{\sum_{b \in I} \langle \lambda_b, a_b \rangle} K_{\xi;0}(a)\, \mathrm{d}a = \mathfrak{n}_\xi \sum_{\mathcal{P} \text{ compatible}} \prod_{D \in \mathcal{P}} k^{\mathrm{f}}_{\xi,D}(\lambda) \,, \tag{3.78}$$

by Lemma 3.18 applied to $\widetilde{\mathbf{V}}_{\xi,\lambda}$, where $k^{\mathrm{f}}_{\xi,D}(\lambda)$ denotes the activity formed from that family; it is entire in λ (a finite sum of integrals of entire functions of λ with integrable majorants on compact sets of λ) and $k^{\mathrm{f}}_{\xi,D}(0) = k^{\mathrm{f}}_\xi(D)$.

Lemma 3.21 (Probe Measures)**.** *On every branch, for every connected final domain D there is a finite complex Borel measure $\nu^{\mathrm{f}}_{\xi,D}$ on $\overline{\mathcal{D}}_{\mathsf{L}_D}$ with*

$$k^{\mathrm{f}}_{\xi,D}(\lambda) = \int e^{\sum_{b \in \mathsf{L}_D} \langle \lambda_b, a_b \rangle}\, \mathrm{d}\nu^{\mathrm{f}}_{\xi,D}(a) \quad (\lambda \in \mathfrak{g}_{\mathbb{C}}^I), \qquad \nu^{\mathrm{f}}_{\xi,D}(\overline{\mathcal{D}}_{\mathsf{L}_D}) = k^{\mathrm{f}}_\xi(D) \,; \tag{3.79}$$

in particular $k^{\mathrm{f}}_{\xi,D}(\lambda)$ depends on λ only through $\lambda|_{\mathsf{L}_D}$. Explicitly, splitting each probe label from its host's ordinary label, a selection becomes $\ell = (\mathcal{D}, \mathsf{J}, \mathsf{P}, Z)$ with $\mathsf{J} \subseteq \mathsf{L}_Z$ the selected probes, $Y_0 = \bigcup_{Y \in \mathcal{D}} Y \cup \bigcup_{b \in \mathsf{J}} Y_b$, and seed, E_0 and Z_0' formed from this Y_0 and P; the Mayer variable of a probe integrates exactly,

$$\int_0^1 \langle \lambda_b, B_b \rangle e^{t_b \langle \lambda_b, B_b \rangle}\, \mathrm{d}t_b = e^{\langle \lambda_b, B_b \rangle} - 1 \,, \tag{3.80}$$

and with $\Lambda^{\mathrm{f}}_{\xi;\ell}$ the complex measure on $\mathfrak{g}^{E_0}$ defined by the right side of (3.68) *for $\mathbf{W} = \mathbf{V}_\xi$, with Y_0, Z_0, E_0, Z_0' the enlarged data of ℓ throughout (the factors χ_b over $E(Y_0)$, the conditional law and the weakening over this seed), with the Mayer factors $\int_0^1 \mathrm{d}t_Y\, \mathbf{W}(Y, B) e^{t_Y \mathbf{W}(Y,B)}$ for $Y \in \mathcal{D}$ only and no probe factor,*

$$\nu^{\mathrm{f}}_{\xi,D} = \sum_{\ell \,|\, Z = D,\ Z_0 \neq \varnothing} \int \Big[\bigotimes_{b \in \mathsf{J}} (\delta_{B_b} - \delta_0) \otimes \bigotimes_{b \in \mathsf{L}_D \setminus \mathsf{J}} \delta_0 \Big]\, \mathrm{d}\Lambda^{\mathrm{f}}_{\xi;\ell}(B) \,. \tag{3.81}$$

Proof. At a host Y carrying the ordinary potential $\mathbf{V}_\xi(Y)$ and the probe terms $F_b = \langle \lambda_b, \bullet \rangle$, b with $Y_b = Y$, write $e^{\widetilde{\mathbf{V}}_{\xi,\lambda}(Y)} = e^{\mathbf{V}_\xi(Y)} \prod_b e^{F_b} = \prod_{\ell'} (1 + (e^{F_{\ell'}} - 1))$ over the labels ℓ' of Y (the ordinary label with $F_{\ell'} = \mathbf{V}_\xi(Y)$ and one label per probe); the sum over nonempty label subsets A of $\prod_{\ell' \in A} (e^{F_{\ell'}} - 1)$ equals $e^{\widetilde{\mathbf{V}}_{\xi,\lambda}(Y)} - 1$. Every term with $A \neq \varnothing$ has the same active host Y, so the split-label expansion has the same Y_0, P, seed, cover and kernel as the one-potential-per-host expansion of Lemma 3.18, and the two activities agree term by term after the Mayer variables t_b of the probes are integrated by (3.80). The Laplace transform of $\delta_{B_b} - \delta_0$ in the variable λ_b is $e^{\langle \lambda_b, B_b \rangle} - 1$, and an unselected label contributes the factor one, the transform of δ_0; this proves (3.79) for the measure (3.81). Each $\Lambda^{\mathrm{f}}_{\xi;\ell}$ is a finite complex measure: it is the image under $(X, B, s, \sigma, t) \mapsto B$ of a complex density whose absolute value is integrable by Lemma 3.16(ii) and the boundedness of each $\mathbf{V}_\xi(Y, \bullet)$ on its local box, the Cauchy kernels and the cutoff factors; the Dirac tensor has total variation at most $2^{|\mathsf{J}|}$, and there are finitely many ℓ at fixed volume. The support lies in the closed box because $\chi_b(B)$, $b \in E(Y_0) \supseteq \mathsf{J}$, forces $|B_b| < r$ on the selected probes. At $\lambda = 0$ every term with $\mathsf{J} \neq \varnothing$ has zero mass, so $\nu^{\mathrm{f}}_{\xi,D}(\overline{\mathcal{D}}_{\mathsf{L}_D}) = k^{\mathrm{f}}_{\xi,D}(0) = k^{\mathrm{f}}_\xi(D)$. *q.e.d.*

For a compatible family $\mathcal{P}$ let $\mathsf{e}_{\mathcal{P}}$ embed $\prod_{D\in\mathcal{P}}\mathfrak{g}^{\mathsf{L}_D}$ into $\mathfrak{g}^I$ by zero on the remaining labels (the L_D are pairwise disjoint).

Lemma 3.22 (Measure Identity)**.** *On every branch,*

$$\frac{K_{\xi;0}(a)\,\mathrm{d}a}{\mathfrak{n}_\xi}=\sum_{\mathcal{P}\ compatible}(\mathsf{e}_{\mathcal{P}})_*\bigotimes_{D\in\mathcal{P}}\nu^{\mathrm{f}}_{\xi,D}\ . \tag{3.82}$$

Proof. Both sides are finite complex measures with compact support (finitely many families at fixed volume), and by (3.78) and (3.79) they have the same Laplace transform on $\mathfrak{g}^I_{\mathbb{C}}$; restricted to imaginary λ this is the Fourier transform, which determines a finite measure. *q.e.d.*

Let δ_{0,L_D} be the Dirac mass at $0\in\mathfrak{g}^{\mathsf{L}_D}$ and

$$\widehat{\nu}^{\mathrm{f}}_{\xi,D}=\nu^{\mathrm{f}}_{\xi,D}-k^{\mathrm{f}}_\xi(D)\delta_{0,\mathsf{L}_D}\ , \tag{3.83}$$

a measure of zero mass (the subtracted term is the sum of the selections with $\mathsf{J}=\varnothing$). For a nonempty compatible family $\mathcal{A}$ and bounded Borel f on $\mathfrak{g}^I$ put

$$\mathfrak{Q}_\xi(\mathcal{A};f)=\int f(\mathsf{e}_{\mathcal{A}}a)\,\mathrm{d}\Big(\bigotimes_{D\in\mathcal{A}}\widehat{\nu}^{\mathrm{f}}_{\xi,D}\Big)(a),\qquad q_\xi(\mathcal{A})=\frac{(\Xi^{\mathrm{f}}_\xi)^{[\cup\mathcal{A}]}}{\Xi^{\mathrm{f}}_\xi}\ . \tag{3.84}$$

Proposition 3.23 (Centred Expectation)**.** *On every branch, for bounded Borel f on $\mathfrak{g}^I$,*

$$\int f\,\mathrm{d}\nu_{\xi;0}=\frac{\int f(a)K_{\xi;0}(a)\,\mathrm{d}a}{I_{\xi;0}}=f(0)+\sum_{\varnothing\neq\mathcal{A}\ compatible}q_\xi(\mathcal{A})\mathfrak{Q}_\xi(\mathcal{A};f)\ , \tag{3.85}$$

and, with Lemmas 3.9 and 3.10,

$$\mathcal{J}^{\mathrm{a}}_\xi[f]=e^{c_\xi}I_{\xi;1}\Big[f(0)+\sum_{\varnothing\neq\mathcal{A}\ compatible}q_\xi(\mathcal{A})\mathfrak{Q}_\xi(\mathcal{A};f)+\mathcal{B}_\xi[f]\Big]\ . \tag{3.86}$$

Proof. Integrate f against (3.82) and expand each $\nu^{\mathrm{f}}_{\xi,D}$ as $\widehat{\nu}^{\mathrm{f}}_{\xi,D}+k^{\mathrm{f}}_\xi(D)\delta_{0,\mathsf{L}_D}$. A term is indexed by $\mathcal{P}$ and the subfamily $\mathcal{A}\subseteq\mathcal{P}$ carrying the zero-mass factors; $f(\mathsf{e}_{\mathcal{P}}a)$ with $a=0$ on the labels of $\mathcal{P}\setminus\mathcal{A}$ is $f(\mathsf{e}_{\mathcal{A}}a)$. Summing over $\mathcal{P}\supseteq\mathcal{A}$ gives $(\Xi^{\mathrm{f}}_\xi)^{[\cup\mathcal{A}]}\mathfrak{Q}_\xi(\mathcal{A};f)$, and $\mathcal{A}=\varnothing$ gives $\Xi^{\mathrm{f}}_\xi f(0)$. Divide by $I_{\xi;0}=\mathfrak{n}_\xi\Xi^{\mathrm{f}}_\xi$. The second display is $\mathcal{J}^{\mathrm{a}}_\xi[f]=e^{c_\xi}I_{\xi;1}\int f\,\mathrm{d}\nu_{\xi;1}$ (3.49) with (3.50). *q.e.d.*

With $f=f_\xi$ (extended by zero off the enlarged box; the measures live on the closed box, inside the enlarged box, where f_ξ is the Wilson function (3.46)), the first display is $F_\xi(0)$, and (3.86) is the exact decomposition of the conditional Wilson numerator on every branch: the loop at zero young field, the sum of closed insertions paired with the signed probe measures, and the exterior response.

3.8 Joint Extraction of the Scalar Reference

Let r_n be the positive representative of traversal $n\in\{1,\dots,n_C\}$ of C, the traversals numbered cyclically from a fixed base point. For a contour bond r that is not a bond of Ω_1 put $\mathcal{F}_r=\varnothing$: its link is $V_0(r)$ by (3.26), independent of a. By Lemma 3.5 and this convention,

the link $U_\xi(a)(r_n)$ depends on the young variable only through $a|_{\mathcal{F}_{r_n}\cap I}$. For a connected final domain D the contact set is

$$\mathsf{T}_D = \{n \,|\, \mathcal{F}_{r_n} \cap \mathsf{L}_D \neq \varnothing\} \,, \tag{3.87}$$

and D is marked when $\mathsf{T}_D \neq \varnothing$. If a family $\mathcal{A}$ contains an unmarked domain D, then for a in the closed box $f_\xi(\mathsf{e}_\mathcal{A} a) = W_\chi(C; U_\xi(\mathsf{e}_\mathcal{A} a))$ does not depend on $a|_{\mathsf{L}_D}$ (every contour link depends on $a|_{\mathcal{F}_{r_n}\cap I}$, and $\mathcal{F}_{r_n} \cap \mathsf{L}_D = \varnothing$ for every n), and $\mathfrak{Q}_\xi(\mathcal{A}; f_\xi) = 0$ because $\widehat{\nu}^{\mathrm{f}}_{\xi,D}$ has zero mass and the measures are supported in the closed box; so only families of marked domains contribute to (3.85) for $f = f_\xi$. Expand $q_\xi(\mathcal{A})$ by Lemma 3.20,

$$q_\xi(\mathcal{A}) = \sum_{k\geqslant 0} \frac{(-1)^k}{k!} \sum_{\mathfrak{c}_1,\dots,\mathfrak{c}_k \text{ each hits } \mathcal{A}} \prod_{j=1}^{k} \Phi^{\mathrm{v}}_\xi(\mathfrak{c}_j) \,, \tag{3.88}$$

the sum being absolutely convergent: $\sum_{\mathfrak{c} \text{ hits } \mathcal{A}} |\Phi^{\mathrm{v}}_\xi(\mathfrak{c})| \leqslant \frac{1}{8} \sum_{D\in\mathcal{A}} N_1(D)$ by Lemma 3.20, so the k-fold sums are dominated by the exponential series. A tuple whose incompatibility graph is disconnected has $\varphi^T(\mathfrak{c}) = 0$ (it has no connected spanning subgraph), so the terms containing such a tuple vanish and are dropped: every cluster occurrence $\mathfrak{c}_j$ below has a connected incompatibility graph. For each term form the graph on the marked domains of $\mathcal{A}$ and the k cluster occurrences: a marked vertex is joined to each cluster that hits it, two clusters are joined when some constituent domains are incompatible. Every component has a marked root; its contacts are the union of the contact sets of its roots. The hull of a nonempty set of traversals is its shortest cyclic interval when that interval has fewer than $n_C/2$ positions, and the whole contour otherwise; a shortest interval of fewer than $n_C/2$ positions is unique (two such intervals containing the set have together fewer than n_C positions, so they intersect in a cyclic interval, which contains the set and is shorter than one of them unless they coincide), and if $S \subseteq S'$ and the hull of S' is short then the hull of S is contained in it (the hull of S' is a short interval containing S, so the hull of S is short and, by the uniqueness just proved, is the shortest interval containing S; the intersection of the two short hulls is again an interval, their union having fewer than n_C positions, and it contains S, so it is the hull of S). Let $\approx$ be the transitive closure of the relation "the hulls of the contacts intersect" on the components, and for a class K let $\mathcal{H}(K)$ be the hull of the union of the contacts of its members. If the union of the members' hulls has fewer than $n_C/2$ positions, $\mathcal{H}(K)$ equals that union: the union is a cyclic interval (the members' hulls are chained by intersections, and two intersecting intervals with together fewer than n_C positions have an interval union), it contains the union of the contacts, so it contains the hull $\mathcal{H}(K)$ by minimality; conversely each member's contacts lie in the union of the contacts, whose hull is short, so by monotonicity each member's hull lies in $\mathcal{H}(K)$. If the union of the members' hulls has at least $n_C/2$ positions, $\mathcal{H}(K)$ is the whole contour: a short $\mathcal{H}(K)$ would contain every member's hull by monotonicity. The groups are the classes if every $\mathcal{H}(K)$ is short, and the single class of all components otherwise; the intervals of distinct groups are disjoint, since intersecting intervals would relate their classes.

Definition 3.24 (Groups)**.** *A group W consists of its nonempty compatible marked-root set $\mathcal{A}(W)$, its ordered list of cluster occurrences $(\mathfrak{c}_1, \dots, \mathfrak{c}_{k(W)})$, $k(W) \geqslant 0$, each hitting a root of W, and its interval $\mathcal{H}(W)$, either a cyclic interval of fewer than $n_C/2$ traversals or the whole contour read from the base point; the data arise from a term of* (3.88) *as above; conversely, the term of* (3.88) *with root family $\mathcal{A}(W)$ and cluster list $(\mathfrak{c}_1, \dots, \mathfrak{c}_{k(W)})$ has W as its only group (its graph has the components of W, which form one class with the interval $\mathcal{H}(W)$). Its weight is*

$$w_\xi(W) = \frac{(-1)^{k(W)}}{k(W)!} \prod_{j=1}^{k(W)} \Phi^{\mathrm{v}}_\xi(\mathfrak{c}_j) \,. \tag{3.89}$$

Two groups are compatible when all their constituent domains (roots and cluster members) are mutually compatible and their intervals are disjoint; a family containing a whole-contour group is a singleton. For a compatible group family $\mathfrak{B}$ and $\mathcal{S} \subseteq \mathfrak{B}$,

$$\mathcal{A}(\mathcal{S}) = \bigcup_{W \in \mathcal{S}} \mathcal{A}(W), \qquad A_\xi(\mathcal{S}) = \Big(\prod_{W \in \mathcal{S}} w_\xi(W)\Big) \mathfrak{Q}_\xi(\mathcal{A}(\mathcal{S}); f_\xi), \qquad A_\xi(\varnothing) = f_\xi(0)\ . \quad (3.90)$$

Lemma 3.25 (Grouped Source)**.** *On every branch,*

$$F_\xi(0) = \sum_{\mathfrak{B}\ compatible} A_\xi(\mathfrak{B})\ , \quad (3.91)$$

an absolutely convergent sum over compatible group families (including the empty family; there are finitely many root families at fixed volume but infinitely many cluster lists).

Proof. Substitute (3.88) into (3.85) for $f = f_\xi$. Each term, indexed by a family $\mathcal{A}$ of marked domains and an ordered list of k cluster occurrences, is cut into its groups; the groups are pairwise compatible (roots in different groups are compatible as members of $\mathcal{A}$, clusters in different groups are compatible with each other and with the other groups' roots because the graph has no edge between components, and the intervals are disjoint by construction). The regrouping is reversible: a compatible group family determines $\mathcal{A} = \mathcal{A}(\mathfrak{B})$ and the multiset of cluster occurrences, and the term's graph has as components exactly the components of the groups' own graphs (no edges between groups, by compatibility), with the same hulls, so the classes and intervals are recovered; identical occurrences cannot split between groups because their domains are self-incompatible, so a group's occurrences are those of its term. The ordered list of the term is an interleaving of the groups' lists, and the interleavings number $k!/\prod_W k(W)!$, which turns the factor $(-1)^k/k!$ into $\prod_W w_\xi(W)/\prod_W \Phi$-products. The rearrangement is of an absolutely convergent sum: there are finitely many root families $\mathcal{A}$ at fixed volume, $|\mathfrak{Q}_\xi(\mathcal{A}; f_\xi)| \leqslant \|f_\xi\|_\infty \prod_{D\in\mathcal{A}} \|\widehat{\nu}^{\mathrm{f}}_{\xi,D}\|$ with the total variations finite, and $\sum_k \frac{1}{k!}\big(\sum_{\mathfrak{c}\text{ hits }\mathcal{A}} |\Phi^{\mathrm{v}}_\xi(\mathfrak{c})|\big)^k \leqslant \exp(\frac{1}{8}\sum_{D\in\mathcal{A}} N_1(D))$; hence the term sum converges absolutely, and so does the group-family sum, being a rearrangement of it. *q.e.d.*

A superscript $\circ$ denotes the reference data: the same Ω_1, Λ_1, I, J, cubes, hosts, constants and cutoffs, with the background $U_1^\circ = \mathbf{1}$ on every bond, $V_0^\circ = \mathbf{1}$, zero retained field $A_{\mathrm{e}}^\circ = 0$ and $\Psi_\xi^\circ = \mathbf{1}$; thus $H_\xi^\circ = \mathsf{H}_{\mathbf{1}}^{\Omega_1}$, the reference phase is $q_{\mathbf{1}}$, the reference primitives are those of Section 3.3 at $U = \mathbf{1}$, $f_\xi^\circ(a) = W_\chi(C; e^{iq_{\mathbf{1}}(a)})$ with the links off Ω_1 equal to $\mathbf{1}$, and $f_\xi^\circ(0) = \chi(\mathbf{1}) = 1$ since $q_{\mathbf{1}}(0) = 0$. These data need not be a branch of the history sum. Lemmas 3.2, 3.3, 3.5 and 3.7 apply to $U = \mathbf{1}$ directly (Section 3.3, opening paragraph; Lemma 3.2 at $A = 0$; the row solutions are those of Lemma 3.6 in its reference form, on the row polydiscs $\mathcal{U}_c$, and (B4) is not invoked), and the estimate lemmas used below (Lemmas 3.4, 3.6, 3.16, 3.19, 3.20) hold for the reference data by Appendix B, Lemma B.13; in particular the reference near primitives are real on the open box (Lemma 3.6: the analytic branches of the logarithms are real at real arguments, the row solution being real), so $K^\circ_{\xi;0} > 0$ there and $I^\circ_{\xi;0} > 0$. Every lemma of Sections 3.5–3.8 stated on every branch (Lemmas 3.13, 3.14, 3.15, 3.18, 3.21, 3.22, 3.25 and the first display of Proposition 3.23) holds verbatim for the reference data: their proofs use only the field-independent geometry and the lemmas just listed. Every object of Sections 3.5–3.8 formed from the reference data carries the superscript $\circ$; in particular $K^\circ_{\xi;0}$, $I^\circ_{\xi;0}$, $k^{\mathrm{f},\circ}_\xi$, $\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}$, $w^\circ_\xi(W)$. The index set of groups is the same for the branch and for the reference data: roots, cluster lists (the clusters are tuples of domains), contact sets and intervals are field-independent data (the sets $\mathcal{F}_{r_n}$ and L_D do not involve the background), and only the weights w_ξ, w°_ξ and the measures $\widehat{\nu}^{\mathrm{f}}_{\xi,D}$, $\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}$ differ. For a group W

let $P^\circ_W(a)$ be the ordered product of the reference links along the consecutive traversals of $\mathcal{H}(W)$, in the cyclic order of the contour starting from the first traversal of the interval (for a whole-contour group, from the base point), the link of a traversal being $e^{iq_{\mathbf{1}}(\mathsf{e}_{\mathcal{A}(W)}a)(r)}$ when its bond r is a bond of Ω_1 and $\mathbf{1}$ otherwise, as in f°_ξ (oriented representatives inverted), with the young variable $\mathsf{e}_{\mathcal{A}(W)}a$, zero off the labels of $\mathcal{A}(W)$; then $P^\circ_W(0)=\mathbf{1}$. Put

$$a_W = w^\circ_\xi(W)\int \chi\big(P^\circ_W(a)\big)\,\mathrm{d}\Big(\bigotimes_{D\in\mathcal{A}(W)}\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}\Big)(a)\;. \tag{3.92}$$

Lemma 3.26 (Schur Identity)**.** *For $h\in G$,*

$$w^\circ_\xi(W)\int \chi\big(P^\circ_W(a)h\big)\,\mathrm{d}\Big(\bigotimes_{D\in\mathcal{A}(W)}\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}\Big)(a) = a_W\chi(h)\;. \tag{3.93}$$

Consequently

$$S^\circ_{\xi,C} = \sum_{\mathfrak{B}\ \mathit{compatible}}\prod_{W\in\mathfrak{B}} a_W = \frac{\int f^\circ_\xi(a)K^\circ_{\xi;0}(a)\,\mathrm{d}a}{I^\circ_{\xi;0}}\;. \tag{3.94}$$

Proof. For constant $u\in G$ let $(R_ua)_b=\mathrm{Ad}_u\,a_b$ on $\mathfrak{g}^I$, an orthogonal map for the invariant inner product. At the identity background the constant gauge transformation u fixes the background, so Lemma 3.3 gives $H^\circ_\xi = R_uH^\circ_\xi R_u^{-1}$; hence all walk coefficients, $C^\circ_{E_0}$, $\Gamma^\circ_{E_0}(\sigma)$ (the domains and weights do not involve the fields) and the Gaussian density (3.66) are invariant under $(X,B)\mapsto(R_uX,R_uB)$, and μ_0, $\mu_{C^\circ_{E_0}}$ are invariant. The cutoff factors are invariant because the bond norm is Ad-invariant ((5.4)). The reconstruction is covariant: by the intertwinings in the proof of Lemma 3.3 at $U=\mathbf{1}$, $\mathcal{N}_{\mathbf{1}}\circ R_u = R_u^{(1)}\mathcal{N}_{\mathbf{1}}$, $h_{\mathbf{1}}R_u^{(1)} = R_uh_{\mathbf{1}}$ and $\mathsf{C}_{\mathbf{1}}R_{u,*} = R_u\mathsf{C}_{\mathbf{1}}$, so the unique small solution on the row polydiscs of Lemma 3.6 (reference version; the polydiscs are R_u-invariant because the bond norm is Ad-invariant) satisfies $\widetilde{D}_{\mathbf{1}}(R_uA)=R_u^{(1)}\widetilde{D}_{\mathbf{1}}(A)$ and $q_{\mathbf{1}}(R_uB)=R_uq_{\mathbf{1}}(B)$, that is, every reconstructed link is conjugated by u; the primitives are invariant, being traces of products of links, Haar densities of conjugated phases (σ is conjugation invariant) and determinants of conjugated operators. The probe term transforms as $L_Y(R_u\lambda,R_ua)=L_Y(\lambda,a)$. Hence $k^{\mathrm{f},\circ}_{\xi,D}(R_u\lambda)=k^{\mathrm{f},\circ}_{\xi,D}(\lambda)$, and by Laplace uniqueness $(R_u)_*\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}=\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}$. Since the links are conjugated, $P^\circ_W(R_ua)=uP^\circ_W(a)u^{-1}$. Let π be the irreducible unitary representation with $\chi=\dim(\pi)^{-1}\operatorname{tr}\pi$ (the normalised character, $\chi(\mathbf{1})=1$). The left side of (3.93), as a function of h, is $\dim(\pi)^{-1}\operatorname{tr}(\Upsilon\pi(h))$ for the matrix $\Upsilon = w^\circ_\xi(W)\int\pi(P^\circ_W(a))\,\mathrm{d}(\bigotimes\widehat{\nu}^\circ)(a)$; by the invariance of the measure and the covariance of P°_W, $\Upsilon=\pi(u)\Upsilon\pi(u)^{-1}$ for all $u\in G$, so Υ is a multiple of the identity by Schur's lemma, and $h=\mathbf{1}$ identifies the multiple as a_W. For (3.94) apply Lemma 3.25 to the reference data (its sum converges absolutely, as shown there). Let $\mathfrak{B}=\{W_1,\dots,W_m\}$ be a compatible family. If $m=1$ and W_1 is a whole-contour group, the reference Wilson function at $\mathsf{e}_{\mathcal{A}(W_1)}a$ is $\chi(P^\circ_{W_1}(a))$. Otherwise all intervals are short and pairwise disjoint; number the groups in cyclic order starting from W_1, and use the cyclicity of the trace to start the ordered link product at the first traversal of $\mathcal{H}(W_1)$. The reference Wilson function at $\mathsf{e}_{\mathcal{A}(\mathfrak{B})}a$ is then $\chi(P^\circ_{W_1}(a)\cdots P^\circ_{W_m}(a))$: a traversal outside every interval is outside every contact set, so its link is $e^{iq_{\mathbf{1}}(0)}=\mathbf{1}$ (or $\mathbf{1}$ by convention when its bond is not a bond of Ω_1); a traversal in $\mathcal{H}(W_j)$ lies in no contact set of a root of another group, so its link depends only on $a|_{\mathcal{A}(W_j)}$, and the links of the consecutive traversals of $\mathcal{H}(W_j)$ multiply to $P^\circ_{W_j}(a)$ in the order of the product. The integral against $\bigotimes_j\bigotimes_{D\in\mathcal{A}(W_j)}\widehat{\nu}^\circ_D$ of this character times $\prod_j w^\circ(W_j)$ is evaluated by Fubini one group at a time: integrating over the variables of W_1 with $h=P^\circ_{W_2}(a)\cdots P^\circ_{W_m}(a)$ fixed gives $a_{W_1}\chi(P^\circ_{W_2}(a)\cdots P^\circ_{W_m}(a))$ by (3.93), and so

on, ending with $\prod_j a_{W_j}\chi(\mathbf{1}) = \prod_j a_{W_j}$. Since $|\chi| \leqslant 1$, $|a_W| \leqslant |w^\circ_\xi(W)| \prod_{D\in\mathcal{A}(W)} \|\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}\|$, and the sum in (3.94) converges absolutely by the bound of Lemma 3.25. *q.e.d.*

Proposition 3.27 (Joint Extraction)**.** *For a compatible group family $\mathfrak{B}$ define the joint defect*

$$R_\xi(\mathfrak{B}) = \sum_{\mathcal{S}\subseteq\mathfrak{B}} (-1)^{|\mathfrak{B}\setminus\mathcal{S}|} \Big(\prod_{W\in\mathfrak{B}\setminus\mathcal{S}} a_W \Big) A_\xi(\mathcal{S}) \,, \qquad R_\xi(\varnothing) = f_\xi(0) \,. \tag{3.95}$$

Then, with $S^{\circ,[\mathfrak{B}]}_{\xi,C}$ the reference sum over group families compatible with every member of $\mathfrak{B}$,

$$F_\xi(0) = S^\circ_{\xi,C} f_\xi(0) + \sum_{\varnothing\neq\mathfrak{B}\ \mathit{compatible}} S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B}) \,, \tag{3.96}$$

and on every branch

$$\mathcal{J}^{\mathrm{a}}_\xi[f_\xi] = e^{c_\xi} I_{\xi;1} \Big[S^\circ_{\xi,C} f_\xi(0) + \sum_{\varnothing\neq\mathfrak{B}\ \mathit{compatible}} S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B}) + \mathcal{B}_\xi[f_\xi] \Big] \,. \tag{3.97}$$

Proof. Boolean inversion of (3.95) gives $A_\xi(\mathfrak{B}) = \sum_{\mathcal{S}\subseteq\mathfrak{B}} (\prod_{\mathfrak{B}\setminus\mathcal{S}} a_W) R_\xi(\mathcal{S})$; substitute into (3.91) and sum over $\mathfrak{B} \supseteq \mathcal{S}$: the families $\mathfrak{B}\setminus\mathcal{S}$ range over the group families compatible with every member of $\mathcal{S}$, giving $S^{\circ,[\mathcal{S}]}_{\xi,C}$. The interchange of the sums is justified by absolute convergence. Put $\beta_W = (|w_\xi(W)| + |w^\circ_\xi(W)|) \prod_{D\in\mathcal{A}(W)} \max(\|\widehat{\nu}^{\mathrm{f}}_{\xi,D}\|, \|\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}\|)$; then $|a_W| \leqslant \beta_W$, $|A_\xi(\mathcal{S})| \leqslant \|f_\xi\|_\infty \prod_{W\in\mathcal{S}} \beta_W$ and $|R_\xi(\mathcal{S})| \leqslant \|f_\xi\|_\infty \prod_{W\in\mathcal{S}} 2\beta_W$. The sum $\sum_W \beta_W$ over all groups is finite: a group is determined by its root family, its cluster list and its interval, there are finitely many root families at fixed volume, and $\sum_{W\,|\,\mathcal{A}(W)=\mathcal{A}} |w_\xi(W)| \leqslant \exp(\frac{1}{8}\sum_{D\in\mathcal{A}} N_1(D))$ by Lemma 3.20, likewise for w°_ξ by Lemma B.13. A compatible group family has at most N_{m} members, the number of marked domains (its roots are pairwise distinct marked domains), so $\sum_{\mathfrak{B}} \sum_{\mathcal{S}\subseteq\mathfrak{B}} \prod_{W\in\mathfrak{B}\setminus\mathcal{S}} |a_W|\, |R_\xi(\mathcal{S})| \leqslant \|f_\xi\|_\infty \sum_{\mathfrak{B}} \prod_{W\in\mathfrak{B}} 3\beta_W \leqslant \|f_\xi\|_\infty \sum_{n\leqslant N_{\mathrm{m}}} \big(3\sum_W \beta_W\big)^n < \infty$; in particular $S^\circ_{\xi,C}$ and every $S^{\circ,[\mathcal{S}]}_{\xi,C}$ converge absolutely. Then (3.86) with $f = f_\xi$. *q.e.d.*

At the reference data every nonempty defect vanishes: there $A^\circ_\xi(\mathcal{S}) = \prod_{W\in\mathcal{S}} a_W$ for every compatible $\mathcal{S}$ (proof of Lemma 3.26), so $R^\circ_\xi(\mathfrak{B}) = \prod_{W\in\mathfrak{B}} a_W \sum_{\mathcal{S}\subseteq\mathfrak{B}} (-1)^{|\mathfrak{B}\setminus\mathcal{S}|} = 0$ for $\mathfrak{B} \neq \varnothing$. The identity uses no division by a character value and no norm of an open transport.

Lemma 3.28 (Covered Branches)**.** *Call ξ covered when every bond of the contour is a bond of Ω_1 and every free-input stencil F_γ of the contour* (3.32) *lies in I, that is, $\mathcal{F}_r \subseteq I$ for every contour bond r, with $\mathcal{F}_r$ of* (3.13) *(the convention $\mathcal{F}_r = \varnothing$ off Ω_1 of Section 3.8 is not needed here, the first condition placing every contour bond in Ω_1). On a covered branch $U_\xi(0)(r) = U_1(r)$ for every contour bond r, hence $f_\xi(0) = W_\chi(C; U_1)$.*

Proof. At $a = 0$ the free vector is $B = \Psi_\xi(A_{\mathrm{e}})$, which vanishes on I. For a contour bond r, a bond of Ω_1 on a covered branch, $U_\xi(0)(r) = e^{iq_U(B)(r)} U_1(r)$ by (3.26), and by Lemma 3.5 $q_U(B)(r)$ depends on $B|_{\mathcal{F}_r}$ only and vanishes when $B|_{\mathcal{F}_r} = 0$ (for a free bond directly, for $r = b_0(c)$ because $\mathcal{F}_r = \mathcal{E}_c \cap E_*$). Since $\mathcal{F}_r \subseteq I$ on a covered branch, $U_\xi(0) = U_1$ on the contour. *q.e.d.*

This ends the construction. Appendix B proves Lemmas 3.4, 3.6, 3.16, 3.19 and 3.20, and Lemma B.13 there transfers them to the reference data; Appendix D bounds $\mathcal{B}_\xi[f_\xi]$ and the uncovered branches; Section 9 identifies $S^\circ_{\xi,C}$ with the flat coefficient $\kappa_C e^{E^\circ_\xi}$ defined there

and bounds the defects. That section does not work at the carrier I directly: it adds a second cutoff test on the collar of Λ_1, retreats Λ_1 and I to a labelled carrier $D(\Lambda) \subseteq I$ on the branches where the test fails, writes eq:c-one-step-branch-identity for each labelled term, and identifies the labelled coefficient S°_{Λ} with $\kappa_C e^{E^{\circ}_{\Lambda}}$ at the retreated carrier, with the defects of every labelled term bounded relative to κ_C up to a label-average error paid by one-point witness probabilities.

4 The Flat Regular-Region Calculation

The flat evaluation sets the reconstructed background to the identity. On an all-regular periodic lattice it gives the integral specified in this section. Section 7 treats the flat straight-side ratio in a proper conditional integration region. The covariance, constraint substitution, Haar density and cutoff are fixed together; the coefficient depends on this prescription. Use the fixed faithful unitary realisation $G \subset U(N)$. Identify its Lie algebra $\mathfrak{g}$ with Hermitian matrices, so its group exponential is e^{iX}. Write $\mathrm{tr}_N = N^{-1}\,\mathrm{Tr}$ and use $\langle X, Y\rangle = \mathrm{tr}_N(XY)$ as the real inner product. The observable is exactly (1.1), with the normalised character $\chi(\mathbf{1}) = 1$ specified in Section 1. Put $d = \dim \mathfrak{g}$. The first derivative of χ vanishes: the trace of its Lie algebra representation vanishes on commutators, and a semisimple Lie algebra is its own commutator algebra.

4.1 The Covariance and the Normalised Insertion

Use fine-lattice units in the operators in this section. Let E be the positively oriented bonds of T, let $T^{(1)}$ be its coarse centres, and let E_1 be the positively oriented coarse bonds. Take the odd blocking factor $L > 11$ used by Bałaban ([4, p. 251]; the lemmas of this paper that need only $L \geqslant 3$ are stated so). For the covariance bounds below, $T = \prod_{\mu=1}^{4}(\mathbb{Z}/N_\mu\mathbb{Z})$ with $N_\mu = LK_\mu$ and $K_\mu \geqslant 3$. The block $\mathcal{B}_y$ consists of the fine sites at displacement n from $y \in T^{(1)}$, with $n \in \mathbb{Z}^4$ and $|n_\mu| \leqslant (L-1)/2$ in each direction. For $x \in \mathcal{B}_y$, the path family $\mathcal{G}(y,x)$ traverses each nonzero coordinate displacement in one uninterrupted run, in every ordering of those runs; the path is empty for $x = y$. These are the centred blocks and path family of [4, Section 0].

The averaging operation is (1.4); the formula below is [4, (0.4)], its value on the set $\mathcal{O}_c$ where every logarithm has its argument in the chart domain (Lemma 1.1). [3, p. 243] permits this choice in the paragraph following (0.1); no identification with the different average [4, (0.12)] is made. For a positive coarse bond $c = (y, y + Le_\mu)$ and $x \in \mathcal{B}_y$, put $x' = x + Le_\mu$. The vectors e_μ are the fine-lattice unit vectors, $[x, x']$ is the straight fine path, and $P_c(U)$ is the transport along the straight fine path from c_- to c_+. On the regular logarithm domain, [4, (0.4)] is

$$\beta_c(U) = L^{-4} \sum_{x \in \mathcal{B}_{c_-}} \frac{1}{|\mathcal{G}(c_-,x)|\,|\mathcal{G}(c_+,x')|} \sum_{\substack{\gamma \in \mathcal{G}(c_-,x) \\ \gamma' \in \mathcal{G}(c_+,x')}} \frac{1}{i} \log\left(U(\gamma)U([x,x'])U(\gamma')^{-1}P_c(U)^{-1}\right) ,$$

$$M(U,c) = e^{i\beta_c(U)}P_c(U) \qquad (U \in \mathcal{O}_c) . \tag{4.1}$$

Every logarithm is applied to a closed relative transport based at c_-. Define its logarithmic constraint map and derivative by

$$\mathcal{Q}(q) = \frac{1}{i}\log M(e^{iq}), \qquad Q = \frac{1}{L}D\mathcal{Q}(0), \qquad \mathcal{N}(q) = \mathcal{Q}(q) - LQq\,. \tag{4.2}$$

The maps $\mathcal{Q}$ and $\mathcal{N}$ take a neighbourhood of zero in $\mathfrak{g}^E$ to $\mathfrak{g}^{E_1}$. The designated bond $b_0(c)$ is the fine bond on the straight coarse bond c crossing between its two blocks. Put $E_0 = \{b_0(c) \,|\, c \in E_1\}$ and $E_* = E \setminus E_0$. Let ι_0 and ι_* extend fields on these two label sets by zero to E, and put $Q_0 = Q\iota_0$. The designated coefficient matrix Q_0 is invertible in this regular construction. The maps are

$$h = \frac{1}{L}\iota_0 Q_0^{-1}, \qquad \mathsf{C} = (I - \iota_0 Q_0^{-1} Q)\iota_* \,. \tag{4.3}$$

Thus $h\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^E$, $\mathsf{C}\colon \mathfrak{g}^{E_*} \to \mathfrak{g}^E$, $LQh = I$ and $Q\mathsf{C} = 0$. The free variable is $B \in \mathfrak{g}^{E_*}$, and $\mathsf{C}B$ agrees with B on E_*. These are the designated-bond construction in [4] following (2.10) and the elimination in [3, (1.14)–(1.15)].

The gauge-fixing functional is the exponential one in [3, (1.7)], not a delta-function axial gauge. For the transports along $\mathcal{G}(y,x)$, let $\mathcal{U}_U(y,x)$ be their local group mean: it is the unique solution near their diagonal value of

$$\sum_{\gamma \in \mathcal{G}(y,x)} \log\bigl(U(\gamma)\mathcal{U}_U(y,x)^{-1}\bigr) = 0 \,. \tag{4.4}$$

This is [4, (0.10)–(0.11)]. The gauge-fixing functional and the action used in the constrained reference integral are

$$\begin{aligned} \mathscr{G}(U) &= \sum_{y \in T^{(1)}} \sum_{x \in \mathcal{B}_y \setminus \{y\}} (1 - \operatorname{Re}\operatorname{tr}_N \mathcal{U}_U(y,x)) \,, \\ \mathscr{A}(U) &= S(U) + \mathscr{G}(U) \,. \end{aligned} \tag{4.5}$$

Here ∂p is the oriented boundary of the plaquette p. All group means and logarithms are used in the regular chart containing the identity.

For $a, b \in \mathfrak{g}^E$, the flat Hessian is defined by

$$\langle a, \Delta_1 b\rangle = \frac{\partial^2}{\partial s\, \partial t} \mathscr{A}(e^{i(sa+tb)})\Big|_{s=t=0} \,. \tag{4.6}$$

The first derivative is zero at the identity. This is the $J_1 = 0$ specialisation of [3, (1.16)]. [3, (1.12)] states strict positivity of the second differential at the regular constrained minimum. With the exponential gauge fixing and the constrained extension above, it makes the quadratic form on the finite free carrier strictly positive; no zero-mode subtraction is introduced. The quadratic form and its inverse on that carrier are

$$\mathsf{H} = \mathsf{C}^* \Delta_1 \mathsf{C}, \qquad \mathsf{K}_B = \mathsf{H}^{-1}, \qquad \mathsf{K} = \mathsf{C}\mathsf{K}_B\mathsf{C}^* \,. \tag{4.7}$$

The adjoint and Lebesgue measure use the stated real inner products, summed over the respective bond labels. Thus K is the covariance of the linearised fine field $\mathsf{C}B$, rather than an unconditioned lattice propagator. The normalised Gaussian measure on the free carrier is denoted by μ_{K_B}. In an orthonormal basis of real dimension $D = d|E_*|$, it has density

$$\mathrm{d}\mu_{\mathsf{K}_B}(B) = (2\pi)^{-D/2} (\det \mathsf{K}_B)^{-1/2} e^{-\frac{1}{2}\langle B, \mathsf{H}B\rangle}\, \mathrm{d}B \,. \tag{4.8}$$

Bałaban's nonlinear constraint solution gives the fine phase

$$q_{g_0}(B) = g_0 \mathsf{C}B - h\widetilde{D}(g_0 \mathsf{C}B), \qquad U_{g_0}(b) = e^{i q_{g_0}(B)(b)} \,. \tag{4.9}$$

The correction $\widetilde{D}\colon \mathfrak{g}^E \to \mathfrak{g}^{E_1}$ is defined near zero by the unique small analytic solution

$$\widetilde{D}(A) = \mathcal{N}(A - h\widetilde{D}(A)), \qquad \widetilde{D}(0) = 0\ . \tag{4.10}$$

Its construction is that of [4] following (2.10). Since $\mathcal{N}(0) = D\mathcal{N}(0) = 0$, its expansion starts at degree two. Equation (4.9) therefore solves the nonlinear averaging constraint and has $q_{g_0}(B) = g_0 \mathsf{C} B + O(g_0^2)$ at fixed B.

Fix $\sigma \in \{\mathrm{op}, \mathrm{HS}\}$ and use the corresponding matrix norm for both group-matrix and Lie-algebra cutoff tests:

$$|X|_{\mathrm{op}} = \sup_{|v|_2=1} |Xv|_2, \qquad |X|_{\mathrm{HS}} = (\mathrm{tr}_N(X^*X))^{\frac{1}{2}}\ . \tag{4.11}$$

Here $|v|_2$ is the Euclidean vector norm. The dependence on the fixed σ is suppressed in ϑ_{g_0} and $\mathcal{W}_{g_0}$ below.

In the all-regular branch, $U_1 = U_{1,\square'} = \mathbf{1}$. [3, (1.19)] removes the approximate relative-field tests on the good fluctuation cubes. The other retained chart tests in [3, (1.25)] have support in $S_1 = \varnothing$. The guard statement following [3, (1.10)] removes the earlier plaquette and gauge cutoffs under its prescribed parameter hierarchy. With that hierarchy retained, [3, (1.18)–(1.19), (1.25)–(1.26)] gives the following bond product when its bond norm is read as $|\bullet|_\sigma$:

$$\vartheta_{g_0}(B) = \prod_{b \in E_*} \mathbf{1}_{\{\,|B(b)|_\sigma < r(g_0)\,\}}, \qquad r(g_0) = A_1(\log g_0^{-2})^{p_0}\ . \tag{4.12}$$

Here $A_1 > 0$ and $p_0 \geqslant 5\mathfrak{r}$ ($\mathfrak{r} \geqslant 2$ the logarithmic exponent of the macro-cube sizes, the convention of Section 3; Section B.8) are the first-step cutoff parameters, and $\delta_0 = g_0 r(g_0)$. We use the inductive normalisation following [3, (3.4)], $\delta_k = g_k A_1/A_0\, p_0(g_k)$ as printed at p.265, together with [3, (2.4)], also at the first step:

$$\begin{aligned} p_0(g_k) &= A_0(\log g_k^{-2})^{p_0}\ , \\ \varepsilon_k &= g_k p_0(g_k)\ , \\ \delta_k &= g_k \frac{A_1}{A_0} p_0(g_k) = \frac{A_1}{A_0}\varepsilon_k\ . \end{aligned} \tag{4.13}$$

For the first step this gives $\delta_0/g_0 = A_1(\log g_0^{-2})^{p_0}$.

For the application to [3], assume its compatible cube partitions and admissible first-step parameters: sufficiently large cutoff constants, sufficiently small A_1/A_0 and coupling, and $p_0 \geqslant 5\mathfrak{r}$, where $\mathfrak{r} \geqslant 2$ is the logarithmic exponent defining the macro-cube sizes (Section 3). Under these hypotheses the guard statement following [3, (1.10)] removes χ_0 and χ_{Ax} in the regular region. The reference integral below is specified separately for each of the two matrix-norm prescriptions; its identification with a conditional kernel uses that same prescription.

Let $\sigma_H(q)$ be the density of normalised Haar measure in exponential coordinates relative to the stated Lie-algebra Lebesgue measure, and put $j_H(q) = \sigma_H(q)/\sigma_H(0)$. Define the Jacobian and residual exponent directly by the measure and action pullback:

$$\begin{aligned} J_{g_0}(B) &= \Big|\det_{\mathfrak{g}^E}\big(I - hD\widetilde{D}(g_0\mathsf{C}B)\big)\Big| \prod_{b\in E} j_H(q_{g_0}(B)(b))\ , \\ \mathcal{P}_{g_0}(B) &= \log J_{g_0}(B) - \frac{1}{g_0^2}\mathscr{A}(U_{g_0}) + \frac{1}{2}\langle B, \mathsf{H}B\rangle\ . \end{aligned} \tag{4.14}$$

The determinant is on the real fine carrier. The nonlinear change of variables is made before the linear constraints are eliminated, as in [3, (1.14)]. The subsequent linear-constraint factor,

Haar density at zero, coupling rescaling, gauge normaliser and vacuum scalar are common to the numerator and denominator in this branch. After the cutoff reductions above, (4.14) retains the action, Haar and Jacobian factors of [3, (1.14)–(1.16)], with the same exponential gauge fixing. Its constant coefficient at $g_0 = 0$ is zero.

The regular sourced ratio is

$$\mathcal{W}_{g_0}(C) = \frac{\int \vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} W_\chi(C; U_{g_0}) \, \mathrm{d}\mu_{\mathsf{K}_B}(B)}{\int \vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} \, \mathrm{d}\mu_{\mathsf{K}_B}(B)} \, . \tag{4.15}$$

Only common scalar prefactors of this same regular integration cancel here. The denominator is not replaced by an independently chosen Gaussian normaliser.

5 The Straight-Edge Coefficient from Gaussian Marginals

The normalised flat integral has a connected expansion. Its extensive contribution along a long straight side defines δm; the contributions remaining near the corners define z_{c}. We construct these terms using Gaussian principal marginals and the local interaction (C.2). Section 7 then compares the straight-side coefficient with the conditional kernel in a proper regular region.

For a square C_n on the compatible finite torus T, the flat reference numerator and denominator are

$$\begin{aligned} N_{n,T} &= \int \vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} W_\chi(C_n; U_{g_0}) \, \mathrm{d}\mu_{\mathsf{K}_B}(B) \, , \\ D_T &= \int \vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} \, \mathrm{d}\mu_{\mathsf{K}_B}(B) \, . \end{aligned} \tag{5.1}$$

These are the identity-background regular integrals. In the two-norm construction below, read the cutoff as $\vartheta^\sigma_{g_0}$ and write $N_{n,T} = N^\sigma_{C_n,T}$, $D_T = D^\sigma_T$ for the chosen prescription σ. They are distinct from the full physical partition function $\mathcal{Z}_0$.

5.1 Connected Terms of the Normalised Integral

Let E_* be the free-bond carrier and let $C_T = \mathsf{K}_B$ on a compatible finite torus T. The flat gap and exponential decay proved in Appendix A give positive constants λ, Λ, A, μ, independent of T, such that

$$\lambda I \leqslant C_T \leqslant \Lambda I, \qquad \|C_T(b,c)\| \leqslant A e^{-\mu d_T(b,c)} \, . \tag{5.2}$$

For positive bonds $b = (x,i)$ and $c = (y,j)$, fix $d_T(b,c) = d_{\infty,T}(x,y) + \mathbf{1}_{\{i \neq j\}}$, where the first term is periodic maximum-norm site distance in fine-lattice units. The same bounds hold for every principal marginal. One may take $\lambda = h_L^{-1}$ and $\Lambda = \gamma_L^{-1}$ from (A.25) and (A.27). The free-inverse Neumann series (A.28) applies on the compatible tori as well, with the same gap and row-sum bound. Let $d_{\mathrm{block}}(b,c)$ be the periodic maximum-norm distance between the coarse indices of the blocks containing the initial sites of b, c. The block-distance decay converts to the fine distance using $d_T(b,c) \leqslant L d_{\mathrm{block}}(b,c) + 2L$; decreasing the decay exponent by the factor L and enlarging A suffices. The metric on each Lie algebra factor is the fixed invariant real metric. All constants below may depend on this metric, G, L, the fixed local reconstruction and χ, but not on the square size or torus volume.

Throughout this construction, $r_0(g_0) = r(g_0) = A_1(\log g_0^{-2})^{p_0}$ is exactly the threshold in (4.12). Restoring the fixed norm index gives the two cutoff prescriptions

$$\vartheta^{\sigma}_{g_0}(B) = \prod_{b\in E_*} \mathbf{1}_{\{\,|B_b|_\sigma < r_0(g_0)\,\}} \,. \tag{5.3}$$

The result holds separately for each prescription, with that cutoff retained in both numerator and denominator. With the bond tests of [3] interpreted in the same norm, the regular-branch reduction above identifies its reference kernel. For the following estimates only the stated norm and its adjoint invariance are used. For unitary u, the required invariance follows from

$$\begin{aligned} |uXu^{-1}|_{\mathrm{op}} &= \sup_{|v|_2=1} |uXu^{-1}v|_2 = \sup_{|w|_2=1} |Xw|_2 = |X|_{\mathrm{op}} \,, \\ |uXu^{-1}|^2_{\mathrm{HS}} &= \mathrm{tr}_N\big(uX^*Xu^{-1}\big) = \mathrm{tr}_N(X^*X) = |X|^2_{\mathrm{HS}} \,. \end{aligned} \tag{5.4}$$

Thus each good- or bad-bond test is invariant for the same fixed norm and threshold; a supremum over the bonds of a cube and products of such tests preserve this invariance. In the regular branch the starred carrier excludes the designated bonds, $A = \mathsf{C}B$ agrees with B on that carrier, and $g_0^{-1}\delta_0 = r_0$ by (4.13). Repeated cube tests reduce to the bond product (4.12) by idempotence, as in the stated regular-branch reduction. This dictionary does not rescale a cutoff ball. The norm-comparison constants used for Gaussian estimates likewise do not change the cutoff set. The Schur argument requires the invariant cutoff itself.

The numerator and denominator under consideration are

$$\begin{aligned} N^{\sigma}_{C,T} &= \int \vartheta^{\sigma}_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} W_\chi(C; U_{g_0})\, \mathrm{d}\mu_{C_T}(B) \,, \\ D^{\sigma}_{T} &= \int \vartheta^{\sigma}_{g_0}(B) e^{\mathcal{P}_{g_0}(B)}\, \mathrm{d}\mu_{C_T}(B) \,, \qquad \mathcal{W}^{\circ,\sigma}_T(C) = N^{\sigma}_{C,T}/D^{\sigma}_T \,. \end{aligned} \tag{5.5}$$

For the proof, fix σ and suppress its index on the cutoffs, integrals and cluster weights. Write $r = r_0(g_0)$ and $\vartheta_b = \mathbf{1}_{\{|B_b|_\sigma < r\}}$. Let $\mathcal{V}_T$ index the local terms v in (C.2), with supports Λ_v. Define bounded measurable functions on the full real carrier by

$$f_v(B) = \left(\prod_{b\in\Lambda_v} \vartheta_b(B)\right)(e^{v(B)} - 1), \qquad h_b(B) = -(1 - \vartheta_b(B)) \,. \tag{5.6}$$

The first function is zero where its cutoff vanishes, so the local analytic function is never evaluated outside its chart. On the original cutoff the exact identity is

$$\vartheta_{g_0}(B) e^{\mathcal{P}_{g_0}(B)} = \prod_{b\in E_*} (1 + h_b(B)) \prod_{v\in\mathcal{V}_T} (1 + f_v(B)) \,. \tag{5.7}$$

For a square $C = (b_a)_{a\in\mathcal{I}_C}$, let $\mathcal{I}_C$ be its ordered finite traversal set and let $U_0 = \mathbf{1}$. For $R \subseteq \mathcal{I}_C$, let $H_R(B)$ be the ordered transport around C using $U_{g_0}(b_a)$ on R and $U_0(b_a)$ elsewhere. The closed source differences are

$$G_S(B) = \sum_{R\subseteq S} (-1)^{|S|-|R|} \chi\left(H_R(B)(\mathrm{Hol}_{U_0} C)^{-1}\right), \qquad \sum_{S\subseteq\mathcal{I}_C} G_S(B) = W_\chi(C; U_{g_0}) \,. \tag{5.8}$$

The last equality is Boolean inversion and uses $\mathrm{Hol}_{U_0} C = \mathbf{1}$. Every summand is a closed contraction, invariant under simultaneous gauge transformation of field and reference. The identity-gauge representative is used to evaluate its flat reference integral. A selected traversal a has its fixed local free-input support Λ_a. Multiplying G_S by $\prod_{a\in S}\prod_{b\in\Lambda_a}\vartheta_b$ leaves its

contribution in the original cutoff integral unchanged. Define this product to be zero off that protected box, evaluating only the reconstruction rows selected by the mixed word.

A label selection is $J = (D, P, S)$, with $D \subseteq \mathcal{V}_T$, $P \subseteq E_*$ and S a subset of contour traversals. Its active carrier and scalar local factor are

$$\begin{aligned} A_J &= \bigcup_{v \in D} \Lambda_v \cup P \cup \bigcup_{a \in S} \Lambda_a \ , \\ F_J(B) &= \prod_{v \in D} f_v(B) \prod_{b \in P} h_b(B) \prod_{a \in S} \prod_{b \in \Lambda_a} \vartheta_b(B) \ . \end{aligned} \tag{5.9}$$

The corresponding numerator term is exactly $\int F_J G_S \, \mathrm{d}\mu_{(C_T)_{A_J}}$. This is a marginal of the original Gaussian: its coordinates in $E_* \setminus A_J$ have been integrated out, while every interaction and cutoff is still present in the sum over J. No retained exterior field is given a new law. There are fixed integers s_*, q_* and a fixed range R_* such that every label support has at most s_* bonds and diameter at most R_*, and at most q_* possible labels meet any bond. Enlarge R_* so that every bond in Λ_a is within R_* of the source traversal a; for a reversed traversal use the free-input stencil of its positive representative. These are consequences of the bounded overlap of the primitive supports and the fact that a square has no repeated traversal. The source cutoff is inserted before expanding the full product of cutoff factors in (5.7); an individual expanded term need not satisfy the full cutoff.

For a finite carrier A, put $C_A = (C_T)_A$. Introduce one scalar variable $x_{bc} = x_{cb}$ for each pair of distinct bonds of A, and define $C_A(x)$ by keeping its diagonal blocks and multiplying its bc block by x_{bc}. For a forest F on A and parameters $w \in [0,1]^F$, let w^F_{bc} be the minimum of the parameters on the unique forest path from b to c, or zero if there is no path; its diagonal is one. Put $\partial_F = \prod_{\{b,c\} \in F} \partial_{x_{bc}}$. The Taylor forest formula gives

$$\int F_J G_S \, \mathrm{d}\mu_{C_{A_J}} = \sum_{F \text{ forest on } A_J} \int_{[0,1]^F} \int F_J(B) G_S(B) \, \partial_F p_{C_{A_J}(x)}(B) \Big|_{x = w^F} \, \mathrm{d}B \, \mathrm{d}w \ . \tag{5.10}$$

For a covariance on $\mathfrak{g}^A$, the density used here is

$$p_M(B) = (2\pi)^{-d|A|/2} (\det M)^{-1/2} \exp\left(-\frac{1}{2} B^{\mathsf{T}} M^{-1} B \right) \ . \tag{5.11}$$

The determinant square root is positive for positive real M and is continued analytically on the accretive covariance domain. Let P_π be the matrix whose bc entry is one when b, c belong to the same block of a partition π of A, and zero otherwise. Let $\mathcal{K}_A$ be the convex hull of these partition matrices. If π_t is the partition into components of the forest edges with $w_e \geqslant t$, then $w^F = \int_0^1 P_{\pi_t} \, \mathrm{d}t \in \mathcal{K}_A$. Block restriction and convexity give $\lambda I \leqslant C_A(x) \leqslant \Lambda I$ throughout $\mathcal{K}_A$.

For the fixed integrand $F_J G_S$, its Gaussian integral is smooth on the open set $\{x \,|\, C_A(x) > (\lambda/2) I\}$: the protected local factors are bounded, and derivatives of the density are polynomials times a uniformly integrable Gaussian on compact subsets. This set contains the compact set $\mathcal{K}_A$. A smooth multiplier with compact support inside this open set and equal to one near $\mathcal{K}_A$ gives a smooth extension by zero to the full interpolation space. The Taylor forest formula applies to this extension. At the all-one matrix and at every w^F, its value and all the required derivatives agree with those of the original Gaussian integral. The extension therefore gives exactly (5.10); its cutoff and derivatives enter none of the estimates. No hard cutoff or Wilson function is differentiated. This is the Taylor forest construction of Abdesselam–Rivasseau, Theorems III.1 and IV.5 and Section V.A [2].

Join two bonds also when they belong to the same selected label support, and denote this finite-range graph by Q_J. A connected component of $F \cup Q_J$, together with its selected labels and forest edges, is a component label γ. Let $A_\gamma, D_\gamma, P_\gamma, S_\gamma, F_\gamma$ be these restrictions. Define its scalar integration functional by

$$\mathcal{I}_\gamma[H] = \int_{[0,1]^{F_\gamma}} \int F_{J_\gamma}(B) H(B)\, \partial_{F_\gamma} p_{C_{A_\gamma}(x)}(B)\Big|_{x=w^{F_\gamma}} \, \mathrm{d}B \, \mathrm{d}w \, . \tag{5.12}$$

No norm of an open Wilson object is assigned to this functional. The covariance and every differentiated edge in (5.10) are block diagonal across distinct components of $F \cup Q_J$. Their normalised densities, including determinants, consequently factor exactly into (5.12). Every component uses the principal marginal of the same C_T; it is independent of the labels outside its carrier.

At the flat reference C_T commutes with simultaneous constant conjugation. Principal restriction, scalar covariance interpolation, inversion and differentiation preserve this law. Every local scalar factor in F_J is invariant as well. Thus each component integration is equivariant without any assertion about seed-adapted auxiliary inverses. For $S_\gamma \neq \varnothing$, let h_γ be the shortest cyclic interval of traversal positions containing S_γ, resolving ties by the first initial position encountered from the fixed contour base in its orientation. A singleton has an interval of length one. On a line, use the ordinary interval from the first to the last selected traversal. Merge components whose intervals intersect into webs; the web interval is their union, possibly the whole circle. This union is an interval because its interval-intersection graph is connected. Distinct web intervals are disjoint. A web is integrated as the single closed source $G_{\cup_{\gamma\in W} S_\gamma}$ against the product of its component functionals. Its scalar activity is

$$z_W = \left(\bigotimes_{\gamma \in W} \mathcal{I}_\gamma \right) \left[G_{\cup_{\gamma\in W} S_\gamma} \right] \, . \tag{5.13}$$

The vacuum activity is $z_\gamma = \mathcal{I}_\gamma[1]$ when $S_\gamma = \varnothing$. Components inside a web have disjoint carriers. Distinct webs must also have disjoint contour intervals. Vacuum objects are incompatible only through intersection of their carriers with another object's carrier. Independent constant conjugation in the component integrals and irreducibility give the integrated Schur factorisation for distinct webs. It is applied inside the closed Wilson contraction. To specify the logarithm domain, attach a complex variable t_a to each contour traversal and replace G_S by $(\prod_{a\in S} t_a) G_S$. The source polydisc is $|t_a| \leqslant 2$. Its factor $2^{|S|}$ is included in the constants of the source bounds below. At zero source the numerator is D_T, and at $t_a = 1$ it is $N_{C,T}$. The finite forest expansion and the unique component and web partitions consequently give

$$N_{C,T} = \Xi_{C,T}, \qquad D_T = \Xi_{0,T}, \qquad \mathcal{W}_T^\circ(C) = \frac{\Xi_{C,T}}{\Xi_{0,T}} \, . \tag{5.14}$$

Both gases are sums over unordered compatible families of the activities just defined, with the empty family contributing one. There is no additional vacuum ratio to identify.

For the converse in this identification, take a compatible family of vacuum components and webs. Their constituent carriers are disjoint, so their label selections and forests have unambiguous unions. Every support edge and forest edge remains within its specified component. The component decomposition of this union recovers the original components, and the interval-intersection decomposition recovers precisely the webs, since distinct web intervals are disjoint. Thus the reassembly is inverse to the decomposition. If a web interval crosses the contour base, cyclicity of the trace moves that base between two web intervals before the Schur factorisation; the resulting scalar is unchanged.

5.2 The Integrated Density Estimate

Fix a forest on A, let $n = |A|$, and let d_b be its vertex degrees. For an edge $e = \{b, c\}$ with nonzero covariance block, use the complex Cauchy radius

$$r_e = \frac{\lambda}{8 d_b d_c \|C_T(b,c)\|} \, . \tag{5.15}$$

A zero covariance block gives a zero differentiated term. Perturb only the forest variables about w^F. The sum of the norms of the perturbation blocks in row b is at most $\lambda/8$, since $\sum_{c:\{b,c\}\in F} (d_b d_c)^{-1} \leqslant 1$. Thus the covariance perturbation has operator norm at most $\lambda/8$. If M is any covariance on these Cauchy circles, then

$$\operatorname{Re} M \geqslant \frac{7\lambda}{8} I, \qquad \|M\| \leqslant \Lambda + \frac{\lambda}{8}, \qquad \operatorname{Re} M^{-1} \geqslant \frac{7\lambda/8}{(\Lambda + \lambda/8)^2} I = \kappa_G I \, . \tag{5.16}$$

The inverse bound follows from $\operatorname{Re} M^{-1} = M^{-*} (\operatorname{Re} M) M^{-1}$. The determinant branch is continued from the positive real covariance. Its modulus satisfies $|\det M| \geqslant (7\lambda/8)^{dn}$. Consequently

$$|p_M(B)| \leqslant (2\pi)^{-dn/2} (7\lambda/8)^{-dn/2} e^{-\kappa_G \|B\|^2/2} \, . \tag{5.17}$$

This keeps the normalising determinant and supplies a product Gaussian majorant. Norm equivalence and a Gaussian exponential moment give a factor Ce^{-cr^2} for each distinct selected bad bond. Each regular vertex is bounded on its own cutoff by $Cg_0(1+r)^3$, using (C.3). The closed source with q selected traversals has bound $(C_\chi g_0 r)^q$; for $q = 1$ it has the improved bound $C_\chi g_0^2 r^2$, since $\mathrm{d}\chi_{\mathbf{1}} = 0$. These bounds are applied to the closed source inside the density integral.

Cauchy's formula and (5.17) therefore bound the integrated term for a fixed component by

$$K^{|A_\gamma|} \prod_{\{b,c\}\in F_\gamma} \left(\frac{8 d_b d_c}{\lambda} \|C_T(b,c)\| \right) \tau^{|D_\gamma| + |P_\gamma| + |S_\gamma| + \mathbf{1}_{\{|S_\gamma|=1\}}}, \qquad \tau = C \left[g_0 (1+r)^3 + e^{-cr^2} \right] \, . \tag{5.18}$$

Here and below g_0 is small enough that $\tau < 1$ and the original chart condition holds. The same majorants multiply for a web. To justify the extra power for a component with exactly one source mark inside a larger web, integrate that component first inside the closed contraction. Its matrix factor is a scalar by constant conjugation and irreducibility, and its scalar is the closed one-mark character difference. Components with two or more marks already supply two source powers. This uses an exact integrated identity; it assigns no activity norm to an open matrix.

The degree factors in (5.18) are paid by spatial decay. The free-bond lattice has uniformly bounded polynomial volume growth of degree four, including on its finite tori. For d distinct neighbours of a fixed bond, their distances have sum at least $c_1 d^{5/4} - c_2 d$. Hence, for each $\zeta > 0$, a constant K_ζ satisfies

$$d^d \leqslant K_\zeta^d \exp\left(\zeta \sum_{j=1}^{d} d_T(b, c_j) \right) \, . \tag{5.19}$$

Indeed $d \log d$ is bounded by any fixed positive multiple of $d^{5/4}$ plus a linear function of d. Set $0^0 = 1$ here. Taking ζ smaller than $\mu/4$ absorbs $\prod_b d_b^{d_b}$ into part of the covariance decay. The remaining forest edge weights are bounded by $Ke^{-\mu d_T(b,c)/2}$, with a further positive part of the decay available for support-length weights.

Here is the connected summation. On a carrier of n bonds there are at most $2^{q_* n}$ choices of elementary labels. A selection covering that carrier contains at least $\lceil n/s_* \rceil$ labels. The edges of Q_J have bounded range and bounded degree. For fixed J and a fixed graph $G = F \cup Q_J$, the edges of F outside Q_J are determined; its edges inside Q_J have at most $2^{|Q_J|}$ choices. The support counts give $|Q_J| \leqslant q_*(s_* - 1)n/2$. Retain this exponential multiplicity in K_1 below. Each forest edge weight is bounded by the following pair weight, and this weight is at least one on every edge of Q_J. Thus the forest sum is bounded, with that multiplicity, by a sum over connected graphs with pair weights

$$v(b,c) = \mathbf{1}_{\{d_T(b,c) \leqslant R_*\}} + K e^{-\mu d_T(b,c)/4}, \qquad \sup_b \sum_{c \neq b} v(b,c) = V_* < \infty \,. \tag{5.20}$$

The unused graph edges cost at most $\prod_{b<c}(1 + v(b,c)) \leqslant e^{V_* n/2}$. Keeping one spanning tree gives the remaining tree sum. For $n \geqslant 2$ and a fixed root bond b_0, summing its possible distinct vertices gives

$$\sum_{\substack{A \ni b_0 \\ |A| = n}} \sum_{T \text{ tree on } A} \prod_{\{b,c\} \in T} v(b,c) \leqslant \frac{n^{n-2}}{(n-1)!} V_*^{n-1} \,. \tag{5.21}$$

The factorial accounts for unordered nonroot vertices; the bound follows by summing their positions along a rooted labelled tree. The right-hand side is bounded exponentially in n. Combining these payments yields convergent series bounded by constants times

$$\sum_{n \geqslant 1} K_1^n \tau^{\lceil n/s_* \rceil} = O(\tau), \qquad \sum_{n \geqslant 1} K_1^n \tau^{\max(2, \lceil n/s_* \rceil)} = O(\tau^2) \,. \tag{5.22}$$

The first applies to vacuum components and the second to source components, rooted either at a spatial bond or at a specified source traversal. Choose the coupling after the constants so that $K_1^{s_*}\tau$ is sufficiently small. The argument leaves exponential weights in the carrier size and in the sum of covariance-edge lengths.

5.3 Extent Weights and the Square Geometry

Define the extent cost of a component, a web and a cluster respectively by

$$\ell(\gamma) = |A_\gamma| + \sum_{\{b,c\} \in F_\gamma} d_T(b,c), \qquad \ell(W) = \sum_{\gamma \in W} \ell(\gamma), \qquad \ell(\Gamma) = \sum_{p \in \Gamma} \ell(p) \,. \tag{5.23}$$

A vacuum object uses its component cost. Write A_p for an object's carrier and H_p for its contour interval length, with $H_p = 0$ for vacuum objects. On a straight line, connectivity of $F_\gamma \cup Q_{J_\gamma}$ gives $H_\gamma \leqslant c_h \ell(\gamma)$, since the support edges have length at most R_* and the source positions are within R_* of their supports; c_h is fixed at (5.24) below, where the square is treated, and the same constant covers the line, the interval length there being at most the spatial span of the marks plus one. (H_p, the contour interval length, is also called the contact-interval length below.)

Here the square C_n has vertices $0, nLe_1, nL(e_1 + e_2), nLe_2$ in a fixed lift, with all other coordinates zero. Its side length is nL fine bonds, its physical perimeter is $4nL\varepsilon$, and its corners are coarse sites. The finite torus has fine periods LK_i, with $K_i \geqslant 3(n+2)$ and the native grid divisibilities. The limits taken below (the joint limit in n and the periods of (5.39), the torus limit at fixed n and the iterated limit of the corner section) are over such compatible

integers n and periods. Between points of this square the periodic coordinate distances equal their distances in the stated lift. This fixes one coordinate plane, one position and one orientation. An oriented square contour C as chosen at the start of the paper, with coarse corners and sides along two coordinate directions, is a coarse translate of the square with vertices $0, nLe_\mu, nL(e_\mu + e_\nu), nLe_\nu$, traversed in this cyclic order or in the reverse one (the base point does not affect $\chi(\mathrm{Hol})$). The permutation of the coordinate axes carrying (e_μ, e_ν) to (e_1, e_2), with the periods permuted accordingly, is a symmetry of the fine lattice, of the Wilson measure $\rho_0(U)\,\mathrm{d}U$ and of every hypothesis on the periods (all are imposed on every period), so $\mathbb{E}_\varepsilon[W_\chi(C)]$ equals the expectation of a coarse translate of C_n, which is the contour admitted in Section 9.6.1 and in (9.39)(i). Reversing the orientation inverts the holonomy, and $\chi(U^{-1}) = \overline{\chi(U)}$ for the character of a unitary representation, so the expectation of the reversed square is the complex conjugate of that of the original. Every statement below about $\mathbb{E}_\varepsilon[W_\chi(C_n)]$ therefore holds for every such square, up to complex conjugation for the reversed orientation.

Let D be the maximum-norm diameter of a nonempty set of square contacts and put $s = nL$. If $D < s/2$, its contacts occupy one side or two adjacent sides near their common corner, and an arc of length at most $4D + 1$ contains them. If $D \geqslant s/2$, the whole contour has length $4s \leqslant 8D$. A connected support network for the component has length at most $R_*|A_\gamma| + \sum_{F_\gamma} d_T(b, c)$, and each contact is within R_* of that network. With the stated bond-distance convention, $c_h = 32(1 + R_*)$ (the interval constant, italic subscript; distinct from the group constant $c_{\mathrm{h}} = \sqrt{N}$ of Appendix B) therefore gives

$$H_\gamma \leqslant c_h \ell(\gamma), \qquad H_W \leqslant \sum_{\gamma \in W} H_\gamma \leqslant c_h \ell(W)\ . \tag{5.24}$$

Let d_C count the shortest number of traversal steps between two positions, cyclically for a square. For $\theta > 0$, a position a in h_γ satisfies

$$\mathbf{1}_{\{a \in h_\gamma\}} \leqslant \sum_{b \in S_\gamma} e^{-\theta d_C(a,b)} e^{\theta H_\gamma}\ . \tag{5.25}$$

The sum of $e^{-\theta d_C(a,b)}$ over all traversal positions is bounded by $1 + 2/(e^\theta - 1)$, uniformly in the contour length. The already proved traversal-root load, with the extra extent weight, therefore gives an interval-root load of order τ^2.

For a source component let u_γ be the positive majorant on the right-hand side of (5.18), with the constants fixed there, and let $H_\gamma = |h_\gamma|$ be its contact-interval length. To sum webs, majorise the indicator that their interval graph is connected by its spanning-tree sum. The rooted-tree recursion is bounded by $u_\gamma e^{aH_\gamma}$ whenever the interval-root load, with this extra weight, is at most a. This follows by iterating $T_\gamma = u_\gamma \exp(\sum_{\gamma' : h_{\gamma'} \cap h_\gamma \neq \varnothing} T_{\gamma'})$ from $T_\gamma = u_\gamma$, since the sum in the exponent is bounded by aH_γ. Spatial incompatibilities only reduce this majorant. The resulting whole-web loads are $O(\tau^2)$. Together with the vacuum load $O(\tau)$, they satisfy the hard-core polymer convergence criterion [1], with size equal to carrier size plus a small positive multiple of contour interval length. This proves absolute convergence of the connected logarithm of (5.14) for sufficiently small fixed g_0. In particular the numerator has no zero on the source-scaling polydisc used by that expansion. Its logarithm is fixed by continuation from zero source, where numerator and denominator agree.

More explicitly, choose $\beta = \mu/(1024(1+c_h))$ and $\eta = \mu/1024$, and put $a(p) = |A_p| + \beta H_p$. The density and graph payments above remain valid after multiplying the component majorants by $e^{4|A_\gamma| + 8\beta H_\gamma + 8\eta\ell(\gamma)}$: the added edge exponent is smaller than the unused covariance decay,

and the added carrier exponent only changes K_1. Use $a = \beta$ in the web recursion, reserving the remaining weights for its interval-root conversion. It then gives a constant C_* such that

$$\begin{aligned} \sup_b \sum_{p \,|\, b \in A_p} |z_p| e^{2a(p)+2\eta\ell(p)} &\leqslant C_* \tau \ , \\ \sup_a \sum_{W \,|\, a \in h_W} |z_W| e^{2a(W)+2\eta\ell(W)} &\leqslant C_* \tau^2 \ . \end{aligned} \tag{5.26}$$

The source-only version of the first sum is also bounded by $C_*\tau^2$. Choose g_0 after these constants so that $C_*\tau \leqslant 1/4$ and $C_*\tau^2 \leqslant \beta/4$. Carrier intersection and interval intersection then give the criterion

$$\sum_{q \not\sim p} |z_q| e^{a(q)+\eta\ell(q)} \leqslant C_*\tau |A_p| + C_*\tau^2 H_p \leqslant \frac{1}{4} a(p) \ . \tag{5.27}$$

A cluster $\Gamma = (p_1, \dots, p_j)$, with $j \geqslant 1$, is an ordered tuple of vacuum components or source webs with connected incompatibility graph; repetitions are allowed and an object is incompatible with itself. Define its scalar weight by

$$\Phi_C(\Gamma) = \frac{1}{j!} \left(\sum_{H \text{ connected on } \{1,\dots,j\}} \prod_{\{r,s\} \in H} (-\mathbf{1}_{\{p_r \not\sim p_s\}}) \right) \prod_{r=1}^{j} z_{p_r} \ . \tag{5.28}$$

For precision, the pinned version used here retains an activity factor. Put $\widetilde{z}_p = z_p e^{\eta\ell(p)}$, and let $m_p(\Gamma)$ count the occurrences of p in the ordered cluster tuple. Then

$$\sum_{\Gamma} m_p(\Gamma) |\Phi_C(\Gamma)| e^{\eta\ell(\Gamma)} \leqslant |z_p| e^{\eta\ell(p)+a(p)} \ . \tag{5.29}$$

Here is a direct justification. For a fixed incompatibility graph on labelled vertices, partition its connected subgraphs by their breadth-first spanning tree rooted at vertex one, choosing the least-labelled parent at the previous level. A tree's interval of subgraphs allows all edges within a level and the edges to the previous level whose other endpoint is no earlier than the chosen parent. These intervals partition the connected graphs. The alternating sum on each interval is zero or has modulus one. Therefore the absolute Ursell coefficient is bounded by the sum over its spanning trees. The positive generating sum of rooted trees with root species p and vertex weights $|\widetilde{z}_q|$ is the monotone limit of

$$T_p^{(0)} = |\widetilde{z}_p|, \qquad T_p^{(k+1)} = |\widetilde{z}_p| \exp\left(\sum_{q \not\sim p} T_q^{(k)} \right) \ . \tag{5.30}$$

Permutation symmetry of the ordered tuple gives $\sum_\Gamma m_p(\Gamma) F(\Gamma) = j \sum_{\Gamma \,|\, p_1 = p} F(\Gamma)$ at each fixed length j for its symmetric weight F. Hence the factor $1/j!$ becomes $1/(j-1)!$, including repeated species. The exponential in (5.30) assembles an unordered family of child trees, with its factorial for repeated choices; $T_p^{(k)}$ enumerates rooted trees of height at most k. Equation (5.27) implies inductively $T_p^{(k)} \leqslant |\widetilde{z}_p| e^{a(p)}$, proving (5.29). For a countable polymer family, first restrict the species to a finite subset. The same upper bound holds for every such restriction, and monotone convergence of the positive majorants proves the pinned bound for the countable family. No total infinite-volume vacuum partition function is asserted.

Apply this bound to source objects meeting a fixed coarse contour segment x. Their weighted activity sum is bounded by the source-only loads in (5.26). Thus

$$\sum_{\Gamma \,|\, x \in J_\Gamma} |\Phi_C(\Gamma)| e^{\eta\ell(\Gamma)} \leqslant C\tau^2 \ . \tag{5.31}$$

Here J_Γ consists of the coarse segments containing its source marks. A coarse segment contains L fine traversal positions, which changes only the constant. The same proof applies on the infinite lattice, using its finite principal covariance marginals.

A spanning tree of $F_\gamma \cup Q_{J_\gamma}$ connects each component carrier. Its metric length is at most $R_*|A_\gamma| + \sum_{F_\gamma} d_T(b,c)$. Paths of length at most R_* attach its source contacts; there are at most $q_*|A_\gamma|$ contacts. A web becomes connected after adjoining its contour intervals, whose total length is at most $c_h \ell(W)$ by (5.24). In a cluster, incompatible objects meet through carriers or contour intervals, so these networks form a connected union. The carrier bonds themselves add at most $|A_\gamma|$ per component, and conversion from maximum-norm distances to lattice path lengths costs at most four. Thus its length and diameter are bounded by $C_{\mathrm{net}}\ell(\Gamma)$ with $C_{\mathrm{net}} = 8(1+R_*)(1+q_*+c_h)$. This remains an upper bound with repeated objects. Equation (5.31) therefore controls spatial extent as well as contour span.

5.4 The Perimeter Coefficient

Use the infinite-lattice covariance constructed from the same periodic block grid. Its finite principal marginals satisfy the same spectral and decay bounds, so the density, counting and weighted KP estimates above apply with the same constants. For an orientation $v \in \mathscr{V}_\square = \{e_1, e_2, -e_1, -e_2\}$, use the straight line through the coarse origin in that orientation and divide it into half-open segments of L fine traversals, indexed by $\mathbb{Z}$. Every side of C_n is a coarse translation of the corresponding oriented segment of this line. Each square traversal belongs to exactly one side and one such segment. The lattice spacing ε is the fixed physical length of a fine bond.

Write Φ_v for the connected weight (5.28) on the straight line in orientation v. For a finite set S of traversal positions on the line, take any finite oriented arc I containing it. For $R \subseteq S$, let $u_a^R = U_{g_0}(b_a)$ on R and $u_a^R = U_0(b_a)$ otherwise, with $U_0 = \mathbf{1}$. The closed relative source on the line is explicitly

$$G_S^v(B) = \sum_{R \subseteq S} (-1)^{|S|-|R|} \chi\left(\left(\overrightarrow{\prod_{a \in I}} u_a^R(B) \right) P_{U_0}(I)^{-1} \right) . \tag{5.32}$$

The arrow means order along the arc. The reference reverse transport closes the word, and extending the arc beyond its selected marks adds only identity factors in the reference gauge. This is the line version of the finite closed source definition used above. Let $J_\Gamma \subset \mathbb{Z}$ be the set of coarse line segments containing at least one source mark of the cluster. The vacuum subtraction in $\log \Xi_C - \log \Xi_0$ retains exactly the clusters with $J_\Gamma \neq \varnothing$. Translation by one coarse segment preserves the reference construction. Define the absolutely convergent line density and its coefficient by

$$\boxed{\kappa_v(g_0) = \sum_{\Gamma \,|\, 0 \in J_\Gamma} \frac{\Phi_v(\Gamma)}{|J_\Gamma|}, \qquad \delta m_v(g_0, \varepsilon) = -\frac{\kappa_v(g_0)}{L\varepsilon}} . \tag{5.33}$$

The factor $|J_\Gamma|^{-1}$ allocates each translated connected contribution once. The weights contain the full local exponentials and exact hard-cutoff factors. The weighted rooted estimate (5.31) proves absolute convergence and $|\kappa_v| \leqslant C\tau^2$.

Length Allocation and Corners. Put $I_m = \{0, \dots, m-1\}$. The infinite-lattice logarithm for the finite straight relative word of length mL is

$$F_v(m) = \sum_{\Gamma \,|\, \varnothing \neq J_\Gamma \subseteq I_m} \Phi_v(\Gamma) \,. \tag{5.34}$$

It is the bulk limit of the logarithm of the same finite-word numerator divided by D_T. For fixed m, this follows by truncating (5.31), taking the limit of finitely many covariance-marginal integrals, and then removing the truncation. No infinite-volume probability measure is used. Translation invariance gives the exact difference

$$F_v(m) - m\kappa_v = -\sum_{x \in I_m} \sum_{\substack{\Gamma \,|\, x \in J_\Gamma \\ J_\Gamma \not\subseteq I_m}} \frac{\Phi_v(\Gamma)}{|J_\Gamma|} \,. \tag{5.35}$$

A cluster in the inner sum joins a source in segment x to a source outside I_m. With $R_{\mathrm{end}} = 2L + 4R_*$, the network bound gives $C_{\mathrm{net}}\ell(\Gamma) \geqslant (L \min(x+1, m-x) - R_{\mathrm{end}})_+$, where $t_+ = \max(t, 0)$. Applying (5.31) and summing that geometric tail yields

$$|F_v(m) - m\kappa_v| \leqslant C\tau^2 \,. \tag{5.36}$$

For the square in the infinite-lattice reference, clusters whose source marks all lie on one side have exactly the corresponding line weights: their contact intervals stay on that side, their closed words have the same ordered nonidentity factors, and every local primitive is translated by a multiple of L. The Gaussian marginal is unchanged by this translation. All other source clusters touch at least two sides. Allocate each such cluster over its source segments using $|J_\Gamma|^{-1}$. A root segment at distance j from its nearest corner can be joined to another side only by a cluster with $C_{\mathrm{net}}\ell(\Gamma) \geqslant (Lj - R_{\mathrm{end}})_+$. Thus the sum over mixed-side clusters is bounded by $C\tau^2 \sum_{j \geqslant 0} e^{-c'j}$, with $c' = \eta L / C_{\mathrm{net}}$ (the extent weight $e^{\eta\ell(\Gamma)}$ of (5.31); the factor $e^{\eta R_{\mathrm{end}}/C_{\mathrm{net}}}$ is absorbed in C). With $\widehat{F}_\square(n)$ denoting its absolutely convergent infinite-lattice connected sum, this proves

$$\left| \widehat{F}_\square(n) - \sum_{v \in \mathscr{V}_\square} F_v(n) \right| \leqslant C\tau^2, \qquad \left| \widehat{F}_\square(n) - n \sum_{v \in \mathscr{V}_\square} \kappa_v \right| \leqslant C\tau^2 \,. \tag{5.37}$$

This retains the bounded corner contribution, whose extraction is carried out in Section 6.

The Buffered Finite-Torus Limit. The perimeter coefficient also follows along the stated buffered tori, without a separate uniform norm on differences of all activities. Fix an integer extent cutoff R. A cluster rooted at a specified source segment with $\ell(\Gamma) \leqslant R$ has all its sites and marks in a neighbourhood of radius $C_{\mathrm{net}}R + L + 2R_*$ in fine-lattice units around that segment. There are finitely many such cluster shapes, including multiplicities: each object has extent at least one, so a tuple has at most R objects. Away from the square corners by more than $\lceil \frac{C_{\mathrm{net}}R + 2L + 2R_*}{L} \rceil$ coarse segments, these are the same shapes as on the corresponding straight line.

Translate each such root to segment zero. Let $\epsilon_R(T)$ be the sum, over the four orientations and these finitely many shapes, of $|\Phi_v^T(\Gamma) - \Phi_v(\Gamma)|/|J_\Gamma|$. The finite-torus weight here is the same finite Gaussian-marginal integral on the translated sites. For large enough periods the shape embeds without wraparound. Periodisation gives entrywise convergence of all its covariance blocks, uniformly under the allowed coarse translations. The accretive density

bound supplies a common integrable majorant for its Cauchy derivatives, so each integral and each finite product defining the cluster weight converges. Consequently $\epsilon_R(T) \to 0$ as the shortest period tends to infinity, for every fixed R.

Allocate the finite-square logarithm over its source segments. The clusters omitted by the extent truncation cost at most $C\tau^2 e^{-\eta R}$ per root. There are at most cR corner-near roots per side, and each has load at most $C\tau^2$. Thus

$$\left| \frac{1}{4n} \log \frac{N_{C_n,T}}{D_T} - \frac{1}{4} \sum_{v \in \mathscr{V}_\square} \kappa_v \right| \leqslant C\tau^2 \left(\frac{R}{n} + e^{-\eta R} \right) + \epsilon_R(T) \, . \tag{5.38}$$

First let n and the compatible buffered periods tend to infinity with R fixed, then let $R \to \infty$. Every logarithm in this limit is the analytic branch continued along $t_a = t$, $0 \leqslant t \leqslant 1$, from the value zero of the logarithm of the ratio at $t = 0$. Restore the cutoff index: κ_v^σ is the absolutely convergent line sum (5.33) with the prescription (5.3). For each $\sigma \in \{\mathrm{op}, \mathrm{HS}\}$, the flat square perimeter coefficient exists and is

$$\boxed{\delta m^\sigma(g_0, \varepsilon) = -\frac{1}{4L\varepsilon} \sum_{v \in \mathscr{V}_\square} \kappa_v^\sigma(g_0) = -\lim_{n \to \infty} \frac{1}{4nL\varepsilon} \log \frac{N^\sigma_{C_n,T_n}}{D^\sigma_{T_n}}} \, . \tag{5.39}$$

No equality of the full coefficients in different orientations is presumed. Reversing a finite relative path conjugates its character, and translation by its coarse length identifies it with the oppositely oriented reference path. All other factors of the weights (the Gaussian density and its forest derivatives, f_v, h_b and θ) are real, the reindexing $a \mapsto -a-1$ carries the $-v$ line onto the reverse of the v line with the stencils, protected boxes, intervals and incompatibilities in bijection, and κ_v is an absolutely convergent sum, so $\kappa_{-v} = \overline{\kappa_v}$. Hence the square coefficient is real.

6 The Corner Coefficient in the Same First Step

Fix the lattice spacing, character, reconstruction and cutoff prescription σ as in Section 5. Choose the coupling below the common minimum of the straight and bent-path smallness thresholds of this section and the regional thresholds of Proposition 7.3, after fixing all chart radii and majorant constants. All limits in this section vary reference contours and enclosing tori at this one RG step; neither the coupling nor the RG scale is iterated. The coefficients below use exactly $N^\sigma_{C,T}$ and D^σ_T in (5.5). Suppress σ until the conclusion. Write $C_\infty = \mathsf{H}_\infty^{-1}$ for the free-field covariance on $\ell^2(E_{*,\infty}; \mathfrak{g})$, constructed in (A.28); it is distinct from the reconstructed fine-field covariance $\mathsf{K}_\infty = \mathsf{C}_\infty C_\infty \mathsf{C}_\infty^*$. The countable primitive index family $\mathcal{V}_\infty$ consists of labels $(0, \alpha)$, $(1, b)$ and $(2, c)$ for the functions v_α, j_b, d_c in (C.1). Here b ranges over positive fine bonds, c over positive coarse bonds, and α over positive plaquettes and the mean-transport pairs (y, x) with $y \in L\mathbb{Z}^4$ and $x \in \mathcal{B}_y \setminus \{y\}$. These are exactly the infinite-lattice translates of the finitely many local templates modulo the coarse grid.

6.1 A Local Corner Subtraction

Let $v, w \in \mathscr{V}_\square$ be perpendicular incoming and outgoing orientations. For $a \in \mathbb{Z}$, the bent reference path has fine vertices and oriented bonds

$$x_a^{v,w} = \begin{cases} av, & a < 0, \\ aw, & a \geqslant 0, \end{cases} \qquad b_a^{v,w} = (x_a^{v,w}, x_{a+1}^{v,w}) \, . \tag{6.1}$$

Its coarse segment $j \in \mathbb{Z}$ contains the fine traversal indices $jL, \dots, (j+1)L-1$. Thus the bend is at a coarse site, between segments -1 and 0.

For every finite set of selected traversals, use the closed relative source (5.32) on the corresponding finite arc of this bent path. Keep the same local vertices, protected cutoffs and Gaussian principal marginals. Define its components and webs by (5.12)–(5.13), with ordinary intervals in the ordered traversal indices, and its connected weight $\Phi_{v,w}(\Gamma)$ by (5.28). The set $J_\Gamma \subset \mathbb{Z}$ again records the coarse segments containing source marks. Vacuum-only clusters are excluded. These definitions involve finite closed contractions and finite-dimensional integrals only, even on the infinite reference path.

The density, label-counting and KP estimates of Section 5 apply to this path with enlarged fixed constants. Indeed, it has no repeated traversal, and the path distance between two of its points is at most twice their maximum-norm spatial distance. Each ray is part of one of the two straight lines, so at most $2q_*$ primitive or source labels meet any bond; the support sizes s_* and ranges R_* are unchanged. Thus the original counting proof applies with $q'_* = 2q_*$, and the interval/extent bound holds with $c'_h = 2c_h$. The resulting rooted estimate is

$$\sum_{\Gamma \,|\, j \in J_\Gamma} |\Phi_{v,w}(\Gamma)| e^{\eta' \ell(\Gamma)} \leqslant C\tau^2 \,. \tag{6.2}$$

Take $\beta' = \mu/(2048(1+c'_h))$ and $\eta' = \eta/2$. The index-distance conversion (5.25) is unchanged, since the traversal indices still form $\mathbb{Z}$ and the sum of its geometric kernel is $1 + 2/(e^\theta - 1)$. The carrier-root and interval-root loads therefore obey (5.26) with the bent path, size $a'(p) = |A_p| + \beta' H_p$, exponent η' and a fixed enlarged constant C'_* obtained from the same label and positive-tree sums with q'_*. Choose the coupling so that $C'_*\tau \leqslant 1/4$ and $C'_*\tau^2 \leqslant \beta'/4$. The incompatibility sum is then at most $a'(p)/4$, and the pinned proof (5.29)–(5.30) gives (6.2). Both cutoff choices satisfy this argument. No open transport receives an activity norm: the relative word transforms by conjugation at its base when field and reference are transformed together, and its character, component-integrated closed contractions and scalar cluster weights are gauge invariant.

Define the rooted density and its straight reference by

$$a_{v,w}(j) = \sum_{\Gamma \,|\, j \in J_\Gamma} \frac{\Phi_{v,w}(\Gamma)}{|J_\Gamma|}, \qquad k_{v,w}(j) = \begin{cases} \kappa_v, & j < 0, \\ \kappa_w, & j \geqslant 0. \end{cases} \,. \tag{6.3}$$

The straight densities are exactly (5.33), not newly fitted constants. The proposed local corner coefficient is

$$\boxed{\zeta_{v,w} = \sum_{j \in \mathbb{Z}} \left(a_{v,w}(j) - k_{v,w}(j)\right), \qquad z_{v,w} = \exp(\zeta_{v,w})} \,. \tag{6.4}$$

Here is the cancellation and the quantitative payment for this subtraction. A cluster with all source marks at negative indices has exactly the weight of the same cluster on the incoming straight line: its ordered nonidentity factors, source stencils, ordinary intervals, local vacuum factors and Gaussian marginals agree. The same holds on the positive half-line with orientation w. Vacuum carriers may cross the bend; they are not cut or discarded in this comparison. Consequently only clusters with source marks on both sides of index zero survive the difference of the rooted sums. On the straight side of the subtraction these are the straight clusters straddling the same cut. A surviving cluster rooted in segment j has extent at least $c|j| - C_0$, with $c = L/(2C_{\mathrm{net}})$ and $C_0 = R_*/C_{\mathrm{net}}$, since its connected support/interval

network reaches the other half-line, so has length at least $L|j| - 2R_*$, while the bent-path network constant (C_{net} with q'_*, c'_h) is at most $2C_{\mathrm{net}}$. The rooted bound, with part of its extent weight retained, therefore gives

$$|a_{v,w}(j) - k_{v,w}(j)| \leqslant C\tau^2 e^{-c|j|}, \qquad |\zeta_{v,w}| \leqslant C\tau^2, \qquad |z_{v,w} - 1| \leqslant C\tau^2 \,. \tag{6.5}$$

This proves absolute convergence and localisation at the corner, rather than merely bounding a fixed square. As a direct check, for the trivial character every nonempty closed source difference is zero. The source cluster sums, straight densities and corner logarithms then vanish, giving $N_{C,T} = D_T$ and $z_{v,w} = 1$, as required.

An explicit cluster formula makes the endpoint allocations visible. Let $J_\Gamma^- = J_\Gamma \cap \mathbb{Z}_{<0}$ and $J_\Gamma^+ = J_\Gamma \cap \mathbb{Z}_{\geqslant 0}$. In each of the following three sums, the cluster has both $J_\Gamma^- \neq \varnothing$ and $J_\Gamma^+ \neq \varnothing$, and its weight is that of the indicated bent or straight path. Then

$$\begin{aligned} \zeta_{v,w} = {} & \sum_{\Gamma}^{v,w} \Phi_{v,w}(\Gamma) - \sum_{\Gamma}^{v} \frac{|J_\Gamma^-|}{|J_\Gamma|} \Phi_v(\Gamma) \\ & - \sum_{\Gamma}^{w} \frac{|J_\Gamma^+|}{|J_\Gamma|} \Phi_w(\Gamma) \,. \end{aligned} \tag{6.6}$$

Indeed, summing the rooted allocations gives one for a mixed bent cluster and the displayed fractions for the two missing straight half-line contributions. Each sum is absolutely convergent by (6.2) and the distance-to-cut payment. In particular the corner factor includes the straight-edge endpoint deficits as well as interactions joining the two perpendicular rays.

6.2 Identification with the Square Constant

First make the fixed-square bulk limit explicit. For a fixed positive integer n, let $\widehat{F}_\square(n)$ be the infinite-lattice connected sum over clusters with source marks on C_n, using its fixed cyclic traversal order and base. Write $F_{\square,\infty}(n)$ for the bulk-limit logarithm whose existence will be proved below. Allocate the connected sum over the $4n$ coarse source segments by $|J_\Gamma|^{-1}$. Its absolute convergence follows from (5.31). The same bound holds uniformly on all compatible tori, which satisfy $K_i \geqslant 3(n+2)$, so the square geometry and weighted KP payment used to prove that bound have their original constants, including at the corners.

Truncate both connected sums to $\ell(\Gamma) \leqslant R$. Once the shortest fine period exceeds a fixed multiple of $nL + R + R_*$, each such rooted cluster has a unique lift through its source root to the fixed neighbourhood of the lifted square. Its connected support/interval network has length at most a fixed multiple of R, so a network winding around a torus cannot occur in this truncation. The primitive functions and supports agree after lifting. The cyclic interval rule and its tie-breaking base depend only on the fixed ordered list of square traversals, so they agree as well, including intervals crossing the base. There are finitely many lifted shapes, labels and multiplicities: the root belongs to a finite contour, the extent bounds all spatial displacements and the number of component and cluster occurrences, and each site has finitely many primitive labels. Covariance entries converge by the periodisation of H_∞^{-1} (Appendix A: the periodisation of the free inverse solves the torus inverse equation, the paragraph preceding (A.31), and the image bound (A.33), displayed there for K, holds for H_∞^{-1} by the same argument from the decay of its entries established before (A.30)). For each shape, the differentiated Gaussian density majorant (5.17) is integrable and uniform in the torus. Dominated convergence therefore gives convergence of every component integral, web

activity and finite product in its cluster weight. Let $\mathscr{G}_{n,R}$ be this finite set of lifted ordered clusters with at least one source mark. Define

$$\epsilon_{n,R}(T) = \sum_{\Gamma \in \mathscr{G}_{n,R}} |\Phi^T_\square(\Gamma) - \Phi^\infty_\square(\Gamma)| \, . \tag{6.7}$$

For the domination just used, every fixed shape has $r_e^{-1} \leqslant 8 d_b d_c A/\lambda$ in (5.15), uniformly in T. Thus the Cauchy factors multiplying (5.17) are uniformly bounded for that shape. The differentiated density is analytic, and hence continuous, in its covariance entries on a common accretive neighbourhood. This gives pointwise convergence as well as the stated integrable majorant, including when a covariance edge tends to zero. It follows that $\epsilon_{n,R}(T) \to 0$ at fixed n, R.

The omitted clusters, including any winding clusters, have total allocated weight at most $Cn\tau^2 e^{-\eta R}$ in each sum. Consequently

$$\left| \log \frac{N_{C_n,T}}{D_T} - \widehat{F}_\square(n) \right| \leqslant Cn\tau^2 e^{-\eta R} + \epsilon_{n,R}(T) \, . \tag{6.8}$$

Here the finite-torus logarithm is the connected expansion of the ratio, on its branch continued from zero source; the vacuum subtraction and original denominator are unchanged. First taking $T \to \infty$ at fixed n, R and then $R \to \infty$ proves

$$F_{\square,\infty}(n) = \widehat{F}_\square(n) = \lim_{T \to \infty} \log \frac{N_{C_n,T}}{D_T} \, . \tag{6.9}$$

Thus the infinite connected sum and the bulk-limit logarithm used below are the same object. This argument needs no convergence of an infinite-volume vacuum partition function.

List the four incoming/outgoing pairs of the counterclockwise square as

$$\mathscr{C}_\square = \{(e_1, e_2), (e_2, -e_1), (-e_1, -e_2), (-e_2, e_1)\} \, . \tag{6.10}$$

Use the branch of $F_{\square,\infty}(n)$ continued from zero source. The same closed-cluster allocation now gives positive constants C, c, independent of n, such that

$$\left| F_{\square,\infty}(n) - n \sum_{v \in \mathscr{V}_\square} \kappa_v - \sum_{(v,w) \in \mathscr{C}_\square} \zeta_{v,w} \right| \leqslant C\tau^2 (n+1) e^{-cn} \, . \tag{6.11}$$

To verify this estimate, allocate every square cluster over its source segments by $|J_\Gamma|^{-1}$. Divide each side into the segments nearest its initial corner and those nearest its terminal corner, breaking a midpoint tie consistently. A corner then receives the last $\lfloor n/2 \rfloor$ incoming segments and the first $n - \lfloor n/2 \rfloor$ outgoing segments. Translate that corner to the coarse origin. At the base corner $(-e_2, e_1)$, first use trace cyclicity to put the incoming factors before the outgoing factors, exactly as in the cyclic reassembly after (5.14). For a root in its assigned half-sides, every cluster of extent less than $c_1 n - C_1$ is identical to the corresponding bent-path cluster: all marked traversals and their stencils stay away from the other three corners, cyclic intervals become the ordinary local intervals, and the same global Gaussian principal marginals and vacuum labels occur. Any failure of this identification costs extent at least $c_1 n - C_1$ and hence at most $C\tau^2 e^{-c_2 n}$ per root by (5.31) and (6.2). There are $4n$ roots. Extending the four truncated rooted differences to all $j \in \mathbb{Z}$ costs another $C\tau^2 e^{-c_3 n}$ by (6.5). This proves (6.11), after enlarging constants for the finitely many short squares. Here $c_1 = L/(4C_{\mathrm{net}})$ and $C_1 = 2R_*/C_{\mathrm{net}}$ (a cluster rooted in its assigned half-sides and reaching within R_* of another corner has network length at least $nL/2 - 2R_*$, and the bent-path

network constant is at most $2C_{\mathrm{net}}$), $c_2 = \eta' c_1$ and $c_3 = c/2$ with the c of (6.5) (the extension runs over $|j| \geqslant n/2$), the factors $e^{\eta' C_1}$ and the like being absorbed in C.

Restore the cutoff index and define the square corner matching factor by

$$\boxed{z_{\mathrm{c}}^{\sigma}\Big(g_0, \frac{\pi}{2}\Big) = \exp\left(\frac{1}{4} \sum_{(v,w)\in\mathscr{C}_{\square}} \zeta_{v,w}^{\sigma}(g_0)\right)}. \tag{6.12}$$

This definition fixes the fourth-root branch by the exponential of the convergent corner logarithm. It is the geometric mean of the four oriented corner factors. Equality of those four factors is not needed or assumed; if the reference construction preserves the corresponding rotations, they agree and (6.12) is their common value. For a real-valued character all the real Gaussian integrals, cluster weights and corner logarithms above are real, so these factors are positive. No reality symmetry for a general complex character is imposed.

For each fixed n, the inner limit in the following expression is the limit over compatible tori with their shortest period tending to infinity. Then

$$\boxed{\log z_{\mathrm{c}}^{\sigma}\Big(g_0, \frac{\pi}{2}\Big) = \frac{1}{4} \lim_{n\to\infty} \left(\lim_{T\to\infty} \log \frac{N_{C_n,T}^{\sigma}}{D_T^{\sigma}} + 4nL\varepsilon\delta m^{\sigma}\right)}. \tag{6.13}$$

The denominator is the original vacuum integral throughout the inner limit, whose existence is (6.9). The outer limit follows from (6.11) and (5.39). The order of limits is part of this reference prescription; a constant-term limit on every diagonally growing finite torus is not asserted here.

Equivalently, write $\mathcal{W}_{\infty}^{\circ,\sigma}(C_n) = \lim_{T\to\infty} N_{C_n,T}^{\sigma}/D_T^{\sigma}$. There is an error E_n^{σ} such that

$$\begin{aligned} \mathcal{W}_{\infty}^{\circ,\sigma}(C_n) &= e^{-4nL\varepsilon\delta m^{\sigma}} \Big(z_{\mathrm{c}}^{\sigma}\Big(g_0, \frac{\pi}{2}\Big)\Big)^4 e^{E_n^{\sigma}}, \\ |E_n^{\sigma}| &\leqslant C\tau^2(n+1)e^{-cn}. \end{aligned} \tag{6.14}$$

The corner coefficient uses the same flat integral and cutoff prescription as the perimeter coefficient. Its identification in a conditional regular region, with boundary and periodic errors, is Proposition 7.3.

7 Identification in a Regular Integration Region

The coefficient in (5.33) was computed with the translation-invariant flat covariance. A conditional integration in the first RG step instead uses the inverse of a restricted Hessian. We compare the two connected expansions, retaining the action terms that cross the integration boundary. This provides a quantitative identification of their straight-side coefficients.

7.1 The Wilson Source before the Flat Evaluation

Let $\mathcal{I}_C$ be the ordered set of traversals of C. For $R \subseteq \mathcal{I}_C$, the mixed field U^R uses the integration link $U(b) = U'(b)U_1(b)$ on a traversal in R and the background link $U_1(b)$ on every other traversal. Here $U_1 = U_1(\Omega_1, V)$ is the reconstruction before the boundary split

(with the exterior determining field retained). Reversed traversals use inverse links. Define

$$\begin{aligned} \mathcal{G}_S(U,U_1) &= \sum_{R\subseteq S} (-1)^{|S|-|R|}\chi(\mathrm{Hol}_{U^R}\, C)\ , \\ \mathcal{W}(t;U,U_1) &= \sum_{S\subseteq \mathcal{I}_C} \left(\prod_{a\in S} t_a\right) \mathcal{G}_S(U,U_1)\ . \end{aligned} \tag{7.1}$$

Here $t=(t_a)_{a\in\mathcal{I}_C}$ is a family of scalar parameters. Boolean inversion gives

$$\mathcal{W}(1;U,U_1)=W_\chi(C;U), \qquad \mathcal{W}(0;U,U_1)=W_\chi(C;U_1)\ . \tag{7.2}$$

Every term is a closed contraction, invariant under simultaneous gauge transformations of U and U_1.

The Wilson insertion starts in (1.3). Its gauge invariance preserves it under the Faddeev–Popov insertion in [3, (1.5)–(1.6)]. It is then pulled back under $U=U'U_1$ and the constraint linearisation leading to (1.14). At that stage we introduce (7.1), whose value at $t=1$ is the pulled-back Wilson insertion. All subsequent changes of variables are applied to every fluctuating link in this polynomial, including the boundary change (1.22).

Fix the domain labels, the coarse field, the exterior fine field and the exterior free fluctuation field in this construction. Denote this collection by ξ. Define $K_\xi(t)$ to be the conditional fluctuation integral in [3, (1.26)], with the pulled-back polynomial inserted and the change-of-variables density from (1.22)–(1.24) retained once. The same integral with insertion 1 is $K_{\xi,\mathrm{vac}}$. These definitions precede the separation of the effective interaction into its regular and boundary terms. The exterior integrations and their prefactors are those of (1.15), (1.25) and (1.29).

The flat projection of this conditional kernel sets its coarse and exterior determining fields to the identity and its exterior free fluctuation field to zero. The domain labels are held fixed. This evaluation is applied to K_ξ, rather than to the exterior indicators or history weights. In particular, a large-field indicator is not evaluated at zero. At the projection point $\mathcal{W}(0)=1$, so the source logarithm is continued from $K_\xi(0)=K_{\xi,\mathrm{vac}}$. No logarithm of $W_\chi(C;U_1)$ at general background is required.

7.2 Regional Integrals and Their Connected Densities

Use fine-lattice units. Write $H=\mathsf{H}_\infty$ for (A.26), and $C_\infty=H^{-1}$. For a finite set $D\subset E_{*,\infty}$, let i_D extend a free field by zero and put

$$H_D=i_D^* H i_D, \qquad C_D=H_D^{-1}\ . \tag{7.3}$$

The inverse exists by (A.27). Let $\mathcal{V}_D$ consist of the primitive labels from (C.1) whose supports meet D. Retain their original labels and set $v_D(B)=v(i_DB)$. Every primitive missing D vanishes at this field. The cutoff and interaction are

$$\Theta_D^\sigma(B)=\prod_{b\in D}\mathbf{1}_{\{|B_b|_\sigma<r\}}, \qquad P_D(B)=\sum_{v\in\mathcal{V}_D} v(i_DB)\ . \tag{7.4}$$

The threshold is the same $r=r_0(g_0)$ as in (5.3).

Let I be an interval of m coarse segments on a coarse translate of the oriented line $v\in\mathscr{V}_\square$. Let $P_U(I)$ denote the ordered transport along its fine traversals. Its closed relative source

uses the inverse reference transport, as in (5.32). The regional numerator and denominator are

$$\begin{aligned} N^{\sigma}_{D,I} &= \int \Theta^{\sigma}_{D}(B) e^{P_D(B)} \chi\big(P_{U_{g_0}(i_D B)}(I) P_{U_0}(I)^{-1}\big)\, \mathrm{d}\mu_{C_D}(B)\ , \\ D^{\sigma}_{D} &= \int \Theta^{\sigma}_{D}(B) e^{P_D(B)}\, \mathrm{d}\mu_{C_D}(B)\ . \end{aligned} \tag{7.5}$$

Here $U_0 = \mathbf{1}$ in the chosen frame; retaining its reverse transport specifies the gauge-invariant closed contraction.

Apply the protected-label forest expansion and the web construction of Section 5, replacing every free-input support Λ by $\Lambda \cap D$. A source mark with empty restricted stencil is zero. The other labels keep their original identities. Denote the connected weights by $\Phi_{D,I}(\Gamma)$ and their marked coarse-segment set by J_Γ. The density assigned to segment j is

$$\kappa_{D,I}(j) = \sum_{\Gamma \,|\, j \in J_\Gamma} \frac{\Phi_{D,I}(\Gamma)}{|J_\Gamma|}\ . \tag{7.6}$$

We suppress σ in these weights and densities. It is fixed throughout each comparison.

For positive bonds $b = (x, i)$ and $c = (y, k)$, use $d(b, c) = \|x - y\|_\infty + \mathbf{1}_{\{i \neq k\}}$. Let R_* bound the label diameters and source-to-stencil distances, and let R_H be the range of H in this metric. With the interval constant c_h of (5.24), set

$$R_0 = 4(1 + R_* + R_H + L), \qquad c_{\mathrm{reg}} = 16(1 + R_* + c_h + L)\ . \tag{7.7}$$

Number the coarse segments of I by $0, \dots, m-1$. Let K_j be the union of their fine-traversal input stencils on segment j, and define

$$d_j = \max\big(0, \min\big(d(K_j, D^c), L \min(j, m-1-j)\big) - R_0\big)\ . \tag{7.8}$$

Proposition 7.1. *For either fixed cutoff prescription σ, and for sufficiently small g_0 after the constants in the comparison have been chosen, there are constants $C, c > 0$, independent of D, I and j, such that*

$$|\kappa_{D,I}(j) - \kappa_v| \leqslant C\tau^2 e^{-c d_j}\ . \tag{7.9}$$

Here τ is defined in (5.18) *and κ_v in* (5.33)*. If every K_j is at distance at least b from D^c, then*

$$\left| \log \frac{N^{\sigma}_{D,I}}{D^{\sigma}_{D}} - m\kappa^{\sigma}_{v} \right| \leqslant C\tau^2 (1 + m e^{-cb})\ . \tag{7.10}$$

The logarithm is continued from zero source.

Proof. Principal restriction preserves $\gamma_L I \leqslant H_D \leqslant h_L I$. The finite-range Neumann series in (A.28) therefore gives exponential decay for C_D, uniformly in the shape of D. Projecting $HC_\infty = I$ onto the D rows gives

$$C_D - (C_\infty)_{D,D} = C_D H_{D,D^c} (C_\infty)_{D^c,D}\ . \tag{7.11}$$

Let $\mu > 0$ be a common inverse-decay exponent in the fine-bond metric. A term on the right joins $u \in D$ and $w \in D^c$ with $d(u, w) \leqslant R_H$. For $s = d(b, u) + d(w, c)$,

$$d(b, D^c) + d(c, D^c) + d(b, c) \leqslant 2s + 2R_H\ . \tag{7.12}$$

Half of $e^{-\mu s}$ pays this distance with exponent $a = \mu/4$; the remaining half is summable over u and its bounded number of neighbours w. Polynomial volume growth gives

$$\|C_D(b, c) - C_\infty(b, c)\| \leqslant C e^{-a(d(b,D^c) + d(c,D^c) + d(b,c))}\ . \tag{7.13}$$

There is no factor proportional to the area of the boundary.

On D, interpolate between $(C_\infty)_{D,D}$ and C_D, and denote their difference by $\mathcal{D}$. Every interpolated covariance and each of its principal marginals has spectrum in $[h_L^{-1}, \gamma_L^{-1}]$, uniform exponential decay and scalar colour blocks. Thus the protected-source bounds and the Schur cancellation used in (5.18) hold throughout the interpolation.

Consider a forest term on carrier A. If $M_t = C_t \circ w^F$ is its interpolated covariance and $E_e(C_t)$ the off-diagonal direction associated with edge e, its density derivative has the product-rule decomposition

$$\begin{aligned}\partial_t D^{|F|} p_{M_t}[(E_e(C_t))_{e\in F}] &= D^{|F|+1} p_{M_t}[\mathcal{D}_A \circ w^F, (E_e(C_t))_{e\in F}] \\ &\quad + \sum_{e\in F} D^{|F|} p_{M_t}[E_e(\mathcal{D}), (E_f(C_t))_{f\in F\setminus\{e\}}] \,.\end{aligned} \tag{7.14}$$

Each tuple lists the directions of the symmetric multilinear derivative of the density with respect to its covariance.

The two kinds of terms use separate Cauchy estimates. In the first, add the direction $s(\mathcal{D}_A \circ w^F)$ to the covariance centre while leaving every forest direction fixed. The positive Schur mask contracts operator norm. With $\lambda = h_L^{-1}$, the radius $\lambda/(32\|\mathcal{D}_A\|)$ uses at most $\lambda/32$ of the accretivity margin; a zero direction gives zero derivative. Reduce the forest radii by a fixed factor to keep the total below $\lambda/8$. In the second kind of term, replace one forest direction by $E_e(\mathcal{D})$ and use (7.13). Choose its radius from the exponential entry majorant, without dividing by an entry of C_t. The normalised density, including its determinant, is bounded by (5.17).

Since $\|\mathcal{D}_A\| \leqslant C|A|e^{-ad(A,D^c)}$, the integrated differentiated term is bounded by the positive forest majorant, with enlarged constants and reduced edge decay, times

$$C|A|^2 e^{-ad(A,D^c)} \,. \tag{7.15}$$

The exponent of τ is unchanged. In particular the differentiated functional is still equivariant under constant conjugation, so a component with one source mark retains its quadratic character gain.

For each component γ, let u_γ be its positive fixed-forest bound with the uniform exponential entry majorants. This definition applies to vacuum and source components. For a web W, put $u_W = \prod_{\gamma\in W} u_\gamma$; summing these numbers over webs gives the web load. The positive majorants themselves satisfy the polymer loads. Restricting a support to D preserves its size bound s_*, overlap bound q_* and range R_*. On n bonds, label choices cost at most $2^{q_* n}$, the forest-to-graph multiplicity costs at most $2^{q_*(s_*-1)n/2}$, unused graph edges at most $e^{V_1 n/2}$, and the rooted spatial tree sum at most $n^{n-2}V_1^{n-1}/(n-1)!$, where V_1 is the finite row sum of the reduced-decay graph weights. Thus a constant $K_2 \geqslant 1$ bounds the vacuum and source root loads by

$$\sum_{n\geqslant 1} K_2^n \tau^{\lceil n/s_*\rceil}, \qquad \sum_{n\geqslant 1} K_2^n \tau^{\max(2,\lceil n/s_*\rceil)} \,. \tag{7.16}$$

Grouping n into intervals of s_* gives bounds $C\tau/(1-K_2^{s_*}\tau)$ and $C\tau^2/(1-K_2^{s_*}\tau)$, respectively. These sums bound the positive forest majorants before any signed activity is used. The interval-tree recursion then bounds the web loads.

Let $\mu_1 > 0$ be an exponent remaining after the degree payment, no greater than $\min(\mu, a)/8$. Choose

$$\alpha_* = \frac{\mu_1}{1024(1+c_h)}, \qquad \eta_1 = \frac{\mu_1}{1024} \,. \tag{7.17}$$

The same sums allow the reserve $e^{4|A|+8\alpha_* H+8\eta_1\ell}$. Let C_{load} bound their weighted spatial and interval root loads after division by τ and τ^2, respectively. Choose g_0 so that $C_{\mathrm{load}}\tau \leqslant 1/4$ and $C_{\mathrm{load}}\tau^2 \leqslant \alpha_*/4$. The incompatibility sum is then bounded by one quarter of $|A_p|+\alpha_* H_p$, so the pinned tree bound applies to the positive majorants u_p.

For a cluster Γ, let $U(\Gamma)$ be its absolute Ursell coefficient, including its factorial, multiplied by the product of these majorants. The product rule, counting repeated activity occurrences, yields

$$|\partial_t \Phi_t(\Gamma)| \leqslant C\ell(\Gamma)^2 e^{-ad(A_\Gamma, D^c)} U(\Gamma), \qquad \sum_{\Gamma \,|\, j\in J_\Gamma} U(\Gamma) e^{\eta_1 \ell(\Gamma)} \leqslant C\tau^2 \,. \tag{7.18}$$

The union of carriers is A_Γ. The inequality follows from $\sum_p \ell(p)^2 \leqslant \ell(\Gamma)^2$ and the positive-majorant KP bound. Half the extent weight absorbs the polynomial factor.

For completeness, connect a component's carrier using its forest and at most $|A|-1$ stencil edges, each of length at most R_*. Attach its contour interval at one selected mark by a path of length at most R_*. The interval costs at most $c_h\ell$. Carrier or interval incompatibility joins these networks for a connected cluster. Their total length is at most $(1+2R_*+c_h)\ell(\Gamma)$; adding the root segment costs at most L. The chosen c_{reg} bounds the result.

For $d_j \leqslant 2(R_0+R_*+c_{\mathrm{reg}})$, the rooted bounds prove the comparison after enlarging C. For larger d_j, clusters with $\ell(\Gamma) \geqslant d_j/(2c_{\mathrm{reg}})$ are bounded by the extent tail in each gas. Every shorter cluster rooted at j lies away from both the integration boundary and the ends of I. Its original stencils lie inside D, so its labels, closed sources, local functions and incompatibilities coincide in the two expansions. Only its covariance changes, and every carrier is at distance at least $d_j/2-R_0$ from D^c. The fixed loss is absorbed in the comparison constant. Integrating the differentiated bound gives (7.9). Division by $|J_\Gamma|$ cannot enlarge it. Finally, allocation and the endpoint geometric series give

$$\log \frac{N^\sigma_{D,I}}{D^\sigma_D} = \sum_{j=0}^{m-1} \kappa_{D,I}(j), \qquad \sum_{j=0}^{m-1} e^{-cd_j} \leqslant C(1+me^{-cb}) \,, \tag{7.19}$$

which proves (7.10). *q.e.d.*

7.3 Periodic Regions and Bałaban's Kernel

Let $E_{*,T}$ be the periodic free carrier, H_T the Hessian in (4.7), and $i_{D,T}$ zero extension from $D \subseteq E_{*,T}$. The periodic primitives, written v_T collectively (the terms v_α, j_b and d_c of (C.1) on T, all three kinds), and the reconstructed links $U_{g_0,T}$ of (4.9) on T are those of Appendix C. Define

$$H_{D,T} = i^*_{D,T} H_T i_{D,T}, \qquad C_{D,T} = H^{-1}_{D,T}, \qquad P_{D,T}(B) = \sum_{v_T \,|\, \operatorname{supp} v_T \cap D \neq \varnothing} v_T(i_{D,T}B) \,. \tag{7.20}$$

The primitives of Section 3.3 at the reference background $U=\mathbf{1}$ (written $p_{\xi,\alpha}$, $p^{\mathrm{H}}_{\xi,b}$, $p^{\mathrm{J}}_{\xi,c}$ as there, with the reference background in place of the branch data) and the local terms of (C.1) are the same functions, label by label.

Lemma 7.2 (Primitives at the Identity Background)**.** *Let the background be $U=\mathbf{1}$ on the torus T with region $\Omega = T$ (Section 3.2), so that every row $c \in T^{[1]}$ and every centre $y \in T^{(1)}$ occurs, and let $B \in \mathfrak{g}^{E_*}$ be real with $g_0 C_{\mathsf{C}} \max_b |B(b)| \leqslant \vartheta_*/2$ and $g_0 \max_b |B(b)| < a_L$, with C_{C}, ϑ_* of Lemma B.7 and a_L the chart radius of Appendix C. Here $|\bullet|$ is the cutoff norm (so $h \leqslant |\bullet|$, the conventions of Section 3), $\|\bullet\|$ the Hilbert norm of Lemma B.7 and*

$\|A\|_\infty = \max_b h(A_b)$ *the phase norm, in which* a_{ph} *and* a_L *are measured; the row solutions, the phase* $q_{\mathbf{1}}$ *and the primitives* $p_{\xi,\alpha}$, $p^{\mathrm{H}}_{\xi,b}$, $p^{\mathrm{J}}_{\xi,c}$ *at the background* $\mathbf{1}$ *are those given by the formulas of Sections 3.2 and 3.3 with* $\Omega = T$ *and* $U = \mathbf{1}$*, the row solution of every row* c *being the unique small solution* φ_c *of Lemma B.7(ii), which applies to every row of the identity background (proof of (i)). Then:*

1. $\mathcal{Q}_{\mathbf{1}} = \mathcal{Q}$, $Q_{\mathbf{1}} = Q$, $\mathcal{N}_{\mathbf{1}} = \mathcal{N}$ (4.2), $\mathsf{C}_{\mathbf{1}} = \mathsf{C}$ *and* $h_{\mathbf{1}}(c) = L^3\mathbf{1}$ *for every row* (4.3)*; every row solution* $\widetilde{D}_{\mathbf{1},c}(g_0\mathsf{C}B)$ *is defined, is the unique small solution of Lemma B.7(ii), and equals the row* c *of* $\widetilde{D}(g_0\mathsf{C}B)$ *of* (4.9)*, the two functions agreeing on the convex open set* $\{A \in \mathfrak{g}^{\mathcal{E}_c} \text{ real} : \|A\| < \vartheta_*,\ \|A\|_\infty < a_{\mathrm{ph}}\}$ *of row arguments; hence* $q_{\mathbf{1}}(B) = q_{g_0}(B)$.
2. $p_{\xi,p}(B) = v_p(B)$ *for every plaquette* p*, and* $p_{\xi,y}(B) = \sum_{x\in\mathcal{B}_y\setminus\{y\}} v_{(y,x)}(B)$ *for every centre* y.
3. $p^{\mathrm{H}}_{\xi,b}(B) = j_b(B)$ *for every bond* b.
4. $p^{\mathrm{J}}_{\xi,c}(B) = d_c(B)$ *for every row* c.

Consequently $\sum_\alpha p_{\xi,\alpha}(B) + \sum_b p^{\mathrm{H}}_{\xi,b}(B) + \sum_c p^{\mathrm{J}}_{\xi,c}(B) = \mathcal{P}_{g_0}(B)$ (C.2)*; and for every bond set* $D \subseteq E_*$ *and every real* $B \in \mathfrak{g}^D$ *with* $g_0 C_{\mathsf{C}} \max_b |B(b)| \leqslant \vartheta_*/2$ *and* $g_0 \max_b |B(b)| < a_L$*, the sum of the primitives whose free-input sets meet* D*, evaluated at* $i_{D,T}B$*, equals* $P_{D,T}(B)$.

Proof. (i) At $U = \mathbf{1}$ every transport is $\mathbf{1}$, so $\beta_c(\mathbf{1}) = 0$ and $\mathsf{M}(\mathbf{1})(c) = P_c(\mathbf{1}) = \mathbf{1}$ in (4.1); hence $\mathcal{Q}_{\mathbf{1},c}(q) = \frac{1}{i}\log \mathsf{M}(e^{iq})(c)$ is the row c of $\mathcal{Q}(q)$ in (4.2), and $Q_{\mathbf{1}} = Q$, $\mathcal{N}_{\mathbf{1}} = \mathcal{N}$ follow from the definitions (3.9). By Lemma 3.2, $Q_0 = Q\iota_0$ is diagonal with entries $\alpha_c(\mathbf{1}) = L^{-4}\mathbf{1}$, so $h_{\mathbf{1}}(c) = L^{-1}\alpha_c(\mathbf{1})^{-1} = L^3\mathbf{1}$ is the row c of $h = L^{-1}\iota_0 Q_0^{-1}$; and since row c of Q is supported on $\mathcal{E}_c$ (Lemma 3.2), $(\mathsf{C}_{\mathbf{1}}B)(b_0(c)) = -\alpha_c(\mathbf{1})^{-1}\sum_{b\in\mathcal{E}_c\cap E_*} Q(c,b)B(b) = -(Q_0^{-1}Q\iota_* B)(c) = (\mathsf{C}B)(b_0(c))$, while both maps are the identity on free bonds; so $\mathsf{C}_{\mathbf{1}} = \mathsf{C}$. The row equation (3.24) at $U = \mathbf{1}$ is the row c of the equation defining $\widetilde{D}$ after (4.9). Lemma B.7 applies to the identity background on every row (its gauge hypothesis holds with $u = \mathbf{1}$, $\phi = 0$), and by its part (i) the hypothesis on B gives $\|g_0\mathsf{C}B|_{\mathcal{E}_c}\| \leqslant g_0 C_{\mathsf{C}} \max_b |B(b)| \leqslant \vartheta_*/2$ for every row, so its part (ii) provides the unique small solution $\varphi_c(g_0\mathsf{C}B|_{\mathcal{E}_c})$, which is the row solution of Section 3.3 for the reference background, analytic on the real ball $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ and agreeing with the germ of Lemma 3.5 at $A = 0$. The row c of $\widetilde{D}$ of Appendix C is real analytic on $\{\|A\|_\infty < a_{\mathrm{ph}}\}$, a function of $A|_{\mathcal{S}_c}$, hence of $A|_{\mathcal{E}_c}$ ($\mathcal{S}_c$, the positive fine bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, is contained in $\mathcal{E}_c$, Section 3.2), and agrees with the same germ at $A = 0$ (the paragraph "Locality of the first-step interaction" there). The set of real $A|_{\mathcal{E}_c}$ with $\|A|_{\mathcal{E}_c}\| < \vartheta_*$ and $\|A\|_\infty < a_{\mathrm{ph}}$ is convex, hence connected, open, and contains $g_0\mathsf{C}B|_{\mathcal{E}_c}$ by the hypothesis on B ($\|g_0\mathsf{C}B\|_\infty \leqslant \|\mathsf{C}\|_{\infty\to\infty} g_0 \max_b |B(b)| < a_{\mathrm{ph}}$ by the definition of a_L and $h \leqslant |\bullet|$); the identity theorem identifies the two analytic functions on it. With $h_{\mathbf{1}} = h$ and $\mathsf{C}_{\mathbf{1}} = \mathsf{C}$, (3.25) becomes (4.9): $q_{\mathbf{1}}(B) = g_0\mathsf{C}B - h\widetilde{D}(g_0\mathsf{C}B) = q_{g_0}(B)$.

(ii) Write $\psi(A) = A - h\widetilde{D}(A)$ as in Appendix C, so that $q_{g_0}(B) = \psi(g_0\mathsf{C}B)$, $\psi(0) = 0$ and $D\psi(0) = \mathbf{1}$ because $D\widetilde{D}(0) = 0$ (Lemma B.7(ii)). For a plaquette p, $s_{p,\mathbf{1}}(q) = 1 - \mathrm{Re}\,\mathrm{tr}_N(e^{iq}(\partial p)) = f_p(q)$ is the p-summand of $S(e^{iq})$, the function f_p of Appendix C, and $s_{p,\mathbf{1}}(0) = 0$ since $\mathrm{tr}_N(\mathbf{1}) = 1$; so $E_{\xi,p}(B) = -g_0^{-2} f_p(\psi(g_0\mathsf{C}B))$ by (i). Each $f_p \geqslant 0$ vanishes at $q = 0$, so $Df_p(0) = 0$. The chain rule then gives $DE_{\xi,p}(0)[B] = -g_0^{-2} Df_p(0)[D\psi(0)g_0\mathsf{C}B] = 0$ and

$$\begin{aligned} D^2E_{\xi,p}(0)[B,B] &= -g_0^{-2}\big(D^2f_p(0)[D\psi(0)g_0\mathsf{C}B, D\psi(0)g_0\mathsf{C}B] \\ &\qquad + Df_p(0)[D^2\psi(0)[g_0\mathsf{C}B, g_0\mathsf{C}B]]\big) \\ &= -D^2f_p(0)[\mathsf{C}B, \mathsf{C}B]\ , \end{aligned}$$

so by (3.29)

$$p_{\xi,p}(B) = -\frac{1}{g_0^2} f_p(\psi(g_0 \mathsf{C} B)) + \frac{1}{2} D^2 f_p(0)[\mathsf{C}B, \mathsf{C}B] = v_p(B) \,. \tag{7.21}$$

For a centre y: at $U = \mathbf{1}$ every path transport is $\mathbf{1}$, so the local group mean $\mathcal{U}_{\mathbf{1}}(y,x)$, the unique solution near the diagonal value of the equation after (4.3), is $\mathbf{1}$; at $q = q_{g_0}(B)$ the mean $\mathcal{U}_{e^{iq}}(y,x)$ exists by Lemma B.7(iv) at the identity background, since for a row c with endpoint block $\mathcal{B}_y$ one has $\mathcal{E}_y \subseteq \mathcal{E}_c$ and $\|q_{g_0}(B)|_{\mathcal{E}_c}\| \leqslant \|A\| + 2H_*K_*\|A\|^2 \leqslant \vartheta_*$ for $A = g_0 \mathsf{C} B|_{\mathcal{E}_c}$, $\|A\| \leqslant \vartheta_*/2$, by (ii) of that lemma, $\|h_{\mathbf{1}}(c)\| \leqslant H_*$ and $4K_*H_*\vartheta_* < 1$; and the y-summand of (3.14) is $\mathscr{G}_{y,\mathbf{1}}(q) = \sum_{x \in \mathcal{B}_y \setminus \{y\}} (1 - \operatorname{Re} \operatorname{tr}_N \mathcal{U}_{e^{iq}}(y,x)) = \sum_{x \in \mathcal{B}_y \setminus \{y\}} f_{(y,x)}(q)$, the sum of the (y,x)-summands of $\mathscr{G}(e^{iq})$ in (4.5). So $E_{\xi,y}(B) = -g_0^{-2} \mathscr{G}_{y,\mathbf{1}}(q_{g_0}(B)) = \sum_x \big(-g_0^{-2} f_{(y,x)}(\psi(g_0 \mathsf{C} B))\big)$. Each $f_{(y,x)} \geqslant 0$ vanishes at $q = 0$, so $D f_{(y,x)}(0) = 0$, and the computation just made for f_p applies to each $f_{(y,x)}$; since the passage from a function to its degree-two Taylor residual at $B = 0$ in (3.29) is linear, $p_{\xi,y}(B) = \sum_{x \in \mathcal{B}_y \setminus \{y\}} v_{(y,x)}(B)$.

(iii) In (3.30), $\sigma(q)\,\mathrm{d}q$ is the pullback under $q \mapsto e^{iq}$ of the Haar measure of [3, (1.11)], and in (4.14) σ_H is the density of normalised Haar measure in the same coordinates relative to the same Lebesgue measure; the Haar measure of the compact group being unique up to a positive constant, $\sigma = \kappa \sigma_H$ with $\kappa > 0$, both are positive on the chart ball (Lemma B.7(iii) takes the real logarithm of $\sigma(q)/\sigma(0)$ there), and $\sigma(q)/\sigma(0) = \sigma_H(q)/\sigma_H(0) = j_H(q)$. With $q_{\mathbf{1}}(B) = q_{g_0}(B)$ from (i), $p^{\mathrm{H}}_{\xi,b}(B) = \log j_H(q_{g_0}(B)(b)) = j_b(B)$, both sides being real logarithms of positive numbers: the branch of j_b in (C.1) is the real logarithm on the convex domain of (i), on which the argument stays positive.

(iv) By (i) (the row solution being φ_c, which agrees with the row of $\widetilde{D}$ on an open set containing $g_0 \mathsf{C} B|_{\mathcal{E}_c}$, so that their derivatives agree there, and $h_{\mathbf{1}}(c) = L^3 \mathbf{1}$), $p^{\mathrm{J}}_{\xi,c}(B) = \log |\det_{\mathfrak{g}}(\mathbf{1} - L^3 \partial_{A(b_0(c))} \widetilde{D}_c(g_0 \mathsf{C} B))|$. The determinant is positive on $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$ (Lemma B.7(iii)), so the absolute value is redundant and the expression is $d_c(B)$, whose branch in (C.1) is the real logarithm of the positive determinant (the sentence after that display).

The first consequence is (ii)–(iv) summed over all labels, with (C.2). For the second, every primitive and every term vanishes at zero input: $p_{\xi,\alpha}(0) = E_{\xi,\alpha}(0) = 0$ ($s_{p,\mathbf{1}}(0) = 0$, $\mathscr{G}_{y,\mathbf{1}}(0) = 0$), $p^{\mathrm{H}}_{\xi,b}(0) = 0$, $p^{\mathrm{J}}_{\xi,c}(0) = 0$ ($\widetilde{D}(0) = 0$, $D\widetilde{D}(0) = 0$), and $v_\alpha(0) = j_b(0) = d_c(0) = 0$ likewise. A primitive depends on B only through its free-input set (Lemma 3.5, the list after (3.31)), a term of (C.1) only through its support. For $B \in \mathfrak{g}^D$ as in the statement, $i_{D,T} B$ satisfies the hypothesis of the lemma (the maximum norm does not increase), a primitive whose free-input set misses D has zero input at $i_{D,T}B$ and vanishes, and a term whose support misses D vanishes there as well; so the sum of the primitives whose free-input sets meet D and $P_{D,T}(B)$ are both the full sums at $i_{D,T}B$, which agree by the first consequence. *q.e.d.*

Define $N^{\sigma}_{D,I,T}$ and $D^{\sigma}_{D,T}$ by (7.5) with C_D, P_D, i_D, U_{g_0} replaced by $C_{D,T}, P_{D,T}, i_{D,T}, U_{g_0,T}$. Their cutoff is still the product over D. Apply the same forest, web and connected-weight definitions to these periodic integrals, obtaining weights $\Phi_{D,I,T}(\Gamma)$, and set

$$\kappa_{D,I,T}(j) = \sum_{\Gamma \,|\, j \in J_\Gamma} \frac{\Phi_{D,I,T}(\Gamma)}{|J_\Gamma|} \,. \tag{7.22}$$

Assume that I has a lift with $mL < N_{\min}/3$, where $N_{\min}$ is the smallest fine period. For both periodic open-segment gases the positive cluster majorants obey a uniform rooted tail. Let $U_{D,I,T}(\Gamma)$ be the absolute Ursell coefficient, including its factorial, times the positive object majorants, and use $U_{I,T}$ for $D = E_{*,T}$. There is $\eta_0 > 0$, independent of the region and

periods, such that for every marked segment j,

$$\sum_{\Gamma \,|\, j \in J_\Gamma} U_{D,I,T}(\Gamma) e^{\eta_0 \ell(\Gamma)} \leqslant C\tau^2 \ . \tag{7.23}$$

The majorants dominate the absolute connected weights. The proof is the positive-majorant argument of Proposition 7.1: restricting supports preserves the label overlap bounds, and the periodic shell sum is bounded by the infinite-lattice shell sum. Under this lift hypothesis, index separation along the segment is its periodic separation. Thus (5.24) holds on the lifted segment. Its ordinary intervals have the straight-line interval tree sum. Hence the same load inequalities and extent reserve prove (7.23) for the full and restricted periodic gases, including the covariance interpolations used below.

On a compatible torus with fine periods N_μ, put $N_{\min} = \min_\mu N_\mu$ and assume that I has a lift with $mL < N_{\min}/3$. The set D may wrap around the torus. Define $\kappa_{I,T}(j)$ by the same allocation with $D = E_{*,T}$. In this paragraph d_j is (7.8) with d_T in place of d, including its endpoint term $L\min(j, m-1-j)$. For bounded $d_j \leqslant 2(R_0 + R_* + c_{\mathrm{reg}})$, (7.23) and $|\kappa_v| \leqslant C\tau^2$ give (7.27) directly after enlarging C. For larger d_j, the periodic version of (7.11) and its differentiated cluster estimate compare $\kappa_{D,I,T}(j)$ with $\kappa_{I,T}(j)$. Clusters of extent at least $d_j/(2c_{\mathrm{reg}})$ have total rooted weight at most $C\tau^2 e^{-cd_j}$. For smaller clusters the labels agree, and their carriers are at distance at least $d_j/2 - R_0$ from D^c. The covariance derivative therefore has the same bound, after absorbing the fixed factor e^{aR_0}. This proves

$$|\kappa_{D,I,T}(j) - \kappa_{I,T}(j)| \leqslant C\tau^2 e^{-cd_j} \ . \tag{7.24}$$

To compare the second density with the infinite straight-line density, lift the root segment to the chosen copy of I. A cluster network of length $c_{\mathrm{reg}}\ell + R_0 < N_{\min}/4$ lifts uniquely and preserves carrier intersections. The distinct translates of I are separated by more than $2N_{\min}/3$ because $mL < N_{\min}/3$, so all its source marks lie on that copy. If also $c_{\mathrm{reg}}\ell + R_0 < L\min(j, m-1-j)$, the network avoids the two ends and has the same labels and ordered source factors as the infinite straight-line expansion. Clusters failing either restriction are bounded by the rooted extent tail. For the remaining clusters only the covariance marginals change. Periodisation and lifted separation $d(b,c) \leqslant N_{\min}/4$ give

$$\|C_T(b,c) - C_\infty(b,c)\| \leqslant Ce^{-\mu(N_{\min} - d(b,c))} \leqslant Ce^{-\mu N_{\min}/2} e^{-\mu d(b,c)} \ , \tag{7.25}$$

after expressing the decay exponent in fine-bond units. The free-inverse periodisation follows by applying H_T to the absolutely convergent periodised kernel of H^{-1}, as in the proof of (A.31). The covariance interpolation and the density-direction estimates in the proof of Proposition 7.3 apply with $h_A = e^{-\mu N_{\min}/2}$. They retain edge decay and the quadratic source gain; their polynomial carrier factors are absorbed by the reserved extent weight. With $\iota_T = \max(0, N_{\min}/4 - R_0)$, summing the short-cluster derivatives and the two tails gives

$$|\kappa_{I,T}(j) - \kappa_v| \leqslant C\tau^2 \left[e^{-c(L\min(j,m-1-j) - R_0)_+} + e^{-c\iota_T} \right] \ . \tag{7.26}$$

Here $x_+ = \max(x, 0)$. Absolute convergence also gives $\log(N^\sigma_{D,I,T}/D^\sigma_{D,T}) = \sum_{j=0}^{m-1} \kappa_{D,I,T}(j)$, since summing the allocation weights $|J_\Gamma|^{-1}$ over the marked segments gives one for each source cluster. Adding the two comparisons proves

$$|\kappa_{D,I,T}(j) - \kappa_v| \leqslant C\tau^2 (e^{-cd_j} + e^{-c\iota_T}) \ . \tag{7.27}$$

For the identification with K_ξ, take the averaging prescription [4, (0.4)], the bond cutoff $|\bullet|_\sigma$, and the threshold $\delta_0 = g_0 r_0(g_0)$ on both sides. The comparison applies separately to either

specified σ; it does not determine an otherwise unspecified norm convention. Let I consist of consecutive traversals of one straight side of C. Let $E_{\Omega,*}$ be the free carrier after the designated-bond elimination in [3, (1.14)–(1.15)]. Let D_- consist of these bonds with both endpoints in Λ_1, and D_+ of those with at least one endpoint there. The comparison holds for every $D_- \subseteq D \subseteq D_+$, assigning the complementary free variables to the exterior integral. It therefore covers either endpoint convention for the integration boundary.

We make the geometric parameter choices in this identification explicit. In the notation of [3] let M_2, M, R_1 be the cube-scale integers, so the test cubes have fine side $L^2M_2R_1$ and the macro cubes have fine side LMR_1; here R_1 denotes the scale integer, not the large-fluctuation region also denoted by R_1 in [3]. Take aligned nested partitions, with all periods divisible by LMR_1, and choose the powers of L so that

$$M \geqslant 8LM_2, \qquad LMR_1 > 4R_0 + 4L^2M_2R_1 + 8\,. \tag{7.28}$$

Here $R_0 = 4(1 + R_* + R_H + L)$ is our fixed range constant, distinct from the scale integer in [3]. These are explicit choices within the compatible-partition and large-M prescription of [3, Section 1]. The collar inequality is included among the common parameter conditions. One may enforce it by first choosing M so that $LM - 4L^2M_2 > 4R_0 + 8$; it then holds for every scale integer $R_1 \geqslant 1$. For macro regions we use the collar-distance statement following [3, (1.10)]; it supplies the metric separation, including near cube corners. For test-cube enlargements we need only a conservative reach bound: each added layer has width at most $L^2M_2R_1$. This bound also covers a layer of the L-fold finer cubes in the enlargement convention of [3, Section 1]; no equality between test and macro enlargements is used. The parallelepiped modification after (1.20) removes macro cubes from Λ_1; the remaining region is still a union of macro cubes and hence of test cubes.

Using (1.20) and that collar-distance statement, a site within distance $LMR_1 - 1$ of Λ_1 lies in $\Omega_1^{\sim -1} \setminus R_1'$. Every stencil meeting $D \subseteq D_+$ therefore remains in the regular region. A bond with an endpoint in Λ_1 is tested by the test cube containing that endpoint, which is contained in Λ_1 and is good. A bad test or a test in S_1 uses only bonds within $2L^2M_2R_1 + 1$ of its cube. The collar and (7.28) keep those bonds outside D_+, including the fixed stencil enlargement. At identity determining data the critical field is the identity: it satisfies the constraints, has zero action and gauge-fixing functional, and is unique in the regular chart by [3, (1.12)]. Its force is zero. The Hessian on D is consequently $H_{D,T}$, including the diagonal contributions of plaquettes crossing its boundary.

The nonlinear constraint in [4, (2.4)] becomes $M(e^{iB'})$ at identity background. Its principal logarithm is $\mathcal{Q}(B')$ from (4.2). The linearisation following (2.10) has linear part $L\widetilde{Q}B'$, hence $\widetilde{Q} = Q$ there. Its designated-row right inverse satisfies $LQh = I$. Uniqueness of the small solution for $\widetilde{D}$ gives exactly (4.9). Thus the action, Haar and Jacobian terms become $v_T(i_{D,T}B)$ with all primitive labels meeting D retained.

The split reconstruction in [3, (1.21)] also equals the identity at identity data, and $Q_1^{s*}\mathbf{1} = \mathbf{1}$ by (1.3). Its logarithmic argument is zero. Equation (1.21) then gives $e^{iL^{-1}\mathbf{H}_{1,Ax}(0)} = 1$; the small principal chart implies $\mathbf{H}_{1,Ax}(0) = 0$. Since the BCH remainder satisfies $\mathbf{F}(X, 0) = 0$, the boundary change (1.22) is the identity, its Jacobian is one and its additional exponent is zero at this projection.

For the cutoff, every free bond in D has an endpoint in a compatible good cube contained in Λ_1. The twice-enlarged starred test in [3, (1.18)] covers it. The designated-bond exclusion is the verbal convention following (1.8). A good cube outside Λ_1 can also test bonds in D: split its product of bond indicators into its D and exterior factors. The D factors are redundant with the covered interior cutoff by idempotence; all exterior factors stay in the outer integral.

Bad tests and changed background tests are separated from D by the collar. At zero exterior free field the conditional cutoff is therefore Θ_D^σ.

The same norm is used on both sides. The threshold is $r = \delta_0/g_0$ by the normalisation convention (4.13). The exterior quadratic cross term and its completion-of-square compensator vanish at the projection. Gaussian and constraint normalisations cancel between the two conditional kernels. They are not removed from the exterior integration.

Finally set $t_a = 1$ on I and $t_a = 0$ elsewhere. At the flat projection, (7.1) is precisely the closed relative source in (7.5). Denoting the projected conditional kernels by K_ξ^{flat}, we have

$$\frac{K_\xi^{\mathrm{flat}}(1_I)}{K_{\xi,\mathrm{vac}}^{\mathrm{flat}}} = \frac{N_{D,I,T}^\sigma}{D_{D,T}^\sigma} \,. \tag{7.29}$$

If $d_T(K_j, D^c) \geqslant b$ for every source segment j, then (7.27) gives

$$\left| -\frac{1}{mL\varepsilon} \log \frac{K_\xi^{\mathrm{flat}}(1_I)}{K_{\xi,\mathrm{vac}}^{\mathrm{flat}}} - \delta m_v^\sigma \right| \leqslant \frac{C\tau^2}{L\varepsilon} \left(\frac{1}{m} + e^{-cb} + e^{-c\iota_T} \right) \,. \tag{7.30}$$

For every domain label ξ for which the buffer fits inside the torus and $mL < N_{\min}/3$, (7.30) bounds the distance between the per-length scalar and δm_v^σ by $C\tau^2(1/m + e^{-cb} + e^{-c\iota_T})/(L\varepsilon)$ on the fixed torus; along any family of such labels with m, b and ι_T tending to infinity this bound tends to zero, so the per-length scalar converges to δm_v^σ, which identifies it with (5.33). The limit is the one defining the coefficient, as in (5.39); the estimate (7.30) itself is a statement on each fixed torus. The four oriented straight-side coefficients give the square coefficient by the average in (5.39). This comparison is made inside the conditional kernels; the outer history weights and the physical denominator (1.7) retain their definitions.

7.4 The Regional Square Scalar

Fix either cutoff prescription σ. Let C_n be a coarse translate of the counterclockwise square with vertices $0, nLe_1, nL(e_1+e_2), nLe_2$ in fine-lattice units, and assume $K_\mu \geqslant 3(n+2)$ for every coarse period. Let D be the free bonds integrated in a covered conditional kernel: every free-input stencil of the contour lies in D. Retain the original primitive labels, including those crossing the boundary of D, and define

$$S_{D,C_n,T}^\circ = \frac{\displaystyle\int \Theta_D^\sigma(B) e^{P_{D,T}(B)} \chi\big(\mathrm{Hol}_{U_{g_0,T}(i_{D,T}B)}\, C_n\big)\, \mathrm{d}\mu_{C_{D,T}}(B)}{\displaystyle\int \Theta_D^\sigma(B) e^{P_{D,T}(B)}\, \mathrm{d}\mu_{C_{D,T}}(B)} \,, \quad C_{D,T} = (i_{D,T}^* H_T i_{D,T})^{-1} \,. \tag{7.31}$$

Here H_T is the full flat free-field precision on the torus, and $P_{D,T}$ is (7.20); Θ_D^σ is the bond cutoff of (7.4). For a conditional history ξ, write $S_{\xi,C_n}^\circ = S_{D,C_n,T}^\circ$ with its young integration carrier D. This is a ratio of conditional reference integrals; the exterior indicators and the physical history weight are not evaluated at the identity.

Index the square's $4n$ coarse segments cyclically by j. Let K_j be the union of the free-input stencils of its fine traversals. With complements taken in the torus free carrier and with the periodic fine-bond metric of (7.8), put

$$d_j = \max\big(0, d_T(K_j, D^c) - R_0\big), \qquad \iota_T = \max\big(0, N_{\min}/4 - R_0\big) \,. \tag{7.32}$$

The fixed R_0 includes the label, reconstruction and precision ranges. Distance to the empty set is infinite, with $e^{-\infty} = 0$. There is no distance to a side endpoint in d_j.

Proposition 7.3. *Under these hypotheses, the collar conditions of the conditional kernel, and the common small-coupling thresholds of the flat and regional expansions, the scalar* (7.31) *is nonzero. Its logarithm is continued from zero in the traversal-source polynomial* (7.1)*. There are $C, c > 0$, independent of D, n and the compatible periods, and an error $E^{\circ}_{D,C_n,T}$ such that*

$$S^{\circ}_{D,C_n,T} = e^{-\delta m^{\sigma} P(C_n)} \big(z^{\sigma}_{\mathrm{c}}(g_0, \pi/2)\big)^4 e^{E^{\circ}_{D,C_n,T}} , \tag{7.33}$$

$$|E^{\circ}_{D,C_n,T}| \leqslant C\tau^2 \left[\sum_{j=0}^{4n-1} e^{-cd_j} + 4ne^{-c\iota_T} + (n+1)e^{-cn} \right] . \tag{7.34}$$

The coefficients are exactly (5.39) *and* (6.12)*, with the same cutoff prescription. The parameter τ is* (5.18)*; fixed enlargements of its majorant constants are absorbed in C and the coupling threshold. The collar hypotheses identify this integral with the conditional scalar of [3]. The estimates for the defined flat integral use its covered stencils, spectral bounds and small-coupling conditions; the collar enters at the identification, not in the subsequent cluster comparison.*

Proof. Write $F_{D,C,T} = \log S^{\circ}_{D,C,T}$, with its branch specified below, $F_{C,T} = \log(N^{\sigma}_{C_n,T}/D^{\sigma}_{T})$, and $F_{C,\infty}(n) = F_{\square,\infty}(n)$. We compare the first two quantities, then the latter two, before applying the bulk corner extraction. This retains errors from changing the kernel and summation domain, as in [4, (4.35)–(4.37)]; the exterior integration of [3, (1.29)–(1.30)] is unchanged.

At the conditional reference data the background and exterior determining fields are the identity and the retained free field is zero. The row precision entries are the flat torus entries on D, including the diagonal terms from crossing plaquettes. The collar contains every stencil touching D, so the precision is $i^{*}_{D,T} H_T i_{D,T}$. The primitives agree with the terms of (C.1) label by label, with the designated inverse L^3 at the identity and the gauge terms of a centre summed over its pairs (y, x), by Lemma 7.2, whose hypothesis on B is the pair of small-coupling conditions fixed in the next paragraph; by its second consequence the sum of the primitives whose free-input sets meet D, at $i_{D,T}B$, is $P_{D,T}(B)$, terms whose support misses D vanishing at $i_{D,T}B$. The cutoff is the same product of bond tests. Thus the numerator and denominator in (7.31) are those of the conditional reference scalar, with their normalisers retained in the ratio.

The local implicit solutions used in the two representations have the same analytic germ. To fix their common domain, let C_{C} and ϑ_* be the constants of Lemma B.7 (C_{C} bounds the fixed row map from the maximum free-bond norm to the row Hilbert norm, ϑ_* is its implicit-solution phase radius), and let a_L be the flat chart radius in Appendix C. The row map is finite dimensional with finitely many translated types, so $C_{\mathsf{C}} < \infty$; the analytic implicit-function theorem gives a uniform positive phase radius. Reduce a_L if necessary to lie within those row charts and their nonvanishing Jacobian domains. Choose the coupling so that

$$g_0 C_{\mathsf{C}}(2 + 4r) < \frac{\vartheta_*}{2}, \qquad g_0(2 + 4r) < a_L . \tag{7.35}$$

The factor $2 + 4r$ specifies a larger free-variable polydisc containing the integration box $\|B\|_{\infty} \leqslant r$ with a positive margin. The convex intersection of this polydisc with the phase balls lies in the common analytic charts. The identity theorem therefore identifies the solutions throughout the integration box. The common chart radius is fixed before the coupling; reducing it preserves all previous estimates by restriction to a smaller domain. These small-coupling requirements and all polymer-load inequalities are imposed together, which is possible because $g_0(1 + r)^3 \to 0$ and $\tau \to 0$.

Apply the protected-label expansion with supports intersected with D, keeping every original label. Zero extension commutes with constant conjugation, so the restricted primitives, cutoffs and closed source differences are invariant. The restricted support graph and forest partition the labels into components. Merge source components using the manuscript's shortest cyclic intervals, with its fixed tie-breaking base; a web interval is their union, possibly the whole circle. Compatible components and webs have disjoint carriers and disjoint web intervals, so their union reverses this decomposition without losing or duplicating labels. If a web crosses the base, cyclicity of the trace moves the base between web intervals before Schur factorisation. This proves the regional counterpart of (5.14). The alternative grouping of the conditional kernel need not agree term by term: both constructions give the same integral ratio.

The required quantitative bounds hold uniformly for these restricted labels. Principal restriction and inversion preserve the spectral bounds, and the finite-range Neumann series gives uniform exponential covariance decay in the periodic metric. Let $\mu_1 > 0$ be a remaining edge-decay exponent after the degree payment, no larger than one eighth of the covariance and comparison exponents. Set

$$\alpha' = \frac{\mu_1}{1024(1+c_h)}, \qquad \eta' = \frac{\mu_1}{1024}, \qquad a'(p) = |A_p| + \alpha' H_p \, . \tag{7.36}$$

Restriction preserves the support-size, overlap and range bounds s_*, q_*, R_*. The same labelled-tree count as in (5.22), with a fixed enlarged K_2, bounds vacuum and source loads by

$$\sum_{k \geqslant 1} K_2^k \tau^{\lceil k/s_* \rceil}, \qquad \sum_{k \geqslant 1} K_2^k \tau^{\max(2, \lceil k/s_* \rceil)} \, . \tag{7.37}$$

The bounds remain valid with the reserve $e^{4|A_\gamma| + 8\alpha' H_\gamma + 8\eta' \ell(\gamma)}$. Contacts remain within R_* of the restricted support network, so (5.24) applies to the square, including at its corners. The cyclic conversion (5.25) has the same uniform geometric sum $1 + 2/(e^{\alpha'} - 1)$. The web-tree recursion therefore gives positive majorants u_p with

$$\begin{aligned} \sup_b \sum_{p \,|\, b \in A_p} u_p e^{2a'(p) + 2\eta' \ell(p)} &\leqslant C'_* \tau \, , \\ \sup_x \sum_{W \,|\, x \in h_W} u_W e^{2a'(W) + 2\eta' \ell(W)} &\leqslant C'_* \tau^2 \, . \end{aligned} \tag{7.38}$$

The source-only spatial load is also $O(\tau^2)$. Choose g_0 last so that $K_2^{s_*} \tau \leqslant 1/2$, $C'_* \tau \leqslant 1/4$ and $C'_* \tau^2 \leqslant \alpha'/4$. The incompatibility sum is then at most $a'(p)/4$. The pinned-tree proof of (5.29) yields

$$\sum_{\Gamma \,|\, j \in J_\Gamma} U(\Gamma) e^{\eta' \ell(\Gamma)} \leqslant C \tau^2 \, , \tag{7.39}$$

where U is the absolute Ursell coefficient times the positive object majorants. The factor $2^{|S|}$ for source variables $|t_a| \leqslant 2$ is included in K_2. Thus the regional source logarithm exists on that polydisc and vanishes at zero source. Its exponential is the ratio (7.31), proving nonvanishing.

For the boundary comparison, the periodic resolvent identity is

$$C_{D,T} - (C_T)_{D,D} = C_{D,T} (H_T)_{D,D^c} (C_T)_{D^c,D} \, . \tag{7.40}$$

If $u \in D$, $w \in D^c$ and $d_T(u,w) \leqslant R_H$, put $s = d_T(b,u) + d_T(w,c)$. The periodic triangle inequalities give

$$d_T(b, D^c) + d_T(c, D^c) + d_T(b,c) \leqslant 2s + 2R_H \, . \tag{7.41}$$

Half the inverse-decay exponent pays these distances; the remaining convolution is bounded by a fixed neighbour count times $\sup_b \sum_u e^{-\mu d_T(b,u)/2}$. Periodic shells have at most the infinite-lattice shell counts, so this sum is uniform even for wrapping or disconnected D. Therefore

$$\|C_{D,T}(b,c) - C_T(b,c)\| \leqslant Ce^{-a(d_T(b,D^c)+d_T(c,D^c)+d_T(b,c))} . \tag{7.42}$$

Interpolate the principal covariances as in (7.14). Here is explicit control of the two derivative terms. On a carrier A of size k, use a common positive edge majorant $B_{bc} = A_0 e^{-\mu_0 d_T(b,c)}$ and write $h_A = e^{-ad_T(A,D^c)}$. The covariance direction satisfies $\|\mathcal{D}_A\| \leqslant Ckh_A$ and $\|\mathcal{D}(b,c)\| \leqslant Ch_A B_{bc}$. For forest degrees d_b, use radii

$$r_{\{b,c\}} = \frac{\lambda}{32 d_b d_c B_{bc}}, \qquad r_0 = \frac{\lambda}{32 Ckh_A} . \tag{7.43}$$

The forest perturbation has norm at most $\lambda/32$, and the extra centre direction $\mathcal{D}_A \circ w^F$ has the same bound on its circle, since a positive Schur mask with unit diagonal is contractive. The total is below $\lambda/8$, so the normalised density has the Gaussian majorant (5.17). Replacing one edge direction by $E_e(\mathcal{D})$ instead uses radius $\lambda/(32 d_b d_c C h_A B_{bc})$ and costs Ch_A against the same forest majorant. Summing the centre and edge derivatives costs at most $C(k+|F|+1)^2 h_A$. Zero directions give zero derivatives; no covariance entry is divided by. Constant-conjugation invariance survives differentiation. A single marked component therefore retains its quadratic character gain, also when it crosses a corner. The product rule over repeated object occurrences and (7.39) give the differentiated cluster bound used in the regional proposition, with the same τ^2 gain.

Allocate each source cluster to its marked coarse segments with weight $|J_\Gamma|^{-1}$. A network joins its carrier, stencil edges and contour intervals with length at most $C_{\mathrm{net}}\ell(\Gamma) + R_0$. The interval may turn a corner, but its fine-path length is still bounded by $c_h \ell$; anchor it at the source mark in segment j, hence within R_0 of K_j. For $d_j > 2R_* + 2R_0$, split at $\ell = d_j/(2C_{\mathrm{net}})$. Short clusters have their original stencils inside D, identical labels and cyclic intervals in both expansions, and carriers at distance at least $d_j/2 - R_0$ from D^c, whose fixed loss is absorbed in the constant. Their covariance derivative is bounded by the preceding estimate. Long clusters are paid by (7.39). Bounded d_j are covered by the same rooted bound after enlarging C. This proves

$$|F_{D,C,T} - F_{C,T}| \leqslant C\tau^2 \sum_{j=0}^{4n-1} e^{-cd_j} . \tag{7.44}$$

No side-endpoint terms occur because both contours are the same closed square.

For the periodic comparison, the Neumann estimate for the free inverse gives $\|C_\infty(e,f)\| \leqslant \gamma_L^{-1} e^{-\mu_L \|\ell(e)-\ell(f)\|_\infty}$. Since $d_T(e,f) \leqslant L\|\ell(e)-\ell(f)\|_\infty + 2L$ for fixed lifts, it gives exponential decay in fine units with exponent $\nu = \mu_L/L$. Periodising this free inverse solves the torus inverse equation. For lifted separation $d \leqslant N_{\min}/4$, summing the image shells gives

$$\|C_T(e,f) - C_\infty(e,f)\| \leqslant Ce^{-\nu(N_{\min}-d)} \leqslant Ce^{-\nu N_{\min}/2} e^{-\nu d} . \tag{7.45}$$

The same density-direction radii apply, with $h_A = e^{-\nu N_{\min}/2}$ and a reduced common decay exponent. A rooted cluster with $C_{\mathrm{net}}\ell + R_0 < N_{\min}/4$ lifts uniquely. Projection is injective on its network, preserves carrier intersections, and preserves the fixed ordered square traversal list and its cyclic intervals. Distinct square images are farther than $2N_{\min}/3$ under the stated period hypothesis, so all its source marks lift to the chosen square. Thus the short torus and bulk clusters correspond with identical combinatorics; only their covariance marginals differ.

The other clusters, including winding ones, are paid by their extent tails. Summing over $4n$ roots gives

$$|F_{C,T} - F_{C,\infty}(n)| \leqslant C\tau^2(4n)e^{-c\iota_T} \ . \tag{7.46}$$

Finally use (6.11). The straight sum equals $-\delta m^\sigma P(C_n)$ and the four oriented corner logarithms sum to $4\log z_{\mathrm{c}}^\sigma$, by their definitions. Define

$$E^\circ_{D,C_n,T} = (F_{D,C,T} - F_{C,T}) + (F_{C,T} - F_{C,\infty}(n)) + E^\sigma_n \ . \tag{7.47}$$

Equations (7.44), (7.46) and (6.14) give the result. Every logarithm is the connected source branch, so the exponential identity has no branch ambiguity. The adjacent-side clusters and straight endpoint deficits remain in the existing corner coefficient. *q.e.d.*

For fixed geometry, E°_{ξ,C_n} depends only on the regional labels and is therefore measurable under the original history sum. Its bound is finite at fixed cutoff. The scalar factorisation consequently leaves the exterior operation, its conditional normalisers and $\mathcal{Z}_0$ in place. The boundary sum in (7.34) can be of order n; Proposition 7.3 does not assert that e^{E° is close to one for a loop running near the integration boundary.

8 The Insertion Retained after One Step

The first step produces a function of the coarse field through (1.6). Its representation below retains the closed flux insertions generated by the block average, together with the other source terms. The scalar factor extracted from the reference integral is

$$\kappa_C = e^{-\delta m^\sigma P(C)}\big(z_{\mathrm{c}}^\sigma(g_0, \pi/2)\big)^4 \ . \tag{8.1}$$

We use the same cutoff prescription and the counterclockwise (e_1, e_2) square and its coarse translates, with the period, collar and coupling hypotheses of Proposition 7.3, throughout this section. The physical conditional cutoff and its flat projection use the same fixed σ (the cutoff norm $|\bullet|_\sigma$ of (4.12), not the Haar density σ) and threshold $r = \delta_0/g_0$. The scale parameters satisfy (7.28) and the standing conditions (3.1). Closeness to the integration boundary remains in E°_ξ, the error of Proposition 7.3 at the carrier I, defined at the end of this subsection.

The first step is the operator (1.3) with the averaging operation (1.4), Borel and gauge covariant by Lemma 1.1, and ρ_1, N_1 are the measures (1.5). Besides Bałaban's construction [3] as recorded in (B1)–(B4), the representation below uses the Borel choices of minimal configurations of Lemma 3.1, the Borel background U_1 and row solutions of the measurability paragraph of Section 3.1, the bijection $\Psi_\xi \colon \mathcal{D}_\xi \to \mathcal{A}_\xi$ with invertible differential of Lemma 3.8, which is the (1.22) part of (B4)(ii) (its hypotheses: $\delta_0 \leqslant \delta_G$ is listed among the further conditions of Section B.8 and holds for small g_0, as $\delta_0 = g_0 r(g_0) \to 0$; the bound (3.40) on $\mathbf{H}_{1,Ax}$ ([3, p. 251], Lemma D.18) at the tolerance $\delta_0/40$ is the condition "MR_1 large" of item 5 there, met for small g_0 at fixed M since $R_1 \to \infty$, with $\varepsilon_1/\delta_0 \leqslant (1+\beta_0)A_0/A_1$ by [3, (2.8)] and the constants of that lemma fixed before g_0), the invertible differential of $B \mapsto U(B)$ (the (1.14) part of (B4)(ii)) and the nonvanishing Jacobian of (1.14) ((B4)(iii); its positivity on $\mathcal{A}_\xi$, Lemma B.8(iv), is used below only to match the Tr log of [3] with $\log|\det|$), Lemma 3.11(ii),(iv) and Lemma 3.9 for the a-independence of the exterior factors and the box form of the young fibre, Lemmas 3.6 and 3.9 for the boundedness of the integrand on the box, the row factorisation of the Jacobian of (1.14) (proof of Lemma 3.7, from Lemmas 3.2 and 3.5), the list of operations after (3.7) with (1.7), the disintegration theorem for standard Borel

spaces [14, (17.35), Theorem 15.6], and, for the coverage event, the stencils of Appendix C, Lemma 3.28 and Proposition 7.3; these are the facts about the unrestricted first step that the formulas below need, and each is proved or cited where it is named.

8.1 The Conditional Integral

With these inputs, this section uses the unrestricted transformation of [3] (0.1) and the first-step identities (1.11)–(1.15) with the cutoff (1.25), as recorded in Section 3.1 under (B1)–(B4), with the averaging choice permitted after (0.1); [3, (1.26)–(1.29)] are not used, except that (1.27) defines the first-step coupling $g_1(g_0)$ entering R_1 and ε_1 (Section 3.1). The outer data ξ consist of the region labels, the coarse field V, the exterior fine field V_0 and the exterior free fluctuation A_{e} in those identities. For the integrations, partition positive fine bonds into $E_\Omega = \{b \in E \,|\, b_- \in \Omega_1 \text{ or } b_+ \in \Omega_1\}$ and $E \setminus E_\Omega$. This disjoint integration convention assigns crossing bonds to E_Ω and specifies V_0 only on bonds with both endpoints outside Ω_1. Let $E_{\Omega,*}$ be the free bonds remaining after the designated-bond elimination on Ω_1, and let $D = I = \Lambda_1^*$ be the bonds integrated in Λ_1, the carrier D_+ of Section 7 (the comparison there holds for every $D_- \subseteq D \subseteq D_+$; this section is at $D = D_+ = I$, the carrier of Section F.1, which the definition of the young fibre (8.4) as a section of $\mathcal{D}_\xi \subseteq \mathfrak{g}^J \times \mathfrak{g}^I$ requires). Put $J = E_{\Omega,*} \setminus D$. The variable $a \in \mathfrak{g}^D$ is integrated and $A_{\mathrm{e}} \in \mathfrak{g}^J$ is held fixed. With i_D and i_J denoting extension by zero into $\mathfrak{g}^{E_{\Omega,*}}$, set $z_\xi(a) = i_D a + i_J A_{\mathrm{e}}$. Lebesgue measure $\mathrm{d}a$ is the product measure from the fixed Lie-algebra inner product.

Write $U_1(\xi)$ for the critical configuration of [3, (1.12)], in its axial mean gauge ((B1)). Let C_ξ be the linear solution map from free fields to fields satisfying the linearised constraints in [3, (1.14)–(1.15)], and set $\mathsf{H}_\xi = \mathsf{C}_\xi^* \Delta_{1,\xi} \mathsf{C}_\xi$, with $\Delta_{1,\xi}$ given by [3, (1.16)]. Let h_ξ be the designated-bond right inverse of the linearised constraint and $\widetilde{D}_\xi$ the analytic correction of [3, (1.14)], as identified in (B4)(i) (the corresponding construction for Bałaban's own step is the one following [4, (2.10), p. 267]): they are h_{U_1} and $\widetilde{D}_{U_1}$ of Section 3.3, with the row solutions and their domains defined there. The inverse boundary change Ψ_ξ is [3, (1.22)], acting on the free fields in $\mathfrak{g}^{E_{\Omega,*}}$ and equal to the identity off S_1^*. Define

$$\begin{aligned} q_\xi(s) &= s - h_\xi \widetilde{D}_\xi(s) \,, \\ U_\xi(a,b) &= \exp\bigl(i q_\xi(g_0 \mathsf{C}_\xi \Psi_\xi(z_\xi(a)))(b)\bigr) U_1(\xi,b) \,. \end{aligned} \tag{8.2}$$

The second formula applies on E_Ω; on $E \setminus E_\Omega$, $U_\xi = V_0$. The map C_ξ takes $\mathfrak{g}^{E_{\Omega,*}}$ into $\mathfrak{g}^{E_\Omega}$. The variable a on I, and A_{e} on $J \setminus S_1^*$, are the principal fluctuation variables of [3, (1.13)], the chart logarithms of $UU_1(\xi)^{-1}$ divided by g_0 on those free bonds (Ψ_ξ is the identity there; $I \cap S_1^* = \varnothing$ by (3.39)); on $S_1^* \subseteq J$, A_{e} is the variable of (1.22), the (1.13) variable there being $\psi_b(A_{\mathrm{e}}(b))$; and the domain of the a-integration is specified in these variables below. With $\chi_0'(\Omega_1)$ denoting the product of the group tests (1.8) retained in (1.11), define

$$\begin{aligned} \mathcal{U}_\xi &= \bigl\{U \in G^E \,|\, U|_{E\setminus E_\Omega} = V_0, M(U)|_{\Omega_1^{[1]}} = V|_{\Omega_1^{[1]}}, \mathcal{I}_\xi(U) > 0\bigr\} \,, \\ t_\xi(U) &= \frac{1}{i} \log(UU_1(\xi)^{-1})|_{E_\Omega} \,, \\ \widehat{\mathcal{A}}_\xi &= \bigl\{g_0^{-1} t_\xi(U)|_{E_{\Omega,*}} \,|\, U \in \mathcal{U}_\xi,\ UU_1(\xi)^{-1} \text{ in the chart domain on } E_\Omega\bigr\} \,. \end{aligned} \tag{8.3}$$

Here $\mathcal{I}_\xi(U)$ is the nonnegative factor multiplying the group Dirac factors in [3, (1.11)], including χ_1, ζ, χ_0' and the gauge-fixing and action exponential, and t_ξ is defined where $UU_1(\xi)^{-1}$ lies in the chart domain bond by bond on E_Ω, as the chart logarithm of [3, (1.13)]. Thus $\mathcal{U}_\xi \supseteq D^{(11)}(V, V_0)$, the fibre of the full integration domain of [3] (Section 3.1), the fibre defined

by all factors of (1.11) and not by the starred group tests χ_0' alone, with equality at every outer datum satisfying the exterior constraint $V(c) = M(V_0, c)$ on the rows c of Ω_1^c ($\bar{V}_0 = V$, Section 3.1), the one condition of $D^{(11)}(V, V_0)$ absent from $\mathcal{U}_\xi$ (the outer measure ω defined below is carried by that set; only the inclusion is used), when $\mathcal{I}_\xi(U) > 0$ is read as including that the averages and means in (1.11) are defined (the domain convention recorded in (B1)); the fluctuations UU_1^{-1} of $D^{(11)}(V, V_0)$ lie in the chart domain by (B1). The linearising change (1.14) is the map $A \mapsto A - h_\xi \widetilde{D}_\xi(A)$ on the open set $\mathcal{N}_\Theta$ of Lemma B.8(iv), which contains $g_0 \mathsf{C}_\xi \mathcal{A}_\xi$, whose Jacobian is the determinant of (B4)(iii); restricted to the constraint surface, the range of $g_0 \mathsf{C}_\xi$, it is $z \mapsto q_\xi(g_0 \mathsf{C}_\xi z)$, and on $\mathcal{A}_\xi$ it is q_{U_1} of (3.25): $q_\xi(g_0 \mathsf{C}_\xi B) = q_{U_1}(B)$ for $B \in \mathcal{A}_\xi$, with the row solutions of (B4)(i) and $\mathsf{C}_\xi = \mathsf{C}_{U_1}$ (Section 3.1). The values of q_ξ are needed on $g_0 \mathsf{C}_\xi \mathcal{A}_\xi$ only, and its Jacobian needs $\widetilde{D}_\xi$ on a neighbourhood of that set in $\mathfrak{g}^{E_\Omega}$, the set $\mathcal{N}_\Theta$, on which the row solutions are defined and analytic ((B4)(i), Lemma B.8(iv)); data that are not branches have zero contribution.

The boundary change is Ψ_ξ of (3.38), [3, (1.22)] with the cutoff g on the bonds of $(R_1^{\sim 2} \cap S_1)^*$, the set defined at p.251, and its second form (g replaced by 1, the form of p.252) on the other bonds of S_1^*, a set differing from the one printed at p.252 as recorded in Section 3.4, from new free variables to old free variables. The change of variables on these branches is Lemma 3.8 (the (1.22) part of (B4)(ii)): Ψ_ξ maps $\mathcal{D}_\xi$ of (3.3) bijectively onto $\mathcal{A}_\xi$ of (3.2), and $\mathcal{A}_\xi \subseteq \widehat{\mathcal{A}}_\xi$, since for $B \in \mathcal{A}_\xi$ the field $U(B)$ lies in $D^{(11)}(V, V_0) \subseteq \mathcal{U}_\xi$ with $t_\xi(U(B)) = q_{U_1}(B)$ chart-valued and $g_0^{-1} q_{U_1}(B)|_{E_{\Omega,*}} = B$. The young fibre over A_e is

$$\mathcal{D}_\xi(A_\mathrm{e}) \coloneqq \left\{ a \in \mathfrak{g}^D \,|\, (A_\mathrm{e}, a) \in \mathcal{D}_\xi \right\} , \tag{8.4}$$

the open box $\{\Theta_\xi = 1\}$ on a branch (Lemma 3.11(iv)). The real coordinate changes and their inverses are restricted to these domains; no injectivity on the whole Lie algebra is required. Their transformed indicators are retained in Θ_D^σ and the exterior factors (1.25). Values outside the admitted domains contribute zero.

Let v_ξ, V_ξ be the functions in [3, (1.15)–(1.16)]: v_ξ contains the Haar logarithms and the logarithm of the Jacobian of the linearising substitution (1.14) ([3, p. 249] says that v is the sum of the first two terms under the second exponential "in (1.15)"; as printed this points to (1.15) itself, whose second exponential begins with the quadratic form and v; the display meant is (1.14), whose second exponential begins with $\log \sigma(A - h\widetilde{D}(A)) + \operatorname{Tr} \log(I - h(\frac{\delta}{\delta A}\widetilde{D})(A))$, and v is read as the normalised Haar logarithm plus $\log|\det(\mathbf{1} - h_\xi \frac{\delta}{\delta A}\widetilde{D}_\xi)|$, Section 3.1 and the paragraph following (8.6)), and V_ξ contains the remaining Taylor residual of the action and gauge fixing. These are the full functions, before the boundary change. For z in their real domain put

$$p_\xi(z) = \exp\left[-\frac{1}{2}\langle z, \mathsf{H}_\xi z\rangle + v_\xi(g_0 \mathsf{C}_\xi z) - g_0^{-2} V_\xi(g_0 \mathsf{C}_\xi z) \right] . \tag{8.5}$$

With $\Theta_D^\sigma(a) = \prod_{b \in D} \mathbf{1}_{\{|a(b)|_\sigma < r\}}$, in the bond norm of [3, (1.18)], written $|\bullet|_\sigma$ as in (4.12) with the fixed σ of this section, which is the norm of the young cutoff (3.42) (so at $D = I$, $\Theta_D^\sigma = \Theta_\xi$ on $\mathfrak{g}^I$, the young fibre (8.4) is the box $\{\Theta_D^\sigma = 1\}$ on a branch, and $\mathbf{1}_{\mathcal{D}_\xi(A_\mathrm{e})}\Theta_D^\sigma = \mathbf{1}_{\mathcal{D}_\xi(A_\mathrm{e})}$ there), the conditional density and integral are

$$\begin{aligned} \mathcal{K}_\xi(a) &= \mathbf{1}_{\mathcal{D}_\xi(A_\mathrm{e})}(a) \Theta_D^\sigma(a) p_\xi(\Psi_\xi(z_\xi(a)))| \det D\Psi_\xi(z_\xi(a))| , \\ \mathcal{J}_\xi[f] &= \int f(U_\xi(a)) \mathcal{K}_\xi(a) \, \mathrm{d}a . \end{aligned} \tag{8.6}$$

Here $\mathcal{J}_\xi$ and $\mathcal{K}_\xi$ are the $\mathcal{J}_\xi^{\mathrm{a}}$ and $\mathcal{K}_\xi$ of the measurability paragraph of Section 3.1, written in the variable a of the young fibre. The determinant inside v_ξ, the Jacobian of [3, (1.14)] (not

$|\det D\Psi_\xi|$, the Jacobian of the boundary change in (8.6)), is that of the map $A \mapsto A - h_\xi \widetilde{D}_\xi(A)$ on $\mathfrak{g}^{E_\Omega}$, the phases on all bonds of Ω_1; it equals the determinant of its designated block, the product over the rows of the determinants in (3.31) (proof of Lemma 3.7: h_ξ has outputs on the designated bonds only, so the free rows of $\mathbf{1} - h_\xi \frac{\delta}{\delta A}\widetilde{D}_\xi$ are identity rows, $b_0(c)$ is the only designated bond of $\mathcal{E}_c$, Lemma 3.2, and $\widetilde{D}_{\xi,c}$ depends on $A|_{\mathcal{E}_c}$ only, Lemma 3.5). Every change of variables uses its absolute real Jacobian: v_ξ contains $\log|\det(\mathbf{1} - h_\xi \frac{\delta}{\delta A}\widetilde{D}_\xi)|$, the sum over the rows of (3.31), and the logarithms of the positive normalised Haar density, so $p_\xi > 0$ where it is defined and $\mathcal{K}_\xi \geqslant 0$; the determinant is positive on $\mathcal{A}_\xi$ ((B4)(iii), Lemma B.8(iv)) and the real logarithm written in [3] agrees with this; the identities of this section use the absolute Jacobian and need only the nonvanishing. The definition retains $p_\xi \circ \Psi_\xi$ without a truncation of the boundary terms described after [3, (1.24)].

The exterior expression of [3] is the sum over region labels and integral over (V_0, A_e), with the constraint $\delta(\bar{V}_0 V^{-1})$, the retained region indicators, $\zeta(\Omega_1^c)$, Haar factors and the prefactors of (1.15), including $z^{(0)}$. Its exterior factors are evaluated at their transformed arguments. After splitting the good tests into their D and exterior bond factors, the factors remaining from (1.25) are independent of a: Lemma 3.11(ii), under the standing conditions (3.1) (its collar hypothesis is their condition $LMR_1 > 4R_\mathrm{c} + 4L^2 M_2 R_1 + 8$; the collar (7.28) is that of Proposition 7.3), separates the bad and changed tests from D, and factors of good tests on D are redundant by idempotence (the separation of [3, (1.26), (1.29)] is not used). Those exterior factors, including any dependence on A_e, remain in $\mathcal{O}_V$. With (8.6), no boundary interaction or normaliser is discarded.

The maps in these integrals are jointly Borel in their parameters and integration variables, by the measurability paragraph of Section 3.1: the background U_1 is Borel on the Borel set of admissible pairs (the uniqueness statement of [3] for the critical point in the domain of integration in (1.11), (B1), and the theorems for Borel graphs with singleton sections); pairs that are not admissible have $D^{(11)}(V, V_0) = \varnothing$, hence zero source weight. The configuration of [3, (1.21)] is the configuration described at p.251, on its prescribed determining data, the notation $M^\bullet(W)$ being the level-wise collection $M^j(W)$ of [3, (2.11)], at the first step W and $M(W)$; the measurable choices of $U_{1,\square'}(V)$ and of $U_1(\Omega_1 \cap \Lambda_1^c, V)$ are the Borel choices of Lemma 3.1, fixed in (B1). The value of $\mathbf{H}_{1,Ax}$ used in (1.22) is $L\frac{1}{i}\log(U_1 U_1(\Omega_1 \cap \Lambda_1^c, V)^{-1})$ by (1.21), the chart logarithm of a relative configuration lying in the chart domain on S_1^* (Lemma D.18), and is Borel in (V, V_0). The maps Q_1^{s*} in [3, (1.3)] and the averages on these regular data are Borel as well.

The matrix coefficients of C_ξ, h_ξ are obtained by inversion of the invertible designated coefficient operators $\alpha_c(U_1)$, so they are Borel in the background. The row solutions $\widetilde{D}_\xi$ of Section 3.3 are Borel on $\mathcal{A}_\xi$ jointly with the branch data, by the Lusin–Novikov argument of the measurability paragraph of Section 3.1 (the set P of pairs (B, d) with d a chart-class solution family at $g_0 \mathsf{C}_{U_1} B$ for which $U_d(B) \in D^{(11)}(V, V_0)$ with the cutoff (1.25) nonzero is Borel with sections of at most one point, Lemma B.8(iii)), and so is $\frac{\delta}{\delta A}\widetilde{D}_\xi$ (measurability paragraph). In [3, (1.22)], the cutoff g is the fixed C^∞ function of the paragraph defining (3.38), each ψ_b is C^∞ on its open domain $\mathcal{W}_b$ (the chart logarithm and the exponential are analytic there), and the boundary data have just been shown Borel. Hence $\Psi_\xi(z)$ and its Jacobian are jointly Borel, and C^∞ in z. The sets $\mathcal{A}_\xi$ and $\mathcal{D}_\xi$, and the young fibres (8.4), sections of the Borel set $\mathcal{D}_\xi$, are Borel (measurability paragraph). The Hessian coefficients, v_ξ, V_ξ and the constraint factor $z^{(0)}$ are Borel: each is a finite sum or product of the Haar logarithm, the plaquette and gauge terms of the action and gauge fixing, the row determinants and the coefficients of $\alpha_c(U_1)^{-1}$, evaluated at the background, the phase and the row solutions, that is a continuous function of $(U_1, A, \widetilde{D}_\xi(A), \frac{\delta}{\delta A}\widetilde{D}_\xi(A))$ and of the Taylor coef-

ficients at $A = 0$ of the germs of Lemma 3.5, themselves continuous in U_1 by the implicit function theorem, composed with maps that are Borel jointly in the branch data and a (the background, the row solutions and their derivative, measurability paragraph of Section 3.1). The vacuum exponential is a constant, and $\zeta(\Omega_1^c)$ is a finite product of small-field tests, indicators of sets defined by inequalities between continuous functions of (V, V_0) on the open set where the local means entering them are defined, and of gauge-fixing exponentials of continuous functions of (V, V_0) on that set, every factor being set to zero where a local mean it contains is undefined (the convention of this paper for $\zeta(\Omega_1^c)$, matching the chart convention of Section 3 for the tests); so $\zeta(\Omega_1^c)$ is Borel as well. Together with the indicators, these facts prove joint Borel measurability of $U_\xi(a)$, $\mathcal{K}_\xi(a)$ and the outer weights. For an empty carrier, its Lie-algebra space is a point with unit measure; empty sums equal zero, and empty products and zero-dimensional determinants equal one. If Ω_1 is empty, set $U_1 = U_\xi = V_0$ on every bond.

The exterior group Dirac factors have a kernel interpretation as well. Let ω be the joint outer measure in [3, (1.15), (1.25)], retaining V, the region labels, V_0 and A_e: in the notation of (3.4), $\int F \,\mathrm{d}\omega = \int \mathcal{O}[F](V) \,\mathrm{d}V$ for nonnegative Borel F of the outer data, in the sense fixed in Section 3.1 (the integral of F against the measure in V that the outer operation defines; $\mathcal{O}[F](V)$ is not a density). It contains exactly the exterior factors specified above. Read its group Dirac factors by first integrating them against the corresponding coarse bond variables, with Haar measure on the remaining coarse variables. This defines a positive joint measure, rather than a pointwise density on a fibre. Its weighted measure $\bar\eta = \mathcal{J}_\xi[1]\omega$ is finite and its V marginal is the measure ρ_1 of (1.5): the operations in [3, (1.1)–(1.25)], performed as listed after (3.7) with a bounded Borel test $h(V)$ retained (signed; $h(V) = h(\mathsf{M}(U))$ is invariant under the gauge transformations u of (1.5), which are the identity at the block centres (the printed "$z \neq y$" in [3, (1.5)] read as $x \neq y$, as in (1.7)), Lemma 1.1(ii), so the Faddeev–Popov step applies with it inserted; every integral below is finite: $\int \mathcal{J}_\xi[1] \,\mathrm{d}\omega = \mathcal{Z}_0 < \infty$ is the case $h = 1$ of the first identity below, both sides being the integral of one nonnegative Borel integrand over all variables by Tonelli, and $|\mathcal{J}_\xi[W_\chi]| \leqslant \mathcal{J}_\xi[1]$) and with insertion 1 or W_χ, give

$$\begin{aligned} \int h(V) \,\mathrm{d}\rho_1(V) &= \int h(V) \mathcal{J}_\xi[1] \,\mathrm{d}\omega(V, \xi) \,, \\ \int h(V) \,\mathrm{d}N_1(V) &= \int h(V) \mathcal{J}_\xi[W_\chi(C; \bullet)] \,\mathrm{d}\omega(V, \xi) \,. \end{aligned} \tag{8.7}$$

Taking $h = 1$ in the first equality gives $\bar\eta$ the total mass $\mathcal{Z}_0 < \infty$. Let Q_V be the regular conditional probability of $\bar\eta$ on the outer data given V, defined for ρ_1-almost every V (disintegration of the finite measure $\bar\eta$ with respect to its marginal ρ_1 along the coordinate projection $\xi \mapsto V$: the measure disintegration theorem for standard Borel spaces, [14, (17.35)], stated there as an exercise with the standard proof, through a Borel isomorphism ϕ [14, Theorem 15.6] of the outer space with $[0, 1]$ and the Radon–Nikodym derivatives with respect to ρ_1 of the countably many measures $B \mapsto \bar\eta(\{\xi : V(\xi) \in B,\ \phi(\xi) \leqslant t\})$, t rational, regularised to a Borel map $V \mapsto Q_V$ into the probability measures; the outer space, a finite union over the region labels of products of the compact groups G^{E_1}, $G^{E \setminus E_\Omega}$ and the Euclidean space $\mathfrak{g}^J$, is Polish, hence standard Borel, and on the branches $\bar\eta$ is carried by a bounded set, $g_0 A_\mathrm{e}(b)$ lying in the chart ball for every $b \in J$, so the compact case would suffice. The theorem gives: Q_V is a probability measure for every V off a Borel ρ_1-null set, $V \mapsto Q_V(E)$ is Borel for every Borel E, $\bar\eta(E) = \int Q_V(E) \,\mathrm{d}\rho_1(V)$, Q_V is concentrated on the fibre $\{\xi : V(\xi) = V\}$ for ρ_1-almost every V, and two such maps agree for ρ_1-almost every V; the concentration is what replaces $h(V(\xi))$ by $h(V)$ in the identity displayed below, and the uniqueness makes $\mathcal{O}_V$ below independent of the version off a ρ_1-null set). For a nonnegative measurable outer

function F, the exterior operator is

$$\mathcal{O}_V[F] = \int_{\{0<\mathcal{J}_\xi[1]<\infty\}} \frac{F(\xi)}{\mathcal{J}_\xi[1]}\,\mathrm{d}Q_V(\xi)\,. \tag{8.8}$$

This defines a positive kernel for ρ_1-almost every V. It gives ω back on finite-positive-mass outer data: $\int h(V)\mathcal{O}_V[F]\,\mathrm{d}\rho_1(V) = \int h(V)F(\xi)\mathbf{1}_{\{0<\mathcal{J}_\xi[1]<\infty\}}\,\mathrm{d}\omega(V,\xi)$ for nonnegative Borel F and bounded Borel $h \geqslant 0$, since $\mathrm{d}\bar{\eta} = \mathcal{J}_\xi[1]\,\mathrm{d}\omega$, and for signed bounded Borel h whenever $\int F\mathbf{1}_{\{0<\mathcal{J}_\xi[1]<\infty\}}\,\mathrm{d}\omega < \infty$; so every exterior weight is retained. Equation (8.8) is the definition of $\mathcal{O}_V$ throughout this section; the identity just displayed is the sense in which it represents the exterior expression of [3]: integrated against bounded Borel $h(V)$, it reproduces the ω-integral of every outer function supported on the branches, and no pointwise identity of kernels is asserted. Set $\mathcal{O}_V = 0$ on the Borel ρ_1-null set where Q_V is not defined, so that $V \mapsto \mathcal{O}_V[F]$ is Borel. Extend it linearly to complex F whenever $\mathcal{O}_V[|F|] < \infty$. In particular $\mathcal{O}_V[\mathcal{J}_\xi[1]] = Q_V(\{0 < \mathcal{J}_\xi[1] < \infty\}) = 1$ for ρ_1-almost every V, by the next paragraph. The changes of variables are those of the unrestricted construction of [3], recorded after (3.7).

Only outer data with $0 < \mathcal{J}_\xi[1] < \infty$ need a conditional quotient; these are exactly the branches in the sense of Section 3.1: $\mathcal{J}_\xi[1] = 0$ when the young fibre $\mathcal{D}_\xi(A_\mathrm{e})$ is empty, that is off the branches of that section, while on a branch $\mathcal{J}_\xi[1] < \infty$ (measurability paragraph) and $\mathcal{J}_\xi[1] > 0$, since $\mathcal{K}_\xi > 0$ on the nonempty open box $\mathcal{D}_\xi(A_\mathrm{e})$ ($p_\xi > 0$, $|\det D\Psi_\xi| > 0$ by Lemma 3.8, and $\Theta^\sigma_D = 1$ there). Zero-mass data are $\bar{\eta}$-null, hence Q_V-null for ρ_1-almost every V. Data with infinite conditional mass are ω-null, since $\int \mathcal{J}_\xi[1]\,\mathrm{d}\omega = \mathcal{Z}_0 < \infty$ (the case $h = 1$ of (8.7), with $\mathcal{Z}_0 = \rho_1(G^{E_1})$ by (1.7)), hence $\bar{\eta}$-null and Q_V-null for ρ_1-almost every V. So Q_V is carried by the branches for ρ_1-almost every V. All branch contractions below are set to zero off the branches. This convention loses no physical mass and asserts no pointwise finiteness at exceptional outer data.

Coverage uses the fixed geometric stencils Λ_b of Appendix C: $\Lambda_b = \{b\}$ for a free bond, and $\Lambda_{b_0(c)} = \mathcal{S}_c \cap E_*$ for a designated bond, where $\mathcal{S}_c$ contains the bonds with both endpoints in the two endpoint blocks of c. Let A_cov consist of the branches on which every fine contour bond is a bond of Ω_1 and $\Lambda_b \subseteq D$ for every such bond. The geometric condition depends only on region labels, and is precisely the coverage used in Proposition 7.3; at the carrier $D = I$ of Section F.1 it is the covered event of Lemma 3.28 and of Theorem 1.2, since $\Lambda_b = \mathcal{F}_b$ for every fine bond ($\mathcal{S}_c \cap E_* = \mathcal{E}_c \cap E_*$, both being the free positive bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, (3.10) and (3.13)). The period and coupling conditions are fixed throughout this section, and (7.28) supplies the collar condition for every compatible set of region labels. Under the hypotheses of Proposition 7.3 fixed at the start of this section (its period, collar and coupling hypotheses, with (7.28) supplying the collar condition at the carrier $D = I(\xi)$ of every branch), that proposition consequently defines E°_ξ on A_cov.

8.2 The Closed Flux Insertions

Let $\bar{C} = (c_1, \ldots, c_m)$ be the oriented coarse square, with $m = 4n$, and let C be the concatenation of its straight fine paths. Fix a covered conditional history ξ and write its constrained background as $U_1 = U_1(\xi)$. For each positive coarse bond traversed by $\bar{C}$, the first and last fine bonds of its straight path lie inside its initial and final blocks, respectively, since $L \geqslant 3$. Coverage puts these bonds in Ω_1. As Ω_1 is a union of blocks, both endpoint blocks lie there. In particular c is a bond of the coarse region $\Omega_1^{(1)}$, so [3, (1.12)] imposes $M(U_1, c) = V(c)$. Here a bond of a region has at least one endpoint in that region, as in [7, Introduction]; both endpoints belong in this case. Reversed traversals use the constraint on the underlying positive bond. All relative logarithms are in the regular chart used in (4.1).

For a positive coarse bond c, define Y_c by the first formula below. The averaging constraint and (4.1) then give the second:

$$Y_c(\xi) = e^{-i\beta_c(U_1)}, \qquad P_c(U_1) = Y_c(\xi)V(c)\ . \tag{8.9}$$

Thus Y_c depends on the closed relative contours inside the two endpoint blocks of c. Set $V(c^{-1}) = V(c)^{-1}$. For a reversed traversal use

$$Y_{c^{-1}}(\xi) = V(c)^{-1}Y_c(\xi)^{-1}V(c), \qquad P_{c^{-1}}(U_1) = Y_{c^{-1}}(\xi)V(c)^{-1}\ . \tag{8.10}$$

The second identity follows by inverting $P_c = Y_cV(c)$. The factor is therefore based at the initial vertex of the reversed traversal as well.

Define the closed contraction

$$\mathcal{L}_\xi(C;V) = \chi\left(Y_{c_1}(\xi)V(c_1)\cdots Y_{c_m}(\xi)V(c_m)\right) = W_\chi(C;U_1(\xi))\ . \tag{8.11}$$

This equality follows by substituting (8.9) and (8.10) in the ordered fine holonomy. Under a site gauge u, the closed logarithms in (4.1) and their exponentials are conjugated at c_-. Consequently

$$Y_{c_i} \longmapsto u(c_{i,-})Y_{c_i}u(c_{i,-})^{-1}, \qquad V(c_i) \longmapsto u(c_{i,-})V(c_i)u(c_{i,+})^{-1}\ . \tag{8.12}$$

The intermediate gauge factors cancel in the ordered product and its character is invariant. All subsequent insertions in this section are closed contractions. No separate activity norm is assigned to a factor Y_c or to an open transport. The dependence on ξ is retained: different histories at the same V can have different constrained backgrounds and different insertions.

8.3 The Numerator Carried to the Next Transformation

The exterior operator $\mathcal{O}_V$ of (8.8) and the conditional integral $\mathcal{J}_\xi$ defined before it give, for ρ_1-almost every V,

$$\mathcal{O}_V[\mathcal{J}_\xi[1]] = 1, \qquad W_1(C;V) = \mathcal{O}_V[\mathcal{J}_\xi[W_\chi(C;\bullet)]]\ , \tag{8.13}$$

that is, $\mathrm{d}N_1 = \mathcal{O}_V[\mathcal{J}_\xi[W_\chi(C;\bullet)]]\,\mathrm{d}\rho_1$.

The identities in (8.13) are identities of measures on G^{E_1}, tested against bounded Borel functions. Let $\mathfrak{M}$ be the image of $\rho_0(U)\,\mathrm{d}U$ under $U \mapsto (M(U),U)$, a finite positive measure on $G^{E_1}\times G^E$ (the blocking kernel (0.1) with bare weight $\rho_0\,\mathrm{d}U$, in the reading of (1.3)). Its V marginal is ρ_1. Disintegration on these standard Borel spaces supplies conditional probabilities ν_V on fine fields, for ρ_1-almost every V. For bounded Borel functions f on G^E and h on G^{E_1},

$$\int h(V)f(U)\,\mathrm{d}\mathfrak{M}(V,U) = \int h(M(U))f(U)\rho_0(U)\,\mathrm{d}U = \int h(V)\left(\int f(U)\,\mathrm{d}\nu_V(U)\right)\mathrm{d}\rho_1(V)\ . \tag{8.14}$$

In particular, with $f = W_\chi(C;\bullet)$ and (1.5), $\int h\,\mathrm{d}N_1 = \int h(V)\int W_\chi(C;U)\,\mathrm{d}\nu_V(U)\,\mathrm{d}\rho_1(V)$, so $W_1(C;V) = \int W_\chi(C;U)\,\mathrm{d}\nu_V(U)$ for ρ_1-almost every V, the conditional form of (1.6); the operations below act on the middle member of (8.14), and ν_V is not used further. The averaging operation is (1.4), Borel and gauge covariant by Lemma 1.1; (4.1) is its formula on $\mathcal{O}_c$, which contains every configuration of $D^{(11)}$ on the rows of Ω_1 (the domain convention recorded in (B1)).

For $f = 1$ or $W_\chi(C;\bullet)$, apply the operations of [3, (1.1)–(1.25)] listed after (3.7) with $h(V)$ retained. The partitions are pointwise decompositions of unity. The Faddeev–Popov step (1.5)–(1.6) is applied to the operator (0.1) with the averaging choice (1.4), whose covariance on all of G^E is Lemma 1.1(ii); the closed Wilson character ($|W_\chi| \leqslant 1$, χ being a normalised irreducible character) is invariant under its fine gauges. The changes of variables pull that character back to $W_\chi(C;U_\xi(a))$, and $h(V)$ stays fixed: the gauge transformations of (1.5) are the identity at the block centres, so $M(U)$, hence $h(M(U))$, is unchanged by them (Lemma 1.1(ii)), and the later changes of variables are at fixed V. Cutoff factors are removed only where [3] states they equal one, and the boundary change is retained as (8.6). Thus the operations supply (8.7) against every bounded Borel h; since $|\mathcal{J}_\xi[W_\chi]| \leqslant \mathcal{J}_\xi[1]$ (nonnegativity of (8.6)), $\mathcal{J}_\xi[W_\chi]$ vanishes where $\mathcal{J}_\xi[1] = 0$, and $\{\mathcal{J}_\xi[1] = \infty\}$ is ω-null, so both ω-integrals are integrals over the branches, and with (8.8) they give $\int h\,\mathrm{d}\rho_1 = \int h(V)\mathcal{O}_V[\mathcal{J}_\xi[1]]\,\mathrm{d}\rho_1(V)$ and $\int h\,\mathrm{d}N_1 = \int h(V)\mathcal{O}_V[\mathcal{J}_\xi[W_\chi(C;\bullet)]]\,\mathrm{d}\rho_1(V)$. Equality of these finite measures gives the ρ_1-almost-everywhere identities (8.13), the second by the uniqueness of the Radon–Nikodym derivative (1.6).

Nonnegativity of (8.6) gives $|\mathcal{J}_\xi[W_\chi]| \leqslant \mathcal{J}_\xi[1]$, hence $|\mathcal{O}_V[\mathcal{J}_\xi[W_\chi]]| \leqslant \mathcal{O}_V[\mathcal{J}_\xi[1]] = 1$, the bound $|W_1| \leqslant 1$ of Section 1.

For the covered branches A_{cov} defined above, set

$$\mathcal{L}_\xi^{(0)}(C;V) = \begin{cases} \kappa_C e^{E_\xi^\circ}\mathcal{L}_\xi(C;V), & \xi \in A_{\mathrm{cov}}, \\ 0, & \xi \notin A_{\mathrm{cov}}. \end{cases} \tag{8.15}$$

Proposition F.2 (Appendix F) identifies its coefficient with $S^\circ_{\xi,C}$ on covered histories, with $E^\circ_\xi = E^\circ_{I,C_n,T}$ at the carrier $D = I = D_+$ (all free bonds with an endpoint in Λ_1, the convention of Section F.1). Its period and coupling hypotheses are fixed for the square; its boundary-distance terms allow arbitrary covered regional labels. The error E°_ξ depends only on those discrete labels; the background and branch indicator are Borel by the preceding construction, so this leading term is measurable. The other source terms are retained as the following conditional integral, with the same denominator:

$$\mathcal{H}_\xi(C;V) = \frac{1}{\mathcal{J}_\xi[1]} \int \Big[W_\chi(C;U_\xi(a)) - \mathcal{L}_\xi^{(0)}(C;V)\Big]\,\mathcal{K}_\xi(a)\,\mathrm{d}a\,. \tag{8.16}$$

This definition retains the difference between the conditional Wilson integral and $\mathcal{L}_\xi^{(0)}$, with the boundary dependence of (8.6) unchanged. On an uncovered branch, $\mathcal{L}_\xi^{(0)} = 0$ and it is the normalised conditional Wilson integral. No bound on $\mathcal{H}_\xi$ beyond the trivial $|\mathcal{H}_\xi| \leqslant 1 + B_C$ of the next paragraph is asserted here; its history average enters (1.9) through the defect and boundary terms E_{def}, E_{bdry} of Corollary F.3, which Theorem 1.2(ii) bounds.

For this fixed square, (7.34) and $|\chi| \leqslant 1$ give $|\mathcal{L}_\xi^{(0)}| \leqslant B_C$, where $B_C = |\kappa_C| \exp(C\tau^2(9n + 1)) < \infty$ (the constant C of (7.34), not the contour). Consequently $\mathcal{J}_\xi[1]|\mathcal{H}_\xi| \leqslant (1+B_C)\mathcal{J}_\xi[1]$. The numerator terms below are therefore integrable for the complex extension of (8.8) at ρ_1-almost every V.

Substitution in (8.13) gives the following first-step identity for ρ_1-almost every V:

$$W_1(C;V) = \mathcal{O}_V\Big[\mathcal{J}_\xi[1]\big(\mathcal{L}_\xi^{(0)}(C;V) + \mathcal{H}_\xi(C;V)\big)\Big]\,. \tag{8.17}$$

More explicitly, let $\nu_\xi = (U_\xi)_*(\mathcal{K}_\xi(a)\,\mathrm{d}a/\mathcal{J}_\xi[1])$ be the probability measure on fine fields carried by the branch. Under a fine gauge transformation u (acting on V at the block centres, Lemma 1.1(ii)), use its pushforward $(U \mapsto U^u)_*\nu_\xi$ together with U_1^u and V^u. Both

closed characters in (8.16) are invariant, E°_ξ depends only on the unchanged region labels, and change of variables preserves the normalised conditional integral. This is the gauge-invariant meaning of each branch contraction; an axial-gauge representative and its conditional measure are transported together. The outer operation retains the corresponding history weights. The physical denominator is still $\mathcal{Z}_0 = \rho_1(G^{E_1})$.

For ρ_1-almost every V, the scalar-extracted outgoing observable is

$$\widehat{W}_1(C;V) = \frac{W_1(C;V)}{\kappa_C} = \frac{\mathcal{O}_V\left[\mathcal{J}_\xi[1]\big(\kappa_C^{-1}\mathcal{L}_\xi^{(0)}(C;V) + \kappa_C^{-1}\mathcal{H}_\xi(C;V)\big)\right]}{\mathcal{O}_V[\mathcal{J}_\xi[1]]} \,. \tag{8.18}$$

The factor κ_C is nonzero: (5.39) defines its finite perimeter exponent and (6.12) writes z_{c} as the exponential of the convergent corner sum. The pair of the complex measure $\kappa_C^{-1}N_1$ and the measure ρ_1 is the input to the next blocking transformation (T acting on measures by the pushforward $T\mu = M_*\mu$ of Section 1). Equation (8.11) displays its leading closed insertion; (8.16) specifies the other source terms carried with it. Iterating this representation requires local estimates for their transformed activities, including correlations between attachments and their boundary dependence. The first step does not require replacing $\mathcal{L}_\xi(C;V)$ by $W_\chi(\bar{C};V)$.

9 The Insertion at the Retreated Carrier

This section bounds the terms of the branch identity (3.97) against the flat coefficient κ_C of Section 8: the source-containing boundary terms relative to $|\kappa_C|$ (Corollary 9.47(a)), and the labelled defect by (9.36), whose far-history part is relative to $|\kappa_C|$ through $|S^\circ_\Lambda| = |\kappa_C||e^{E^\circ_\Lambda}|$ and the exponential moment of Corollary 9.47(c), while its near-history part is an absolute perimeter rate (Remark 1.3(2)). It does not work at the carrier I of Section 3 directly. A second cutoff test is placed on the collar of Λ_1; on the branches where it fails, Λ_1 and I are retreated to a labelled carrier $D(\Lambda) \subseteq I$, the branch identity is written for each labelled term, and the labelled coefficient S°_Λ is identified with $\kappa_C e^{E^\circ_\Lambda}$ at the retreated carrier by Proposition 7.3. The bonds removed by the retreat are paid by the physical measure, through one-point witness probabilities of the kind of Appendix D; no lower bound on the exterior Schur complement is used. The inputs are Sections 3, 7 and 8, and Appendices B, D, E and F. Bałaban's equation and page numbers below are those of [3].

9.1 Setting and Inputs

Interior (young) bonds $D = I = \Lambda_1^*$, the free bonds with an endpoint in Λ_1 (the young carrier of Section 3; Λ_1^* is the starred bond set of the conventions preceding (3.1)), exterior free bonds $J = \Omega_1^* \setminus D$, where Ω_1^* is the free carrier (the $E_{\Omega,*}$ of Section 8), and $z = z_\xi(a) = i_D a + i_J A_{\mathrm{e}}$. The kernel of Section 8 is $K_\xi(a) = \mathbf{1}_{\mathcal{D}_\xi(A_{\mathrm{e}})}(a)\Theta^\sigma_D(a)p_\xi(\Psi_\xi(z_\xi(a)))|\det D\Psi_\xi|$ with $p_\xi(z) = \exp[-\frac{1}{2}\langle z, H_\xi z\rangle + v_\xi(g_0 C_\xi z) - g_0^{-2}V_\xi(g_0 C_\xi z)]$ of (8.5) (written there with $\mathsf{H}_\xi = H_\xi$, $\mathsf{C}_\xi = C_\xi$; H_ξ is the self-adjoint operator of the quadratic form) and $\Theta^\sigma_D(a) = \prod_{b\in D}\mathbf{1}\{|a_b|_\sigma < r\}$, $r = \delta_0/g_0$. Ψ_ξ is the bondwise boundary change [3, (1.22)], the identity off $S_1^* \subset \Omega_1 \cap \Lambda_1^c$, and $D \cap S_1^* = \varnothing$ by (3.39), so it changes A_{e} only: write $A'_{\mathrm{e}} = \Psi_\xi(A_{\mathrm{e}})|_J$, so that $\Psi_\xi(z) = i_D a + i_J A'_{\mathrm{e}}$. Then, with $H_{JD} = H_{DJ}^{\mathsf{T}}$,

$$-\tfrac{1}{2}\langle\Psi_\xi(z), H_\xi\Psi_\xi(z)\rangle = -\tfrac{1}{2}\langle a, H_{DD}a\rangle - \langle a, H_{DJ}A'_{\mathrm{e}}\rangle - \tfrac{1}{2}\langle A'_{\mathrm{e}}, H_{JJ}A'_{\mathrm{e}}\rangle \,, \tag{9.1}$$

with H_{DD}, H_{DJ}, H_{JJ} the blocks of H_ξ. All three terms are inside K_ξ (not in the exterior operator $\mathcal{O}_V$ of (8.8)), by (8.5).

Inputs (Lemma 3.4, stated in Section 3 and proved in Appendix B; Lemmas B.3, B.6 and 3.3); $\| \bullet \|$ on an entry is the operator norm for h:

(I1) entry range: $(H_\xi)_{bb'} = 0$ unless $|z(b) - z(b')|_1 \leqslant 2$, hence unless the initial sites are within ℓ^∞ distance $\rho_L = 4L$ in fine units; each row has at most $164L^4$ nonzero entries;

(I2) row sums: $\sum_{b' \in \Omega_1^*} \|(H_\xi)_{bb'}\| \leqslant m_L$ for every row $b \in D$, the row-sum statement of Lemma 3.4 ($D = I$). Column sums are not used;

(I3) two-sided bound on every branch in the support of the outer weight: $(\gamma_L/2)\mathbf{1} \leqslant H_{\xi,DD} \leqslant m_L \mathbf{1}$, and the same for every principal restriction (Appendix B writes I for the carrier D).

Put $\gamma = \gamma_L/2$ and $h_+ = m_L$. The constants γ_L, m_L, ρ_L depend on L and the group only (Section B.8); the bounds (I2)–(I3) hold under the hypotheses of Lemma 3.4, that is under (3.1) and the coupling thresholds of Appendix B ((B.8), Section B.8). Norms: $|\bullet|_\sigma$, $\sigma \in \{\mathrm{op}, \mathrm{HS}\}$, is the cutoff norm of (4.12), the operator norm or the normalised Hilbert–Schmidt norm h, fixed once (the conventions of Section 3); $h \leqslant |\bullet|_\sigma \leqslant c_{\mathrm{h}} h$ with $c_{\mathrm{h}} = \sqrt{N}$. Section 3 and Appendix B write I for the carrier that Section 7 calls D; in this subsection $D = I$, while the retreated carriers $D(\Lambda) \subseteq I$ of Sections 9.3–9.9 are its principal restrictions: $D' \subset D$ denotes the carrier of a retreated region in Section 9.4.1, and (I3) passes to it by principal restriction while (I2), a bound on the rows of the one operator H_ξ, is unchanged. Bond convention (Section 3): the bonds of a set X are the positive fine bonds with at least one endpoint in X, and the bonds of Ω_1^c are those with both endpoints off Ω_1. $\ell \coloneqq LMR_1$ (the macroscopic scale of Section D.7; the even-period restriction of that section is not imported by this notation).

9.2 The Mean Shift

Put $m = -H_{DD}^{-1} H_{DJ} A'_{\mathrm{e}} \in \mathfrak{g}^D$, so that $-\frac{1}{2}\langle a, H_{DD} a\rangle - \langle a, H_{DJ} A'_{\mathrm{e}}\rangle = -\frac{1}{2}\langle a - m, H_{DD}(a - m)\rangle + \frac{1}{2}\langle m, H_{DD} m\rangle$.

Lemma 9.1 (Entrywise Decay of the Interior Inverse)**.** *On a branch in the support of the outer weight, under the hypotheses of Lemma 3.4 (so that (I1)–(I3) of Section 9.1 hold), for $b, b' \in D$, with $d(b, b')$ the ℓ^∞ distance of the initial sites in fine units,*

$$\|H_{DD}^{-1}(b, b')\| \leqslant \gamma^{-1}(1 - \gamma/h_+)^{\lceil d(b,b')/\rho_L \rceil} \leqslant \gamma^{-1} e^{-\mu_H d(b,b')}\,, \qquad \mu_H \coloneqq \frac{\gamma}{h_+ \rho_L} = \frac{\gamma_L}{8 L m_L}\,.$$

Proof. By (I3), $0 \leqslant K \coloneqq \mathbf{1} - h_+^{-1} H_{DD} \leqslant (1 - \gamma/h_+)\mathbf{1}$, so $H_{DD}^{-1} = h_+^{-1} \sum_{n \geqslant 0} K^n$ (norm-convergent, $\|K\| \leqslant 1 - \gamma/h_+ < 1$). K has the entry range ρ_L of (I1) (the identity is diagonal), so $K^n(b, b') = 0$ when $d(b, b') > n\rho_L$. Hence $H_{DD}^{-1}(b, b') = h_+^{-1} \sum_{n \geqslant \lceil d/\rho_L \rceil} K^n(b, b')$ and $\|K^n(b, b')\| \leqslant \|K^n\| \leqslant (1 - \gamma/h_+)^n$, since the inner product on $\mathfrak{g}^D$ is the bondwise sum of the inner product $\langle X, Y\rangle = \mathrm{tr}_N(XY)$ of $\mathfrak{g}$ (Appendix A), whose norm is h, so that an entry of an operator on $\mathfrak{g}^D$ has h-operator norm at most the norm of the operator; the geometric sum gives $h_+^{-1}(1 - \gamma/h_+)^{\lceil d/\rho_L \rceil}/(\gamma/h_+)$. Finally $(1 - x)^s \leqslant e^{-xs}$ with $x = \gamma/h_+$, $s \geqslant d/\rho_L$. *q.e.d.*

Lemma 9.2 (Sup-Norm Shift Bound)**.** *Under the hypotheses of Lemma 9.1, let $J_1 \coloneqq \{e \in J \mid H(b', e) \neq 0$ for some $b' \in D\}$; by (I1), $J_1 \subset \{e \in J \mid \mathrm{dist}(e, D) \leqslant \rho_L\}$, where $\mathrm{dist}(e, D) = \min_{b \in D} d(e, b)$ is the ℓ^∞ distance of initial sites of (I1). Then m depends on $A'_{\mathrm{e}}|_{J_1}$ only*

(exterior bonds with $\operatorname{dist}(e,D) > \rho_L$ *have a zero column in* H_{DJ}*;* $m = 0$ *if* $J_1 = \varnothing$*), and for every* $b \in D$

$$h(m_b) \leqslant C_H^{\mathrm{near}} \max_{e\in J_1} h(A'_{\mathrm{e}}(e)) \;, \qquad C_H^{\mathrm{near}} \coloneqq \gamma^{-1} m_L N_{\mathrm{lat}}(\mu_H) \;, \\ N_{\mathrm{lat}}(\mu) \coloneqq \sum_{s\geqslant 0} 4(2s+1)^4 e^{-\mu s} < \infty \;. \tag{9.2}$$

Proof. $m_b = -\sum_{b'\in D} H_{DD}^{-1}(b,b') \sum_{e\in J_1} H(b',e) A'_{\mathrm{e}}(e)$. Operator norms of entries are those induced by h (the convention of Appendix B), so $h(H(b',e)A'_{\mathrm{e}}(e)) \leqslant \|H(b',e)\| h(A'_{\mathrm{e}}(e))$. By Lemma 9.1 and the row-sum bound (I2) for the rows $b' \in D$,

$$h(m_b) \leqslant \sum_{b'\in D} \gamma^{-1} e^{-\mu_H d(b,b')} \sum_{e\in J_1} \|H(b',e)\| h(A'_{\mathrm{e}}(e)) \leqslant \gamma^{-1} m_L \max_{e\in J_1} h(A'_{\mathrm{e}}(e)) \sum_{b'\in D} e^{-\mu_H d(b,b')} \;,$$

and $\sum_{b'} e^{-\mu_H d(b,b')} \leqslant \sum_{s\geqslant 0} 4(2s+1)^4 e^{-\mu_H s}$: at most $(2s+1)^4$ sites lie at ℓ^∞ distance s from the initial site of b, each carries four positive bonds, and projection to the torus only identifies sites. Only row sums over $b' \in D$ are used; no column-sum bound for columns $e \in J$ is needed. *q.e.d.*

Remark 9.3 (Norms and the Choice of ϑ). The cutoff norm $|\bullet|_\sigma$ is the operator norm or h (Section 9.1; the conventions of Section 3 and Appendix B), and $h \leqslant |\bullet|_\sigma \leqslant c_{\mathrm{h}} h$ with $c_{\mathrm{h}} = \sqrt{N}$. Hence $|m_b|_\sigma \leqslant c_{\mathrm{h}} h(m_b) \leqslant c_{\mathrm{h}} C_H^{\mathrm{near}} \max_{e\in J_1} h(A'_{\mathrm{e}}(e)) \leqslant c_{\mathrm{h}} C_H^{\mathrm{near}} \max_{e\in J_1} |A'_{\mathrm{e}}(e)|_\sigma$. Put

$$\vartheta \coloneqq \frac{1}{4 c_{\mathrm{h}} C_H^{\mathrm{near}}} \;,$$

a constant depending on L and the group only (through γ_L, m_L, $\rho_L = 4L$ and N); $\vartheta \leqslant 1/16$, since $c_{\mathrm{h}} \geqslant 1$, $N_{\mathrm{lat}} \geqslant 4$ (its $s = 0$ term) and $m_L \geqslant \gamma$ by (I3). Consequently, if $|A'_{\mathrm{e}}(e)|_\sigma < \vartheta r$ for every $e \in J$ with $\operatorname{dist}(e,D) \leqslant \rho_L$ (a set containing J_1), then

$$\sup_{b\in D} |m_b|_\sigma \leqslant r/4 \;. \tag{9.3}$$

No coupling threshold beyond those under which (I2)–(I3) hold (Section 9.1), and no condition on the exterior bonds farther than ρ_L from D, is involved: those bonds do not enter m (Lemma 9.2). The decay of H_{DD}^{-1} (Lemma 9.1) is used only to sum the b'-series; it plays no role in decoupling far exterior bonds, which are decoupled by the entry range (I1) alone.

9.3 The Exterior Gaussian in the History Weight, and the Payment of a Bad Collar Bond

Two things are needed from the exterior Gaussian: the identification of its place inside the history weight, which is the exact identity (9.4), and a payment for a collar bond on which the fluctuation is large. A lower bound $S \geqslant \gamma$ for the Schur complement S of (9.4), which would pay such a bond by $e^{-c\gamma\vartheta^2 r^2}$ ($c = 1/2$ for the Hilbert–Schmidt cutoff, $1/(2N)$ for the operator-norm cutoff), is not supplied by the sources (Remark 9.4), so the payment is taken from the physical measure instead (Section 9.3.2).

9.3.1 Where the Exterior Gaussian Sits

By (9.1) and completing the square (which needs only H_{DD} invertible, (I3), and $|\det D\Psi_\xi|$ independent of a: $\log|\det D\Psi_\xi(A')|$ depends on $A'|_{S_1^*}$ only, Section 3.4, (3.38)),

$$K_\xi(a) = e^{-\frac{1}{2}\langle A'_{\mathrm{e}}, SA'_{\mathrm{e}}\rangle}\cdot \mathbf{1}_{\mathcal{D}_\xi(A_{\mathrm{e}})}(a)\Theta_D^\sigma(a) e^{-\frac{1}{2}\langle a-m, H_{DD}(a-m)\rangle + v_\xi - g_0^{-2}V_\xi}|\det D\Psi_\xi| \,, \qquad S \coloneqq H_{JJ} - H_{JD}H_{DD}^{-1}H_{DJ} \,, \tag{9.4}$$

using $\frac{1}{2}\langle m, H_{DD}m\rangle - \frac{1}{2}\langle A'_{\mathrm{e}}, H_{JJ}A'_{\mathrm{e}}\rangle = -\frac{1}{2}\langle A'_{\mathrm{e}}, SA'_{\mathrm{e}}\rangle$ and $H_{JD} = H_{DJ}^{\mathsf{T}}$. The factor is written in $A'_{\mathrm{e}} = \Psi_\xi(A_{\mathrm{e}})|_J$, which differs from A_{e} on S_1^* only. Off S_1^* it is the quadratic form Bałaban collects in T_1, [3, (1.29)]; on S_1^* the conventions differ ([3, p. 252] moves first-order boundary corrections into $V^{(0)}(S_1)$, while (8.5) retains $p_\xi \circ \Psi_\xi$ without truncation), and nothing below uses this correspondence. In the convention of Section 3 ($\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$, with c_ξ of Lemma 3.9 carrying $\mathscr{E}_\xi(0)$, the exterior quadratic form, the far primitives and $\log|\det D\Psi_\xi|$, and $K_{\xi;1}$ carrying the II and IJ blocks and the near primitives) the same algebra gives $e^{c_\xi} I_{\xi;1} = e^{-\frac{1}{2}\langle A'_{\mathrm{e}}, SA'_{\mathrm{e}}\rangle} \times e^{c_\xi + \frac{1}{2}\langle A'_{\mathrm{e}}, H_{JJ}A'_{\mathrm{e}}\rangle} \times \int \mathbf{1}_{\mathcal{D}}\Theta\, e^{-\frac{1}{2}\langle a-m, H_{DD}(a-m)\rangle} \times$ (near primitives and cutoffs)$(a)\,\mathrm{d}a$, the far primitives and $\mathscr{E}_\xi(0)$ staying in c_ξ. So in either convention the Schur factor is an exact factor of w_ξ. It is independent of a, hence cancels in the conditional ratio $\mathcal{J}_\xi[f]/\mathcal{J}_\xi[1]$ inside $\mathcal{O}_V$ ((8.8)) and acts only through the history weight.

Remark 9.4 (No Lower Bound for S on $\mathfrak{g}^J$ Is Used). $\langle x, Sx\rangle = \min_{y\in\mathfrak{g}^D}\langle (y,x), H_\xi(y,x)\rangle$ ($H_{DD} > 0$ by (I3)), so $S \geqslant \gamma$ on $\mathfrak{g}^J$ is equivalent to $\langle (y,x), H_\xi(y,x)\rangle \geqslant \gamma|x|^2$ for all $(y,x) \in \mathfrak{g}^{D\cup J}$, a lower bound of H_ξ in the J-directions on all of $\mathfrak{g}^{E_{\Omega,*}}$. The sources supply no such bound: (i) the lower bound of Lemma 3.4 (its proof, local comparison) covers $\operatorname{supp} B$ by patches within $2R_{\mathrm{c}} - 1$ of I, where the plaquette bound of Lemma B.4 (stated for cubes $\square' \subset \Omega_1^{\sim -1} \setminus R_1$ with $\square'^{\sim 4} \subset \Omega_1$) yields a gauge with phases $\leqslant \widehat{\mu}$ and the flat gap γ_L applies after the perturbation $c_L\widehat{\mu} < \gamma_L/4$; (ii) J contains the free bonds of $S_1 = (\Omega_1 \setminus \Omega_1^{\sim -1}) \cup R_1$ ([3, p. 251]): on R_1 the fluctuation test of [3, (1.18)] fails, which says nothing about the background U_1 (the comparison $U_1 \approx U_{1,\square'}$ holds there too, Proposition D.17), but the plaquette bound of Lemma B.4, from which that gauge is built, is proved only for cubes of $\Omega_1^{\sim -1} \setminus R_1$; on $\Omega_1 \setminus \Omega_1^{\sim -1}$ the tests $\chi_1(\Omega_1)$, $\chi_0'(\Omega_1)$ are in force ([3, (1.11), p. 250]) but the comparison $U_1 \approx U_{1,\square'}$ is not available there ($\square'^{\sim 4} \not\subset \Omega_1$), and no near-flat gauge is supplied; (iii) a bound $H_\xi \geqslant \gamma$ on the regular part $D \cup J_1$ alone gives only the compression $S_{J_1J_1} \geqslant \gamma$, because $S_{J_1,J\setminus J_1} = H_{J_1,J\setminus J_1} \neq 0$ (entry range ρ_L). Hence the Gaussian factor (9.4) is not used for any bound; the exterior field on the large-field part is treated as in [3], paid by the cutoff factors that put it there. No counterexample is claimed: what is established is that no proof of $S \geqslant \gamma$ on $\mathfrak{g}^J$ exists in the sources, and none is attempted. Nothing in Sections 9.3–9.9 uses S.

9.3.2 The Payment of a Bad Collar Bond from the Physical Measure

For a region Λ (a union of ℓ-cubes, $\Lambda \subset \Lambda_1$) the carrier, the exterior, and the collar are

$$D(\Lambda) \coloneqq \{b \in I \mid b \text{ has an endpoint in } \Lambda\} \subset I \,, \qquad J(\Lambda) \coloneqq E_{\Omega,*} \setminus D(\Lambda) \supset J \,,$$
$$J_{\mathrm{col}}(\Lambda) \coloneqq \{e \in J(\Lambda) \mid \operatorname{dist}(e, D(\Lambda)) \leqslant \rho_L\}$$

(ℓ^∞ distance of initial sites); the collar contains $J_1(\Lambda)$ of Lemma 9.2 (H_ξ does not depend on Λ). Fix $\vartheta \in (0,1]$. A collar bond e is bad if $|A_{\mathrm{e}}(e)|_\sigma \geqslant \vartheta r$, good otherwise; $A_{\mathrm{e}}(e)$ denotes the fluctuation variable of [3, (1.13)–(1.16)] at e whether e is exterior or interior in the original split (for $e \in I \setminus D(\Lambda)$ it is the young variable a_e of the original split, exterior on the labelled

terms where it is tested, Section 9.4.1). In the construction of Section 9.4.1 a labelled term carries the regions $\Lambda_1 \supset \Lambda_1^{(0)} = \mathrm{Fill}(\Lambda_1) \supset \Lambda_1^{(1)} \supset \cdots \supset \Lambda_1^{\mathrm{fin}}$, unions of ℓ-cubes, $\Lambda_1^{(0)}$ label-independent (Lemma 9.9) and the others determined by the labels $(B^{(0)}, B^{(1)}, \dots)$ by the construction of Section 9.4.1, and the set of bonds tested on it is $T(\mathrm{labels}) \coloneqq \bigcup_k J_{\mathrm{col}}(\Lambda_1^{(k)}) \subset J(\Lambda_1^{\mathrm{fin}})$. The mark of a bond e on a labelled term is

$$M_e \coloneqq \mathbf{1}\{e \in T(\mathrm{labels})\} \cdot \mathbf{1}\{|A_{\mathrm{e}}(e)|_\sigma \geqslant \vartheta r\} \,,$$

a function of the labels and of U (A_{e} is a function of U, of V, the block average of U, and of Ω_1 through $U_1(\Omega_1, V)$); $M_e = 0$ on every labelled term on which e is not tested, so M_e is defined on every history and every labelled term; on the support of a term's indicators a tested bond is bad exactly when it belongs to some $B^{(k)}$, so there $M_e = \mathbf{1}\{e \in \bigcup_k B^{(k)}\}$ is a function of the labels alone. Let $\mathsf{w}_\xi^{\mathrm{lab}}$ be the weight of a labelled term: $\mathsf{w}_\xi = \int_{\mathfrak{g}^I} \rho_\xi(a_{\mathrm{old}})\,\mathrm{d}a_{\mathrm{old}}$ with $\rho_\xi \coloneqq e^{c_\xi} K_{\xi;1}$ the integrand of Lemma 3.9 (the factors already placed in κ_ξ, ζ_ξ, $\chi_{\xi,\mathrm{e}}$ of $\mathcal{O}$ are not duplicated), and $\mathsf{w}_\xi^{\mathrm{lab}} \coloneqq \int_{\mathfrak{g}^I} \rho_\xi(a_{\mathrm{old}}) \mathbf{1}_{\mathrm{lab}}(a_{\mathrm{old}}, A_{\mathrm{e}})\,\mathrm{d}a_{\mathrm{old}}$, where $\mathbf{1}_{\mathrm{lab}}$ is the product of the term's collar indicators (functions of A_{e} and of $a_{\mathrm{old}}|_{I\setminus D(\Lambda_1^{\mathrm{fin}})}$, Section 9.4.1); inside this integral the variables $a_{\mathrm{old}}|_{I\setminus D(\Lambda_1^{\mathrm{fin}})}$ are exterior for the re-split of Lemma 9.7 and $a_{\mathrm{old}}|_{D(\Lambda_1^{\mathrm{fin}})}$ is the young variable, so $\mathsf{w}_\xi^{\mathrm{lab}} = \int_{\mathfrak{g}^{I\setminus D^{\mathrm{fin}}}} \mathbf{1}_{\mathrm{lab}} e^{c_\xi^{\Lambda^{\mathrm{fin}}}} \big(\int_{\mathfrak{g}^{D^{\mathrm{fin}}}} K_{\xi;1}^{\Lambda^{\mathrm{fin}}}\,\mathrm{d}a\big)\,\mathrm{d}a_{\mathrm{old}}|_{I\setminus D^{\mathrm{fin}}}$. Since the indicators are nonnegative measurable functions summing to one (Section 9.4.2(c)), $\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} = \mathsf{w}_\xi$ (Tonelli). The marked physical probability is

$$p_e^\vartheta \coloneqq \mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} M_e\Big](V)\,\mathrm{d}V \,,$$

with the history sum inside $\mathcal{O}$ as in (D.23); since M_e is a function of the labels alone on the support of $\mathbf{1}_{\mathrm{lab}}$, $\mathsf{w}_\xi^{\mathrm{lab}} M_e = \int_{\mathfrak{g}^I} \rho_\xi \mathbf{1}_{\mathrm{lab}} M_e\,\mathrm{d}a_{\mathrm{old}}$, so p_e^ϑ is well defined. For $e \in J$ this equals $\mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi \sum_{\mathrm{lab}} \mathbf{1}_{\mathrm{lab}} M_e](V)\,\mathrm{d}V$ with $\sum_{\mathrm{lab}} \mathbf{1}_{\mathrm{lab}} M_e$ independent of a_{old}, because $T(\mathrm{labels}) \cap J = J_{\mathrm{col}}(\Lambda_1^{(0)}) \cap J$ is label-independent ($D^{(k)} \subset D^{(0)}$); for $e \in I$ the mark involves a variable that is young in the original split, and the labelled-term form is the definition.

Lemma 9.5 (Collar Bonds Are Regular, Eligible and in the Chart)**.** *Let $e \in J_{\mathrm{col}}(\Lambda)$ for some $\Lambda \subset \Lambda_1$ as above. Then: (a) $d(e, S_1^*) \geqslant \ell - 2 - \rho_L > 0$, so Ψ_ξ is the identity at e and $A'_{\mathrm{e}}(e) = A_{\mathrm{e}}(e)$ ($d(I, S_1^*) \geqslant LMR_1 - 2$, (3.39); Ψ_ξ bondwise with support S_1^*); also $e \in \mathcal{G}_\xi \setminus S_1^*$ by Lemma 3.11(i), since $\rho_L = 4L \leqslant 3R_{\mathrm{c}}$. (b) On the branch $|A_{\mathrm{e}}(e)|_\sigma < r$ (for $e \in J$ by Lemma 3.11(iii); for $e \in I \setminus D(\Lambda)$ by the box factor $\mathbf{1}\{|a_e|_\sigma < r\}$ of the retained cutoff, Lemma 3.11(ii) with $I \subset \mathcal{G}_\xi$; both from the good test of [3, (1.25)] $\prod_{b\in\mathcal{G}_\xi} \mathbf{1}\{|A(b)| < r\}$), so a bad collar bond has $\vartheta r \leqslant |A_{\mathrm{e}}(e)|_\sigma < r$ and $g_0 A_{\mathrm{e}}(e)$ lies in the principal chart (chart condition $h(g_0 A_{\mathrm{e}}(e)) < \pi/\sqrt{N}$: $h \leqslant |\bullet|_\sigma$ and $|g_0 A_{\mathrm{e}}(e)|_\sigma < \delta_0$, with $\delta_0 < \pi/\sqrt{N}$ for small coupling). (c) The $L^2 M_2 R_1$-cube $\square'$ of [3, (1.18)] containing the initial vertex of e lies within $3R_{\mathrm{c}} + 1 + L^2 M_2 R_1 < \ell - 1$ of Λ_1, hence in $\Omega_1^{\sim -1} \setminus R_1'$ (Lemma 3.11(i)), and $e \in (\square'^{\sim 2})^*$ (the carrier written $(Q_w^{[2]})^*$ in Proposition D.17 and Lemma D.11(c), with $Q = \square'$ and $w = L^2 M_2 R_1$, Section D.7): it is a good, eligible cube in the sense of Proposition D.17, and the comparison $\sup_{b\in(\square'^{\sim 2})^*} |U_1(b) U_{1,\square'}(b)^{-1} - \mathbf{1}|_\sigma \leqslant C\varepsilon_1 e^{-cLM_2R_1}$ holds for it ([3, p. 250] after (1.19) with Lemma B.4).*

Proof. (a) and (b) are the cited statements of Section 3, (c) those of Lemma B.4 and Proposition D.17, applied to e using $D(\Lambda) \subset I$ and $\mathrm{dist}(e, D(\Lambda)) \leqslant \rho_L$. *q.e.d.*

Lemma 9.6 (A Bad Collar Bond Forces a Witness at Threshold θ_ϑ)**.** *Put*

$$\theta_\vartheta \coloneqq \min\big(\varepsilon_0, \varepsilon_1/(4B_3 N^{3/2} L^2), \vartheta\delta_0/(2C_Q)\big) \,, \qquad C_Q = (9K_L + 2)(1 + 2B_3 N^{3/2} L^2) \,,$$

with C_Q the constant of (D.26) *($k_N = \sqrt{N}$ as in Lemma D.11). Since $\vartheta \leqslant 1$, $\theta_\vartheta \geqslant \vartheta\theta$ with $\theta = \min(\varepsilon_0, \varepsilon_1/(4B_3N^{3/2}L^2), \delta_0/(2C_Q))$ the threshold* (D.26)*. For g_0 below a threshold depending on L, M_1, M_2, the group, the source constants and ϑ (those of Lemma D.12 and of Lemma 9.5(b)): on every labelled term and for every tested bond e with $M_e = 1$, some plaquette p or some active mean-transport variable w within the neighbourhood $N(\square')$ of Lemma D.11 (the 6ℓ-neighbourhood of $\square'$, whose side $L^2M_2R_1 \leqslant \ell/8$ by $M \geqslant 8LM_2$; its sites are within 7ℓ of the initial vertex of e, the gauge-label centres within $O(L)$ more, hence within $K\ell$, $K = 256$) satisfies $|U(\partial p) - \mathbf{1}|_\sigma \geqslant \theta_\vartheta$ or $|w - \mathbf{1}|_\sigma \geqslant \theta_\vartheta$.*

Proof. This is a re-run of the R_1-step of Lemma D.12 with δ_0 replaced by $\vartheta\delta_0 \leqslant \delta_0$; it uses the comparison (D.28) of Lemma D.11(c), which is linear in the threshold, with the constant C_Q and its smallness hypotheses. By Lemma 9.5(b), $|g_0A_{\mathrm{e}}(e)|_\sigma \geqslant \vartheta\delta_0$ at the free bond e, and the unitary logarithm inequality gives $|\widetilde{U}(e)U_1(e)^{-1} - \mathbf{1}| \geqslant (2/\pi)\vartheta\delta_0$ for the gauge-fixed field $\widetilde{U}$. The comparison of Lemma 9.5(c) is $< \vartheta\delta_0/10$ for small coupling: ε_1/δ_0 is bounded by fixed source constants ([3, (2.6), (2.8)]), and $e^{-cLM_2R_1} \to 0$ as $R_1 \to \infty$. Hence $|\widetilde{U}(e)U_{1,\square'}(e)^{-1} - \mathbf{1}| \geqslant (2/\pi - 1/10)\vartheta\delta_0 > \vartheta\delta_0/2$. If every plaquette and every active gauge-variable witness in $N(\square')$ were below θ_ϑ, Lemma D.11(c) at the threshold $\theta_\vartheta \leqslant \theta$ (its final sentence; $\square'$ is R_1-eligible by Lemma 9.5(c)) would give $|\widetilde{U}(e)U_{1,\square'}(e)^{-1} - \mathbf{1}| \leqslant C_Q\theta_\vartheta \leqslant \vartheta\delta_0/2$, a contradiction. The smallness hypotheses of Lemma D.11 ($\theta \leqslant \eta_0$, $2B_3k_N^3L^2\theta \leqslant \eta_0$, $k_N\theta \leqslant c_2'$, $C_0L^2k_N\theta \leqslant 1$, $\alpha = 2k_N^2L^2\theta \leqslant a_1$) are all upper bounds on the threshold and are preserved by $\theta_\vartheta \leqslant \theta$; the P_0, P_1 cutoffs at ε_0, ε_1 in the neighbourhood and the local determining-data problem are those of the branch, unchanged by the collar labels. The conclusion holds on the original averaging-constraint and earlier-cutoff support, which is the support on which it is used below. *q.e.d.*

Lemma 9.7 (The History Weight of a Retreated Carrier)**.** *Let $\Lambda \subset \Lambda_1$ be a union of ℓ-cubes, $D(\Lambda) \subset I$, $J(\Lambda) \supset J$; write $a \in \mathfrak{g}^{D(\Lambda)}$ for the young variable of the retreated split and $A_{\mathrm{e}}^\Lambda \in \mathfrak{g}^{J(\Lambda)}$ for its exterior variable, so $A_{\mathrm{e}}^\Lambda = (A_{\mathrm{e}}, a_{\mathrm{old}}|_{I\setminus D(\Lambda)})$ in the original variables, and $A^{\Lambda,\Psi} \coloneqq \Psi_\xi(A_{\mathrm{e}}^\Lambda) = (\Psi_\xi(A_{\mathrm{e}}), a_{\mathrm{old}}|_{I\setminus D(\Lambda)})$, Ψ_ξ being the identity on $I \setminus D(\Lambda) \subset I$. Keep S_1, R_1, Ψ_ξ, $U_1(\Omega_1 \cap \Lambda_1^{\mathrm{c}}, V)$ and $\chi(\Omega_1 \cap \Lambda_1^{\mathrm{c}}, S_1)$ as built from the original Λ_1 (determined by (Ω_1, R_1), hence recoverable from the labels); call a primitive near if its free-input set meets $D(\Lambda)$, far otherwise (the near primitives of Section 3.3 are those whose free-input set meets I, so near relative to $D(\Lambda)$ implies near relative to I). With $\mathscr{E}_\xi(B) = \mathscr{E}_\xi(0) - \frac{1}{2}\langle B, H_\xi B\rangle + \sum_\alpha \mathfrak{p}_\alpha(B)$ the pre-boundary-change exponent of [3, (1.15)] in the primitives of Section 3.3 (*(3.34), written through the primitive sum (3.33) of Lemma 3.7; α runs over the primitives of all three kinds), put*

$$c_\xi^\Lambda \coloneqq \mathscr{E}_\xi(0) - \tfrac{1}{2}\langle A^{\Lambda,\Psi}, H_{J(\Lambda)J(\Lambda)}A^{\Lambda,\Psi}\rangle + \sum_{\alpha\ \mathrm{far}} \mathfrak{p}_\alpha(A^{\Lambda,\Psi}) + \log|\det D\Psi_\xi(A_{\mathrm{e}})| \,,$$

$$K_{\xi;1}^\Lambda(a) \coloneqq \mathbf{1}_{\mathcal{D}^\Lambda}(a) \exp\Big[-\tfrac{1}{2}\langle a, H_{D(\Lambda)D(\Lambda)}a\rangle - \langle a, H_{D(\Lambda)J(\Lambda)}A^{\Lambda,\Psi}\rangle + \sum_{\alpha\ \mathrm{near}} \mathfrak{p}_\alpha(a + A^{\Lambda,\Psi})\Big] \,,$$

where $\mathbf{1}_{\mathcal{D}^\Lambda}(a) \coloneqq \mathbf{1}\{\Psi_\xi(a + A_{\mathrm{e}}^\Lambda) \in \mathcal{A}_\xi\}$ is the indicator of the young fibre of the retreated split (for $\Lambda = \Lambda_1$ it is $\mathbf{1}_{\mathcal{D}_\xi(A_{\mathrm{e}})}(a)$ of (8.4)*, by the definition* (3.3) *of $\mathcal{D}_\xi$; the clause of that definition that Ψ_ξ be defined depends on A_{e} only and is carried by $\chi_{\xi,\mathrm{e}}$). Then, pointwise,*

$$e^{c_\xi^\Lambda} K_{\xi;1}^\Lambda(a) = \mathbf{1}_{\mathcal{D}^\Lambda}(a)|\det D\Psi_\xi(A_{\mathrm{e}})| \exp \mathscr{E}_\xi(\Psi_\xi(a + A_{\mathrm{e}}^\Lambda)) \,,$$

and the composed retained cutoffs and ζ multiply both sides. The box factors $\mathbf{1}\{|a_b|_\sigma < r\}$ of the bonds $b \in I \setminus D(\Lambda)$ are factors of the retained cutoff (Lemma 3.11(ii), $I \subset \mathcal{G}_\xi$), now functions of exterior variables. For $\Lambda = \Lambda_1$ these are the c_ξ, $K_{\xi;1}$ of Lemma 3.9 and

the identity (D.32) *of Lemma D.13.* *(*$|\det D\Psi_\xi(A_{\rm e})| = |\det D\Psi_\xi(A^\Lambda_{\rm e})|$*, the determinant depending on the* S_1^**-components only.)*

Proof. $\Psi_\xi(a + A^\Lambda_{\rm e}) = a + A^{\Lambda,\Psi}$, because Ψ_ξ is bondwise with support S_1^* and $d(I, S_1^*) \geqslant \ell - 2 > 0$ ((3.39)), $D(\Lambda) \subset I$. Write $z = a + A^{\Lambda,\Psi}$. Expanding $-\frac12\langle z, H_\xi z\rangle$ into the $D(\Lambda)D(\Lambda)$, $D(\Lambda)J(\Lambda)$, $J(\Lambda)J(\Lambda)$ blocks and splitting $\sum_\alpha \mathfrak{p}_\alpha(z)$ into near and far primitives, where a far primitive depends on z only through $z|_{J(\Lambda)} = A^{\Lambda,\Psi}$ (its free-input set misses $D(\Lambda)$; Lemma 3.5), gives $\mathscr{E}_\xi(z) = \mathscr{E}_\xi(0) - \frac12\langle a, H_{D(\Lambda)D(\Lambda)}a\rangle - \langle a, H_{D(\Lambda)J(\Lambda)}A^{\Lambda,\Psi}\rangle - \frac12\langle A^{\Lambda,\Psi}, H_{J(\Lambda)J(\Lambda)}A^{\Lambda,\Psi}\rangle + \sum_{\rm near}\mathfrak{p}_\alpha(z) + \sum_{\rm far}\mathfrak{p}_\alpha(A^{\Lambda,\Psi})$; multiplying by $|\det D\Psi_\xi(A_{\rm e})|\mathbf{1}_{\mathcal{D}^\Lambda}(a)$ gives the identity. The density $\mathbf{1}_{\mathcal{D}^\Lambda}|\det D\Psi_\xi|\exp\mathscr{E}_\xi(\Psi_\xi(a + A^\Lambda_{\rm e}))$ is the density of [3, (1.15)] in the transformed variables whatever free bonds are called interior; the split only decides which factors are collected in c^Λ_ξ and which in $K^\Lambda_{\xi;1}$. *q.e.d.*

Proposition 9.8 (Normalised Payment)**.** *On compatible even fine periods (the standing restriction of Section D.7), for* $0 < g_0 < g_*(\vartheta)$*, a threshold depending on* L*,* M_1*,* M_2*,* M*, the group, the source constants and* ϑ *(below the thresholds of Lemmas 9.5(b), 9.6, D.12 and D.14, and small enough for* (9.6)*), and for every bond* e*,*

$$p_e^\vartheta \leqslant (\mu_0 \otimes q_g^{\otimes\mathcal{I}} \otimes \mathrm{Haar}^{\otimes T})(W_e^\vartheta) \leqslant q_\vartheta \coloneqq n_W \max(z_P(\theta_\vartheta), z_G(\theta_\vartheta)) \,, \tag{9.5}$$

with $W_e^\vartheta \coloneqq \{$*a plaquette or auxiliary gauge-variable witness* $\geqslant \theta_\vartheta$ *within* $K\ell$ *of* $e\}$*,* $z_P(\bullet)$*,* $z_G(\bullet)$ *the tail bounds* (D.37)–(D.38) *of Lemma D.14, stated there for every threshold in* $(0, \varepsilon_0]$ *and here applied at* $\theta_\vartheta \leqslant \varepsilon_0$*:* $z_P(\theta_\vartheta) = \exp[-b_G\theta_\vartheta^2/(4g_0^2) + 4d_G\log g_0^{-1} + 4\log c_H^{-1} + 48]$*,* $z_G(\theta_\vartheta) = e^{1/2}(c_H g_0^{d_G})^{-1}e^{-b_G\theta_\vartheta^2/g_0^2}$*,* $b_G = 1/(2N)$*,* $n_W = 7(2K\ell + 7)^4$*. Moreover, with* $\pi_* = \min(p_0(g_0), p_0(g_1))$ *and* $\theta \geqslant c_\theta g_0\pi_*$ *(Proposition D.15, from [3, (2.6)–(2.9)] and* $\delta_0 = (A_1/A_0)\varepsilon_0$*),*

$$q_\vartheta \leqslant \exp[-c_q\vartheta^2\pi_*^2] \,, \qquad c_q \coloneqq b_G c_\theta^2/8 \,, \qquad \text{for } 0 < g_0 < g_*(\vartheta) \,. \tag{9.6}$$

Proof. Fix a history and a labelled term. By Lemma 9.7 (with $\Lambda = \Lambda_1^{\rm fin}$ of the term) its weight $\mathsf{w}_\xi^{\rm lab}$ is the integral of the density of [3, (1.15)] in transformed variables, times the term's collar indicators, and the mark M_e is a nonnegative function of the labels and of U. The reversal in the proof of Lemma D.13 is a change of variables for integrals of (density $\times$ nonnegative function of U and the labels), valid for every such integrand, so it applies to $\mathsf{w}_\xi^{\rm lab}M_e$ exactly as it applies to $\mathsf{w}_\xi\mathbf{1}\{z \in E(B_\xi)\}$ there: undo Ψ_ξ, the constraint elimination (it changes designated bonds only; e is free), the scaling, the linearisation, the chart and the translation by U_1. After the reversal M_e equals $\mathbf{1}\{e \in T(\text{labels})\} \cdot \mathbf{1}\{|\frac1i\log(\widetilde{U}(e)U_1(\Omega_1, V)(e)^{-1})|_\sigma \geqslant \vartheta\delta_0\}$ (Ψ_ξ is the identity at e by Lemma 9.5(a); the relevant background is $U_1(\Omega_1, V)$, as $e \notin S_1^*$). By Lemma 9.6, $M_e \leqslant \mathbf{1}_{W_e^\vartheta}$ on the support of the term's cutoffs, and W_e^ϑ is label-free (independent q_g variables are added for inactive indices, Haar variables for centres and unused sites, as in Lemma D.13). All labelled terms are nonnegative and their partitions (P_0, P_1, Q_1, R_1 and the collar labels, Section 9.4.2(c)) sum to one, so $\sum_{\rm hist}\sum_{\rm lab}(\text{term} \cdot M_e) \leqslant \sum_{\rm hist}\sum_{\rm lab}(\text{term} \cdot \mathbf{1}_{W_e^\vartheta}) \leqslant (\mu_0 \otimes q_g^{\otimes\mathcal{I}} \otimes \mathrm{Haar}^{\otimes T})(W_e^\vartheta) \cdot \mathcal{Z}_0$ (Lemmas D.13 and D.14; only "$\leqslant$" is needed), which gives the first inequality of (9.5); the second is the union bound of Lemma D.14 over the n_W witness labels with the single-witness tails z_P, z_G at threshold θ_ϑ. For (9.6): $\theta_\vartheta/g_0 \geqslant \vartheta\theta/g_0 \geqslant \vartheta c_\theta\pi_*$, so $-\log z_P(\theta_\vartheta) \geqslant b_G c_\theta^2\vartheta^2\pi_*^2/4 - 4d_G\log g_0^{-1} - C$ and $-\log z_G(\theta_\vartheta) \geqslant b_G c_\theta^2\vartheta^2\pi_*^2 - d_G\log g_0^{-1} - C$; $\log n_W \leqslant C\log\log g_0^{-2}$ since $\ell \leqslant L^2M(2\log g_0^{-2})^{\mathfrak{r}}$ (from [3, (2.5)–(2.6)], Section 9.4.2); and $\pi_*^2 \geqslant A_0^2(\frac12\log g_0^{-2})^{2p_0}$ with $p_0 \geqslant 5\mathfrak{r} \geqslant 10$ dominates every logarithm for small g_0, leaving the exponent $b_G c_\theta^2\vartheta^2\pi_*^2/8$. *q.e.d.*

The Wilson insertion does not enter p_e^{ϑ}: as in Appendix D, the insertion is bounded inside the conditional estimate ($|\mathcal{B}_\xi[f_\xi]| \leqslant C_{\mathrm{resp},\chi} g_0 r D_\xi$ on covered branches under (D.2), Lemma D.6), and the history average of D_ξ is insertion-free.

9.4 The Retreat, Its Termination and Its Costs

Cubes are the aligned ℓ-cubes of [3, (1.10), (1.20)]; Λ_1 is a union of such cubes, $D = D(\Lambda_1)$, $J = J(\Lambda_1)$, J_{col} as in Section 9.3.2; for the regions $\Lambda_1^{(k)}$ and Λ_1^{fin} defined below, $D^{(k)} \coloneqq D(\Lambda_1^{(k)})$, $J^{(k)} \coloneqq J(\Lambda_1^{(k)})$, $D^{\mathrm{fin}} \coloneqq D(\Lambda_1^{\mathrm{fin}})$ and $J^{\mathrm{fin}} \coloneqq J(\Lambda_1^{\mathrm{fin}})$ (the carrier and exterior of Section 9.3.2); ϑ as in Remark 9.3 ($\vartheta \leqslant 1/16$). All tests use $|\bullet|_\sigma$.

9.4.1 The Collar Partition, the Retreat and the Filling

Connected components of unions of ℓ-cubes are taken with respect to a fixed adjacency of ℓ-cubes (that of [3]; the arguments below use only that a component of $X \cup Y$ is either a component of X not adjacent to Y or a set meeting Y). $\mathrm{Fill}_1(\Lambda)$ denotes one pass of the modification after [3, (1.20), p. 251]: every connected component of Λ^c contained in a cube of side 100ℓ is replaced by the smallest rectangular parallelepiped of ℓ-cubes containing it; $\mathrm{Fill}(\Lambda)$ is the fixed point of iterating Fill_1 (each pass removes at least one cube from Λ or stops, so the iteration ends after finitely many passes). A region Λ is filled if $\mathrm{Fill}_1(\Lambda) = \Lambda$, i.e. every component of Λ^c contained in a cube of side 100ℓ is a rectangular parallelepiped; this is the condition [3, (2.3), p. 255] ($j = 1$: $100MR_1$ in the L^{-1}-lattice is 100ℓ fine sites; the side-100ℓ version is used so that filled regions satisfy (2.3) literally, while p.251 fills components in cubes of side $< 100\ell$). Adjacency: two ℓ-cubes are adjacent when they share a face, so that a rectangular parallelepiped of ℓ-cubes is connected (used in the equivalence "filled $\Leftrightarrow \mathrm{Fill}_1(\Lambda) = \Lambda$"); [3, p. 251] does not specify its adjacency, nor does the filling induction in the proof of Lemma D.12, and the proofs below hold for any symmetric adjacency under which parallelepipeds are connected. Standing hypothesis of this section and of Sections 9.5–9.9, which import it, additional to (3.1) (every fine period a multiple of LMR_1 and at least $8M$) and to the compatible even fine periods of Appendix D: every fine period exceeds 200ℓ. It is coupling-dependent through $\ell = LMR_1(g_0)$ (on a fixed torus it fails as $g_0 \to 0$), so every statement using it holds, at a fixed small g_0, on every compatible torus whose periods exceed 200ℓ: a lower bound on the periods in terms of the coupling, not a limit (Section B.8). Under it a set contained in a cube of side 100ℓ lifts isometrically to $\mathbb{Z}^4$, and its smallest containing parallelepiped of ℓ-cubes is well defined. Whether the Λ_1 of [3] is filled is not used: round 0 fills it.

Lemma 9.9 (Round 0)**.** *Put $\Lambda_1^{(0)} \coloneqq \mathrm{Fill}(\Lambda_1)$ and $\mathfrak{B}_0 \coloneqq (\Lambda_1^{(0)})^c \supset \Lambda_1^c$ (a set of sites; the bad bond sets of the labels are $B^{(k)}$). (a) $\Lambda_1^{(0)} \subset \Lambda_1$ is a filled union of ℓ-cubes, determined by (Ω_1, R_1) like Λ_1, so label-independent; if Λ_1 already satisfies (2.3) then $\Lambda_1^{(0)} = \Lambda_1$. (b) Every point of $\mathfrak{B}_0$ is within 100ℓ of a point of the unfilled bad region $Q_1'^{\sim 4} \cup (B(P_1^1)^c)^{\sim 5} \cup R_1'^{\sim 1}$ of [3, (1.10), (1.20)], so the witness argument of Lemma D.12 applies to $\mathfrak{B}_0$ without change: every $z \in E(\mathfrak{B}_0)$ forces a witness $\geqslant \theta$ within $K\ell$ of z, $K = 256$, with the same n_W, and $p_z^{(0)} \coloneqq \mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi \mathbf{1}\{z \in E(\mathfrak{B}_0)\}](V)\,\mathrm{d}V \leqslant q$ on compatible even fine periods, for g_0 below the thresholds of Lemmas D.12 and D.14 ($g_0 \leqslant g_H$ there).*

Proof. (a) is the definition ($\mathrm{Fill}(\Lambda) \subset \Lambda$, and the output is filled). (b) The induction of Lemma D.12 (every current component contains an original bad point, and each fill is performed inside a cube of side less than 100ℓ containing that component; this also holds if

small merged components are filled again) is run with the fill of [3, p. 251] as the base and the passes of Fill_1 as further steps: a Fill_1 pass changes only components contained in a cube of side 100ℓ, each containing an original bad point, and two points of such a cube are within $100\ell - 1 < 100\ell$ of each other, so every added point is within 100ℓ of an original bad point of its component; mergers preserve the presence of an original point. A point of the unfilled region is within 10ℓ of a bad test (proof of Lemma D.12), whose witness lies in the witness neighbourhood of the test cube (the 6ℓ-neighbourhood $N(\square')$ of Lemma D.11, as in Lemma 9.6, or the defining enlargement of P_0), within 7ℓ of the cube, whose side is less than ℓ; so the witness is within $118\ell + O(L) + 1 < K\ell$ of z. The mark $\mathbf{1}\{z \in E(\mathfrak{B}_0)\}$ depends on the regions only, so Lemmas D.13 and D.14 apply verbatim with the same n_W, giving $p_z^{(0)} \leqslant q$. *q.e.d.*

On a labelled term with current region $\Lambda_1^{(k)}$ ($k = 0$: $\Lambda_1^{(0)} = \mathrm{Fill}(\Lambda_1)$, Lemma 9.9; the bonds of $D \setminus D^{(0)}$ are exterior on every term, with their box factors as exterior good tests, Lemma 9.7) insert, inside the a-independent part of the integrand, $1 = \prod_e [\mathbf{1}\{|A_{\mathrm{e}}(e)|_\sigma < \vartheta r\} + \mathbf{1}\{|A_{\mathrm{e}}(e)|_\sigma \geqslant \vartheta r\}]$ over the untested bonds e of $J_{\mathrm{col}}(\Lambda_1^{(k)})$ (collar bonds have $A'_{\mathrm{e}}(e) = A_{\mathrm{e}}(e)$, Lemma 9.5(a), so the factors are a-independent, and they remain exterior factors on every later term, since $J(\Lambda)$ only grows as Λ shrinks), and expand. A term is labelled by its bad set $B^{(k)}$. If $B^{(k)} = \varnothing$ the iteration stops with $\Lambda_1^{\mathrm{fin}} = \Lambda_1^{(k)}$. Otherwise put

$$\Lambda_1^{(k+1)} \coloneqq \mathrm{Fill}\Big(\Lambda_1^{(k)} \setminus \bigcup_{e \in B^{(k)}} \mathcal{Q}(e)\Big)\,,$$

$$\mathcal{Q}(e) \coloneqq \text{the union of the } \ell\text{-cubes meeting the closed } \ell^\infty\text{-ball of radius } \rho_L + 1 \text{ around the initial site } x_e \text{ of } e.$$

Since $2(\rho_L + 1) + 1 = 8L + 3 < \ell$ (the standing condition $LMR_1 > 4R_{\mathrm{c}} + 4L^2 M_2 R_1 + 8$ of (3.1)), $\mathcal{Q}(e)$ consists of at most $2^4 = 16$ cubes and every point of $\mathcal{Q}(e)$ is within $\ell + \rho_L + 1 \leqslant 2\ell$ of x_e. The bonds of $D^{(k)} \setminus D^{(k+1)}$ become exterior variables of the term; they are the same bond variables as before (the chart [3, (1.13)–(1.16)] and U_1 depend on Ω_1 and V, not on Λ_1: [3, (1.12)–(1.16)] are defined on Ω_1, and Λ_1 first enters in (1.20)), and their box factors $\mathbf{1}\{|a_b|_\sigma < r\}$ become exterior good tests (Lemma 9.7). Convention (Lemma 9.7): S_1, R_1, Ψ_ξ, $U_1(\Omega_1 \cap \Lambda_1^c, V)$ and $\chi(\Omega_1 \cap \Lambda_1^c, S_1)$ are those of the original Λ_1 throughout (consistent, since $\Omega_1 \cap \Lambda_1^c \subset \Omega_1 \cap (\Lambda_1^{(k)})^c$); the retreated term is the same integral with a different split of the free bonds into integrated and exterior ones, and Ψ_ξ remains an invertible change of variables with its Jacobian retained ((8.4)). The weight $\mathsf{w}_\xi^{\mathrm{lab}}$ of Section 9.3.2 is the integral of $\rho_\xi \times$(the term's indicators) over all of $\mathfrak{g}^I$, the variables of $I \setminus D^{\mathrm{fin}}$ being integrated as exterior variables of the re-split.

Lemma 9.10 (Termination, Final State, Counts)**.** *(a) The iteration stops after at most $|\Lambda_1|/\ell^4$ rounds with $B^{(k)} \neq \varnothing$, followed by one terminal round. (b) On every final term, every $e \in J_{\mathrm{col}}(\Lambda_1^{\mathrm{fin}})$ is good: $|A_{\mathrm{e}}(e)|_\sigma < \vartheta r$. (c) Every region $\Lambda_1^{(k)}$ is filled, and every point of $\Lambda_1^{(0)} \setminus \Lambda_1^{\mathrm{fin}}$ is within 102ℓ of x_e for some bond e found bad on the term. (d) The bonds with an endpoint within 102ℓ of a given site number at most $N_{\mathrm{ret}} \coloneqq 4(207\ell)^4$; hence at most N_{ret} bonds are removed on account of one bad bond e, and for a given bond z at most N_{ret} bonds e have x_e within $102\ell + 1$ of an endpoint of z (that count is $4(204\ell + 4)(204\ell + 3)^3 \leqslant N_{\mathrm{ret}}$ for $\ell \geqslant 2$).*

Proof. (a) A bad $e \in J_{\mathrm{col}}(\Lambda_1^{(k)})$ has $\mathrm{dist}(e, D^{(k)}) \leqslant \rho_L$, so some bond of $D^{(k)}$ has an endpoint in $\Lambda_1^{(k)}$ within $\rho_L + 1$ of x_e; the cube containing that endpoint lies in $\mathcal{Q}(e) \cap \Lambda_1^{(k)}$ and is

removed. So each non-terminal round removes at least one cube. (b) $\Lambda_1^{\mathrm{fin}} = \Lambda_1^{(k)}$ for the terminal round k, whose untested collar bonds were all good; the collar bonds tested earlier were either good (the same variable, still good) or bad, and a bad bond e cannot lie in $J_{\mathrm{col}}(\Lambda_1^{\mathrm{fin}})$: $\Lambda_1^{\mathrm{fin}} \subset \Lambda_1^{(j)} \setminus \mathcal{Q}(e)$ for the round j in which e was found bad (Fill only removes cubes), so every $b \in D^{\mathrm{fin}}$ has an endpoint in Λ_1^{fin}, disjoint from the closed ball of radius $\rho_L + 1$ around x_e; if that endpoint is x_b then $|x_b - x_e|_\infty \geqslant \rho_L + 2$, and if it is $x_b + e_\mu$ then $|x_b - x_e|_\infty \geqslant \rho_L + 1$; in both cases $\mathrm{dist}(b, e) > \rho_L$. (c) Induction over the rounds and over the passes of Fill, with the invariant: the current region is filled, and every point removed so far lies within 100ℓ of a point of $\bigcup \mathcal{Q}(e)$ over the bonds e found bad so far. The invariant holds for $\Lambda_1^{(0)}$ (filled by Lemma 9.9(a); nothing removed yet). In a round, removing the cubes of $\mathcal{Q}(B^{(k)})$ gives a complement whose components are either components of $(\Lambda_1^{(k)})^c$ not adjacent to any removed cube, or sets meeting $\bigcup_{e \in B^{(k)}} \mathcal{Q}(e)$. A component of the first kind contained in a cube of side 100ℓ is a parallelepiped ($\Lambda_1^{(k)}$ is filled), so a Fill_1 pass does not change it; a pass changes only components of the second kind, each inside a cube of side 100ℓ that contains a point of some $\mathcal{Q}(e)$, so every point added lies within 100ℓ of that point. After the pass the components are again either untouched components of the first kind or sets meeting $\bigcup \mathcal{Q}(e)$ (a merger involving a filled hull contains a $\mathcal{Q}$-point). Iterating to the fixed point keeps the invariant, and the output is filled by definition. Since every point of $\mathcal{Q}(e)$ is within 2ℓ of x_e, every removed point is within 102ℓ of x_e for some bad e. (d) Sites within 102ℓ of a given site number at most $(204\ell + 1)^4$; the bonds with an endpoint among them have their initial site within $102\ell + 1$, at most $4(204\ell + 3)^4 \leqslant 4(207\ell)^4$ bonds. Sites within $102\ell + 1$ of one of the two endpoints of z lie in a box of $(204\ell + 4)(204\ell + 3)^3$ sites, each the initial site of four bonds, and $204\ell + 4 \leqslant 207\ell$ for $\ell \geqslant 2$. *q.e.d.*

Remark 9.11 (What the Final State Gives). On a final term every $e \in J^{\mathrm{fin}}$ within ρ_L of D^{fin} is good (Lemma 9.10(b)) and $A'_{\mathrm{e}}(e) = A_{\mathrm{e}}(e)$ there (for $e \in J$ by Lemma 9.5(a); for $e \in I \setminus D^{\mathrm{fin}}$ because $I \cap S_1^* = \varnothing$), so (9.3) gives $\sup_{b \in D^{\mathrm{fin}}} |m_b|_\sigma \leqslant r/4$ with the same C_H^{near} and ϑ: (I1) is unchanged, (I2) is a statement about the rows $b \in I \supset D^{\mathrm{fin}}$ with the sum over all b', and (I3) passes to D^{fin} by principal restriction. This is the input for Sections 9.5 and 9.6, with margin $3r/4$.

Lemma 9.12 (The Regions and the Retreated Fibre)**.** *(i) $\Lambda_1^{\mathrm{fin}} \subset \Lambda_1^{(0)} \subset \Lambda_1 \subset \Omega_1^{\sim -2} \cap (R_1'^{\sim})^c$ is a filled union of ℓ-cubes (Lemmas 9.9(a), 9.10(c)), so it satisfies [3, (2.3)] with the adjacency of Section 9.4.1. (ii) Lemma 3.11(i) (every site within $\ell - 1$ of Λ_1^{fin} lies in $\Omega_1^{\sim -1} \setminus R_1'$; every free bond of Ω_1 within $3R_{\mathrm{c}}$ of D^{fin} lies in $\mathcal{G}_\xi \setminus S_1^*$) and (3.39) ($d(D^{\mathrm{fin}}, S_1^*) \geqslant \ell - 2$) hold by the inclusions $\Lambda_1^{\mathrm{fin}} \subset \Lambda_1$, $D^{\mathrm{fin}} \subset I$. (iii) $|A_{\mathrm{e}}^{\mathrm{fin}}(b)|_\sigma < r$ for every $b \in J^{\mathrm{fin}}$ with $d(b, D^{\mathrm{fin}}) \leqslant 3R_{\mathrm{c}}$: for $b \in J$ by Lemma 3.11(iii) ($d(b, I) \leqslant d(b, D^{\mathrm{fin}})$), for $b \in I \setminus D^{\mathrm{fin}}$ by the retained box factor $\mathbf{1}\{|a_b|_\sigma < r\}$ (Lemma 9.7), nonzero on the term. (iv) On a final term with exterior data $A_{\mathrm{e}}^{\mathrm{fin}}$ for which the young fibre $\{a \in \mathfrak{g}^{D^{\mathrm{fin}}} \mid \Psi_\xi(a + A_{\mathrm{e}}^{\mathrm{fin}}) \in \mathcal{A}_\xi\}$ is nonempty, the fibre is the open box $\{a \mid |a_b|_\sigma < r \text{ for all } b \in D^{\mathrm{fin}}\}$ (otherwise it is empty and the term vanishes). Ω_1, R_1, S_1, Ψ_ξ and U_1 are those of the original Λ_1 by the convention of Section 9.4.1. The history label acquires the finite sequence $(B^{(0)}, B^{(1)}, \dots)$ of bad sets; its bookkeeping in the history sum of Section 3 is Section 9.8.*

Proof of (iv). The proof of Lemma 3.11(iv) is re-run with I replaced by D^{fin}; its inputs are checked one by one. Let a_0 be a point of the fibre (nonempty on a term that survives) and a any point of the box; $B_0 = \Psi_\xi(a_0 + A_{\mathrm{e}}^{\mathrm{fin}}) \in \mathcal{A}_\xi$ and $B = a + \Psi_\xi(A_{\mathrm{e}}^{\mathrm{fin}})$ (Lemma 9.7). The cutoff [3, (1.25)] at B factorises as $\Theta_{D^{\mathrm{fin}}}(a) \times$(a function of $\Psi_\xi(A_{\mathrm{e}}^{\mathrm{fin}})$ and the branch data): the proof of Lemma 3.11(ii) uses only $D^{\mathrm{fin}} \subset \mathcal{G}_\xi$, the bondwise identity $\Psi_\xi(a + A_{\mathrm{e}}^{\mathrm{fin}}) = a + \Psi_\xi(A_{\mathrm{e}}^{\mathrm{fin}})$, and that the bonds of the factors $\chi^{(0)c}(R_1)$, $\chi'_0(S_1)$ have no endpoint in $\Lambda_1 \supset \Lambda_1^{\mathrm{fin}}$; so (1.25)

at B equals (1.25) at B_0 with $\Theta_{D^{\mathrm{fin}}}(a_0)=1$ replaced by $\Theta_{D^{\mathrm{fin}}}(a)=1$, nonzero. Every row equation has a solution at $g_0\mathsf{C}_U B$: on the rows within $2R_{\mathrm{c}}+2L$ of D^{fin} (near rows for I as well, $D^{\mathrm{fin}}\subset I$) by Lemma 3.6, whose polydisc condition $|B(b)|<2+4r$ on $\mathcal{E}_c$ holds by (iii), $|a_b|<r$ and $B=a+\Psi_\xi(A_{\mathrm{e}}^{\mathrm{fin}})$ with Ψ_ξ the identity on $\mathcal{E}_c$ (the bonds of $\mathcal{E}_c$ are within $2R_{\mathrm{c}}+4L-1\leqslant 3R_{\mathrm{c}}$ of D^{fin}, hence in $\mathcal{G}_\xi\setminus S_1^*$ by (ii)); on the other rows by Lemma 3.5, since $\mathcal{E}_c$ misses D^{fin} and $B=B_0$ off D^{fin}, where the solution exists because $B_0\in\mathcal{A}_\xi$. The constraint $\mathsf{M}(U(B))=V$ on every row and the restriction V_0 off Ω_1 follow as in the proof of Lemma 3.11(iv) (they use the row equations only). The tests of [3, (1.11)] at $U(B)$: $\chi_1(\Omega_1)$ (a function of V) and $\zeta(\Omega_1^{\mathrm{c}})$ ((3.6)) do not depend on a, so they take at $U(B)$ their values at $U(B_0)$, which are nonzero; $\chi_0'(S_1)$ is nonzero by the first condition; every other cube $\square'\subset\Omega_1$ lies in $\Omega_1^{\sim-1}\setminus R_1$ and its factor is the second factor of (1.19), equal to one by (B3)(ii). Hence $B\in\mathcal{A}_\xi$. Conversely a point of the fibre has nonzero (1.25), hence $\Theta_{D^{\mathrm{fin}}}(a)=1$. *q.e.d.*

9.4.2 Costs of the Retreat and the Comparison

(a) Coverage and attachment. Appendix D defines $D_\xi=\sum_{b\in I_C}e^{-\mu d(b,J)/2}$ ((D.17)), with $I_C\subset I$ the young bonds in the free-input stencils of the contour traversals, and the covered event A_{cov} (contour bonds in Ω_1, free inputs in I; Section D.6). For a final term with carrier D^{fin} the corresponding quantities are $D_\xi^{\mathrm{fin}}\coloneqq\sum_{b\in I_C\cap D^{\mathrm{fin}}}e^{-\mu d(b,J^{\mathrm{fin}})/2}$ and $A_{\mathrm{cov}}^{\mathrm{fin}}\coloneqq\{$all contour bonds lie in Ω_1 and every free input lies in $D^{\mathrm{fin}}\}$; both depend on the labels only through the regions. Put $B_\xi^{\mathrm{fin}}\coloneqq(\Lambda_1^{\mathrm{fin}})^c$ (the bad region B_ξ of Section D.7 is that of the original Λ_1). Pointwise on the term, with $c=\mu/2$,

$$D_\xi^{\mathrm{fin}}\leqslant\sum_{b\in\Gamma_C}\sum_{z}e^{-cd(b,z)}\mathbf{1}\{z\in E(B_\xi^{\mathrm{fin}})\}\,,\qquad \mathbf{1}_{(A_{\mathrm{cov}}^{\mathrm{fin}})^c}\leqslant\sum_{b\in\Gamma_C}\mathbf{1}\{b\in E(B_\xi^{\mathrm{fin}})\}:$$

a free bond not in D^{fin} has no endpoint in Λ_1^{fin}, so $J^{\mathrm{fin}}\subset E(B_\xi^{\mathrm{fin}})$, and each $b\in I_C\cap D^{\mathrm{fin}}\subset\Gamma_C$ contributes $e^{-cd(b,J^{\mathrm{fin}})}\leqslant\sum_{z\in J^{\mathrm{fin}}}e^{-cd(b,z)}$; a contour bond outside Ω_1 or a free input outside D^{fin} meets B_ξ^{fin} (the argument of Lemma D.9 with I replaced by D^{fin}). These two inequalities are combinatorial. The statement that the response of the retreated conditional integral is bounded by $C_{\mathrm{resp},\chi}g_0rD_\xi^{\mathrm{fin}}$ is the conditional estimate for the retreated carrier (Section 9.5, with Remark 9.11 and Lemma 9.12 as its inputs). For the history average: by Lemma 9.10(c)–(d), on every final term

$$\mathbf{1}\{z\in E(B_\xi^{\mathrm{fin}})\}\leqslant\mathbf{1}\{z\in E(\mathfrak{B}_0)\}+\sum_{e\,:\,x_e\text{ within }102\ell+1\text{ of an endpoint of }z}M_e \tag{9.7}$$

(a bond z meets $(\Lambda_1^{\mathrm{fin}})^c$ either because it meets $\mathfrak{B}_0=(\Lambda_1^{(0)})^c$ or because one of its endpoints lies in $\Lambda_1^{(0)}\setminus\Lambda_1^{\mathrm{fin}}$, which is within 102ℓ of x_e for a bond e found bad on the term, $M_e=1$). Multiply by $\rho_\xi\mathbf{1}_{\mathrm{lab}}$, integrate over $\mathfrak{g}^I$, sum over labels and histories, apply $\mathcal{O}$ and divide by $\mathcal{Z}_0$ (all integrands nonnegative, finitely many labels per history, Tonelli): the first term is label-independent, $\sum_{\mathrm{lab}}\mathbf{1}_{\mathrm{lab}}=1$ reduces it to $\mathcal{Z}_0^{-1}\int\mathcal{O}[\mathsf{w}_\xi\mathbf{1}\{z\in E(\mathfrak{B}_0)\}]\,\mathrm{d}V=p_z^{(0)}\leqslant q$ (Lemma 9.9(b)); the second gives $\sum_{e\text{ near }z}p_e^{\vartheta}\leqslant N_{\mathrm{ret}}q_\vartheta$ (Lemma 9.10(d), Proposition 9.8). Hence, on compatible even fine periods exceeding 200ℓ and for $0<g_0<g_*(\vartheta)$ (Proposition 9.8; this threshold also covers Lemma 9.9(b)),

$$\begin{aligned}\mathcal{Z}_0^{-1}\int\mathcal{O}\Big[\sum_{\mathrm{lab}}\mathsf{w}_\xi^{\mathrm{lab}}D_\xi^{\mathrm{fin}}\Big]\mathrm{d}V&\leqslant C_LS_cn_C(q+N_{\mathrm{ret}}q_\vartheta)\,,\\ \mathcal{Z}_0^{-1}\int\mathcal{O}\Big[\sum_{\mathrm{lab}}\mathsf{w}_\xi^{\mathrm{lab}}\mathbf{1}_{(A_{\mathrm{cov}}^{\mathrm{fin}})^c}\Big]\mathrm{d}V&\leqslant C_Ln_C(q+N_{\mathrm{ret}}q_\vartheta)\,.\end{aligned} \tag{9.8}$$

This is the bound (D.25) with q replaced by $q + N_{\mathrm{ret}} q_\vartheta$, which has the same normalised form. The scalar κ_C of (8.1) does not enter the history average: the ratio $\kappa_C^{-1} \mathcal{B}_\xi[f_\xi]$ is formed in Section 9.9 from the response bound and this average.

(b) Boundary activities. The bonds that change status are $D \setminus D^{(0)}$ (label-independent, round 0, Lemma 9.9) and, on a final term, $D^{(0)} \setminus D^{\mathrm{fin}}$ (young in the original split, exterior on the term): at most N_{ret} of the latter per bad bond e, each with an endpoint within 102ℓ of x_e (Lemma 9.10(c)–(d)), and $p_e^\vartheta \leqslant q_\vartheta$. How the connected boundary terms of the retreated conditional integral are localised relative to these bonds, and their bound, are Section 9.6. If that cost on a term is a sum over the bonds $b \in D^{(0)} \setminus D^{\mathrm{fin}}$ of quantities bounded by $C e^{-cd(b,\Gamma_C)}$ with C, c independent of the labels, the fields and r, then it is linear in the marks, (c) applies, and its history average is at most $C N_{\mathrm{ret}} \sum_b e^{-cd(b,\Gamma_C)} \sup_e p_e^\vartheta \leqslant C N_{\mathrm{ret}} S_c C_L n_C q_\vartheta$, uniform in the volume; without such a localisation the sum over all testable bonds is proportional to the volume, and if the cost were multiplicative in the bad bonds, (c) would not apply. Section 9.6 establishes the additive form: in (9.27) the source-containing connected boundary terms are bounded by sums over $e \in J(\Lambda)$ of $e^{-\mu' \operatorname{dist}(e,C)}$ (and over $e \in J_1(\Lambda)$ of $e^{-\mu_H \operatorname{dist}(e,I_C)/2}$) with prefactors uniform in the exterior data within the term's support, and Section 9.9 takes their history average through (9.7) and (c). The bonds $D \setminus D^{(0)}$ carry no marks; they form the label-independent boundary of $\Lambda_1^{(0)}$, belong to $J(\Lambda)$ on every final term, and enter (9.27) through the sum over $J(\Lambda)$ like the bonds of J.

(c) No label entropy for the coverage cost. The collar labels are a partition of unity on each term: $\sum_{B^{(k)}} \prod_{e \in B^{(k)}} \mathbf{1}\{e \text{ bad}\} \prod_{e \notin B^{(k)}} \mathbf{1}\{e \text{ good}\} = 1$ pointwise, round by round, so by induction from the terminal rounds (termination: Lemma 9.10(a)) $\sum_{\mathrm{lab}} \mathbf{1}_{\mathrm{lab}} = 1$ pointwise on $\mathfrak{g}^I \times \mathfrak{g}^J$, and $\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} = \int_{\mathfrak{g}^I} \rho_\xi \sum_{\mathrm{lab}} \mathbf{1}_{\mathrm{lab}} \, \mathrm{d}a_{\mathrm{old}} = \mathsf{w}_\xi$ (Tonelli; the labels are indicators of finitely many measurable sets). Every term is nonnegative and the cost in (a) is bounded by $\sum_e M_e \times$(a local quantity), linear in the marks; summing the labelled terms back therefore gives $\sum_e M_e \times$(local quantity) under the history average, with no product over labels and no factor depending on the number of tested bonds. This is established for the cost (a) only; for (b) see the conditional sentence of (b) and Section 9.6.

Comparison. $r = \delta_0/g_0 = A_1(\log g_0^{-2})^{p_0}$ ([3, pp. 246–247]: $\varepsilon_0 = g_0 A_0 (\log g_0^{-2})^{p_0}$, $\delta_0 = (A_1/A_0)\varepsilon_0$; r is not used in (9.9)–(9.10)). [3, (2.6), p. 255] ($n = 1$, $m = 0$): $g_1 \leqslant (1 + \beta_0) g_0$ and $g_0 \leqslant (1 + \beta_0) g_1$, so for small g_0, $\frac{1}{2} \log g_0^{-2} \leqslant \log g_1^{-2} \leqslant 2 \log g_0^{-2}$. R_1 is the smallest power of L with $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$ (p.246), so $R_1 \leqslant L (\log g_1^{-2})^{\mathfrak{r}} \leqslant L (2 \log g_0^{-2})^{\mathfrak{r}}$ and

$$\ell = LMR_1 \leqslant L^2 M (2 \log g_0^{-2})^{\mathfrak{r}} , \qquad \text{hence} \qquad \log N_{\mathrm{ret}} \leqslant C + 4\mathfrak{r} \log(2 \log g_0^{-2}) \qquad (9.9)$$
$$\text{and} \qquad C\ell^4 \leqslant C (L^2 M)^4 2^{4\mathfrak{r}} (\log g_0^{-2})^{4\mathfrak{r}} .$$

With $\pi_* \geqslant A_0 (\frac{1}{2} \log g_0^{-2})^{p_0}$ and $p_0 \geqslant 5\mathfrak{r}$ (p.246),

$$\log(1/q_\vartheta) \geqslant c_q \vartheta^2 A_0^2 2^{-2p_0} (\log g_0^{-2})^{2p_0} \gg C (L^2 M)^4 2^{4\mathfrak{r}} (\log g_0^{-2})^{4\mathfrak{r}} \geqslant C\ell^4 , \qquad (9.10)$$

since $2p_0 \geqslant 10\mathfrak{r} > 4\mathfrak{r}$. Hence for g_0 below a threshold depending on L, M, M_1, M_2, the group, the source constants (the A_0, A_1, β_0, p_0, $\mathfrak{r}$ of [3] among them; M_1 enters through $g_*(\vartheta)$) and ϑ, $N_{\mathrm{ret}} q_\vartheta \leqslant e^{-c_q \vartheta^2 \pi_*^2/2}$; and for each C independent of g_0 there is a threshold, depending also on C, below which $e^{C\ell^4} q_\vartheta \leqslant e^{-c_q \vartheta^2 \pi_*^2/2}$ (C a constant of L, M, M_2, the group and the source constants; a per-term cost growing with r, say like $r^2 \propto (\log g_0^{-2})^{2p_0}$, is not covered by this comparison and would have to be paid differently). This is the $p_0 \geqslant 5\mathfrak{r}$ accounting of [3] with the present constants.

9.5 The Selected Majorant for the Translated Box

Fix a history and a labelled term with final region $\Lambda \coloneqq \Lambda_1^{\mathrm{fin}}$ (a filled union of ℓ-cubes, Lemma 9.12(i)), carrier $D \coloneqq D(\Lambda) = D^{\mathrm{fin}} \subset I$, exterior $J(\Lambda) = E_{\Omega,*} \setminus D$ and exterior variable $A^{\Lambda,\Psi} = (\Psi_\xi(A_{\mathrm{e}}), a_{\mathrm{old}}|_{I\setminus D})$ (Lemma 9.7). Write $H \coloneqq H_{D(\Lambda)D(\Lambda)}$, $\ell_\Lambda \coloneqq H_{D(\Lambda)J(\Lambda)} A^{\Lambda,\Psi} \in \mathfrak{g}^D$ and, for $t \in [0,1]$,

$$P_t^\Lambda(a) \coloneqq \sum_{\mathfrak{p}\ \text{near}} \mathfrak{p}(a + tA^{\Lambda,\Psi}) \,, \qquad K_{\xi;t}^\Lambda(a) \coloneqq \mathbf{1}_{\mathcal{D}^\Lambda}(a) \exp\Big[-\tfrac{1}{2}\langle a, Ha\rangle - t\langle a, \ell_\Lambda\rangle + P_t^\Lambda(a)\Big] \,,$$

$$I_t^\Lambda \coloneqq \int_{\mathfrak{g}^D} K_{\xi;t}^\Lambda \,\mathrm{d}a \,, \qquad \mathrm{d}\nu_t^\Lambda \coloneqq K_{\xi;t}^\Lambda \,\mathrm{d}a / I_t^\Lambda \,,$$

the retreated split's version of the exterior interpolation (3.45): near means near relative to $D(\Lambda)$ (Lemma 9.7), $\mathfrak{p}$ runs over the near primitives of all three kinds, and for $\Lambda = \Lambda_1$ these are the $P_{\xi,t}$, $K_{\xi;t}$, $I_{\xi;t}$, $\nu_{\xi;t}$ of Section 3 (Ψ_ξ is the identity on the bonds read by the near primitives and by H_{IJ}, Lemma 3.11(i)); at $t = 1$, $K_{\xi;1}^\Lambda$ is that of Lemma 9.7. On the support of the term's exterior cutoffs (the branch indicator ι_ξ, the collar indicators and the box factors on $I\setminus D(\Lambda)$, Lemma 9.7) the retreated young fibre is nonempty and is the box: by Lemma 3.11(iv) the original fibre over the branch data is the box over I, and the box factors on $I \setminus D(\Lambda)$ place $(a, a_{\mathrm{old}}|_{I\setminus D(\Lambda)})$ in it for every a with $|a_b|_\sigma < r$, $b \in D(\Lambda)$, so the fibre of the retreated split over $A^{\Lambda,\Psi}$ is nonempty and Lemma 9.12(iv) gives $\mathbf{1}_{\mathcal{D}^\Lambda}(a) = \Theta_D(a) \coloneqq \prod_{b\in D} \chi_b(a)$, $\chi_b(a) \coloneqq \mathbf{1}\{|a_b|_\sigma < r\}$; off that support the term vanishes, and nothing is claimed for it. (This nonemptiness is used for $I_t^\Lambda > 0$ in Lemma 9.13(c), for $\mathbf{1}_{\mathcal{D}^\Lambda}(B + tm) = \Theta^t(B)$ in (9.13), and for the identity hard cutoff in Lemma 9.17.) The mean shift of the retreated split is $m \coloneqq -H^{-1}\ell_\Lambda$ (Section 9.2 with $D(\Lambda)$, $J(\Lambda)$), and by Lemma 9.2 (its proof uses (I1) and (I3) for the principal block $H_{D(\Lambda)D(\Lambda)}$, which hold by principal restriction, and (I2) for the rows $b \in D(\Lambda) \subset I$, Section 9.1 and Remark 9.11) m depends on $A^{\Lambda,\Psi}|_{J_1(\Lambda)}$ only, $J_1(\Lambda) \subset J_{\mathrm{col}}(\Lambda)$. The final collar is good (Lemma 9.10(b)), $|A^{\Lambda,\Psi}(e)|_\sigma = |A_{\mathrm{e}}^\Lambda(e)|_\sigma < \vartheta r$ for every $e \in J_{\mathrm{col}}(\Lambda)$ (Ψ_ξ is the identity there, Lemma 9.5(a); on $I \setminus D(\Lambda)$ by (3.39)), so (9.3) of Remark 9.3 (the choice $c_{\mathrm{h}} C_H^{\mathrm{near}} \vartheta \leqslant 1/4$), applied at $D(\Lambda)$ by Remark 9.11, gives

$$\sup_{b\in D(\Lambda)} |m_b|_\sigma \leqslant r/4 \,, \qquad \text{hence} \qquad \sup_b t|m_b|_\sigma \leqslant r/4 \quad \text{for every } t \in [0,1]. \tag{9.11}$$

Completing the square (Section 9.2): $-\frac{1}{2}\langle a, Ha\rangle - t\langle a, \ell_\Lambda\rangle = -\frac{1}{2}\langle a - tm, H(a - tm)\rangle + \frac{1}{2}t^2\langle m, Hm\rangle$.

Lemma 9.13 (The Final Term Is a Branch for the Estimate Lemmas)**.** *(a) $H = H_{D(\Lambda)D(\Lambda)}$ satisfies* (3.56)*, $\gamma\,\mathbf{1} \leqslant H \leqslant h_+\mathbf{1}$ with $\gamma = \gamma_L/2$, $h_+ = m_L$, and the entry range $\rho_L \leqslant R$: this is (I1), (I3) for the principal restriction (Section 9.1). Hence Lemmas 3.13, 3.14, 3.15 and 3.16 hold for H with the constants n_*, R_*, C_w, δ_M, A_*, C_d, α_G of Section 3 and Appendix B (these lemmas use only* (3.56)*, the range and the partitions, as the proof of Lemma B.13 records). (b) Retained inputs near the carrier. Let $r_{\mathrm{e}}^\Lambda \coloneqq \max\{|A^{\Lambda,\Psi}(b)|_\sigma : b \in J(\Lambda),\ \mathrm{dist}(b, D(\Lambda)) \leqslant 3R_{\mathrm{c}}\}$ (zero if the set is empty), the retreated split's $r_{\mathrm{e},\xi}$ (*(3.52)*, Section D.1). Then $r_{\mathrm{e}}^\Lambda < r$: a bond $b \in J$ within $3R_{\mathrm{c}}$ of $D(\Lambda) \subset I$ is within $3R_{\mathrm{c}}$ of I, so by Lemma 3.11(i),(iii) for the original term it lies in $\mathcal{G}_\xi \setminus S_1^*$, Ψ_ξ is the identity at b and $|A_{\mathrm{e}}(b)|_\sigma < r$ on the branch; a bond $b \in I \setminus D(\Lambda)$ carries $a_{\mathrm{old}}(b)$ with $|a_{\mathrm{old}}(b)|_\sigma < r$ (its box factor is a factor of the retained cutoff of the term, Lemma 9.7) and Ψ_ξ is the identity on I (*(3.39)*). So $s^\Lambda \coloneqq 1 + r + r_{\mathrm{e}}^\Lambda \leqslant 1 + 2r$. On the collar $J_{\mathrm{col}}(\Lambda)$ (range $\rho_L = 4L \leqslant 3R_{\mathrm{c}}$) the sharper bound $< \vartheta r$ of Lemma 9.10(b) holds; it is used only through* (9.11)*. (c) Charts and primitive bounds. A row within $2R_{\mathrm{c}} + 2L$ of $D(\Lambda)$ is within $2R_{\mathrm{c}} + 2L$ of I, so the applicability paragraph of Appendix B (Lemmas B.4, B.5, B.7, which need only that the row's two-block set lies within $3R_{\mathrm{c}}$ of I and the retained*

inputs there are below r, supplied by (b)) applies to it; hence Lemma B.9 holds for the near primitives of the retreated split under threshold item 3 of Section B.8, $4C_{\mathsf{C}}g_0(1+2r) \leqslant \vartheta_*/2$, with s_ξ replaced by $s^\Lambda \leqslant 1+2r$: every near primitive $\mathfrak{p}$ is, at the input $a+tA^{\Lambda,\Psi}$, the restriction to real arguments of a function analytic in (a,t) on a complex neighbourhood of the enlarged box $\{|a_b|_\sigma \leqslant 2r \ \forall b \in D(\Lambda)\} \times [0,1]$, with $|\mathfrak{p}(a+tA^{\Lambda,\Psi})| \leqslant C_p g_0 (s^\Lambda)^3$ for the plaquette and gauge primitives and the smaller bounds $C_p g_0^2 (s^\Lambda)^2$, $C_p g_0 s^\Lambda$ for the Haar and Jacobian primitives ($g_0 \leqslant 1 \leqslant s^\Lambda$), and the derivative bounds (B.19) *with s^Λ; and Lemma 3.6 holds at $D(\Lambda)$, so the exponent of $K^\Lambda_{\xi;t}$ is real and finite on the open box and $K^\Lambda_{\xi;t} > 0$ there, $I^\Lambda_t > 0$. (d) Hosts. The host $Y_\mathfrak{p}$ of a primitive* (3.36) *and the probe host Y_b of a bond* (3.75) *are defined from the lifted stencil and M alone; the localisation family of the retreated split is $\mathbf{D}_0(\Lambda) \coloneqq \{Y_\mathfrak{p} : \mathfrak{p}$ near relative to $D(\Lambda)\} \cup \{Y_b : b \in D(\Lambda)\}$, with $E(Y) \coloneqq \{b \in D(\Lambda) : b \cap Y \neq \varnothing\}$, and the free-input set of a primitive assigned to Y lies in Y (*(3.36)*), its young part in $E(Y)$. The counts of Lemma B.10, at most N_h primitives per host and $d_0(Y) \leqslant B_h + 4$, are counts over templates and anchor positions, independent of the carrier. Hence, for $Y \in \mathbf{D}_0(\Lambda)$, $t \in [0,1]$ and a in the local box $\{|a_b|_\sigma < r \ \forall b \in E(Y)\}$,*

$$\mathbf{V}^\Lambda_{\xi,t}(Y,a) \coloneqq \sum_{\mathfrak{p} \ near,\, Y_\mathfrak{p}=Y} \mathfrak{p}(a+tA^{\Lambda,\Psi}) \quad \textit{satisfies}$$

$$|\mathbf{V}^\Lambda_{\xi,t}(Y,a)| \leqslant N_h C_p g_0 (s^\Lambda)^3 \leqslant \omega_t e^{-\kappa d_0(Y)}\,, \qquad \omega_t \coloneqq C_0 g_0 (1+2r)^3\,, \tag{9.12}$$

by (c), $d_0(Y) \leqslant B_h + 4$ and $C_0 = N_h C_p e^{\kappa(B_h+4)}$ ((B.20)*); $\mathbf{V}^\Lambda_{\xi,t}(Y,\bullet) \coloneqq 0$ off the local box and on hosts carrying no near primitive. For $t = 0$ and every Λ the input is a alone, with $|a_b|_\sigma < r$ on the local box, so (c) gives the sharper bound $|\mathbf{V}^\Lambda_{\xi,0}(Y,a)| \leqslant \omega(g_0) e^{-\kappa d_0(Y)}$ with $\omega(g_0) = C_0 g_0 (1+r)^3 \leqslant \omega_t$ ($s = 1+r$ in Lemma B.9); for $\Lambda = \Lambda_1$ this is the $\mathbf{V}_\xi$ of Section 3 with its bound $\omega(g_0)$. The factor $(1+2r)^3$ in place of $(1+r)^3$ is the retained input, which enters for $t > 0$.*

9.5.1 The Translated Objects

Let $B \coloneqq a - tm$ be the centred variable, $a = B + tm$. Put, for $b \in D(\Lambda)$ and $Y \in \mathbf{D}_0(\Lambda)$,

$$\chi^t_b(B) \coloneqq \chi_b(B+tm) = \mathbf{1}\{|B_b + tm_b|_\sigma < r\}\,, \qquad \Theta^t(B) \coloneqq \prod_{b \in D(\Lambda)} \chi^t_b(B)\,,$$

$$\mathbf{W}_t(Y,B) \coloneqq \mathbf{V}^\Lambda_{\xi,t}(Y, B+tm);$$

χ^t_b is a Borel function of B_b alone with values in $\{0,1\}$, $\mathbf{W}_t(Y,\bullet)$ depends on B through $B|_{E(Y)}$ only (m is a fixed vector), and $\mathbf{W}_t(Y,B) = 0$ unless $|B_b + tm_b|_\sigma < r$ for all $b \in E(Y)$ (the translated local box), where $|\mathbf{W}_t(Y,B)| \leqslant \omega_t e^{-\kappa d_0(Y)}$ by (9.12). Translation invariance of Lebesgue measure gives, for every bounded Borel F on $\mathfrak{g}^{D(\Lambda)}$,

$$\begin{aligned}\int F(a) K^\Lambda_{\xi;t}(a)\,\mathrm{d}a = e^{\frac12 t^2 \langle m, Hm\rangle} \int F(B+tm)\,\Theta^t(B) \\ \times \exp\Big[-\tfrac12\langle B, HB\rangle + \sum_{Y \in \mathbf{D}_0(\Lambda)} \mathbf{W}_t(Y,B)\Big]\mathrm{d}B\,,\end{aligned} \tag{9.13}$$

since $\mathbf{1}_{\mathcal{D}^\Lambda}(B+tm) = \Theta^t(B)$ and, where $\Theta^t(B) = 1$, $\sum_Y \mathbf{W}_t(Y,B) = \sum_{\mathfrak{p}\ \mathrm{near}} \mathfrak{p}(B + tm + tA^{\Lambda,\Psi}) = P^\Lambda_t(B+tm)$ (each near primitive is assigned to exactly one host). The constant $e^{\frac12 t^2\langle m, Hm\rangle}$ is the same in the numerator and the denominator of every conditional expectation,

$$\int F\,\mathrm{d}\nu^\Lambda_t = \frac{\int F(B+tm)\,\Theta^t e^{-\frac12\langle B,HB\rangle + \sum_Y \mathbf{W}_t(Y,B)}\,\mathrm{d}B}{\int \Theta^t e^{-\frac12\langle B,HB\rangle + \sum_Y \mathbf{W}_t(Y,B)}\,\mathrm{d}B}\,, \tag{9.14}$$

and is never estimated: its logarithm is at most $\frac{1}{2}h_+\sum_b h(m_b)^2 \leqslant h_+|D(\Lambda)|r^2/32$ ($h \leqslant |\bullet|_\sigma$, $t \leqslant 1$), a constant of the order of the carrier volume, which disappears in (9.14). Every use of the translated expansion below is through ratios of the form (9.14) or through quantities normalised by $\Xi_\Lambda^{f,t}$ (Corollary 9.16).

Lemma 9.14 (Translated Polymer Representation)**.** *With* $\mathfrak{n}_\Lambda := (2\pi)^{d_G|D(\Lambda)|/2}\det(H)^{-1/2}$,

$$\int \Theta^t(B)\exp\Big[-\tfrac{1}{2}\langle B,HB\rangle+\sum_Y \mathbf{W}_t(Y,B)\Big]\mathrm{d}B = \mathfrak{n}_\Lambda\,\Xi_\Lambda^{f,t}\,,\quad \Xi_\Lambda^{f,t} := \sum_{\mathcal{P}\ compatible}\prod_{Z\in\mathcal{P}} k_\Lambda^{f,t}(Z)\,,$$
$$k_\Lambda^{f,t}(Z) := \sum_{(\mathcal{D},\mathsf{P})\,:\,Z_0\neq\varnothing,\ Z_0'\subseteq Z} \mathcal{A}^{f,t}_{(\mathcal{D},\mathsf{P},Z)}[1]\,,$$

where the selections $\mathfrak{s}=(\mathcal{D},\mathsf{P},Z)$ *are those of Definition 3.17 for the carrier* $D(\Lambda)$ *and the family* $\mathbf{D}_0(\Lambda)$ *(*$\mathcal{D}\subseteq\mathbf{D}_0(\Lambda)$ *finite,* $\mathsf{P}\subseteq D(\Lambda)\setminus E(Y_0)$ *finite, seed* Z_0*,* $E_0=E(Z_0)$*, enlargement* Z_0'*, final domain* $Z\supseteq Z_0'$ *a union of LM-cubes), and the translated selected integral of a bounded Borel* F *on* $\mathfrak{g}^{E_0}$ *is* (3.68) *with* χ_b *replaced by* χ_b^t *and* $\mathbf{W}$ *by* $\mathbf{W}_t$*:*

$$\begin{aligned}\mathcal{A}_{\mathfrak{s}}^{f,t}[F] = \prod_{\Delta\in\pi_1(Z)\setminus\pi_1(Z_0')}\int_0^1 \mathrm{d}s_\Delta \oint_{|\sigma_\Delta|=e^{\kappa_1}}\frac{\mathrm{d}\sigma_\Delta}{2\pi i(\sigma_\Delta-s_\Delta)^2}\\ \times\prod_{Y\in\mathcal{D}}\int_0^1\mathrm{d}t_Y\int\mathrm{d}\mu_0(X)\int\mathrm{d}\mu_{C_{E_0}}(B)\\ \times e^{-\frac{1}{2}\langle\Gamma_{E_0}(\sigma)X,C_{E_0}\Gamma_{E_0}(\sigma)X\rangle}e^{-\langle B,\Gamma_{E_0}(\sigma)X\rangle}(-1)^{|\mathsf{P}|}\prod_{b\in E(Y_0)}\chi_b^t(B)\prod_{b\in\mathsf{P}}(1-\chi_b^t(B))\\ \times\prod_{Y\in\mathcal{D}}\mathbf{W}_t(Y,B)e^{t_Y\mathbf{W}_t(Y,B)}\,F(B)\,,\end{aligned}$$

with C_{E_0}*,* $\Gamma_{E_0}(\sigma)$*,* μ_0*,* $\mu_{C_{E_0}}$ *built from* $H=H_{D(\Lambda)D(\Lambda)}$*. Consequently*

$$\Xi_\Lambda^{f,t} = e^{-\frac{1}{2}t^2\langle m,Hm\rangle}I_t^\Lambda/\mathfrak{n}_\Lambda > 0\,.$$

For each selection $\mathfrak{s}$ *and each bounded Borel* F *on* $\mathfrak{g}^{E_0}$ *the translated selected integral* $\mathcal{A}_{\mathfrak{s}}^{f,t}[F]$ *is a well-defined finite complex number. No representation of the numerator of* (9.14) *for an insertion is asserted in this lemma: the insertion of a function of the young variables requires the probe-host (forced-seed) expansion of Section 3.7 for the t-family, Section 9.6.*

Proof. The proof of Lemma 3.18 is reproduced with $\chi_b\to\chi_b^t$, $\mathbf{W}\to\mathbf{W}_t$, $I\to D(\Lambda)$, $\mathbf{D}_0\to\mathbf{D}_0(\Lambda)$. The properties of the cutoff and of the potential family that proof uses are: (i) $\Theta=\prod_b\chi_b$ with each χ_b a Borel function of a_b alone with values in $[0,1]$: in the expansion $\Theta=\prod_{E(Y_0)}\chi_b\prod_{I\setminus E(Y_0)}(1-(1-\chi_b))$ with the sign $(-1)^{|\mathsf{P}|}$, and in the factorisation over the components of Z, where a bad bond lies in the component containing its incident cubes and a cutoff factor is a function of one bond variable; χ_b^t has these properties. (ii) $\mathbf{W}(Y,\bullet)$ depends on a through $a|_{E(Y)}$, with $E(Y)\subseteq E_0$ for $Y\in\mathcal{D}$, and is bounded where $\prod_{b\in E(Y)}\chi_b\neq 0$: in the reduction to $a|_{E_0}$ that lets Lemma 3.14 produce (3.66), and in the Fubini steps (bounded factors times an absolutely integrable density); $\mathbf{W}_t$ has these properties, with the translated local box in place of the local box, by Section 9.5.1. (iii) Everything else, the walk series (Lemma 3.13), the conditioning identity and the weakened density (Lemma 3.14, (3.66)), the block structure (Lemma 3.15), the integrability of the absolute density uniformly on the closed polydisc (Lemma 3.16(ii)), the Cauchy representation of the σ-derivatives and the vanishing of unseeded components, and the geometry of seeds, enlargements and the coarse

cover, involves H, the partitions π_0, π_1 of the torus and σ only, and holds for $H = H_{D(\Lambda)D(\Lambda)}$ by Lemma 9.13(a); the host and probe geometry is that of Lemma 9.13(d). The expansion of $\exp(\sum_Y \mathbf{W}_t)$ into Mayer factors $\int_0^1 \mathbf{W}_t e^{t_Y \mathbf{W}_t}\,\mathrm{d}t_Y$ is the same algebra. Positivity: by (9.13) with $F = 1$, $\mathfrak{n}_\Lambda \Xi_\Lambda^{f,t} = e^{-\frac{1}{2}t^2\langle m, Hm\rangle} I_t^\Lambda$, and $I_t^\Lambda > 0$ by Lemma 9.13(c). Well-definedness of $\mathcal{A}_{\mathfrak{s}}^{f,t}[F]$: the integrand is a bounded function ($|F| \leqslant \sup|F|$, cutoffs $\leqslant 1$, $|\mathbf{W}_t e^{t_Y \mathbf{W}_t}| \leqslant \omega_t e^{\omega_t}$ by (9.12)) times the density (3.66), whose modulus is integrable uniformly on the closed polydisc by Lemma 3.16(ii) for H (Lemma 9.13(a)); the finitely many s-, σ- and t_Y-integrals are over compact sets. *q.e.d.*

Lemma 9.15 (Translated Selected Majorant)**.** *Let g_0 satisfy the thresholds of Section B.8, items 1–5. Two of their consequences are used here:*

$$C_0 g_0 (1+2r)^3 \leqslant \omega_* \quad \textit{(i.e. } \omega_t \leqslant \omega_*\textit{; threshold item 4(ii) gives } \omega_t \leqslant \omega_*/2\textit{)}, \tag{9.15}$$

$$\tfrac{9}{16} a_{\mathrm{bad}} \geqslant \max\bigl(160(\kappa + \log(8M^4+8)+1),\, 64\log(1/\eta_*)\bigr)\,, \qquad a_{\mathrm{bad}} = c_n \alpha_G r^2/4$$
$$\textit{(}(\mathrm{B}.29)\textit{ for } \tfrac{9}{16} a_{\mathrm{bad}}\textit{; threshold item 4(i), which states it for } \tfrac{9}{128} a_{\mathrm{bad}}\textit{)}. \tag{9.16}$$

Both are conditions on g_0 alone ($g_0(1+r)^3 \to 0$ and $a_{\mathrm{bad}} \to \infty$ as $g_0 \to 0$, $r = A_1(\log g_0^{-2})^{p_0}$), with thresholds depending on L, the group, M, κ, α_G, ω_* and the cutoff parameters A_1, p_0 only (through $r(g_0)$). Then on every labelled term with final region Λ, for exterior data in the support of the term's collar indicators and exterior cutoffs (where (9.11) and $r_{\mathrm{e}}^\Lambda < r$ hold, Lemmas 9.10(b) and 9.12(iii); the term vanishes elsewhere), for every $t \in [0,1]$, every selection $\mathfrak{s} = (\mathcal{D}, \mathsf{P}, Z)$ for $D(\Lambda)$ and every bounded Borel F on $\mathfrak{g}^{E_0}$,

$$|\mathcal{A}_{\mathfrak{s}}^{f,t}[F]| \leqslant \sup|F| \cdot e^{-(\kappa_1-1)N_1(Z\setminus Z_0') + N_1(Z) - \frac{9}{16}a_{\mathrm{bad}}|\mathsf{P}|} \prod_{Y\in\mathcal{D}} \omega_* e^{1/2} e^{-\kappa d_0(Y)}\,. \tag{9.17}$$

The right side depends neither on t, nor on m or the exterior data $A^{\Lambda,\Psi}$ within that support, nor on the labels; bondwise, the reserve is $(1 - t|m_b|_\sigma/r)^2 a_{\mathrm{bad}} \geqslant \frac{9}{16} a_{\mathrm{bad}}$.

Proof. The proof of Lemma B.12 with the substitutions $\chi_b \to \chi_b^t$, $\mathbf{W} \to \mathbf{W}_t$ and $I \to D(\Lambda)$ with $\mathbf{D}_0 \to \mathbf{D}_0(\Lambda)$. The weakening step is unchanged: for each cube Δ of $Z \setminus Z_0'$ the operation $G \mapsto \int_0^1 \mathrm{d}s \oint G\,\mathrm{d}\sigma_\Delta/(2\pi i(\sigma_\Delta - s)^2)$ on the inner integral as a function of σ_Δ on the circle $|\sigma_\Delta| = e^{\kappa_1}$ is bounded by $e^{-(\kappa_1-1)}$ times the supremum of the inner integral's modulus over the distinguished boundary, by $e^{\kappa_1}(e^{\kappa_1}-1)^{-2} \leqslant e^{-(\kappa_1-1)}$ ($e^{\kappa_1} \geqslant 3$); it needs the continuity of the integrand in σ (the walk series, Lemma 3.13 for H) and the uniform integrability below. On the distinguished boundary, the inner integrand's modulus is bounded factor by factor: $\int_0^1 \mathrm{d}t_Y$ has mass one; $\chi_b^t \leqslant 1$; for $b \in \mathsf{P}$ and real B, by the triangle inequality for the norm $|\bullet|_\sigma$ and (9.11),

$$\begin{aligned} 1 - \chi_b^t(B) &= \mathbf{1}\{|B_b + t m_b|_\sigma \geqslant r\} \leqslant \mathbf{1}\{|B_b|_\sigma \geqslant r - t|m_b|_\sigma\} \\ &\leqslant \mathbf{1}\{|B_b|_\sigma \geqslant (1 - t|m_b|_\sigma/r) r\} \leqslant \mathbf{1}\{|B_b|_\sigma \geqslant 3r/4\}\,, \end{aligned}$$

and on $\{|X|_\sigma \geqslant 3r/4\}$ the group inequality $h(X)^2 \geqslant c_n |X|_\sigma^2$ of Lemma 3.16(iii) ($c_n = 1/N$ serves for both cutoff norms, the conventions of Appendix B) gives $h(X)^2 \geqslant \frac{9}{16} c_n r^2$, so $\mathbf{1}\{|B_b|_\sigma \geqslant 3r/4\} \leqslant \exp[(\gamma_2/2)(h(B_b)^2 - \frac{9}{16}c_n r^2)]$ for every $\gamma_2 > 0$ (the exponent is nonnegative on the set); with $\gamma_2 = \alpha_G/2$, $a_{\mathrm{bad}} = c_n \gamma_2 r^2/2 = c_n \alpha_G r^2/4$ and $\|B\|^2 = \sum_b h(B_b)^2$,

$$\prod_{b\in\mathsf{P}} (1 - \chi_b^t(B)) \leqslant e^{-\frac{9}{16}a_{\mathrm{bad}}|\mathsf{P}|} e^{(\alpha_G/2)\sum_{b\in\mathsf{P}} h(B_b)^2} \leqslant e^{-\frac{9}{16}a_{\mathrm{bad}}|\mathsf{P}|} e^{(\alpha_G/2)\|B\|^2}$$

(this is Lemma 3.16(iii) at the radius $3r/4$ in place of r; bondwise the same argument gives the reserve $(1 - t|m_b|_\sigma/r)^2 a_{\mathrm{bad}}$); for $Y \in \mathcal{D}$, on the support of $\prod_{b\in E(Y)} \chi_b^t$ the point $B + tm$

lies in the local box of Y, so $|\mathbf{W}_t(Y,B)| \leqslant \omega_t e^{-\kappa d_0(Y)} \leqslant \omega_* e^{-\kappa d_0(Y)}$ by (9.12) and (9.15), and $|\mathbf{W}_t e^{t_Y \mathbf{W}_t}| \leqslant \omega_* e^{\omega_*} e^{-\kappa d_0(Y)} \leqslant \omega_* e^{1/2} e^{-\kappa d_0(Y)}$, since $\omega_* \leqslant 1/2$ (the standing choice of Section B.8); $|F| \leqslant \sup|F|$; and the integral of the modulus of the density (3.66) against $e^{(\alpha_G/2)\|B\|^2}$ is at most $e^{N_1(Z)}$ by Lemma 3.16(ii) for H (Lemma 9.13(a)), under the conditions on δ_M and α_G of Section B.8. Multiplying the bounds and applying the weakening bound once per cube of $Z \setminus Z'_0$ gives (9.17). *q.e.d.*

Corollary 9.16 (Activity Bounds and Vacuum Clusters for the t-Family)**.** *Under the hypotheses of Lemma 9.15, for every $t \in [0,1]$ and every connected final domain Z for $D(\Lambda)$,*

$$|k^{f,t}_\Lambda(Z)| \leqslant \varepsilon_* e^{-\lambda_* \ell_1(Z)} , \qquad \sum_{Z' \not\sim Z} |k^{f,t}_\Lambda(Z')| e^{(1+6\theta)N_1(Z')} \leqslant N_1(Z)/8 ,$$

with the constants λ_*, ε_*, θ of (B.26)–(B.28)*, and Lemma 3.20 holds for the cluster weights $\Phi^{v,t}_\Lambda(\mathfrak{c}) = (\varphi^T(\mathfrak{c})/k!)\prod_{j=1}^k k^{f,t}_\Lambda(D_j)$ of a cluster $\mathfrak{c} = (D_1,\dots,D_k)$ (the symbol m is the shift here): the hitting sum $\sum_{\mathfrak{c}\ hits\ \mathcal{A}} |\Phi^{v,t}_\Lambda(\mathfrak{c})| e^{2\theta\sum_j N_1(D_j)} \leqslant \frac{1}{8}\sum_{D\in\mathcal{A}} N_1(D)$ and the identity $(\Xi^{f,t}_\Lambda)^{[\cup\mathcal{A}]}/\Xi^{f,t}_\Lambda = \exp(-\sum_{\mathfrak{c}\ hits\ \mathcal{A}} \Phi^{v,t}_\Lambda(\mathfrak{c}))$, uniformly in t, m, the exterior data in the support of the term's collar indicators and exterior cutoffs, and the labels.*

Proof. The proof of Lemma 3.19 uses, about the activities, only the majorant (B.31), here (9.17) with $F = 1$, with $\omega_* e^{1/2} e^{-\kappa d_0(Y)} \leqslant \eta_* e^{-\kappa \ell_0(Y)}$ ($\ell_0 = d_0 + 5$ and $\eta_* = 2e^{1/2+5\kappa}\omega_*$), and the number a_{bad} through the conditions it draws from (B.29): $a_{\mathrm{bad}} \geqslant 160(\kappa+1) > 1$ and $64M^4 e^{-a_{\mathrm{bad}}/4} \leqslant 1$ (bad-label sum), $a_{\mathrm{bad}}/32 \geqslant 5\kappa$ (component reserve), $e^{-a_{\mathrm{bad}}/64} \leqslant \eta_* \leqslant t_0$ (the case $k_j = 0$), and $C_g e^{-a_{\mathrm{bad}}/64} \leqslant 162 e^{-5(\kappa+1)/2} \leqslant \kappa/40$ (choice of Y_0). Each of these holds for every number not smaller than one satisfying (B.29); (9.16) says that $\frac{9}{16}a_{\mathrm{bad}}$ satisfies it, so the proof applies verbatim with a_{bad} replaced by $\frac{9}{16}a_{\mathrm{bad}}$ and η_*, t_0, ω_*, κ, δ, α_6, C_g, M, L unchanged (the geometric inputs Lemmas B.1, B.11 and B.2 concern the partitions only; the cover sum is over $\mathbf{D}_0(\Lambda) \subseteq \mathbf{D}_0$, every host of the retreated family being a host of the original one). The proof of Lemma 3.20 (Appendix B.6, Kotecký–Preiss with the pinned-tree bound) uses only the two displayed bounds and the geometry, and nothing else about the branch (proof of Lemma B.13). *q.e.d.*

Lemma 9.17 (The Conditional Estimate at the Retreated Carrier)**.** *Under the thresholds $C_P g_0(1+2r) \leqslant \gamma/2$ and $r \geqslant 1$ of Appendix D (*(D.2)*; Lemma D.3 uses $r \geqslant 1$ in $O(g_0 s^2/r) = O(g_0 s)$; both are conditions on g_0), threshold item 3 of Section B.8 and the hypotheses listed in Lemma 9.13, on every labelled term with final region Λ, for exterior data in the support of the term's collar indicators and exterior cutoffs, and on the covered event $A^{\mathrm{fin}}_{\mathrm{cov}}$ of Section 9.4.2(a) (every contour bond in Ω_1, every free input of the contour in $D(\Lambda)$),*

$$|\mathcal{B}^\Lambda_\xi[f_\xi]| \leqslant C_{\mathrm{resp},\chi} g_0 r D^\Lambda_\xi , \qquad \mathcal{B}^\Lambda_\xi[f] \coloneqq \int f \,\mathrm{d}\nu^\Lambda_1 - \int f \,\mathrm{d}\nu^\Lambda_0 = \int_0^1 \mathrm{Cov}_{\nu^\Lambda_t}(f, L^\Lambda_t)\,\mathrm{d}t ,$$
$$D^\Lambda_\xi \coloneqq \sum_{b \in I_C \cap D(\Lambda)} e^{-(\mu/2)d(b,J(\Lambda))} ,$$

with the constant $C_{\mathrm{resp},\chi}$ and the rate $\mu = R^{-1}\log(1+\gamma/(4M_*))$, $M_* = m_L + \gamma/2$, of Appendix D (Lemmas D.4 and D.6), independent of g_0, the term, the labels, the volume and t; D^Λ_ξ is the D^{fin}_ξ of Section 9.4.2(a), and $L^\Lambda_t \coloneqq -\langle \bullet, \ell_\Lambda\rangle + \partial_t P^\Lambda_t$.

Proof. Sections D.1–D.5 are applied with $(I, J, A_{\mathrm{e}}, H, \ell, P_t, r_{\mathrm{e},\xi})$ replaced by $(D(\Lambda), J(\Lambda), A^{\Lambda,\Psi}, H_{D(\Lambda)D(\Lambda)}, \ell_\Lambda, P^\Lambda_t, r^\Lambda_{\mathrm{e}})$. The inputs they use, each supplied: $\gamma \leqslant H \leqslant m_L$ and the finite range R of H and of the primitive stencils (Lemma 9.13(a); Section D.2 takes $R =$

$8L+8 \geqslant \rho_L$); the primitive bounds on the enlarged box at the inputs $a+tA^{\Lambda,\Psi}$, uniformly in t, with $s^\Lambda \leqslant 1+2r$ (Lemma 9.13(c)), giving $\|D_a^2 P_t^\Lambda\| \leqslant C_P g_0 s^\Lambda$ and the Schur row/column bounds of Lemma D.3; $r_{\mathrm{e}}^\Lambda < r$ for the score bound $h(\nabla_b \partial_t P_t^\Lambda) \leqslant C_S g_0 s^\Lambda r_{\mathrm{e}}^\Lambda \mathbf{1}_{K^\Lambda}(b)$ and $h((\ell_\Lambda)_b) \leqslant m_L r_{\mathrm{e}}^\Lambda \mathbf{1}_{K^\Lambda}(b)$, $K^\Lambda \coloneqq \{b \in D(\Lambda) : \mathrm{dist}(b, J(\Lambda)) \leqslant R\}$ (Lemma 9.13(b) and (I2)); the source gradient bound $\sup h(\nabla_b f_\xi) \leqslant C_{\mathrm{resp},\chi} g_0 \mathbf{1}_{I_C}(b)$ on the enlarged box, from the rowwise reconstruction and the uniform charts for the rows meeting $D(\Lambda)$ (Lemma 9.13(c); the links of the contour depend on $a|_{D(\Lambda)}$ through those rows, the de-younged bonds being constants of the term); the identity $\mathcal{B} = \int_0^1 \mathrm{Cov}\, \mathrm{d}t$, which is the differentiation of $t \mapsto \int f \,\mathrm{d}\nu_t^\Lambda$ under the integral (the exponent of $K_{\xi;t}^\Lambda$ is smooth in t with bounded derivative on the box, Lemma 9.13(c)); the extension T_r of Section D.3, the barrier β_n and the Helffer–Sjöstrand identity of Section D.4, which involve only the box $\{|a_b|_\sigma < r,\ b \in D(\Lambda)\}$, the Hessian bounds and the Schur bounds; and the shell summation of Lemma D.6 with $d(c, J(\Lambda)) \leqslant R$ on K^Λ. On $A_{\mathrm{cov}}^{\mathrm{fin}}$ every free input of the contour lies in $D(\Lambda)$, so $I_C \subseteq D(\Lambda)$ and D_ξ^Λ sums over all of I_C; the intersection in its definition records this. Nothing in Sections D.1–D.5 uses that the carrier is Λ_1^*. *q.e.d.*

Remark 9.18 (Scope of This Subsection). The translated expansion gives the vacuum activities $k_\Lambda^{f,t}$, their clusters and a majorant for selected integrals with a bounded insertion F on $\mathfrak{g}^{E_0}$; the hosts are unchanged and the primitives are evaluated at $a = B + tm$ in the original local box, so the chart condition is threshold item 3 at $s^\Lambda \leqslant 1+2r$, and the enlarged box $\{|a_b|_\sigma \leqslant 2r\}$, not needed for the majorant, is available for analyticity, $B \mapsto B + tm$ mapping $\{|B_b|_\sigma \leqslant 7r/4\}$ into it. Nothing here concerns the source-containing connected terms attached to $J(\Lambda)$: $f_\xi(B+tm)$ is not a function of $B|_{E_0}$ for a selection unrelated to the contour, so (9.17) with F the source is not available, and the insertion of the source through the probe hosts of $\mathbf{D}_0(\Lambda)$, the t-centred reference and the marked connected terms are Section 9.6, whose result is (9.27): $|\mathcal{B}_\xi^\Lambda[f_\xi]|$ is bounded by $|S_\Lambda^\circ| e^{2n_C \bar{\varepsilon}_t}$ times $C_1 g_0 r\, \mathfrak{S}_1(\Lambda) + C_2 \bar{\varepsilon}_t r^{d_G} \mathfrak{S}_2(\Lambda)$, with $\mathfrak{S}_1$, $\mathfrak{S}_2$ the exponentially weighted sums over $J_1(\Lambda)$ and $J(\Lambda)$ defined there; the bookkeeping of the labels in the history sum is Section 9.8. The consequences (9.15), (9.16) of threshold item 4 are the form in which it is used wherever the t-family is used.

9.6 The Source-Containing Connected Boundary Terms

9.6.1 Setting and Target

Throughout this subsection fix a history, a labelled term with final region Λ, carrier $D \coloneqq D(\Lambda)$, exterior $J(\Lambda)$, exterior variable $A^{\Lambda,\Psi}$, shift m, and the t-family $K_{\xi;t}^\Lambda$, ν_t^Λ, $\mathbf{W}_t$, χ_b^t of Section 9.5, with the exterior data in the support of the term's collar indicators and exterior cutoffs (where (9.11), $r_{\mathrm{e}}^\Lambda < r$ and Lemma 9.13 hold; the term vanishes elsewhere, and every bound below is a bound on a factor of the term that is multiplied by those indicators), on the covered event $A_{\mathrm{cov}}^{\mathrm{fin}}$ of Section 9.4.2(a) (every contour bond in Ω_1, every free input of the contour in $D(\Lambda)$; so $I_C \subseteq D(\Lambda)$, and $f_\xi(a) = W_\chi(C; U_\xi(a))$ depends on $a \in \mathfrak{g}^{D(\Lambda)}$ through $a|_{I_C}$ only, the de-younged bonds $I \setminus D(\Lambda)$ being constants of the term, Lemma 9.17). Write

$$F^\Lambda(t) \coloneqq \int f_\xi \,\mathrm{d}\nu_t^\Lambda\,, \qquad \mathcal{B}_\xi^\Lambda[f_\xi] = F^\Lambda(1) - F^\Lambda(0) \quad \text{(Lemma 9.17)},$$

and n_C for the number of traversals of C, I_C for the set of free-input bonds of the contour (each within ℓ^∞ distance $2L+1$ of a contour bond: the forward influence set $\mathcal{R}_b$ of Section 3.7 lies within ℓ^∞ distance $2L+1$ of b, applied to the contour bond r with $b \in \mathcal{F}_r$; hence $\mathrm{dist}(e, I_C) \geqslant \mathrm{dist}(e, C) - (2L+1)$ in the distance convention below), $J_1(\Lambda) \coloneqq \{e \in J(\Lambda) : H(b', e) \neq 0 \text{ for some } b' \in D(\Lambda)\} \subseteq J_{\mathrm{col}}(\Lambda)$ (Lemma 9.2 at the retreated carrier; Section 9.5: m depends

on $A^{\Lambda,\Psi}|_{J_1(\Lambda)}$ only, and $|A^{\Lambda,\Psi}(e)|_\sigma = |A^\Lambda_{\mathrm{e}}(e)|_\sigma < \vartheta r$ on $J_{\mathrm{col}}(\Lambda)$). The target is a bound on $\mathcal{B}^\Lambda_\xi[f_\xi]$ relative to the reference gas, with the perimeter factor extracted inside the connected terms before absolute values, of the form (reference)$\times \sum_{e\in J(\Lambda)} e^{-\mu' \operatorname{dist}(e,C)} \times$(polynomial in r)$\times$(history weights). The reference used here is S°_Λ, the reference group gas at the carrier $D(\Lambda)$ (Lemma 9.21); on $A^{\mathrm{fin}}_{\mathrm{cov}}$ it is identified with the regional square scalar at $D(\Lambda)$ by Lemma 9.42 and with $\kappa_C e^{E^\circ_\Lambda}$ by Corollary 9.43, under (9.39); its comparison with the reference scalar at the carrier I is Lemma 9.38, whose ratio factor is not small on every history (Remark 9.41(1)).

Inputs beyond Sections 9.1–9.5, used as stated at their stated hypotheses: Lemmas 3.21, 3.22, Proposition 3.23 and Section 3.8 (Definition 3.24, Lemma 3.25, the reference data, Lemma 3.26, Proposition 3.27); Lemmas E.2 and E.13 (the dummy comparison (E.10), $\sum_{\ell:\, Z(\ell)=Z,\, \mathsf{J}\neq\varnothing} u_*^{q(\ell)} M_\ell \leqslant m(Z) := \varepsilon_* e^{-\lambda_* \ell_1(Z)}$ with $M_\ell = \sup_{|\zeta|\leqslant R_\Upsilon} \|\Lambda^{\mathrm{f}}_{\xi;\ell,\zeta}\|_{\mathrm{TV}}$ the ζ-uniform total-variation majorant of Section E.4, $u_* = \min(1/4, \omega_* e^{-\kappa s_{\mathrm{P}}}/(8P_M))$ (E.9), proved by the positive host comparison from the selected majorant and the cover/seed sums of Appendix B), Lemma E.7 (the closed-contraction factorisation of $R_\xi(\mathfrak{B})$ and the product inequality $|R_\xi(\mathfrak{B})| \leqslant \prod_W K_\chi \mathfrak{d}_W$, K_χ the constant of the irreducible character); Lemma E.10 and Section B.6 (the reference exclusion ratio $q^\circ(\mathfrak{B}) = S^{\circ[\mathfrak{B}]}/S^\circ = \exp(-\sum_{\Gamma \text{ hits } \mathfrak{B}} \Phi^\circ(\Gamma))$ by Kotecký–Preiss for the reference group gas, and the Kotecký–Preiss theorem in the rooted form of the proof of Lemma 3.20, the recursion following (5.29) with [13, Proposition 5.3, Theorem 5.4]: with activities $|a_W| e^{3\eta s(W)/2}$ and node cost $\eta s(W)/8$ it gives the rooted bound $\sum_{\Gamma\ni W} m_W(\Gamma)|\Phi^\circ(\Gamma)| \leqslant |a_W| e^{13\eta s(W)/8}$ and the exclusion identity); Lemma E.9 (the positive domain/cluster/interval summation giving the footprint and contour pins $\sup_\Delta \sum_{W:\,\Delta\in F(W)} e^{\nu\sigma(W)} M(W) \leqslant C\varepsilon$ and $\sup_j \sum_{W:\, j\in\mathcal{H}(W)} e^{\nu\sigma(W)} M(W) \leqslant C\varepsilon$ for a positive group majorant $M(W)$ with root weights $\rho\, m(Z)$ and the positive cluster tree weights, ν the support reserve, $\sigma(W)$ the size, $F(W)$ the footprint; the positive source-containing group summation with the weights $e^{4\eta s(W)}$ under $C\rho LM \leqslant \eta/8$ and the root-incidence bound of Section E.7.2); Lemma E.8 (the connected envelope $\mathfrak{E}(W)$ of a group has at most $A_E \sigma(W)$ cubes and is connected under maximum-norm adjacency, so a group whose footprint contains the cube Δ has $\sigma(W) \geqslant \operatorname{dist}(\Delta, C)/(A_E LM)$); the fact that incompatible groups have touching footprints or overlapping intervals is the compatibility of groups of Section 3.8 (after Definition 3.24: compatible groups have mutually compatible constituent domains and disjoint intervals) with the contact convention of Section E.11.4; the tree-graph justification following (5.29) ($|\varphi^T| \leqslant$ number of spanning trees of the incompatibility graph, used in Appendix B.6); the far event $\mathsf{N}_b = \{\operatorname{dist}(\Gamma_C, E(\Lambda_1^c)) \leqslant b\}$ of (E.2) (Lemma E.1), with $b = \lfloor LMR_1/16 \rfloor$. The finite-difference bound of Section E.4.1 is not used as an input: Lemma 9.25 proves the form needed here from Lemmas 3.6 and B.7.

Geometry. Throughout Sections 9.6 and 9.8 the square/torus geometry of Section 5.3 is in force, as imposed by Section E.5 and collected with the other torus conditions in Section B.8: C is a coarse square of side $s = nL$ fine bonds, $n_C = 4s$, on fine periods LK_i with $K_i \geqslant 3(n+2)$, together with the period condition of Section 9.4.1 (compatible even periods exceeding 200ℓ; when $n \geqslant 67MR_1$, $K_i \geqslant 3(n+2)$ already gives $LK_i \geqslant 3L(n+2) > 200LMR_1 = 200\ell$, while for the fixed loop as $g_0 \to 0$ the condition on the periods is the binding one). The constants C_{pin}, A_F, A_E and the hull/extent bounds of Section E.5.2 (Lemmas E.8 and E.9) are proved under this geometry, and Lemmas 9.21(d), 9.27, Proposition 9.28 and Corollary 9.29 inherit it as a hypothesis. Distance convention: every distance in Sections 9.6–9.8 is the ℓ^∞ distance in fine units; the distance of a bond is measured from its initial site, that of a cube or host from its nearest point, and the distance to C is the distance to the vertex set of C (so $\operatorname{dist}(b, C) = 0$ for a contour bond b, whose initial site is a vertex, and $\operatorname{dist}(\Delta, C) = 0$ for a cube of $\pi_1(\mathcal{H})$, the cubes containing vertices of the traversals of $\mathcal{H}$, Lemma E.8); the

counts $4(2s+1)^4$ and the conversions by $LM+1$, $4M+LM$, $2L+1$ below are stated in this convention.

9.6.2 The Decay Form of the Mean Shift

Lemma 9.2 bounds $\sup_b h(m_b)$; the dependence on the distance to the exterior is needed.

Lemma 9.19. *For every $b \in D(\Lambda)$,*

$$h(m_b) \leqslant C_m \vartheta r \sum_{e \in J_1(\Lambda)} e^{-\mu_H d(b,e)} , \qquad C_m \coloneqq \gamma^{-1} m_L \cdot 4(2\rho_L+1)^4 e^{\mu_H \rho_L} , \tag{9.18}$$

with $\mu_H = \gamma/(h_+\rho_L)$ of Lemma 9.1 and d the ℓ^∞ distance of initial sites in fine units. Consequently, with $N_{\mathrm{lat}}(\mu) = \sum_{s \geqslant 0} 4(2s+1)^4 e^{-\mu s}$ of Lemma 9.2,

$$\sum_{b \in D(\Lambda)} h(m_b) e^{-\mu \operatorname{dist}(b,X)} \leqslant C_m \vartheta r N_{\mathrm{lat}}(\mu_H/2) \sum_{e \in J_1(\Lambda)} e^{-\min(\mu_H/2,\mu)\operatorname{dist}(e,X)} \tag{9.19}$$

for every set X of bonds or of sites ($\operatorname{dist}(b,X)$ the ℓ^∞ distance from the initial site of b; Proposition 9.28 takes X the vertex set of C) and $\mu \geqslant 0$; in particular $\sum_{b \in I_C} h(m_b) \leqslant C_m \vartheta r N_{\mathrm{lat}}(\mu_H/2) \sum_{e \in J_1(\Lambda)} e^{-\mu_H \operatorname{dist}(e,I_C)/2}$.

Proof. $m_b = -\sum_{b' \in D(\Lambda)} H^{-1}(b,b') \sum_{e \in J_1(\Lambda)} H(b',e) A_e^{\Lambda,\Psi}$ (Section 9.2, Lemma 9.2). By (I1), $H(b',e) \neq 0$ forces $d(b',e) \leqslant \rho_L$, hence $d(b,b') \geqslant d(b,e) - \rho_L$; by Lemma 9.1, $\|H^{-1}(b,b')\| \leqslant \gamma^{-1} e^{-\mu_H d(b,b')} \leqslant \gamma^{-1} e^{\mu_H \rho_L} e^{-\mu_H d(b,e)}$ for such b'; the number of b' with $d(b',e) \leqslant \rho_L$ is at most $4(2\rho_L+1)^4$; each entry has $\|H(b',e)\| \leqslant \sum_{b''} \|H(b',b'')\| \leqslant m_L$ by (I2) for the row $b' \in D(\Lambda)$; and $h(A_e^{\Lambda,\Psi}) \leqslant |A_e^{\Lambda,\Psi}|_\sigma < \vartheta r$ for $e \in J_1(\Lambda) \subseteq J_{\mathrm{col}}(\Lambda)$ (Lemma 9.10(b)). Multiplying gives (9.18). For (9.19): $e^{-\mu_H d(b,e)} e^{-\mu \operatorname{dist}(b,X)} \leqslant e^{-\mu_H d(b,e)/2} e^{-\min(\mu_H/2,\mu)(d(b,e)+\operatorname{dist}(b,X))} \leqslant e^{-\mu_H d(b,e)/2} e^{-\min(\mu_H/2,\mu)\operatorname{dist}(e,X)}$ by the triangle inequality, and $\sum_{b \in D(\Lambda)} e^{-\mu_H d(b,e)/2} \leqslant N_{\mathrm{lat}}(\mu_H/2)$ by the lattice count of Lemma 9.2 (the sum over all bonds dominates the sum over $D(\Lambda)$). The last display is (9.19) with $X = I_C$ and any $\mu \geqslant \mu_H/2$ (so that $\min(\mu_H/2,\mu) = \mu_H/2$), restricted to $b \in I_C \subseteq D(\Lambda)$, where $\operatorname{dist}(b,I_C) = 0$; the terms with $b \notin I_C$ are nonnegative and dropped. Only (I1)–(I3) at the retreated carrier and Lemma 9.10(b) are used; no column-sum bound. *q.e.d.*

9.6.3 Thresholds

In addition to the thresholds of Section B.8, (9.15) and (9.16), assume

$$\begin{aligned} &\text{(i) } u_t \coloneqq C'_\chi g_0(1+2r) \leqslant u_*; \qquad \text{(ii) } \omega_t = C_0 g_0 (1+2r)^3 \leqslant \omega_*/2; \\ &\text{(iii) } \bar{\omega} \coloneqq C_0 g_0 (1+3r)^3 \leqslant \min\bigl(1,\, \omega_* e^{-\kappa(B_h+4)}/(e^{9/8} N_{\mathrm{host}})\bigr); \\ &\text{(iv) } \rho_t \coloneqq u_t/u_* \leqslant \rho_{\mathrm{pin}}; \qquad \text{(v) } \bar{\varepsilon}_t \coloneqq 2K_\chi C_{\mathrm{pin}} \rho_t \leqslant 1, \end{aligned} \tag{9.20}$$

together with the standing choice $\omega_* \leqslant 1/2$ recorded in Section B.8 (used in the positive host comparison of Lemma E.13). Here $C'_\chi \coloneqq 2N_{\mathrm{tr}} C_{\mathrm{link}}$ is the constant of Lemma 9.25 (the finite-difference bound for closed contractions: Cauchy estimates on the row chart with C_{link}, the traversal multiplicity N_{tr}, and the factor 2 for the two s-dependent words of the source slot in Lemma 9.26(d); it depends on L, the group and the character only). $N_{\mathrm{host}} \coloneqq 4^4 \cdot 20^4$ is the host count of Lemma 9.24 and Proposition 9.28 (the number of hosts $Y \in \mathbf{D}_0(\Lambda)$ with

$b \in E(Y)$ for a given bond b, and the number within distance $R_{\mathrm{st}} \leqslant 5M$ of a given bond: a host is a box of M-cubes of side at most 4 cubes, (3.36) and (3.75), determined by its lowest cube, one of at most 20^4 cubes within $9M$ of the bond, and by its side lengths, at most 4^4 choices). ρ_{pin} is the root-weight smallness required by the positive summation of Lemma E.9 (the condition $C\rho LM \leqslant \eta/8$ there, together with the root-incidence bound of Section E.7.2 required by Lemma E.9, a property of the fixed geometry, and $(3^4 A_F + LM + 1)C_{\mathrm{pin}}\rho \leqslant \eta/16$ for the reference gas, Lemma 9.21(d), with their fixed constants; 3^4 is the neighbour factor for footprint contact of Section E.11.4 and Lemma E.8(d)), and C_{pin}, K_χ are the constants of Lemma 9.27(a) and Lemma 9.26(a). The subscript t in u_t, ρ_t, $\bar{\varepsilon}_t$ names the t-family; none of them depends on t. Every condition holds for g_0 small, since $g_0(1+r)^3 \to 0$ ($r = A_1(\log g_0^{-2})^{p_0}$), with thresholds depending on L, the group, M, κ, ω_*, η, B_h, A_1, p_0 and the character only, not on n, the periods or the history. Put $\bar{s} \coloneqq 1 + 2r + r_{\mathrm{e}}^{\Lambda} \leqslant 1 + 3r$.

9.6.4 Translated Probe Measures and the Centred Expectation for the t-Family

The probe hosts Y_b, $b \in D(\Lambda)$, are in $\mathbf{D}_0(\Lambda)$ (Lemma 9.13(d)); for a union R of LM-cubes put $\mathsf{L}_R \coloneqq \{b \in D(\Lambda) : Y_b \subseteq R\}$ ((3.76) at the carrier $D(\Lambda)$), and for $\lambda \in \mathfrak{g}_{\mathbb{C}}^{D(\Lambda)}$ let $L_Y(\lambda, B) \coloneqq \sum_{b\colon Y_b = Y} \langle \lambda_b, B_b \rangle$ (the probe in the centred variable). The family $\mathbf{W}_t(Y, \bullet) + L_Y(\lambda, \bullet)$, $Y \in \mathbf{D}_0(\Lambda)$, has the properties (i)–(iii) of the proof of Lemma 9.14: $L_Y(\lambda, \bullet)$ depends on $B|_{E(Y)}$ ($Y_b = Y$ gives $b \in E(Y)$), and is bounded on the translated local box of Y ($|B_b|_\sigma < r + t|m_b|_\sigma \leqslant 5r/4$ there), and nothing else changes. Hence Lemma 9.14 holds for it, with activities $k^{f,t}_{\Lambda,Z}(\lambda)$ entire in λ and $k^{f,t}_{\Lambda,Z}(0) = k^{f,t}_{\Lambda}(Z)$.

Lemma 9.20 (Probe Measures, Majorant and Dummy Comparison for the t-Family)**.** *Under the hypotheses of Lemma 9.15 and* (9.20)*, for every $t \in [0,1]$ and every connected final domain Z for $D(\Lambda)$:*

(a) There is a finite complex Borel measure $\nu^{f,t}_{\Lambda,Z}$ on $\mathfrak{g}^{\mathsf{L}_Z}$, supported in the closed translated box $\{|B_b + t m_b|_\sigma \leqslant r,\ b \in \mathsf{L}_Z\}$, with $k^{f,t}_{\Lambda,Z}(\lambda) = \int e^{\sum_{b\in\mathsf{L}_Z}\langle \lambda_b, B_b\rangle}\, \mathrm{d}\nu^{f,t}_{\Lambda,Z}(B)$ and $\nu^{f,t}_{\Lambda,Z}(\mathfrak{g}^{\mathsf{L}_Z}) = k^{f,t}_{\Lambda}(Z)$; explicitly, with the split selections $\ell = (\mathcal{D}, \mathsf{J}, \mathsf{P}, Z)$ of Lemma 3.21 ($\mathsf{J} \subseteq \mathsf{L}_Z$ the selected probes, enlarged seed data $Y_0 = \bigcup_{Y\in\mathcal{D}} Y \cup \bigcup_{b\in\mathsf{J}} Y_b$, Z_0, $E_0 = E(Z_0)$, Z_0') and $\Lambda^{f,t}_{\Lambda;\ell}$ the complex measure on $\mathfrak{g}^{E_0}$ defined by the right side of the display of Lemma 9.14 with these enlarged data (cutoffs χ^t_b over $E(Y_0)$, the weakened density and the bad set over this seed, Mayer factors for $Y \in \mathcal{D}$ only, no probe factor), $\nu^{f,t}_{\Lambda,Z} = \sum_{\ell\colon Z(\ell)=Z,\, Z_0\neq\varnothing} \int \big[\bigotimes_{b\in\mathsf{J}}(\delta_{B_b} - \delta_0) \otimes \bigotimes_{b\in\mathsf{L}_Z\setminus\mathsf{J}} \delta_0\big] \mathrm{d}\Lambda^{f,t}_{\Lambda;\ell}(B)$.

(b) (Probe-enlarged majorant.) For every split selection ℓ and every bounded Borel F on $\mathfrak{g}^{E_0}$,

$$\Big|\int F\, \mathrm{d}\Lambda^{f,t}_{\Lambda;\ell}\Big| \leqslant \sup|F| \cdot e^{-(\kappa_1 - 1)N_1(Z\setminus Z_0') + N_1(Z) - \frac{9}{16}a_{\mathrm{bad}}|\mathsf{P}|} \prod_{Y\in\mathcal{D}} \omega_* e^{1/2} e^{-\kappa d_0(Y)}\,, \tag{9.21}$$

uniformly in t, m, the exterior data within the cutoff support and the labels; the probe hosts Y_b, $b \in \mathsf{J}$, carry no factor.

(c) (Dummy comparison for the t-family.) $\sum_{\ell\colon Z(\ell)=Z} u_*^{|\mathsf{J}|} \|\Lambda^{f,t}_{\Lambda;\ell}\|_{\mathrm{TV}} \leqslant m(Z) = \varepsilon_* e^{-\lambda_* \ell_1(Z)}$, the sum over all split selections with final domain Z and nonempty seed $Z_0 \neq \varnothing$ ($\mathsf{J} = \varnothing$ included; the activity sum of Section 3 excludes $Z_0 = \varnothing$, and the $Z_0 = \varnothing$ term has zero measure by the Cauchy argument of Lemma 3.18). The bound holds with the same constants when $\mathbf{W}_t$ is replaced by any potentials $\mathbf{W}'$ with $|\mathbf{W}'(Y, B)| \leqslant A_Y =$

$(\omega_*/2)e^{-\kappa d_0(Y)}$ *on the support of the host's cutoffs, the proof (and that of (b), on which it rests) using* $\mathbf{W}_t$ *only through this bound; in particular for* $\zeta\mathbf{W}_0$*, uniformly in complex* ζ *with* $|\zeta| \leqslant R_\Upsilon$*, wherever* $R_\Upsilon|\mathbf{W}_0(Y,B)| \leqslant A_Y$ *on that support. The bound is termwise: each* $\|\Lambda^{f,t}_{\Lambda;\ell}\|_{\mathrm{TV}}$ *is bounded by a majorant depending on the potentials only through* A_Y *(proof), so the supremum over such* $\mathbf{W}'$*, in particular over* $|\zeta| \leqslant R_\Upsilon$*, may be taken inside the sum over* ℓ*.*

(d) (Centred expectation.) For bounded Borel g *on* $\mathfrak{g}^{D(\Lambda)}$*, with*

$$\mathfrak{Q}^t_\Lambda(\mathcal{A};g) \coloneqq \int g(\mathsf{e}_{\mathcal{A}}B)\,\mathrm{d}\big(\bigotimes_{Z\in\mathcal{A}}\hat\nu^{f,t}_{\Lambda,Z}\big)(B)\;,$$

$$\hat\nu^{f,t}_{\Lambda,Z} \coloneqq \nu^{f,t}_{\Lambda,Z} - k^{f,t}_\Lambda(Z)\delta_0\;, \qquad q^t_\Lambda(\mathcal{A}) \coloneqq (\Xi^{f,t}_\Lambda)^{[\cup\mathcal{A}]}/\Xi^{f,t}_\Lambda\;,$$

$$\frac{\int g(B)\,\Theta^t e^{-\frac12\langle B,HB\rangle+\sum_Y \mathbf{W}_t(Y,B)}\,\mathrm{d}B}{\int \Theta^t e^{-\frac12\langle B,HB\rangle+\sum_Y \mathbf{W}_t(Y,B)}\,\mathrm{d}B} = g(0) + \sum_{\varnothing\neq\mathcal{A}\ \mathit{compatible}} q^t_\Lambda(\mathcal{A})\,\mathfrak{Q}^t_\Lambda(\mathcal{A};g)\;,$$

and, by (9.14) *with* $F = f_\xi$*,*

$$F^\Lambda(t) = f_\xi(tm) + \sum_{\varnothing\neq\mathcal{A}\ \mathit{compatible}} q^t_\Lambda(\mathcal{A})\,\mathfrak{Q}^t_\Lambda(\mathcal{A};f_\xi(\bullet+tm))\;, \tag{9.22}$$

$$\mathfrak{Q}^t_\Lambda(\mathcal{A};f_\xi(\bullet+tm)) = \sum_{(\ell_Z)_{Z\in\mathcal{A}}\,:\,\mathsf{J}_Z\neq\varnothing}\int \Delta_{\mathsf{J}}[f_\xi(\bullet+tm)](B)\,\mathrm{d}\bigotimes_Z \Lambda^{f,t}_{\Lambda;\ell_Z}(B)\;,$$

where $\mathsf{J} = \bigcup_Z \mathsf{J}_Z$ *and* $\Delta_{\mathsf{J}}g(B) \coloneqq \sum_{K\subseteq\mathsf{J}}(-1)^{|\mathsf{J}\setminus K|}g(\mathbf{1}_K B)$ *is the mixed finite difference of* (E.4) *(every label outside* K *set to zero, so the young variable of* f_ξ *in the* K*-term is* $\mathbf{1}_K B + tm$*: the unselected labels sit at* $a = tm$*).*

Proof. (a) is the proof of Lemma 3.21 with $\mathbf{V}_\xi \to \mathbf{W}_t$, $\chi_b \to \chi^t_b$, $I \to D(\Lambda)$: the split-label expansion at a host, the exact integration (3.80) of the probe Mayer variables, the Laplace transform of $\delta_{B_b} - \delta_0$ and the finiteness of each $\Lambda^{f,t}_{\Lambda;\ell}$ (image of a density whose modulus is integrable by Lemma 3.16(ii) for H, times bounded factors: the Cauchy kernels on the circles $|\sigma_\Delta| = e^{\kappa_1}$ are bounded by $e^{\kappa_1}(e^{\kappa_1}-1)^{-2}$, and the integrand is jointly Borel in (X,B,s,σ,t_Y)) use only Lemma 9.14 and its properties (i)–(iii). The support statement: $\chi^t_b(B)$, $b \in E(Y_0) \supseteq \mathsf{J}$, forces $|B_b + tm_b|_\sigma < r$ on the selected probes, and the unselected labels carry δ_0 ($B_b = 0$, and $|tm_b|_\sigma \leqslant r/4 < r$). At $\lambda = 0$ every term with $\mathsf{J} \neq \varnothing$ has zero mass (each $\delta_{B_b} - \delta_0$ has), so the total mass is $k^{f,t}_{\Lambda,Z}(0) = k^{f,t}_\Lambda(Z)$. (b) The proof of Lemma 9.15 is written for a selection $\mathfrak{s} = (\mathcal{D},\mathsf{P},Z)$ and bounds factor by factor; the enlarged seed enters only through $E_0 = E(Z_0)$ in the density (3.66), whose modulus is paid against $e^{(\alpha_G/2)\|B\|^2}$ by Lemma 3.16(ii) for every seed Z_0 and every $Z \supseteq Z'_0$ (the lemma is stated for every seed), through the cutoffs χ^t_b over $E(Y_0)$, each $\leqslant 1$, and through the bad bonds $\mathsf{P} \subseteq D(\Lambda)\setminus E(Y_0)$, each paying $e^{-\frac{9}{16}a_{\mathrm{bad}}}e^{(\alpha_G/2)h(B_b)^2}$ as in Lemma 9.15; the Mayer factors are those of $Y \in \mathcal{D}$ only, each bounded by $\omega_* e^{1/2}e^{-\kappa d_0(Y)}$ on the support of its cutoffs; a probe host contributes no factor; the weakening bound $e^{-(\kappa_1-1)}$ per cube of $Z\setminus Z'_0$ is unchanged. (c) is the positive host comparison of Lemma E.13: at a host Y the sum over nonempty label subsets of the positive weights $v_{\mathrm{ordinary}} = e^{A_Y}-1$, $v_{\mathrm{probe}} = u_*$ is at most $\omega_* e^{1/2}e^{-\kappa d_0(Y)}$, and all split selections with the same active-host set have the same enlarged seed, density, bad set and weakening, so the many-to-one sum is dominated hostwise by the undifferentiated majorant (9.21) with $F = 1$, after which the seed/enlargement/final-domain sums (B.34), (B.35) and the Y'-sum in the proof of Lemma 3.19 give $m(Z)$ exactly as in Corollary 9.16 (whose proof shows that these sums hold with $\frac{9}{16}a_{\mathrm{bad}}$ in place of a_{bad} under (9.16)); the comparison needs

the ordinary factor's modulus $|\int_0^1 \mathbf{W}_t e^{t_Y \mathbf{W}_t}\,\mathrm{d}t_Y| = |e^{\mathbf{W}_t}-1| \leqslant e^{|\mathbf{W}_t|}-1 \leqslant e^{A_Y}-1 = v_{\text{ordinary}}$ with $A_Y = (\omega_*/2)e^{-\kappa d_0(Y)}$, which holds because $|\mathbf{W}_t(Y,B)| \leqslant \omega_t e^{-\kappa d_0(Y)} \leqslant A_Y$ on the support of the host's cutoffs by (9.12) and (9.20)(ii), and the chain's last step uses $\omega_* \leqslant 1/2$ and $n_Y u_* \leqslant (\omega_*/8)e^{-\kappa d_0(Y)}$ ((E.27)). The sum includes the split selections with $\mathsf{J} = \varnothing$ (a host is active through its ordinary label or through probe labels; (E.28) sums over all nonempty label subsets), so (c) holds with $\sum_{\ell\,:\,Z(\ell)=Z}$ in place of $\sum_{\ell\,:\,Z(\ell)=Z,\,\mathsf{J}\neq\varnothing}$. (d) is the proof of Lemma 3.22 and Proposition 3.23 for the translated expansion: both sides of $\mathfrak{n}_\Lambda^{-1}\Theta^t e^{-\frac12\langle B,HB\rangle+\sum \mathbf{W}_t}\,\mathrm{d}B = \sum_{\mathcal{P}}(\mathsf{e}_{\mathcal{P}})_*\bigotimes_{Z\in\mathcal{P}}\nu^{f,t}_{\Lambda,Z}$ are finite complex measures with compact support and the same Laplace transform (Lemma 9.14 for $\mathbf{W}_t + L(\lambda,\bullet)$ and (a)), hence equal; expanding $\nu^{f,t} = \hat\nu^{f,t} + k^{f,t}\delta_0$ and dividing by $\mathfrak{n}_\Lambda \Xi^{f,t}_\Lambda$ gives the first display; the second is (9.14) with $F = f_\xi$ (bounded on $\mathfrak{g}^{D(\Lambda)}$, extended by zero off the enlarged box, which contains the support of every $\nu^{f,t}$: $|B_b + tm_b|_\sigma \leqslant r$ and $|tm_b|_\sigma \leqslant r/4$ give $|B_b|_\sigma \leqslant 5r/4$ and $|(\mathbf{1}_K B + tm)_b|_\sigma \leqslant 3r/2 \leqslant 2r$), and the formula for $\mathfrak{Q}^t$ in terms of Δ_J is the integration of g against the Dirac tensors in (a), zero-mass part (the terms with $\mathsf{J}_Z = \varnothing$ for some Z are removed by the centring). *q.e.d.*

Lemma 9.21 (Joint Extraction for the t-Family)**.** *(a) For a connected final domain Z for $D(\Lambda)$ the contact set $\mathsf{T}_Z \coloneqq \{n : \mathcal{F}_{r_n} \cap \mathsf{L}_Z \neq \varnothing\}$ ((3.87), with the convention $\mathcal{F}_r = \varnothing$ off Ω_1) is field-independent and t-independent; if $\mathcal{A}$ contains an unmarked Z ($\mathsf{T}_Z = \varnothing$) then $\mathfrak{Q}^t_\Lambda(\mathcal{A}; f_\xi(\bullet + tm)) = 0$. The groups $W = (\mathcal{A}(W), (\mathfrak{c}_1,\dots,\mathfrak{c}_{k(W)}), \mathcal{H}(W))$ of Definition 3.24 for the carrier $D(\Lambda)$ form one index set for all $t \in [0,1]$ and for the reference data; their weights are $w^t_\Lambda(W) \coloneqq ((-1)^{k(W)}/k(W)!)\prod_j \Phi^{v,t}_\Lambda(\mathfrak{c}_j)$ with the cluster weights of Corollary 9.16. With $A^t_\Lambda(\mathcal{S}) \coloneqq \prod_{W\in\mathcal{S}} w^t_\Lambda(W)\,\mathfrak{Q}^t_\Lambda(\mathcal{A}(\mathcal{S}); f_\xi(\bullet + tm))$, $A^t_\Lambda(\varnothing) \coloneqq f_\xi(tm)$, Lemma 3.25 holds: $F^\Lambda(t) = \sum_{\mathfrak{B}\ \text{compatible}} A^t_\Lambda(\mathfrak{B})$, absolutely convergent.*

(b) Reference data at the carrier $D(\Lambda)$: the same Ω_1, cubes, hosts $\mathbf{D}_0(\Lambda)$, constants and cutoffs, with $U_1^\circ = \mathbf{1}$, $V_0^\circ = \mathbf{1}$, $A^\circ = 0$, $\Psi^\circ = \mathbf{1}$, exterior variable zero on $J(\Lambda)$ (so $m^\circ = 0$, $\ell^\circ = 0$, and the reference objects are independent of t), $H^\circ = (\mathsf{H}^{\Omega_1}_{\mathbf{1}})_{D(\Lambda)D(\Lambda)}$, $f^\circ(a) = W_\chi(C; e^{iq_1(a)})$ with links off Ω_1 equal to $\mathbf{1}$. Lemma 9.13 holds for these data (its (a) by principal restriction of the reference Hessian of Section 3, which satisfies (3.56) by Lemma B.13; (b) trivially, $r_{\mathrm{e}}^\circ = 0$; (c), (d) as stated), and so do Lemma 9.14, Lemma 9.15, Corollary 9.16 and Lemma 9.20 (their proofs use the listed lemmas only). Define $a^\Lambda_W \coloneqq w^\circ_\Lambda(W)\int \chi(P^\circ_W(a))\,\mathrm{d}\big(\bigotimes_{Z\in\mathcal{A}(W)}\hat\nu^{f,\circ}_{\Lambda,Z}\big)(a)$ as in (3.92), with P°_W the ordered product of reference links along $\mathcal{H}(W)$ at the young variable $\mathsf{e}_{\mathcal{A}(W)}a$. Then Lemma 3.26 holds: $w^\circ_\Lambda(W)\int \chi(P^\circ_W(a)h)\,\mathrm{d}\bigotimes \hat\nu^{f,\circ}_\Lambda = a^\Lambda_W \chi(h)$ for $h \in G$, and $S^\circ_\Lambda \coloneqq \sum_{\mathfrak{B}\ \text{compatible}} \prod_{W\in\mathfrak{B}} a^\Lambda_W = \int f^\circ K^\circ_\Lambda\,\mathrm{d}a / I^\circ_\Lambda$ (K°_Λ the kernel $K^\Lambda_{\xi;0}$ of Section 9.5 at these reference data and $I^\circ_\Lambda \coloneqq \int K^\circ_\Lambda\,\mathrm{d}a$, written out in (9.38)).

(c) With $R^t_\Lambda(\mathfrak{B}) \coloneqq \sum_{\mathcal{S}\subseteq\mathfrak{B}}(-1)^{|\mathfrak{B}\setminus\mathcal{S}|}\big(\prod_{W\in\mathfrak{B}\setminus\mathcal{S}} a^\Lambda_W\big)A^t_\Lambda(\mathcal{S})$ and $S^{\circ[\mathfrak{B}]}_\Lambda$ the reference sum over families compatible with every member of $\mathfrak{B}$,

$$F^\Lambda(t) = S^\circ_\Lambda f_\xi(tm) + \sum_{\varnothing\neq\mathfrak{B}\ \text{compatible}} S^{\circ[\mathfrak{B}]}_\Lambda R^t_\Lambda(\mathfrak{B})\,, \qquad \text{hence} \tag{9.23}$$

$$\mathcal{B}^\Lambda_\xi[f_\xi] = S^\circ_\Lambda[f_\xi(m) - f_\xi(0)] + \sum_{\varnothing\neq\mathfrak{B}\ \text{compatible}} S^{\circ[\mathfrak{B}]}_\Lambda[R^1_\Lambda(\mathfrak{B}) - R^0_\Lambda(\mathfrak{B})]\,.$$

(d) $|a^\Lambda_W| \leqslant M_\Lambda(W)$, the positive group majorant of Lemma 9.26(d) below, $M_\Lambda(W) = (1/k(W)!)\prod_{Z\in\mathcal{A}(W)}\rho_t m(Z)\prod_j |\Phi|^+_\Lambda(\mathfrak{c}_j)$, defined without reference to the source data. Under (9.20)(iv) the reference group gas with the activities a^Λ_W satisfies the Kotecký–Preiss condition, so that $S^\circ_\Lambda \neq 0$, $q^\circ_\Lambda(\mathfrak{B}) \coloneqq S^{\circ[\mathfrak{B}]}_\Lambda / S^\circ_\Lambda = \exp(-\sum_{\Gamma\ \text{hits}\ \mathfrak{B}} \Phi^\circ_\Lambda(\Gamma))$ (the

cluster expansion of the reference group gas, Lemma E.10 and Section B.6, at the carrier $D(\Lambda)$; a cluster hits $\mathfrak{B}$ when it contains a group incompatible with a member of $\mathfrak{B}$), and $|q^\circ_\Lambda(\mathfrak{B})| \leqslant \prod_{W\in\mathfrak{B}} e^{\theta^\circ(W)}$ with $\theta^\circ(W) \coloneqq \sum_{\Gamma \text{ hits } W} |\Phi^\circ_\Lambda(\Gamma)| \leqslant \nu\sigma(W)/16$, where $\sigma(W)$ and ν are the size and the reserve constant of Lemma 9.27; and the rooted bound $\sum_{\Gamma\ni W} |\Phi^\circ_\Lambda(\Gamma)| \leqslant |a^\Lambda_W| e^{13\nu\sigma(W)/8}$ holds. All of (d) holds verbatim for the reference gas at the carrier I (the reference data of Section 3, activities a_W, cluster coefficients Φ°_ξ), since its proof uses, at the carrier, only Lemma 9.25(b) at the reference data (Lemma 3.6 at $U = \mathbf{1}$), Lemma 9.20(a),(c),(d) at the reference data (Lemmas 3.21, 3.22 and E.13 at I), the first display of Lemma 9.27(a) (the summation of Lemma E.9, whose geometry does not involve the carrier) and the Kotecký–Preiss rooted bound of Section B.6; each holds at I, the case $D(\Lambda) = I$ (no retreat) of these statements.

Proof. (a) A traversal's link depends on the young variable through $a|_{\mathcal{F}_{r_n}\cap D(\Lambda)}$ (Lemma 3.5; the de-younged bonds are constants); for unmarked Z the function $B \mapsto f_\xi(\mathbf{1}_K B + tm)$ does not depend on $B|_{\mathsf{L}_Z}$, so the K-sum over the labels of J_Z telescopes to zero in (9.22) (equivalently, $\hat\nu^{f,t}_{\Lambda,Z}$ has zero mass). The groups are built from roots, cluster lists (tuples of domains), contact sets and hulls, none of which involves the fields or t; only w^t_Λ and $\hat\nu^{f,t}_\Lambda$ depend on t. The regrouping proof of Lemma 3.25 is combinatorial, and its absolute convergence uses $|\mathfrak{Q}^t_\Lambda(\mathcal{A}; g)| \leqslant \sup|g| \prod_Z \|\hat\nu^{f,t}_{\Lambda,Z}\|$ (finite by Lemma 9.20(a)) and the hitting sum $\sum_{\mathfrak{c} \text{ hits } \mathcal{A}} |\Phi^{v,t}_\Lambda(\mathfrak{c})| \leqslant \frac18 \sum_{Z\in\mathcal{A}} N_1(Z)$ of Corollary 9.16. (b) The proof of Lemma 3.26 uses the constant gauge R_u: $H^\circ = R_u H^\circ R_u^{-1}$ holds for the principal restriction because R_u acts bondwise; the domains, weights, cutoffs (Ad-invariant norm), the Gaussian density, the probe term and the reconstruction covariance $q_{\mathbf{1}}(R_u B) = R_u q_{\mathbf{1}}(B)$ are as in Section 3, and the hosts and partitions are field-independent; Schur's lemma and the Fubini evaluation of the group product are verbatim. (c) is the Boolean inversion and interchange of absolutely convergent sums of Proposition 3.27, with the bound β_W of its proof formed from w^t_Λ, w°_Λ, $\hat\nu^{f,t}_\Lambda$, $\hat\nu^{f,\circ}_\Lambda$; the t-independence of a^Λ_W and $S^{\circ[\mathfrak{B}]}_\Lambda$ gives the second form. (d) The activity bound: $a^\Lambda_W = w^\circ_\Lambda(W) \int g^\circ_W \,\mathrm{d} \bigotimes_{Z\in\mathcal{A}(W)} \hat\nu^{f,\circ}_{\Lambda,Z}$ with $g^\circ_W(a) \coloneqq \chi(P^\circ_W(a))$, a closed Wilson contraction at the reference data; by Lemma 9.20(a),(d) for the reference data (part (b)), $\int g^\circ_W \,\mathrm{d} \bigotimes \hat\nu^\circ = \sum_{(\ell_Z)_{Z\in\mathcal{A}(W)\,:\,\mathsf{J}_Z\neq\varnothing} \int \Delta_{\mathsf{J}} g^\circ_W \,\mathrm{d} \bigotimes_Z \Lambda^{f,\circ}_{\Lambda;\ell_Z}$; Lemma 9.25(b) at the reference data gives $\sup|\Delta_{\mathsf{J}} g^\circ_W| \leqslant u_t^{|\mathsf{J}|}$ (base point $a_0 = 0$, increments $|B_b|_\sigma \leqslant r \leqslant 5r/4$ on the support of $\nu^{f,\circ}$, $m^\circ = 0$), and Lemma 9.20(c) for the reference data with $(u_t/u_*)^{|\mathsf{J}_Z|} \leqslant \rho_t$ ($\mathsf{J}_Z \neq \varnothing$, and $\rho_t \leqslant 1$ by (9.20)(i)) gives $|\int g^\circ_W \,\mathrm{d} \bigotimes \hat\nu^\circ| \leqslant \prod_{Z\in\mathcal{A}(W)} \rho_t m(Z)$; $|w^\circ_\Lambda(W)| \leqslant (1/k(W)!) \prod_j |\Phi^{v,\circ}_\Lambda(\mathfrak{c}_j)| \leqslant (1/k(W)!) \prod_j |\Phi|^+_\Lambda(\mathfrak{c}_j)$ by Corollary 9.16 for the reference data and the tree-graph inequality (Lemma 9.24(b)). The Kotecký–Preiss condition: by the first display of Lemma 9.27(a), which uses only the majorant M_Λ, $\sup_\Delta \sum_{W\,:\,\Delta\in F(W)} e^{4\nu\sigma(W)} M_\Lambda(W) + \sup_n \sum_{W\,:\,n\in\mathcal{H}(W)} e^{4\nu\sigma(W)} M_\Lambda(W) \leqslant C_{\mathrm{pin}}\rho_t$; two groups are incompatible only if their footprints touch (contact includes adjacency across a face, edge or corner, not only intersection: Section E.11.4) or their intervals overlap (the compatibility of groups, Section 3.8 after Definition 3.24; Section E.5.2); a footprint has at most $A_F\sigma(W)$ cubes (Lemma E.8), so an incompatible W' has a footprint cube in the set $N_1(F(W))$ of cubes touching $F(W)$, of at most $3^4 A_F\sigma(W)$ cubes (the neighbour factor 3^4), or a traversal in $\mathcal{H}(W)$, of at most $LM\sigma(W)$ traversals ($|\mathcal{H}(W)|/(LM) \leqslant \sigma(W)$); hence $\sum_{W' \text{ incompatible with } W} |a^\Lambda_{W'}| e^{3\nu\sigma(W')/2} e^{\nu\sigma(W')/8} \leqslant \sum_{\Delta\in N_1(F(W))} \sum_{W'\,:\,\Delta\in F(W')}(\dots) + \sum_{n\in\mathcal{H}(W)} \sum_{W'\,:\,n\in\mathcal{H}(W')}(\dots) \leqslant (3^4 A_F + LM)\sigma(W) C_{\mathrm{pin}}\rho_t \leqslant \nu\sigma(W)/8$ under (9.20)(iv): this is the Kotecký–Preiss hypothesis for the gas with activities $|a^\Lambda_W| e^{3\nu\sigma(W)/2}$ and node cost $\nu\sigma(W)/8$, and the Kotecký–Preiss theorem in the rooted form of Section B.6 (the recursion following (5.29) applied to this gas, with [13, Proposition 5.3, Theorem 5.4]) gives the absolute

convergence of the cluster series, $S^\circ_\Lambda = \exp(\sum_\Gamma \Phi^\circ_\Lambda(\Gamma)) \neq 0$, the exclusion identity $S^{\circ[\mathfrak{B}]}_\Lambda = \exp(\sum_{\Gamma \text{ hits no member of } \mathfrak{B}} \Phi^\circ_\Lambda(\Gamma))$ (the same with the species incompatible with a member of $\mathfrak{B}$ excluded), and the rooted bound $\sum_{\Gamma \ni W} m_W(\Gamma)|\Phi^\circ_\Lambda(\Gamma)| \leqslant |a^\Lambda_W| e^{13\nu\sigma(W)/8}$, $m_W(\Gamma)$ the occurrence multiplicity. Hence $\theta^\circ(W) \leqslant \sum_{W'=W \text{ or incompatible with } W} \sum_{\Gamma \ni W'} |\Phi^\circ_\Lambda(\Gamma)| \leqslant \sum_{W'=W \text{ or incompatible with } W} |a^\Lambda_{W'}| e^{13\nu\sigma(W')/8} \leqslant (3^4 A_F + LM + 1)\sigma(W) C_{\mathrm{pin}} \rho_t \leqslant \nu\sigma(W)/16$ under (9.20)(iv) ($e^{13\nu\sigma/8} \leqslant e^{4\nu\sigma}$, and the term $W' = W$ is bounded by the footprint pin at a cube of $F(W)$). *q.e.d.*

9.6.5 The Two Terms of (9.23)

The first is the shift of the loop at zero fluctuation; the second is the difference of the joint defects, which is where the source-containing connected boundary terms live.

Lemma 9.22 (The Shift Term)**.**

$$|f_\xi(m) - f_\xi(0)| \leqslant C_{\mathrm{resp},\chi} g_0 \sum_{b \in I_C} h(m_b) \leqslant C_{\mathrm{resp},\chi} C_m \vartheta N_{\mathrm{lat}}(\mu_H/2) g_0 r \sum_{e \in J_1(\Lambda)} e^{-\mu_H \operatorname{dist}(e, I_C)/2} .$$

Proof. $f_\xi(m) - f_\xi(0) = \int_0^1 \sum_{b \in I_C} \langle m_b, \nabla_b f_\xi(sm) \rangle \, \mathrm{d}s$; the points sm lie in the box $|a_b|_\sigma \leqslant r/4$ by (9.11), inside the enlarged box, where $\sup h(\nabla_b f_\xi) \leqslant C_{\mathrm{resp},\chi} g_0 \mathbf{1}_{I_C}(b)$ ((D.8), the labelled source-gradient bound of Section D.2, as used in the proof of Lemma 9.17); then the last clause of Lemma 9.19. *q.e.d.*

Lemma 9.23 (Conditional-Gaussian Shell Bound)**.** *Let $b \in E_0$, $S_b \coloneqq \{x \in \mathfrak{g} : r - |m_b|_\sigma \leqslant |x|_\sigma < r + |m_b|_\sigma\}$, $v \in \mathfrak{g}^{E_0}$ real, and $0 \leqslant \alpha \leqslant \alpha_G$ ($\alpha_G \leqslant \gamma/2$; $h(x)^2 = \langle x, x \rangle$ is the squared norm of the invariant inner product). Write $\mu_{C_{E_0}}(\mathrm{d}B) = \mu_{C'}(\mathrm{d}B') \, N(\mu(B'), c_b)(\mathrm{d}B_b)$ for the disintegration of the centred Gaussian of covariance $C_{E_0} = (\iota^*_{E_0} H \iota_{E_0})^{-1}$ (Section 3.5, at the carrier $D(\Lambda)$) along the coordinate b (B' the other coordinates, $N(\mu, c)$ the Gaussian on $\mathfrak{g}$ with mean μ and covariance c), where $c_b = ((C^{-1}_{E_0})_{bb})^{-1} = (H_{bb})^{-1}$ and $\mu(B')$ is linear in B'. Then for every B',*

$$\begin{aligned} &\int \mathbf{1}_{S_b}(B_b) e^{-\langle B_b, v_b \rangle + (\alpha/2) h(B_b)^2} \, N(\mu(B'), c_b)(\mathrm{d}B_b) \\ &\qquad \leqslant \varpi_b \int e^{-\langle B_b, v_b \rangle + (\alpha/2) h(B_b)^2} \, N(\mu(B'), c_b)(\mathrm{d}B_b) \, , \\ &\varpi_b \coloneqq (h_+/2\pi)^{d_G/2} \operatorname{Leb}(S_b) \leqslant C_V r^{d_G - 1} |m_b|_\sigma \, , \end{aligned} \tag{9.24}$$

$C_V \coloneqq (h_+/2\pi)^{d_G/2} \, 2 d_G V_\sigma (5/4)^{d_G - 1}$, V_σ the volume of the unit $|\bullet|_\sigma$-ball of $\mathfrak{g}$ for the Lebesgue measure of the h-inner product. Consequently $\varpi_b \leqslant C_\varpi r^{d_G} \sum_{e \in J_1(\Lambda)} e^{-\mu_H d(b,e)}$, $C_\varpi \coloneqq C_V c_h C_m \vartheta$.

Proof. The conditional law of B_b given B' under a centred Gaussian of covariance C is Gaussian with covariance $((C^{-1})_{bb})^{-1}$ and mean linear in B'; here $C^{-1} = \iota^* H \iota$, whose bb block is H_{bb}. Completing the square (write μ for $\mu(B')$): the density of $e^{-\langle x, v_b \rangle + (\alpha/2)\langle x, x \rangle} N(\mu, c_b)(\mathrm{d}x)$ is a constant times $\exp(-\frac{1}{2}\langle x, (H_{bb} - \alpha \mathbf{1}) x \rangle + \langle x, H_{bb}\mu - v_b \rangle)$, and $H_{bb} - \alpha \mathbf{1} \geqslant (\gamma - \alpha_G)\mathbf{1} \geqslant (\gamma/2)\mathbf{1} > 0$ ($H_{bb} \geqslant \gamma \mathbf{1}$ as a principal restriction of H, (I3)), so $e^{-\langle x, v_b \rangle + (\alpha/2)\langle x, x \rangle} N(\mu, c_b)(\mathrm{d}x) = Z_\alpha N(\mu_\alpha, c_\alpha)(\mathrm{d}x)$ with $c_\alpha \coloneqq (H_{bb} - \alpha \mathbf{1})^{-1}$, $\mu_\alpha \coloneqq c_\alpha(H_{bb}\mu - v_b)$ and a positive constant Z_α (for $\alpha = 0$: $Z_0 = e^{-\langle \mu, v_b \rangle + \frac{1}{2}\langle v_b, c_b v_b \rangle}$, $\mu_0 = \mu - c_b v_b$); hence the left side of (9.24) equals $Z_\alpha N(\mu_\alpha, c_\alpha)(S_b)$ and the right side's integral equals Z_α; the Gaussian density of $N(\mu_\alpha, c_\alpha)$ is at most $(2\pi)^{-d_G/2} \det(c_\alpha)^{-1/2} =$

$(2\pi)^{-d_G/2}\det(H_{bb}-\alpha\,\mathbf{1})^{1/2}\leqslant(2\pi)^{-d_G/2}\det(H_{bb})^{1/2}\leqslant(h_+/2\pi)^{d_G/2}$ ($H_{bb}\leqslant h_+\mathbf{1}$ by (I3)), whence $N(\mu_\alpha,c_\alpha)(S_b)\leqslant(h_+/2\pi)^{d_G/2}\,\mathrm{Leb}(S_b)$. By homogeneity of the norm, $\mathrm{Leb}(\{|x|_\sigma<s\})=V_\sigma s^{d_G}$, so $\mathrm{Leb}(S_b)=V_\sigma[(r+\epsilon)^{d_G}-(r-\epsilon)^{d_G}]$ with $\epsilon=|m_b|_\sigma\leqslant r/4$, and the mean value theorem gives $\leqslant V_\sigma\cdot 2\epsilon\cdot d_G(5r/4)^{d_G-1}$. The last bound is $|m_b|_\sigma\leqslant c_h h(m_b)$ (Remark 9.3) and (9.18). *q.e.d.*

The point of (9.24) is that the shell is paid by the Gaussian itself, with the tilt $e^{-\langle B,v\rangle}$ (which in the density (3.66) is $e^{-\langle B,\mathrm{Re}\,\Gamma_{E_0}(\sigma)X\rangle}$ after moduli are taken) kept intact: the ratio ϖ_b is of order $r^{d_G-1}|m_b|$ and does not involve the reserve a_{bad}. The alternative sup-over-B_b thin-shell bound would cost $e^{25h_+r^2/32}$ per mark, which $e^{-\frac{9}{16}a_{\mathrm{bad}}}$ cannot pay ($\alpha_G\leqslant\gamma/2<h_+$), and is not used.

Lemma 9.24 (Marked Differences)**.** *Define, for a bond $b\in D(\Lambda)$ and a host $Y\in\mathbf{D}_0(\Lambda)$, the mark weights*

$$\mathfrak{m}_b\coloneqq 2\varpi_b\leqslant 2C_\varpi r^{d_G}\sum_{e\in J_1(\Lambda)}e^{-\mu_H d(b,e)}$$

(one ϖ_b for the bond's own translated cutoff,

one for the cutoffs of the hosts containing b, paid by (9.20)(iii)),

$$\mathfrak{m}_Y\coloneqq 2\bar\omega e^{\bar\omega}\cdot e^{\kappa(B_h+4)}/\omega_*\cdot e^{1/8}$$

if some near primitive assigned to Y has an input in $J(\Lambda)$,

$$\mathfrak{m}_Y\coloneqq N_hC_pg_0\bar s^2(1+\bar\omega)e^{\bar\omega}\cdot e^{\kappa(B_h+4)}/\omega_*\cdot e^{1/8}\sum_{b\in E(Y)}h(m_b)\quad\textit{otherwise,}$$

and for a union Z of LM-cubes $\mathfrak{M}_\Lambda(Z)\coloneqq\sum_{b\in E(Z)}\mathfrak{m}_b+\sum_{Y\in\mathbf{D}_0(\Lambda),\,Y\subseteq Z}\mathfrak{m}_Y$. Under the hypotheses of Lemma 9.20:

(a) For every split selection ℓ with $Z(\ell)=Z$ ($\mathsf{J}=\varnothing$ allowed) and every bounded Borel F,

$$\Big|\int F\,\mathrm{d}(\Lambda^{f,1}_{\Lambda;\ell}-\Lambda^{f,0}_{\Lambda;\ell})\Big|\leqslant\sup|F|\cdot\mathfrak{M}_\Lambda(Z)\cdot e^{-(\kappa_1-1)N_1(Z\setminus Z_0')+N_1(Z)-\frac{9}{16}a_{\mathrm{bad}}|\mathsf{P}|}\prod_{Y\in\mathcal{D}}\omega_*e^{1/2}e^{-\kappa d_0(Y)}$$

(the majorant (9.21) times the sum of the mark weights; the division of a host mark by the host's majorant factor is built into $\mathfrak{m}_Y$); and $\sum_{\ell\colon Z(\ell)=Z}u_*^{|\mathsf{J}|}\|\Lambda^{f,1}_{\Lambda;\ell}-\Lambda^{f,0}_{\Lambda;\ell}\|_{\mathrm{TV}}\leqslant\mathfrak{M}_\Lambda(Z)m(Z)$.

(b) $|k^{f,1}_\Lambda(Z)-k^{f,0}_\Lambda(Z)|\leqslant\mathfrak{M}_\Lambda(Z)m(Z)$, and for a cluster $\mathfrak{c}=(D_1,\dots,D_n)$ with the positive incompatibility-tree majorant $|\Phi|^+_\Lambda(\mathfrak{c})\coloneqq(1/n!)\sum_{T\in\mathrm{Trees}_n}\prod_{\{i,j\}\in T}\mathbf{1}\{D_i\not\sim D_j\}\prod_i m(D_i)$ (the $M_v(\mathfrak{c})$ of Section E.5.2;$|\Phi|^+_\Lambda(\mathfrak{c})\geqslant\max_t|\Phi^{v,t}_\Lambda(\mathfrak{c})|$ by the tree-graph inequality $|\varphi^T(\mathfrak{c})|\leqslant\#\{$spanning trees of the incompatibility graph of $\mathfrak{c}\}$, proved in the justification following (5.29) and used in Appendix B.6, and by $|k^{f,t}_\Lambda(D_i)|\leqslant m(D_i)$, Corollary 9.16), $|\Phi^{v,1}_\Lambda(\mathfrak{c})-\Phi^{v,0}_\Lambda(\mathfrak{c})|\leqslant\big(\sum_i\mathfrak{M}_\Lambda(D_i)\big)|\Phi|^+_\Lambda(\mathfrak{c})$.

Proof. (a) In the display of Lemma 9.14 for the enlarged data, the t-dependent factors are χ^t_b ($b\in E(Y_0)$), $1-\chi^t_b$ ($b\in\mathsf{P}$) and the Mayer factors $M^t_Y\coloneqq\int_0^1\mathbf{W}_t(Y,B)e^{t_Y\mathbf{W}_t(Y,B)}\,\mathrm{d}t_Y$ ($Y\in\mathcal{D}$), with $\mathbf{W}_t(Y,B)=\Theta^t_{E(Y)}(B)\mathbf{P}_t(Y,B)$, $\mathbf{P}_t(Y,B)\coloneqq\sum_{\mathfrak{p}\text{ near},\,Y_\mathfrak{p}=Y}\mathfrak{p}(B+tm+tA^{\Lambda,\Psi})$, $\Theta^t_{E(Y)}\coloneqq\prod_{b\in E(Y)}\chi^t_b$; the density, the Dirac tensor's pushforward (through F) and the s-,

σ-integrations are t-independent. Order these factors (bonds first, then hosts) and telescope: $\prod a_k - \prod b_k = \sum_k (\prod_{i<k} a_i)(a_k - b_k)(\prod_{i>k} b_i)$, a the $t=1$ and b the $t=0$ factors. Each term has exactly one differenced factor, the mark. In every term each bond of $E(Y_0) \cup \mathsf{P}$ carries a factor χ_b^1, χ_b^0, $1-\chi_b^t$ or a difference of two of these, so on the support $|B_b|_\sigma < 5r/4$ for $b \in E(Y_0)$ ($|B_b + tm_b|_\sigma < r$, $|m_b|_\sigma \leqslant r/4$), and every host's primitives are evaluated in the enlarged box (young part of $B + tm + tA^{\Lambda,\Psi}$ of norm $< 3r/2 \leqslant 2r$), where they are analytic with $|\mathfrak{p}| \leqslant C_p g_0 \bar{s}^3$ and $\|D\mathfrak{p}\| \leqslant C_p g_0 \bar{s}^2$ (Lemma 9.13(c), Lemma B.9 on the chart containing the enlarged box under threshold item 3), so $|\mathbf{P}_t(Y,B)| \leqslant N_h C_p g_0 \bar{s}^3 \leqslant \bar{\omega}$ for both t. Bond mark at b: $|\chi_b^1 - \chi_b^0| \leqslant \mathbf{1}_{S_b}(B_b)$ and $|(1-\chi_b^1) - (1-\chi_b^0)| \leqslant \mathbf{1}_{S_b}(B_b)$, since the two indicators differ only where $|B_b|_\sigma$ is within $|m_b|_\sigma$ of r. Host mark at $Y \in \mathcal{D}$: in its term the bond factors are at $t=1$, and $E(Y) \subseteq E(Y_0) \setminus \mathsf{P}$ (Definition 3.17: $\mathsf{P} \cap E(Y_0) = \varnothing$), so $\Theta^1_{E(Y)} = 1$ on the support and

$$M_Y^1 - M_Y^0 = \int_0^1 \big[\mathbf{P}_1 e^{t_Y \mathbf{P}_1} - \mathbf{P}_0 e^{t_Y \mathbf{P}_0}\big] \mathrm{d}t_Y + (1 - \Theta^0_{E(Y)}) \int_0^1 \mathbf{P}_0 e^{t_Y \mathbf{P}_0} \, \mathrm{d}t_Y;$$

the smooth part is bounded on the support by $2\bar{\omega}e^{\bar{\omega}}$ always, and, when no primitive of Y has an input in $J(\Lambda)$, by $\int_0^1 |\partial_t(\mathbf{P}_t e^{t_Y \mathbf{P}_t})| \, \mathrm{d}t \leqslant (1+\bar{\omega})e^{\bar{\omega}} \sup_t |\partial_t \mathbf{P}_t|$ with $\partial_t \mathbf{P}_t(Y,B) = \sum_{\mathfrak{p}} D\mathfrak{p}(B + tm)[m|_{E(Y)}]$ (no exterior input), $|\partial_t \mathbf{P}_t| \leqslant N_h C_p g_0 \bar{s}^2 \sum_{b \in E(Y)} h(m_b)$; the cutoff part satisfies $0 \leqslant 1 - \Theta^0_{E(Y)} \leqslant \sum_{b \in E(Y)} (1 - \chi_b^0)$, and on the support of χ_b^1, $(1-\chi_b^0)\chi_b^1 \leqslant \mathbf{1}_{S_b}(B_b)$, with $|\int_0^1 \mathbf{P}_0 e^{t_Y \mathbf{P}_0} \, \mathrm{d}t_Y| \leqslant \bar{\omega}e^{\bar{\omega}}$: it is a sum over $b \in E(Y)$ of bond marks $\mathbf{1}_{S_b}$ in which the host's factor is replaced by $\bar{\omega}e^{\bar{\omega}}$. Now bound each term. Take moduli; every factor other than the mark and the density is bounded by its supremum as in the proof of Lemma 9.15 (cutoffs $\leqslant 1$; bad bonds by $e^{-\frac{9}{16}a_{\mathrm{bad}}} e^{(\alpha_G/2)h(B_b)^2}$, from $\mathbf{1}\{|B_b|_\sigma \geqslant 3r/4\} \leqslant e^{-\frac{9}{16}a_{\mathrm{bad}}} e^{(\alpha_G/2)h(B_b)^2}$; undifferenced Mayer factors by $\omega_* e^{1/2} e^{-\kappa d_0(Y)}$, valid at either t since $\Theta^t_{E(Y)}$ restricts $B + tm$ to the local box of Y, where (9.12) and (9.20)(ii) give $|\mathbf{W}_t| \leqslant (\omega_*/2) e^{-\kappa d_0(Y)}$; $|F| \leqslant \sup|F|$); the modulus of the density is $e^{-\frac{1}{2}\operatorname{Re}\langle \Gamma(\sigma)X, C\Gamma(\sigma)X\rangle} e^{-\langle B, v\rangle} \, \mathrm{d}\mu_{C_{E_0}}(B) \, \mathrm{d}\mu_0(X)$ with $v := \operatorname{Re}\Gamma_{E_0}(\sigma)X$ real. At a bond mark $b \in E(Y_0)$ ($b \notin \mathsf{P}$): integrate over B_b first; by Lemma 9.23 with $\alpha = 0$ the mark $\mathbf{1}_{S_b}$ is replaced by ϖ_b, leaving $\int e^{-\langle B, v\rangle} e^{(\alpha_G/2)\sum_{b' \in \mathsf{P}} h(B_{b'})^2} \, \mathrm{d}\mu_C \leqslant \int e^{-\langle B, v\rangle} e^{(\alpha_G/2)\|B\|^2} \, \mathrm{d}\mu_C$, which Lemma 3.16(ii) pays with $e^{N_1(Z)}$ together with the X-integral. At a bond mark $b \in \mathsf{P}$: on S_b, $|B_b|_\sigma \geqslant r - |m_b|_\sigma \geqslant 3r/4$, so $\mathbf{1}_{S_b}(B_b) \leqslant \mathbf{1}_{S_b}(B_b) e^{-\frac{9}{16}a_{\mathrm{bad}}} e^{(\alpha_G/2)h(B_b)^2}$, the reserve of b is kept, and Lemma 9.23 with $\alpha = \alpha_G$ (the conditional Gaussian in B_b with the tilt and the factor $e^{(\alpha_G/2)h(B_b)^2}$ on b) replaces the mark by ϖ_b with all factors $e^{(\alpha_G/2)h(B_{b'})^2}$, $b' \in \mathsf{P}$, in place, so the bad-bond payment $e^{-\frac{9}{16}a_{\mathrm{bad}}|\mathsf{P}|}$ is complete and the Gaussian payment is as before. Hence a bond-marked term is bounded by ϖ_b times the majorant (9.21). At a host mark Y, the smooth part is a constant, $2\bar{\omega}e^{\bar{\omega}}$ or $N_h C_p g_0 \bar{s}^2 (1+\bar{\omega}) e^{\bar{\omega}} \sum_{E(Y)} h(m_b)$, in place of the host's majorant factor $\omega_* e^{1/2} e^{-\kappa d_0(Y)} \geqslant \omega_* e^{1/2} e^{-\kappa(B_h+4)}$; the quotient (constant)/$(\omega_* e^{1/2} e^{-\kappa d_0(Y)})$ is at most (constant)$\cdot e^{\kappa(B_h+4)}/\omega_*$ ($d_0(Y) \leqslant B_h + 4$, $e^{1/2} \geqslant 1$), which is $\mathfrak{m}_Y$ without its factor $e^{1/8}$, reserved for the dummy comparison below; the cutoff part gives, for each $b \in E(Y)$, a bond mark $\mathbf{1}_{S_b}$ ($b \notin \mathsf{P}$), paid by ϖ_b as above, with the host's factor $\bar{\omega}e^{\bar{\omega}}$ in place of $\omega_* e^{1/2} e^{-\kappa d_0(Y)}$, a quotient $\leqslant \bar{\omega}e^{\bar{\omega}} e^{\kappa(B_h+4)}/\omega_* \leqslant e\bar{\omega} e^{\kappa(B_h+4)}/\omega_* \leqslant e^{-1/8}/N_{\mathrm{host}} \leqslant 1/N_{\mathrm{host}}$ by (9.20)(iii); the hosts $Y \in \mathcal{D}$ with $b \in E(Y)$ number at most N_{host} (Section 9.6.3), so these contribute at most ϖ_b per bond b, which with the bond's own mark ϖ_b gives $\mathfrak{m}_b = 2\varpi_b$; the rest of the proof of Lemma 9.15 (weakening per cube, Gaussian payment) is unchanged. Summing over the marks gives the first assertion with $\mathfrak{M}_\Lambda(Z)$ (every mark bond lies in $E(Y_0) \cup \mathsf{P} \subseteq E(Z)$, every marked host in $\mathcal{D}$ lies in $Z_0' \subseteq Z$). For the dummy comparison with a mark, repeat the positive host comparison of Lemma 9.20(c) with the marked factor: at the bond's own mark (the differenced factor is χ_b^t or $1 - \chi_b^t$) the hostwise comparison is unchanged (the undifferenced hosts have $v_{\mathrm{ordinary}} = e^{A_Y} - 1$ at either t, as above) and the mark ϖ_b multiplies the whole; at a bond mark generated by the cutoff part of a host

$Y \in \mathcal{D}$, the host Y is active through its ordinary label with its factor replaced by $\bar\omega e^{\bar\omega}$, so the sum over Y's label subsets containing the ordinary label is $\bar\omega e^{\bar\omega}(1+u_*)^{n_Y} \leqslant \bar\omega e^{\bar\omega} e^{1/8}$, whose quotient by Y's majorant factor $\omega_* e^{1/2} e^{-\kappa d_0(Y)}$ is at most $e^{9/8}\bar\omega e^{\kappa(B_h+4)}/\omega_* \leqslant 1/N_{\mathrm{host}}$ by (9.20)(iii) (this is what the factor $e^{9/8}$ in (iii) is for), and the at most N_{host} such hosts again contribute at most ϖ_b per bond, so $\mathfrak{m}_b = 2\varpi_b$ suffices in the dummy comparison as well; at a host mark Y the ordinary weight v_{ordinary} of Y is replaced by the mark's constant, and the sum over the label subsets S containing the ordinary label (which carries the mark) is $(\text{constant}) \cdot \sum_{S' \subseteq \{\text{probe labels of } Y\}} u_*^{|S'|} = (\text{constant}) \cdot (1+u_*)^{n_Y} \leqslant (\text{constant}) \cdot e^{1/8}$ ($n_Y u_* \leqslant \omega_*/8 \leqslant 1/16 < 1/8$), so the host's contribution is $\leqslant \mathfrak{m}_Y \cdot \omega_* e^{1/2} e^{-\kappa d_0(Y)}$, the factor $e^{1/8}$ being the one in $\mathfrak{m}_Y$; the remaining sums give $m(Z)$ as in Lemma 9.20(c), and dropping the constraint that the mark lies in the selection only enlarges the sum. (b) $k_\Lambda^{f,t}(Z) = \sum_{(\mathcal{D},\mathsf{P})} \mathcal{A}^{f,t}_{(\mathcal{D},\mathsf{P},Z)}[1]$ is the $\mathsf{J} = \varnothing$, $F = 1$ case of (a), summed over the selections, which yields $\mathfrak{M}_\Lambda(Z) m(Z)$ by the same seed/enlargement sums; the cluster bound is the telescoping of the product $\prod_i k_\Lambda^{f,t}(D_i)$ with $|k_\Lambda^{f,t}(D_i)| \leqslant m(D_i)$ (Corollary 9.16) and the tree-graph inequality for $|\varphi^T(\mathfrak{c})|$. *q.e.d.*

The mark weights are small where it matters: $\mathfrak{m}_b$ is of order r^{d_G} times the decay from b to $J_1(\Lambda)$; a host carrying an exterior input lies within the stencil range of $J(\Lambda)$ (its near primitive has an input in $J(\Lambda)$ and an input in $D(\Lambda)$), and its mark $\mathfrak{m}_Y \leqslant C_{\mathrm{bd}} g_0 (1+3r)^3$ with $C_{\mathrm{bd}} := 2e \cdot e^{\kappa(B_h+4)} e^{1/8} C_0/\omega_*$ is small in the coupling ($g_0(1+r)^3 \to 0$) though not by a threshold; a host without exterior input has $\mathfrak{m}_Y \leqslant C'_{\mathrm{bd}} g_0 (1+3r)^2 \sum_{E(Y)} h(m_b)$ with $C'_{\mathrm{bd}} := 2e \cdot N_h C_p e^{\kappa(B_h+4)} e^{1/8}/\omega_*$ ($(1+\bar\omega)e^{\bar\omega} \leqslant 2e$ for $\bar\omega \leqslant 1$), which carries the decay of Lemma 9.19.

Lemma 9.25 (The Finite-Difference Bound for Closed Contractions)**.** *Let π be the irreducible unitary representation with $\chi = \dim(\pi)^{-1} \operatorname{tr} \pi$ (π also denotes the Lie-algebra representation, extended complex-linearly to $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g}$, and π denotes as well the holomorphic extension to the complexified group, with $\pi(e^{iX}) = e^{i\pi(X)}$ for $X \in \mathfrak{g}_{\mathbb{C}}$), $h_{\mathbb{C}}(x+iy)^2 := h(x)^2 + h(y)^2$ the Hermitian extension of h to $\mathfrak{g}_{\mathbb{C}}$, $C_\pi := \sup_{0 \neq x \in \mathfrak{g}} \|\pi(x)\|_{\mathrm{op}}/h(x)$, so that $\|\pi(x+iy)\|_{\mathrm{op}} \leqslant C_\pi(h(x)+h(y)) \leqslant \sqrt{2} C_\pi h_{\mathbb{C}}(x+iy)$, $m_* := 1 + \max_n |\mathcal{F}_{r_n}|$ (one more than the largest free-input set of a traversal link, $\mathcal{F}_r$ the full free-input set of the link at r, Section 3.7 and Lemma 3.5; $\mathcal{F}_r \subseteq \mathcal{E}_c \cap E_*$ for the row c of r, so m_* depends on L only), $N_{\mathrm{tr}} := \max_b \#\{\text{traversals } n \text{ of } C : b \in \mathcal{F}_{r_n}\}$ (b lies in the two-block set $\mathcal{E}_c$ of boundedly many rows c, and a traversal with input b is the free bond b itself or the designated bond $b_0(c)$ of such a row, each traversed once by the square, so N_{tr} depends on L only). For a row c within $2R_{\mathrm{c}} + 2L$ of I, Lemmas 3.6 and 3.5 give the phase $q_U(B)|_{\mathcal{E}_c}$ as an analytic function Φ_c of the row phase $A = g_0 \mathsf{C}_U B|_{\mathcal{E}_c}$ on a complex neighbourhood of the closed ball $\|A\| \leqslant \vartheta_*/2$ (the chart statements of Lemma B.9: the row solution and the phase are analytic there), with values in $\mathfrak{g}_{\mathbb{C}}^{\mathcal{E}_c}$ and $h_{\mathbb{C}}(\Phi_c(A)(b)) \leqslant Q_*$ for every $b \in \mathcal{E}_c$ on that ball, $Q_* := C_0^{\mathrm{a}}$ the order-zero bound of Lemma B.7(iii) on the phase, in the Hilbert norm of $\mathfrak{g}_{\mathbb{C}}^{\mathcal{E}_c}$, which dominates $h_{\mathbb{C}}$ of each component (Q_* depends on L and the group only); put $C_\Pi := e^{\sqrt{2} C_\pi Q_*}$ (so $\|\pi(e^{i\Phi_c(A)(b)})\|_{\mathrm{op}} = \|e^{i\pi(\Phi_c(A)(b))}\|_{\mathrm{op}} \leqslant e^{\|\pi(\Phi_c(A)(b))\|} \leqslant e^{\sqrt{2} C_\pi h_{\mathbb{C}}(\Phi_c(A)(b))} \leqslant C_\Pi$ on the ball) and $C_{\mathrm{link}} := \max(C_\pi, 4 m_* C_\Pi C_{\mathsf{C}}/\vartheta_*)$, $C'_\chi := 2 N_{\mathrm{tr}} C_{\mathrm{link}}$; all depend on L, the group and the character only. Under threshold item 3 of Section B.8 ($4 C_{\mathsf{C}} g_0 (1+2r) \leqslant \vartheta_*/2$), on $A_{\mathrm{cov}}^{\mathrm{fin}}$ the row c of every traversal n of C has its free-input set $\mathcal{F}_{r_n} \subseteq D(\Lambda) \subseteq I$ (so the row is within $2L \leqslant 2R_{\mathrm{c}} + 2L$ of I: $\mathcal{F}_{r_n} \subseteq \mathcal{E}_c$ lies in the two endpoint blocks of the row, of ℓ^∞ diameter at most $2L$), and the row phase $A = g_0 \mathsf{C}_U a|_{\mathcal{E}_c}$ depends on $a|_{\mathcal{F}_{r_n}}$ only (Lemma 3.5), so the exterior variable $A^{\Lambda,\Psi}$ (supported on $J(\Lambda)$, disjoint from $D(\Lambda)$) does not enter a traversal link at all; for every real a in the enlarged box $\{|a_b|_\sigma \leqslant 2r,\ b \in D(\Lambda)\}$ the input of the row lies in the row polydisc $\mathcal{U}_c = \{|B(b)| < 2 + 4r \ \forall b \in \mathcal{E}_c \cap E_*\}$ of Lemmas 3.6*

and B.9 (the polydisc is in the norm of the enlarged box of Section 3.3, which is the norm $|\bullet|_\sigma$ of the enlarged box here, Lemma 9.13(c); every input bond has $|a_b|_\sigma \leqslant 2r < 2+4r$), and the chart statements give $\|A\| \leqslant 2C_{\mathsf{C}} g_0(1+2r) \leqslant \vartheta_*/4$ for the row phase. At the reference data (Lemma 9.21(b)) the exterior variable is zero and the same holds. "The enlarged box" below means this.

(a) *(Link derivatives.) For a traversal link $L_n(a) \coloneqq U_\xi(a)(r_n) = e^{iq_U(\Psi_\xi(a+A_{\mathrm{e}}))(r_n)} U_1(r_n)$ ((3.26); it depends on a through $a|_{\mathcal{F}_{r_n}}$ only, Lemma 3.5), every real a in the enlarged box, every $1 \leqslant k \leqslant m_*$ (for $k=0$ the factor is unitary, of norm 1) and every bonds $b_1, \dots, b_k \in \mathcal{F}_{r_n}$ (repetitions allowed) with h-unit directions $e_1, \dots, e_k \in \mathfrak{g}$: $\|\partial_{e_1,b_1} \cdots \partial_{e_k,b_k} \pi(L_n)(a)\|_{\mathrm{op}} \leqslant (C_{\mathrm{link}} g_0)^k$. Consequently $\|\partial_s \pi(L_n(a+sm))\| \leqslant C_{\mathrm{link}} g_0 \sum_{b \in \mathcal{F}_{r_n}} h(m_b)$ and $\|\pi(L_n(a+m)) - \pi(L_n(a))\| \leqslant C_{\mathrm{link}} g_0 \sum_{b \in \mathcal{F}_{r_n}} h(m_b)$ whenever the segment from a to $a+m$ lies in the enlarged box.*

(b) *(Mixed finite differences, with fixed insertions.) Let*

$$g(a) = \dim(\pi)^{-1} \operatorname{tr}(X_1(a) \cdots X_p(a)) \,,$$

where each factor $X_i(a)$ is either a traversal link $\pi(L_{n_i}(a))$ of C or its inverse $\pi(L_{n_i}(a))^{-1} = \pi(L_{n_i}(a))^*$ (the orientation-reversed traversals of the Wilson word; at real points its real-direction derivatives are the adjoints of those of $\pi(L_{n_i})$, with the same norms, so (a) applies to it), with every traversal n of C occurring at most once among the link factors (as in the contour word, its inverse and their sub-words), or a fixed matrix $N_i \in \mathrm{End}(V_\pi)$ not depending on a, and put $\|N\| \coloneqq \prod_{i\,:\,X_i \text{ fixed}} \|N_i\|_{\mathrm{op}}$ ($\|N\| = 1$ when every fixed factor is unitary; the closed contractions $\chi(L_{n_1}(a) \cdots L_{n_j}(a) h_0)$, $h_0 \in G$, with h_0 anywhere in the word, and the words with a fixed unitary insertion such as $\pi(P)^{-1}\pi(h)$ for a word P of links at a fixed argument, are of this kind with $\|N\| = 1$). Let J be a finite set of bonds of $D(\Lambda)$, a_0 real with $|a_{0,b}|_\sigma \leqslant r/4$ for all b (the base point), and real B with $|B_b|_\sigma \leqslant 5r/4$ for $b \in \mathsf{J}$. Then $\Delta_{\mathsf{J}} g(B) \coloneqq \sum_{K \subseteq \mathsf{J}} (-1)^{|\mathsf{J}\setminus K|} g(\mathbf{1}_K B + a_0)$ satisfies

$$|\Delta_{\mathsf{J}} g(B)| \leqslant (N_{\mathrm{tr}} C_{\mathrm{link}} g_0 \cdot 5r/4)^{|\mathsf{J}|} \|N\| \leqslant u_t^{|\mathsf{J}|} \|N\| \,,$$
$$|\Delta_{\mathsf{J}} [\partial_{e,b} g](B)| \leqslant (N_{\mathrm{tr}} C_{\mathrm{link}} g_0 \cdot 5r/4)^{|\mathsf{J}|} N_{\mathrm{tr}} C_{\mathrm{link}} g_0 \|N\| \leqslant u_t^{|\mathsf{J}|} (C'_\chi/2) g_0 \|N\|$$

for every bond b of $D(\Lambda)$ and h-unit e ($u_t = C'_\chi g_0(1+2r) = 2N_{\mathrm{tr}} C_{\mathrm{link}} g_0(1+2r) \geqslant N_{\mathrm{tr}} C_{\mathrm{link}} g_0 \cdot 5r/4$). Both hold at the reference data ($U = \mathbf{1}$, Lemma 3.6) with the same constants.

Proof. (a) Write $\pi(L_n(a)) = \Pi_n(g_0 \mathsf{C}_U a|_{\mathcal{E}_c}) \pi(U_1(r_n))$ (on $A^{\mathrm{fin}}_{\mathrm{cov}}$, where no exterior input enters the row, as in the preamble) with $\Pi_n(A) \coloneqq \pi(e^{i\Phi_c(A)(b_0(c))}) = e^{i\pi(\Phi_c(A)(b_0(c)))}$ for the designated bond $r_n = b_0(c)$ (the component of the row phase at the designated bond), analytic on the neighbourhood of the closed ball $\|A\| \leqslant \vartheta_*/2$ (composition of the analytic Φ_c with the entire exponential of the linear map π on $\mathfrak{g}_{\mathbb{C}}$) with $\|\Pi_n\| \leqslant C_\Pi$ there by the bound of the statement; for a free bond r_n, $q_U(B)(r_n) = g_0 B(r_n)$ (Lemma 3.5), so $\pi(L_n(a)) = e^{ig_0\pi(a_{r_n})} \pi(U_1(r_n))$ depends on a_{r_n} alone, and its k-fold derivative at a real point is bounded by $(C_\pi g_0)^k$ directly (Duhamel's formula: the k-th derivative of $e^{ig_0\pi(a)}$ in the h-unit directions $e_1, \dots, e_k$ is a sum over the $k!$ orderings of the k insertions $ig_0\pi(e_i)$, each of norm $\leqslant C_\pi g_0$, between unitary factors, integrated over a simplex of volume $1/k!$). The directional derivative $\partial_{e,b}$ of $a \mapsto \Pi_n(g_0 \mathsf{C}_U a)$ is $g_0 D\Pi_n(g_0 \mathsf{C}_U a)[\mathsf{C}_U e_b]$ with $\|\mathsf{C}_U e_b\| \leqslant C_{\mathsf{C}}$ (Lemma B.7(i); e_b the vector e in the slot b, of Hilbert norm 1), and the k-fold mixed derivative is $g_0^k D^k \Pi_n(A)[\mathsf{C}_U e_{b_1}, \dots, \mathsf{C}_U e_{b_k}]$. At a real a in the

enlarged box, $\|A\| \leqslant \vartheta_*/4$, so the ball of radius $\vartheta_*/4$ about A lies in the closed ball of radius $\vartheta_*/2$; Cauchy's estimate for the analytic Π_n on that ball bounds the homogeneous polynomial $D^k\Pi_n(A)[h,\dots,h]/k!$ by $C_\Pi(4\|h\|/\vartheta_*)^k$, and polarisation bounds the symmetric k-linear form $D^k\Pi_n(A)/k!$ by $k^k/k!$ times the polynomial's norm; hence $\|D^k\Pi_n(A)[h_1,\dots,h_k]\| \leqslant k^k C_\Pi(4/\vartheta_*)^k \prod\|h_i\| \leqslant (m_* C_\Pi 4/\vartheta_*)^k \prod\|h_i\|$ for $k \leqslant m_*$ ($C_\Pi \geqslant 1$), giving $\|\partial_{e_1,b_1}\cdots\partial_{e_k,b_k}\pi(L_n)\| \leqslant (4m_* C_\Pi C_{\mathsf{C}} g_0/\vartheta_*)^k \leqslant (C_{\mathrm{link}} g_0)^k$; $\pi(U_1(r_n))$ is a constant unitary. The derivative and difference bounds are the chain rule and the mean value inequality along the segment, $\partial_s \pi(L_n(a+sm)) = \sum_{b\in\mathcal{F}_{r_n}} h(m_b)\partial_{m_b/h(m_b),b}\pi(L_n)(a+sm)$, each term of norm $\leqslant h(m_b)C_{\mathrm{link}}g_0$ by the case $k=1$. (b) Write $q \coloneqq |\mathsf{J}|$ and $B^s \coloneqq (s_b B_b)_{b\in\mathsf{J}} + a_0$ for $s \in [0,1]^{\mathsf{J}}$. For each $b \in \mathsf{J}$, the difference in the label b alone is $G(1)-G(0) = \int_0^1 \partial_{s_b} G(s_b)\,\mathrm{d}s_b$ for the function G of s_b at the other labels fixed, and iterating over the labels (the signs $(-1)^{|\mathsf{J}\setminus K|}$ are exactly the iterated differences) gives $\Delta_{\mathsf{J}} g(B) = \int_{[0,1]^{\mathsf{J}}} \partial_{s_{b_1}}\cdots\partial_{s_{b_q}} g(B^s)\,\mathrm{d}s$ with $\partial_{s_b} g(B^s) = \langle B_b, \nabla_b g\rangle(B^s) = h(B_b)\partial_{B_b/h(B_b),b} g(B^s)$, so $|\Delta_{\mathsf{J}} g(B)| \leqslant \prod_{b\in\mathsf{J}} h(B_b)\cdot \sup_s |\partial^{\mathsf{J}} g(B^s)|$, where ∂^{J} is the mixed derivative in the q directions $B_b/h(B_b)$ at the labels $b \in \mathsf{J}$; every B^s is real and in the enlarged box ($|s_b B_b + a_{0,b}|_\sigma \leqslant 5r/4 + r/4 \leqslant 2r$). Product rule inside the normalised trace: the fixed factors N_i do not depend on a, and a link L_{n_i} depends on a_b only if $b \in \mathcal{F}_{r_{n_i}}$, so $\partial^{\mathsf{J}} g = \sum_{\text{assignments}} \dim(\pi)^{-1}\operatorname{tr}(X_1^{(\mathsf{J}_1)}\cdots X_p^{(\mathsf{J}_p)})$, where $X_i^{(\mathsf{J}_i)} \coloneqq \partial^{\mathsf{J}_i}\pi(L_{n_i})$ for a link factor and $X_i^{(\varnothing)} \coloneqq N_i$ for a fixed factor, the sum over the assignments $\mathsf{J} = \bigsqcup_i \mathsf{J}_i$ of each label b to one of the link factors with $b \in \mathcal{F}_{r_{n_i}}$, of which there are at most N_{tr}^q; $|\operatorname{tr}(\prod M_i)| \leqslant \dim(\pi)\prod\|M_i\|_{\mathrm{op}}$, the undifferentiated links are unitary at real points, the fixed factors contribute $\prod\|N_i\| = \|N\|$, and each differentiated link has $|\mathsf{J}_i| \leqslant |\mathcal{F}_{r_{n_i}}| \leqslant m_* - 1$ directions, so by (a) $\sup|\partial^{\mathsf{J}} g| \leqslant (N_{\mathrm{tr}} C_{\mathrm{link}} g_0)^q \|N\|$. With $h(B_b) \leqslant |B_b|_\sigma \leqslant 5r/4$ ($h \leqslant |\bullet|_\sigma$, as in the proof of Lemma 9.19) this gives $|\Delta_{\mathsf{J}} g(B)| \leqslant (N_{\mathrm{tr}} C_{\mathrm{link}} g_0 \cdot 5r/4)^q\|N\| \leqslant (N_{\mathrm{tr}} C_{\mathrm{link}} g_0(1+2r))^q\|N\| \leqslant u_t^q\|N\|$, since $C'_\chi = 2N_{\mathrm{tr}} C_{\mathrm{link}} \geqslant N_{\mathrm{tr}} C_{\mathrm{link}}$. For $\partial_{e,b} g$ the same argument applies to the mixed derivative $\partial^{\mathsf{J}}\partial_{e,b} g$ of order $q+1$ (the extra direction at the label b, possibly repeating a label of J; each link then carries at most m_* directions, within (a)): $|\Delta_{\mathsf{J}}[\partial_{e,b} g](B)| \leqslant (5r/4)^q (N_{\mathrm{tr}} C_{\mathrm{link}} g_0)^{q+1}\|N\| \leqslant u_t^q \cdot N_{\mathrm{tr}} C_{\mathrm{link}} g_0 \|N\| = u_t^q (C'_\chi/2) g_0 \|N\|$. At the reference data Lemma 3.6 holds for $U = \mathbf{1}$ with the same ϑ_*, C_{C}, Q_*, so (a) and (b) hold there verbatim. *q.e.d.*

Lemma 9.25 replaces the finite-difference bound of Section E.4.1 (whose argument it follows: product rule inside the closed character, Cauchy estimates on the row chart, bounded traversal multiplicity) by a proof from Lemmas 3.6 and B.7 at the carrier $D(\Lambda)$; the parity gain of Section E.4.1 is not used.

Lemma 9.26 (The Defect Difference as a Sum of Marked Connected Terms)**.** *Let π be the irreducible unitary representation with $\chi = \dim(\pi)^{-1}\operatorname{tr}\pi$, acting on V_π. For a group W and $t \in \{0,1\}$ let $T_W^t(B)$ be the ordered product of the reconstructed links along the consecutive traversals of $\mathcal{H}(W)$ at the young variable $\mathbf{1}_{\mathcal{A}(W)}B + tm$ (links off Ω_1 equal to their background value), $P_W^t \coloneqq T_W^t(0)$ (young variable tm throughout), and*

$$\Upsilon_W^t \coloneqq w_\Lambda^t(W)\int \pi\big(T_W^t(B)(P_W^t)^{-1}\big)\,\mathrm{d}\Big(\bigotimes_{Z\in\mathcal{A}(W)} \hat\nu_{\Lambda,Z}^{f,t}\Big)(B) - a_W^\Lambda\cdot\mathbf{1}_{V_\pi} \in \mathrm{End}(V_\pi)\,,$$

so that $\dim(\pi)^{-1}\operatorname{tr}(\Upsilon_W^t\pi(h)) = c_W^t(h) - a_W^\Lambda\chi(h)$ with

$$c_W^t(h) \coloneqq w_\Lambda^t(W)\int \chi(T_W^t(B)(P_W^t)^{-1}h)\,\mathrm{d}\bigotimes \hat\nu_\Lambda^{f,t} \quad (\text{(E.18) at the young variable } tm).$$

(a) *(Burnside.) There is K_χ depending on π only with*

$$\|\Upsilon\|_{\mathrm{op}} \leqslant K_\chi \sup_{h\in G} |\dim(\pi)^{-1}\operatorname{tr}(\Upsilon\pi(h))|$$

for every $\Upsilon \in \operatorname{End}(V_\pi)$.

(b) *For a compatible family $\mathfrak{B} = \{W_1,\dots,W_n\}$ with short intervals in cyclic order (every nonempty compatible family is of this kind or a single whole-contour group: compatible groups have disjoint intervals, and a family containing a whole-contour group is a singleton, Section 3.8 after Definition 3.24), $R^t_\Lambda(\mathfrak{B}) = \dim(\pi)^{-1}\operatorname{tr}(\Upsilon^t_{W_1}U^t_1\Upsilon^t_{W_2}U^t_2\cdots\Upsilon^t_{W_n}U^t_n)$, where $U^t_j := \pi(P^t_{W_j}G^t_j)$ with G^t_j the ordered product of the links of the traversals strictly between $\mathcal{H}(W_j)$ and $\mathcal{H}(W_{j+1})$ (cyclically; $G^t_j = \mathbf{1}$ if there are none) at the young variable tm: the t-background word of the interval $\mathcal{H}(W_j)$ followed by the gap to the next interval (for a whole-contour group, $n=1$, $G^t_1 = \mathbf{1}$ and $U^t_1 = \pi(P^t_W)$).*

(c) *Hence*

$$\begin{aligned} R^1_\Lambda(\mathfrak{B}) - R^0_\Lambda(\mathfrak{B}) &= \sum_{j=1}^{n} \dim(\pi)^{-1}\operatorname{tr}\big(\Upsilon^1_{W_1}U^1_1\cdots(\Upsilon^1_{W_j}-\Upsilon^0_{W_j})U^0_j\cdots\Upsilon^0_{W_n}U^0_n\big) \\ &\quad + \sum_j \dim(\pi)^{-1}\operatorname{tr}\big(\cdots\Upsilon^1_{W_j}(U^1_j-U^0_j)\Upsilon^0_{W_{j+1}}\cdots\big)\,, \\ |R^1_\Lambda(\mathfrak{B}) - R^0_\Lambda(\mathfrak{B})| &\leqslant K_\chi^{|\mathfrak{B}|}\Big[\sum_{W_0\in\mathfrak{B}} \mathfrak{d}^{\mathrm{diff}}_{W_0}\prod_{W\neq W_0}\bar{\mathfrak{d}}_W + C'_\chi g_0\Big(\sum_{b\in I_C}h(m_b)\Big)\prod_{W\in\mathfrak{B}}\bar{\mathfrak{d}}_W\Big]\,, \quad (9.25) \end{aligned}$$

with $\bar{\mathfrak{d}}_W := \max_{t\in\{0,1\}}\sup_h|c^t_W(h) - a^\Lambda_W\chi(h)|$ and $\mathfrak{d}^{\mathrm{diff}}_W := \sup_h|c^1_W(h)-c^0_W(h)|$.

(d) *(Marks in $\mathfrak{d}^{\mathrm{diff}}_W$.) With $M^t_\Lambda(W) := (1/k(W)!)\prod_{Z\in\mathcal{A}(W)}\rho_t m(Z)\prod_j|\Phi|^+_\Lambda(\mathfrak{c}_j)$ (the positive group majorant with root weight $\rho_t m(Z)$, the same for $t=0$ and $t=1$ since ρ_t does not depend on t; write $M_\Lambda(W)$ for it; it is the $M(W)$ of Section E.5.2 with $\rho=\rho_t$),*

$$\mathfrak{d}^{\mathrm{diff}}_W \leqslant M_\Lambda(W)\Big[\sum_{Z\in\mathcal{A}(W)}\mathfrak{M}_\Lambda(Z) + \sum_j\sum_{D\in\mathfrak{c}_j}\mathfrak{M}_\Lambda(D) + C'_\chi g_0\sum_{b\in I_{\mathcal{H}(W)}}h(m_b)\Big]\,, \quad (9.26)$$

$$I_{\mathcal{H}(W)} := \textit{free inputs of the traversals of } \mathcal{H}(W),$$

where the last term is the source-shift mark (the root measures contribute their $\prod_Z\rho_t m(Z)$ through the finite-difference factor $u_t^{|\mathsf{J}|}$ exactly as in $M_\Lambda(W)$; the shift adds the factor $C'_\chi g_0\sum h(m_b)$).

(e) *(Bounds for the undifferenced slots.) $\bar{\mathfrak{d}}_W \leqslant M_\Lambda(W) + |a^\Lambda_W| =: \bar{\mathfrak{b}}_W$.*

Proof. (a) $\Upsilon \mapsto (h\mapsto\operatorname{tr}(\Upsilon\pi(h)))$ is linear and injective on $\operatorname{End}(V_\pi)$ ($\pi(G)$ spans $\operatorname{End}(V_\pi)$ for an irreducible representation, Burnside's theorem), so the sup over h is a norm on the finite-dimensional $\operatorname{End}(V_\pi)$, equivalent to the operator norm; K_χ is the equivalence constant. (b) is the factorisation of Lemma E.7(b) at the young variable tm: a traversal in $\mathcal{H}(W_j)$ has no contact with a root of another group, so its link uses W_j's root variables (and tm); a traversal outside every interval has its link at tm; in the expansion of the product, a slot W_j taken from the $\int\pi(T^t_{W_j}(P^t_{W_j})^{-1})$-term contributes $\pi(T^t_{W_j}(B))\pi(G^t_j)$ (the links of $\mathcal{H}(W_j)$ at the root variables, then the gap at tm) and a slot taken from $-a^\Lambda_{W_j}\mathbf{1}$ contributes $-a^\Lambda_{W_j}\pi(P^t_{W_j}G^t_j)$ (the links of $\mathcal{H}(W_j)$ and the gap at

tm), so the normalised trace of the ordered product is $f_\xi(\mathbf{1}_{\mathcal{A}(\mathcal{S})}B + tm)$ for the subfamily $\mathcal{S}$ of slots of the first kind, times $\prod_{W\notin\mathcal{S}}(-a_W^\Lambda)$; expanding $\prod(X_W - a_W\mathbf{1})$ over subsets reproduces the signed sum $R_\Lambda^t(\mathfrak{B})$ of Lemma 9.21(c) by multilinearity of the trace, and the Fubini evaluation of the $\int\pi(\dots)$-slots against the product measure is the one in the proof of Lemma 3.26. (c) Telescoping of the ordered product in its $2n$ factors; then $|\operatorname{tr}(X_1\cdots X_{2n})| \leqslant \dim(\pi)\prod\|X_i\|_{\mathrm{op}}$, $\|U_j^t\| = 1$, $\|\Upsilon_W^t\| \leqslant K_\chi\bar{\mathfrak{d}}_W$, $\|\Upsilon_W^1 - \Upsilon_W^0\| \leqslant K_\chi\mathfrak{d}_W^{\mathrm{diff}}$ by (a), and $\|U_j^1 - U_j^0\| \leqslant \sum_{\text{traversals in }\mathcal{H}(W_j)\text{ and in the gap after it}}\|\pi(\text{link at } m) - \pi(\text{link at } 0)\| \leqslant C'_\chi g_0\sum_{b\in I_C}h(m_b)$ (the links are unitary and the word is their ordered product, so the difference is at most the sum of the link differences; $\|\pi(L_n(m)) - \pi(L_n(0))\| \leqslant C_{\mathrm{link}}g_0\sum_{b\in\mathcal{F}_{r_n}}h(m_b)$ by Lemma 9.25(a), the segment sm lying in the box of radius $r/4$; over j the intervals and gaps partition the traversals of C, each traversal's inputs lie in I_C, and a bond is an input of at most N_{tr} traversals, so the total is at most $N_{\mathrm{tr}}C_{\mathrm{link}}g_0\sum_{b\in I_C}h(m_b) \leqslant C'_\chi g_0\sum_{b\in I_C}h(m_b)$). (d) $c_W^1 - c_W^0$ is telescoped over its t-dependent slots: the $k(W)$ cluster weights $\Phi_\Lambda^{v,t}(\mathfrak{c}_j)$, the $|\mathcal{A}(W)|$ root measures $\hat{\nu}_{\Lambda,Z}^{f,t}$ (through their split selections ℓ_Z, Lemma 9.20(a)), and the source word $g_t(B) \coloneqq \chi(T_W^t(B)(P_W^t)^{-1}h)$. Cluster slot: $|\Phi^{v,1} - \Phi^{v,0}|(\mathfrak{c}_j) \leqslant \sum_{D\in\mathfrak{c}_j}\mathfrak{M}_\Lambda(D)|\Phi|^+(\mathfrak{c}_j)$ (Lemma 9.24(b)), the other clusters at $|\Phi^{v,t}| \leqslant |\Phi|^+$, and the root integral $|\int\Delta_{\mathfrak{J}}g_t\,\mathrm{d}\bigotimes\Lambda_{\ell_Z}^{f,t}| \leqslant \prod_Z u_t^{|\mathfrak{J}_Z|}\|\Lambda_{\ell_Z}^{f,t}\|_{\mathrm{TV}}$ summed with Lemma 9.20(c) and $(u_t/u_*)^{|\mathfrak{J}_Z|} \leqslant \rho_t$ gives $\prod_Z\rho_t m(Z)$. Here $\sup|\Delta_{\mathfrak{J}}g_t| \leqslant u_t^{|\mathfrak{J}|}$ is the first assertion of Lemma 9.25(b) for $g_t(B) = \dim(\pi)^{-1}\operatorname{tr}(\pi(T_W(\mathbf{1}_KB + tm))N_t)$ with the fixed unitary factor $N_t \coloneqq \pi(P_W^t)^{-1}\pi(h)$ ($\|N_t\| = 1$: P_W^t is the word of $\mathcal{H}(W)$ at the fixed argument tm and h a fixed group element) at the base point $a_0 = tm$ ($|tm_b|_\sigma \leqslant r/4$) with the increments $\mathbf{1}_KB$, $|B_b|_\sigma \leqslant 5r/4$ on the support of $\nu^{f,t}$. Root slot: one root Z_0 carries $\Lambda_\ell^{f,1} - \Lambda_\ell^{f,0}$ and Lemma 9.24(a) gives $\sum_\ell u_*^{|\mathfrak{J}|}\|\Lambda_\ell^1 - \Lambda_\ell^0\| \leqslant \mathfrak{M}_\Lambda(Z_0)m(Z_0)$, with the same $u_t^{|\mathfrak{J}|}$ from the finite difference, so the root pays $\mathfrak{M}_\Lambda(Z_0)\rho_t m(Z_0)$. Source slot: all measures and weights at one t, and the insertion is $\Delta_{\mathfrak{J}}[g_1 - g_0](B) = \int_0^1\Delta_{\mathfrak{J}}[\partial_s g_s](B)\,\mathrm{d}s$ with $g_s(a) \coloneqq \dim(\pi)^{-1}\operatorname{tr}(\pi(T_W(a))\pi(P_W^s)^{-1}\pi(h))$, evaluated in the K-term of $\Delta_{\mathfrak{J}}$ at $a = \mathbf{1}_KB + sm$, $T_W(a)$ the word of $\mathcal{H}(W)$ at the young variable a and $P_W^s = T_W(sm)$ (the de-younged bonds are constants, so only the inputs in $I_{\mathcal{H}(W)}\cap D(\Lambda)$ vary; ∂_s below is the derivative of $s\mapsto g_s(\mathbf{1}_KB + sm)$ in each K-term). Two things depend on s: the argument $\mathbf{1}_KB + sm$ of the link word and the fixed factor $N_s \coloneqq \pi(P_W^s)^{-1}\pi(h)$, whose links sit at the argument sm and not at $\mathbf{1}_KB + sm$. Hence $\partial_s g_s = \sum_{b\in I_{\mathcal{H}(W)}\cap D(\Lambda)}h(m_b)\partial_{m_b/h(m_b),b}\tilde{g}_s + \dim(\pi)^{-1}\operatorname{tr}(\pi(T_W(\mathbf{1}_KB + sm))N'_s)$, with $\tilde{g}_s(a) \coloneqq \dim(\pi)^{-1}\operatorname{tr}(\pi(T_W(a))N_s)$ (fixed unitary insertion, $\|N_s\| = 1$) and $N'_s \coloneqq \partial_s[\pi(P_W^s)^{-1}]\pi(h) = -\pi(P_W^s)^{-1}(\partial_s\pi(P_W^s))\pi(P_W^s)^{-1}\pi(h)$, a fixed non-unitary insertion with $\|N'_s\| \leqslant \|\partial_s\pi(P_W^s)\| \leqslant \sum_{n\in\mathcal{H}(W)}\|\partial_s\pi(L_n(sm))\| \leqslant \sum_{n\in\mathcal{H}(W)}C_{\mathrm{link}}g_0\sum_{b\in\mathcal{F}_{r_n}}h(m_b) \leqslant N_{\mathrm{tr}}C_{\mathrm{link}}g_0\sum_{b\in I_{\mathcal{H}(W)}}h(m_b)$ (product rule for the word of unitaries; Lemma 9.25(a) along the segment sm, which lies in the box of radius $r/4$; each bond is an input of at most N_{tr} traversals). The mixed difference of order $|\mathfrak{J}|$ of the first sum is bounded by the second assertion of Lemma 9.25(b) (base point sm, increments $\mathbf{1}_KB$, $\|N_s\| = 1$): $\leqslant u_t^{|\mathfrak{J}|}(C'_\chi/2)g_0\sum_{b\in I_{\mathcal{H}(W)}}h(m_b)$; that of the second term by the first assertion with the insertion N'_s: $\leqslant u_t^{|\mathfrak{J}|}\|N'_s\| \leqslant u_t^{|\mathfrak{J}|}N_{\mathrm{tr}}C_{\mathrm{link}}g_0\sum_{b\in I_{\mathcal{H}(W)}}h(m_b) = u_t^{|\mathfrak{J}|}(C'_\chi/2)g_0\sum_{b\in I_{\mathcal{H}(W)}}h(m_b)$. Together $\sup|\Delta_{\mathfrak{J}}[\partial_s g_s]| \leqslant u_t^{|\mathfrak{J}|}C'_\chi g_0\sum_{b\in I_{\mathcal{H}(W)}}h(m_b)$: the factor 2 in $C'_\chi = 2N_{\mathrm{tr}}C_{\mathrm{link}}$ pays the two s-dependent words. This gives (9.26); the three sums are over the marks of the three kinds. (e) $|c_W^t(h)| \leqslant M_\Lambda^t(W)$ by the same root and cluster bounds (no mark), and $|a_W^\Lambda\chi(h)| \leqslant |a_W^\Lambda|$. *q.e.d.*

Nothing in Lemma 9.26 uses a parity (odd-root) cancellation: the translated cutoffs χ_b^t are not even in B for $t > 0$, so the reversal argument of Section E.4.1 is unavailable and only

the first-order gain $\rho_t = u_t/u_*$ per root is taken; this is why $\rho_t = Cg_0(1+2r)$ rather than $Cg_0^2(1+r)^4$ appears. Section 9.7 recovers the second order for the undifferenced slots at $t \in \{0,1\}$ and for the activities a_W^Λ (Lemma 9.33); the differenced slots and the marked part are unchanged.

Lemma 9.27 (Pins for the t-Family)**.** *Let $F(W)$ be the footprint of W (Section E.5.2: the LM-cubes meeting its roots, cluster constituents, their stencil halos and the stencils of the traversals of $\mathcal{H}(W)$), $\sigma(W) \coloneqq s(W) = n(W) + |\mathcal{H}(W)|/(LM)$ its size (*(E.17)*; $n(W)$ counts the cubes of the roots and cluster constituents with tuple multiplicity, so $\sigma(W)$ is at least the number of domain occurrences in W), $\nu \coloneqq \eta$ the reserve constant of Section E.5.2, and $C_F \coloneqq A_E$ the envelope constant of Lemma E.8 ($|\mathfrak{E}(W)| \leqslant A_E\sigma(W)$). (a) Under* (9.20)*(iv) the positive summation of Lemma E.9 applies to $M_\Lambda(W)$ (its hypotheses hold: $C\rho_t LM \leqslant \eta/8$ is part of $\rho_t \leqslant \rho_{\mathrm{pin}}$, Section 9.6.3, and the root-incidence bound of Section E.7.2 is a property of the fixed geometry, independent of the carrier; it uses only the root weights $\rho\, m(Z)$ with $\rho = \rho_t \leqslant \rho_{\mathrm{pin}}$, the positive incompatibility-tree weights $|\Phi|_\Lambda^+ = M_v$, and the field-independent domain/cluster/interval geometry, none of which involves the carrier beyond the cubes): with C_{pin} the constant of* (E.21)*,*

$$\sup_\Delta \sum_{W\,:\,\Delta\in F(W)} e^{4\nu\sigma(W)} M_\Lambda(W) + \sup_n \sum_{W\,:\,n\in\mathcal{H}(W)} e^{4\nu\sigma(W)} M_\Lambda(W) \leqslant C_{\mathrm{pin}}\rho_t \ .$$

Consequently, with $\bar\varepsilon_t \coloneqq 2K_\chi C_{\mathrm{pin}}\rho_t$,

$$\sup_\Delta \sum_{W\,:\,\Delta\in F(W)} e^{2\nu\sigma(W)} K_\chi e^{\theta^\circ(W)} \bar{\mathfrak{b}}_W \leqslant \bar\varepsilon_t \ , \qquad \sup_n \sum_{W\,:\,n\in\mathcal{H}(W)} e^{2\nu\sigma(W)} K_\chi e^{\theta^\circ(W)} \bar{\mathfrak{b}}_W \leqslant \bar\varepsilon_t \ ,$$
$$\sum_W K_\chi e^{\theta^\circ(W)} \bar{\mathfrak{b}}_W \leqslant n_C \bar\varepsilon_t \ ,$$

the last because every root of a group is marked (Definition 3.24), so $\mathcal{H}(W) \neq \varnothing$ and W is counted in $\sum_n$ for some n. The same statements hold for the reference gas at the carrier I (the reference data of Section 3, activities a_W) with the same constants: the proof uses only the lemmas of Section 3 and Appendix B at the carrier, which hold at I. (b) (Decay from the footprint to the contour.) If $\Delta \in F(W)$ then $\sigma(W) \geqslant \operatorname{dist}(\Delta, C)/(C_F LM)$: the envelope $\mathfrak{E}(W) \supseteq F(W) \cup \pi_1(\mathcal{H}(W))$ is connected under maximum-norm adjacency of LM-cubes and has at most $A_E\sigma(W)$ cubes (Lemma E.8), so its ℓ^∞ diameter in fine units is at most $A_E LM\sigma(W)$, and it contains Δ and a cube meeting the contour. Hence for every cube Δ,

$$\sum_{W\,:\,\Delta\in F(W)} e^{\nu\sigma(W)} K_\chi e^{\theta^\circ(W)} \bar{\mathfrak{b}}_W \leqslant \bar\varepsilon_t e^{-\nu \operatorname{dist}(\Delta,C)/(C_F LM)} \ ,$$
$$\sum_{W\,:\,\Delta\in F(W)} \sigma(W) K_\chi e^{\theta^\circ(W)} \bar{\mathfrak{b}}_W \leqslant \nu^{-1}\bar\varepsilon_t e^{-\nu \operatorname{dist}(\Delta,C)/(C_F LM)} \ .$$

Proof. (a) The first display is the cited summation with ρ_t for the root weights; the second follows from $\bar{\mathfrak{b}}_W \leqslant 2M_\Lambda(W)$ (Lemma 9.21(d) and Lemma 9.26(e)) and $e^{\theta^\circ(W)} \leqslant e^{\nu\sigma(W)/16} \leqslant e^{2\nu\sigma(W)}$ (Lemma 9.21(d)). (b) $e^{2\nu\sigma(W)} \geqslant e^{\nu\sigma(W)} e^{\nu \operatorname{dist}(\Delta,C)/(C_F LM)}$ by the geometric fact, and $\sigma \leqslant \nu^{-1} e^{\nu\sigma}$ ($x \leqslant e^x$). *q.e.d.*

Proposition 9.28 (The Source-Containing Connected Boundary Terms)**.** *In the square/torus geometry of Section 9.6.1, under the hypotheses of Lemmas 9.13–9.17*

and Corollary 9.16, (9.15), (9.16) *and* (9.20)*, on every labelled term with final region* Λ*, for exterior data in the support of the term's collar indicators and exterior cutoffs, on* $A_{\mathrm{cov}}^{\mathrm{fin}}$*,*

$$|\mathcal{B}_\xi^\Lambda[f_\xi]| \leqslant |S_\Lambda^\circ| e^{2n_C\bar\varepsilon_t}\big[C_1 g_0 r\,\mathfrak{S}_1(\Lambda) + C_2\bar\varepsilon_t r^{d_G}\mathfrak{S}_2(\Lambda)\big]\,, \tag{9.27}$$
$$\mathfrak{S}_1(\Lambda) \coloneqq \sum_{e\in J_1(\Lambda)} e^{-\mu_H \operatorname{dist}(e,I_C)/2}\,, \qquad \mathfrak{S}_2(\Lambda) \coloneqq \sum_{e\in J(\Lambda)} e^{-\mu'\operatorname{dist}(e,C)}\,,$$
$$\mu' \coloneqq \min\big(\mu_H/4,\, \nu/(2C_F LM)\big)\,,$$

with $C_1 \coloneqq C_m\vartheta N_{\mathrm{lat}}(\mu_H/2)(C_{\mathrm{resp},\chi} + (1+N_{\mathrm{tr}})C'_\chi)$ *and* $C_2 \coloneqq \nu^{-1}C_{\mathrm{dist}}[4C_\varpi N_{\mathrm{lat}}(\mu_H/2) + N_{\mathrm{host}}e^{\nu R_{\mathrm{st}}/(C_F LM)}(64C_{\mathrm{bd}} + 16C'_{\mathrm{bd}}C_m\vartheta N_{\mathrm{lat}}(\mu_H/2))]$*,* $C_{\mathrm{dist}} \coloneqq e^{7\nu/C_F}$*,* $R_{\mathrm{st}} \coloneqq 4M$ *(*$\geqslant 4L+3M$*, since* $M \geqslant 8LM_2 \geqslant 8L^2 \geqslant 4L$ *by* (3.1)*), constants depending on* L*, the group,* M*,* κ*,* ω_**,* ν*,* ϑ*, the character and the stencil ranges only, in the square/torus geometry of Section 9.6.1;* $g_0 r \to 0$ *and* $\bar\varepsilon_t = 2K_\chi C_{\mathrm{pin}}\rho_t \to 0$ *as* $g_0 \to 0$*.*

Proof. Divide (9.23) by S_Λ° (Lemma 9.21(d)):

$$\mathcal{B}^\Lambda/S_\Lambda^\circ = [f_\xi(m) - f_\xi(0)] + \sum_{\mathfrak{B}\neq\varnothing} q_\Lambda^\circ(\mathfrak{B})[R_\Lambda^1(\mathfrak{B}) - R_\Lambda^0(\mathfrak{B})]\,.$$

The first term is Lemma 9.22: $\leqslant C_{\mathrm{resp},\chi}C_m\vartheta N_{\mathrm{lat}}(\mu_H/2)g_0 r\mathfrak{S}_1(\Lambda)$. For the second, insert (9.25) and $|q_\Lambda^\circ(\mathfrak{B})| \leqslant \prod_W e^{\theta^\circ(W)}$. The background-word part:

$$\begin{aligned} C'_\chi g_0\Big(\sum_{I_C} h(m_b)\Big)\sum_{\mathfrak{B}}\prod_{W\in\mathfrak{B}} K_\chi e^{\theta^\circ(W)}\bar{\mathfrak{d}}_W &\leqslant C'_\chi g_0\Big(\sum_{I_C} h(m_b)\Big)\prod_W(1 + K_\chi e^{\theta^\circ(W)}\bar{\mathfrak{b}}_W) \\ &\leqslant C'_\chi g_0\Big(\sum_{I_C} h(m_b)\Big)e^{n_C\bar\varepsilon_t} \end{aligned}$$

(families are sets of distinct groups; Lemma 9.27(a)), and

$$\sum_{I_C} h(m_b) \leqslant C_m\vartheta N_{\mathrm{lat}}(\mu_H/2) r\mathfrak{S}_1(\Lambda)$$

by the last clause of Lemma 9.19: together with the first term this is

$$C_m\vartheta N_{\mathrm{lat}}(\mu_H/2)(C_{\mathrm{resp},\chi} + C'_\chi)g_0 r\mathfrak{S}_1(\Lambda)e^{n_C\bar\varepsilon_t}\,,$$

the first part of the C_1 term (the second part is (ii) below). The marked part:

$$\sum_{\mathfrak{B}}\sum_{W_0\in\mathfrak{B}}\prod_W e^{\theta^\circ(W)}K_\chi\mathfrak{d}_{W_0}^{\mathrm{diff}}\prod_{W\neq W_0} K_\chi\bar{\mathfrak{d}}_W \leqslant e^{n_C\bar\varepsilon_t}\sum_{W_0} K_\chi e^{\theta^\circ(W_0)}\mathfrak{d}_{W_0}^{\mathrm{diff}}$$

(sum first over $\mathfrak{B}\setminus\{W_0\}$, as above). Insert (9.26). (i) Root and cluster marks. Write $\mathfrak{M}_\Lambda(\Delta) \coloneqq \sum_{b\in E(\Delta)}\mathfrak{m}_b + \sum_{Y\text{ anchored at }\Delta}\mathfrak{m}_Y$ for an LM-cube Δ (a host is anchored at the LM-cube containing its lowest M-cube; a bond meeting two cubes is counted in both), so that $\mathfrak{M}_\Lambda(Z) \leqslant \sum_{\Delta\subseteq Z}\mathfrak{M}_\Lambda(\Delta)$ for every domain Z. The occurrences of a cube Δ among the roots and cluster constituents of W_0 number $\mathrm{mult}_{W_0}(\Delta) \leqslant$ (number of domain occurrences in W_0) $\leqslant \sigma(W_0)$ (Lemma 9.27), so $\sum_{Z\in\mathcal{A}(W_0)}\mathfrak{M}_\Lambda(Z) + \sum_j\sum_{D\in\mathfrak{c}_j}\mathfrak{M}_\Lambda(D) \leqslant \sum_{\Delta\in F(W_0)}\mathrm{mult}_{W_0}(\Delta)\mathfrak{M}_\Lambda(\Delta) \leqslant \sigma(W_0)\sum_{\Delta\in F(W_0)}\mathfrak{M}_\Lambda(\Delta)$, and

$$\begin{aligned} \sum_{W_0} K_\chi e^{\theta^\circ(W_0)}M_\Lambda(W_0)\sigma(W_0)\sum_{\Delta\in F(W_0)}\mathfrak{M}_\Lambda(\Delta) &\leqslant \sum_\Delta \mathfrak{M}_\Lambda(\Delta)\sum_{W_0\,:\,\Delta\in F(W_0)}\sigma(W_0)K_\chi e^{\theta^\circ}\bar{\mathfrak{b}}_{W_0} \\ &\leqslant \nu^{-1}\bar\varepsilon_t\sum_\Delta \mathfrak{M}_\Lambda(\Delta)e^{-\nu\operatorname{dist}(\Delta,C)/(C_F LM)} \end{aligned}$$

by Lemma 9.27(b). Convert cube distances to bond and host distances: for $b \in E(\Delta)$, $\operatorname{dist}(\Delta, C) \geqslant \operatorname{dist}(b, C) - LM - 1$, and for Y anchored at Δ, $\operatorname{dist}(\Delta, C) \geqslant \operatorname{dist}(Y, C) - 4M - LM$ (a host has side at most $4M$), so $e^{-\nu \operatorname{dist}(\Delta,C)/(C_F LM)} \leqslant C_{\mathrm{dist}} e^{-\nu \operatorname{dist}(b,C)/(C_F LM)}$, resp. $\leqslant C_{\mathrm{dist}} e^{-\nu \operatorname{dist}(Y,C)/(C_F LM)}$, with $C_{\mathrm{dist}} \coloneqq e^{7\nu/C_F}$ ($(LM+1)/(LM) \leqslant 2$ and $(4M+LM)/(LM) \leqslant 7/3$, both at most 7). Now the three kinds of $\mathfrak{m}$-weights. Bond marks: a bond lies in $E(\Delta)$ for at most two cubes Δ (no site lies on a cube face), so $\sum_{\Delta} \sum_{b \in E(\Delta)} \mathfrak{m}_b e^{-\nu \operatorname{dist}(\Delta,C)/(C_F LM)} \leqslant 2 C_{\mathrm{dist}} \sum_{b \in D(\Lambda)} \mathfrak{m}_b e^{-\nu \operatorname{dist}(b,C)/(C_F LM)}$, and

$$\sum_{b \in D(\Lambda)} \mathfrak{m}_b e^{-\nu \operatorname{dist}(b,C)/(C_F LM)} \leqslant 2 C_{\varpi} r^{d_G} \sum_{b} \sum_{e \in J_1} e^{-\mu_H d(b,e)} e^{-\nu \operatorname{dist}(b,C)/(C_F LM)}$$
$$\leqslant 2 C_{\varpi} r^{d_G} N_{\mathrm{lat}}(\mu_H/2) \sum_{e \in J_1(\Lambda)} e^{-\mu' \operatorname{dist}(e,C)}$$

(so the bond term of C_2 is $4 C_{\varpi} N_{\mathrm{lat}}(\mu_H/2)$ after the factor C_{dist} is taken out) by the splitting in the proof of (9.19) with $\mu' \leqslant \min(\mu_H/2, \nu/(C_F LM))/2$. Host marks with an exterior input $e \in J(\Lambda)$: the primitive's stencil has ℓ^∞ diameter at most $4L$ and contains its input e, and its host lies in the M-cubes meeting the stencil with one further M-layer ((3.36)), so every point of the host is within $4L + 3M \leqslant 4M = R_{\mathrm{st}}$ of e ($M \geqslant 8L^2 \geqslant 4L$), $\operatorname{dist}(Y, C) \geqslant \operatorname{dist}(e, C) - R_{\mathrm{st}}$, and there are at most N_{host} such hosts per e (Section 9.6.3); their total is $\leqslant N_{\mathrm{host}} e^{\nu R_{\mathrm{st}}/(C_F LM)} C_{\mathrm{bd}} g_0 (1+3r)^3 \sum_{e \in J(\Lambda)} e^{-\mu' \operatorname{dist}(e,C)}$ ($\nu/(C_F LM) \geqslant \mu'$). Host marks without exterior input: $\sum_Y \mathfrak{m}_Y e^{-\nu \operatorname{dist}(Y,C)/(C_F LM)} \leqslant C'_{\mathrm{bd}} g_0 (1+3r)^2 \sum_{b \in D(\Lambda)} h(m_b) \cdot \#\{Y \in \mathbf{D}_0(\Lambda) : b \in E(Y)\} \cdot e^{\nu R_{\mathrm{st}}/(C_F LM)} e^{-\nu \operatorname{dist}(b,C)/(C_F LM)} \leqslant C'_{\mathrm{bd}} N_{\mathrm{host}} e^{\nu R_{\mathrm{st}}/(C_F LM)} g_0 (1+3r)^2 C_m \vartheta r N_{\mathrm{lat}}(\mu_H/2) \sum_{e \in J_1} e^{-\mu' \operatorname{dist}(e,C)}$ by (9.19) ($\operatorname{dist}(Y, C) \geqslant \operatorname{dist}(b, C) - 4M = \operatorname{dist}(b, C) - R_{\mathrm{st}}$ for $b \in E(Y)$, a host having side at most $4M$; at most N_{host} hosts contain b). (ii) Source-shift marks: $\sum_{W_0} K_\chi e^{\theta^\circ} M_\Lambda(W_0) C'_\chi g_0 \sum_{b \in I_{\mathcal{H}(W_0)}} h(m_b) \leqslant C'_\chi g_0 \sum_{b \in I_C} h(m_b) \sum_{n \in n(b)} \sum_{W_0 : n \in \mathcal{H}(W_0)} K_\chi e^{\theta^\circ} \bar{\mathfrak{b}}_{W_0} \leqslant N_{\mathrm{tr}} \bar{\varepsilon}_t C'_\chi g_0 \sum_{b \in I_C} h(m_b)$ (the first step is an inequality: a bond $b \in I_{\mathcal{H}(W_0)}$ is an input of at least one traversal $n \in \mathcal{H}(W_0) \cap n(b)$, so every term on the left occurs at least once on the right, possibly several times when b is an input of several traversals of $\mathcal{H}(W_0)$) $\leqslant N_{\mathrm{tr}} C'_\chi g_0 C_m \vartheta N_{\mathrm{lat}}(\mu_H/2) r \mathfrak{S}_1(\Lambda)$ ($n(b)$ the set of traversals with input b, $|n(b)| \leqslant N_{\mathrm{tr}}$, Lemma 9.25; the contour pin of Lemma 9.27(a) for each $n \in n(b)$; $\bar{\varepsilon}_t \leqslant 1$ by (9.20)(v); then (9.19)); this is the second part of the C_1 term, $C_1 = C_m \vartheta N_{\mathrm{lat}}(\mu_H/2)(C_{\mathrm{resp},\chi} + (1 + N_{\mathrm{tr}}) C'_\chi)$. Collecting, with $e^{n_C \bar{\varepsilon}_t} \leqslant e^{2 n_C \bar{\varepsilon}_t}$, $g_0 \leqslant 1$, $g_0 (1+3r)^3 \leqslant 64 r^{d_G}$ and $(1+3r)^2 r \leqslant 16 r^{d_G}$ for $r \geqslant 1$, $d_G \geqslant 3$, gives (9.27) with $C_2 = \nu^{-1} C_{\mathrm{dist}} [4 C_{\varpi} N_{\mathrm{lat}}(\mu_H/2) + N_{\mathrm{host}} e^{\nu R_{\mathrm{st}}/(C_F LM)} (64 C_{\mathrm{bd}} + 16 C'_{\mathrm{bd}} C_m \vartheta N_{\mathrm{lat}}(\mu_H/2))]$. *q.e.d.*

Corollary 9.29 (The Exterior Sum on the Far Event)**.** *$J(\Lambda) = J \cup (I \setminus D(\Lambda))$ with $J = E_{\Omega,*} \setminus I \subseteq E(\Lambda_1^c)$ (a free bond not in I has both endpoints off Λ_1, by the bond convention of Section 9.1). On the far event N_b^c of* (E.2) *(Section E.2: $\operatorname{dist}(\Gamma_C, E(\Lambda_1^c)) > b$, with $b = \lfloor LMR_1/16 \rfloor$ as in Lemma E.1) every $e \in J$ has $\operatorname{dist}(e, C) \geqslant b - 1$ in the convention of Section 9.6.1 (the distance for N_b may be measured between bond sites rather than to the vertex set of C, which changes a distance by at most 1), and*

$$\sum_{e \in J} e^{-\mu' \operatorname{dist}(e,C)} \leqslant \sum_{s \geqslant b-1} \#\{e : \operatorname{dist}(e, C) \leqslant s\} e^{-\mu' s}$$
$$\leqslant 4 n_C \sum_{s \geqslant b-1} (2s+1)^4 e^{-\mu' s} \leqslant n_C N_{\mathrm{lat}}(\mu'/2) e^{-\mu'(b-1)/2}$$

(at most $4(2s+1)^4$ bonds within ℓ^∞ distance s of each of the n_C vertices of C). Hence on

$A^{\mathrm{fin}}_{\mathrm{cov}} \cap \mathsf{N}_b^c$,

$$|\mathcal{B}^{\Lambda}_{\xi}[f_\xi]| \leqslant |S^{\circ}_{\Lambda}| e^{2n_C\bar{\varepsilon}_t} \big(C_1' g_0 r + C_2 \bar{\varepsilon}_t r^{d_G}\big) \big(n_C N_{\mathrm{lat}}(\mu'/2) e^{-\mu'(b-1)/2} + \mathfrak{D}_\Lambda\big)\,, \tag{9.28}$$
$$\mathfrak{D}_\Lambda \coloneqq \sum_{e \in I \setminus D(\Lambda)} e^{-\mu' \operatorname{dist}(e,C)} \leqslant |I \setminus D(\Lambda)|\,,$$

using $\mathfrak{S}_1(\Lambda) \leqslant e^{\mu_H(2L+1)/2}\mathfrak{S}_2(\Lambda)$ *(*$\operatorname{dist}(e, I_C) \geqslant \operatorname{dist}(e,C) - (2L+1)$*,* $\mu_H/2 \geqslant \mu'$*), with* $C_1' \coloneqq C_1 e^{\mu_H(2L+1)/2}$*. The first summand is volume-independent and, since* $b = \lfloor \ell_{\mathrm{mac}}/16 \rfloor$ *with* $\ell_{\mathrm{mac}} = LMR_1$ *growing as a fixed positive power of* $\log g_1^{-2}$ *(the scale integer* R_1*;* $g_1 \leqslant (1+\beta_0)g_0$ *by [3, (2.6)], as recalled in Section 9.4 before* (9.9)*, so* $\log g_1^{-2} \geqslant \log g_0^{-2} - 2\log(1+\beta_0)$*), beats every polynomial in* r*:* $r^{d_G} e^{-\mu'(b-1)/2} \leqslant e^{-\mu'(b-1)/4}$ *as soon as* $b - 1 \geqslant (4d_G/\mu')\log r$*, which holds for* g_0 *small, since* $b - 1 \geqslant \ell_{\mathrm{mac}}/16 - 2$ *grows as a power of* $\log g_0^{-2}$ *while* $\log r = p_0 \log\log g_0^{-2} + \log A_1$*. The second summand* $\mathfrak{D}_\Lambda$ *is the term's de-younged bonds weighted by their distance to the contour; it is a quantity of the labelled term, bounded by the number* $|I \setminus D(\Lambda)|$ *of de-younged bonds, and its average over the labels is the bookkeeping of Section 9.8 (Section 9.4.2(a)–(b): a de-younged bond* z *of* $D^{(0)} \setminus D^{\mathrm{fin}}$ *is not itself marked; it is attributed to a marked bond* e *(a collar-test failure* M_e*) with* x_e *within* $102\ell + 1$ *of an endpoint of* z*, and Section 9.4.2(a) pays the marked bonds attributed to* z *by* $\sum_e p_e^{\vartheta} \leqslant N_{\mathrm{ret}} q_\vartheta$*, which is the term* $N_{\mathrm{ret}} q_\vartheta$ *of* (9.8)*; the round-0 bonds* $D \setminus D^{(0)}$ *carry no marks and are paid by* $p_z^{(0)} \leqslant q$*, Lemma 9.9(b)).*

Remark 9.30 (Scope of This Subsection). The marked terms are the source-containing connected boundary terms: each is a closed contraction (Lemma 9.26(b)) connected to $J(\Lambda)$ through its mark, a translated cutoff (bond), a potential with an exterior input or with the shift (host), or the shift of the source word, and each mark pays its distance to the contour by the footprint pin (Lemma 9.27(b)) or by the decay of m (Lemma 9.19); the perimeter factor S°_{Λ} is extracted before absolute values. What this subsection does not contain: (1) the identification of S°_{Λ} with $\kappa_C e^{E^{\circ}_{\Lambda}}$ at the retreated carrier, which is Lemma 9.42 (the identity with the regional square scalar at $D(\Lambda)$, through Proposition 7.3 with $D = D(\Lambda)$) and Corollary 9.43 (the form $\kappa_C e^{E^{\circ}_{\Lambda}}$); the reference data at $D(\Lambda)$ differ from those at I not only in the groups whose roots or clusters meet $I \setminus D(\Lambda)$ but also in H° (the principal restriction to $D(\Lambda)$), the eligible labels L_Z, the host set $\mathbf{D}_0(\Lambda)$, the seed sets E_0 and their covariances C_{E_0}, so the activities of the groups near $I \setminus D(\Lambda)$ change as well, and no bound on the ratio $S^{\circ}_{\Lambda}/S^{\circ}_{\xi,C}$ is asserted; (2) the history average of $\mathfrak{D}_\Lambda$, Section 9.8; (3) the complementary events: $(A^{\mathrm{fin}}_{\mathrm{cov}})^c$ is paid by Section 9.4.2(a) in terms of $q + N_{\mathrm{ret}} q_\vartheta$, and N_b is the far-history exclusion (E.2) of Lemma E.1, whose probability is paid there; (4) the inputs listed in Section 9.6.1 (the pin sums of Lemma E.9, the dummy comparison of Lemma E.2, the closed-contraction factorisation of Lemma E.7, the reference-gas cluster expansion, the envelope and footprint bounds, and the far event) are used as stated, at their stated hypotheses, and are not reverified here; a failure of any of them would propagate to (9.27)–(9.28); $\bar{\varepsilon}_t = 2K_\chi C_{\mathrm{pin}} \rho_t$ is the pin bound of Lemma E.9 with ρ_t for the root weights; (5) the labels enter (9.27)–(9.28) additively through $\mathfrak{D}_\Lambda$, $\mathfrak{S}_1(\Lambda)$ and $\mathfrak{S}_2(\Lambda)$ and multiplicatively through $|S^{\circ}_{\Lambda}|$ ($\Lambda = \Lambda_1^{\mathrm{fin}}$ is label-dependent); the multiplicative dependence is divided out in Section 9.8, which bounds $|\mathcal{B}^{\Lambda}|/|S^{\circ}_{\Lambda}|$ and uses $|S^{\circ}_{\Lambda}| \leqslant 1$; (6) the square/torus geometry of Section 9.6.1 is a standing hypothesis of C_{pin}, A_F, A_E and the hull bounds. The factor $e^{2n_C\bar{\varepsilon}_t}$ in (9.27) is the cost of summing over the undifferenced groups of a family, $\prod_W (1 + K_\chi e^{\theta^{\circ}(W)} \bar{\mathfrak{b}}_W) \leqslant e^{n_C \bar{\varepsilon}_t}$; at fixed n_C it is $1 + O(n_C g_0 r)$, but in the square geometry it is a perimeter rate $2L\bar{\varepsilon}_t$ per coarse segment ($2n_C\bar{\varepsilon}_t$ over the $4n$ segments, $n_C = 4nL$) with $\bar{\varepsilon}_t = O(g_0 r)$, first order, whereas $\kappa_C = e^{-\delta m P(C)} z_c^4$ has the rate δm built from one-mark components of weight τ^2 and the tolerated error E° has the rate $C\tau^2$ per segment, $\tau = C[g_0(1+r)^3 + e^{-cr^2}]$. The bound (9.28) is small only while $2n_C\bar{\varepsilon}_t$ stays below the exponents of its two gains, $\mu'(b-1)/4$ for

the far-event summand and, after the label average of $\mathfrak{D}_\Lambda$ (Proposition 9.39), $c_q\vartheta^2\pi_*^2/2$; with $\bar\varepsilon_t = O(g_0 r)$, $b-1 \geqslant \ell/16-2$, $\ell \leqslant L^2M(2\log g_0^{-2})^{\mathfrak{r}}$ (9.9) and $\pi_* \geqslant A_0(\frac12\log g_0^{-2})^{p_0}$ (9.10), this restricts n_C to $O((\log g_0^{-2})^{\mathfrak{r}}/(g_0 r))$ and $O((\log g_0^{-2})^{2p_0}/(g_0 r))$ respectively; the statement is about the fixed loop as $g_0 \to 0$. Section 9.7 brings the rate to second order, $e^{n_C\bar\varepsilon_{\mathrm{ph}}}$ with $\bar\varepsilon_{\mathrm{ph}} = O(g_0^2(1+r)^4) = O(\tau^2)$, the order accepted for E°; a bound uniform in n_C would need the undifferenced slots resummed against the reference exclusions, which is not done.

9.7 Second-Order Bounds for the Undifferenced Slots

Proposition 9.28 carries the family-sum factor $e^{n_C\bar\varepsilon_t}$ (written $e^{2n_C\bar\varepsilon_t}$ in (9.27)), produced by $\prod_W(1+K_\chi e^{\theta^\circ(W)}\bar{\mathfrak{d}}_W) \leqslant \exp(\sum_W K_\chi e^{\theta^\circ(W)}\bar{\mathfrak{d}}_W)$ with the first-order bound $\bar{\mathfrak{d}}_W \leqslant M_\Lambda(W) + |a_W^\Lambda| \leqslant 2M_\Lambda(W)$ of Lemma 9.26(e) and Lemma 9.21(d), whose root weight $\rho_t m(Z)$ has $\rho_t = u_t/u_* = O(g_0 r)$. This subsection proves the second-order bound $\bar{\mathfrak{d}}_W \leqslant 2M_\Lambda^{\mathrm{ph}}(W)$, root weight $\rho_{\mathrm{ph}} m(Z)$ with $\rho_{\mathrm{ph}} = O(g_0^2(1+r)^4)$, for the undifferenced slots at $t \in \{0,1\}$ and for the reference activities, root by root inside the selected integrals, by three mechanisms: (α) a root with at least two selected labels carries $(u_t/u_*)^{|\mathsf{J}_Z|} \leqslant \rho_t^2$ from the finite-difference bound as it stands; (β) a root with a single selected label whose split selection contains an ordinary host ($|\mathsf{J}_Z| = 1$, $\mathcal{D}_Z \neq \varnothing$; a root with two or more labels is in (α) whatever its hosts) carries the factor $2\omega_t/\omega_*$ of its own Mayer factors, through a complex parameter on the selected ordinary primitives, at every t and without parity; (γ) a single-label root without ordinary host ($\mathcal{D}_Z = \varnothing$, $|\mathsf{J}_Z| = 1$) has an even measure at $t = 0$ and at the reference data, so the term of the source linear in its label integrates to zero and the remainder carries one more factor u_t; at $t = 1$ its measure is the $t = 0$ measure plus marked differences of its translated cutoffs, and each mark is paid by the large-field reserve (a mark at a bond outside P costs $e^{-\frac{9}{16}a_{\mathrm{bad}}}$ against the majorant with the reserve $\frac{9}{128}a_{\mathrm{bad}}$, a mark at a bad bond is paid by that bond's own factor, and the whole marked difference of the measure costs $2C_M e^{-\frac{9}{32}a_{\mathrm{bad}}}$, Lemma 9.32(b)), not by ϖ_b, so the marks of the undifferenced slots carry neither exterior decay nor label dependence, and the exponent of the family sum stays a constant times n_C. The differenced slots ($\mathfrak{d}_W^{\mathrm{diff}}$, (9.26)) and the marked part of Proposition 9.28 are unchanged. The results are Lemmas 9.31–9.33 and Proposition 9.34, which restates (9.27), (9.28), (9.33), (9.34) with $e^{n_C\bar\varepsilon_{\mathrm{ph}}}$ in place of $e^{2n_C\bar\varepsilon_t}$.

Thresholds. In addition to the thresholds of Section B.8, (9.15), (9.16), (9.20) and $\omega_* \leqslant 1/2$ assume

$$\begin{aligned}
&\text{(i) } \tfrac{9}{128}a_{\mathrm{bad}} \geqslant \max\bigl(160(\kappa+\log(8M^4+8)+1),\, 64\log(1/\eta_*)\bigr) \quad \text{((9.16) for } a_{\mathrm{bad}}/8);\\
&\text{(ii) } \rho_t + u_t/2 + 2\omega_t/\omega_* + 2C_M e^{-\frac{9}{32}a_{\mathrm{bad}}} \leqslant 1; \qquad (9.29)\\
&\text{(iii) } 2C_M \leqslant (1+2r)^3, \quad \log g_0^{-2} \geqslant 1, \quad (9c_n\alpha_G/64)A_1^2\log g_0^{-2} \geqslant 1,
\end{aligned}$$

where $C_M := 8(4M+1)^4 \geqslant \max_b |E(Y_b)|$ (a probe host is a box of M-cubes of side at most 4 cubes, (3.75), so it has at most $(4M+1)^4$ sites, each an endpoint of eight bonds). Lemmas 9.31–9.33 and the bounds of Proposition 9.34 use (i)–(ii) only; (iii) is used only for the explicit form of the rate (the display $\rho_{\mathrm{ph}} \leqslant C_{\mathrm{ph}} g_0^2(1+2r)^4$ below and the rate bound of Proposition 9.34). All three are conditions on g_0 alone of the kind of (9.16) and (9.20): $a_{\mathrm{bad}} = c_n\alpha_G r^2/4 \to \infty$, $r \to \infty$ and $g_0(1+r)^3 \to 0$ as $g_0 \to 0$, with thresholds depending on L, M, the group, the character, κ, ω_*, α_G, η_*, A_1 and p_0. Put

$$\rho_{\mathrm{ph}} := \rho_t\bigl[\rho_t + u_t/2 + 2\omega_t/\omega_* + 2C_M e^{-\frac{9}{32}a_{\mathrm{bad}}}\bigr], \qquad \bar\varepsilon_{\mathrm{ph}} := 2K_\chi C_{\mathrm{pin}}\rho_{\mathrm{ph}},$$
$$M_\Lambda^{\mathrm{ph}}(W) := \frac{1}{k(W)!}\prod_{Z\in\mathcal{A}(W)}\rho_{\mathrm{ph}} m(Z)\prod_j |\Phi|_\Lambda^+(\mathfrak{c}_j),$$

so that $\rho_{\rm ph} \leqslant \rho_t \leqslant \rho_{\rm pin}$ by (9.29)(ii) and (9.20)(iv), $M_\Lambda^{\rm ph}(W) \leqslant M_\Lambda(W)$, and, with $\rho_t = C'_\chi g_0(1+2r)/u_*$, $u_t = C'_\chi g_0(1+2r)$, $\omega_t = C_0 g_0(1+2r)^3$ and $a_{\rm bad} = c_n \alpha_G r^2/4$,

$$\rho_{\rm ph} \leqslant C_{\rm ph} g_0^2 (1+2r)^4 \,, \qquad C_{\rm ph} \coloneqq (C'_\chi/u_*)(C'_\chi/u_* + C'_\chi/2 + 2C_0/\omega_* + 1) \,,$$
$$\text{hence} \qquad \bar\varepsilon_{\rm ph} \leqslant 32 K_\chi C_{\rm pin} C_{\rm ph} g_0^2 (1+r)^4$$

under (9.29)(iii): the last bracket term needs $2C_M e^{-9c_n\alpha_G r^2/128} \leqslant g_0(1+2r)^3$, which holds once $2C_M \leqslant (1+2r)^3$ and $e^{-9c_n\alpha_G r^2/128} \leqslant g_0$, the latter because $r = A_1(\log g_0^{-2})^{p_0}$ with $p_0 \geqslant 1$ and $\log g_0^{-2} \geqslant 1$ gives, with $c_{\rm ph} \coloneqq 9c_n\alpha_G/128$, $e^{-c_{\rm ph} r^2} \leqslant e^{-c_{\rm ph} A_1^2 (\log g_0^{-2})^2} = g_0^{2c_{\rm ph} A_1^2 \log g_0^{-2}} \leqslant g_0$ as soon as $2c_{\rm ph} A_1^2 \log g_0^{-2} \geqslant 1$, which is the third condition of (9.29)(iii). The first three bracket terms are at most their coefficients times $g_0(1+2r)^3$ without any condition. In $\tau = C[g_0(1+r)^3 + e^{-cr^2}]$ ((5.18); its C, c are the constants of that bound), $\tau^2 \geqslant C^2 g_0^2 (1+r)^6 \geqslant C^2 g_0^2 (1+r)^4$, so $\bar\varepsilon_{\rm ph} \leqslant (32 K_\chi C_{\rm pin} C_{\rm ph}/C^2)\tau^2$: the family-sum rate becomes second order, of the order τ^2 of the bound (7.34) on $E^\circ_{D,C_n,T}$.

9.7.1 Higher Derivatives and the Two-Term Taylor Remainder

Lemma 9.31. *(a) Lemma 9.25(a) holds for every $k \leqslant 2m_*$ with $2C_{\rm link}$ in place of $C_{\rm link}$: $\|\partial_{e_1,b_1} \cdots \partial_{e_k,b_k} \pi(L_n)(a)\|_{\rm op} \leqslant (2C_{\rm link} g_0)^k$ for real a in the enlarged box, $b_i \in \mathcal{F}_{r_n}$ (repetitions allowed), h-unit e_i. (b) Let g be as in Lemma 9.25(b) (link factors, each traversal at most once, fixed insertions N), a_0 real with $|a_{0,b}|_\sigma \leqslant r/4$, J a finite set of bonds of $D(\Lambda)$, and $\varphi(B) \coloneqq g(B + a_0)$ for $B \in \mathfrak{g}^{\mathsf{J}}$ with $|B_b|_\sigma \leqslant 5r/4$ (B extended by zero off J). For $b \in \mathsf{J}$ define, on C^2 functions Ψ of B (D_b makes sense on every function, L_b needs one derivative and R_b two, and all three are applied to the analytic φ only), $(D_b\Psi)(B) \coloneqq \Psi(B) - \Psi(B|_{B_b \to 0})$, $(L_b\Psi)(B) \coloneqq \langle B_b, \nabla_b \Psi(B|_{B_b\to 0})\rangle = h(B_b)\partial_{B_b/h(B_b),b}\Psi(B|_{B_b\to 0})$ (the right side read as 0 when $B_b = 0$), $(R_b\Psi)(B) \coloneqq \int_0^1 (1-s)\partial_s^2[\Psi(B|_{B_b \to sB_b})]\,{\rm d}s$; operators at distinct bonds commute, and $D_b = L_b + R_b$ on C^2 functions. Then $\Delta_{\mathsf{J}} g(B) = (\prod_{b\in\mathsf{J}} D_b\varphi)(B)$, and for every partition $\mathsf{J} = \mathsf{J}' \sqcup \{b_1, \dots, b_p\}$ ($p \geqslant 0$)*

$$\Big| \Big(\prod_{b\in\mathsf{J}'} D_b \prod_{i=1}^p R_{b_i} \varphi \Big)(B) \Big| \leqslant 2^{-p} u_t^{|\mathsf{J}'|+2p} \|N\| \,,$$
$$\prod_{b\in\mathsf{J}} D_b\varphi - \prod_{b\in\mathsf{J}'} D_b \prod_{i=1}^p R_{b_i}\varphi = \sum_{i=1}^p \prod_{b\in\mathsf{J}'} D_b \Big(\prod_{i'<i} D_{b_{i'}} \Big) L_{b_i} \Big(\prod_{i'>i} R_{b_{i'}} \Big) \varphi \,,$$

where the i-th term of the sum is, at fixed values of the coordinates other than B_{b_i}, a linear function of B_{b_i}: $B_{b_i} \mapsto \langle B_{b_i}, c_i(B|_{\mathsf{J}\setminus b_i})\rangle$. Both (a) and (b) hold at the reference data ($U = \mathbf{1}$, base point $a_0 = 0$, the link factors $\pi(L_n)$ at the flat phase) with the same constants, as Lemma 9.25 does: the proofs use only the charts and the Cauchy estimates of Lemma 9.25, which hold there.

Proof. (a) The proof of Lemma 9.25(a) with m_* replaced by $2m_*$: Cauchy's estimate and polarisation give $\|D^k\Pi_n(A)[h_1,\dots,h_k]\| \leqslant k^k C_\Pi (4/\vartheta_*)^k \prod \|h_i\|$ for every k, and $k^k \leqslant (2m_*)^k$ for $k \leqslant 2m_*$, so $\|D^k\Pi_n(A)[h_1,\dots,h_k]\| \leqslant (2m_*)^k C_\Pi (4/\vartheta_*)^k \prod \|h_i\|$ and $\|\partial_{e_1,b_1}\cdots\partial_{e_k,b_k}\pi(L_n)\| \leqslant (8m_* C_\Pi C_{\mathsf{C}} g_0/\vartheta_*)^k \leqslant (2C_{\rm link} g_0)^k$; for a free bond the Duhamel bound $(C_\pi g_0)^k \leqslant (2C_{\rm link} g_0)^k$ holds for every k. (b) $D_b = L_b + R_b$ is Taylor's formula with integral remainder for $s \mapsto \Psi(B|_{B_b \to sB_b})$ on $[0,1]$. Expanding $\prod_{b\in\mathsf{J}} D_b$ (value minus value with the b-th coordinate zeroed) gives $\sum_{K\subseteq\mathsf{J}} (-1)^{|\mathsf{J}\setminus K|} \varphi(\mathbf{1}_K B) = \Delta_{\mathsf{J}} g(B)$. The telescoping identity $\prod_i (L_{b_i} + R_{b_i}) - \prod_i R_{b_i} = \sum_i (\prod_{i'<i}(L+R)_{b_{i'}}) L_{b_i} (\prod_{i'>i} R_{b_{i'}})$ with $(L+R)_b = D_b$,

and the commuting factor $\prod_{\mathsf{J}'} D_b$, give the second display. $L_{b_i}\Psi$ is linear in B_{b_i} with a coefficient not depending on B_{b_i}; the operators at bonds $b \neq b_i$ are linear and act on the other coordinates, so they preserve this form. The bound: $D_b\Psi(B) = \int_0^1 \partial_{s_b}\Psi(B|_{B_b\to s_bB_b})\,\mathrm{d}s_b$ with $\partial_{s_b} = h(B_b)\partial_{B_b/h(B_b),b}$, and R_{b_i} contributes $\int_0^1(1-s_i)\,\mathrm{d}s_i \cdot h(B_{b_i})^2\partial^2_{e_i,b_i}$, $e_i := B_{b_i}/h(B_{b_i})$; iterating, $\prod_{\mathsf{J}'} D_b \prod_i R_{b_i}\varphi(B)$ is an integral over $[0,1]^{|\mathsf{J}'|+p}$ with weights $\prod_i(1-s_i)$ of $\prod_{\mathsf{J}'} h(B_b)\prod_i h(B_{b_i})^2 \cdot (\partial^{\mathsf{J}'}\partial^2_{e_1,b_1}\cdots\partial^2_{e_p,b_p}g)(a_0+B^s)$, every evaluation point real and in the enlarged box ($|s_bB_b + a_{0,b}|_\sigma \leqslant 3r/2 \leqslant 2r$). The mixed derivative of order $Q := |\mathsf{J}'| + 2p$ is bounded as in the proof of Lemma 9.25(b): product rule inside the normalised trace, at most N_{tr}^Q assignments of the Q directions to link factors whose input set contains the direction's bond, fixed factors contributing $\|N\|$, undifferentiated links unitary, and each differentiated link carrying at most two directions per bond of its input set, hence at most $2|\mathcal{F}_{r_{n_i}}| \leqslant 2m_* - 2$ directions, bounded by (a); so $\sup|\partial^{\cdots}g| \leqslant (2N_{\mathrm{tr}}C_{\mathrm{link}}g_0)^Q\|N\|$. With $h(B_b) \leqslant |B_b|_\sigma \leqslant 5r/4$, $\int_0^1(1-s)\,\mathrm{d}s = 1/2$ and $(5r/4)\cdot 2N_{\mathrm{tr}}C_{\mathrm{link}}g_0 \leqslant 2N_{\mathrm{tr}}C_{\mathrm{link}}g_0(1+2r) = u_t$, the bound is $2^{-p}u_t^Q\|N\|$. *q.e.d.*

9.7.2 Evenness, Marks Paid by the Large-Field Reserve, and the Cauchy Parameter

For a split selection $\ell = (\mathcal{D}, \mathsf{J}, \mathsf{P}, Z)$ for the carrier $D(\Lambda)$ (Lemma 9.20(a)) and $x \in \{1/8, 1\}$ put

$$\mathfrak{M}^{(x)}(\ell) := e^{-(\kappa_1-1)N_1(Z\setminus Z_0')+N_1(Z)-x\frac{9}{16}a_{\mathrm{bad}}|\mathsf{P}|}\prod_{Y\in\mathcal{D}}(e^{A_Y}-1)\,, \qquad A_Y := (\omega_*/2)e^{-\kappa d_0(Y)}\,,$$

the right side of (9.21) with the host factor $e^{A_Y} - 1 \leqslant \omega_* e^{1/2}e^{-\kappa d_0(Y)}$ (as in the proof of Lemma 9.20(c)) and the reserve exponent multiplied by x; $\mathfrak{M}^{(1)} \leqslant \mathfrak{M}^{(1/8)}$. ($\mathfrak{M}^{(x)}(\ell)$ is a majorant of a single selection and is not the mark weight $\mathfrak{M}_\Lambda(Z)$ of Section 9.6.)

Lemma 9.32. *Let ℓ be a split selection for $D(\Lambda)$, $\Lambda^{f,t}_{\Lambda;\ell}$ its measure on $\mathfrak{g}^{E_0}$ ($t \in [0,1]$) and $\Lambda^{f,\circ}_{\Lambda;\ell}$ the reference measure (Lemma 9.21(b)); the hypotheses are those of Lemma 9.20.*

(a) (Evenness.) If $\mathcal{D} = \varnothing$ then $\Lambda^{f,0}_{\Lambda;\ell}$ and $\Lambda^{f,\circ}_{\Lambda;\ell}$ are invariant under $B \mapsto -B$ on $\mathfrak{g}^{E_0}$: $\int F(-B)\,\mathrm{d}\Lambda(B) = \int F(B)\,\mathrm{d}\Lambda(B)$ for bounded Borel F. In particular $\int \Psi\,\mathrm{d}\Lambda = 0$ for every bounded Borel Ψ with $\Psi(-B) = -\Psi(B)$, e.g. $\Psi(B) = \langle B_b, c\rangle\mathbf{1}\{|B_b|_\sigma \leqslant 5r/4\}$ for a bond $b \in E(Y_0)$ and a fixed $c \in \mathfrak{g}$ (the truncation is even and equals 1 on the support of the measure, where $|B_b|_\sigma < r$ by the cutoff χ_b^0; the untruncated linear function is not bounded).

(b) (Marked difference, large-field payment.) If $\mathcal{D} = \varnothing$ and $|\mathsf{J}| = 1$, then under (9.29)(i), for bounded Borel F, $|\int F\,\mathrm{d}(\Lambda^{f,1}_{\Lambda;\ell} - \Lambda^{f,0}_{\Lambda;\ell})| \leqslant \sup|F| \cdot 2C_M e^{-\frac{9}{32}a_{\mathrm{bad}}}\mathfrak{M}^{(1/8)}(\ell)$.

(c) (Cauchy parameter.) For $\zeta \in \mathbb{C}$ let $\Lambda^{f,t}_{\Lambda;\ell}(\zeta)$ be defined by the display of Lemma 9.14 for the enlarged data with each Mayer factor $\mathbf{W}_t(Y,B)e^{t_Y\mathbf{W}_t(Y,B)}$, $Y \in \mathcal{D}$, replaced by $\zeta\mathbf{W}_t(Y,B)e^{t_Y\zeta\mathbf{W}_t(Y,B)}$ (every near primitive assigned to a host of $\mathcal{D}$ multiplied by ζ; density, cutoffs, probes unchanged), so that $\Lambda^{f,t}_{\Lambda;\ell}(1) = \Lambda^{f,t}_{\Lambda;\ell}$; put $R_\Upsilon := \omega_*/(2\omega_t) \geqslant 1$ ((9.20)(ii)). For bounded Borel F (complex-valued; the total variation norm below is the supremum over such F with $|F| \leqslant 1$) the function $\zeta \mapsto \int F\,\mathrm{d}\Lambda^{f,t}_{\Lambda;\ell}(\zeta)$ is entire; it vanishes at $\zeta = 0$ if $\mathcal{D} \neq \varnothing$; and $|\int F\,\mathrm{d}\Lambda^{f,t}_{\Lambda;\ell}(\zeta)| \leqslant \sup|F| \cdot \mathfrak{M}^{(1)}(\ell)$ for $|\zeta| \leqslant R_\Upsilon$. Consequently $\|\Lambda^{f,t}_{\Lambda;\ell}\|_{\mathrm{TV}} \leqslant \mathfrak{M}^{(1)}(\ell)$, and if $\mathcal{D} \neq \varnothing$ then $|\int F\,\mathrm{d}\Lambda^{f,t}_{\Lambda;\ell}| \leqslant$

$(2\omega_t/\omega_*)\cdot\sup_{|\zeta|\leqslant R_\Upsilon}|\int F\,\mathrm{d}\Lambda^{f,t}_{\Lambda;\ell}(\zeta)|\leqslant(2\omega_t/\omega_*)\sup|F|\mathfrak{M}^{(1)}(\ell)$. *The same holds at the reference data.*

(d) (Weakened dummy comparison.) Under (9.29)*(i), for every connected final domain* Z *for* $D(\Lambda)$ *and* $x\in\{1/8,1\}$, $\sum_{\ell\colon Z(\ell)=Z}u_*^{|\mathsf{J}|}\mathfrak{M}^{(x)}(\ell)\leqslant m(Z)$, *the sum over all split selections with final domain* Z *and* $Z_0\neq\varnothing$ *(*$\mathsf{J}=\varnothing$ *included).*

Proof. (a) With $\mathcal{D}=\varnothing$ the display of Lemma 9.14 for the enlarged data has, inside the s- and σ-integrations (which do not involve (B,X)), the integrand $\prod_{b\in E(Y_0)}\chi^0_b(B)\prod_{b\in\mathsf{P}}(1-\chi^0_b(B))F(B)$ against the density (3.66)

$$e^{-\frac12\langle\Gamma_{E_0}(\sigma)X,C_{E_0}\Gamma_{E_0}(\sigma)X\rangle}e^{-\langle B,\Gamma_{E_0}(\sigma)X\rangle}\,\mathrm{d}\mu_{C_{E_0}}(B)\,\mathrm{d}\mu_0(X)\,.$$

Substitute $(B,X)\to(-B,-X)$: μ_0 is the standard Gaussian measure and $\mu_{C_{E_0}}$ the centred Gaussian measure of covariance C_{E_0} (Section 3.5), both invariant; the quadratic form in X and the bilinear form $\langle B,\Gamma_{E_0}(\sigma)X\rangle$ are invariant under the joint sign change; $\chi^0_b(-B)=\mathbf{1}\{|-B_b|_\sigma<r\}=\chi^0_b(B)$ and likewise $1-\chi^0_b$ ($|\bullet|_\sigma$ is a norm); there is no Mayer factor. Hence $\int F(-B)\,\mathrm{d}\Lambda^0=\int F\,\mathrm{d}\Lambda^0$. At the reference data the display is the same with H°, $m^\circ=0$ and the cutoffs $\chi_b(a)=\mathbf{1}\{|a_b|_\sigma<r\}$ (Lemma 9.21(b)). For odd Ψ, $\int\Psi\,\mathrm{d}\Lambda=\int\Psi(-B)\,\mathrm{d}\Lambda=-\int\Psi\,\mathrm{d}\Lambda$. (b) With $\mathcal{D}=\varnothing$ and $\mathsf{J}=\{b_0\}$, the t-dependent factors of the display are χ^t_b, $b\in E(Y_0)=E(Y_{b_0})$ (at most C_M bonds), and $1-\chi^t_b$, $b\in\mathsf{P}$ (proof of Lemma 9.24(a)); order them and telescope, $\prod a_k-\prod c_k=\sum_k(\prod_{i<k}a_i)(a_k-c_k)(\prod_{i>k}c_i)$, a the $t=1$ and c the $t=0$ factors: at most $C_M+|\mathsf{P}|$ terms, each with exactly one differenced factor (the mark at a bond b), of modulus $|\chi^1_b-\chi^0_b|\leqslant\mathbf{1}_{S_b}(B_b)\leqslant\mathbf{1}\{|B_b|_\sigma\geqslant r-|m_b|_\sigma\}\leqslant\mathbf{1}\{|B_b|_\sigma\geqslant3r/4\}\leqslant e^{-\frac{9}{16}a_{\mathrm{bad}}}e^{(\alpha_G/2)h(B_b)^2}$ ((9.11) and the proof of Lemma 9.15). Bound the modulus of each term factor by factor as in the proof of Lemma 9.15: undifferenced cutoffs $\leqslant1$; undifferenced bad factors $1-\chi^t_{b'}\leqslant e^{-\frac{9}{16}a_{\mathrm{bad}}}e^{(\alpha_G/2)h(B_{b'})^2}$; the mark as displayed; $|F|\leqslant\sup|F|$; the product of the exponential factors is $e^{-\frac{9}{16}a_{\mathrm{bad}}n}e^{(\alpha_G/2)\|B\|^2}$ with $n\coloneqq|\mathsf{P}|+\mathbf{1}\{b\notin\mathsf{P}\}\geqslant\max(|\mathsf{P}|,1)$, and the density's modulus against $e^{(\alpha_G/2)\|B\|^2}$ integrates to at most $e^{N_1(Z)}$ (Lemma 3.16(ii) for H), the weakening giving $e^{-(\kappa_1-1)N_1(Z\setminus Z'_0)}$. Since $\frac{9}{16}a_{\mathrm{bad}}n\geqslant\frac{9}{128}a_{\mathrm{bad}}|\mathsf{P}|+\frac{63}{128}a_{\mathrm{bad}}\cdot\max(|\mathsf{P}|,1)$, each term is at most $\sup|F|\cdot e^{-\frac{63}{128}a_{\mathrm{bad}}\max(|\mathsf{P}|,1)}\mathfrak{M}^{(1/8)}(\ell)$, and the number of terms is at most $C_M+|\mathsf{P}|\leqslant C_M(1+|\mathsf{P}|)$. For $|\mathsf{P}|=0$ this gives $C_Me^{-\frac{63}{128}a_{\mathrm{bad}}}\leqslant C_Me^{-\frac{9}{32}a_{\mathrm{bad}}}$; for $|\mathsf{P}|=k\geqslant1$, $C_M(1+k)e^{-\frac{63}{128}a_{\mathrm{bad}}k}=C_Me^{-\frac{9}{32}a_{\mathrm{bad}}}\cdot(1+k)e^{-\frac{63k-36}{128}a_{\mathrm{bad}}}\leqslant C_Me^{-\frac{9}{32}a_{\mathrm{bad}}}\cdot(1+k)e^{-\frac{27}{128}a_{\mathrm{bad}}k}\leqslant2C_Me^{-\frac{9}{32}a_{\mathrm{bad}}}$, because $63k-36\geqslant27k$ for $k\geqslant1$ and $(1+k)e^{-ck}\leqslant2e^{-c}\leqslant2$ for $c\coloneqq\frac{27}{128}a_{\mathrm{bad}}\geqslant1$ (implied by (9.29)(i)). (c) For fixed $(X,B,s,\sigma,(t_Y))$ the integrand is $\prod_{Y\in\mathcal{D}}\zeta\mathbf{W}_te^{t_Y\zeta\mathbf{W}_t}$ times a ζ-independent factor, entire in ζ; $\mathbf{W}_t(Y,B)=\mathbf{V}^\Lambda_{\xi,t}(Y,B+tm)$ (Section 9.5.1) vanishes off the translated local box of Y and has $|\mathbf{W}_t(Y,B)|\leqslant\omega_te^{-\kappa d_0(Y)}\leqslant\omega_t$ on it ((9.12)), so on $|\zeta|\leqslant R$ the integrand's modulus is at most $(R\omega_te^{R\omega_t})^{|\mathcal{D}|}\sup|F|$ times the density's modulus, which is integrable (Lemma 3.16(ii)); the s-, σ-, t_Y-integrals are over compact sets. Dominated convergence gives continuity in ζ and, with Morera's theorem (or differentiation under the integral sign), analyticity on every disc: the function is entire. At $\zeta=0$ every Mayer factor vanishes and $\mathcal{D}\neq\varnothing$, so the integral is zero. For $|\zeta|\leqslant R_\Upsilon$, integrate t_Y first: $\int_0^1\zeta\mathbf{W}_te^{t_Y\zeta\mathbf{W}_t}\,\mathrm{d}t_Y=e^{\zeta\mathbf{W}_t}-1$, of modulus $\leqslant e^{|\zeta\mathbf{W}_t|}-1\leqslant e^{R_\Upsilon\omega_te^{-\kappa d_0(Y)}}-1=e^{A_Y}-1$ on the support of $\Theta^t_{E(Y)}$; the other factors are bounded as in the proofs of Lemma 9.15 and Lemma 9.20(b) (cutoffs $\leqslant1$, bad bonds by $e^{-\frac{9}{16}a_{\mathrm{bad}}}e^{(\alpha_G/2)h^2}$, Gaussian payment $e^{N_1(Z)}$, weakening), giving $\sup|F|\mathfrak{M}^{(1)}(\ell)$. The TV bound is the case $\zeta=1$ (for $\mathcal{D}=\varnothing$ there is no ζ, and the bound is the factor-by-factor bound directly). Schwarz's lemma for $\Phi(\zeta)\coloneqq\int F\,\mathrm{d}\Lambda^{f,t}_{\Lambda;\ell}(\zeta)$ on $|\zeta|\leqslant R_\Upsilon$ with $\Phi(0)=0$ and $|\Phi|\leqslant M_F\coloneqq\sup|F|\mathfrak{M}^{(1)}(\ell)$ gives $|\Phi(\zeta)|\leqslant M_F|\zeta|/R_\Upsilon$, and $\zeta=1$ lies in the disc since $R_\Upsilon\geqslant1$. At the reference data (9.12)

holds by Lemma 9.13(d) for these data (Lemma 9.21(b)). (d) The proof of Lemma 9.20(c): the hostwise comparison bounds the sum over the nonempty label subsets of a host of the weights $e^{A_Y}-1$ (ordinary) and u_* (probe) by $\omega_* e^{1/2} e^{-\kappa d_0(Y)}$, so $\sum_\ell u_*^{|\mathsf{J}|}\mathfrak{M}^{(x)}(\ell)$ is at most the sum over the ordinary selections $\mathfrak{s}=(\mathcal{D},\mathsf{P},Z)$ of the majorant (9.17) with $x\frac{9}{16}a_{\mathrm{bad}}$ in place of $\frac{9}{16}a_{\mathrm{bad}}$; the seed, enlargement and final-domain sums of Appendix B bound this by $m(Z)$ exactly as in Corollary 9.16, whose proof applies to every number $\geqslant 1$ in the role of a_{bad} that satisfies (B.29), which $\frac{9}{128}a_{\mathrm{bad}}$ does by (9.29)(i) (and $\frac{9}{16}a_{\mathrm{bad}}$ by (9.16)). *q.e.d.*

9.7.3 The Second-Order Group Bound

Lemma 9.33. *Under the hypotheses of Lemma 9.20,* (9.29) *and the setting of Section 9.6.1 (on $A_{\mathrm{cov}}^{\mathrm{fin}}$), for every group W (Lemma 9.21(a)), every $h\in G$ and $t\in\{0,1\}$: $|c_W^t(h)|\leqslant M_\Lambda^{\mathrm{ph}}(W)$ and $|a_W^\Lambda|\leqslant M_\Lambda^{\mathrm{ph}}(W)$. Hence $\bar{\mathfrak{d}}_W\leqslant 2M_\Lambda^{\mathrm{ph}}(W)=:\bar{\mathfrak{b}}_W^{\mathrm{ph}}$ (the bound $\bar{\mathfrak{d}}_W\leqslant M_\Lambda(W)+|a_W^\Lambda|$ of Lemma 9.26(e) with both terms improved), and Lemma 9.27 holds with $(M_\Lambda^{\mathrm{ph}},\rho_{\mathrm{ph}},\bar\varepsilon_{\mathrm{ph}})$ in place of $(M_\Lambda,\rho_t,\bar\varepsilon_t)$:*

$$\sup_\Delta\sum_{W\,:\,\Delta\in F(W)} e^{2\nu\sigma(W)}K_\chi e^{\theta^\circ(W)}\bar{\mathfrak{b}}_W^{\mathrm{ph}}\leqslant\bar\varepsilon_{\mathrm{ph}}\,,\qquad \sup_n\sum_{W\,:\,n\in\mathcal{H}(W)} e^{2\nu\sigma(W)}K_\chi e^{\theta^\circ(W)}\bar{\mathfrak{b}}_W^{\mathrm{ph}}\leqslant\bar\varepsilon_{\mathrm{ph}}\,,$$
$$\sum_W K_\chi e^{\theta^\circ(W)}\bar{\mathfrak{b}}_W^{\mathrm{ph}}\leqslant n_C\bar\varepsilon_{\mathrm{ph}}\,,$$

and (b) of Lemma 9.27 likewise with $\bar\varepsilon_{\mathrm{ph}}$. The differenced bound (9.26)*, its majorant $M_\Lambda(W)$ and $\theta^\circ(W)\leqslant\nu\sigma(W)/16$ are unchanged.*

Proof. Fix W, h and $t\in\{0,1\}$. By Lemma 9.20(a),(d) and Lemma 9.21(a), $c_W^t(h)=w_\Lambda^t(W)\sum_{(\ell_Z)_{Z\in\mathcal{A}(W)}:\mathsf{J}_Z\neq\varnothing}\int\Delta_{\mathsf{J}}g_t\,\mathrm{d}\bigotimes_Z\Lambda^{f,t}_{\Lambda;\ell_Z}$, with $g_t(a)=\dim(\pi)^{-1}\operatorname{tr}(\pi(T_W(a))N_t)$, $N_t:=\pi(P_W^t)^{-1}\pi(h)$ fixed unitary, base point $a_0=tm$, $\mathsf{J}=\bigsqcup_Z\mathsf{J}_Z$, and $|w_\Lambda^t(W)|\leqslant(1/k(W)!)\prod_j|\Phi|_\Lambda^+(\mathfrak{c}_j)$ (Corollary 9.16 and Lemma 9.24(b), as in the proof of Lemma 9.21(d)). The integrand $\Delta_{\mathsf{J}}g_t$ depends on B through the labels B_{J} only, and the labels of the root Z are its $\mathsf{J}_Z\subseteq E_0(\ell_Z)$. Fix a tuple (ℓ_Z) and sort its roots into three classes: (α) $|\mathsf{J}_Z|\geqslant 2$; (β) $|\mathsf{J}_Z|=1$ and $\mathcal{D}_Z\neq\varnothing$; (γ) $|\mathsf{J}_Z|=1$ and $\mathcal{D}_Z=\varnothing$, with single label b_Z.

Step 1 (splitting the class-(γ) measures; $t=1$ only). For each class-(γ) root write $\Lambda^{f,1}_{\Lambda;\ell_Z}=\Lambda^{f,0}_{\Lambda;\ell_Z}+\delta\Lambda_{\ell_Z}$ and expand the product measure: $\bigotimes_Z\Lambda^{f,1}_{\ell_Z}=\sum_{\mathcal{S}\subseteq(\gamma)}[\bigotimes_{Z\in\mathcal{S}}\delta\Lambda_{\ell_Z}]\otimes[\bigotimes_{Z\in(\gamma)\setminus\mathcal{S}}\Lambda^{f,0}_{\ell_Z}]\otimes[\bigotimes_{Z\notin(\gamma)}\Lambda^{f,1}_{\ell_Z}]$. At $t=0$ only $\mathcal{S}=\varnothing$ occurs. Fix $\mathcal{S}$; the roots of $(\gamma)\setminus\mathcal{S}$ are the parity roots, $Z_1,\dots,Z_p$ with labels $b_1,\dots,b_p$, and $\mathsf{J}':=\mathsf{J}\setminus\{b_1,\dots,b_p\}$.

Step 2 (parity). By Lemma 9.31(b) with $\varphi_t:=g_t(\bullet+tm)$ ($|tm_b|_\sigma\leqslant r/4$ by (9.11); $|B_b|_\sigma\leqslant 5r/4$ on the supports of the measures, Lemma 9.20(a); every function of the labels below, φ_t, the L- and R-terms, $g^{(p)}$, is multiplied by the even truncation $\prod_{b\in\mathsf{J}}\mathbf{1}\{|B_b|_\sigma\leqslant 5r/4\}$, which equals 1 on the supports of the measures and makes it bounded Borel on $\mathfrak{g}^{E_0}$, so that Lemma 9.32 applies), $\Delta_{\mathsf{J}}g_t=g^{(p)}+\sum_{i=1}^p(\text{term linear in }B_{b_i})$, $g^{(p)}:=\prod_{\mathsf{J}'}D_b\prod_i R_{b_i}\varphi_t$. The i-th linear term depends on the variables of the root Z_i through B_{b_i} alone ($\mathsf{J}_{Z_i}=\{b_i\}$) and is odd under $B_{E_0(Z_i)}\to-B_{E_0(Z_i)}$ at fixed values of the other roots' variables; the measures are finite and the integrand is bounded Borel, so by Fubini and Lemma 9.32(a) ($\Lambda^{f,0}_{\ell_{Z_i}}$ is even, $\mathcal{D}_{Z_i}=\varnothing$) its integral against the product measure of Step 1 vanishes. Hence $\int\Delta_{\mathsf{J}}g_t\,\mathrm{d}(\text{product for }\mathcal{S})=\int g^{(p)}\,\mathrm{d}(\text{product for }\mathcal{S})$, and $\sup|g^{(p)}|\leqslant 2^{-p}u_t^{|\mathsf{J}'|+2p}$ ($\|N_t\|=1$).

Step 3 (Cauchy parameter). Let

$$\Phi((\zeta_Z)_{Z\in(\beta)}) \coloneqq \int g^{(p)}\,\mathrm{d}\Big[\bigotimes_{Z\in(\beta)} \Lambda^{f,t}_{\Lambda;\ell_Z}(\zeta_Z) \otimes (\text{the other factors of the product for } \mathcal{S})\Big]\,.$$

Φ is analytic in each ζ_Z separately on the open polydisc $\{|\zeta_Z| < R_\Upsilon\}$: at fixed values of the other roots' variables, the integral over the variables of the root Z is, for every bounded Borel integrand, entire in ζ_Z with the uniform bound $\sup|\text{integrand}|\cdot\mathfrak{M}^{(1)}(\ell_Z)$ on $|\zeta_Z|\leqslant R_\Upsilon$ (Lemma 9.32(c), a single-root statement), and this uniform bound carries analyticity through the integrals over the other roots (finite measures, Fubini; dominated convergence and Morera's theorem for the outer integral). Analyticity on the open polydisc with a bound on the closed one is all Schwarz's lemma below needs. Φ vanishes whenever some $\zeta_Z = 0$, and on the polydisc $\{|\zeta_Z|\leqslant R_\Upsilon\ \forall Z\in(\beta)\}$ satisfies $|\Phi|\leqslant M_\Phi \coloneqq \sup|g^{(p)}|\cdot\prod_{Z\in(\beta)}\mathfrak{M}^{(1)}(\ell_Z)\cdot\prod_{Z\in\mathcal{S}}\|\delta\Lambda_{\ell_Z}\|_{\mathrm{TV}}\cdot\prod_{Z\in(\alpha)\cup((\gamma)\setminus\mathcal{S})}\|\Lambda_{\ell_Z}\|_{\mathrm{TV}}$ (the TV norms of the $t=0$ measures of the parity roots, of the $t=1$ measures of class (α)). Schwarz's lemma one variable at a time, from the inside out: write $(\beta)=\{Z_1,\dots,Z_q\}$; for each fixed $(\zeta_{Z_1},\dots,\zeta_{Z_{q-1}})$ in the polydisc, $\zeta_{Z_q}\mapsto\Phi(\zeta_{Z_1},\dots,\zeta_{Z_{q-1}},\zeta_{Z_q})$ vanishes at 0 and is bounded by M_Φ on $|\zeta_{Z_q}|\leqslant R_\Upsilon$, so $|\Phi(\zeta_{Z_1},\dots,\zeta_{Z_{q-1}},1)|\leqslant M_\Phi/R_\Upsilon$ on the polydisc of the remaining $q-1$ variables; the function $(\zeta_{Z_1},\dots,\zeta_{Z_{q-1}})\mapsto\Phi(\dots,1)$ is again analytic in each variable on the open polydisc, vanishes when any variable is 0 and is bounded by M_Φ/R_Υ, so the same step in $\zeta_{Z_{q-1}}$ gives M_Φ/R_Υ^2 at $\zeta_{Z_{q-1}}=1$, and after q steps $|\Phi(1,\dots,1)|\leqslant R_\Upsilon^{-q}M_\Phi = (2\omega_t/\omega_*)^{|(\beta)|}M_\Phi$.

Step 4 (collecting). With $\|\Lambda_{\ell_Z}\|_{\mathrm{TV}}\leqslant\mathfrak{M}^{(1)}(\ell_Z)\leqslant\mathfrak{M}^{(1/8)}(\ell_Z)$ (Lemma 9.32(c)) and $\|\delta\Lambda_{\ell_Z}\|_{\mathrm{TV}}\leqslant 2C_M e^{-\frac{9}{32}a_{\mathrm{bad}}}\mathfrak{M}^{(1/8)}(\ell_Z)$ (Lemma 9.32(b)), the summand for $((\ell_Z),\mathcal{S})$ is at most

$$\prod_{Z\in(\alpha)} u_t^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)\cdot\prod_{Z\in(\beta)} u_t(2\omega_t/\omega_*)\mathfrak{M}^{(1/8)}(\ell_Z) \times\prod_{Z\in(\gamma)\setminus\mathcal{S}}(u_t^2/2)\mathfrak{M}^{(1/8)}(\ell_Z)\cdot\prod_{Z\in\mathcal{S}} u_t\cdot 2C_M e^{-\frac{9}{32}a_{\mathrm{bad}}}\mathfrak{M}^{(1/8)}(\ell_Z)\,.$$

Summing over $\mathcal{S}\subseteq(\gamma)$, a root of each class contributes

$$\begin{aligned}
(\gamma):\quad & [u_t^2/2+2C_M u_t e^{-\frac{9}{32}a_{\mathrm{bad}}}]\mathfrak{M}^{(1/8)}(\ell_Z) = \rho_t[u_t/2+2C_M e^{-\frac{9}{32}a_{\mathrm{bad}}}]u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)\\
&\qquad\leqslant\rho_{\mathrm{ph}}u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)\,,\\
(\beta):\quad & \rho_t(2\omega_t/\omega_*)u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)\leqslant\rho_{\mathrm{ph}}u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)\,,\\
(\alpha):\quad & \rho_t^{|\mathsf{J}_Z|}u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)\leqslant\rho_t^2u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)\leqslant\rho_{\mathrm{ph}}u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)
\end{aligned}$$

($\rho_t\leqslant 1$ by (9.20)(i)). Summing over the tuples,

$$\sum_{(\ell_Z)}\prod_Z\rho_{\mathrm{ph}}u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z) = \prod_Z\Big[\rho_{\mathrm{ph}}\sum_{\ell_Z:\,Z(\ell_Z)=Z}u_*^{|\mathsf{J}_Z|}\mathfrak{M}^{(1/8)}(\ell_Z)\Big]\leqslant\prod_Z\rho_{\mathrm{ph}}m(Z)$$

by Lemma 9.32(d) (the sum over all split selections dominates the sum over those with $\mathsf{J}_Z\neq\varnothing$). Multiplying by $|w_\Lambda^t(W)|$ gives $|c_W^t(h)|\leqslant M_\Lambda^{\mathrm{ph}}(W)$.

The reference activity $a_W^\Lambda = w_\Lambda^\circ(W)\int g_W^\circ\,\mathrm{d}\bigotimes_Z\hat\nu_{\Lambda,Z}^{f,\circ}$ (Lemma 9.21(b),(d)) is treated by the same four steps at the reference data, where Lemma 9.32 holds (Lemma 9.21(b)): base point 0, increments $|B_b|_\sigma\leqslant r\leqslant 5r/4$, $\mathcal{S}=\varnothing$ throughout (the reference cutoffs are even, so every class-(γ) root is a parity root), g_W° a closed contraction with $\|N\|=1$, and $|w_\Lambda^\circ(W)|\leqslant(1/k(W)!)\prod_j|\Phi|_\Lambda^+(\mathfrak{c}_j)$.

Lemma 9.26(e) improved: $\bar{\mathfrak{d}}_W = \max_t \sup_h |c_W^t(h) - a_W^\Lambda \chi(h)| \leqslant M_\Lambda^{\rm ph}(W) + |a_W^\Lambda| \leqslant 2M_\Lambda^{\rm ph}(W)$ by the two bounds just proved. The pins: the first display of Lemma 9.27(a) is the positive summation with a root weight $\rho\, m(Z)$, and the inherited summation bound (Lemma E.9) holds in the form "sum $\leqslant C_{\rm pin}\rho$ for every $\rho \leqslant \rho_{\rm pin}$" (its smallness conditions on ρ are monotone, so every $\rho \leqslant \rho_{\rm pin}$ satisfies them, and the bound is linear in ρ at each such ρ; monotonicity alone would only give $C_{\rm pin}\rho_t$); applied at $\rho = \rho_{\rm ph} \leqslant \rho_{\rm pin}$ it gives $C_{\rm pin}\rho_{\rm ph}$. Alternatively, $M_\Lambda^{\rm ph}(W) = (\rho_{\rm ph}/\rho_t)^{|\mathcal{A}(W)|} M_\Lambda(W)$ with $|\mathcal{A}(W)| \geqslant 1$ (Definition 3.24, nonempty root set), so the first-order pin at ρ_t gives $C_{\rm pin}\rho_t(\rho_{\rm ph}/\rho_t) = C_{\rm pin}\rho_{\rm ph}$ as well; this argument is independent of the first. The second display uses $\bar{\mathfrak{b}}_W \leqslant 2M_\Lambda(W)$ and $e^{\theta^\circ(W)} \leqslant e^{2\nu\sigma(W)}$, now with $\bar{\mathfrak{b}}_W^{\rm ph} = 2M_\Lambda^{\rm ph}(W)$ and the unchanged $\theta^\circ(W) \leqslant \nu\sigma(W)/16$ (Lemma 9.21(d), which uses the first-order M_Λ, $|a_W^\Lambda| \leqslant M_\Lambda(W)$ and (9.20)(iv), all still in force); Lemma 9.27(b) is the geometric fact applied to the new pin. *q.e.d.*

9.7.4 The Repaired Bounds

Proposition 9.34. *Under the hypotheses of Proposition 9.28 and* (9.29)*, on every labelled term with final region Λ, for exterior data in the support of the term's collar indicators and exterior cutoffs, on $A_{\rm cov}^{\rm fin}$,*

$$|\mathcal{B}_\xi^\Lambda[f_\xi]| \leqslant |S_\Lambda^\circ| e^{n_C \bar{\varepsilon}_{\rm ph}} \big[C_1 g_0 r\, \mathfrak{S}_1(\Lambda) + C_2 \bar{\varepsilon}_t r^{d_G} \mathfrak{S}_2(\Lambda) \big] , \tag{9.30}$$

with C_1, C_2, μ', $\mathfrak{S}_1$, $\mathfrak{S}_2$ of Proposition 9.28; on $A_{\rm cov}^{\rm fin} \cap N_b^c$, (9.28) *holds with $e^{n_C\bar{\varepsilon}_{\rm ph}}$ in place of $e^{2n_C\bar{\varepsilon}_t}$; and, under the hypotheses of Proposition 9.39 (compatible even fine periods exceeding 200ℓ, g_0 below the thresholds of Lemma 9.6, of Proposition 9.8 and of* (9.6)*, and for the closing form its further threshold),* (9.33)*,* (9.34) *and the closing form hold with $e^{n_C\bar{\varepsilon}_{\rm ph}}$ in place of $e^{2n_C\bar{\varepsilon}_t}$. In the square geometry of Section 9.6.1 ($n_C = 4nL$, $4n$ coarse segments) the family-sum factor has, under* (9.29)*(iii), the perimeter rate $L\bar{\varepsilon}_{\rm ph} \leqslant 32LK_\chi C_{\rm pin} C_{\rm ph} g_0^2 (1+r)^4$ per coarse segment, second order in $g_0(1+r)^2$, of the order τ^2 of the bound* (7.34) *on $E_{D,C_n,T}^\circ$; the first-order constant $\bar{\varepsilon}_t$ survives only as the prefactor $C_2\bar{\varepsilon}_t$ of $\mathfrak{S}_2(\Lambda)$ (through Corollary 9.29 it multiplies $n_C N_{\rm lat}(\mu'/2) e^{-\mu'(b-1)/2} + \mathfrak{D}_\Lambda$) and no longer appears in an exponent.*

Proof. The proof of Proposition 9.28 with one change. In (9.25) the undifferenced slots carry $\bar{\mathfrak{d}}_W$, bounded by Lemma 9.33 instead of Lemma 9.26(e); so the two family sums $\prod_W (1 + K_\chi e^{\theta^\circ(W)} \bar{\mathfrak{d}}_W)$, in the background-word part, and over $\mathfrak{B} \setminus \{W_0\}$ in the marked part, are at most $\exp(\sum_W K_\chi e^{\theta^\circ(W)} \bar{\mathfrak{b}}_W^{\rm ph}) \leqslant e^{n_C\bar{\varepsilon}_{\rm ph}}$ (the contour pin of Lemma 9.33 summed over the n_C traversals, every group having a contact with the contour). The differenced slot $\mathfrak{d}_{W_0}^{\rm diff}$ keeps (9.26) with the first-order $M_\Lambda(W_0)$ and the pins of Lemma 9.27 with $\bar{\varepsilon}_t$, so (i) and (ii) of that proof and the constants C_1, C_2 are unchanged, and the closing step $e^{n_C\bar{\varepsilon}_t} \leqslant e^{2n_C\bar{\varepsilon}_t}$ is dropped. Corollary 9.29 and Proposition 9.39 use (9.27) only through its right side. The rate: $n_C\bar{\varepsilon}_{\rm ph} = 4n \cdot L\bar{\varepsilon}_{\rm ph}$, and $\bar{\varepsilon}_{\rm ph} \leqslant 32K_\chi C_{\rm pin} C_{\rm ph} g_0^2 (1+r)^4$ by the display above under (9.29)(iii). *q.e.d.*

Remark 9.35 (Scope of This Subsection). The mechanisms (parity of the kernel, the parameter ζ on the ordinary primitives) are those of Section E.4.1, re-proved here from the display of Lemma 9.14, the factor bounds of Lemma 9.15 and Schwarz's lemma, and not imported. The differenced slots and the marked part keep the first-order prefactor $\bar{\varepsilon}_t$ ((9.26)), harmless because it multiplies the exterior sum and not n_C. The family-sum factor remains $e^{n_C\bar{\varepsilon}_{\rm ph}} \geqslant 1$ and is not uniform in n_C: like $e^{E_{D,C_n,T}^\circ}$ (the bound (7.34) is of order $\tau^2 n$), it may grow with the perimeter at a second-order rate; a bound linear in n_C in place of this exponential factor is not proved here. The inherited inputs of Section 9.6.1 (the pin sums for every $\rho \leqslant \rho_{\rm pin}$,

Lemma E.9; the host comparison of Section E.6; the reference-gas expansion; the tree-graph inequality; the square geometry) remain inherited from the sections cited there and are not re-verified in this subsection.

9.8 The Labelled History Sum

The history sum of Section 3 is the sum over the branch data $\xi = (\Omega_1, R_1, V, V_0, A_{\mathrm{e}})$ inside the outer operation $\mathcal{O}$, each term carrying the weight $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1} = \int_{\mathfrak{g}^I} \rho_\xi \, \mathrm{d}a_{\mathrm{old}}$, with $\rho_\xi \coloneqq e^{c_\xi} K_{\xi;1}$ the marked density (D.32) of Lemma D.13 (the composed retained cutoff $\chi_{\xi,\mathrm{e}}$ and ζ_ξ multiply it), and the conditional Wilson expectation $F_\xi(1) = \mathcal{J}^a_\xi[f_\xi]/\mathcal{J}^a_\xi[1]$ (Lemma 3.9, (3.49)), and the leading term, the defects and the exterior response are read off (3.97). After Section 9.4 every history term is the finite sum of its labelled terms ($\mathrm{lab} = (B^{(0)}, B^{(1)}, \dots)$, Lemma 9.10(a)), with $\sum_{\mathrm{lab}} \mathbf{1}_{\mathrm{lab}} = 1$ pointwise (Section 9.4.2(c)); a labelled term has the final region $\Lambda \coloneqq \Lambda_1^{\mathrm{fin}}(\mathrm{lab})$, carrier $D(\Lambda)$, exterior $J(\Lambda) = J \cup (I \setminus D(\Lambda))$, and the objects of Lemma 9.7 and Sections 9.5–9.7. Write $\mathcal{J}^{a,\mathrm{lab}}_\xi[f] \coloneqq \int_{\mathfrak{g}^I} \rho_\xi \mathbf{1}_{\mathrm{lab}} f \, \mathrm{d}a_{\mathrm{old}}$ for bounded Borel f on $\mathfrak{g}^I$; then $\mathcal{J}^a_\xi[f] = \sum_{\mathrm{lab}} \mathcal{J}^{a,\mathrm{lab}}_\xi[f]$ (Tonelli for $f \geqslant 0$, then by linearity), and the extra resummation index is lab.

Lemma 9.36 (The Labelled Branch Identity)**.** *On every history and every labelled term, with the convention that $e^{c^\Lambda_\xi} I^\Lambda_1 F^\Lambda(1)$ and $e^{c^\Lambda_\xi} I^\Lambda_1$ denote the integrals $\int_{\mathfrak{g}^{D(\Lambda)}} \rho_\xi(a, a_{\mathrm{old}}|_{I\setminus D(\Lambda)}) f_\xi(a) \, \mathrm{d}a$ and $\int_{\mathfrak{g}^{D(\Lambda)}} \rho_\xi(a, a_{\mathrm{old}}|_{I\setminus D(\Lambda)}) \, \mathrm{d}a$ over the retreated fibre (here $f_\xi(a)$ stands for $f_\xi(a, a_{\mathrm{old}}|_{I\setminus D(\Lambda)})$; both integrals vanish where the fibre is empty, i.e. where $|a_{\mathrm{old},b}|_\sigma \geqslant r$ for some $b \in I \setminus D(\Lambda)$, since ρ_ξ carries the box factor $\mathbf{1}_{\mathcal{D}_\xi}$, and also wherever the composed retained cutoff or ζ_ξ vanishes, these being factors of ρ_ξ which multiply both sides of the identity of Lemma 9.7; c^Λ_ξ and $F^\Lambda(1)$ themselves are used only where the fibre is nonempty and these cutoffs are positive),*

$$\mathcal{J}^{a,\mathrm{lab}}_\xi[f_\xi] = \int_{\mathfrak{g}^{I\setminus D(\Lambda)}} \mathbf{1}_{\mathrm{lab}} e^{c^\Lambda_\xi} I^\Lambda_1 F^\Lambda(1) \, \mathrm{d}a_{\mathrm{old}}|_{I\setminus D(\Lambda)} \,, \qquad \mathsf{w}^{\mathrm{lab}}_\xi = \int_{\mathfrak{g}^{I\setminus D(\Lambda)}} \mathbf{1}_{\mathrm{lab}} e^{c^\Lambda_\xi} I^\Lambda_1 \, \mathrm{d}a_{\mathrm{old}}|_{I\setminus D(\Lambda)} \,, \tag{9.31}$$

and, on $A^{\mathrm{fin}}_{\mathrm{cov}}$ (an event depending on the labels through the region only), on the support of $\mathbf{1}_{\mathrm{lab}}$ (which carries the collar condition $|A_{\mathrm{e}}|_\sigma < \vartheta r$ on $J_{\mathrm{col}}(\Lambda)$ required by Section 9.5 and Lemma 9.15; every use of (9.32) below is multiplied by $\mathbf{1}_{\mathrm{lab}}$), where the fibre is nonempty, under the hypotheses of Lemma 9.15 and (9.20) (through Lemma 9.21, whose absolute convergence and reference gas rest on the inputs listed in Section 9.6.1),

$$F^\Lambda(1) = S^\circ_\Lambda W_\chi(C; U_1) + \sum_{\varnothing \neq \mathfrak{B} \ \mathit{compatible}} S^{\circ[\mathfrak{B}]}_\Lambda R^0_\Lambda(\mathfrak{B}) + \mathcal{B}^\Lambda_\xi[f_\xi] \,, \tag{9.32}$$

the retreated carrier's version of (3.97), with S°_Λ, $S^{\circ[\mathfrak{B}]}_\Lambda$, $R^0_\Lambda(\mathfrak{B})$ of Lemma 9.21 and $\mathcal{B}^\Lambda_\xi[f_\xi]$ of Lemma 9.17, bounded by Propositions 9.28 and 9.34 (the latter under (9.29)); here $f_\xi(0) = W_\chi(C; U_1)$ on $A^{\mathrm{fin}}_{\mathrm{cov}}$ because the free inputs of the contour lie in $D(\Lambda)$, where the free vector vanishes at $a = 0$ (proof).

Proof. By Lemma 9.7, pointwise $e^{c^\Lambda_\xi} K^\Lambda_{\xi;1}(a) = \mathbf{1}_{\mathcal{D}^\Lambda}(a) |\det D\Psi_\xi(A_{\mathrm{e}})| \exp \mathscr{E}^\circ_\xi(\Psi_\xi(a + A^\Lambda_{\mathrm{e}})) = \rho_\xi(a_{\mathrm{old}})$ at $a_{\mathrm{old}} = (a, a_{\mathrm{old}}|_{I\setminus D(\Lambda)})$ ($\rho_\xi = e^{c_\xi} K_{\xi;1}$ by definition, (D.32) of Lemma D.13: the right side of (D.32) and that of the identity of Lemma 9.7 are the same function of a_{old} written in the two splits, since $\Psi_\xi(a + A^\Lambda_{\mathrm{e}}) = \Psi_\xi(a_{\mathrm{old}} + A_{\mathrm{e}})$ in the original variables and the cutoff factors are common), and f_ξ, $\mathbf{1}_{\mathrm{lab}}$ are functions of the same variables ($\mathbf{1}_{\mathrm{lab}}$ of A_{e} and $a_{\mathrm{old}}|_{I\setminus D(\Lambda)}$ only); integrate over $a \in \mathfrak{g}^{D(\Lambda)}$ first (Fubini; $\rho_\xi \geqslant 0$, f_ξ and $\mathbf{1}_{\mathrm{lab}}$ bounded):

$\int K^{\Lambda}_{\xi;1} f_\xi \,\mathrm{d}a = I^{\Lambda}_1 F^{\Lambda}(1)$ by the definition of ν^{Λ}_1 in Section 9.5 where the fibre is nonempty (both sides vanish otherwise), and with $f = 1$ the weight. For (9.32): $F^{\Lambda}(1) = F^{\Lambda}(0) + \mathcal{B}^{\Lambda}_{\xi}[f_\xi]$ (Lemma 9.17), $F^{\Lambda}(0) = S^{\circ}_{\Lambda} f_\xi(0) + \sum_{\mathfrak{B}} S^{\circ[\mathfrak{B}]}_{\Lambda} R^0_{\Lambda}(\mathfrak{B})$ (Lemma 9.21(c) at $t = 0$), and $f_\xi(0) = W_\chi(C; U_1)$ on $A^{\mathrm{fin}}_{\mathrm{cov}}$: at $a = 0$ the free vector is (0 on $D(\Lambda)$, $A^{\Lambda,\Psi}$ on $J(\Lambda)$), every contour bond is a bond of Ω_1 and its free inputs lie in $D(\Lambda)$, so the argument of Lemma 3.28 with I replaced by $D(\Lambda)$ gives $U_\xi(0) = U_1$ on the contour (the de-younged bonds, though nonzero, are not inputs of the contour on $A^{\mathrm{fin}}_{\mathrm{cov}}$). *q.e.d.*

9.8.1 The Species and Their Locality

The species of the labelled term are the final domains Z for the carrier $D(\Lambda)$ with their activities $k^{f,t}_{\Lambda}(Z)$ and probe measures $\nu^{f,t}_{\Lambda,Z}$ (Lemmas 9.14, 9.20), the groups W with their weights $w^t_{\Lambda}(W)$ and reference activities a^{Λ}_W (Lemma 9.21), and the marked-polymer objects built from them (the pins of Lemma 9.27). They depend on the labels through $D(\Lambda)$ only. Let $\mathcal{N}_{\mathrm{lab}}$ be the set of LM-cubes meeting a bond of $I \setminus D(\Lambda)$ (no site lies on a cube face, so a bond meets exactly the one or two LM-cubes containing its endpoints, and $|\mathcal{N}_{\mathrm{lab}}| \leqslant 2|I \setminus D(\Lambda)|$).

Lemma 9.37 (Locality of the Species)**.** *Under the hypotheses of Lemma 9.20 (Lemma 9.15 and (9.20)), let Z be a union of LM-cubes containing no cube of $\mathcal{N}_{\mathrm{lab}}$, equivalently: every bond of I with an endpoint in Z lies in $D(\Lambda)$. Then for every selection and split selection with final domain Z, the selected integrals at the carrier $D(\Lambda)$ and at the carrier I coincide at $t = 0$: $k^{f,0}_{\Lambda}(Z) = k^{f}_{\xi}(Z)$, $\nu^{f,0}_{\Lambda,Z} = \nu^{f}_{\xi,Z}$, and likewise for the reference data ($k^{f,\circ}_{\Lambda}(Z) = k^{f,\circ}_{\xi}(Z)$, $\hat{\nu}^{f,\circ}_{\Lambda,Z} = \hat{\nu}^{f,\circ}_{\xi,Z}$). Consequently, for a group W none of whose roots and cluster constituents contains a cube of $\mathcal{N}_{\mathrm{lab}}$ (W is then a group for both carriers: the probe label set L_Z of each of its domains is the same at both carriers, since $Y_b \subseteq Z$ gives $b \in E(Y_b) \subseteq E(Z) \subseteq D(\Lambda)$, and its contact sets and hulls are determined by its domains and the contour), $w^0_{\Lambda}(W) = w_\xi(W)$, $a^{\Lambda}_W = a_W$, and on $A^{\mathrm{fin}}_{\mathrm{cov}}$ the pairings $\mathfrak{Q}^0_{\Lambda}(\mathcal{A}(\mathcal{S}); f_\xi) = \mathfrak{Q}_{\xi}(\mathcal{A}(\mathcal{S}); f_\xi)$ for families $\mathcal{S}$ of such groups, so $A^0_{\Lambda}(\mathcal{S}) = A_\xi(\mathcal{S})$ and $R^0_{\Lambda}(\mathcal{S}) = R_\xi(\mathcal{S})$. A species that differs between the two carriers has a root or constituent containing a cube of $\mathcal{N}_{\mathrm{lab}}$.*

Proof. Fix a selection $\mathfrak{s} = (\mathcal{D}, \mathsf{P}, Z)$ (or a split selection) at the carrier I with final domain Z; it is a selection at the carrier $D(\Lambda)$ and conversely: the hosts $Y \in \mathcal{D}$ lie in Z ($Y \subseteq Z_0 \subseteq Z'_0 \subseteq Z$), their primitives have free-input sets in Y, whose young part lies in $E(Y) = \{b \in \text{carrier} : b \cap Y \neq \varnothing\}$, the same set for both carriers by hypothesis, so a primitive assigned to Y is near relative to I exactly when it is near relative to $D(\Lambda)$, and $\mathbf{D}_0(\Lambda)$-hosts inside Z are the $\mathbf{D}_0$-hosts inside Z (probe hosts $Y_b \subseteq Z$ have $b \in E(Y_b)$, the same set); the bad bonds P meet Z (their incident cubes are in $Z_0 \subseteq Z$, Definition 3.17), so $\mathsf{P} \subseteq D(\Lambda) \setminus E(Y_0)$ and $\mathsf{P} \subseteq I \setminus E(Y_0)$ are the same condition; the seed and Z'_0 are formed from $\mathcal{D}$, P alone, and $E_0 = E(Z_0)$ (Section 3.5) is the same set for both carriers by hypothesis. The integrand of (3.68) at $t = 0$: the potentials $\mathbf{W}_0(Y, B) = \sum_{\mathfrak{p}\,:\, Y_{\mathfrak{p}} = Y} \mathfrak{p}(B)$ (exterior input zero in both splits) and the cutoffs are the same functions of $B|_{E_0}$; $C_{E_0} = (\iota^*_{E_0} H \iota_{E_0})^{-1}$ involves the entries of H between bonds of E_0, which are in both carriers; $\Gamma_{E_0}(\sigma) = \iota^*_{E_0} H \Pi_{\mathrm{carrier} \setminus E_0} S_{Z_0}(\sigma)$ involves, for a row $e \in E_0$, the entries $H_{eb'}$ with $d(e_-, b'_-) \leqslant R$, so $b' \subset \operatorname{int} D \subset Z$ for the component D of e (proof of Lemma 3.15), and the walk series $S_{Z_0}(\sigma)_{b'b}$ with $\sigma_\Delta = 0$ off Z, whose nonvanishing walks have $D(\omega) \subset Z$ (every cube meeting $D(\omega)$ is in Z, by the weight (3.61)); the blocks $(\mathsf{A}^r_H)_{b_0 b_1}$, $(\mathsf{P}_H)_{b_j b_{j+1}}$ of a walk are entries of powers of $\mathbf{1} - H/s_H$ restricted to the carrier, and each depends only on the carrier's H-coefficients within distance $(n+1)R$ of its first bond, which lie in $D(\omega) \subset Z$ by the interior margin of D_b (Section 3.5: D_b contains the bonds and coefficient stencils of every block starting at b with an interior margin); hence every bond

entering $S_{Z_0}(\sigma)$ and $\Gamma_{E_0}(\sigma)$ is a bond of I meeting Z, in $D(\Lambda)$ by hypothesis, and every coefficient is an entry of H between two such bonds, the same for both carriers (principal restrictions of the one Hessian H_ξ), so the two carriers give the same $\Gamma_{E_0}(\sigma)$ and the same C_{E_0}; the X-integral is over $\mathfrak{g}^{\text{carrier}}$ with the standard Gaussian, but $\Gamma_{E_0}(\sigma)X$ involves X only on bonds of Z (the columns of Γ outside $\bigsqcup J_D$ vanish, Lemma 3.15), so the remaining coordinates integrate to one in both cases. The Dirac tensors of a split selection are the same. Hence $\mathcal{A}^{f,0}_{\Lambda;\mathfrak{s}}[F] = \mathcal{A}^{f}_{\xi;\mathfrak{s}}[F]$ for every F, giving the activities and measures; the reference case is the same argument with H°. The group statements follow: $w^0_\Lambda(W)$ is the product of the activities of its constituents with the combinatorial factors of its structure (the Ursell coefficients of its cluster constituents and $1/k(W)!$), which depend on the incompatibility graph only, a^Λ_W an integral of the reference source word (the same function, the reference links depending on the root variables only) against the same measures, and $\mathfrak{Q}$ involves $f_\xi(\mathsf{e}_{\mathcal{A}}B)$, which on $A^{\text{fin}}_{\text{cov}}$ depends on $B|_{I_C}$ with $I_C \subseteq D(\Lambda)$ and is the same function for both carriers. *q.e.d.*

Lemma 9.38 (The Reference Gas at the Retreated Carrier)**.** *Under* (9.20)*(iv),* $S^\circ_\Lambda = S^\circ_{\xi,C} \cdot \exp(\eta_{\text{lab}})$ *with*

$$|\eta_{\text{lab}}| \leqslant \sum_{\Gamma \text{ containing a group } W \text{ with } F(W)\cap\mathcal{N}_{\text{lab}}\neq\varnothing} \big(|\Phi^\circ_\Lambda(\Gamma)| + |\Phi^\circ_\xi(\Gamma)|\big)$$
$$\leqslant \sum_{\Delta\in\mathcal{N}_{\text{lab}}} \sum_{\Gamma \text{ hits } \Delta} \big(|\Phi^\circ_\Lambda(\Gamma)| + |\Phi^\circ_\xi(\Gamma)|\big) \leqslant 2|I \setminus D(\Lambda)| \cdot \varepsilon^\circ_g \,,$$

where

$$\varepsilon^\circ_g := \sup_\Delta \sum_{\Gamma \text{ hits } \Delta} (|\Phi^\circ_\Lambda(\Gamma)| + |\Phi^\circ_\xi(\Gamma)|) \leqslant 2C_{\text{pin}}\rho_t :$$

a cluster Γ *hits the cube* Δ *when one of its groups* W *has* $\Delta \in F(W)$*; the rooted bound of Lemma 9.21(d),* $\sum_{\Gamma\ni W}|\Phi^\circ(\Gamma)| \leqslant |a_W|e^{13\nu\sigma(W)/8}$*, gives* $\sum_{\Gamma \text{ hits } \Delta}|\Phi^\circ(\Gamma)| \leqslant \sum_{W:\Delta\in F(W)}|a_W|e^{13\nu\sigma(W)/8} \leqslant \sum_{W:\Delta\in F(W)} e^{4\nu\sigma(W)}M_\Lambda(W) \leqslant C_{\text{pin}}\rho_t$ *by* $|a_W| \leqslant M_\Lambda(W)$ *and the first display of Lemma 9.27(a), at the carrier* $D(\Lambda)$ *and, by the closing remarks of Lemma 9.21(d) and Lemma 9.27(a), at the carrier* I*. Moreover* $|S^\circ_\Lambda| \leqslant 1$*.*

Proof. Under (9.20)(iv) the reference group gas satisfies the Kotecký–Preiss condition at both carriers (Lemma 9.21(d); the reference group gas at I is that of Section 3), so its cluster expansion (Appendix B.6) gives $\log S^\circ = \sum_\Gamma \Phi^\circ(\Gamma)$, absolutely convergent, at both carriers, and $\log S^\circ_\Lambda - \log S^\circ_{\xi,C} = \sum_\Gamma[\Phi^\circ_\Lambda(\Gamma) - \Phi^\circ_\xi(\Gamma)]$; a cluster all of whose groups have roots and constituents containing no cube of $\mathcal{N}_{\text{lab}}$ has $\Phi^\circ_\Lambda(\Gamma) = \Phi^\circ_\xi(\Gamma)$ by Lemma 9.37 (the Ursell coefficient depends on the incompatibility graph only, the group activities coincide), and the groups present for one carrier and not the other contain a cube of $\mathcal{N}_{\text{lab}}$ as well (a root or constituent met by a de-younged bond). $|S^\circ_\Lambda| \leqslant 1$: $S^\circ_\Lambda = \int f^\circ K^\circ_\Lambda \,\mathrm{d}a/I^\circ_\Lambda$ with $K^\circ_\Lambda > 0$ on the box (Lemma 9.13(c) for the reference data) and $|f^\circ| \leqslant 1$ (χ is the normalised character, $|\chi| \leqslant \chi(\mathbf{1}) = 1$). *q.e.d.*

9.8.2 The Accounting

Three quantities acquire the index lab: the history weight ($\mathsf{w}_\xi = \sum_{\text{lab}} \mathsf{w}^{\text{lab}}_\xi$, (9.31)), the exterior-response/uncovered term E_{bdry}, and the defect term E_{def}. Define

$$E^{\text{lab}}_{\text{bdry}} := \mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\sum_{\text{lab}} \int_{\mathfrak{g}^{I\setminus D(\Lambda)}} \mathbf{1}_{\text{lab}} e^{c^\Lambda_\xi} I^\Lambda_1\big(\mathcal{B}^\Lambda_\xi[f_\xi]\mathbf{1}_{A^{\text{fin}}_{\text{cov}}} + F^\Lambda(1)\mathbf{1}_{(A^{\text{fin}}_{\text{cov}})^c}\big)\mathrm{d}a_{\text{old}}\Big](V)\,\mathrm{d}V \,,$$

the labelled version of the E_{bdry} of Appendix D (the exterior response on covered terms, the whole conditional expectation on uncovered terms, $|F^{\Lambda}(1)| \leqslant 1$; the integrands are read with the convention of Lemma 9.36, as $\int_{\mathfrak{g}^{D(\Lambda)}} \rho_{\xi}(\dots)\,\mathrm{d}a$, so they are defined and vanish where the fibre is empty), and $E^{\mathrm{lab}}_{\mathrm{def}}$ the same with $\sum_{\mathfrak{B}} S^{\circ[\mathfrak{B}]}_{\Lambda} R^{0}_{\Lambda}(\mathfrak{B}) \mathbf{1}_{A^{\mathrm{fin}}_{\mathrm{cov}}}$ in place of the bracket; the leading term is $\mathcal{Z}_0^{-1} \int \mathcal{O}[\sum_{\mathrm{lab}} \int \mathbf{1}_{\mathrm{lab}} e^{c^{\Lambda}_{\xi}} I^{\Lambda}_{1} S^{\circ}_{\Lambda} \mathbf{1}_{A^{\mathrm{fin}}_{\mathrm{cov}}}\,\mathrm{d}a_{\mathrm{old}} \cdot W_{\chi}(C;U_1)]\,\mathrm{d}V$. By (9.31)–(9.32) these three sum to the labelled Wilson numerator, exactly.

Proposition 9.39 (The Boundary Accounting with Labels)**.** *On compatible even fine periods exceeding 200ℓ, under the hypotheses of Proposition 9.28, for g_0 below the thresholds of Lemma 9.6, of Proposition 9.8 ($g_*(\vartheta)$) and of* (9.6)*, with q, q_ϑ, N_{ret}, C_L of Sections 9.3–9.4 $\bar{\varepsilon}_t$, C_1, C_2, μ' of Section 9.6, and $N_{\mathrm{lat}}(\mu') = \sum_{s\geqslant 0} 4(2s+1)^4 e^{-\mu' s}$ of Lemma 9.2 (the labelled integrands S°_{Λ}, $F^{\Lambda}(t)$, $\mathcal{B}^{\Lambda}_{\xi}[f_\xi]$ are Borel in the branch data: the labels take finitely many values, and on each labelled term these are ratios and differences of the Borel integrals of the measurability paragraph of Section 3):*

$$\begin{aligned} &\mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\sum_{\mathrm{lab}} \int_{\mathfrak{g}^{I\setminus D(\Lambda)}} \mathbf{1}_{\mathrm{lab}} e^{c^{\Lambda}_{\xi}} I^{\Lambda}_{1} \mathbf{1}_{A^{\mathrm{fin}}_{\mathrm{cov}}} \frac{|\mathcal{B}^{\Lambda}_{\xi}[f_\xi]|}{|S^{\circ}_{\Lambda}|}\,\mathrm{d}a_{\mathrm{old}}\Big]\mathrm{d}V \\ &\leqslant e^{2n_C\bar{\varepsilon}_t}\big(C_1 g_0 r + C_2 \bar{\varepsilon}_t r^{d_G}\big)\big(1+e^{\mu_H(2L+1)/2}\big) N_{\mathrm{lat}}(\mu')\, n_C (q + N_{\mathrm{ret}} q_\vartheta)\,, \qquad (9.33) \\ &\mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\sum_{\mathrm{lab}} \mathsf{w}^{\mathrm{lab}}_{\xi} \mathbf{1}_{(A^{\mathrm{fin}}_{\mathrm{cov}})^c}\Big]\mathrm{d}V \leqslant C_L n_C (q + N_{\mathrm{ret}} q_\vartheta) \quad \text{((9.8))}, \end{aligned}$$

hence, with $|S^{\circ}_{\Lambda}| \leqslant 1$ (Lemma 9.38),

$$|E^{\mathrm{lab}}_{\mathrm{bdry}}| \leqslant n_C(q + N_{\mathrm{ret}} q_\vartheta)\Big[e^{2n_C\bar{\varepsilon}_t}\big(C_1 g_0 r + C_2 \bar{\varepsilon}_t r^{d_G}\big)\big(1+e^{\mu_H(2L+1)/2}\big) N_{\mathrm{lat}}(\mu') + C_L\Big]\,, \quad (9.34)$$

and for g_0 below a further threshold depending on L, M, M_2, the group, the source constants, ϑ and the character, $r^{d_G} N_{\mathrm{ret}}(q + N_{\mathrm{ret}} q_\vartheta) \leqslant e^{-c_q \vartheta^2 \pi_*^2/2}$ (the factor N_{ret} is not needed for (9.34)*; it costs only a logarithm, proof, and is kept for the form of the statement), so that, with $g_0 \leqslant 1 \leqslant r$, $d_G \geqslant 1$ and $\bar{\varepsilon}_t \leqslant 1$ ((9.20)(v)), $C_1 g_0 r + C_2 \bar{\varepsilon}_t r^{d_G} \leqslant (C_1+C_2) r^{d_G}$ and $|E^{\mathrm{lab}}_{\mathrm{bdry}}| \leqslant C n_C e^{2n_C\bar{\varepsilon}_t} e^{-c_q\vartheta^2\pi_*^2/2}$ with $C = (C_1+C_2)(1+e^{\mu_H(2L+1)/2}) N_{\mathrm{lat}}(\mu') + C_L$: the form $|E_{\mathrm{bdry}}| \leqslant C n_C (1+g_0 r) e^{-c_* \pi_*^2}$ of Appendix D with the exponent $c_q\vartheta^2/2$ in place of c_* and the family-sum factor $e^{2n_C\bar{\varepsilon}_t}$ of* (9.27) *(it is not a history weight; Proposition 9.34 replaces it by $e^{n_C \bar{\varepsilon}_{\mathrm{ph}}}$ under* (9.29)*). Both perimeter factors grow with n_C: the bound is a statement about the fixed loop C as $g_0 \to 0$, the factor $e^{2n_C\bar{\varepsilon}_t} \leqslant e^{2n_C}$ being a constant of that loop.*

Proof. (9.33): on $A^{\mathrm{fin}}_{\mathrm{cov}}$, (9.27) gives $|\mathcal{B}^{\Lambda}|/|S^{\circ}_{\Lambda}| \leqslant e^{2n_C\bar{\varepsilon}_t}[C_1 g_0 r \mathfrak{S}_1(\Lambda) + C_2 \bar{\varepsilon}_t r^{d_G} \mathfrak{S}_2(\Lambda)]$ with $\mathfrak{S}_1(\Lambda) \leqslant e^{\mu_H(2L+1)/2}\mathfrak{S}_2(\Lambda)$ (Corollary 9.29), and $\mathfrak{S}_2(\Lambda) = \sum_{e\in J(\Lambda)} e^{-\mu'\,\mathrm{dist}(e,C)} \leqslant \sum_e e^{-\mu'\,\mathrm{dist}(e,C)} \mathbf{1}\{e \in E(B^{\mathrm{fin}}_{\xi})\}$, $B^{\mathrm{fin}}_{\xi} := (\Lambda^{\mathrm{fin}}_1)^c$, because a bond not in $D(\Lambda)$ has no endpoint in Λ^{fin}_1 (Section 9.4.2(a)). The prefactor is a constant (independent of the history, the labels and the fields: (9.27) is uniform in the exterior data within the term's support), and $\mathfrak{S}_2(\Lambda)$, S°_{Λ} and $A^{\mathrm{fin}}_{\mathrm{cov}}$ depend on the labels through Λ only, so the integral over $a_{\mathrm{old}}|_{I\setminus D(\Lambda)}$ of $\mathbf{1}_{\mathrm{lab}} e^{c^{\Lambda}_{\xi}} I^{\Lambda}_{1} \mathbf{1}_{A^{\mathrm{fin}}_{\mathrm{cov}}} |\mathcal{B}^{\Lambda}|/|S^{\circ}_{\Lambda}|$ is at most (prefactor)$\cdot \mathfrak{S}_2(\Lambda) \cdot \mathsf{w}^{\mathrm{lab}}_{\xi} \mathbf{1}_{A^{\mathrm{fin}}_{\mathrm{cov}}}$. By (9.7), $\mathbf{1}\{e \in E(B^{\mathrm{fin}}_{\xi})\} \leqslant \mathbf{1}\{e \in E(\mathfrak{B}_0)\} + \sum_{e' : x_{e'} \text{ within } 102\ell+1 \text{ of an endpoint of } e} M_{e'}$ (Lemma 9.10(c)–(d)); multiply by $\mathsf{w}^{\mathrm{lab}}_{\xi} \geqslant 0$, sum over the labels and histories, apply $\mathcal{O}$, divide by $\mathcal{Z}_0$ (Tonelli, finitely many labels per history): the first term gives $p^{(0)}_e \leqslant q$ (Lemma 9.9(b)), the second $\sum_{e'} p^{\vartheta}_{e'} \leqslant N_{\mathrm{ret}} q_\vartheta$ (Lemma 9.10(d), Proposition 9.8). Then $\sum_e e^{-\mu'\,\mathrm{dist}(e,C)} \leqslant n_C N_{\mathrm{lat}}(\mu')$ (at most $4n_C(2s+1)^4$ bonds at ℓ^∞ distance $\leqslant s$ from the contour's initial sites, as in Corollary 9.29). The second display is (9.8). (9.34) adds the two with $|\mathcal{B}^{\Lambda}| \leqslant |S^{\circ}_{\Lambda}| \cdot (\dots)$ and

$|F^\Lambda(1)| \leqslant 1$. The comparison: $\log(r^{d_G} N_{\mathrm{ret}}) \leqslant d_G(\log A_1 + p_0 \log\log g_0^{-2}) + C + 4\mathfrak{r}\log(2\log g_0^{-2})$ by (9.9) and $r = A_1(\log g_0^{-2})^{p_0}$, while $\log(1/q_\vartheta) \geqslant c_q \vartheta^2 \pi_*^2 \geqslant c_q \vartheta^2 A_0^2 2^{-2p_0} (\log g_0^{-2})^{2p_0}$ by (9.6) and (9.10), and $q \leqslant q_\vartheta$ (proof of Lemma 9.6: $\theta_\vartheta \leqslant \theta$ for $\vartheta \leqslant 1$, and z_P, z_G are decreasing in θ at fixed n_W), so $q + N_{\mathrm{ret}} q_\vartheta \leqslant (1 + N_{\mathrm{ret}}) q_\vartheta$ (alternatively, without comparing q and q_ϑ: Proposition D.15 gives $q \leqslant e^{-c_* \pi_*^2} = e^{-c_q \pi_*^2} \leqslant e^{-c_q \vartheta^2 \pi_*^2}$ for $g_0 < g_*$ ((9.6) at $\vartheta = 1$), and $q_\vartheta \leqslant e^{-c_q \vartheta^2 \pi_*^2}$ by (9.6) itself, so $q + N_{\mathrm{ret}} q_\vartheta \leqslant (1 + N_{\mathrm{ret}}) e^{-c_q \vartheta^2 \pi_*^2}$, which serves equally); every logarithm is dominated for small g_0, leaving $e^{-c_q \vartheta^2 \pi_*^2 / 2}$. The polynomial prefactor r^{d_G} and the factor N_{ret} in front are thus paid by the same log comparison ($\log r = O(\log\log g_0^{-2})$, $\log N_{\mathrm{ret}} = O(\log\log g_0^{-2})$ by (9.9)); they are prefactors, not the cost in the exponent growing with r that the comparison of Section 9.4.2 set aside. *q.e.d.*

9.8.3 The Labelled Defect Term

Proposition 9.40 (The Labelled Defect Bound)**.** *Under the hypotheses of Proposition 9.39,* (9.20) *and the hypotheses of Lemma 9.15 (under which Corollary 9.16 and Lemma 9.20 are stated), and those of Proposition E.22 (the hypotheses stated at the start of Section E.11, g_0 below the thresholds of Lemmas E.1, E.9 and E.10, $R_\Upsilon \geqslant 2$ and $u \leqslant u_*$, $v \leqslant 1$ of Lemma E.3, and the small-coupling conditions of Sections E.9.1 and E.10.5, the joint radii τ_{b}, θ_0 and the source specifications used in Section E.11), with ρ of* (E.11) *satisfying $\rho \leqslant \rho_{\mathrm{pin}}$ (a condition on g_0, since $\rho = O(g_0^2(1+r)^4)$), $\varepsilon_1 = g_1 p_0(g_1)$, $b = \lfloor LMR_1/16 \rfloor$, $\pi_* = \min(p_0(g_0), p_0(g_1))$, the event N_b of* (E.2) *and the constants C_L', c_* of* (E.3) *and C_χ, c of* (E.24)*, write $\mathscr{R}^\Lambda := \sum_{\varnothing \neq \mathfrak{B} \text{ compatible}} S_\Lambda^{\circ[\mathfrak{B}]} R_\Lambda^0(\mathfrak{B})$ for the labelled defect of* (9.32)*.*

(a) On every labelled term with final region Λ, for exterior data in the support of the term's collar indicators and exterior cutoffs, $\mathscr{R}^\Lambda = F^\Lambda(0) - S_\Lambda^\circ f_\xi(0)$ satisfies $|\mathscr{R}^\Lambda| \leqslant 2$, and on $A_{\mathrm{cov}}^{\mathrm{fin}}$ it depends on the term through $D(\Lambda)$ and on the history through U_1, Ω_1, Λ_1 only (functions of V, V_0 and the regions), not on $a_{\mathrm{old}}|_{I \setminus D(\Lambda)}$ or A_{e}; of this, only the independence from $a_{\mathrm{old}}|_{I \setminus D(\Lambda)}$ is used, in (b). On $A_{\mathrm{cov}}^{\mathrm{fin}} \cap \mathsf{N}_b^c$,

$$|\mathscr{R}^\Lambda| \leqslant |S_\Lambda^\circ| \big(e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1 \big) \,. \tag{9.35}$$

(b) Consequently

$$\begin{aligned} |E_{\mathrm{def}}^{\mathrm{lab}}| &\leqslant C_L' n_C e^{-(c_*/2)\pi_*^2} + \mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}} \cap \mathsf{N}_b^c} |S_\Lambda^\circ| \\ &\qquad \cdot \big(e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1 \big) \Big] \mathrm{d}V \\ &\leqslant C_L' n_C e^{-(c_*/2)\pi_*^2} + e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1 \,, \end{aligned} \tag{9.36}$$

the bounds (E.78)–(E.79) *of Proposition E.22 with the same constants; S_Λ° is retained and not identified here.*

Proof. (a) By Lemma 9.21(c) at exterior parameter $t = 0$, $F^\Lambda(0) = S_\Lambda^\circ f_\xi(0) + \mathscr{R}^\Lambda$, with $\mathscr{R}^\Lambda$ the defect term of (c), the series over the countable index set of nonempty compatible families (Lemma 9.21(a) gives the absolute convergence of the series $\sum_{\mathfrak{B}} A^t(\mathfrak{B})$ and (c) the identity; the absolute summability of the series defining $\mathscr{R}^\Lambda$ on $A_{\mathrm{cov}}^{\mathrm{fin}} \cap \mathsf{N}_b^c$ is established by the termwise bounds of steps (ii)–(iii) below, the only place the series is used, and on the whole support of the exterior cutoffs it follows from Lemma 9.26(a),(b) with Lemma 9.21(d) and Lemma 9.27(a); elsewhere only the closed form $F^\Lambda(0) - S_\Lambda^\circ f_\xi(0)$ and the bound $|\mathscr{R}^\Lambda| \leqslant 2$

enter); the inequality $\sum_{\mathfrak{B}\neq\varnothing}\prod_{W\in\mathfrak{B}}x_W\leqslant\exp(\sum_W x_W)-1$ used below holds for countable sums. At $t=0$ the kernel $K^{\Lambda}_{\xi;0}$ of Section 9.5 has no linear term and its primitives are evaluated at a alone (exterior variable zero), so, on $A^{\mathrm{fin}}_{\mathrm{cov}}$, ν^{Λ}_0 and $F^{\Lambda}(0)$ depend on U_1, Ω_1 and $D(\Lambda)$ only (its box indicator is the product cutoff Θ over $D(\Lambda)$ on the support of the exterior cutoffs, Lemma 3.11(iv) and Lemma 9.12(iv)); S°_{Λ} is built from the reference data at $D(\Lambda)$ (Lemma 9.21(b)); and on $A^{\mathrm{fin}}_{\mathrm{cov}}$ every contour link is reconstructed from $a|_{D(\Lambda)}$ at the background U_1 (preamble of Lemma 9.25: the free inputs of the contour lie in $D(\Lambda)$, and the de-younged bonds are not inputs), so f_ξ and $f_\xi(0)=W_\chi(C;U_1)$ (Lemma 9.36) involve, besides the branch datum U_1, neither $a_{\mathrm{old}}|_{I\setminus D(\Lambda)}$ nor A_{e}. $|\mathscr{R}^{\Lambda}|\leqslant 2$: ν^{Λ}_0 is a probability measure ($K^{\Lambda}_{\xi;0}>0$ on the open box and $I^{\Lambda}_0>0$, Lemma 9.13(c)) and $|f_\xi|\leqslant 1$, so $|F^{\Lambda}(0)|\leqslant 1$; $|S^{\circ}_{\Lambda}|\leqslant 1$ (Lemma 9.38) and $|f_\xi(0)|\leqslant 1$.

For (9.35) the three steps of the proof of Lemma E.12 are taken at the carrier $D(\Lambda)$. (i) Exclusion ratios: by Lemma 9.21(d), under (9.20)(iv), $S^{\circ}_{\Lambda}\neq 0$ and $|S^{\circ[\mathfrak{B}]}_{\Lambda}/S^{\circ}_{\Lambda}|=|q^{\circ}_{\Lambda}(\mathfrak{B})|\leqslant\prod_{W\in\mathfrak{B}}e^{\theta^{\circ}(W)}$ with $\theta^{\circ}(W)\leqslant\nu\sigma(W)/16\leqslant\eta\mathfrak{s}(W)/8$ ($\nu=\eta$ and $\sigma(W)=\mathfrak{s}(W)$, Lemma 9.27), the retreated carrier's form of (E.23). (ii) Factorisation: by Lemma 9.26(b) at exterior parameter zero, $R^0_{\Lambda}(\mathfrak{B})=\dim(\pi)^{-1}\operatorname{tr}(\Upsilon^0_{W_1}U^0_1\cdots\Upsilon^0_{W_n}U^0_n)$, and the elementary trace bound $|\dim(\pi)^{-1}\operatorname{tr}(X_1\cdots X_m)|\leqslant\prod_j\|X_j\|$ (as in the proof of Lemma 9.26(c)) with $\|U^0_j\|=1$ and $\|\Upsilon^0_W\|\leqslant K_\chi\mathfrak{d}^0_W$ (Lemma 9.26(a)), $\mathfrak{d}^0_W:=\sup_{h\in G}|c^0_W(h)-a^{\Lambda}_W\chi(h)|$ (the $\mathfrak{d}_W(U_1)$ of (E.19) at the carrier $D(\Lambda)$; the superscript 0 marks the exterior parameter zero at the physical background U_1, as in c^0_W, whereas the quantities of the coefficient path below carry its parameter t), gives $|R^0_{\Lambda}(\mathfrak{B})|\leqslant\prod_{W\in\mathfrak{B}}K_\chi\mathfrak{d}^0_W$; hence, families being sets of distinct groups,

$$|\mathscr{R}^{\Lambda}|\leqslant|S^{\circ}_{\Lambda}|\sum_{\varnothing\neq\mathfrak{B}}\prod_{W\in\mathfrak{B}}K_\chi e^{\theta^{\circ}(W)}\mathfrak{d}^0_W\leqslant|S^{\circ}_{\Lambda}|\Big[\exp\Big(\sum_W K_\chi e^{\theta^{\circ}(W)}\mathfrak{d}^0_W\Big)-1\Big]\,.$$

(iii) The one-group sum: on $A^{\mathrm{fin}}_{\mathrm{cov}}\cap\mathsf{N}^c_b$,

$$\sum_W K_\chi e^{\theta^{\circ}(W)}\mathfrak{d}^0_W\leqslant C_\chi n_C\rho\big(\varepsilon_1+e^{-cb/(LM)}\big)\,,\tag{9.37}$$

Lemma E.11 at $D(\Lambda)$. Its proof (Section E.11.2) rests on (E.74) for the small interior groups (E.43), on (E.75) for the others and on the weighted pins; the carrier enters these only through the following inputs, each supplied at $D(\Lambda)$.

Hessian along the coefficient path. For a small interior group with comparison rectangle Q_W and path $U_{W,t}=e^{itB_W}$ (Section E.8.1) put $H_t:=(\mathsf{H}^{\Omega_1}_{U_{W,t}})_{D(\Lambda)D(\Lambda)}$. Lemma E.15 holds for this compression, a compression of the I-compression of Section E.8.1 since $D(\Lambda)\subseteq I$ (Lemma 9.7): the two-sided bound passes to every compression by the min–max principle, the flat lower bound γ_L included, and the Schur-test bound $c_L\tau_{\mathrm{b}}$ on the entry differences holds for a principal submatrix, whose row and column sums are at most those of the full matrix; so $\frac34\gamma_L\leqslant H_t\leqslant m_L$ and the R-locality of the entries (Lemma 3.3) hold for every $t\in[0,1]$. This, with the Schur bound c_H on the row and column norms of the entry derivatives $\mathscr{D}_\omega H_t/\|\omega\|_\infty$ (Section E.9.2, supplied by the finite analytic entries of Section E.9.1, which the principal submatrix inherits), is all that Sections E.9.2 and E.9.3 use of the background: the differentiated walk series (E.54), the derivatives of C_{E_0} and Γ_{E_0} (built from H_t with $\Pi_{D(\Lambda)\setminus E_0}$, as in Lemma 9.14) and the Gaussian moment payment (E.58), whose inputs $H_E-\alpha_G\geqslant\gamma/2$, $\|T_\alpha\|\leqslant 1/4$ and $\|\Gamma_1\|\leqslant A_*$ are those of Lemma 3.16 for H_t, supplied by its proof in Appendix B.4 ((B.24); that proof uses only (3.56), the two-sided bound with $\gamma=\gamma_L/2\leqslant\frac34\gamma_L$, $h_+=m_L$ as in Lemma 9.13(a), the range and the partitions), and whose $d\leqslant 8d_G(LM+1)^4N_1(Z)$ is a count over the partitions.

Charts and primitives. Throughout this proof the exterior parameter is zero; the parameter $t \in [0,1]$ of this and the preceding paragraph is that of the coefficient path $U_{W,t}$. Section E.9.1 is stated per stencil type; at the rows meeting $D(\Lambda)$ it applies because the joint chart condition $\|Z\| < 2\tau_{\mathrm{b}}$ holds along the path $U_{W,t}$ at every such row for every $t \in [0,1]$, by the phase test (E.65): $\|tB_W\|_\infty \leqslant \tau_{\mathrm{b}}/2$ on Q_W and $B_W = 0$ off Q_W (the path runs from the identity to e^{iB_W}, not through U_1), so that the condition is that of the coefficient path at I (Section E.8.1); with zero exterior variable ($s = 1 + r \leqslant s^\Lambda$); the hosts are those of $\mathbf{D}_0(\Lambda)$ (Lemma 9.13(d)), defined from the lifted stencils and M alone, with the counts of Lemma B.10. Hence the parts of Lemmas 9.13–9.15, Corollary 9.16 and Lemma 9.20(a)–(c) used along the path, namely the selected-integral representation and its finiteness, the activity bound $|k| \leqslant m$ and the hitting-sum bound of Corollary 9.16, and the majorants of Lemma 9.20(a)–(c), hold at exterior parameter zero at the background $U_{W,t}$ for every $t \in [0,1]$ with the same constants (the normaliser statements, $K^\Lambda > 0$ and $I^\Lambda > 0$ of Lemma 9.13(c), the identity of Lemma 9.14 for $\Xi_\Lambda^{f,t}$, the ratio identity of Corollary 9.16 and part (d) of Lemma 9.20, which divide by $\Xi_\Lambda^{f,t}$ or concern the physical kernel, are not used along the path): their proofs use the background only through Lemma 9.13(a) (the Hessian bounds and range, supplied for H_t by the preceding paragraph), (c) (the row charts and primitive bounds, supplied along the path by Section E.9.1 at these rows) and (d) (the hosts). The dummy comparison (E.10) at $D(\Lambda)$ is Lemma 9.20(c) at exterior parameter zero, and it holds with the ordinary potentials $\zeta\mathbf{W}_0$, $|\zeta| \leqslant R_\Upsilon$, with the supremum over ζ inside the sum over the selections, as (E.10) requires (each term is bounded by a majorant independent of ζ, Lemma 9.20(c)): the proof of Lemma 9.20(c), and of the majorant (b) on which it rests, uses the potentials only through the host budget $|\zeta\mathbf{W}_0(Y,B)| \leqslant A_Y = (\omega_*/2)e^{-\kappa d_0(Y)}$ (as stated in (c)), and at exterior parameter zero $|\mathbf{W}_0(Y,B)| \leqslant C_0 g_0 (1+r)^3 e^{-\kappa d_0(Y)}$ at U_1 for every Λ (Lemma 9.13(d), whose potentials $\mathbf{V}^\Lambda_{\xi,0}(Y,\bullet)$ are the $\mathbf{W}_0(Y,\bullet)$ of Lemma 9.14: the input is a alone), while along the path the Taylor-subtracted potentials of Section E.8.1 have the host bound $C_0^\partial g_0 (1+r)^3 e^{-\kappa d_0(Y)}$ (Section E.8.1); in both cases $R_\Upsilon C_0 g_0 (1+r)^3 \leqslant R_\Upsilon C_0^\partial g_0 (1+r)^3 = \omega_*/2$, since $R_\Upsilon = \omega_*/(2C_0^\partial g_0 (1+r)^3)$ with $C_0^\partial \geqslant C_0$ ((E.26) and the sentence following it). The positive host comparison of Lemma E.13(a) is a numerical inequality free of the carrier, and Lemma E.13(b),(c) is the same comparison with the differentiated host bound of Section E.9.1 and the moment payment above, with at most $H_M N_1(D)$ active hosts (a count over templates).

Parity and the source. At exterior parameter zero the cutoffs of $K^\Lambda_{\xi;0}$ are the χ_b, even (the translated χ_b^t enter for positive exterior parameter only), the exterior variable is zero, and the density (3.66) for H_t is even in (B,X) with its determinant and compensator; the selected ordinary potentials carry the parameter ζ (Section E.4.1). So Lemmas E.14 and E.3 hold at $D(\Lambda)$ at exterior parameter zero, at U_1 and along the path, $|\mathfrak{Q}_\Lambda(\mathcal{A};f)| \leqslant \prod_{D\in\mathcal{A}} \rho\, m(D)$ at either background with ρ of (E.11): the source remainder bounds of Section E.9.4 hold on $A^{\mathrm{fin}}_{\mathrm{cov}}$, where every contour row has its free inputs in $D(\Lambda)$ and the exterior variable does not enter a link (Lemma 9.25), with the row charts of Section E.9.1 along the path (preceding paragraph). Along the path the quantities used are the selected integrals and their majorants, Corollary 9.16 and Lemma 9.20(b),(c) at the background $U_{W,t}$, together with (E.34) for $m(D)$ and Lemma E.13(c), which hold at $D(\Lambda)$ by the preceding paragraph; the joint ratio $q^t_\Lambda(\mathcal{A})$ of Lemma 9.20(d) and Lemma E.5 are not invoked along the path. This gives (E.72)–(E.73) at $D(\Lambda)$: $|c^0_W(h)| \leqslant M^{(\rho)}_\Lambda(W)$, $|a^\Lambda_W| \leqslant M^{(\rho)}_\Lambda(W)$ and $|\mathscr{D}_\omega c_W(U_{W,t};h)| \leqslant C_\chi \|\omega\|_\infty (n(W) + |\mathcal{H}(W)|) M^{(\rho)}_\Lambda(W)$, where $M^{(\rho)}_\Lambda(W)$ is the majorant $M_\Lambda(W)$ of Lemma 9.26(d) with root weight $\rho\, m(Z)$ in place of $\rho_t m(Z)$. The majorant holds for $|c_W(U_{W,t};h)|$ along the path and for $|c^0_W(h)|$ and $|a^\Lambda_W|$ at the endpoints: at the physical background U_1 by the same argument (the parity argument applies verbatim at U_1, the cutoffs being the even χ_b and the density

(3.66) for $\mathsf{H}_{D(\Lambda)D(\Lambda)}$ being even, Lemmas 9.14 and 9.20(a), and the source remainders at U_1 come from the U_1 charts, Lemma 9.13(c) and Section E.4.1), and at the reference data (Lemma 9.21(b)); on $A_{\mathrm{cov}}^{\mathrm{fin}}$ the source is the carrier-I source with zero on $I \setminus D(\Lambda)$. The endpoint majorants are what (E.75), and the failed-phase-test case of (E.74), use for the groups without a path.

Locality, covariance and endpoints. The gauge action R_g preserves each bond subspace and commutes with geometric restrictions, so the principal restriction to $D(\Lambda)$ is conjugated bondwise, R_g acting on each bond subspace separately (Section E.10.2); the support statements of Section E.9.2 (the seed inverse on E_0, the walks inside their domains, the stencils in Q_W) concern H_t and the partitions. Hence (E.70) and the endpoint identities (E.71) hold at $D(\Lambda)$: $c_W(U_{W,1}; h) = c_W^0(h)$, and $c_W(U_{W,0}; h) = a_W^\Lambda \chi(h)$, the $t = 0$ recipe being the reference data of Lemma 9.21(b) (the same $D(\Lambda)$, Ω_1, $\mathbf{D}_0(\Lambda)$ and cutoffs) with its Schur identity; the coefficient path (E.46) (Lemma E.19) follows by dominated convergence with the majorants above; at $t = 1$ the Taylor-subtracted primitives of the path are the near primitives relative to $D(\Lambda)$ of Section E.8.1, which agree primitive by primitive with the compensated primitives of the retreated split (the locality statement of Section E.8.1, free of the carrier, with the near set selected relative to $D(\Lambda)$ by Lemma 9.7). The small interior groups (E.43), the rectangle Q_W, its injective lift (Section E.10.4) and, on N_b^c, the inclusion of Q_W in Λ_1 outside S_1 (Section E.7.4) concern the group's domains, the contour and Λ_1, not the carrier (the groups at $D(\Lambda)$ have the footprints and hulls of their domains, Lemma 9.27); the plaquette estimate $\delta_{Q_W}(U_1) \leqslant C\varepsilon_1$ (Section E.10.5) is a statement about U_1 near $I \supseteq D(\Lambda)$. So (E.74) holds for $\mathfrak{d}_W^0$ of a small interior group, whether or not it passes the phase test, and (E.75) for the others.

Pins. The first display of Lemma 9.27(a) is the positive summation of Lemma E.9 at $D(\Lambda)$ with root weight ρ_t; it holds with the bound $C_{\mathrm{pin}}\rho'$ for every root weight ρ' satisfying the smallness conditions of Lemma E.9 (which states its bound in the form "$\leqslant C\rho'$ for every ρ' below the threshold", the conditions being monotone in the root weight), in particular for $\rho' = \rho$ under the thresholds of Lemma E.9 assumed through Proposition E.22. With $e^{\theta^\circ(W)} \leqslant e^{\eta\mathfrak{s}(W)/8}$ the computation (E.76)–(E.77) is verbatim: $\sum_W K_\chi e^{\theta^\circ(W)} \mathfrak{d}_W^0 \leqslant C_\chi(\varepsilon_1 + e^{-2\eta T_b}) \sum_W M_\Lambda^{(\rho)}(W) e^{4\eta\mathfrak{s}(W)} \leqslant C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})$, the contour pin summed over the n_C traversals (every group has a nonempty interval). This is (9.37), and with (i)–(ii) it gives (9.35).

(b) By (a) and Lemma 9.36, on each labelled term the integrand of $E_{\mathrm{def}}^{\mathrm{lab}}$ integrates over $a_{\mathrm{old}}|_{I\setminus D(\Lambda)}$ to $\mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}} \mathscr{R}^\Lambda$ ($A_{\mathrm{cov}}^{\mathrm{fin}}$ depends on the labels through Λ only). Split by N_b, a history event ((E.2); Λ_1 is a function of the history, not of the labels). On N_b use $|\mathscr{R}^\Lambda| \leqslant 2$ and $\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} = \mathsf{w}_\xi$ ((9.31)): the contribution is at most $2\mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi \mathbf{1}_{\mathsf{N}_b}]\, \mathrm{d}V = 2\mathbb{P}_{\mathrm{h}}(\mathsf{N}_b)$, the probability of N_b under the original history law (D.20), in which (E.3) is proved (the labels split the weight w_ξ without changing the law, and N_b is defined by the original Λ_1), so it is at most $8C_L n_C (2b+3)^4 q_{\mathrm{pt}} \leqslant C_L' n_C e^{-(c_*/2)\pi_*^2}$ by (E.3). On N_b^c use (9.35); this is the middle expression of (9.36), and $|S_\Lambda^\circ| \leqslant 1$, $\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}} \cap \mathsf{N}_b^c} \leqslant \mathsf{w}_\xi$ and $\mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi]\, \mathrm{d}V = 1$ ((D.20) at $X = 1$) give the last bound. Not used: the inclusion $D_- \subseteq D(\Lambda)$ (Remark 9.48(3); S_Λ° is not identified here) and the locality of Lemma 9.37 (every object is taken at $D(\Lambda)$ throughout). *q.e.d.*

Remark 9.41 (Scope of This Subsection). The locality of the species at $t = 0$ (Lemma 9.37) means that every activity, probe measure, group weight and reference activity of a labelled term at $t = 0$, and on $A_{\mathrm{cov}}^{\mathrm{fin}}$ every defect $R_\Lambda^0(\mathfrak{B})$, coincides with that of Section 3 unless its domains meet the de-younged bonds, so the extra resummation index touches only the species whose domains contain a cube of $\mathcal{N}_{\mathrm{lab}}$ (such domains may extend far from $I \setminus D(\Lambda)$; what is

local is the marking, not the domain); the $t > 0$ species, which enter $\mathcal{B}^\Lambda$ only, depend on m and the exterior data and are bounded, not identified, by Section 9.6. Every term of (9.33) is a one-point witness probability times a lattice sum, uniform in the volume. Not contained here: (1) Lemma 9.38 gives $S^\circ_\Lambda = S^\circ_{\xi,C} e^{\eta_{\mathrm{lab}}}$ with η_{lab} marked by the de-younged bonds, but $|I \setminus D(\Lambda)|$ counts the round-0 bonds $D \setminus D^{(0)}$ (Lemma 9.9; zero when Λ_1 is filled in the side-100ℓ sense) and up to N_{ret} bonds per failed collar test, numbers controlled in probability only ($p_z^{(0)} \leqslant q$, $p_e^\vartheta \leqslant q_\vartheta$ per bond), so $e^{\eta_{\mathrm{lab}}} - 1$ is not small on every history, and averaging it over the labels would need a bound on η_{lab} in terms of $|I \setminus D(\Lambda)|$, which is not established, together with exponential moments of $|I \setminus D(\Lambda)|$ under the history measure, which the joint witness bound of Lemma 9.45 would supply (Sections 9.3–9.4 themselves give one-point probabilities only); this route is not pursued, and the leading coefficient is instead identified at Λ_1^{fin} directly rather than compared to the carrier I: Lemma 9.42 identifies S°_Λ with the regional square scalar at $D(\Lambda)$ and Corollary 9.43 gives $S^\circ_\Lambda = \kappa_C e^{E^\circ_\Lambda}$ with a label-dependent E°_Λ, so that $e^{E^\circ_\Lambda}$ stays inside $\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} S^\circ_\Lambda$ as a label-dependent factor (the form of (8.15), Remark 9.48(5)), whose label average is bounded in first moment only (Corollary 9.47(b)). (2) $E^{\mathrm{lab}}_{\mathrm{def}}$ is bounded in Proposition 9.40, the reading of Proposition E.22 ((E.78), (E.79)) at the carrier $D(\Lambda)$ ($I \to D(\Lambda)$, $S^\circ \to S^\circ_\Lambda$, $R_\xi \to R^0_\Lambda$); that reading takes the inputs of Sections E.1–E.11 at their stated status and does not re-verify them (as in (4)), and it does not use the locality of Lemma 9.37, whose content is that the species of the two carriers coincide away from $\mathcal{N}_{\mathrm{lab}}$. (3) The marked-polymer induction of Section 5 (the pinned recursion (5.30)) is unchanged in form; its species are the groups at the carrier of the term, and Lemma 9.37 is what makes the induction's inputs label-independent away from the de-younged bonds; writing the label index into that induction is not done here. (4) The inputs of Section 9.6.1 (the pin sums of Lemma E.9 for every $\rho \leqslant \rho_{\mathrm{pin}}$, the host comparison of Section E.6, the reference-gas expansion of Lemma 9.21(d), the tree-graph inequality and the square geometry) enter this subsection through Lemmas 9.21 and 9.27 and Propositions 9.28 and 9.34; they are used as stated there and not re-verified here, and a failure of any of them would propagate to (9.32)–(9.34) and Lemma 9.38.

9.9 Identification of the Labelled Leading Coefficient

The labelled leading coefficient S°_Λ of (9.32) is identified with $\kappa_C e^{E^\circ_\Lambda}$, $\kappa_C := e^{-\delta m P(C)} z_c^4$ (the $e^{-\delta m^\sigma P(C_n)} (z_c^\sigma(g_0, \pi/2))^4$ of (8.1) in the cutoff prescription σ fixed throughout), on $A^{\mathrm{fin}}_{\mathrm{cov}}$, with a bounded E°_Λ, and the bound (9.33) (in the form of Proposition 9.34) and the leading term of (9.31)–(9.32) are restated relative to κ_C (Corollaries 9.47(a) and 9.43(b)). The objects: $S^\circ_\Lambda = \sum_{\mathfrak{B}\ \mathrm{compatible}} \prod_{W \in \mathfrak{B}} a^\Lambda_W = \int f^\circ K^\circ_\Lambda \, \mathrm{d}a / I^\circ_\Lambda$ (Lemma 9.21(b); proof of Lemma 9.38), the reference data at $D(\Lambda)$ being $H^\circ = (\mathsf{H}_{\mathbf{1}}^{\Omega_1})_{D(\Lambda)D(\Lambda)}$, exterior variable zero on $J(\Lambda)$, $m^\circ = \ell^\circ = 0$, $f^\circ(a) = W_\chi(C; e^{iq_{\mathbf{1}}(a)})$, so that, by the definition of $K^\Lambda_{\xi;t}$ (Section 9.5) at $t = 0$ and these data,

$$K^\circ_\Lambda(a) = \mathbf{1}_{\mathcal{D}^\Lambda}(a) \exp\Big[-\tfrac{1}{2}\langle a, H^\circ a\rangle + \sum_{\mathfrak{p}\ \text{near rel. } D(\Lambda)} \mathfrak{p}(a) \Big] , \quad a \in \mathfrak{g}^{D(\Lambda)} \text{ (exterior variable zero)}, \tag{9.38}$$

$$I^\circ_\Lambda = \int K^\circ_\Lambda \, \mathrm{d}a \, , \qquad S^\circ_\Lambda = \frac{\int f^\circ K^\circ_\Lambda \, \mathrm{d}a}{I^\circ_\Lambda} \, ,$$

the near primitives being those of all three kinds (action, Haar, constraint Jacobian) whose free-input set meets $D(\Lambda)$. Inputs, used at their stated status: (I1) the regional square scalar

(7.31),

$$S^{\circ}_{D,C_n,T} = \frac{\int \Theta^{\sigma}_{D}(B) e^{P_{D,T}(B)} \chi(\mathrm{Hol}_{U_{g_0,T}(i_{D,T}B)}\, C_n)\, \mathrm{d}\mu_{C_{D,T}}(B)}{\int \Theta^{\sigma}_{D}(B) e^{P_{D,T}(B)}\, \mathrm{d}\mu_{C_{D,T}}(B)}\,, \qquad C_{D,T} = (i^{*}_{D,T} H_T i_{D,T})^{-1}\,,$$

defined there for the free bonds D integrated in a covered conditional kernel, i.e. every free-input stencil of the contour lies in D (the free-input stencil of a traversal link r being the coverage stencil Λ_r of Section 8, which is the coverage of Proposition 7.3), with the original primitive labels retained including those crossing the boundary of D; the proposition's closing sentence states that its estimates use only the covered stencils, the spectral bounds and the small-coupling conditions of the defined flat integral, so that they apply to every covered $D \subseteq D_+$ with the collar, in particular to $D(\Lambda)$ (Lemma 9.42); (I2) Proposition 7.3: under coverage, $K_\mu \geqslant 3(n+2)$, the collar conditions (7.28) ($M \geqslant 8LM_2$, $LMR_1 > 4R_0 + 4L^2M_2R_1 + 8$, $R_0 = 4(1 + R_* + R_H + L)$, enforceable by choosing M with $LM - 4L^2M_2 > 4R_0 + 8$ before the coupling) and the common small-coupling thresholds of the flat and regional expansions, $S^{\circ}_{D,C_n,T} = \kappa_C e^{E^{\circ}_{D,C_n,T}}$ with $|E^{\circ}_{D,C_n,T}| \leqslant C\tau^2[\sum_{j=0}^{4n-1} e^{-cd_j} + 4ne^{-c\iota_T} + (n+1)e^{-cn}]$ ((7.34)), $d_j = \max(0, d_T(K_j, D^c) - R_0)$, $\iota_T = \max(0, N_{\min}/4 - R_0)$, K_j the union of the free-input stencils of the fine traversals of coarse segment j, complements in the torus free carrier, d_T the periodic fine-bond metric ($d_T(b,c) = d_{\infty,T}(x_b, x_c) + \mathbf{1}\{\text{directions differ}\}$, Section 5, after (5.2)), C, c independent of D, n and the periods; (I3) the identification paragraphs of Section 7 between (7.27) and (7.29): the comparison holds for every $D_- \subseteq D \subseteq D_+$ ($D_\pm$ the free bonds with both/at least one endpoint in Λ_1), assigning the complementary free variables to the exterior integral, with the following facts at identity data, numbered here for reference: (1) every stencil meeting $D \subseteq D_+$ stays in the regular region and the Hessian on D is $H_{D,T} = i^{*}_{D,T} H_T i_{D,T}$ including the diagonal terms of crossing plaquettes; (2) the constraint linearisation gives the flat phase (4.9), so that the reconstructed links at identity data are $e^{iq_{\mathbf{1}}}$, and the action, Haar and Jacobian terms become $v_T(i_{D,T}B)$ with all primitive labels meeting D retained; (3) the boundary change [3, (1.22)] is the identity at identity data, with Jacobian one and zero additional exponent; (4) at zero exterior free field the conditional cutoff is Θ^{σ}_{D} with $r = \delta_0/g_0$; (5) the exterior quadratic cross term and its compensator vanish at the projection. Each of (1)–(5) is argued there for $D \subseteq D_+$ with the collar (7.28): the arguments use that every bond of D has an endpoint in Λ_1, so that it is tested by a good test cube inside Λ_1 and its stencils stay in the regular region, and that bad tests and changed-background tests are separated from D_+ by the collar; the lower inclusion $D_- \subseteq D$ is not used in them. The primitives of the conditional reference scalar are those of (7.20), $P_{D,T}(B) = \sum_{v_T\,:\,\mathrm{supp}\, v_T \cap D \neq \varnothing} v_T(i_{D,T}B)$, the terms whose free support misses D vanishing at $i_{D,T}B$; (I4) Lemma 9.21(b) (the reference data at $D(\Lambda)$), $A^{\mathrm{fin}}_{\mathrm{cov}}$ of Section 9.4.2(a), used in Lemma 9.36 (every contour bond in Ω_1 with its free-input set $\mathcal{F}_r \subseteq D(\Lambda)$), Section 9.4.2(a) (a free bond not in $D(\Lambda)$ has no endpoint in Λ_1^{fin}, so lies in $E(B_\xi^{\mathrm{fin}})$, $B_\xi^{\mathrm{fin}} \coloneqq (\Lambda_1^{\mathrm{fin}})^c$), Proposition 9.39 and Section 9.7. Thresholds, in addition to items 1–5 of Section B.8 (with the collar item 4), (9.15), (9.16), (9.20), (9.29) and the period

hypothesis of Section 9.4.1:

(i) the collar conditions (7.28): $M \geqslant 8LM_2$ (already in the order of choice of Section B.8) and $LMR_1 > 4R_0 + 4L^2M_2R_1 + 8$ with $R_0 = 4(1 + R_* + R_H + L)$ (implied by $LM - 4L^2M_2 > 4R_0 + 8$ since $R_1 \geqslant 1$; threshold item 1 of Section B.8 is the same condition with R_{c} in place of R_0; both are imposed), with aligned nested partitions and all periods divisible by LMR_1 ((3.1)) and the contour a coarse translate of the counterclockwise (e_1, e_2) square of side nL (Section 9.6.1);
(ii) the common small-coupling thresholds of the flat and regional expansions ($g_0 C_{\mathsf{C}}(2 + 4r) < \vartheta_*/2$, $g_0(2 + 4r) < a_L$, and the polymer-load inequalities of the proof of Proposition 7.3), conditions on g_0 alone;
(iii) $K_\mu \geqslant 3(n + 2)$ for every coarse period (Section 9.6.1), and, for the n-uniform form of Corollary 9.43(c) only, $4ne^{-c\iota_T} \leqslant 1$ (a condition on the torus: $N_{\min} \geqslant 4R_0 + (4/c)\log(4n)$). (9.39)

9.9.1 The Identification at the Retreated Carrier

Lemma 9.42. *Under* (9.39)*, on every labelled term with final region* Λ*, for exterior data in the support of the term's collar indicators and exterior cutoffs, on* $A^{\mathrm{fin}}_{\mathrm{cov}}$*:* $S^{\circ}_{\Lambda} = S^{\circ}_{D(\Lambda),C,T}$*, the regional square scalar (I1) at* $D = D(\Lambda)$*.*

Proof. Both sides are ratios of two integrals over $\mathfrak{g}^{D(\Lambda)}$ with the same denominator structure; it suffices to identify the four factors of the integrands, numerator and denominator alike, the Gaussian normaliser of $\mu_{C_{D(\Lambda),T}}$ being common to numerator and denominator on the side of (I1) and absent on both on the side of (9.38). First, $D(\Lambda)$ qualifies: $D(\Lambda) \subseteq I = D_+(\Lambda_1)$ (the carrier of Section 3 consists of all free bonds with an endpoint in Λ_1, the bond convention of Section 9.1; the retreat only removes bonds), so every stencil meeting $D(\Lambda)$ stays in the regular region and every bond of $D(\Lambda)$ has an endpoint in a good cube of Λ_1, which is all the facts (I3) use (the lower inclusion $D_- \subseteq D$ of the comparison statement is not used in the arguments for the facts (1)–(5), which are made for $D \subseteq D_+$ with the collar, (I3)); the de-younged bonds $I \setminus D(\Lambda)$ are complementary free variables assigned to the exterior integral, which is what the zero exterior variable on $J(\Lambda)$ of Lemma 9.21(b) does; and $D(\Lambda)$ is covered on $A^{\mathrm{fin}}_{\mathrm{cov}}$, since every contour bond lies in Ω_1 with $\mathcal{F}_r \subseteq D(\Lambda)$, and $\mathcal{F}_r$ contains the coverage stencil Λ_r of Section 8: $\Lambda_b = \{b\} = \mathcal{F}_b$ for a free bond, and for a designated bond $\Lambda_{b_0(c)} = \mathcal{S}_c \cap E_*$ with $\mathcal{S}_c$ the positive fine bonds with both endpoints in the two endpoint blocks $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$ (Appendix C), while $\mathcal{F}_{b_0(c)} = \mathcal{E}_c \cap E_*$ (proof of Lemma 3.28) with $\mathcal{E}_c = \{r : \{r_-, r_+\} \subset \mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}\}$ (Section 3.2), so $\mathcal{S}_c \subseteq \mathcal{E}_c$ and $\Lambda_r \subseteq \mathcal{F}_r$; what (I2) needs is $\Lambda_r \subseteq D(\Lambda)$ (Remark 9.48(2)). (F1) Precision: $H^{\circ} = (\mathsf{H}_{\mathbf{1}}^{\Omega_1})_{D(\Lambda)D(\Lambda)}$ is the Hessian at identity of the regular-region action (the constrained quadratic form of [3], $\mathsf{H}_{U_1}^{\Omega_1} = \mathsf{C}^*\Delta_1\mathsf{C}$, Section 3.2, as used in the proof of Lemma 3.7) restricted to $D(\Lambda) \times D(\Lambda)$; the proof of Lemma B.13 states, by Lemma B.6 at the identity background, that the entries of H° with a label in I are $f_{bb'}(0)$, the entries of the flat Hessian of (4.7) on the branch's torus, so the principal restriction to $D(\Lambda) \subseteq I$ is the flat torus precision on $D(\Lambda)$ including the diagonal contributions of the plaquettes crossing the boundary of $D(\Lambda)$, the same statement as (I3) fact 1 at $D = D(\Lambda)$, which holds because every stencil meeting $D(\Lambda)$ lies in the regular region, where the action at identity is the flat quadratic form; the flat Hessian of Appendix B and H_T are the same operator (4.7) on the same torus. So $\exp(-\frac{1}{2}\langle a, H^{\circ}a\rangle)\,\mathrm{d}a$ and $\mathrm{d}\mu_{C_{D(\Lambda),T}}(B)$ agree up to the common normaliser, with $a = B$. (F2) Cutoff: at the reference data the cutoff of K°_{Λ} is the product of the bond tests $\prod_{b \in D(\Lambda)} \mathbf{1}\{|a_b|_\sigma < r\}$ by the reference-data convention (Lemma 9.21(b), the same cutoffs, the reference data not being a branch; Lemma 9.13(c)

for these data, $K^\circ_\Lambda > 0$ on the open box, proof of Lemma 9.38; the $K^\circ_{\xi;0} = \mathbf{1}_{\mathcal{D}_I}\exp(\dots)$ of Section 3.8, positive on the open box at the carrier I by Lemma B.13); on a branch the corresponding statement is Lemma 9.12(iv), which is not what is used here. Θ^σ_D is the same product of strict bond tests at the same threshold $r = \delta_0/g_0$ in the same norm $|\bullet|_\sigma$ ((I3) fact (4)). (F3) Primitives: $\sum_{\mathfrak{p}\text{ near rel. }D(\Lambda)}\mathfrak{p}(a)$ at zero exterior variable is the sum of the local primitives of Section 3.3 at identity background whose free-input set meets $D(\Lambda)$, evaluated at $a = i_{D(\Lambda),T}B$; $P_{D(\Lambda),T}(i_{D(\Lambda),T}B)$ is the sum of the periodic primitives v_T of (7.20) (those of (C.1) on T) whose supports meet $D(\Lambda)$ (crossing labels retained), the others vanishing at $i_{D(\Lambda),T}B$. Both sums are labelled by the plaquettes, gauge centres, bonds and rows of the regular region, and they agree once the primitives agree label by label, the gauge terms as the sum over a centre: the primitives of Section 3.3 are, by Lemma 3.7, the degree-two Taylor remainders of the action and gauge-fixing terms, the Haar logarithms and the row Jacobian logarithms at the phase $q_{\mathbf{1}}(a)$, and Lemma 7.2 gives this agreement label by label (energy residuals plaquette by plaquette and, for the gauge terms, centre by centre as the sum over the pairs (y,x); Haar and constraint-Jacobian primitives bond by bond and row by row) for every B with $g_0 C_{\mathsf{C}}\max_b|B(b)| \leqslant \vartheta_*/2$ and $g_0\max_b|B(b)| < a_L$, which the small-coupling conditions of the proof of Proposition 7.3 (part of (9.39)) guarantee on the integration box; its second consequence is exactly the equality of the two sums at $i_{D(\Lambda),T}B$ (Remark 9.48(1)); the far primitives contribute their value at zero exterior variable, a constant common to numerator and denominator of (9.38), so whether they are counted is immaterial. (F4) Wilson factor: $f^\circ(a) = W_\chi(C; e^{iq_{\mathbf{1}}(a)})$ is χ of the holonomy along C of the links $e^{iq_{\mathbf{1}}(a)_r}$; by (I3) fact 2 the reconstructed links at identity data are $U_{g_0,T}(i_{D(\Lambda),T}B) = e^{iq_{\mathbf{1}}}$ evaluated at the same free field, and on $A^{\mathrm{fin}}_{\mathrm{cov}}$ the link of each contour bond r depends on the free field through $\mathcal{F}_r \subseteq D(\Lambda)$ only, so the exterior variable (zero) does not enter and $f^\circ(a) = \chi(\mathrm{Hol}_{U_{g_0,T}(i_{D(\Lambda),T}B)}C)$. There is no fifth factor: at $t=0$ and $m^\circ = \ell^\circ = 0$ the kernel $K^\Lambda_{\xi;0}$ has no linear term and no shift, and facts 3 and 5 of (I3) (boundary change, cross term) concern the identification of the conditional scalar of [3] with $S^\circ_{D,C,T}$, which is the statement of Section 7 itself and is not needed to compare the two explicit integrals (9.38) and (I1). Numerator and denominator agree factor by factor, so the ratios agree. *q.e.d.*

Corollary 9.43. *Under* (9.39) *and, for (b), the hypotheses of Proposition 9.34 (those of Proposition 9.28 and* (9.29)*; the collar condition on the exterior data is carried by* $\mathbf{1}_{\mathrm{lab}}$*), on every labelled term, on* $A^{\mathrm{fin}}_{\mathrm{cov}}$*, with* $E^\circ_\Lambda := E^\circ_{D(\Lambda),C,T}$ *(the error of (I2) at* $D = D(\Lambda)$*, with the branch of the logarithm of Proposition 7.3, continued from zero in the traversal-source polynomial* (7.1)*) and* $d_j(\Lambda) := \max(0, d_T(K_j, D(\Lambda)^c) - R_0)$*:*

(a)

$$S^\circ_\Lambda = \kappa_C e^{E^\circ_\Lambda}\,, \qquad |E^\circ_\Lambda| \leqslant C\tau^2\Big[\sum_{j=0}^{4n-1} e^{-cd_j(\Lambda)} + 4ne^{-c\iota_T} + (n+1)e^{-cn}\Big] \leqslant C\tau^2(9n+1)\,, \tag{9.40}$$

E°_Λ depends on the labels through $D(\Lambda)$ only (a simple function on the finitely many labels of a history), $\kappa_C \neq 0$ (finite perimeter exponent, z_c the exponential of a convergent sum, Section 8; also $S^\circ_\Lambda \neq 0$ by (I2)), and $|\kappa_C|e^{-C\tau^2(9n+1)} \leqslant |S^\circ_\Lambda| \leqslant |\kappa_C|e^{C\tau^2(9n+1)}$ (so $S^\circ_\Lambda \neq 0$ on $A^{\mathrm{fin}}_{\mathrm{cov}}$, consistent with $|S^\circ_\Lambda| \leqslant 1$ of Lemma 9.38);

(b) the leading term of (9.31)–(9.32) *is* $\kappa_C\sum_{\mathrm{lab}} e^{E^\circ_\Lambda}\int_{\mathfrak{g}^{I\setminus D(\Lambda)}}\mathbf{1}_{\mathrm{lab}}e^{c^\Lambda_\xi}I^\Lambda_1\mathbf{1}_{A^{\mathrm{fin}}_{\mathrm{cov}}}W_\chi(C;U_1)\,\mathrm{d}a_{\mathrm{old}}$ *($e^{E^\circ_\Lambda}$ and $\mathbf{1}_{A^{\mathrm{fin}}_{\mathrm{cov}}}$ depend on the labels only and leave the a_{old}-integral), and on $A^{\mathrm{fin}}_{\mathrm{cov}}$, by*

(9.30),

$$|\kappa_C|^{-1}|\mathcal{B}_\xi^\Lambda[f_\xi]| \leqslant e^{|E_\Lambda^\circ|}e^{n_C\bar\varepsilon_{\rm ph}}\big[C_1 g_0 r\mathfrak{S}_1(\Lambda)+C_2\bar\varepsilon_t r^{d_G}\mathfrak{S}_2(\Lambda)\big] \tag{9.41}$$
$$\leqslant e^{C\tau^2(9n+1)}e^{n_C\bar\varepsilon_{\rm ph}}\big[C_1 g_0 r\mathfrak{S}_1(\Lambda)+C_2\bar\varepsilon_t r^{d_G}\mathfrak{S}_2(\Lambda)\big]\ ;$$

(c) (refined split) $|E_\Lambda^\circ| \leqslant C\tau^2(2+1/c)+C\tau^2\sum_j e^{-cd_j(\Lambda)}$ *under* $4ne^{-c\iota_T}\leqslant 1$ *((9.39)(iii); for* $n\geqslant 1$ *this is* $N_{\min}\geqslant 4R_0+(4/c)\log(4n)$*, since* $(n+1)e^{-cn}\leqslant 1+\sup_{x\geqslant 0}xe^{-cx}=1+1/(ec)\leqslant 1+1/c$*: the only part of the error that grows with* n *at fixed* τ *is the boundary sum* $\sum_j e^{-cd_j(\Lambda)}\leqslant 4n$*, each of whose terms is at most 1 and exponentially small in* $d_T(K_j,D(\Lambda)^c)-R_0$ *(the sum is zero only when* $D(\Lambda)^c=\varnothing$*, by the convention* $e^{-\infty}=0$*); it is the quantity averaged in Lemma 9.44.*

Proof. (a) Lemma 9.42 and (I2) at $D=D(\Lambda)$, whose hypotheses are (9.39) and coverage ($A_{\rm cov}^{\rm fin}$); the crude bound uses $e^{-cd_j}\leqslant 1$, $e^{-c\iota_T}\leqslant 1$ and $(n+1)e^{-cn}\leqslant n+1$ (as in the constant B_C following (8.16)); $D(\Lambda)$ is a function of the labels. (b) The leading term of (9.32) is $S_\Lambda^\circ W_\chi(C;U_1)$ integrated as in (9.31); substitute (a). The bound: (9.30) is $|\mathcal{B}^\Lambda|\leqslant |S_\Lambda^\circ|e^{n_C\bar\varepsilon_{\rm ph}}[\dots]$ and $|S_\Lambda^\circ|=|\kappa_C|e^{\operatorname{Re}E_\Lambda^\circ}\leqslant|\kappa_C|e^{|E_\Lambda^\circ|}$. (c) The stated elementary inequality. *q.e.d.*

9.9.2 The Label Average of the Boundary Distances

Lemma 9.44. *In the setting of Proposition 9.39 (square geometry of Section 9.6.1, compatible even fine periods exceeding* 200ℓ*, the hypotheses of Sections 9.3–9.4), for every* $c>0$*,*

$$\mathcal{Z}_0^{-1}\int\mathcal{O}\Big[\sum_{\rm lab}\mathsf{w}_\xi^{\rm lab}\mathbf{1}_{A_{\rm cov}^{\rm fin}}\sum_{j=0}^{4n-1}e^{-cd_j(\Lambda)}\Big]\mathrm{d}V\leqslant e^{c(R_0+\varrho_K)}N_{\rm lat}(c)\,n_C(q+N_{\rm ret}q_\vartheta)\ , \tag{9.42}$$
$$\varrho_K:=\max_j\max_{k\in K_j}\min_{r\in C_j}d_T(k,r)\leqslant 2L+1\ ,$$

with $N_{\rm lat}(c)=\sum_{s\geqslant 0}4(2s+1)^4e^{-cs}$ *of Lemma 9.2,* C_j *the contour bonds of coarse segment* j*,* q *and* q_ϑ *of* (9.6) *and Lemma 9.6,* $N_{\rm ret}$ *of Lemma 9.10: uniform in the volume, and of the same form as* (9.33)*, a lattice sum times one-point witness probabilities.*

Proof. Pointwise on a term: $d_j(\Lambda)\geqslant d_T(K_j,D(\Lambda)^c)-R_0$, so $e^{-cd_j(\Lambda)}\leqslant e^{cR_0}e^{-cd_T(K_j,D(\Lambda)^c)}\leqslant e^{cR_0}\sum_{b\notin D(\Lambda)}e^{-cd_T(K_j,b)}$ (the maximum over $b\in D(\Lambda)^c$ of $e^{-cd_T(K_j,b)}$ is at most the sum; if $D(\Lambda)^c=\varnothing$ the left side is zero by the convention $e^{-\infty}=0$). For $k\in K_j$ and b, the triangle inequality for d_T gives $d_T(k,b)\geqslant\min_{r\in C_j}d_T(r,b)-\varrho_K$, hence $e^{-cd_T(K_j,b)}\leqslant e^{c\varrho_K}\max_{r\in C_j}e^{-cd_T(r,b)}\leqslant e^{c\varrho_K}\sum_{r\in C_j}e^{-cd_T(r,b)}$, and $\sum_j\sum_{r\in C_j}=\sum_{r\in C}$ because the segments partition the contour bonds. A free bond $b\notin D(\Lambda)$ (the complement is taken in the torus free carrier, (7.32)) has no endpoint in $\Lambda_1^{\rm fin}$ (for $b\in I\setminus D(\Lambda)$ by Section 9.4.2(a); for $b\notin I$ because $I=D_+(\Lambda_1)$ is the set of all free bonds with an endpoint in $\Lambda_1\supseteq\Lambda_1^{\rm fin}$, the bond convention of Section 9.1; the equality, not only $D(\Lambda)\subseteq I$, is used here), so $\mathbf{1}\{b\notin D(\Lambda)\}\leqslant\mathbf{1}\{b\in E(B_\xi^{\rm fin})\}\leqslant\mathbf{1}\{b\in E(\mathfrak{B}_0)\}+\sum_{e'\,:\,x_{e'}\text{ within }102\ell+1\text{ of an endpoint of }b}M_{e'}$ ((9.7), Lemma 9.10(c)–(d)). Altogether $\sum_j e^{-cd_j(\Lambda)}\leqslant e^{c(R_0+\varrho_K)}\sum_{r\in C}\sum_b e^{-cd_T(r,b)}[\mathbf{1}\{b\in E(\mathfrak{B}_0)\}+\sum_{e'\text{ near }b}M_{e'}]$, every term nonnegative and label-independent except through the marks. Multiply by $\mathsf{w}_\xi^{\rm lab}\mathbf{1}_{A_{\rm cov}^{\rm fin}}\leqslant\mathsf{w}_\xi^{\rm lab}$, sum over the labels and histories, apply $\mathcal{O}$ and divide by $\mathcal{Z}_0$ (Tonelli; $\mathsf{w}_\xi^{\rm lab}\geqslant 0$ by (9.31) with $\rho_\xi\geqslant 0$, and $\mathcal{O}$ is positive; finitely many labels per history; $\sum_{\rm lab}\mathsf{w}_\xi^{\rm lab}=\mathsf{w}_\xi$, Section 9.4.2(c)): for each fixed b the first term gives $p_b^{(0)}\leqslant q$ (Lemma 9.9(b))

and the second $\sum_{e' \text{ near } b} p_{e'}^{\vartheta} \leqslant N_{\mathrm{ret}} q_\vartheta$ (Lemma 9.10(d), Proposition 9.8), exactly as in the proof of Proposition 9.39. Finally $\sum_b e^{-cd_T(r,b)} \leqslant \sum_{s\geqslant 0} 4(2s+1)^4 e^{-cs} = N_{\mathrm{lat}}(c)$ for each contour bond r ($d_T(r,b) \geqslant \|x_r - x_b\|_\infty$, and at most $4(2s+1)^4$ bonds have initial site within ℓ^∞ distance s of x_r), and $\sum_{r\in C} 1 = n_C$. The bound on ϱ_K: for a free contour bond r, $\Lambda_r = \{r\}$ and $d_T(r,r) = 0$; for a designated $r = b_0(c)$, the stencil bonds and r itself have both endpoints in the two endpoint blocks of c ($\Lambda_{b_0(c)} = \mathcal{S}_c \cap E_*$ with $\mathcal{S}_c$ the bonds with both endpoints in the two endpoint blocks of c, Section 8; $r = b_0(c)$ is a bond of the row c), two adjacent blocks of side L, any two sites of which are within ℓ^∞ distance $2L-1$ of each other, so $d_T(k,r) \leqslant 2L-1+1 = 2L \leqslant 2L+1$ ($d_T(b,c) = d_{\infty,T}(x_b, x_c) + \mathbf{1}\{\text{directions differ}\}$, (I2)). *q.e.d.*

9.9.3 Joint Witness Bounds and the Exponential Moment of the Boundary Sum

Lemma 9.45 (Joint Witness Bound)**.** *In the setting of Proposition 9.8 and Lemma 9.9 (compatible even fine periods, $0 < g_0 < g_*(\vartheta)$), call a mark either M_e for a bond e (Section 9.3.2) or $a_b := \mathbf{1}\{b \in E(\mathfrak{B}_0)\}$ for a bond b (Lemma 9.9), located at the initial site of its bond. Put $\sigma_K := 2K\ell + 4$ and $C_K := 4(2\sigma_K+1)^4 = 4(4K\ell+9)^4$, an upper bound on the number of bonds whose initial site lies within ℓ^∞ distance σ_K of a given site. For every finite set S of marks,*

$$\mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \prod_{m\in\mathsf{S}} m\Big] \mathrm{d}V \leqslant q_\vartheta^{|\mathsf{S}'|} \leqslant q_\vartheta^{|\mathsf{S}|/(2C_K)} , \tag{9.43}$$

where the first inequality holds for every subset S' of S whose marks are at pairwise ℓ^∞ distance $> \sigma_K$, and the second for a maximal such subset (which has $|\mathsf{S}'| \geqslant |\mathsf{S}|/(2C_K)$: every mark of S lies within σ_K of a mark of S', at most C_K bonds lie within σ_K of a given site, and a bond carries at most two marks; the count C_K is an upper bound on every torus, so this holds also on periods at most $2\sigma_K+1$, where a separated subset has at most one mark and the second inequality carries only the rate $q_\vartheta^{1/(2C_K)}$). The same holds, without labels and with q in place of q_ϑ, for the marks $\mathbf{1}\{z \in E(B_\xi)\}$ of Section D.7 ($B_\xi = \Lambda_1^c$, (D.23)), for g_0 below the thresholds of Lemmas D.12 and D.14, on compatible even fine periods (Section D.7): for every finite set S of bonds and every subset S' of S at pairwise ℓ^∞ distance $> \sigma_K$,

$$\begin{gathered} \mathbb{P}_{\mathrm{h}}\big(\mathsf{S} \subseteq E(B_\xi)\big) = \mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\mathsf{w}_\xi \prod_{z\in\mathsf{S}} \mathbf{1}\{z \in E(B_\xi)\}\Big] \mathrm{d}V \leqslant q^{|\mathsf{S}'|} , \quad \text{and} \\ \mathbb{P}_{\mathrm{h}}\big(\mathsf{S} \subseteq E(B_\xi)\big) \leqslant q^{|\mathsf{S}|/C_K} . \end{gathered} \tag{9.44}$$

The second inequality of (9.43) *uses $q_\vartheta \leqslant 1$: $q_\vartheta \leqslant e^{-c_q \vartheta^2 \pi_*^2}$ for $g_0 < g_*(\vartheta)$ by* (9.6)*. The second bound of* (9.44) *follows from the first when $q \leqslant 1$, with S' a maximal separated subset (one mark per bond, so $|\mathsf{S}'| \geqslant |\mathsf{S}|/C_K$, and $q^{|\mathsf{S}'|} \leqslant q^{|\mathsf{S}|/C_K}$ for $q \leqslant 1$), in particular for $g_0 < g_*$, the threshold of Proposition D.15, where $q \leqslant e^{-c_* \pi_*^2}$ with $c_* = c_q$ (this is* (9.6) *at $\vartheta = 1$, where $\theta_1 = \theta$ and $q_1 = q$). When $q > 1$ it holds trivially, the left side being a probability, at most $1 \leqslant q^{|\mathsf{S}|/C_K}$ (with equality only for $\mathsf{S} = \varnothing$); the two bounds of* (9.44) *are then not a chain ($q^{|\mathsf{S}'|} > q^{|\mathsf{S}|/C_K}$ when $|\mathsf{S}'| > |\mathsf{S}|/C_K$), which is why they are stated separately. So both bounds of* (9.44) *hold for every g_0 below the thresholds of Lemmas D.12 and D.14, the second being informative when $q \leqslant 1$.*

Proof. Fix a history and a labelled term. $\prod_{m\in\mathsf{S}} m$ is a nonnegative function of U, of the regions and of the labels (a_b depends on the regions only, M_e on the labels and on U), and $\prod_{\mathsf{S}} m \leqslant \prod_{\mathsf{S}'} m$, the marks taking the values 0 and 1. The reversal of Lemma D.13, applied in the proof of Proposition 9.8 to $\mathsf{w}_\xi^{\mathrm{lab}} M_e$ and in Lemma 9.9(b) to $\mathsf{w}_\xi a_b$, is a change of

variables for integrals of (density $\times$ nonnegative function of U, the regions and the labels), so it applies to $\mathsf{w}_\xi^{\mathrm{lab}} \prod_{\mathsf{S}'} m$. After the reversal, on the support of the term's cutoffs, each M_e with $e \in \mathsf{S}'$ is dominated by $\mathbf{1}_{W_e^\vartheta}$ (Lemma 9.6: a plaquette or gauge-variable witness at threshold θ_ϑ within $K\ell$ of e) and each a_b by $\mathbf{1}_{W_b} \leqslant \mathbf{1}_{W_b^\vartheta}$ (Lemma 9.9(b): a witness at threshold $\theta \geqslant \theta_\vartheta$ within $K\ell$ of b; the event at the larger threshold is contained in the event at the smaller one). The events W^ϑ are label-free, so, as in the proof of Proposition 9.8, summing the nonnegative labelled terms and histories (their partitions sum to one) gives $\mathcal{Z}_0^{-1} \int \mathcal{O}[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \prod_{\mathsf{S}} m] \,\mathrm{d}V \leqslant (\mu_0 \otimes q_g^{\otimes \mathcal{I}} \otimes \mathrm{Haar}^{\otimes T})(\bigcap_{m \in \mathsf{S}'} W_m^\vartheta)$. Here W_m^ϑ is taken as the union, over the plaquette labels p and the gauge-variable labels w all of whose sites (the four corners of p; the centre of w and the sites of its variable) lie within ℓ^∞ distance $K\ell + 1$ of the initial site x_m of the mark, of the single-witness events $\{|U(\partial p) - \mathbf{1}|_\sigma \geqslant \theta_\vartheta\}$ and $\{|w - \mathbf{1}|_\sigma \geqslant \theta_\vartheta\}$. The dominations above hold with this union: the proofs of Lemmas D.12 and 9.6 place every site of the witnessing label within a distance of the bond that is a fixed multiple of ℓ far below $K\ell$ (the witness lies in the 6ℓ-neighbourhood $N(\square')$ of a cube of side at most $\ell/8$, its sites within 7ℓ of the initial vertex of the bond and the gauge-label centres within $O(L) \leqslant \ell$ more, Lemma 9.6; the neighbourhood of Lemma D.12 is larger by a fixed multiple of ℓ, still below $K\ell$ with $K = 256$), so the witnessing label is among those of the union. The number of labels in the union is at most $7(2K\ell + 3)^4 \leqslant n_W$: at most $(2K\ell + 3)^4$ sites lie within $K\ell + 1$ of x_m, a plaquette is fixed by its base site and one of six plane orientations, and a gauge label by the noncentral site of its variable, which lies within $K\ell + 1$ of x_m and determines the centre (the centre of its block); a centre alone carries $L^4 - 1$ labels and would not fix one. So the intersection is contained in a union of at most $n_W^{|\mathsf{S}'|}$ intersections of single-witness events, one label per mark of S'. Fix such a choice. Two marks of S' are at distance $> \sigma_K = 2(K\ell+1)+2$, so their labels have all their sites at distance > 2 from each other: the chosen gauge variables are distinct, and the chosen plaquettes have distinct base cubes C_x (the closed unit cube whose minimal corner x is the plaquette's base site). Under the product measure the probability of the intersection factorises into the μ_0-probability of the plaquette events and the product of the q_g-probabilities of the gauge-variable events, each of the latter at most $z_G(\theta_\vartheta)$ (D.38). For the plaquette events apply the chessboard estimate (D.40) with f_x, for each chosen base cube C_x, the indicator $\mathbf{1}\{|U(\partial p_0) - \mathbf{1}|_\sigma \geqslant \theta_\vartheta\}$ of the plaquette p_0 of C_0 whose transport $f_x^{[x]}$ is the indicator of the chosen plaquette of C_x, and $f_x = 1$ for the other cubes: every plaquette of C_x is the transport of a plaquette of C_0 of the same plane orientation, because a reflection in direction ρ carries the transverse coordinate $t \in \{0, 1\}$ of a face to $1 - t$, so the face of C_0 with transverse coordinates (t_ρ) is carried to the face of C_x with transverse coordinates $x_\rho + t_\rho$ or $x_\rho + 1 - t_\rho$ according to the parity of x_ρ, and the indicator does not depend on the orientation or the base point of the plaquette. Each root $(\int \prod_y f_x^{[y]} \,\mathrm{d}\mu_0)^{1/V_T}$ of a disseminated expectation is at most $z_P(\theta_\vartheta)$ by the computation in the proof of Lemma D.14, which applies to every plaquette of C_0 (its dissemination contains $V_T/4$ distinct plaquettes of its orientation, each with $s_G \geqslant b_G \theta_\vartheta^2$ on the event, and (D.36) bounds $\mathcal{Z}_0$ below). Hence the fixed choice has probability at most $\max(z_P(\theta_\vartheta), z_G(\theta_\vartheta))^{|\mathsf{S}'|}$, the union bound over the choices gives $(n_W \max(z_P(\theta_\vartheta), z_G(\theta_\vartheta)))^{|\mathsf{S}'|} = q_\vartheta^{|\mathsf{S}'|}$ (9.5), and $q_\vartheta \leqslant 1$ with $|\mathsf{S}'| \geqslant |\mathsf{S}|/(2C_K)$ gives the second inequality. As in Lemma D.13, the product measure is a representation of the integral before the history conditions; no independence of the marks under the history measure is asserted, the factorisation is that of the physical measure after the reversal, and the correlation of distinct plaquettes under μ_0 is paid by the chessboard estimate. For (9.44) the same argument runs without labels: the reversal of Lemma D.13 with the product of the marks $\mathbf{1}\{z \in E(B_\xi)\}$, $z \in \mathsf{S}'$, inserted (each a function of the regions and the deterministic filling), each mark dominated by $\mathbf{1}_{W_z}$ (Lemma D.12: a witness at threshold θ within $K\ell$ of z; (D.34)), the intersection $\bigcap_{z \in \mathsf{S}'} W_z$, with W_z now the union of the

single-witness events at threshold θ over the labels all of whose sites lie within $K\ell+1$ of the initial site of z (it contains the witnessing label of Lemma D.12 by the geometry above, so the domination holds with it), a union of at most $n_W^{|\mathsf{S}'|}$ intersections of single-witness events at threshold θ with the same separation, and the chessboard and product bounds as above at threshold θ, giving $(n_W \max(z_P, z_G))^{|\mathsf{S}'|} = q^{|\mathsf{S}'|}$. *q.e.d.*

Lemma 9.46 (Exponential Moment of the Boundary Sum)**.** *In the setting of Lemma 9.44, for $c > 0$ put $R_0' \coloneqq \lceil R_0 \rceil + 2L + 1$ (an integer, at least $R_0 + \varrho_K$ since $\varrho_K \leqslant 2L+1$ (Lemma 9.44), and independent of the contour and of n; the lattice distances $d_T(r,b)$ exceeding R_0' are then at least $R_0'+1$) and, for a bond b and a bond e,*

$$\begin{aligned} \varphi_c(b) &\coloneqq \sum_{r \in C} e^{-c(d_T(r,b)-R_0')_+} \leqslant \Phi_c \coloneqq 4(2R_0'+1)(1+1/c) + 8/c^2 \,, \\ \Phi_c(e) &\coloneqq \sum_{b\,:\,x_e \text{ within } 102\ell+1 \text{ of an endpoint of } b} \varphi_c(b) \leqslant N_{\mathrm{ret}} \Phi_c \,, \end{aligned} \tag{9.45}$$

and $N'_{\mathrm{lat}}(c) \coloneqq \sum_{s \geqslant 0} 4(2s+1)^4 e^{-c(s-R_0')_+} < \infty$. Then, on the support of the indicators $\mathbf{1}_{\mathrm{lab}}$ of every labelled term with final region Λ (there every bond found bad on the term has $M_e = 1$; every use below carries the factor $\mathsf{w}_\xi^{\mathrm{lab}}$, which contains $\mathbf{1}_{\mathrm{lab}}$),

$$\sum_{j=0}^{4n-1} e^{-c d_j(\Lambda)} \leqslant \sum_b \varphi_c(b)\, a_b + \sum_e \Phi_c(e)\, M_e \,, \tag{9.46}$$

and for every $\lambda > 0$

$$\begin{aligned} \mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}} \exp\Big(\lambda \sum_{j=0}^{4n-1} e^{-c d_j(\Lambda)} \Big) \Big] \mathrm{d}V \\ \leqslant \exp\big[n_C\, \lambda (1 + N_{\mathrm{ret}}) N'_{\mathrm{lat}}(c)\, e^{\lambda N_{\mathrm{ret}} \Phi_c}\, q_\vartheta^{1/(2C_K)} \big] \,, \end{aligned} \tag{9.47}$$

uniform in the volume; the bound (9.47) *holds with $\mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}}$ omitted, the proof using the indicator only through $\mathbf{1} \leqslant 1$. At the original carrier ($D = I = D_+(\Lambda_1)$, the d_j of* (7.32) *at $D = I$, which are those of Proposition F.2; no labels; compatible even fine periods, Section D.7; otherwise in the setting of Lemma 9.44: the square C with its n_C contour bonds, the distances d_T, the constant R_0 behind R_0' and the bound $\varrho_K \leqslant 2L+1$ used in the proof, and φ_c, Φ_c, $N'_{\mathrm{lat}}(c)$ as defined above), pointwise on every branch and for every $\lambda > 0$, and again with the second bound holding when $\mathbf{1}_{A_{\mathrm{cov}}}$ is omitted,*

$$\begin{aligned} \sum_{j=0}^{4n-1} e^{-c d_j} &\leqslant \sum_b \varphi_c(b) \mathbf{1}\{ b \in E(B_\xi) \} \,, \\ \mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}}} \exp\Big(\lambda \sum_{j=0}^{4n-1} e^{-c d_j} \Big) \Big] \mathrm{d}V &\leqslant \exp\big[n_C\, \lambda N'_{\mathrm{lat}}(c)\, e^{\lambda \Phi_c}\, q^{1/C_K} \big] \,, \end{aligned} \tag{9.48}$$

for g_0 below the thresholds of Lemmas D.12 and D.14.

Proof. The bounds in (9.45): the initial sites of the contour bonds lie on the four lines of the square, each meeting an ℓ^∞ ball of radius s in at most $2s+1$ sites, so at most $4(2s+1)$ contour bonds have $d_T(r,b) \leqslant s$ ($d_T(r,b) \geqslant \|x_r - x_b\|_\infty$); hence $\varphi_c(b) \leqslant 4(2R_0'+1) + \sum_{u \geqslant 1} 4(2(R_0'+u)+1) e^{-cu} \leqslant 4(2R_0'+1)(1+1/c) + 8/c^2$, using $\sum_{u \geqslant 1} e^{-cu} \leqslant 1/c$ and $\sum_{u \geqslant 1} u e^{-cu} \leqslant 1/c^2$. The bonds b with an endpoint within $102\ell+1$ of x_e have their initial site within

$102\ell+2$ of x_e and number at most $4(204\ell+5)^4 \leqslant 4(207\ell)^4 = N_{\mathrm{ret}}$ for $\ell \geqslant 2$ (an elementary count; N_{ret} is the constant of Lemma 9.10(d)). (9.46): $d_j(\Lambda) = (d_T(K_j, D(\Lambda)^c) - R_0)_+$ with $d_T(K_j, D(\Lambda)^c) = \min_{b \notin D(\Lambda)} d_T(K_j, b)$, and $x \mapsto e^{-c(x-R_0)_+}$ is nonincreasing, so $e^{-cd_j(\Lambda)} \leqslant \sum_{b \notin D(\Lambda)} e^{-c(d_T(K_j,b)-R_0)_+}$ (zero if $D(\Lambda)^c = \varnothing$); for $k \in K_j$, $d_T(k,b) \geqslant \min_{r \in C_j} d_T(r,b) - \varrho_K$ (proof of Lemma 9.44), so $e^{-c(d_T(K_j,b)-R_0)_+} \leqslant e^{-c(d_T(C_j,b)-R_0')_+} \leqslant \sum_{r \in C_j} e^{-c(d_T(r,b)-R_0')_+}$, and $\sum_j \sum_{r \in C_j} = \sum_{r \in C}$. Thus $\sum_j e^{-cd_j(\Lambda)} \leqslant \sum_b \varphi_c(b) \mathbf{1}\{b \notin D(\Lambda)\}$, the sum over the free bonds of the torus free carrier, which is enlarged below to the sum over all bonds b of the torus (the marks a_b and M_e are defined for every bond, on every history and labelled term, Section 9.3.2 and Lemma 9.9, so the index set of the marks is the same for every history, as the exchange of the history sum with the expansion below requires; the added terms are nonnegative, and the counts below run over all bonds), and $\mathbf{1}\{b \notin D(\Lambda)\} \leqslant a_b + \sum_{e\,:\,x_e \text{ within } 102\ell+1 \text{ of an endpoint of } b} M_e$ on every labelled term (proof of Lemma 9.44, from (9.7)); exchanging the two sums gives (9.46). (9.47): by (9.46) and $\mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}} \leqslant 1$ the integrand is at most $\mathsf{w}_\xi^{\mathrm{lab}} \prod_b (1 + (e^{\lambda\varphi_c(b)} - 1)a_b) \prod_e (1 + (e^{\lambda\Phi_c(e)} - 1)M_e)$, since $e^{\lambda w m} = 1 + (e^{\lambda w} - 1)m$ for $m \in \{0,1\}$; the products are finite. Expand them into the sum over the sets S of marks of $\prod_{m \in \mathsf{S}} (e^{\lambda w_m} - 1) \prod_{m \in \mathsf{S}} m$, w_m the weight $\varphi_c(b)$ or $\Phi_c(e)$ of the mark; every term is nonnegative, so the history average (positive; finitely many labels per history; Tonelli) is at most $\sum_{\mathsf{S}} \prod_{m \in \mathsf{S}} (e^{\lambda w_m} - 1)\, q_\vartheta^{|\mathsf{S}|/(2C_K)}$ by Lemma 9.45, which equals $\prod_m (1 + (e^{\lambda w_m} - 1)\hat{q})$ with $\hat{q} \coloneqq q_\vartheta^{1/(2C_K)}$, and $\prod_m (1 + (e^{\lambda w_m} - 1)\hat{q}) \leqslant \exp(\hat{q} \sum_m (e^{\lambda w_m} - 1))$. Finally $e^x - 1 \leqslant x e^x$ for $x \geqslant 0$, $w_m \leqslant N_{\mathrm{ret}} \Phi_c$ for every mark, and $\sum_m w_m = \sum_b \varphi_c(b) + \sum_e \Phi_c(e) \leqslant (1 + N_{\mathrm{ret}}) \sum_b \varphi_c(b) \leqslant (1 + N_{\mathrm{ret}})\, n_C N'_{\mathrm{lat}}(c)$: $\sum_e \Phi_c(e) = \sum_b \varphi_c(b)\, \#\{e : x_e \text{ within } 102\ell+1 \text{ of an endpoint of } b\} \leqslant N_{\mathrm{ret}} \sum_b \varphi_c(b)$ (Lemma 9.10(d)), and $\sum_b \varphi_c(b) = \sum_{r \in C} \sum_b e^{-c(d_T(r,b)-R_0')_+} \leqslant n_C N'_{\mathrm{lat}}(c)$, since at most $4(2s+1)^4$ bonds have initial site within ℓ^∞ distance s of x_r (as in the proof of Lemma 9.44). (9.48): the first steps are the same with $D(\Lambda)$ replaced by I; a free bond $b \notin I$ has no endpoint in Λ_1 (I is the set of all free bonds with an endpoint in Λ_1, the bond convention of Section 9.1), so both its endpoints lie in $B_\xi = \Lambda_1^c$ and $\mathbf{1}\{b \notin I\} \leqslant \mathbf{1}\{b \in E(B_\xi)\}$, which gives the pointwise bound; the expansion is the same with the marks $\mathbf{1}\{b \in E(B_\xi)\}$, one per bond with weight $\varphi_c(b) \leqslant \Phi_c$ and $\sum_b \varphi_c(b) \leqslant n_C N'_{\mathrm{lat}}(c)$, the history average $\mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi \bullet]\, \mathrm{d}V$ in place of the labelled one, and (9.44) in place of (9.43). *q.e.d.*

Corollary 9.47 (Bounds Relative to κ_C)**.** *Under* (9.39) *and the hypotheses of Proposition 9.34 and Lemma 9.44:*

(a) (crude form)

$$\begin{aligned}
&\mathcal{Z}_0^{-1} \int \mathcal{O}[\sum_{\mathrm{lab}} \int \mathbf{1}_{\mathrm{lab}} e^{c_\xi^\Lambda} I_1^\Lambda \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}} |\kappa_C|^{-1} |\mathcal{B}_\xi^\Lambda[f_\xi]|\, \mathrm{d}a_{\mathrm{old}}]\, \mathrm{d}V \\
&\quad \leqslant e^{C\tau^2(9n+1)} \times \textit{(the right side of (9.33) with } e^{n_C \bar{\varepsilon}_{\mathrm{ph}}}\textit{)} \\
&\quad = e^{C\tau^2(9n+1)} e^{n_C \bar{\varepsilon}_{\mathrm{ph}}} (C_1 g_0 r + C_2 \bar{\varepsilon}_t r^{d_G})(1 + e^{\mu_H(2L+1)/2}) N_{\mathrm{lat}}(\mu') n_C (q + N_{\mathrm{ret}} q_\vartheta)\,,
\end{aligned}$$

by (9.41) *and the statement* (9.33) *of Proposition 9.39 (monotonicity of its integrand) with* $|\mathcal{B}^\Lambda|/|S_\Lambda^\circ|$ *replaced by* $|\kappa_C|^{-1}|\mathcal{B}^\Lambda| \leqslant e^{C\tau^2(9n+1)} |\mathcal{B}^\Lambda|/|S_\Lambda^\circ|$*; by the comparison of Proposition 9.39 (*$r^{d_G} N_{\mathrm{ret}}(q + N_{\mathrm{ret}} q_\vartheta) \leqslant e^{-c_q \vartheta^2 \pi_*^2/2}$ *for* g_0 *below a further threshold;* N_{ret} *and* r^{d_G} *themselves diverge as* $g_0 \to 0$*) this is at most* $C' n_C e^{n_C \bar{\varepsilon}_{\mathrm{ph}}} e^{C\tau^2(9n+1)} e^{-c_q \vartheta^2 \pi_*^2/2}$ *with the constant* C' *of the closing form of Proposition 9.39, and* $e^{C\tau^2(9n+1)}$ *is the factor carried in* $B_C = |\kappa_C| \exp(C\tau^2(9n+1))$ *following* (8.16)*.*

(b) (first moment of the leading error) $\mathcal{Z}_0^{-1} \int \mathcal{O}[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}} |E_\Lambda^\circ|]\, \mathrm{d}V \leqslant C\tau^2(2 + 1/c) + C\tau^2 e^{c(R_0 + \varrho_K)} N_{\mathrm{lat}}(c) n_C (q + N_{\mathrm{ret}} q_\vartheta)$ *under* (9.39)*(iii), by Corollary 9.43(c), Lemma 9.44*

and the normalisation $\mathcal{Z}_0^{-1}\int \mathcal{O}[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}}]\,\mathrm{d}V \leqslant \mathcal{Z}_0^{-1}\int \mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V = 1$ *(*$\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} = \mathsf{w}_\xi$*, Section 9.4.2(c);* $\mathcal{Z}_0 = \int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]]\,\mathrm{d}V = \int \mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V$ *by the measurability paragraph of Section 3.1 and* (3.49)*;* $\mathcal{O}$ *is positive and* $\mathsf{w}_\xi^{\mathrm{lab}} \geqslant 0$*, as in the proof of Proposition 9.39): the labelled leading error has, in label average, a part* $O(\tau^2)$ *uniform in* n *and a part linear in* n_C *paid by the one-point witness probabilities (small for* g_0 *below the threshold of the comparison of Proposition 9.39, at fixed* n_C*); the first moment is not uniform in* n*. The exponential moment is (c).*

(c) (exponential moment of the leading error) Under (9.39)*(iii), with the constants* C*,* c *of* (9.40)*,* $N'_{\mathrm{lat}}(c)$*,* Φ_c*,* C_K *of Lemmas 9.45 and 9.46, and*

$$\delta_q \coloneqq C\tau^2(1+N_{\mathrm{ret}})N'_{\mathrm{lat}}(c)\,e^{C\tau^2 N_{\mathrm{ret}}\Phi_c}\,q_\vartheta^{1/(2C_K)}\,, \tag{9.49}$$

$$\mathcal{Z}_0^{-1}\int \mathcal{O}\Big[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}} e^{|E_\Lambda^\circ|}\Big]\mathrm{d}V \leqslant \exp\big[C\tau^2(2+1/c)+n_C\delta_q\big]\,, \tag{9.50}$$

by Corollary 9.43(c) ($|E_\Lambda^\circ| \leqslant C\tau^2(2+1/c) + C\tau^2\sum_j e^{-cd_j(\Lambda)}$*) and Lemma 9.46 at* $\lambda = C\tau^2$*. Consequently the leading term of* (9.31)–(9.32) *in the form of Corollary 9.43(b) satisfies*

$$\begin{aligned} &|\kappa_C|^{-1}\Big|\mathcal{Z}_0^{-1}\int \mathcal{O}\Big[\kappa_C\sum_{\mathrm{lab}} e^{E_\Lambda^\circ}\int \mathbf{1}_{\mathrm{lab}} e^{c_\xi^\Lambda} I_1^\Lambda \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}} W_\chi(C;U_1)\,\mathrm{d}a_{\mathrm{old}}\Big]\mathrm{d}V\Big| \\ &\quad\leqslant \exp\big[C\tau^2(2+1/c)+n_C\delta_q\big] \end{aligned} \tag{9.51}$$

(the left side is the modulus of the leading term of Corollary 9.43(b) divided by $|\kappa_C|$*;* $|W_\chi| \leqslant 1$*,* $|e^{E_\Lambda^\circ}| \leqslant e^{|E_\Lambda^\circ|}$*,* E_Λ° *and* $\mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}}$ *depend on the labels only, and the* a_{old}*-integral of* $\mathbf{1}_{\mathrm{lab}} e^{c_\xi^\Lambda} I_1^\Lambda$ *is* $\mathsf{w}_\xi^{\mathrm{lab}}$*, Section 9.3.2), and* $\mathcal{Z}_0^{-1}\int \mathcal{O}[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}} |e^{E_\Lambda^\circ}-1|]\,\mathrm{d}V \leqslant \exp[C\tau^2(2+1/c)+n_C\delta_q]-1$*: pointwise on every labelled term,* $\mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}}|e^{E_\Lambda^\circ}-1| \leqslant \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}}(e^{|E_\Lambda^\circ|}-1) \leqslant e^{C\tau^2(2+1/c)}\exp\big(C\tau^2\sum_j e^{-cd_j(\Lambda)}\big)-1$ *by Corollary 9.43(c) and* $\mathbf{1} \leqslant 1$ *(the last member is nonnegative), and the label average of the last member is at most* $\exp[C\tau^2(2+1/c)+n_C\delta_q]-1$ *by* (9.47) *with the indicator omitted and the exact normalisation* $\mathcal{Z}_0^{-1}\int \mathcal{O}[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}}]\,\mathrm{d}V = 1$ *of (b). (The indicator is not subtracted:* $\mathcal{Z}_0^{-1}\int \mathcal{O}[\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} \mathbf{1}_{A_{\mathrm{cov}}^{\mathrm{fin}}}]\,\mathrm{d}V$ *may be less than one.) Comparison: since* $\tau < 1$ *(fixed after* (5.18)*),* $\log\delta_q \leqslant \log(\tau^{-2}\delta_q) \leqslant \log C + \log(1+N_{\mathrm{ret}}) + \log N'_{\mathrm{lat}}(c) + C\tau^2 N_{\mathrm{ret}}\Phi_c - c_q\vartheta^2\pi_*^2/(2C_K)$*, where* $N'_{\mathrm{lat}}(c)$ *and* Φ_c *are constants of* c *and* $R'_0 = \lceil R_0\rceil + 2L + 1 \leqslant R_0 + 2L + 2$ *(*R_0 *the fixed range constant of* (7.28)*),* $\log N_{\mathrm{ret}} = O(\log\log g_0^{-2})$ (9.9)*,* $\tau^2 N_{\mathrm{ret}} \to 0$ *as* $g_0 \to 0$ *(*$\tau = C[g_0(1+r)^3 + e^{-cr^2}]$*,* $r = A_1(\log g_0^{-2})^{p_0}$*), and* $C_K = 4(4K\ell+9)^4 \leqslant 4(4K+9)^4\ell^4 \leqslant 4(4K+9)^4(L^2M)^4 2^{4\mathfrak{r}}(\log g_0^{-2})^{4\mathfrak{r}}$ *while* $\pi_*^2 \geqslant A_0^2 2^{-2p_0}(\log g_0^{-2})^{2p_0}$ *with* $2p_0 - 4\mathfrak{r} \geqslant 6\mathfrak{r} \geqslant 12$ *((*9.9*)–(*9.10*)); so for* g_0 *below a further threshold depending on* L*,* M*,* M_1*,* M_2*, the group, the character (through the constant* C *of* (9.40)*), the source constants,* ϑ *and* c*,*

$$\delta_q \leqslant \exp\big[-c_q\vartheta^2\pi_*^2/(4C_K)\big] \leqslant \exp\big[-c'(\log g_0^{-2})^{2p_0-4\mathfrak{r}}\big]\,, \tag{9.52}$$

smaller than every power of g_0*, with* $c' > 0$ *depending on* L*,* M*,* M_1*,* M_2*, the group, the character, the source constants and* ϑ*. The* n*-dependence of the label average of* $e^{|E_\Lambda^\circ|}$ *is thus the perimeter-rate factor* $e^{n_C\delta_q}$*, of the same kind as the factor* $n_C(q + N_{\mathrm{ret}}q_\vartheta)$ *of (a) and of* (9.34)*; a bound independent of* n *is not claimed, and the perimeter rate* δ_q *is governed by the* $2C_K$*-th root* $q_\vartheta^{1/(2C_K)}$ *of the one-point rate* q_ϑ *of bad collar bonds and round-0 bonds along the contour, the separation costing the factor* $2C_K$*, of order*

$(\log g_0^{-2})^{4\mathfrak{r}}$, *in the exponent. At the original carrier* $D = I$, *under the conventions of Section F.1, on compatible even fine periods (Section D.7) and for* $g_0 < g_*$ *(Proposition D.15), the same holds for the identity* (F.5) *of Corollary F.3, with* $E^\circ_\xi = E^\circ_{I,C_n,T}$ *of Proposition F.2 (whose bound has the form of* (9.40)*, with constants* C, c *fixed before the coupling, so Corollary 9.43(c) applies verbatim under* $4ne^{-c\iota_T} \leqslant 1$*), the history expectation* $\mathbb{E}_{\mathrm{h}}$ *of that corollary (weight* w_ξ*, physical* $\mathcal{Z}_0$*) and* (9.48) *at* $\lambda = C\tau^2$*:*

$$\begin{aligned} \mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}} e^{|E^\circ_\xi|}\big] &\leqslant \exp\big[C\tau^2(2+1/c) + n_C\delta_q^I\big]\,, \\ \delta_q^I &\coloneqq C\tau^2 N'_{\mathrm{lat}}(c)\, e^{C\tau^2\Phi_c}\, q^{1/C_K} \leqslant \exp\big[-c_q\pi_*^2/(2C_K)\big]\,, \end{aligned} \tag{9.53}$$

hence the leading term $\kappa_{C_n}\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)]$ *of* (F.5) *has modulus at most* $|\kappa_{C_n}|\exp[C\tau^2(2+1/c)+n_C\delta_q^I]$*, that is* $|\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^\circ_\xi} W_\chi(C_n;U_1)]| \leqslant \exp[C\tau^2(2+1/c)+n_C\delta_q^I]$*, and* $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^\circ_\xi}-1|] \leqslant \exp[C\tau^2(2+1/c)+n_C\delta_q^I]-1$ *by the pointwise bound* $\mathbf{1}_{A_{\mathrm{cov}}}|e^{E^\circ_\xi}-1| \leqslant e^{C\tau^2(2+1/c)}\exp(C\tau^2\sum_j e^{-cd_j})-1$ *(Corollary 9.43(c) for* E°_ξ *and* $\mathbf{1}\leqslant 1$*), (9.48) with the indicator omitted and* $\mathbb{E}_{\mathrm{h}}[1]=1$ *(*$\mathbb{E}_{\mathrm{h}}$ *is a probability expectation,* $\mathcal{Z}_0 = \int \mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V$*); the last inequality of* (9.53) *holds for* g_0 *below a further threshold, depending on* L*,* M*,* M_1*,* M_2*, the group, the character, the source constants and* c*, by the comparison above with* $q \leqslant e^{-c_*\pi_*^2} = e^{-c_q\pi_*^2}$ *for* $g_0 < g_*$ *(Proposition D.15,* $c_* = c_q$*; this is* (9.6) *at* $\vartheta = 1$*, where* $\theta_1 = \theta$ *and* $q_1 = q$*, as in the proof of Proposition 9.39);* g_* *is the threshold named in (H2) of the main theorem.*

Remark 9.48 (Scope of This Subsection). The identity $S^\circ_\Lambda = S^\circ_{D(\Lambda),C,T}$ on $A^{\mathrm{fin}}_{\mathrm{cov}}$ (Lemma 9.42) is a factorwise comparison of the two explicit integrals (precision, cutoff, primitives, Wilson factor) using the facts (I3) at $D = D(\Lambda) \subseteq D_+(\Lambda_1)$; the identification deferred in the closing remark of Section 3 (after Lemma 3.28) is, at the retreated carrier, reduced to these four factor identities. (1) Proposition 7.3 and the facts of the identification paragraph are used at their stated status; in particular (F3) is the label-by-label agreement of Lemma 7.2 between the primitives of Section 3.3 at the identity background (the exponent of Section 3, [3, (1.14)–(1.16)], Lemma 3.7) and the terms of (C.1) (energy residuals plaquette by plaquette, gauge terms centre by centre as sums over the pairs, Haar and constraint-Jacobian primitives bond by bond and row by row), the same lemma that the second paragraph of the proof of Proposition 7.3 invokes for its own kernel. (2) The inclusion $\Lambda_r \subseteq \mathcal{F}_r$ (the coverage stencil of Section 8 inside the free-input set of the traversal link), which is what makes $A^{\mathrm{fin}}_{\mathrm{cov}}$ imply the coverage of (I2), follows, for a free bond r, from $\Lambda_r = \{r\} = \mathcal{F}_r$ and, for a designated bond, from $\mathcal{F}_{b_0(c)} = \mathcal{E}_c \cap E_*$ (3.13) and $\Lambda_{b_0(c)} = \mathcal{S}_c \cap E_*$, i.e. from $\mathcal{S}_c \subseteq \mathcal{E}_c$, which holds by the two definitions (Appendix C: $\mathcal{S}_c$ consists of the positive fine bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$; Section 3.2: $\mathcal{E}_c$ consists of all bonds with both endpoints there), as recorded in the proof of Lemma 9.42. (3) The comparison statement of Section 7 is for $D_- \subseteq D \subseteq D_+$; $D(\Lambda) \subseteq D_+$ but $D_- \subseteq D(\Lambda)$ fails whenever a bond with both endpoints in Λ_1 is de-younged; Lemma 9.42 uses only $D(\Lambda) \subseteq D_+$ with the collar, which is what the arguments for the facts (1)–(5) summarised in (I3) use. (4) The label average of $e^{|E^\circ_\Lambda|}$ is bounded by $\exp[C\tau^2(2+1/c)+n_C\delta_q]$ with δ_q smaller than every power of g_0 (Corollary 9.47(c), from the joint witness bound of Lemma 9.45, which is the chessboard estimate (D.40) applied to several plaquettes at once after the reversal of Lemma D.13); its n-dependence is a perimeter rate, and a bound independent of n is not claimed. (5) $E^{\mathrm{lab}}_{\mathrm{def}}$ is bounded in Proposition 9.40 with S°_Λ retained ($|S^\circ_\Lambda| \leqslant 1$) and is untouched by this subsection; (8.15) already keeps $e^{E^\circ_\xi}$ inside the history expectation as a label-dependent factor, which is the form Corollary 9.43(b) gives at the retreated carrier. (6) (9.39)(i) adds the R_0-collar condition to the R_c-condition of threshold item 1 of Section B.8 (listed there among the conditions depending on the coupling through R_1); both are implied by conditions on M

alone, $LM - 4L^2M_2 > 4R_0 + 8$ and $LM - 4L^2M_2 > 4R_{\mathrm{c}} + 8$, since $R_1 \geqslant 1$, and are compatible (choose M after both; $R_0 = 4(1 + R_* + R_H + L)$ is fixed before M in the order of Section B.8, R_* and R_H being the ranges of Section 5); whether $R_0 \leqslant R_{\mathrm{c}}$ or conversely is not needed. (7) Further inputs used as stated: the reference-data cutoff convention behind (F2); $I = D_+(\Lambda_1)$ in Lemma 9.44; the geometry of $\Lambda_{b_0(c)}$ behind ϱ_K; the square orientation and $M \geqslant 8LM_2$ of (9.39)(i); the normalisation $\mathcal{Z}_0 = \int \mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V$ in Corollary 9.47(b); (7.20), (4.9), (C.1), (5.18) (τ), Lemma 3.2 ($\mathcal{E}_c$), (3.13), and Lemmas 9.2 and 9.13(c).

10 Two Disjoint Squares

This section extends the extraction of Section 3.8 and Appendices E and F from one square to the product of the Wilson functions of two vertex-disjoint squares in planes parallel to that of e_1, e_2. Nothing in the single-square sections is changed; the objects of this section are defined for the pair and compared with the single-square objects of each square. Section 10.7 records the corresponding statements at the retreated carrier of Section 9; Theorem 1.8 does not use them.

10.1 Setting

Let C_1, C_2 be coarse translates of the counterclockwise squares with vertices $0, n_iLe_1, n_iL(e_1+e_2), n_iLe_2$, $i = 1, 2$, in fine-lattice units (so both lie in planes parallel to that of e_1, e_2); let $V(C_i)$ be the set of fine sites of C_i, the endpoints of its bonds, and assume $V(C_1) \cap V(C_2) = \varnothing$. Let χ_1, χ_2 be normalised characters of irreducible unitary representations π_1, π_2 of G, $\chi_i = \dim(\pi_i)^{-1} \operatorname{tr} \pi_i$. The reversed orientation of C_i is the case $\bar{\chi}_i$, since $\chi_i(U^{-1}) = \overline{\chi_i(U)}$. Put $C = C_1 \sqcup C_2$ and

$$W_{\chi_1,\chi_2}(C; U) = \chi_1\big(\mathrm{Hol}_U\, C_1\big)\, \chi_2\big(\mathrm{Hol}_U\, C_2\big)\,. \tag{10.1}$$

The traversal set is $\mathcal{I}_C = \mathcal{I}_{C_1} \sqcup \mathcal{I}_{C_2}$, the traversals of each C_i numbered cyclically from the base point used for the single-square objects of C_i; $n_{C_i} = 4n_iL$ and $n_C = n_{C_1} + n_{C_2}$. Each square satisfies the square/torus geometry of Section 5.3 with its own side, $K_\mu \geqslant 3(n_i + 2)$ for every coarse period and $i = 1, 2$; no condition relating the positions of the two squares is imposed beyond $V(C_1) \cap V(C_2) = \varnothing$. For $a \in \mathcal{I}_{C_i}$ let δ_a be the periodic ℓ^∞ fine distance from the initial site of the traversal a to $V(C_{3-i})$, and put

$$\Delta_\times = \sum_{a \in \mathcal{I}_C} e^{-c_\times \delta_a/(LM)} \leqslant n_C e^{-c_\times d/(LM)}\,, \qquad d = \min_{a \in \mathcal{I}_C} \delta_a\,, \tag{10.2}$$

with the constant $c_\times > 0$ of Lemma 10.7, fixed before the coupling.

A superscript or subscript i marks the single-square objects of the square C_i with the character χ_i: the contact sets $\mathsf{T}_D^i = \mathsf{T}_D \cap \mathcal{I}_{C_i}$, the groups of Definition 3.24 for the contour C_i alone, their reference scalars $a_W^{(i)}$ of (3.92), the reference gas S°_{ξ,C_i} of (3.94), and the single-square lemmas of Appendix E with $C = C_i$. All of them are formed on the same branch ξ with the same carrier, cubes, hosts, measures and cluster occurrences: the domains, probe measures $\widehat{\nu}^{\mathrm{f}}_{\xi,D}$, vacuum clusters and reference data of Sections 3.5–3.8 do not involve the observable, which enters only through the contact sets, f_ξ, f_ξ° and the reference scalars a_W. The coupling is below the thresholds of the single-square results for χ_1 and for χ_2 and below those of this section.

Covered and far are taken for both squares: $\xi \in A_{\mathrm{cov}}$ means that ξ is covered (Lemma 3.28) for the contour $C = C_1 \sqcup C_2$, which, the conditions being imposed bond by bond, is the intersection of the covered events of C_1 and C_2; a history is far when $\mathrm{dist}(\Gamma_C, E(\Lambda_1^c)) > b$ with $\Gamma_C = \Gamma_{C_1} \cup \Gamma_{C_2}$, that is far for both squares.

10.2 Groups for Two Squares

For the pair, $f_\xi(a) = W_{\chi_1,\chi_2}(C; U_\xi(a))$, and the contact set T_D of (3.87) is taken over $n \in \mathcal{I}_C$. The argument preceding Definition 3.24 applies verbatim up to the formation of the components: unmarked domains contribute zero, $q_\xi(\mathcal{A})$ is expanded by (3.88), and the graph on the marked roots and the cluster occurrences of a term has components γ, each with a marked root and the contact set $\mathsf{T}_\gamma \subseteq \mathcal{I}_C$, the union of the contact sets of its roots. For $i = 1, 2$ let $H^i_\gamma = \varnothing$ if $\mathsf{T}_\gamma \cap \mathcal{I}_{C_i} = \varnothing$; otherwise H^i_γ is the hull of $\mathsf{T}_\gamma \cap \mathcal{I}_{C_i}$ in the cyclic set $\mathcal{I}_{C_i}$, defined as in Section 3.8: the shortest cyclic interval containing it when that interval has fewer than $n_{C_i}/2$ positions, and $\mathcal{I}_{C_i}$ otherwise. Put $H_\gamma = H^1_\gamma \sqcup H^2_\gamma$. Two components overlap when $H^i_\gamma \cap H^i_{\gamma'} \neq \varnothing$ for some i; the overlap classes are the connected components of the overlap graph on the components of the term. A set K of components is long on C_i when $\big|\bigcup_{\gamma\in K} H^i_\gamma\big| \geqslant n_{C_i}/2$.

Definition 10.1 (Groups for Two Squares)**.** *A partition $\mathfrak{P}$ of the components of a term is admissible if (A1) every overlap class lies in one block, and (A2) a block that is long on C_i contains every component γ with $\mathsf{T}_\gamma \cap \mathcal{I}_{C_i} \neq \varnothing$. The groups of the term are the blocks of its finest admissible partition (Lemma 10.2(a)). A group W consists of its root family $\mathcal{A}(W)$, its ordered list of cluster occurrences, and its interval set $\mathcal{H}(W) = \mathcal{H}_1(W) \sqcup \mathcal{H}_2(W)$, where $\mathcal{H}_i(W) = \mathcal{I}_{C_i}$ if W is long on C_i, read from the base point of C_i, and $\mathcal{H}_i(W) = \bigcup_{\gamma\in W} H^i_\gamma$ otherwise, a finite union of cyclic intervals (a cyclic interval or $\mathcal{I}_{C_i}$ for a C_i-group, by Lemma 10.2(d)); as in Definition 3.24, these data arise from a term, and conversely the term with root family $\mathcal{A}(W)$ and cluster list of W has W as its only group, with the same interval set (Lemma 10.2(a)). Its weight $w_\xi(W)$ is* (3.89)*. W is a C_i-group if $\mathcal{H}_{3-i}(W) = \varnothing$ and a cross group if $\mathcal{H}_1(W) \neq \varnothing \neq \mathcal{H}_2(W)$. Every group has $\mathcal{H}(W) \neq \varnothing$ (each component has a marked root), so it is exactly one of a C_1-group, a C_2-group and a cross group. Two groups are compatible when all their constituent domains are mutually compatible and $\mathcal{H}_i(W) \cap \mathcal{H}_i(W') = \varnothing$ for $i = 1, 2$. A compatible group family is a finite set of pairwise compatible groups; the singleton clause of Definition 3.24 is not imposed (for families of C_i-groups it is implied, Lemma 10.2(c)). For a compatible group family, $\mathcal{A}(\mathcal{S})$ and $A_\xi(\mathcal{S})$ are* (3.90)*.*

Lemma 10.2 (Two-Square Grouping)**.** *(a) Every term has a finest admissible partition; each of its blocks that is long on neither square is an overlap class. If no component of the term has a contact on C_{3-i}, it is the grouping of Section 3.8 for the square C_i. For a group W of a term, the term T_W with root family $\mathcal{A}(W)$ and the cluster list of W has W as its only group, with the same interval set. (b) The groups of a term are pairwise compatible, and Lemma 3.25 holds for the pair: $F_\xi(0) = \sum_{\mathfrak{B}\ \mathit{compatible}} A_\xi(\mathfrak{B})$, absolutely convergent. (c) A C_i-group of the pair is a group of the single square C_i (Definition 3.24 with $C = C_i$) none of whose roots has a contact on C_{3-i}; conversely every such single-square group is a C_i-group of the pair, with the same interval, weight and footprint (the footprint of a group of the pair is defined in Section 10.5). Two C_i-groups are compatible for the pair if and only if they are compatible for the square C_i; a C_1-group and a C_2-group are compatible if and only if their constituent domains are mutually compatible. (d) If W is a C_i-group that is not long on C_i, then $\mathcal{H}_i(W)$ is a cyclic interval of fewer than $n_{C_i}/2$ traversals.*

Proof. (a) The one-block partition is admissible. If $\mathfrak{P}$ and $\mathfrak{Q}$ are admissible, so is their

common refinement, whose blocks are the nonempty sets $P \cap Q$: (A1) holds because every overlap class lies in one block of each; for (A2), if $P \cap Q$ is long on C_i, then so are $P \supseteq P \cap Q$ and $Q \supseteq P \cap Q$ (the union of hulls is monotone), so both contain every component with a contact on C_i, and so does $P \cap Q$. There are finitely many components, so the common refinement of all admissible partitions is admissible and is the finest. If a block P of the finest partition is long on neither square and contains two overlap classes, replace P by its overlap classes: (A1) still holds, the new blocks are long on neither square (they are subsets of P), and the other blocks are unchanged, so the new partition is admissible and strictly finer, a contradiction. If no component has a contact on C_{3-i}, condition (A2) on C_{3-i} is void; if no overlap class is long the overlap classes form an admissible partition, hence the finest, and these are the classes of Section 3.8 for C_i with the intervals $\mathcal{H}(K)$ there, which equal the unions of the members' hulls; if some overlap class is long, every admissible partition has a long block containing it, which by (A2) contains every component (each has a contact on C_i), so the finest partition is the single class of all components, the whole-contour group of Section 3.8. The term of a group: let $\mathfrak{P}$ be the finest admissible partition of a term T and $W \in \mathfrak{P}$. Every cluster occurrence of W hits a root of T, which is joined to it by an edge and so lies in its component, hence in W; so T_W is a term of (3.88). Edges are defined pairwise, so the graph of T_W is the subgraph of that of T induced on W, and W being a union of components of T, the components of T_W are those of T in W, with the same contact sets, hulls and long-ness; by (A1) W is a union of overlap classes of T, which are the overlap classes of T_W. Let $\mathfrak{P}'$ be an admissible partition of T_W and $\mathfrak{P}'' = \mathfrak{P}' \cup (\mathfrak{P} \setminus \{W\})$. (A1) holds for $\mathfrak{P}''$, and (A2) for its blocks outside W; if $Q \in \mathfrak{P}'$ is long on C_i, then $W \supseteq Q$ is long on C_i, so W contains every component of T with a contact on C_i by (A2) for $\mathfrak{P}$, and all of them lie in Q by (A2) for $\mathfrak{P}'$. So $\mathfrak{P}''$ is admissible for T, $\mathfrak{P}$ refines it, and $\mathfrak{P}' = \{W\}$. Long-ness and hulls being the same in T and T_W, the interval set is the same.

(b) Distinct groups of a term have disjoint intervals: if neither is long on C_i, their C_i-parts are unions of hulls of components in different blocks, hence of different overlap classes by (A1), and such hulls are disjoint; if one is long on C_i, it contains every component with a contact on C_i, so the other has $\mathcal{H}_i = \varnothing$. Domains in different groups are compatible as in the proof of Lemma 3.25 (no edge of the term's graph joins different components). Every component's hull satisfies $H^i_\gamma \subseteq \mathcal{H}_i(W)$ for its group W: trivially if W is long on C_i, and by definition otherwise. Reversibility: let $\mathfrak{B} = \{W_1, \dots, W_m\}$ be a compatible family, each W_j being the only group of its own term by (a), and let $\mathfrak{P}$ be the finest admissible partition of the union term, whose components are those of the W_j (compatibility excludes edges between groups). $\mathfrak{B}$ is admissible for the union term: components of different W_j have hulls in disjoint intervals, so they do not overlap and (A1) holds; if W_j is long on C_i, then $\mathcal{H}_i(W_j) = \mathcal{I}_{C_i}$, so every other W_k has $\mathcal{H}_i(W_k) = \varnothing$, no contact on C_i, and W_j contains every component with a contact on C_i. Hence $\mathfrak{P}$ refines $\mathfrak{B}$. The restriction of $\mathfrak{P}$ to the components of W_j is admissible for the term of W_j alone: its overlap classes are those of the union term inside W_j, and a block of $\mathfrak{P}$ inside W_j that is long on C_i contains every component of the union term with a contact on C_i, in particular those of W_j. Since $\{W_j\}$ is the finest admissible partition of that term, $\mathfrak{P}$ restricted to W_j is the single block W_j, so $\mathfrak{P} = \mathfrak{B}$, with the same intervals. The rest of the proof of Lemma 3.25 (interleaving count, multiplicities of identical occurrences, absolute convergence) does not involve the intervals and applies verbatim.

(c) Let W be a C_i-group of the pair. Its roots are marked and $\mathcal{H}_{3-i}(W) = \varnothing$, so every root D has $\mathsf{T}_D = \mathsf{T}^i_D \neq \varnothing$, and the components, their hulls on C_i, the overlap graph and the long-ness on C_i of the term T_W of W are the same for the pair and for the square C_i. So T_W is a term of (3.88) for the square C_i (its roots are marked for C_i, and its clusters hit $\mathcal{A}(W)$ by (a)), no component of which has a contact on C_{3-i}; by (a) its finest admissible partition

$\{W\}$ is its grouping of Section 3.8 for C_i, with $\mathcal{H}(W) = \mathcal{H}_i(W)$. Conversely, if W is a group of C_i whose roots have no contact on C_{3-i}, its term (Definition 3.24) is a term of the pair, with $\mathsf{T}_D = \mathsf{T}_D^i \neq \varnothing$ for its roots, and by (a) its pair grouping is its grouping for C_i, the single group W, with $\mathcal{H}_i(W) = \mathcal{H}(W)$ and $\mathcal{H}_{3-i}(W) = \varnothing$. The weight (3.89) and the footprint of Section E.5 depend on the roots, the cluster list and the interval only. Compatibility for the pair of two C_i-groups is compatibility of domains and disjointness of the C_i-intervals, which is Definition 3.24 for C_i (its singleton clause for a whole-contour group being implied by the disjointness, since every group has a nonempty interval); for a C_1-group and a C_2-group the interval condition is void. (d) W is not long on C_{3-i}, since $\mathcal{H}_{3-i}(W) = \varnothing$, so by (a) it is an overlap class; its members have hulls on C_i only, so they are chained by intersections of these hulls, and $\mathcal{H}_i(W)$ is a union of chained cyclic intervals with fewer than $n_{C_i}/2$ positions in total, which is a cyclic interval by the argument of Section 3.8. *q.e.d.*

10.3 Reference Scalars and the Joint Extraction

For a C_i-group W let $P^\circ_W(a)$ be the ordered product of the reference links along the consecutive traversals of $\mathcal{H}_i(W)$, in the cyclic order of C_i starting from the first traversal of the interval (from the base point of C_i if W is long on C_i), as in Section 3.8, and define a_W by (3.92) with χ_i in place of χ; $\mathcal{H}_i(W)$ is a cyclic interval or $\mathcal{I}_{C_i}$ by Lemma 10.2(d), and $a_W = a_W^{(i)}$ by Lemma 10.2(c), since the reference links and the measures $\widehat{\nu}^{\mathrm{f},\circ}_{\xi,D}$ do not involve the observable (Lemma 3.21). For a cross group put $a_W \coloneqq 0$. With these numbers define, for the pair,

$$S^\circ_{\xi,C} = \sum_{\mathfrak{B}\ \text{compatible}} \prod_{W\in\mathfrak{B}} a_W \ , \tag{10.3}$$

in which a family containing a cross group contributes zero, the restricted sums $S^{\circ,[\mathfrak{B}]}_{\xi,C}$ over the families compatible with every member of $\mathfrak{B}$, and the joint defect $R_\xi(\mathfrak{B})$ by (3.95).

Lemma 10.3 (Schur Identity and Joint Extraction for Two Squares)**.** *(a) For a C_i-group W and $h \in G$,* (3.93) *holds with χ_i in place of χ. (b) Proposition 3.27 holds for the pair:* (3.96) *and* (3.97) *hold with $S^\circ_{\xi,C}$ of* (10.3)*, the sums converging absolutely. (c) At the reference data, $R^\circ_\xi(\mathfrak{B}) = 0$ for every nonempty compatible family $\mathfrak{B}$ without cross groups.*

Proof. (a) The proof of Lemma 3.26 for the square C_i: it uses only the invariance of the reference measures under the constant gauge rotations R_u, the covariance $P^\circ_W(R_u a) = u P^\circ_W(a) u^{-1}$ and Schur's lemma for π_i. (b) The proof of Proposition 3.27 is the Boolean inversion of (3.95) and an absolutely convergent rearrangement; it uses only $|a_W| \leqslant \beta_W$, which holds for $a_W = 0$, the finiteness of $\sum_W \beta_W$ (a group is determined by its root family, cluster list and interval, Lemma 3.20 and Lemma B.13) and Lemma 10.2(b) in place of Lemma 3.25. (c) Let $\mathcal{S}$ be a compatible family of C_1- and C_2-groups. The contacts on C_i of the roots of a C_i-group W lie in $\mathcal{H}_i(W)$ (Lemma 10.2(b)), and the roots of a C_{3-i}-group have no contact on C_i. So, as in the proof of Lemma 3.26, a traversal outside every interval of $\mathcal{S}$ has the reference link $\mathbf{1}$, and a traversal in $\mathcal{H}_i(W)$ has at $\mathsf{e}_{\mathcal{A}(\mathcal{S})}a$ its link at $\mathsf{e}_{\mathcal{A}(W)}a$, since it lies in no contact set of a root of another group; the links of $\mathcal{H}_i(W)$ multiply to $P^\circ_W(a)$. Number the C_i-groups of $\mathcal{S}$ in the cyclic order of C_i; by cyclicity of the trace the reference Wilson function at $\mathsf{e}_{\mathcal{A}(\mathcal{S})}a$ is $\prod_{i=1,2} \chi_i\big(\prod_{W\in\mathcal{S},\ C_i\text{-group}} P^\circ_W(a)\big)$, the product over the C_i-groups in that order (an empty product being $\mathbf{1}$), and a family containing a group long on C_i has no other C_i-group. Integrate one group at a time by Fubini, with (a) for the group integrated, the product of the other P° of the same square as h, and the character of the other square as a constant factor: $A^\circ_\xi(\mathcal{S}) = \prod_{W\in\mathcal{S}} a_W$. Hence $R^\circ_\xi(\mathfrak{B}) = \prod_{W\in\mathfrak{B}} a_W \sum_{\mathcal{S}\subseteq\mathfrak{B}} (-1)^{|\mathfrak{B}\setminus\mathcal{S}|} = 0$ for $\mathfrak{B} \neq \varnothing$. *q.e.d.*

For a family containing a cross group, $R^\circ_\xi(\mathfrak{B})$ need not vanish (for instance $R^\circ_\xi(\{W\}) = A^\circ_\xi(\{W\})$ for a cross group W); the defects of such families are bounded directly (Lemmas 10.5 and 10.7). In place of the second equality of (3.94), Lemma 10.2(b) at the reference data (its proof uses only the field-independent geometry, as in Section 3.8) and the Fubini step of (c) give

$$\frac{\int f^\circ_\xi K^\circ_{\xi;0}\,\mathrm{d}a}{I^\circ_{\xi;0}} = S^\circ_{\xi,C} + \sum_{\mathfrak{B}\text{ compatible, with a cross group}} A^\circ_\xi(\mathfrak{B})\ .$$

The choice $a_W \coloneqq 0$ is made for every cross group, and no reduction of the reference integrals of cross families to products of numbers is used. The pair scalar is identified with the single-square scalars by Lemma 10.7(b).

10.4 The Defects of a Family with Cross Groups

For a C_i-group W the closed contraction $c_W(U_1;h)$ of (E.18), its defect d_W and $\mathfrak{d}_W(U_1)$ of (E.19) are formed with χ_i and the interval $\mathcal{H}_i(W)$; by Lemma 10.2(c) they are the single-square quantities of C_i. For every group, cross groups included, $M(W)$ is the positive majorant of Section E.5.2: the product of the root weights $\rho\, m(D)$, the cluster weights $M_v(\mathfrak{c}_j)$ and $1/k(W)!$, with ρ of Lemma 10.4; for C_i-groups this only enlarges the single-square majorant, so every single-square bound by $M(W)$ still holds. Put $K_\chi = \max(K_{\chi_1}, K_{\chi_2})$ with the constants of Lemma E.7(a).

Lemma 10.4 (Sources with Two Closed Words)**.** *Let $f(a) = \prod_{i=1,2} \dim(\pi_i)^{-1} \operatorname{tr} Y_i(a)$, where $Y_i(a)$ is an ordered product of the matrices $\pi_i(U_\xi(a)(r))^{\pm 1}$ of a set of reconstructed contour links of C_i, each traversal used at most once, and of fixed unitary matrices of V_{π_i}. Then* (E.6), (E.60)–(E.61)*, the undifferentiated bound* (E.31) *and the last statement of Lemma E.14, and Lemma E.3 hold for f, with C_χ enlarged once to a constant $C^{\mathrm{src}}_{\chi_1\chi_2}$ of (χ_1,χ_2), so that $u = C^{\mathrm{src}}_{\chi_1\chi_2} g_0(1+r)$, and g_0 so small that $u \leqslant u_*$ and $v \leqslant 1$; in particular $|\mathfrak{Q}_\xi(\mathcal{A};f)| \leqslant \prod_{D\in\mathcal{A}} \rho\, m(D)$ for every nonempty compatible root family $\mathcal{A}$, with ρ of* (E.11) *formed with this u.*

Proof. f is the normalised trace of $Y_1(a)\otimes Y_2(a)$ on $V_{\pi_1}\otimes V_{\pi_2}$, and $|f| \leqslant 1$; since $\operatorname{tr}(A\otimes B) = \operatorname{tr} A \operatorname{tr} B$, the normalised trace on $V_{\pi_1}\otimes V_{\pi_2}$ satisfies $|\dim^{-1}\operatorname{tr}(M_1\cdots M_k)| \leqslant \prod_j \|M_j\|_{\mathrm{op}}$. The proof of (E.60) uses only this inequality, the unitarity of the untouched factors, the row locality and the fixed derivative constants of the rows. For the row locality: $V(C_1)\cap V(C_2) = \varnothing$, so the bond sets of C_1 and C_2 are disjoint, each bond carries at most one traversal of $C_1\cup C_2$, and the number of reconstructed links a free label affects is the geometric count of bonds r with the label in $\mathcal{F}_r$, at most K as for one square, each traversal being used at most once in the two words together; the row input count, which fixes the derivative order of (E.52), is unchanged. Then the product rule distributes the derivatives over the links of both words, with at most K^q row assignments, and every untouched factor, a reconstructed link or a fixed matrix, is unitary in $V_{\pi_1}\otimes V_{\pi_2}$. The remainder bounds (E.61) follow by the same Taylor argument. The proofs of (E.31), of the last statement of Lemma E.14 and of Lemma E.3 use, besides these bounds, only properties of the selected measures (the reversal of a root with empty ordinary selection, the Cauchy difference in ζ_j, the majorants M_ℓ and (E.10)) and the vanishing of $\Delta_\jmath f$ when a selected label does not influence the words; none of them uses the form of f otherwise. *q.e.d.*

Lemma 10.5 (Defect Factorisation for Two Squares)**.** *On a branch covered for $C_1 \sqcup C_2$*

(Section 10.1), for every compatible family $\mathfrak{B}$ with cross groups $\mathfrak{B}_\times \subseteq \mathfrak{B}$,

$$|R_\xi(\mathfrak{B})| \leqslant \prod_{W\in\mathfrak{B}\setminus\mathfrak{B}_\times} K_\chi \mathfrak{d}_W(U_1) \prod_{W\in\mathfrak{B}_\times} M(W)\,. \tag{10.4}$$

Proof. As in the proof of Lemma E.7(b), a traversal of $\mathcal{H}(W)$, $W \in \mathfrak{B}$, has no contact with a root of another group of $\mathfrak{B}$ (the intervals are disjoint and contain the contacts), so its reconstructed link depends only on the root variables of W, and a traversal outside every interval has its background link U_1 (Lemma 3.28, Section E.7.1). For a C_i-group $W \in \mathfrak{B}$ let $X_W = w_\xi(W) \int \pi_i(T_W P_W^{-1})\,\mathrm{d} \bigotimes_{D\in\mathcal{A}(W)} \widehat{\nu}^{\mathrm{f}}_{\xi,D}$ and $\Upsilon_W = X_W - a_W \mathbf{1}$, with T_W, P_W of Section E.5.1. Since $a_W = 0$ for cross groups, the term of $\mathcal{S}$ in (3.95) vanishes unless $\mathcal{S} \supseteq \mathfrak{B}_\times$. Fix the root variables $a_\times$ of the cross groups. For $i = 1, 2$ let Y_i be the ordered product, in the cyclic order of C_i starting at the first traversal of the interval of a C_i-group of $\mathfrak{B}$ if there is one and at the base point of C_i otherwise (the trace is cyclic; as in the proof of Lemma E.7(b)), of the following factors: for a C_i-group $W \in \mathfrak{B}$, the matrix $\Upsilon_W \pi_i(P_W)$ at the position of $\mathcal{H}_i(W)$; for a traversal in $\mathcal{H}_i(W)$ of a cross group W, its reconstructed link at $a_\times$; for every other traversal, its background link. Expanding each $\Upsilon_W = X_W - a_W\mathbf{1}$ and using Fubini over the product measures of the C_1- and C_2-groups, exactly as in the proof of Lemma E.7(b), the term of $\mathcal{S} \supseteq \mathfrak{B}_\times$ is reproduced, because f_ξ at $\mathsf{e}_{\mathcal{A}(\mathcal{S})}a$ is the product of the two normalised traces and each X_W enters exactly one of them; hence

$$R_\xi(\mathfrak{B}) = \Big(\prod_{W\in\mathfrak{B}_\times} w_\xi(W)\Big) \int \prod_{i=1,2} \frac{\operatorname{tr} Y_i(a_\times)}{\dim \pi_i}\,\mathrm{d} \bigotimes_{W\in\mathfrak{B}_\times} \bigotimes_{D\in\mathcal{A}(W)} \widehat{\nu}^{\mathrm{f}}_{\xi,D}(a_\times)\,.$$

The integrand is multilinear in the matrices Υ_W, $W \in \mathfrak{B}\setminus\mathfrak{B}_\times$. Write $\Upsilon_W = \|\Upsilon_W\|_{\mathrm{op}} T'_W$ with $\|T'_W\|_{\mathrm{op}} \leqslant 1$, and each contraction $T' = V|T'|$ (polar decomposition, V unitary) as $T' = \frac{1}{2}(Ve^{iA} + Ve^{-iA})$ with $A = \arccos|T'|$, an average of two unitary matrices. The integrand is then $\prod_W \|\Upsilon_W\|_{\mathrm{op}}$ times an average of $2^{|\mathfrak{B}\setminus\mathfrak{B}_\times|}$ functions of the form of Lemma 10.4, depending on $a_\times$ only through the links of the cross groups' traversals. If $\mathfrak{B}_\times \neq \varnothing$, that lemma applied to the compatible root family $\mathcal{A}(\mathfrak{B}_\times)$ bounds each such integral by $\prod_{D\in\mathcal{A}(\mathfrak{B}_\times)} \rho\, m(D)$ in modulus; if $\mathfrak{B}_\times = \varnothing$, there is no integral, and each of the unitary words has normalised traces of modulus at most one (for $\mathfrak{B} = \varnothing$ the bound is $|f_\xi(0)| \leqslant 1$). Further $|w_\xi(W)| \leqslant \prod_j M_v(\mathfrak{c}_j)/k(W)!$: by (3.73) the cluster weight is $\varphi^T/m!$ times the product of the activities, $|\varphi^T|$ is at most the number of spanning trees of the incompatibility graph (proof of Lemma 3.20, Section B.6), and the activities are bounded by $m(D)$ by (3.72) of Lemma 3.19. Together with $\|\Upsilon_W\|_{\mathrm{op}} \leqslant K_{\chi_i}\mathfrak{d}_W(U_1) \leqslant K_\chi \mathfrak{d}_W(U_1)$ for a C_i-group (Lemma E.7(a) for the square C_i) this is (10.4); the bound holds with K_{χ_i} on the C_i-groups. *q.e.d.*

10.5 Pins for the Groups of Two Squares

For a group W of the pair, $n(W)$, $\mathfrak{s}(W) = n(W) + |\mathcal{H}(W)|/(LM)$, the footprint $F(W)$, $X(W)$, $\pi_1(\mathcal{H}(W))$ and the envelope $\mathfrak{E}(W)$ are defined as in Section E.5 and Lemma E.8, with $\mathcal{H}(W) = \mathcal{H}_1(W) \sqcup \mathcal{H}_2(W)$ and the traversals of both squares; for components, t_γ, n_γ, X_γ, $F(\gamma)$ and $\mathfrak{s}(\gamma) = n_\gamma + |H_\gamma|/(LM)$ are those of Section E.7.2 with $H_\gamma = H^1_\gamma \sqcup H^2_\gamma$, and $\sigma_K = \sum_{\gamma\in K} \mathfrak{s}(\gamma)$. Put $A'_E = 17(2R_{\mathrm{e}} + 1)^4$.

Lemma 10.6 (Pins for Two Squares)**.** *Under the hypotheses of Lemma E.9, with its constant C enlarged by a fixed factor: (a) Lemma E.8 holds for every group W of the pair with $16n(W) + |\mathcal{H}(W)|/(LM)$ and $17\mathfrak{s}(W)$ in place of the bounds $8 + |\mathcal{H}(W)|/(LM)$ and $9\mathfrak{s}(W)$ of its part (a), A'_E in place of A_F and A_E, and $\operatorname{dist}(\Delta, C)$ the distance to $V(C_1) \cup V(C_2)$;*

if W is a cross group, $\pi_1(\mathcal{H}(W)) \subseteq \mathfrak{E}(W)$ contains a cube containing a site of $V(C_1)$ and a cube containing a site of $V(C_2)$. (b) Lemma E.17 holds for the components of the pair, with $|H^i_\gamma| \leqslant A_{\rm h} LMn_\gamma$ for $i = 1, 2$, the hull pin (E.39) *for $a \in \mathcal{I}_{C_i}$ taken with the contour distance d_{C_i} of C_i, and the sum over $a' \in \mathcal{I}_C$ in* (E.40)*. (c) Lemma E.18(a) holds for the overlap classes of the pair. A class long on C_i satisfies $\sigma_K \geqslant n_{C_i}/(2LM)$ and $\sum_{K \text{ long on } C_i} t_K e^{34\eta\sigma_K} \leqslant C\rho n_{C_i} e^{-15\eta n_{C_i}/(LM)}$. The groups long on some square have total weighted mass $\sum_{W \text{ long}} M(W) e^{4\eta\mathfrak{s}(W)} \leqslant 2C\rho\big(n_{C_1} e^{-10\eta n_{C_1}/(LM)} + n_{C_2} e^{-10\eta n_{C_2}/(LM)}\big) \leqslant C'\rho$ with $C' = 2CLM/(5e\eta)$, uniformly in n_1, n_2. (d) The pins* (E.21) *hold for the groups of the pair, with $\mathcal{I}_C$ the traversals of both squares; for every group W of the pair, cross groups included, $\sum_{W' \gtrdot W} |a_{W'}| e^{3\eta\mathfrak{s}(W')} \leqslant \frac{1}{8}\eta\mathfrak{s}(W)$, the sum over the groups of the pair; and Lemma E.10 and Lemma E.20 hold for the reference gas* (10.3)*, with $C\rho n_C$ in* (E.48)*.*

Proof. (a) Count. On each square, $\mathcal{H}_i(W)$ is empty, or $\mathcal{I}_{C_i}$, or, if W is not long on C_i, a nonempty union of member hulls with fewer than $n_{C_i}/2$ positions in total; in the second case each member hull is a cyclic interval (none is the whole of $\mathcal{I}_{C_i}$), each maximal cyclic interval of the union has fewer than $n_{C_i}/2$ positions because the union does, and there are at most $|W|$ of them, where $|W|$, the number of components of W, is at most $n(W) = \sum_\gamma n_\gamma$ since $n_\gamma \geqslant 1$. As in the proof of Lemma E.8(a), a cyclic interval of fewer than $n_{C_i}/2$ traversals meets each side in one run and lies in at most $8 + t/(LM)$ cubes for t traversals, and so does $\mathcal{I}_{C_i}$; summing over at most $2n(W)$ intervals, $|\pi_1(\mathcal{H}(W))| \leqslant 16n(W) + |\mathcal{H}(W)|/(LM)$, so $|X(W) \cup \pi_1(\mathcal{H}(W))| \leqslant 17\mathfrak{s}(W)$ and $|F(W)| \leqslant |\mathfrak{E}(W)| \leqslant A'_E \mathfrak{s}(W)$. Connectivity. For a component γ, X_γ is connected (proof of Lemma E.8(b)); $\pi_1(H^i_\gamma)$ is connected, H^i_γ being a cyclic interval or $\mathcal{I}_{C_i}$, and it contains the cube of the initial vertex of a contact of γ on C_i, which is at index distance at most one from a cube of a root of γ; so $X_\gamma \cup \pi_1(H_\gamma)$ is connected. Two overlapping components share a traversal, hence the cube of its initial vertex. So $X(W) \cup \pi_1(\mathcal{H}(W))$ is connected if W is an overlap class, which by Lemma 10.2(a) covers every group long on neither square. Let W be long on C_i. Each overlap class in W has a component with a contact on C_i, or W is also long on C_{3-i} and the class has a contact on C_{3-i}: otherwise splitting this class off W gives an admissible partition (the remainder is nonempty, since it contains the components of W with a contact on C_i; it keeps every component with a contact on a square on which W is long and keeps the same union of hulls there, the class is long on no square, being a subset of W without contacts on those squares and with the union of its hulls on the other square inside that of W), contradicting the finest choice. If W is long on C_i only, every class in W meets the connected set $\pi_1(\mathcal{I}_{C_i})$, so the union is connected. If W is long on both squares and no class in W has contacts on both, then splitting W into its classes with contacts on C_1 and those with contacts on C_2 is admissible (each part contains every component with a contact on the square on which it is long, and has no contact on the other square), a contradiction; so some class of W meets both $\pi_1(\mathcal{I}_{C_1})$ and $\pi_1(\mathcal{I}_{C_2})$, and the union is connected. Hence $\mathfrak{E}(W)$ is connected. The statements (c) and (d) of Lemma E.8 then follow by its proof. For a cross group both $\mathcal{H}_1(W)$ and $\mathcal{H}_2(W)$ are nonempty, and $\pi_1(\mathcal{H}(W))$ contains the cubes of their vertices.

(b) The proof of Lemma E.17. Part (a) there does not involve hulls. In (b) the contacts of γ on C_i lie within $2L$ of X_γ, and the square geometry of C_i (Section 5.3) gives $|H^i_\gamma| \leqslant A_{\rm h} LMn_\gamma$ and $d_{C_i}(a, a') \leqslant A_{\rm h} LMn_\gamma$ for $a \in H^i_\gamma$ and a contact a' of γ on C_i; the contour sum is over $\mathcal{I}_{C_i}$. In (E.40) the sum over the traversals of each square is bounded as there, and there are two squares. For part (c) of Lemma E.17: for the pair $\mathfrak{s}(\gamma) \leqslant (1 + 2A_{\rm h})n_\gamma \leqslant 2A_{\rm s} n_\gamma$, so $100\eta\mathfrak{s}(\gamma) \leqslant 200\eta A_{\rm s} n_\gamma < \varpi n_\gamma/40$ by $1000\eta A_{\rm s} < \varpi/8$, and the pins with the weight $e^{100\eta\mathfrak{s}(\gamma)}$ are dominated by those of (b) with the weight $e^{\varpi n_\gamma/4}$, with a margin $e^{\varpi n_\gamma/8}$; η is not changed. In the footprint pin, at most CLM traversals of each square have stencils meeting a given

cube, so at most $2CLM$ in all, and the footprint pin holds with the constant doubled.

(c) The proof of Lemma E.18(a) applies with $H_\gamma = H^1_\gamma \sqcup H^2_\gamma$: the components whose hulls meet H_γ have weighted mass at most $C\rho|H_\gamma|$ by the hull pins of (b) at the traversals of H_γ, and an overlap class long on neither square has the interval $\bigcup_\gamma H_\gamma$, of size at most $\sum_\gamma |H_\gamma| \leqslant LM\sigma_K$. A class long on C_i has $\sigma_K \geqslant \sum_\gamma |H^i_\gamma|/(LM) \geqslant n_{C_i}/(2LM)$; anchoring it at a traversal of $\bigcup_\gamma H^i_\gamma$ (n_{C_i} choices) and using the class pin with the weight $e^{64\eta\sigma_K}$ gives the stated sum, the exponent $30\eta\sigma_K$ being at least $15\eta n_{C_i}/(LM)$. Long groups: if no overlap class of a term is long, the overlap classes are admissible and no group is long. Let W be long; it contains an overlap class K_0 long on some square C_{i_0}: otherwise replacing W by its overlap classes keeps (A1), the new blocks are long on no square, and the other blocks are unchanged, so the new partition is admissible, and it is strictly finer unless W is a single class, which is then not long; the first contradicts the finest choice, the second the assumption that W is long. W is long on C_{i_0}, and by the splitting argument of (a) it is the union of the overlap classes with a contact on a square on which it is long. As in Lemma E.18(c), the sum of $M(W)$ over the groups with a given set of components is at most $\prod_\gamma t_\gamma$, and $\mathfrak{s}(W) \leqslant \sum_{\gamma\in W} \mathfrak{s}(\gamma) + \sum_{i:\, W \text{ long on } C_i} n_{C_i}/(LM)$. If W is long on C_{i_0} only, the classes other than K_0 are anchored at traversals of C_{i_0} and sum to at most $e^{C\rho n_{C_{i_0}}}$ with their weights $e^{64\eta\sigma}$, the interval weight is $e^{4\eta n_{C_{i_0}}/(LM)}$, and K_0 contributes $C\rho n_{C_{i_0}} e^{-15\eta n_{C_{i_0}}/(LM)}$; with $C\rho LM \leqslant \eta$ the product is at most $C\rho n_{C_{i_0}} e^{-10\eta n_{C_{i_0}}/(LM)}$. If W is long on both squares, let $j \neq i_0$; since $\sum_{\gamma\in W} |H^j_\gamma| \geqslant n_{C_j}/2$ and $|H^j_\gamma| \leqslant LM\mathfrak{s}(\gamma)$, $1 \leqslant e^{-15\eta n_{C_j}/(LM)} \prod_{\gamma\in W} e^{30\eta\mathfrak{s}(\gamma)}$; inserting this factor, the classes other than K_0 are anchored at traversals of C_1 or C_2 and sum to at most $e^{C\rho n_C}$ with the weights $e^{64\eta\sigma}$, the interval weight is $e^{4\eta n_C/(LM)}$, and K_0 contributes $C\rho n_{C_{i_0}} e^{-15\eta n_{C_{i_0}}/(LM)}$ with the weight $e^{34\eta\sigma_{K_0}}$; the product is at most $C\rho n_{C_{i_0}} e^{-10\eta n_C/(LM)} \leqslant C\rho n_{C_{i_0}} e^{-10\eta n_{C_{i_0}}/(LM)}$. The groups long on C_i only thus contribute at most $C\rho n_{C_i} e^{-10\eta n_{C_i}/(LM)}$, and the groups long on both squares at most that for each $i_0 = i$; summing over the four pairs (set of long squares, i_0) gives the bound, and $xe^{-10\eta x/(LM)} \leqslant LM/(10e\eta)$ for $x = n_{C_1}, n_{C_2}$ gives $C' = 2CLM/(5e\eta)$.

(d) The pins over groups long on neither square are the class pins of (c), as in Lemma E.18: such a group is an overlap class K (Lemma 10.2(a)), the sum of $M(W)$ over the groups with class K is at most t_K, $\mathfrak{s}(W) \leqslant \sigma_K$ and $F(W) \subseteq \bigcup_{\gamma\in K} F(\gamma)$; the long groups are bounded by their total mass (c), so (E.21) holds with the constant $C + C'$. For the second statement, a group $W' \not\sim W$ has a footprint cube in $\mathfrak{N}_1(F(W))$ or a traversal in $\mathcal{H}(W)$ by (a) and Lemma E.8(d), so the pins give at most $C\rho(3^4|F(W)| + |\mathcal{H}(W)|) \leqslant C\rho(3^4 A'_E + LM)\mathfrak{s}(W)$, which is at most $\eta\mathfrak{s}(W)/8$ under $C\rho LM \leqslant \eta/8$ with the constant C of (c) replaced by $C(1 + 2LM/(5e\eta))(1 + 3^4 A'_E/(LM))$; this factor, together with the doubling in (b), is the enlargement of the lemma, fixed with η before the coupling (and $e^{3\eta\mathfrak{s}} \leqslant e^{4\eta\mathfrak{s}}$, $|a_{W'}| \leqslant M(W')$). The proofs of Lemmas E.10 and E.20 then apply verbatim; the species of the reference gas are the groups with $a_W \neq 0$, all of them C_1- or C_2-groups, and the exclusion ratio (E.23) holds for every compatible family $\mathfrak{B}$ of groups of the pair, cross groups included, because the criterion was proved for every W. *q.e.d.*

10.6 The Cross Terms

Covered and far histories are those of Section 10.1. A single-square group W of C_i touches C_{3-i} if some root D of W has $\mathsf{T}_D \cap \mathcal{I}_{C_{3-i}} \neq \varnothing$, with the contact set T_D of the pair. Put $c_\times = 2\eta/A'_E$.

Lemma 10.7 (The Cross Terms)**.** *Under the hypotheses of Lemma 10.6 and, for (c), those of Lemma E.11 for each square, with a constant $C_\times$ fixed before the coupling and independent of n_1, n_2 and the positions of the squares, and with $C_\chi = \max_i C_{\chi_i}$ and $c = \min_i c_i$ formed from the constants C_{χ_i}, c_i of* (E.24) *for the square C_i: (a) On every branch ξ, for every cross group W of the pair and every $a \in \mathcal{H}(W)$, $\mathfrak{s}(W) > \delta_a/(A'_E LM)$; and $\sum_{W \text{ cross}} M(W) e^{\mathfrak{a}_W/8} \leqslant C_\times \rho \Delta_\times$. (b) Let $\mathscr{L}$ and $\mathscr{L}_i$ be the connected-cluster logarithms of the reference gas* (10.3) *of the pair and of the reference gas of the single square C_i (Lemma E.20), and $E^\times_\xi = \mathscr{L} - \mathscr{L}_1 - \mathscr{L}_2$. Then, on every branch ξ,*

$$S^\circ_{\xi,C} = S^\circ_{\xi,C_1} S^\circ_{\xi,C_2} e^{E^\times_\xi} , \qquad |E^\times_\xi| \leqslant C_\times \rho \Delta_\times ; \tag{10.5}$$

in particular $|S^\circ_{\xi,C}| \leqslant e^{C_\times \rho \Delta_\times}$ on every branch. (c) On covered far histories, with $\mathscr{R}_\xi$ the family sum (E.1) *of the pair,*

$$|\mathscr{R}_\xi| \leqslant |S^\circ_{\xi,C}| \Big[\exp\Big(C_\chi n_C \rho \big(\varepsilon_1 + e^{-cb/(LM)}\big) + C_\times \rho \Delta_\times \Big) - 1 \Big] . \tag{10.6}$$

Proof. Distance from an envelope. Let $\mathfrak{X}$ be a finite set of cubes, connected under adjacency, with $|\mathfrak{X}| \leqslant k$. As in the proof of Lemma E.8(c), any two sites in cubes of $\mathfrak{X}$ are at periodic fine ℓ^∞ distance less than kLM. Hence if $\mathfrak{X}$ contains the cube of the initial vertex of a traversal $a \in \mathcal{I}_{C_i}$ and a cube containing a site of $V(C_{3-i})$, then $\delta_a < kLM$.

(a) For a cross group W, $\mathfrak{E}(W)$ is connected, has at most $A'_E \mathfrak{s}(W)$ cubes and contains the cubes of the initial vertices of the traversals of $\mathcal{H}(W)$ and a cube containing a site of $V(C_1)$ and one containing a site of $V(C_2)$ (Lemma 10.6(a)); so $\delta_a < A'_E LM \mathfrak{s}(W)$ for every $a \in \mathcal{H}(W)$. Every group has a nonempty interval, so, pinning a cross group at a traversal of its interval and using $\mathfrak{a}_W/8 \leqslant \eta \mathfrak{s}(W)$, $e^{-3\eta \mathfrak{s}(W)} \leqslant e^{-3\eta \delta_a/(A'_E LM)} \leqslant e^{-c_\times \delta_a/(LM)}$ and the pin (E.21) of the pair (Lemma 10.6(d)),

$$\sum_{W \text{ cross}} M(W) e^{\mathfrak{a}_W/8} \leqslant \sum_{a \in \mathcal{I}_C} e^{-c_\times \delta_a/(LM)} \sum_{W \,|\, a \in \mathcal{H}(W)} M(W) e^{4\eta \mathfrak{s}(W)} \leqslant C \rho \Delta_\times .$$

The pins of Lemma 10.6 and (E.21), the criterion (10.7) below and Lemma E.20 use only $|a_W| \leqslant M(W)$ (Lemma E.3 at the reference data, Lemma B.13), the smallness of ρ and the field-independent geometry; coverage and farness enter only Lemmas 10.5 and E.11, so (a) and (b) hold on every branch.

(b) The three gases. The species of the pair gas with nonzero activity are its C_1- and C_2-groups; the species of the gas of C_i are the single-square groups of C_i, with the activities $a^{(i)}_W$. By Lemma 10.2(c) and the equality $a_W = a^{(i)}_W$ of Section 10.3, the C_i-groups of the pair are the single-square groups of C_i that do not touch C_{3-i}, with the same activities, and the compatibility relation of the pair restricted to them is that of the square C_i. Each gas satisfies the Kotecký–Preiss criterion in the following weighted form: for every species W of that gas,

$$\sum_{W' \not\sim W} |a_{W'}| e^{3\eta \mathfrak{s}(W')} \leqslant \frac{1}{8} \eta \mathfrak{s}(W) , \tag{10.7}$$

for the pair by Lemma 10.6(d), and for the square C_i by the proof of (E.22), which bounds $e^{\mathfrak{a}_{W'}}$ by $e^{4\eta \mathfrak{s}(W')}$ and so bounds $e^{3\eta \mathfrak{s}(W')}$ in the same way. This is the criterion of Section B.6 for the weights $\widetilde{z}_W = |a_W| e^{2\eta \mathfrak{s}(W)}$ and sizes $\eta \mathfrak{s}(W)$, so the rooted-tree argument there gives, for every species W of each gas,

$$\sum_{\mathfrak{c}} m_W(\mathfrak{c}) |\Psi(\mathfrak{c})| e^{2\eta \sum_{W' \in \mathfrak{c}} \mathfrak{s}(W')} \leqslant |a_W| e^{3\eta \mathfrak{s}(W)} , \tag{10.8}$$

the sum over the cluster tuples $\mathfrak{c}$ of that gas, $\Psi(\mathfrak{c}) = \frac{\varphi^T(\mathfrak{c})}{m!}\prod_{W'\in\mathfrak{c}} a_{W'}$, $m_W(\mathfrak{c})$ the multiplicity of W in $\mathfrak{c}$. The logarithms $\mathscr{L}$, $\mathscr{L}_i$ are the absolutely convergent series $\sum_{\mathfrak{c}}\Psi(\mathfrak{c})$ of the three gases, $S^{\circ}_{\xi,C} = e^{\mathscr{L}}$ and $S^{\circ}_{\xi,C_i} = e^{\mathscr{L}_i}$ (Lemma E.20, for the pair by Lemma 10.6(d)), which gives the identity in (10.5).

Cancellation. A cluster tuple of the pair gas all of whose members are C_i-groups is a cluster tuple of the gas of C_i with the same Ψ: the Ursell coefficient depends only on the incompatibility graph of the tuple, which is the same for both gases, and the activities agree. Conversely a tuple of the gas of C_i none of whose members touches C_{3-i} is such a tuple of the pair gas. Hence $E^{\times}_{\xi} = \sum_{\mathfrak{c}\in\mathcal{X}_0}\Psi(\mathfrak{c}) - \sum_{i=1,2}\sum_{\mathfrak{c}\in\mathcal{X}_i}\Psi(\mathfrak{c})$, where $\mathcal{X}_0$ is the set of tuples of the pair gas containing a C_1-group and a C_2-group, and $\mathcal{X}_i$ the set of tuples of the gas of C_i containing a group that touches C_{3-i}; the rearrangement is justified by absolute convergence.

Extent of the bridging tuples. Let $\mathfrak{c}\in\mathcal{X}_0\cup\mathcal{X}_1\cup\mathcal{X}_2$, let $W\in\mathfrak{c}$ and $a\in\mathcal{H}(W)$, $a\in\mathcal{I}_{C_i}$. The union $\mathfrak{X}$ of the envelopes of the members of $\mathfrak{c}$ is connected: each envelope is connected (Lemma 10.6(a) for the pair, Lemma E.8(b) for a single square), the incompatibility graph of a tuple with $\varphi^T(\mathfrak{c})\neq 0$ is connected, and two incompatible members have footprints at index distance at most one or a common traversal in their intervals (Lemma E.8(d), for the pair by Lemma 10.6(a)), the footprints and the cubes of the traversal vertices lying in the envelopes. $\mathfrak{X}$ contains the cube of the initial vertex of a, and a cube containing a site of $V(C_{3-i})$: for $\mathfrak{c}\in\mathcal{X}_0$, the cube of a traversal vertex of the interval of a member that is a C_{3-i}-group; for $\mathfrak{c}\in\mathcal{X}_j$ (then $i=j$), the cube of the initial vertex of a contact $n\in\mathsf{T}_D\cap\mathcal{I}_{C_{3-i}}$ of a root D of a member W_0 touching C_{3-i}, which is within $2L$ of D (proof of Lemma E.8: the stencil bond of the contact has its endpoints in D), hence at index distance at most one from a cube of $X(W_0)$ and in $\mathfrak{E}(W_0)$, since $R_{\mathrm{e}}\geqslant 1$. With $|\mathfrak{X}|\leqslant A'_E\sum_{W'\in\mathfrak{c}}\mathfrak{s}(W')$ ($A_E\leqslant A'_E$), the distance bound gives $\sum_{W'\in\mathfrak{c}}\mathfrak{s}(W') > \delta_a/(A'_E LM)$.

Bound. Every member of a tuple has a nonempty interval. Pinning each bridging tuple at a member W and a traversal $a\in\mathcal{H}(W)$, by (10.8) and the extent bound,

$$\begin{aligned}\sum_{\mathfrak{c}\in\mathcal{X}_0}|\Psi(\mathfrak{c})| &\leqslant \sum_{a\in\mathcal{I}_C} e^{-2\eta\delta_a/(A'_E LM)} \sum_{W\,|\,a\in\mathcal{H}(W)}\sum_{\mathfrak{c}} m_W(\mathfrak{c})|\Psi(\mathfrak{c})|e^{2\eta\sum_{W'\in\mathfrak{c}}\mathfrak{s}(W')}\\ &\leqslant \sum_{a\in\mathcal{I}_C} e^{-c_\times\delta_a/(LM)} \sum_{W\,|\,a\in\mathcal{H}(W)} M(W)e^{3\eta\mathfrak{s}(W)}\,,\end{aligned}$$

the inner sums over the species of the pair gas, and the pin (E.21) of the pair bounds the last sum by $C\rho\Delta_\times$. The same computation in the gas of C_i, with the single-square pin (E.21) for C_i (with $|a^{(i)}_W|\leqslant M_i(W)$ and $\rho_i\leqslant\rho$, since (E.11) is increasing in u) and the traversals of $\mathcal{I}_{C_i}$, bounds $\sum_{\mathfrak{c}\in\mathcal{X}_i}|\Psi(\mathfrak{c})|$ by $C\rho\sum_{a\in\mathcal{I}_{C_i}} e^{-c_\times\delta_a/(LM)}$. Summing, $|E^{\times}_{\xi}|\leqslant 2C\rho\Delta_\times$. The series are those of the countable index sets; their absolute convergence is Lemma E.20, and no truncation is needed. Finally $|S^{\circ}_{\xi,C_i}|\leqslant 1$ on every branch, by the second equality of (3.94) and $K^{\circ}_{\xi;0}>0$ (Section E.8.4: a normalised integral of a normalised character against a positive kernel), so $|S^{\circ}_{\xi,C}|\leqslant e^{|E^{\times}_{\xi}|}\leqslant e^{C_\times\rho\Delta_\times}$.

(c) By Lemma 10.3(b) the family sum of the pair is (E.1), $\mathscr{R}_\xi = \sum_{\mathfrak{B}\neq\varnothing} S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B})$ over the compatible families of groups of the pair. By the ratio bound (E.23) for the pair (Lemma 10.6(d), valid for families containing cross groups) and Lemma 10.5,

$$\big|S^{\circ,[\mathfrak{B}]}_{\xi,C}R_\xi(\mathfrak{B})\big| \leqslant |S^{\circ}_{\xi,C}| \prod_{i=1,2}\prod_{W\in\mathfrak{B},\ C_i\text{-group}} K_{\chi_i}\mathfrak{d}_W(U_1)e^{\mathfrak{a}_W/8} \prod_{W\in\mathfrak{B},\ \text{cross}} M(W)e^{\mathfrak{a}_W/8}\,,$$

with K_{χ_i} on the C_i-groups (the last sentence of the proof of Lemma 10.5). The C_i-groups of the pair are single-square groups of C_i with the same $\mathfrak{d}_W$ and $\mathfrak{a}_W$, and the pair's ρ dominates

that of the square C_i, as in (b), so Lemma E.11 for C_i (on histories covered and far for both squares) bounds the sum of the first factors over the C_i-groups by $C_{\chi_i} n_{C_i} \rho(\varepsilon_1 + e^{-c_i b/(LM)})$; the sum over all non-cross groups is at most $C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})$, and (a) bounds the sum of the second factors. Dropping the compatibility constraints, the sum over nonempty families is at most the exponential of the one-group sum minus one, as in the proof of Lemma E.12. The constant $C_\times$ is the largest of those above. *q.e.d.*

Corollary 10.8 (The Scalar of the Pair)**.** *Under the hypotheses of Lemma 10.7 and of Proposition F.2 for each square, on* A_{cov}*,*

$$S^\circ_{\xi,C} = \kappa_{C_1}\kappa_{C_2}\, e^{E^\circ_{\xi,1}+E^\circ_{\xi,2}+E^\times_\xi}\ , \tag{10.9}$$

with $E^\circ_{\xi,i}$ *the error* (F.4) *of the square* C_i*, bounded by* (F.3) *with* n_i*,* J_{C_i} *and the carrier distances* d_j *of the segments of* C_i*, and* $|E^\times_\xi| \leqslant C_\times \rho \Delta_\times$*.*

Proof. On A_{cov} the branch is covered for each square, so Proposition F.2 gives $S^\circ_{\xi,C_i} = \kappa_{C_i} e^{E^\circ_{\xi,i}}$; insert this in (10.5). *q.e.d.*

10.7 The Retreated Carrier

On a labelled term with final region Λ (Section 9) the carrier is $D(\Lambda)$ of Section 9.3.2, a function of Λ and I. The labels, hence $\Lambda = \Lambda_1^{\mathrm{fin}}$, come from the collar tests of Section 9.4.1, which test the fluctuation variables of collar bonds only, and the labelled weight (Lemma 9.7) contains no observable; so the labelled terms, their carriers and their weights do not depend on the contour, and both squares share them. The groups of the pair at $D(\Lambda)$ are those of Definition 10.1 formed from the domains for $D(\Lambda)$ and the contact sets T_Z of Lemma 9.21(a), taken over $n \in \mathcal{I}_C$; a^Λ_W is the reference scalar of Lemma 9.21(b) with χ_i for a C_i-group and $a^\Lambda_W := 0$ for a cross group, and $S^\circ_{\Lambda,C} := \sum_{\mathfrak{B}\ \text{compatible}} \prod_{W\in\mathfrak{B}} a^\Lambda_W$. S°_{Λ,C_i} is the labelled scalar of the square C_i, and $\mathscr{R}^\Lambda = \sum_{\varnothing\neq\mathfrak{B}\ \text{compatible}} S^{\circ[\mathfrak{B}]}_{\Lambda,C} R^0_\Lambda(\mathfrak{B})$ is the labelled defect (9.32) of the pair. $A^{\mathrm{fin}}_{\mathrm{cov}}$ (Section 9.4.2) and N_b (E.2) are taken for $C = C_1 \sqcup C_2$; the first is defined bond by bond, so it is the intersection of the single-square events, and the second, defined through $\Gamma_C = \Gamma_{C_1} \cup \Gamma_{C_2}$, is their union, so N^c_b is the intersection of the single-square far events.

Lemma 10.9 (The Pair at the Retreated Carrier)**.** *Under the hypotheses of Proposition 9.40 and of* (9.20) *for each square, and for* g_0 *below the thresholds of Lemmas 10.4 and 10.6, with their constants, on every labelled term, for exterior data in the support of the term's collar indicators and exterior cutoffs: (a)* $S^\circ_{\Lambda,C} = S^\circ_{\Lambda,C_1} S^\circ_{\Lambda,C_2} e^{E^\times_\Lambda}$ *with* $|E^\times_\Lambda| \leqslant C_\times \rho\Delta_\times$*, and* $|S^\circ_{\Lambda,C}| \leqslant e^{C_\times \rho \Delta_\times}$*; (b) on* $A^{\mathrm{fin}}_{\mathrm{cov}} \cap \mathsf{N}^c_b$*,* $\mathscr{R}^\Lambda$ *satisfies* (10.6) *with* $S^\circ_{\Lambda,C}$ *in place of* $S^\circ_{\xi,C}$*; (c) under* (9.39) *for each square, on* $A^{\mathrm{fin}}_{\mathrm{cov}}$*,* $S^\circ_{\Lambda,C} = \kappa_{C_1}\kappa_{C_2} e^{E^\circ_{\Lambda,1}+E^\circ_{\Lambda,2}+E^\times_\Lambda}$*, with* $E^\circ_{\Lambda,i}$ *the error of Corollary 9.43 for the square* C_i*.*

Proof. (i) Lemma 10.2 and Lemma 10.6(a) do not involve the carrier: they use the contact sets, which are field- and t-independent (Lemma 9.21(a)), and the footprints and hulls of the domains (Lemma 9.27). (ii) Lemma 10.3 at $D(\Lambda)$: the Schur identity at the reference data of Lemma 9.21(b) for each π_i, as in the proof of Lemma 10.3(a), and the Boolean inversion with $a^\Lambda_W := 0$ for cross groups, Lemma 9.21(c) with Lemma 10.2(b) in place of Lemma 3.25. (iii) Lemma 10.4 at $D(\Lambda)$, at exterior parameter zero, on $A^{\mathrm{fin}}_{\mathrm{cov}}$: its proof uses $|f| \leqslant 1$, the normalised-trace inequality, the row locality, the unitarity of the untouched factors and properties of the selected measures; the last are supplied at $D(\Lambda)$ by the paragraphs

"Charts and primitives" and "Parity and the source" of the proof of Proposition 9.40 (even cutoffs χ_b and density (3.66) at exterior parameter zero, the dummy comparison (E.10) as Lemma 9.20(c), the majorants, and the source remainder bounds of Section E.9.4 on $A_{\mathrm{cov}}^{\mathrm{fin}}$, where every contour row has its free inputs in $D(\Lambda)$ and no exterior variable enters a link, Lemma 9.25); those paragraphs use the source only through the properties listed in the proof of Lemma 10.4. So $|\mathfrak{Q}_\Lambda^0(\mathcal{A};f)| \leqslant \prod_{Z\in\mathcal{A}} \rho\, m(Z)$ for the two-word f. (iv) Lemma 10.5 at $D(\Lambda)$: on $A_{\mathrm{cov}}^{\mathrm{fin}}$ the contour links are reconstructed from the young variable on $D(\Lambda)$ at the background U_1 (proof of Proposition 9.40(a), Lemma 9.36), so a traversal outside the intervals has the link U_1 and one in $\mathcal{H}(W)$ depends on the roots of W only; the weights obey $|w_\Lambda^0(W)| \leqslant \prod_j |\Phi|_\Lambda^+(\mathfrak{c}_j)/k(W)!$ (Corollary 9.16, Lemma 9.24); with (iii) the proof gives (10.4) with $\mathfrak{d}_W^0$ and $M_\Lambda^{(\rho)}(W)$. (v) Lemma 10.6(b)–(d) at $D(\Lambda)$: these are the summations of Lemmas E.17 and E.18 with the anchoring of the pair, and they use only the root weights $\rho\, m(Z)$, the tree weights $|\Phi|_\Lambda^+ = M_v$ and the field-independent domain, cluster and interval geometry, which is the reason Lemma 9.27(a) gives for one square; so they hold for $M_\Lambda^{(\rho)}$ with the constants of Lemma 10.6, and, since $\sigma(W) = \mathfrak{s}(W)$, $\nu = \eta$ and $C_F = A_E$ there, $c_\times$ is unchanged. In particular the weighted criterion (10.7) holds at $D(\Lambda)$ for the pair, from $|a_W^\Lambda| \leqslant M_\Lambda^{(\rho)}(W)$ and the pins with $3^4 A'_E + LM$ in place of $3^4 A_F + LM$, and for each square as in the proof of Lemma 10.7(b). (vi) With (i)–(v), the proof of Lemma 10.7(a),(b) gives (a), with $|S^\circ_{\Lambda,C_i}| \leqslant 1$ (Lemma 9.38) in place of $|S^\circ_{\xi,C_i}| \leqslant 1$; for (b), the proof of Lemma 10.7(c) uses (9.37) for each square (Lemma E.11 at $D(\Lambda)$ in the form with $e^{\mathfrak{a}W/8}$, proved in the paragraph "Pins" of the proof of Proposition 9.40). (c) is (a) with Corollary 9.43(a) for each square, $S^\circ_{\Lambda,C_i} = \kappa_{C_i} e^{E^\circ_{\Lambda,i}}$ on $A_{\mathrm{cov}}^{\mathrm{fin}}$. *q.e.d.*

Lemma 10.9 is not used in the proof of Theorem 1.8, which works at the carrier I throughout (Section 10.8), as Theorem 1.2 does. For the pair it gives, in (c), the identification that Remark 1.3(2) records for one square (Corollary 9.43(a)); in (a), the factorisation and the bound of $S^\circ_{\Lambda,C}$; and in (b), the far labelled family sum, the pair inputs of (9.36). The labelled defect bound (9.36) and Corollary 9.47 are not restated for the pair.

10.8 Proof of Theorem 1.8

The single-contour statements used by Theorem 1.2 outside the extraction (Lemmas D.1, D.3, D.6, Corollary D.8, Lemmas D.9, D.14, Proposition D.15, Lemma E.1, (E.35) and Lemma 9.46) involve the contour through its observable f_ξ, the set I_C of its free inputs (D.6), the set Γ_C of its contour bonds and free-input templates (Section D.7) and its segments, and, in Lemma E.1 and (E.35), through the bound $|S^\circ_{\xi,C}| \leqslant 1$ of (3.94), which for the pair is replaced by $|S^\circ_{\xi,C}| \leqslant e^{C_\times \rho\Delta_\times}$ of (b) of Lemma 10.10. For the pair, f_ξ is (10.1) at the reconstructed field, $I_C = I_{C_1} \cup I_{C_2}$, $\Gamma_C = \Gamma_{C_1} \cup \Gamma_{C_2}$, so that $|\Gamma_C| \leqslant C_L n_{C_1} + C_L n_{C_2} = C_L n_C$, and $J_C = J_{C_1} \sqcup J_{C_2}$ is the set of the $4n_1 + 4n_2$ coarse segments of the two squares, d_j being the distance (F.1) of the square of the segment j at the common carrier I. The constants C, c are those of Theorem 1.8.

Lemma 10.10 (Single-Contour Inputs for the Pair)**.** *Under the hypotheses of Theorem 1.8: (a)* (D.8) *holds for the pair with $C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2}$ in place of $C_{\mathrm{resp},\chi}$, and so do Lemma D.3, Lemma D.6 and Corollary D.8;* (D.41) *holds for the pair with $C_{\mathrm{resp},\chi_1\chi_2}$. (b) On A_{cov}, $f_\xi(0) = W_{\chi_1,\chi_2}(C;U_1)$ and $|S^\circ_{\xi,C}| \leqslant e^{C_\times \rho\Delta_\times}$, and* (E.35) *holds for the pair, every term being integrable against the history weight. (c) On A_{cov}, $|\mathscr{R}_\xi| \leqslant 1 + e^{C_\times \rho\Delta_\times}$; and $\mathbb{P}_{\mathrm{h}}(\mathsf{N}_b) \leqslant 4C_L n_C (2b+3)^4 q_{\mathrm{pt}}$ and $|E_{\mathrm{def}}^{\mathrm{near}}| \leqslant e^{C_\times \rho\Delta_\times} C'_L n_C e^{-(c_*/2)\pi_*^2}$, with N_b and $E_{\mathrm{def}}^{\mathrm{near}}$ of Section E.2 formed with Γ_C and $\mathscr{R}_\xi$ of the pair. (d) Put $\varphi_c(b) := \sum_{r\in C_1\sqcup C_2} e^{-c(d_T(r,b)-R'_0)_+}$, the sum of the functions φ_c of*

(9.45) *of the two squares. Pointwise on every branch,* $\sum_{j\in J_C} e^{-cd_j} \leqslant \sum_b \varphi_c(b)\mathbf{1}\{b \in E(B_\xi)\}$, *with* $\varphi_c(b) \leqslant 2\Phi_c$ *and* $\sum_b \varphi_c(b) \leqslant n_C N'_{\mathrm{lat}}(c)$*; and for every* $\lambda > 0$*, also with* $\mathbf{1}_{A_{\mathrm{cov}}}$ *omitted,*

$$\mathcal{Z}_0^{-1} \int \mathcal{O}\Big[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}}} \exp\Big(\lambda \sum_{j\in J_C} e^{-cd_j}\Big)\Big] \mathrm{d}V \leqslant \exp\big[n_C \lambda N'_{\mathrm{lat}}(c)\, e^{2\lambda\Phi_c}\, q^{1/C_K}\big] \,.$$

(e) For g_0 *below a further threshold depending on* L*,* M*,* M_1*,* M_2*, the group, the characters, the source constants and* c*,*

$$\delta_q^{I,2} := C\tau^2 N'_{\mathrm{lat}}(c)\, e^{2C\tau^2\Phi_c}\, q^{1/C_K} \leqslant \exp\big[-c_q\pi_*^2/(2C_K)\big] \,. \tag{10.10}$$

Proof. (a) On the enlarged box the real reconstruction gives group-valued links (proof of Lemma D.3), so $|\chi_i(Y_i)| \leqslant 1$ for the ordered link product Y_i of C_i and $|f_\xi| \leqslant 1$. By the product rule, $\nabla_b f_\xi = (\nabla_b\chi_1(Y_1))\chi_2(Y_2) + \chi_1(Y_1)\nabla_b\chi_2(Y_2)$, and on a branch covered for the pair, which is covered for each square (Section 10.1), the proof of (D.8) for the square C_i gives $h(\nabla_b\chi_i(Y_i)) \leqslant C_{\mathrm{resp},\chi_i} g_0 \mathbf{1}_{I_{C_i}}(b)$, so $h(\nabla_b f_\xi) \leqslant |\chi_2(Y_2)|\, h(\nabla_b\chi_1(Y_1)) + |\chi_1(Y_1)|\, h(\nabla_b\chi_2(Y_2)) \leqslant (C_{\mathrm{resp},\chi_1}\mathbf{1}_{I_{C_1}}(b) + C_{\mathrm{resp},\chi_2}\mathbf{1}_{I_{C_2}}(b)) g_0 \leqslant (C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2}) g_0 \mathbf{1}_{I_C}(b)$, which is (D.8) for the pair. Lemma D.3 uses the observable through $\widetilde{f} = f_\xi \circ T$, $|\widetilde{f}| \leqslant 1$ and (D.8) composed with DT; Lemma D.6 uses it through Lemma D.3 and the support I_C; Corollary D.8 is that lemma averaged. The enlargement in the proof of Lemma D.6 multiplies the constant of (D.8) by a factor independent of the character, so the enlarged pair constant is the sum of the enlarged constants of χ_1 and χ_2. The proof of Proposition D.15 combines Corollary D.8 with Lemmas D.9 and D.14, whose bounds are sums of one-point marks over the members of Γ_C; with $|\Gamma_C| \leqslant C_L n_C$ it gives (D.41) with $C_L \max((C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2})S_c, 1) = C_{\mathrm{resp},\chi_1\chi_2}$.

(b) Lemma 3.28 is proved bond by bond, so on A_{cov}, $U_\xi(0)(r) = U_1(r)$ for every bond r of C_1 and of C_2, and $f_\xi(0) = W_{\chi_1,\chi_2}(C; U_1)$. By Lemma 10.7(b), $|S^\circ_{\xi,C}| \leqslant |S^\circ_{\xi,C_1}||S^\circ_{\xi,C_2}| e^{|E^\times_\xi|} \leqslant e^{C_\times \rho\Delta_\times}$, where $|S^\circ_{\xi,C_i}| \leqslant 1$ because it is a normalised integral of a normalised character against the positive reference kernel (proof of Proposition E.22). The derivation of (E.35) in Section E.7.1 multiplies (3.97), which holds for the pair by Lemma 10.3(b), by $\mathbf{1}_{A_{\mathrm{cov}}}$ and writes $\mathcal{J}^{\mathrm{a}}_\xi[f_\xi] = \mathsf{w}_\xi F_\xi(1)$ on $A^{\mathrm{c}}_{\mathrm{cov}}$; (3.7) holds with $W_{\chi_1,\chi_2}(C)$ in place of $W_\chi(C)$, f_ξ being jointly Borel with $|f_\xi| \leqslant 1$ (the measurability paragraph of Section 3.1). Its integrability uses $|F_\xi(s)| \leqslant 1$ ($F_\xi(s) = \int f_\xi \,\mathrm{d}\nu_{\xi;s}$ with $|f_\xi| \leqslant 1$), $|\mathcal{B}_\xi[f_\xi]| \leqslant (C_{\mathrm{resp},\chi_1} + C_{\mathrm{resp},\chi_2}) g_0 r |I_C|$ on A_{cov} by (a), and a bound on $S^\circ_{\xi,C} f_\xi(0)$, which is now $|S^\circ_{\xi,C} f_\xi(0)| \leqslant e^{C_\times\rho\Delta_\times}$ in place of 3.

(c) On A_{cov}, $\mathscr{R}_\xi = F_\xi(0) - S^\circ_{\xi,C} f_\xi(0)$ by (3.96) for the pair (Lemma 10.3(b)), so $|\mathscr{R}_\xi| \leqslant 1 + e^{C_\times\rho\Delta_\times} \leqslant 2e^{C_\times\rho\Delta_\times}$ by (b). The union bound of the proof of Lemma E.1, over the at most $C_L n_C$ members of Γ_C, gives the bound on $\mathbb{P}_{\mathrm{h}}(\mathsf{N}_b)$. Positivity of $\mathbb{E}_{\mathrm{h}}$ then gives $|E^{\mathrm{near}}_{\mathrm{def}}| \leqslant 2e^{C_\times\rho\Delta_\times} \cdot 4C_L n_C (2b+3)^4 q_{\mathrm{pt}}$, and the last step of that proof gives $8C_L(2b+3)^4 q_{\mathrm{pt}} \leqslant C'_L e^{-(c_*/2)\pi_*^2}$.

(d) The first inequality of (9.48) is proved segment by segment, the term of a segment being bounded through the contour bonds of its own square (proof of Lemma 9.46). The two squares' inequalities at the common carrier add up to the pointwise bound. Each square's φ_c is at most Φ_c and has sum at most $n_{C_i} N'_{\mathrm{lat}}(c)$, by the counts of that proof. The rest of that proof applies with the weights $w_b = \varphi_c(b) \leqslant 2\Phi_c$: the expansion over the marks $\mathbf{1}\{b \in E(B_\xi)\}$, one per bond, and the joint bound (9.44), which holds for every finite set of bonds. They bound the average by $\exp(\hat{q}\sum_b (e^{\lambda w_b} - 1))$ with $\hat{q} = q^{1/C_K}$. Since $e^x - 1 \leqslant xe^x$ for $x \geqslant 0$, this is at most $\exp(\hat{q}\lambda e^{2\lambda\Phi_c}\sum_b w_b)$, and the indicator enters only through $\mathbf{1} \leqslant 1$.

(e) Since $\tau < 1$ and $q \leqslant e^{-c_*\pi_*^2} = e^{-c_q\pi_*^2}$ (Proposition D.15), $\log\delta_q^{I,2} \leqslant \log(CN'_{\mathrm{lat}}(c)) + 2C\Phi_c - c_q\pi_*^2/C_K$. Moreover $\pi_*^2/C_K \geqslant c'(\log g_0^{-2})^{2p_0-4\mathfrak{r}}$ with $2p_0 - 4\mathfrak{r} \geqslant 12$ (the arithmetic

preceding (9.52), Remark 1.3(1)). So $c_q\pi_*^2/(2C_K) \geqslant \log(CN'_{\rm lat}(c)) + 2C\Phi_c$ for small g_0, which is (10.10). *q.e.d.*

Proof of Theorem 1.8. (i) By Lemma 10.10(b),

$$\mathbb{E}_\varepsilon[W_{\chi_1,\chi_2}(C)] = \mathbb{E}_{\rm h}[\mathbf{1}_{A_{\rm cov}} S^\circ_{\xi,C} W_{\chi_1,\chi_2}(C;U_1)] + E_{\rm def} + E_{\rm bdry} \,,$$

with all terms integrable. On $A_{\rm cov}$ substitute (10.9) (Corollary 10.8). Its hypotheses are those of Lemma 10.7 and, for each square, those of Proposition F.2, which (H2) and (H3) for that square imply as in the proof of Theorem 1.2. After the substitution the first integrand is still exactly $\mathbf{1}_{A_{\rm cov}} S^\circ_{\xi,C} f_\xi(0)$. The statements on κ_{C_i} are Theorem 1.2(i) for (C_i, χ_i).

(ii) The pair's $A_{\rm cov}$ is contained in the covered event of each square, on which Proposition F.2 bounds $E^\circ_{\xi,i}$ with the constants C_i, c_i of χ_i; these bounds are at most the stated ones, since $C_i \leqslant C$ and $e^{-c_i x} \leqslant e^{-cx}$ for $x \geqslant 0$. The elementary bounds give $C\tau^2(9n_i+1)$, as in Theorem 1.2. The bound on $E^\times_\xi$ is Lemma 10.7(b) with (10.2). Split $E_{\rm def} = E^{\rm near}_{\rm def} + E^{\rm far}_{\rm def}$ as in Section E.2, both terms being integrable by Lemma 10.10(c). The near term is bounded in Lemma 10.10(c). The histories of $A_{\rm cov} \cap \mathsf{N}_b^c$ are covered and far for the pair. On them (10.6) and Lemma 10.10(b) give $|\mathscr{R}_\xi| \leqslant e^{C_\times \rho \Delta_\times}(e^{X + C_\times \rho \Delta_\times} - 1)$ with $X = C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})$. So $|E^{\rm far}_{\rm def}| \leqslant e^{C_\times \rho \Delta_\times}(e^{X + C_\times \rho \Delta_\times} - 1)$, by the positivity of $\mathbb{E}_{\rm h}$ and $\mathbb{E}_{\rm h}[1] = 1$. The bound on $E_{\rm bdry}$ is (D.41) for the pair (Lemma 10.10(a)).

(iii) The first bound follows from the positivity of $\mathbb{E}_{\rm h}$, $\mathbb{E}_{\rm h}[1] = 1$, $|W_{\chi_1,\chi_2}| \leqslant 1$ and $|e^z| \leqslant e^{|z|}$ with (ii). Under the condition of (H3) for (iii) for the square C_i, $4n_i e^{-c_i \iota_T} \leqslant 1$ ((9.39)(iii)). So Corollary 9.43(c), applied to $E^\circ_{\xi,i}$ as in the proof of Theorem 1.2, gives $|E^\circ_{\xi,i}| \leqslant C_i\tau^2(2+1/c_i) + C_i\tau^2 \sum_{j \in J_{C_i}} e^{-c_i d_j} \leqslant C\tau^2(2+1/c) + C\tau^2 \sum_{j \in J_{C_i}} e^{-cd_j}$ on $A_{\rm cov}$. Hence $\mathbf{1}_{A_{\rm cov}} e^{|E^\circ_{\xi,1}| + |E^\circ_{\xi,2}|} \leqslant e^{2C\tau^2(2+1/c)} \mathbf{1}_{A_{\rm cov}} \exp(C\tau^2 \sum_{j \in J_C} e^{-cd_j})$, and Lemma 10.10(d) at $\lambda = C\tau^2$, with (e), gives (1.20). The modulus of the leading term is at most $|\kappa_{C_1}\kappa_{C_2}| e^{C_\times \rho \Delta_\times} \mathbb{E}_{\rm h}[\mathbf{1}_{A_{\rm cov}} e^{|E^\circ_{\xi,1}| + |E^\circ_{\xi,2}|}]$, by $|W_{\chi_1,\chi_2}| \leqslant 1$ and (ii). Pointwise on $A_{\rm cov}$, with $z = E^\circ_{\xi,1} + E^\circ_{\xi,2} + E^\times_\xi$, $|e^z - 1| \leqslant e^{|z|} - 1 \leqslant e^{2C\tau^2(2+1/c) + C_\times \rho \Delta_\times} \exp(C\tau^2 \sum_{j \in J_C} e^{-cd_j}) - 1$, a nonnegative function. Its $\mathbb{E}_{\rm h}$-average without the indicator is at most $\exp[2C\tau^2(2+1/c) + C_\times \rho \Delta_\times + n_C \delta_q^{I,2}] - 1$, by Lemma 10.10(d) and $\mathbb{E}_{\rm h}[1] = 1$. *q.e.d.*

11 Towards the Next Step

The statement of this section is a conjecture; nothing in the other sections uses it. It concerns loops whose side is a power of the block size, $n = L^k$ with an integer $k \geqslant 0$, after $k+1$ steps (the letter K is the exponent of [4] in (H3) and the index of σ_K, C_K in Lemma 9.45). On a torus of (H3) the lattice $L^{k+1}\mathbb{Z}^4$ and its plaquettes are defined ($K_\mu = 2L^{m-1} \geqslant 3(n+2)$ forces $m \geqslant k+2$, so L^{k+1} divides every period $N_\mu = 2L^m$), and a translation of C_n does not change $\mathbb{E}_\varepsilon[W_\chi(C_n)]$ (the Wilson measure $\rho_0(U)\,\mathrm{d}U$ on the periodic torus is invariant under lattice translations, its action being a sum over all plaquettes and the Haar measure a product measure; Section 5.3 uses the remaining lattice symmetries), so we take the corners of C_n in $L^{k+1}\mathbb{Z}^4$; then C_n, of side L^{k+1} fine units, is a contour of the lattice of spacing $L^j\varepsilon$ for every $j \leqslant k+1$ and the boundary of a single plaquette of the lattice of spacing $L^{k+1}\varepsilon$. The scales τ and π_* are those of the notation paragraph of Section 1.2, at the bare coupling g_0.

Conjecture 11.1 (The Loop along the Scales)**.** *Let G, χ be as in (H1) and fix the cutoff prescription σ and the constants as in (H2). There are $C, c > 0$, depending on L, the group, the character, σ and the constants fixed in (H2) only, thresholds $g_j^* > 0$ ($j \geqslant 1$) and, for*

$0 < g_0 < g_j^*$, *real numbers* δm_j *and complex numbers* ζ_j, *depending on* g_0, ε, j, L, *the group, the character,* σ *and the fixed prescriptions listed in Theorem 1.2(i) only, with* $z_j := e^{\zeta_j}$ *and*

$$\delta m_1 = \delta m^\sigma(g_0,\varepsilon)\,, \qquad \zeta_1 = \frac{1}{4}\sum_{(v,w)\in\mathscr{C}_\square} \zeta^\sigma_{v,w}(g_0)\,, \qquad |\delta m_j|\, L^j\varepsilon \leqslant C\tau^2\,, \qquad |\zeta_j| \leqslant C\tau^2\,, \tag{11.1}$$

so that $z_1 = z_{\mathrm{c}}^\sigma(g_0,\pi/2)$ (6.12), *such that for every integer* $k \geqslant 0$, *with* $n = L^k$, *there is* $g_{\mathrm{it}}(k) \leqslant \min_{j\leqslant k+1} g_j^*$ *such that for* $0 < g_0 < g_{\mathrm{it}}(k)$, *on every torus of (H3) whose exponent* m *exceeds a bound depending on* k *and* g_0,

$$\mathbb{E}_\varepsilon[W_\chi(C_n)] = \kappa_{C_n}^{(k+1)}(1+R) + R'\,, \qquad \kappa_{C_n}^{(k+1)} = \exp\Bigl(-P(C_n)\sum_{j=1}^{k+1}\delta m_j\Bigr)\prod_{j=1}^{k+1} z_j^4\,, \tag{11.2}$$
$$|R| \leqslant C\bigl[(k+2)\tau^2 + n_C e^{-c\pi_*^2}\bigr]\,, \qquad |R'| \leqslant C n_C e^{-c\pi_*^2}\,.$$

Remark 11.2 (Content of the Conjecture and What a Proof Needs). (1) The j-th coefficients do not depend on the size of the loop: the perimeter part is a sum over the scales of perimeter coefficients, each of size τ^2 in units of its own lattice spacing, so $|P(C_n)\sum_j \delta m_j| \leqslant 4C\tau^2 n(1+L^{-1}+L^{-2}+\cdots)$, while the corner factors contribute one factor z_j^4 per scale, $k+1 = \log_L(nL)$ in all. For $k=0$ the statement is Theorem 1.2, with the relative form of the far-history bound of Remark 1.3(1), which places the second term of the bound on E_{def} in $\kappa_{C_1}R$, together with the claim that the leading expectation $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} e^{E_\xi^\circ} W_\chi(C_1;U_1)]$ of the single-plaquette loop is $1 + O(\tau^2 + n_C e^{-c\pi_*^2})$; the first step gives the bounds (1.10), the upper bounds of (iii) and, by Corollaries 1.4 and 1.5 at $n=1$, $1 + O(\tau^2 + \varepsilon_1 + (L^2 + LC_L)q + n_C\delta_q^\sharp)$, in which the area term ε_1 of the bound, of order $g_0(\log g_0^{-2})^{p_0}$ by [3, (2.6)] as quoted in Section B.8, is not $O(\tau^2)$; whether the leading expectation itself is $1+O(\tau^2)$ is not decided here. (2) For $j \geqslant 2$ the coefficients δm_j, z_j are not defined in this paper; they are to be the perimeter coefficient and the corner factor of the j-th step, computed from the density after $j-1$ steps as (5.39) and (6.12) compute them from the flat reference integrals $N_{C,T}^\sigma$, D_T^σ of (5.5), in which case the conjecture says that the coefficients of every step have the size that Sections 5 and 6 prove for the first. At the stated precision the corner factors are not separately determined: $|z_j^4 - 1| = |e^{4\zeta_j} - 1| \leqslant e^{4C\tau^2} - 1$, so $|\prod_{j\leqslant k+1} z_j^4 - 1| \leqslant e^{4C(k+1)\tau^2} - 1 \leqslant 8C(k+1)\tau^2$ once $4C(k+1)\tau^2 \leqslant 1$, and replacing every z_j by 1 replaces $1+R$ by $\prod_j z_j^4(1+R)$, with $|\prod_j z_j^4(1+R) - 1| \leqslant 8C(k+1)\tau^2(1+|R|) + |R|$, which a larger C in the bound on R absorbs below a threshold depending on k; the z_j record where the term $(k+2)\tau^2$ of R comes from. With the threshold $g_{\mathrm{it}}(k)$ depending on k, R' is not needed either: $|\kappa_{C_n}^{(k+1)}| \geqslant e^{-8C\tau^2(n+k+1)}$ ($L \geqslant 2$), so once $8C\tau^2(n+k+1) \leqslant 1$ the bound on R' is at most $eCn_Ce^{-c\pi_*^2}|\kappa_{C_n}^{(k+1)}|$ and R' can be moved into $\kappa_{C_n}^{(k+1)}R$ at the cost of replacing C by $(1+e)C$; it is substantive in the form with a threshold independent of k. Since $\delta m_1 = \delta m^\sigma$ is given, what the statement determines beyond the first step and beyond $O((k+2)\tau^2)$ is $P(C_n)\sum_{j\geqslant 2}\delta m_j$, of size at most $4C\tau^2(n-1)/(L-1)$, modulo $O((k+2)\tau^2)$. For $k \leqslant 1$ this is at most $4C\tau^2$, so nothing beyond the first step is determined; in general the scales j with $L^{k+1-j} \leqslant k+2$ contribute at most $4C\tau^2(k+2)L/(L-1)$, also of order $(k+2)\tau^2$, and the content lies in the scales $j \geqslant 2$ with $L^{k+1-j} > k+2$. (3) A proof by $k+1$ applications of the first step needs the following ((c) only for a threshold independent of k), none of which is proved here beyond the first step. (a) A local form of the leading term: on the covered event, $W_\chi(C_n;U_1)e^{E_\xi^\partial}$ as a function of the coarse field V near the loop, a coarse Wilson loop times the exponential of a sum over the coarse bonds of the loop of terms depending on V within a fixed distance of the bond, analytic on the small-field chart, gauge covariant and bounded

by $C\tau^2$, with the history labels absorbed into the large-field expansion of ρ_1; this is what turns the output of the step, the pair of the complex measure $\kappa_{C_n}^{-1}N_1$ and the measure ρ_1 (Remark 1.3(3)), into an input of the same type. (b) General-step analogues of (B1)–(B4), which Section 2 of [3] does not state as such: it formulates the inductive assumption with the operations corresponding to $\mathbf{T}_1$ left undefined, only their general properties being formulated (p.254), their inductive definition being deferred to its next section and forthcoming papers (pp.254–255), and with modifications connected with new terms obtained by the **R**-operations (p.254), among them the admitted case $\Lambda_j = \Omega_j$ (p.255); the sets $S_j \subset \Omega_j \cap \Lambda_j^c$ and the functions $\chi(\Omega_j \cap \Lambda_j^c, S_j)$, $\zeta(\Omega_j^c)$ are the general-step forms of the first-step objects of [3, (1.29), pp. 254–255]; the geometry and the scales are [3, (2.1)–(2.9)], with the couplings of the first $k+1$ scales compared through the first inequality $g_n \leqslant (1 + g_n^2\beta'(n-m))^{1/2} g_m$ of [3, (2.6), p. 255], $g_{k+1}^2\beta'(k+1)$ being small for $g_0 < g_{\mathrm{it}}(k)$ (Section B.8 quotes (2.6) for the first step). (c) The defect: the far-history part of (E.78) is relative to $|\kappa_{C_n}|$ but of order $x = C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})$ when $x \leqslant 1$, polynomially small per contour bond (Remark 1.3(1)); for a threshold independent of k it must be shown to be a perimeter-local correction, absorbed in δm_2 and z_2 at the next step. Below a threshold depending on k it is at most $C\tau^2$ relative to $|\kappa_{C_n}|$ ($\rho \leqslant C\tau^2$, $\varepsilon_1 \to 0$ and $e^{-cb/(LM)} \to 0$ at fixed n_C) and goes into R, as at $k = 0$. (d) The absolute terms E_{bdry} and the near-history part of E_{def} at the j-th step, of order $4L^{k+2-j}e^{-c\pi_{*,j}^2}$ for the loop of $4L^{k+2-j}$ bonds of the lattice of spacing $L^{j-1}\varepsilon$, with $\pi_{*,j} = \min(p_0(g_{j-1}), p_0(g_j))$ the scale of that step, $\pi_{*,j}^2 \geqslant \pi_*^2/2$ once $g_i^2\beta' i \leqslant 1$ for $i \in \{j-1, j\}$ and $(1 - \log 2/\log g_0^{-2})^{2p_0} \geqslant 1/2$ (the first inequality of [3, (2.6)] with $m = 0$, $n = i \geqslant 1$ gives $g_i^2 \leqslant (1 + g_i^2\beta' i)g_0^2 \leqslant 2g_0^2$, hence $\log g_i^{-2} \geqslant \log g_0^{-2} - \log 2$ and $p_0(g_i)^2 \geqslant p_0(g_0)^2/2 \geqslant \pi_*^2/2$; for $i = 0$ this is $p_0(g_0) \geqslant \pi_*$), and entering multiplied by the leading coefficients of the steps $1, \dots, j-1$, whose product must be bounded; so that their sum over the steps is at most $Cn_C e^{-c\pi_*^2}$; their relative size is discussed in Remark 1.3(1). (e) At every step the remainder of Corollary 1.4, which at the first step is at most $e^{n_C\delta_q^\sharp} - 1$, and the torus and corner part $E_{T,n}^\circ$, at most $C\tau^2(2 + 1/c)$, one source of the term $(k+2)\tau^2$. (f) The endpoint: the leading expectation of the single-plaquette loop at the last step is $1 + O(\tau^2)$; at the first step Corollary 1.5 gives $1 + O(\tau^2 + \varepsilon_1 + (L^2 + LC_L)q + n_C\delta_q^\sharp)$ only, and its area term ε_1, which comes from the deterministic bound on the plaquettes of the background, must be replaced by a bound of order τ^2. (g) The terms of the **R**-operations at every step, by which the general representation modifies the first-step representation [3, (1.28)–(1.30), p. 254].

A The Flat Covariance

The bulk coefficient uses the flat fluctuation operator on the infinite lattice with the same periodic block grid and constraint extension. We first obtain a uniform lower bound on compatible finite tori, then construct that operator and compare its inverse with the finite-torus inverses. This is a quadratic first-step calculation at the flat background.

The spatial dimension is four throughout this comparison. For a site field λ, set $(D\lambda)(x, x') = \lambda(x') - \lambda(x)$. For an oriented plaquette p, set $(\operatorname{curl} a)(p) = a(\partial p)$, where $a(\gamma)$ is the signed sum along an oriented path γ. All squared norms in this comparison sum the Lie-algebra inner product over positive bonds, plaquettes or sites of the indicated carrier. Define the mean path sum by

$$(\Phi a)(y, x) = \frac{1}{|\mathcal{G}(y, x)|} \sum_{\gamma \in \mathcal{G}(y,x)} a(\gamma)\ . \tag{A.1}$$

[4, (0.8) and (2.5)] identify $G^{(2)}(a) = \|\Phi a\|^2$. Differentiating [4, (0.4)], with $x' = x + Le_\mu$ for $c = (y, y + Le_\mu)$, gives

$$\begin{aligned}(Q_1 a)(c) &= L^{-5} \sum_{x \in \mathcal{B}_y} a([x, x'])\ , \\ (Qa)(c) &= (Q_1 a)(c) + L^{-5} \sum_{x \in \mathcal{B}_y} ((\Phi a)(y, x) - (\Phi a)(y + Le_\mu, x'))\ .\end{aligned} \tag{A.2}$$

The operator Q_1 is the straight-path average in [6, (2.125)]; it is not identified with Q. Telescoping these paths gives $Q(D\lambda)(c) = (\lambda(c_+) - \lambda(c_-))/L$.

For each block choose its lower corner $o_y = y - \frac{L-1}{2} \sum_{\mu=1}^4 e_\mu$. Let $\kappa_{y,x}$ traverse the coordinate displacements from o_y to x in the order $1, \dots, 4$. The union of these paths is a spanning tree. Define

$$\lambda(x) = a(\kappa_{y,x}) - a(\kappa_{y,y}), \qquad b = a - D\lambda, \qquad E(y, x) = (\Phi a)(y, x) - \lambda(x)\ . \tag{A.3}$$

The block partition makes λ a periodic site field, and $\lambda(y) = 0$. Every tree bond of b vanishes. Moreover $Qb = Qa$ and $\operatorname{curl} b = \operatorname{curl} a$.

The Lie-algebra element $E(y, x)$ is a mean of the signed sums of a around the closed walks obtained by adjoining a mean path from y to x to the reverse tree path. Each walk lies inside its block and has length at most $12(L - 1)$. To fill it, commute adjacent steps in different directions and cancel opposite steps in each direction. Every commutation adds one unit plaquette. It remains inside the block, because exchanging two different directions preserves the intermediate coordinate ranges. Sorting the directions takes at most the square of the walk length commutations. Thus a filling has integer plaquette coefficients with total absolute sum at most $J_L = 144(L - 1)^2$. Cauchy–Schwarz and averaging give

$$\|E\|^2 \leqslant K_L \|\operatorname{curl} a\|^2, \qquad K_L = L^4 J_L^2\ . \tag{A.4}$$

There are L^4 site tests per block and the interior plaquettes of different blocks are disjoint, so this constant is independent of their number. Also $\|\lambda\|^2 \leqslant 2\|\Phi a\|^2 + 2\|E\|^2$ and $\|D\lambda\|^2 \leqslant 16\|\lambda\|^2$.

For $Qa = 0$, equation (A.2) applied to b has $\Phi b = E$. Jensen's inequality and the eight incidences of a block with coarse bonds imply

$$\|Q_1 b\|^2 \leqslant 16 L^{-6} \|E\|^2\ . \tag{A.5}$$

For the block-tree gauge, the following periodic bound holds with $A_L = 384 L^5$:

$$\|b\|^2 \leqslant A_L(\|\operatorname{curl} b\|^2 + L^2 \|Q_1 b\|^2)\ . \tag{A.6}$$

For the proof, the coordinate permutation $R(x_1, x_2, x_3, x_4) = (x_4, x_3, x_2, x_1)$ reverses the block-tree order. Its pushforward on positive bonds is $(R_* b)(Rx, Rx + e_{5-\mu}) = b(x, x + e_\mu)$. It preserves the bond and curl norms, and (A.2) gives $(Q_1 R_* b)(Rc) = (Q_1 b)(c)$. Thus it suffices to prove the same bound for the reversed order $4, \dots, 1$, after translating a block corner to the origin. The inner product remains the fixed tr_N inner product throughout.

Direct Face Estimates in the Positive-Bond Norm. The fine lattice has $N_\mu = LK_\mu$ sites in direction μ, with odd $L \geqslant 3$ and $K_\mu \geqslant 3$. The preceding axis reversal permutes the periods to (N_4, N_3, N_2, N_1) and reduces the proof to the tree order $4, \dots, 1$. Its inverse

restores the original order at the end. A local calculation uses the translated block $B_0 = \{0, \dots, L-1\}^4$. All norms and sums below count positive bond or plaquette labels once, with the fixed $\mathfrak{g}$ inner product.

For a block B, let $\mathcal{E}_B^+$ be its positive fine bonds with both endpoints in B, and let $\mathcal{P}_B^+$ be its positive plaquettes with all vertices in B. For a boundary face $\Sigma \subset B$, let $\mathcal{E}_\Sigma^+$ be its positive fine bonds with both endpoints in Σ. Define

$$I_B = \sum_{r \in \mathcal{E}_B^+} \|b(r)\|_{\mathfrak{g}}^2, \qquad T_\Sigma = \sum_{r \in \mathcal{E}_\Sigma^+} \|b(r)\|_{\mathfrak{g}}^2, \qquad F_B = \sum_{p \in \mathcal{P}_B^+} \|(\operatorname{curl} b)(p)\|_{\mathfrak{g}}^2 . \tag{A.7}$$

These are scalar quadratic forms in the existing tree-gauged field. For the flat comparison, a linear gauge change $a \mapsto a + D\omega$ preserving its coarse average has constant values $\omega(y)$ on the connected coarse lattice. Equation (A.3) then leaves b unchanged. Constant colour conjugation sends b to $\mathrm{Ad}_g\, b$ and preserves every inner product below. Only the flat reference covariance is estimated here.

The Block-Face Trace Bound. Write $b_\nu(x) = b(x, x+e_\nu)$ for an internal positive bond of B_0, so $x_\nu \leqslant L-2$. The source tree condition makes $b_\nu(x) = 0$ whenever $x_1 = \cdots = x_{\nu-1} = 0$; in particular $b_1 = 0$. For $k < \nu$ and integers $0 \leqslant t < x_k$, define

$$u^{(k)}(x,t) = (0, \dots, 0, t, x_{k+1}, \dots, x_4) . \tag{A.8}$$

Here the t is the kth component. Let $F_{k\nu}(u)$ be the curl on the positive (k,ν) plaquette based at u. On the plane $u_1 = \cdots = u_{k-1} = 0$, both k-direction sides of that plaquette vanish by the tree condition. Successive telescoping therefore gives the identity used before [6, (2.123)]:

$$b_\nu(x) = \sum_{k=1}^{\nu-1} \sum_{t=0}^{x_k-1} F_{k\nu}(u^{(k)}(x,t)) . \tag{A.9}$$

Every displayed plaquette lies inside B_0. The number of terms is at most $3(L-1)$.

Fix a face $x_\mu = \alpha$, where $\alpha \in \{0, L-1\}$, and a tangential component $\nu \neq \mu$. For a fixed (k,ν) plaquette in (A.9), its multiplicity in the face sum is bounded by

$$\begin{array}{r|l} \mu < k & L^{k-2}\, L \leqslant L^2 \\ \mu = k & L^{k-1} \leqslant L^2 \\ \mu > k,\ \mu \neq \nu & L^{k-1}\, L \leqslant L^2 \end{array} . \tag{A.10}$$

In the first row, one of the $k-1$ otherwise free earlier components is fixed by the face. In the second row, x_k is fixed, so a given t occurs at most once. In the third row, the two distinct indices μ, ν exceed k, hence $k \leqslant 2$. In all rows, later components are fixed by the plaquette label. Without the face restriction the multiplicity is at most $L^{k-1}L \leqslant L^3$. For different ν, the plaquette pairs (k,ν) are disjoint. Cauchy–Schwarz in (A.9), followed by these counts, proves

$$I_B \leqslant 3(L-1)L^3 F_B \leqslant 4L^4 F_B, \qquad T_\Sigma \leqslant 3(L-1)L^2 F_B \leqslant 4L^3 F_B . \tag{A.11}$$

The first estimate is the positive-label version of [6, (2.123)]. The second is the required face restriction of its telescoping calculation.

The Interface Poincare Estimate. Let $F = \{0, \dots, L-1\}^3$ and let $\mathcal{E}_F^+$ be its positive nearest-neighbour edges. The Neumann graph Laplacian $\mathcal{L}_F \colon \mathfrak{g}^F \to \mathfrak{g}^F$ acts componentwise. For $s \in F$, it is given by

$$(\mathcal{L}_F v)(s) = \sum_{\substack{s' \in F \\ |s'-s|_1 = 1}} (v(s) - v(s')) \,. \tag{A.12}$$

It has quadratic form $\sum_{(s,s') \in \mathcal{E}_F^+} \|v(s') - v(s)\|^2$. For $m \in \{0, \dots, L-1\}^3$, its real scalar eigenbasis and eigenvalues are

$$v_m(s) = \prod_{i=1}^{3} \cos\left(\frac{\pi m_i (s_i + 1/2)}{L}\right), \qquad \lambda_m = 4 \sum_{i=1}^{3} \sin^2\left(\frac{\pi m_i}{2L}\right) \,. \tag{A.13}$$

The one-dimensional cosine identity verifies the eigenvalue at interior and endpoint vertices; finite cosine orthogonality gives a basis. Only $m = 0$ is constant. Since $\sin t \geqslant 2t/\pi$ for $0 \leqslant t \leqslant \pi/2$, every nonzero eigenvalue is at least $4/L^2$. Apply this to each orthonormal colour component. For $\bar{v} = L^{-3} \sum_{s \in F} v(s)$, it follows that

$$\sum_{s \in F} \|v(s) - \bar{v}\|^2 \leqslant \frac{L^2}{4} \sum_{(s,s') \in \mathcal{E}_F^+} \|v(s') - v(s)\|^2 \,. \tag{A.14}$$

Crossing Bonds and the Exact Average. For a positive coarse bond c in direction μ, write its two endpoint blocks as $B_- = o + \{0, \dots, L-1\}^4$ and $B_+ = B_- + L e_\mu$. The transverse label set is $F_c = \{0, \dots, L-1\}^{\{\nu \neq \mu\}}$, identified with F by increasing transverse index. In that label set, e_ν denotes its transverse coordinate unit vector. For $s \in F_c$, put

$$x_c(s) = o + (L-1)e_\mu + \sum_{\nu \neq \mu} s_\nu e_\nu, \qquad t_c(s) = b(x_c(s), x_c(s) + e_\mu), \qquad J_c = \sum_{s \in F_c} \|t_c(s)\|^2 \,. \tag{A.15}$$

Let Σ_- and Σ_+ be the facing boundary faces in B_- and B_+. For $\nu \neq \mu$ and $s, s + e_\nu \in F_c$, the plaquette based at $x_c(s)$ has curl

$$t_c(s) - t_c(s + e_\nu) + b_\nu(x_c(s) + e_\mu) - b_\nu(x_c(s)) \,. \tag{A.16}$$

For $\nu < \mu$ its sign is opposite to the positive plaquette convention; its squared norm is unchanged. Let F_c^{cr} be the sum of these squared curls, each transverse edge once. The inequality $\|u+v+w\|^2 \leqslant 3(\|u\|^2 + \|v\|^2 + \|w\|^2)$ gives

$$\sum_{\nu \neq \mu} \sum_{\substack{s \in F_c \\ s + e_\nu \in F_c}} \|t_c(s + e_\nu) - t_c(s)\|^2 \leqslant 3(F_c^{\mathrm{cr}} + T_{\Sigma_-} + T_{\Sigma_+}) \,. \tag{A.17}$$

Define $I_{B,\mu}$ as the part of I_B in direction μ, and set $\bar{t}_c = L^{-3} \sum_{s \in F_c} t_c(s)$. The straight-path formula (A.2) has the exact decomposition

$$\begin{aligned} (Q_1 b)(c) &= L^{-1} \bar{t}_c + q_c \,, \\ q_c &= L^{-5} \sum_{s \in F_c} \sum_{i=0}^{L-2} (i+1) b_\mu \left(o + i e_\mu + \sum_{\nu \neq \mu} s_\nu e_\nu \right) \\ &\quad + L^{-5} \sum_{s \in F_c} \sum_{j=0}^{L-2} (L-1-j) b_\mu \left(o + (L+j) e_\mu + \sum_{\nu \neq \mu} s_\nu e_\nu \right) \,. \end{aligned} \tag{A.18}$$

Indeed, an internal lower-block bond at position i belongs to $i+1$ starting paths, and an internal upper-block bond at position j belongs to $L-1-j$ paths. Each crossing bond belongs to L paths. This is the positive-label content of [6, (2.125)]. There are at most $2L^4$ internal bonds in this sum, and every coefficient is at most L^{-4}. Cauchy–Schwarz therefore yields

$$\|q_c\|^2 \leqslant 2L^{-4}(I_{B_-,\mu} + I_{B_+,\mu}), \qquad L^3\|\bar{t}_c\|^2 \leqslant 2L^5\|(Q_1 b)(c)\|^2 + 4L(I_{B_-,\mu} + I_{B_+,\mu}) \,. \quad \text{(A.19)}$$

Combining the exact mean/variance decomposition of J_c with (A.14), (A.17) and (A.19) gives

$$J_c \leqslant \frac{3L^2}{4}(F_c^{\mathrm{cr}} + T_{\Sigma_-} + T_{\Sigma_+}) + 2L^5\|(Q_1 b)(c)\|^2 + 4L(I_{B_-,\mu} + I_{B_+,\mu}) \,. \quad \text{(A.20)}$$

The Periodic Sum and Its Payments. All sums over c run over the positive coarse bonds E_1, and $\mu(c)$ is the direction of c. Let $\mathscr{I} = \sum_B I_B$, $\mathscr{F} = \|\operatorname{curl} b\|^2$ and $\mathscr{Q} = \|Q_1 b\|^2$. Every positive fine bond is either internal to exactly one block or crosses exactly one positive coarse interface, so $\|b\|^2 = \mathscr{I} + \sum_c J_c$. Every plaquette counted by F_c^{cr} crosses only the μ-interface: its transverse edge remains inside the face. It is therefore counted for at most one c. Plaquettes crossing two coarse interfaces are not needed. Each block has eight incident faces, and an internal μ-bond enters the last term of (A.20) only for the incoming and outgoing μ-interfaces. Consequently,

$$\begin{aligned} \mathscr{I} &\leqslant 4L^4\mathscr{F} \,, \\ \sum_c F_c^{\mathrm{cr}} &\leqslant \mathscr{F} \,, \\ \sum_c (T_{\Sigma_-} + T_{\Sigma_+}) &\leqslant 32L^3\mathscr{F} \,, \\ \sum_{c\in E_1} (I_{B_-,\mu(c)} + I_{B_+,\mu(c)}) &= 2\mathscr{I} \,. \end{aligned} \quad \text{(A.21)}$$

Thus summing (A.20) gives the explicit payment

$$\begin{aligned} \|b\|^2 &\leqslant (1+8L)\mathscr{I} + \frac{3L^2}{4}\mathscr{F} + 24L^5\mathscr{F} + 2L^5\mathscr{Q} \\ &\leqslant \left(4L^4 + 56L^5 + \frac{3L^2}{4}\right)\mathscr{F} + 2L^5\mathscr{Q} \\ &\leqslant 384L^5(\mathscr{F} + L^2\mathscr{Q}) \,. \end{aligned} \quad \text{(A.22)}$$

The last inequality uses $L \geqslant 3$; its coefficients are bounded by $61L^5$ and $2L^5$, respectively. This proves (A.6).

Combining the proved bound (A.6) with the constraint-error bound gives $\|b\|^2 \leqslant A_L(1 + 16L^{-4}K_L)\|\operatorname{curl} a\|^2$. Since $a = b + D\lambda$, the bound $\|a\|^2 \leqslant 2\|b\|^2 + 2\|D\lambda\|^2$ proves, for $Qa = 0$,

$$\|a\|_2^2 \leqslant C_L\big(\|\operatorname{curl} a\|_2^2 + G^{(2)}(a)\big) \,. \quad \text{(A.23)}$$

An explicit admissible constant is

$$C_L = \max(64,\, 2A_L(1 + 16L^{-4}K_L) + 64K_L) \,. \quad \text{(A.24)}$$

At the flat background, differentiating the original action and exponential gauge functional in (4.5) gives $\langle a, \Delta_1 a\rangle = \|\operatorname{curl} a\|^2 + \|\Phi a\|^2$. Substituting $a = \mathsf{C}B$ in (A.23) uses $Q\mathsf{C}B = 0$ and $\|\mathsf{C}B\| \geqslant \|B\|$, because the free bonds are unchanged. Consequently the free Hessian satisfies

$$\mathsf{H} \geqslant \gamma_L I, \qquad \gamma_L = C_L^{-1} \,. \quad \text{(A.25)}$$

Construction on the Infinite Lattice. Let $E_\infty = \mathbb{Z}^4 \times \{1,2,3,4\}$ denote positive fine bonds, with (x,μ) representing $(x, x+e_\mu)$. The coarse centres are $L\mathbb{Z}^4$, with the centred blocks and path families specified above. Remove the designated bonds to obtain $E_{*,\infty}$. The square-summable field spaces use the same real Lie-algebra inner product as the finite-torus calculation. Initially on finitely supported fields, define Φ_∞ and Q_∞ by (A.1)–(A.2), and C_∞ by (4.3). The designated coefficient is $Q_{0,\infty} = L^{-4} I$, so the inverse in that definition is fibrewise. Put

$$\Delta_\infty = \operatorname{curl}^* \operatorname{curl} + \Phi_\infty^* \Phi_\infty, \qquad \mathsf{H}_\infty = \mathsf{C}_\infty^* \Delta_\infty \mathsf{C}_\infty \,. \tag{A.26}$$

The coefficient $Q_{0,\infty}$ can be checked directly in (A.2). The within-block paths entering Φ contain no designated bond. For $b_0(c)$, precisely L of the translated straight paths in the sum for $Q_1(c)$ traverse that bond: their transverse coordinates equal those of the coarse centres, and their longitudinal initial positions range over the L sites of the block. No translated path for another coarse bond contains it. Hence its coefficient is $L^{-5}L = L^{-4}$ and all off-diagonal coefficients vanish.

The action normalisation also agrees with the finite-torus Hessian. At the identity, the first plaquette phase is $\operatorname{curl} a$ and the first mean-transport phase is Φa. The mixed second derivative of $1 - \operatorname{Re}\operatorname{tr}_N e^{i(sX+tY)}$ at zero is $\operatorname{tr}_N(XY)$, the prescribed real inner product. The terms from second derivatives of the phases multiply the zero first derivative of this class function. Summing over the plaquettes and mean transports gives $\Delta_1 = \operatorname{curl}^* \operatorname{curl} + \Phi^* \Phi$ with the same normalisation on every torus.

Each map has finite range, bounded coefficients and finitely many entry types modulo $L\mathbb{Z}^4$. Its row and column absolute sums are bounded, so it extends uniquely to a bounded operator on its square-summable carrier. In particular H_∞ is bounded and self-adjoint on all of $\ell^2(E_{*,\infty}; \mathfrak{g})$.

For every finitely supported B, the support of $\mathsf{C}_\infty B$ and all action terms meeting that latter support embed without wraparound in a sufficiently large compatible torus. Their quadratic form and norm are unchanged. Equation (A.25) therefore applies to this same field. Density of finitely supported fields gives

$$\gamma_L I \leqslant \mathsf{H}_\infty \leqslant h_L I, \qquad h_L = \max_{e \in E_{*,\infty}/L\mathbb{Z}^4} \sum_{f \in E_{*,\infty}} \|\mathsf{H}_\infty(e,f)\| \,. \tag{A.27}$$

The maximum uses representatives of the finitely many free-bond types. Symmetry gives the same column bound. Thus h_L is finite and independent of the torus and coupling, with $h_L \geqslant \gamma_L > 0$. Set $b_L = h_L + \gamma_L$ and $\vartheta_L = 1 - \gamma_L / b_L$. Then $0 < \vartheta_L < 1$ and $\|I - \mathsf{H}_\infty / b_L\| \leqslant \vartheta_L$. The inverse is

$$\mathsf{H}_\infty^{-1} = \frac{1}{b_L} \sum_{j=0}^{\infty} (I - \mathsf{H}_\infty / b_L)^j \,. \tag{A.28}$$

The series converges in operator norm. The fine-field covariance is the bounded positive operator $\mathsf{K}_\infty = \mathsf{C}_\infty \mathsf{H}_\infty^{-1} \mathsf{C}_\infty^*$. At the flat background every colour component has the same scalar operator; denote its scalar bond kernel by $K_\infty(b,b')$.

Decay with Finite Constants. Let $\ell(b) \in \mathbb{Z}^4$ be the coarse index of the block containing the initial point of b, and put $d(b,b') = \|\ell(b) - \ell(b')\|_\infty$. Let R_H be the maximum of 1 and the distances $d(e,f)$ for nonzero entries of H_∞. Let R_C be the maximum of $d(b,e)$ for nonzero entries of C_∞. These maxima can be taken over the finitely many entry types modulo coarse

translation. Define

$$c_L^{\mathsf{C}} = \max_{b \in E_\infty / L\mathbb{Z}^4} \sum_{e \in E_{*,\infty}} \|\mathsf{C}_\infty(b,e)\|, \qquad \mu_L = \frac{-\log \vartheta_L}{R_H}, \qquad A_L^{\ker} = \frac{(c_L^{\mathsf{C}})^2}{\gamma_L} e^{2\mu_L R_C} \ . \tag{A.29}$$

The jth power in (A.28) has zero (e,f) entry if $jR_H < d(e,f)$, and its norm is at most ϑ_L^j. Summing the remaining terms bounds the inverse entry by $\gamma_L^{-1} e^{-\mu_L d(e,f)}$. Using the two finite rows of C_∞ and $d(e,f) \geqslant d(b,b') - 2R_C$ yields

$$|K_\infty(b,b')| \leqslant A_L^{\ker} e^{-\mu_L d(b,b')} \ . \tag{A.30}$$

This constructs the inverse and decay in the exponential-gauge free carrier.

Comparison with the Periodic Covariance. Take a compatible torus with fine periods $N_\alpha = LK_\alpha$ and $K_\alpha \geqslant 3$. For $z \in \mathbb{Z}^4$, write $Nz = (N_1 z_1, \dots, N_4 z_4)$. The local torus definitions of $Q, \Phi, \mathsf{C}, \Delta_1$ are the periodisations of their infinite-lattice definitions. Periodisation preserves their finite-range products. The absolutely convergent periodisation of the free inverse therefore solves the finite-torus inverse equation: applying H_T gives the periodised Kronecker delta. Uniqueness of the inverse and multiplication by the two constraint-extension maps give

$$K_T(b,b') = \sum_{z \in \mathbb{Z}^4} K_\infty(b, b' + Nz) \ . \tag{A.31}$$

Bonds on the right have fixed lifts and unchanged orientations. Let $K_{\min} = \min_\alpha K_\alpha$ and define

$$B_L^{\mathrm{img}} = \sum_{z \in \mathbb{Z}^4 \setminus \{0\}} e^{-\mu_L(\|z\|_\infty - 1)} < \infty \ . \tag{A.32}$$

The shell of maximum norm j has $(2j+1)^4 - (2j-1)^4$ points. Since $d(b, b' + Nz) \geqslant K_{\min}\|z\|_\infty - d(b,b')$, the image sum satisfies

$$|K_T(b,b') - K_\infty(b,b')| \leqslant A_L^{\ker} B_L^{\mathrm{img}} e^{-\mu_L(K_{\min} - d(b,b'))} \ . \tag{A.33}$$

Thus every fixed finite collection of covariance entries converges as all coarse periods tend to infinity. The limiting connected-cluster construction uses the infinite-lattice covariance, not a periodically repeated finite-volume covariance.

B Background Estimates at the First Step

This appendix proves the estimates stated in Section 3: Lemmas 3.4, 3.6, 3.16, 3.19 and 3.20. It fixes every constant named there that Section 3.5 does not fix itself, and places all of them in the one order of choice stated in Section B.8: L and the group with the group constants $c_{\mathrm{h}} = \sqrt{N}$, $c_{\mathrm{n}} = 1/N$, $\varrho_G = \pi/\sqrt{N}$, c, r_G, c_G below; the constants of the lemmas of this appendix in the order of their proofs; M_1, M_2 and M with the constants chosen after them, including the collar constants of item 4 there; the coupling g_0, through the thresholds collected there; and last the torus, whose periods satisfy the conditions of (3.1) (multiples of LMR_1, at least $8M$; the scale R_1 depends on g_0, so the period conditions are imposed after g_0). Group constants: c is a constant with $\|e^{iX} - \mathbf{1}\|_{\mathrm{op}} \leqslant c\, h(X)$ for all $X \in \mathfrak{g}$ ($c = \sqrt{N}$ serves, as $\|e^{iX} - \mathbf{1}\|_{\mathrm{op}} \leqslant \|X\|_{\mathrm{op}} \leqslant \sqrt{N} h(X)$). For every $N \times N$ matrix Y put $h(Y) = (N^{-1} \operatorname{tr} Y^* Y)^{1/2}$, which extends the normalised Hilbert–Schmidt norm of $\mathfrak{g}$ to all matrices; then $\|Y\|_{\mathrm{op}} \leqslant (\operatorname{tr} Y^* Y)^{1/2} = \sqrt{N}\, h(Y)$, so $\|Y\|_{\mathrm{op}} \leqslant c_{\mathrm{h}} |Y|$ for every matrix Y and

either cutoff norm $|\bullet|$ (the operator norm or h), not only for Hermitian Y; this is used below for differences of unitaries. The exponential map restricts to an analytic diffeomorphism from the ball $\{X \in \mathfrak{g} \,|\, h(X) < \varrho_G\}$ onto an open neighbourhood N_G of the identity in G (the conventions of Section 3), with inverse $\log_G$; since G is compact and $\|g-\mathbf{1}\|_{\mathrm{op}}$ vanishes on G only at the identity, there is $r_G \in (0,1]$ with $\{g \in G \,|\, \|g-\mathbf{1}\|_{\mathrm{op}} \leqslant r_G\} \subset N_G$. The constant c_G with $h(\log_G g) \leqslant c_G \|g-\mathbf{1}\|_{\mathrm{op}}$ on that set exists: $\log_G$ is analytic, so Lipschitz with some constant c' on $\{\|g-\mathbf{1}\|_{\mathrm{op}} \leqslant r'\}$ for some $r' \in (0, r_G]$, and on $r' \leqslant \|g-\mathbf{1}\|_{\mathrm{op}} \leqslant r_G$ one has $h(\log_G g) < \varrho_G \leqslant (\varrho_G/r')\|g-\mathbf{1}\|_{\mathrm{op}}$; take $c_G = \max(c', \varrho_G/r')$. The radius r_M of the local group mean is fixed in Lemma B.7(iv). The phase norm of Lemmas 3.2 and 3.3 is $\|A\|_\infty = \max_b h(A_b)$, and operator norms of entries and their derivatives are taken with respect to the Hilbert norms induced by h. Throughout, a branch is a branch in the sense of Section 3.1 (an admissible pair with $(A_{\mathrm{e}}, a) \in \mathcal{D}_\xi$ for some a); the reference data of Section 3.8 are treated in Section B.7.

B.1 Cubes, Localisation Domains and Tree Lengths

The partitions π_0 (side M) and π_1 (side LM) of Section 3.5 have cubes $Q_z = Mz + [-M/2, M/2]^4$ and $\Delta_z = LMz + [-LM/2, LM/2]^4$, reduced modulo the periods, with M_1 and M odd. Cube faces therefore lie at half-integer fine coordinates: every lattice site lies in the interior of exactly one cube of each partition, a fine bond meets at most two cubes of each partition, and a fine bond lies in the interior of a union of cubes exactly when both its endpoints lie in cubes of that union. A cube has M^4 sites and meets at most $8M^4$ positive fine bonds. Two cubes of the same partition are wall adjacent when they share a three-dimensional face; a localisation domain of π_i is a finite nonempty union of cubes of π_i that is connected under wall adjacency ([4, p. 257]). A localisation domain Y of π_i has $N_i(Y)$ cubes and, following [4, p. 257], tree length

$$d_i(Y) = \inf\{\, |\tau| \,|\, \tau \subset Y \text{ a connected finite union of segments meeting every cube of } Y \,\}, \tag{B.1}$$

the Euclidean length $|\tau|$ being measured in units of the cube side. Put $\ell_i(Y) = d_i(Y) + 5$. Two domains are compatible when they share neither a cube nor a wall ([5, (2.11)]); a domain is incompatible with itself.

Lemma B.1 (Geometry of Localisation Domains)**.** *Let $Y, Y_1, \dots, Y_m$ be localisation domains of one partition and $C_{\mathrm{g}} = 162$.*

1. *$N_i(Y) \leqslant C_{\mathrm{g}} \ell_i(Y)$ and $\ell_i(Y) \leqslant N_i(Y) + 4$.*
2. *If $\bigcup_j Y_j$ is a localisation domain, then $\ell_i(\bigcup_j Y_j) \leqslant \sum_j \ell_i(Y_j)$.*
3. *If Z is a localisation domain containing $Y_1, \dots, Y_k$ ($k \geqslant 0$, pairwise disjoint) and $Z \setminus \bigcup_j Y_j$ consists of n cubes, then $\ell_i(Z) \leqslant \sum_j \ell_i(Y_j) + 5n$.*
4. *The number of localisation domains with n cubes containing a given cube is at most 64^{n-1}, and for $s > C_{\mathrm{g}} \log 64$*

$$\sum_{Y \ni Q} e^{-s\ell_i(Y)} \leqslant B(s) = \frac{e^{-s/C_{\mathrm{g}}}}{1 - 64 e^{-s/C_{\mathrm{g}}}} \tag{B.2}$$

for every cube Q, on every torus. For $s \geqslant C_{\mathrm{g}}(\log 64 + \log 2)$, $B(s) \leqslant 2e^{-s/C_{\mathrm{g}}}$.

Proof. (i) Let τ realise (B.1) up to $\varepsilon \in (0, 1/2)$. A closed traversal of τ (each segment twice) has length at most $2(d_i(Y) + \varepsilon)$; place sample points along it at arclength spacing one, at most $2d_i(Y) + 2$ of them. Every cube of Y contains a point of τ, which is within arclength $1/2$, hence Euclidean and ℓ^∞ distance $1/2$, of a sample; the centre of that cube is then within ℓ^∞ distance one of the sample. Cube centres form a unit lattice, so at most $3^4 = 81$ cubes are assigned to each sample, and $N_i(Y) \leqslant 81(2d_i(Y) + 2) \leqslant 162\,\ell_i(Y)$. For the second inequality join the centres of wall-adjacent cubes along a spanning tree of the wall-adjacency graph; each of the $N_i(Y) - 1$ edges has length one and lies in Y, so $d_i(Y) \leqslant N_i(Y) - 1$.

(ii) Take trees $\tau_j \subset Y_j$ with $|\tau_j| \leqslant d_i(Y_j) + \varepsilon$. The graph on $\{Y_j\}$ with an edge when $Y_j \cap Y_k \neq \varnothing$ is connected: a wall-adjacent chain of cubes of the union passes from Y_j to Y_k through consecutive cubes that share a wall, hence through domains that intersect. For each edge of a spanning tree pick $x \in Y_j \cap Y_k$, a cube $Q_j \ni x$ of Y_j and a point $p_j \in \tau_j \cap Q_j$, and add the segment $[p_j, x] \subset Q_j$ (length at most the diameter 2 of a unit cube in $\mathbb{R}^4$), and likewise for k. The result is a connected union of segments inside $\bigcup Y_j$ meeting every cube, of length at most $\sum_j (d_i(Y_j) + \varepsilon) + 4(m-1)$. Letting $\varepsilon \to 0$, $d_i(\bigcup Y_j) + 5 \leqslant \sum_j d_i(Y_j) + 4m + 1 \leqslant \sum_j (d_i(Y_j) + 5)$ for $m \geqslant 1$.

(iii) Apply (ii) to the family consisting of $Y_1, \dots, Y_k$ and the n single cubes, each of which has $\ell_i = 5$.

(iv) A depth-first traversal of a spanning tree of the wall-adjacency graph of a domain with n cubes, started at the given cube, is a word of length $2(n-1)$ over the eight wall directions, and determines the domain; this gives 64^{n-1}. By (i), $e^{-s\ell_i(Y)} \leqslant e^{-sN_i(Y)/C_{\mathrm{g}}}$, so the rooted sum is at most $\sum_{n \geqslant 1} 64^{n-1} e^{-sn/C_{\mathrm{g}}} = B(s)$. The last claim is $64e^{-s/C_{\mathrm{g}}} \leqslant 1/2$. *q.e.d.*

Lemma B.2 (Scale Comparison)**.** *Let Z be a localisation domain of π_0, $\widetilde{Z}$ the union of the M-cubes meeting Z, and Z' the union of the LM-cubes meeting* $\operatorname{int} \widetilde{Z}$*. This is the enlargement rule of Section 3.5, which takes the LM-cubes meeting* $\operatorname{int} Q^\sim$ *for some cube Q of Z: indeed $\widetilde{Z} = \bigcup_{Q \subseteq Z} Q^\sim$, an LM-cube meeting* $\operatorname{int} Q^\sim$ *meets* $\operatorname{int} \widetilde{Z}$*, and conversely an LM-cube Δ meeting* $\operatorname{int} \widetilde{Z}$ *contains a nonempty open subset of $\widetilde{Z}$, a finite union of the closed sets $Q^\sim$, one of which therefore contains a nonempty open subset of Δ (a finite union of closed sets with empty interiors has empty interior), so Δ meets* $\operatorname{int} Q^\sim$ *for that Q. The same argument shows, for a union Z_0 of M-cubes with wall components $Z_{0,j}$, that $Z_0' = \bigcup_j Z_{0,j}'$. Then $Z \subset \operatorname{int} \widetilde{Z}$, $\widetilde{Z} \subset Z'$, Z' is a localisation domain of π_1, every point of $\widetilde{Z}$ is within ℓ^∞ distance $2M$ of every tree realising $d_0(Z)$ up to any $\varepsilon > 0$, and*

$$\ell_1(Z') \leqslant a_L^{\mathrm{sc}} \ell_0(Z) + b_L^{\mathrm{sc}}, \qquad a_L^{\mathrm{sc}} = \frac{2049}{L}, \quad b_L^{\mathrm{sc}} = 5 + \frac{1024}{L}\,. \tag{B.3}$$

Proof. Every M-cube meeting a point of Z belongs to $\widetilde{Z}$, so $Z \subset \operatorname{int} \widetilde{Z}$. An M-cube $\widetilde{Q}$ of $\widetilde{Z}$ lies in a single LM-cube, which meets $\operatorname{int} \widetilde{Q} \subset \operatorname{int} \widetilde{Z}$; hence $\widetilde{Z} \subset Z'$. If $\widetilde{Q}$ meets $Q \subset Z$ only at a lower-dimensional face, the cubes between them sharing walls also meet Q, so $\widetilde{Z}$ is wall connected, $\operatorname{int} \widetilde{Z}$ is connected, and the LM-cubes meeting a connected open set are wall connected (a path avoiding lower-dimensional faces passes between consecutive cubes through a common wall). Let $\tau \subset Z$ meet every M-cube of Z with $|\tau| \leqslant M(d_0(Z) + \varepsilon)$ in fine units. For $x \in \widetilde{Z}$ pick $\widetilde{Q} \ni x$, a cube $Q \subset Z$ meeting $\widetilde{Q}$, $y \in \widetilde{Q} \cap Q$ and $p \in \tau \cap Q$; then $|x - p|_\infty \leqslant |x - y|_\infty + |y - p|_\infty \leqslant 2M$.

For (B.3), place samples along a closed traversal of τ at arclength spacing LM; there are at most $2(d_0(Z) + \varepsilon)/L + 1$ of them, and every point of τ is within Euclidean distance $LM/2$ of a sample. For $\Delta \subset Z'$ pick $x_\Delta \in \Delta \cap \operatorname{int} \widetilde{Z}$; by the preceding paragraph it is within ℓ^∞ distance $2M + LM/2 \leqslant 7LM/6$ of a sample ($L \geqslant 3$), so the centre of Δ is within ℓ^∞

distance $5LM/3 < 2LM$ of that sample. The centres of LM-cubes form a lattice of spacing LM, and an open ℓ^∞ ball of radius $2LM$ contains at most $4^4 = 256$ of its points. Hence $N_1(Z') \leqslant 256(2(d_0(Z)+\varepsilon)/L+1)$. Now build a connected union of segments in Z' meeting every LM-cube of Z': start from $\tau \subset Z \subset Z'$ and, for each $\Delta \subset Z'$ not met by τ, add the segments $[x_\Delta, y] \subset \widetilde{Q} \subset Z'$ and $[y,p] \subset Q \subset Z'$ chosen above, each of length at most $2M$. Its length is at most $M(d_0(Z)+\varepsilon) + 4MN_1(Z')$, so

$$d_1(Z') \leqslant \frac{d_0(Z)+\varepsilon}{L} + \frac{4}{L}\cdot 256\Big(\frac{2(d_0(Z)+\varepsilon)}{L}+1\Big) \leqslant \frac{2049}{L}(d_0(Z)+\varepsilon) + \frac{1024}{L}\,.$$

Let $\varepsilon \to 0$, add 5 to both sides and use $d_0(Z) \leqslant \ell_0(Z)$. *q.e.d.*

B.2 The Regular-Background Bounds

Lemma B.3 (Entry Range and Stencils)**.** *For free bonds b, b' let $z(b)$ be the coarse label of the block containing b_- and $f_{bb'}(U)$ the entry function* (3.18) *with its entry stencil $\mathcal{S}_{bb'}$ (the union of the background sets of the terms of* (3.17) *whose free-input set contains both b and b'). If $|z(b)-z(b')|_1 > 2$ then $\mathcal{T}_{bb'} = \varnothing$ and $f_{bb'} = 0$. The stencil of a nonzero entry consists of bonds e with $|z(e)-z(b)|_1 \leqslant 2$; in fine units all bonds and sites of $\mathcal{S}_{bb'}$, and both endpoints of b and b', lie within ℓ^∞ distance $3L$ of b_-, so $\rho_L = 4L$ is an admissible entry range in Lemma 3.3. Each row has at most $164L^4$ nonzero entries.*

Proof. By (3.17) and its support sets, a term containing the free input b is $\mathfrak{a}_p$ with $b \in \mathcal{F}_r$ for a boundary bond r of p, $\mathfrak{g}_y$ with $b \in \mathcal{E}_y$, or $\mathfrak{t}_c$ with $b \in \mathcal{E}_c \cap E_*$. The free inputs of a bond r have block labels among the endpoint blocks of r (Lemma 3.2): $z(r)$ for a free bond, $\{c_-, c_+\}$ for $r = b_0(c)$. The four vertices of a fine plaquette in directions μ, ν have block labels differing only in the μ and ν coordinates, by at most one each, so these labels, and the endpoint blocks of its boundary bonds, lie in a set of ℓ^1 diameter at most two. The free inputs of $\mathfrak{g}_y$ lie in one block, those of $\mathfrak{t}_c$ in the two blocks of c. Hence two free inputs of one term have $|z(b)-z(b')|_1 \leqslant 2$, and the background bonds of that term (plaquette bonds, the sets $\mathcal{E}_c$, and for $\mathfrak{t}_c$ the plaquettes meeting $b_0(c)$, whose vertices lie in the two blocks of c) have block labels within ℓ^1 distance two of $z(b)$. A site of a block with label within ℓ^1 distance two of $z(b)$ is within ℓ^∞ distance $2L + (L-1) < 3L$ of b_-; the endpoints of b and b' are such sites or within one of them. The ℓ^1 ball of radius two in $\mathbb{Z}^4$ has 41 points and a block has at most $4L^4$ positive bonds with initial site in it. *q.e.d.*

Lemma B.4 (Plaquette Deviation on the Collar)**.** *Let B_3, δ_{c} and the $O(1)$ be the constants of the comparison following [3, (1.19)], as recorded in (B3)(iii): for every $L^2M_2R_1$-cube $\square' \subseteq \Omega_1^{\sim -1} \setminus R_1$ (the $\square' \subset R_1^c$ of [3], for which $\square'^{\sim 4} \subseteq \Omega_1$, p.250), $|U_1(e)U_{1,\square'}(e)^{-1} - 1| < O(1)B_3 e^{-\delta_{\mathrm{c}} LM_2R_1}\varepsilon_1$ on the bonds e of $\square'^{\sim 2}$ (p.250). [3] does not state the dependence of these constants, which are Bałaban's ((H4)): B_3 is that of [7, Theorem 1, p. 279] and the δ of [3] the rate of the exponential decay property [7, (190)], which Bałaban invokes in [3, p. 266]; Bałaban reuses the symbols B_3 and δ at every step (p.268, the regularity bound $|\partial U_{k+1} - 1| < 2B_3\varepsilon_{k+1}(L^{-1}\eta)^2$ and the comparison with $B_3\exp(-\delta LM_2R_{k+1})44d^2B_3\varepsilon_{k+1}$) and derives the comparison (3.19) from these facts and the bounds (106)–(108), (159)–(163) of [8] (p.268). This paper uses one property of them only, recorded in (B3)(iii) and (H4): B_3, δ_{c} and the $O(1)$ do not depend on the coupling g_0 or on the branch, so that at fixed L and M_2 the factor $e^{-\delta_{\mathrm{c}} LM_2R_1}$ tends to zero as $g_0 \to 0$. The δ of [3] is written δ_{c} here to distinguish it from the Mayer constant δ of Section B.5. Reduce the coupling threshold until $4\,O(1)B_3L^2e^{-\delta_{\mathrm{c}} LM_2R_1} \leqslant 1$, which holds for small g_0 because the constants do not depend on g_0 and $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}} \to \infty$ as $g_0 \to 0$ ([3, (2.5) and (2.6), p. 255]: $g_1 \leqslant (1+\beta_0)g_0$ and*

$g_0 \leqslant (1+\beta_0)g_1$, so $g_1 \to 0$ with g_0), and let R_{c} satisfy Lemma 3.11. Here $\varepsilon_1 = g_1 p_0(g_1)$ is the constant of the same name in [3, p. 246, (2.4)] and g_1 the first-step coupling. On every branch, every fine plaquette p having a bond within distance $3R_{\mathrm{c}}$ of I satisfies

$$\|U_1(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant 2c_{\mathrm{h}}\varepsilon_1 L^{-2} \,. \tag{B.4}$$

Proof. Let b be such a bond and $\square'$ the cube of the partition of [3, (1.18)] into cubes of side $L^2M_2R_1$ that contains b_- (a partition, so the cube is unique). By Lemma 3.11(i) (stated for $d(b,I) \leqslant 3R_{\mathrm{c}}$; the standing condition $LMR_1 > 4R_{\mathrm{c}} + 4L^2M_2R_1 + 8$ gives $3R_{\mathrm{c}} + 1 + L^2M_2R_1 < LMR_1 - 1$), $b_- \in \Omega_1^{\sim -1}$ and $\square'$ is a good cube of [3, (1.18)], $\square' \subseteq \Omega_1^{\sim -1} \setminus R_1' \subseteq \Omega_1^{\sim -1} \cap R_1^c$; since $M \geqslant 8LM_2$, the layer of LMR_1-cubes around $\Omega_1^{\sim -1}$ inside Ω_1 contains $\square'^{\sim 4}$, as [3] states after (1.19). The outer weight (3.4) contains the background test $\chi_1(\Omega_1)$ of [3, (2.17)] at $k=1$, a function of V, and it is nonzero on a branch: the pair is admissible, so $D^{(11)}(V,V_0) \neq \varnothing$ (Section 3.1), and $\chi_1(\Omega_1) \neq 0$ at every point of $D^{(11)}$. That test is written on the lattice T_η, $\eta = L^{-1}$, whose unit is the block side, so its cubes of side LM_2R_1 have fine side $L^2M_2R_1$ and the plaquette threshold $\varepsilon_1\eta^2$ is $\varepsilon_1 L^{-2}$ ([3, (1.4), p. 246], prints the step-one test of the $L^2M_2R_1$-cubes with the threshold $\varepsilon_1 L^{-2}$ directly); its partition is "compatible with all other partitions" ([3, p. 257]; the term is not defined there, pp.245 and 257, and in the sense of [9, p. 98], (H4), the cubes of the coarser of two compatible partitions are unions of cubes of the finer), in particular with the partition of (1.18) into cubes of the same side, and two compatible partitions into cubes of equal side coincide. Hence on the branch $\sup_{p' \subset \square'^{\sim}} |U_{1,\square'}(\partial p') - 1| < \varepsilon_1 L^{-2}$ for the local minimum $U_{1,\square'}$ of [3, (2.16)]; a gauge transformation conjugates the plaquette holonomy and $|\bullet|$ is Ad-invariant, so the bound holds for the axial-gauge representative of $U_{1,\square'}$ (p.247) to which the comparison (B3)(iii) refers. The plaquette p contains b, hence lies in $\square'^{\sim}$. The comparison (B3)(iii), stated for the cubes $\square' \subset R_1^c$, for which $\square'^{\sim 4} \subset \Omega_1$ and U_1, $U_{1,\square'}$ are determined by the same averages on $\square'^{\sim 3}$, applies to this $\square'$ and gives $|U_1(e)U_{1,\square'}(e)^{-1} - 1| < O(1)B_3 e^{-\delta_{\mathrm{c}} LM_2R_1}\varepsilon_1$ on the bonds e of $\square'^{\sim 2}$, hence $\|U_1(e) - U_{1,\square'}(e)\|_{\mathrm{op}} = \|U_1(e)U_{1,\square'}(e)^{-1} - \mathbf{1}\|_{\mathrm{op}} \leqslant c_{\mathrm{h}}|U_1(e)U_{1,\square'}(e)^{-1} - \mathbf{1}| \leqslant c_{\mathrm{h}} O(1)B_3 e^{-\delta_{\mathrm{c}} LM_2R_1}\varepsilon_1 \leqslant c_{\mathrm{h}}\varepsilon_1 L^{-2}/4$ there (unitary invariance of the operator norm, the norm comparison of the preamble, valid for every matrix, and the threshold above). Writing the plaquette holonomy as an ordered product of four unitaries and telescoping, $\|U_1(\partial p) - U_{1,\square'}(\partial p)\|_{\mathrm{op}} \leqslant \sum_{e \subset \partial p} \|U_1(e) - U_{1,\square'}(e)\|_{\mathrm{op}} \leqslant c_{\mathrm{h}}\varepsilon_1 L^{-2}$. With $\|U_{1,\square'}(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant c_{\mathrm{h}}|U_{1,\square'}(\partial p) - 1| < c_{\mathrm{h}}\varepsilon_1 L^{-2}$ this gives $\|U_1(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant 2c_{\mathrm{h}}\varepsilon_1 L^{-2}$. *q.e.d.*

Lemma B.5 (Local Gauge on a Rectangle)**.** *Let $\mathfrak{P}$ be a product of fine integer intervals of side at most $s_{\mathfrak{P}}$ (a lifted rectangle, injective modulo the periods), and let U be a configuration whose plaquettes inside $\mathfrak{P}$ satisfy $\|U(\partial p) - \mathbf{1}\|_{\mathrm{op}} \leqslant 2c_{\mathrm{h}}\varepsilon_1$ with $6c_{\mathrm{h}} s_{\mathfrak{P}}\varepsilon_1 \leqslant r_G$. There is a site gauge u on $\mathfrak{P}$ such that every bond $b \subset \mathfrak{P}$ has $U^u(b) = e^{iA_b}$ with $A_b \in \mathfrak{g}$ and*

$$h(A_b) \leqslant \widehat{\mu}_{\mathfrak{P}} = 6c_G c_{\mathrm{h}} s_{\mathfrak{P}}\varepsilon_1 \,. \tag{B.5}$$

Proof. Fix the corner x_0 of $\mathfrak{P}$ with minimal coordinates. For $x \in \mathfrak{P}$ let $P(x)$ be the lattice path from x_0 to x that runs first in direction e_1, then e_2, e_3, e_4; it stays in $\mathfrak{P}$. These paths form a spanning tree; define $u(x_0) = \mathbf{1}$ and $u(x') = u(x)U(x,x')$ along tree bonds oriented away from the root. Then $U^u(x,x') = u(x)U(x,x')u(x')^{-1} = \mathbf{1}$ on tree bonds, and for any bond $(x,x') \subset \mathfrak{P}$, $U^u(x,x') = U(P(x))U(x,x')U(P(x'))^{-1}$ is the holonomy of the closed loop $P(x)\cdot(x,x')\cdot P(x')^{-1}$ based at x_0. Let (x,x') have direction e_μ, so $x' = x + e_\mu$. The two tree paths share their runs in directions $\nu < \mu$ and their first $x_\mu - x_{0,\mu}$ steps in direction μ; after conjugation by the holonomy of that common initial segment, which preserves $\|\bullet - \mathbf{1}\|_{\mathrm{op}}$, the loop becomes $B\cdot(x,x')\cdot B'^{-1}\cdot(y+e_\mu,y)$, where $B = (b_1,\dots,b_k)$ is the staircase from $y = (x_1,\dots,x_\mu,x_{0,\mu+1},\dots,x_{0,4})$ to x in directions $\nu > \mu$, B' its translate

by e_μ, and $k \leqslant 3s_{\mathfrak{P}}$. Writing $\mathcal{L}_j$ for the loop built from $(b_1, \dots, b_j)$, one checks $\mathcal{L}_j = (b_1 \cdots b_{j-1})\, \partial p_j\, (b_1 \cdots b_{j-1})^{-1} \cdot \mathcal{L}_{j-1}$, where $\partial p_j = b_j e_\mu b_j'^{-1} e_\mu^{-1}$ is the boundary of the plaquette spanned by b_j and e_μ, which lies in $\mathfrak{P}$. Hence the holonomy of the loop is an ordered product of k conjugates of plaquette holonomies, and telescoping gives $\|U^u(x,x') - \mathbf{1}\|_{\mathrm{op}} \leqslant 2kc_{\mathrm{h}}\varepsilon_1 \leqslant 6c_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_1 \leqslant r_G$. The element $g = U^u(x,x') \in G$ therefore lies in N_G, and $A_b = \log_G g \in \mathfrak{g}$ is the unique X in the ball $h(X) < \varrho_G$ with $e^{iX} = g$ (the group chart of the conventions), with $h(A_b) \leqslant c_G \|g - \mathbf{1}\|_{\mathrm{op}} \leqslant 6c_G c_{\mathrm{h}} s_{\mathfrak{P}} \varepsilon_1$. *q.e.d.*

Where it is applied with Lemma B.4 below, the plaquette hypothesis holds because $2c_{\mathrm{h}}\varepsilon_1 L^{-2} \leqslant 2c_{\mathrm{h}}\varepsilon_1$. The proof uses ε_1 only through the plaquette hypothesis and the condition $6c_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_1 \leqslant r_G$; so the lemma holds with any $\varepsilon > 0$ in place of ε_1, with the bound $6c_G c_{\mathrm{h}} s_{\mathfrak{P}}\varepsilon$ in (B.5). Lemma B.8 uses it with $\varepsilon = \varepsilon_0/2$.

Here are the fixed geometric choices. Reduce τ_L of Lemma 3.2 so that

$$\tau_L \leqslant \min\Big(1,\ 2c_G r_G,\ \frac{r_G}{2(L-1)c},\ \frac{r_M}{4(L-1)c\,c_G}\Big)\,, \tag{B.6}$$

with c, r_G, c_G the group constants of the preamble (depending on the group only) and r_M the radius of Lemma B.7(iv) (depending on L and the group); this is the final value of τ_L, used in every statement. Put

$$\begin{aligned} c_{1,L} &= 320000\, m_L, \qquad R_{\mathrm{p}} = \max\Big(2, \Big\lceil 2\sqrt{c_{1,L}/\gamma_L}\Big\rceil\Big), \\ s_{\mathfrak{P}} &= L(2R_{\mathrm{p}} + 5) + 1, \qquad C_{L,R_{\mathrm{c}}} = 6c_G c_{\mathrm{h}} s_{\mathfrak{P}}\,, \end{aligned} \tag{B.7}$$

and choose the collar range $R_{\mathrm{c}} \geqslant \max(2R_{\mathrm{loc}} + 1,\ s_{\mathfrak{P}} + 2)$. Require every coarse period to exceed $2R_{\mathrm{p}} + 8$, so that the coarse ℓ^∞ ball of radius $R_{\mathrm{p}} + 3$ about any coarse label lifts injectively, and impose on ε_1

$$\widehat{\mu} = C_{L,R_{\mathrm{c}}}\varepsilon_1 < \frac{\tau_L}{2}, \qquad \max(1, c_L)\widehat{\mu} < \frac{\gamma_L}{4}\,. \tag{B.8}$$

Since $\tau_L \leqslant 2c_G r_G$, the first condition gives $6c_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_1 = \widehat{\mu}/c_G < r_G$, the hypothesis of Lemma B.5.

Lemma B.6 (Regional Entries)**.** *Let Ω be a region and U a background on which the regional form H_U^Ω is defined (Section 3.2: $\mathsf{M}(U)$ defined and $\alpha_c(U)$ invertible on the rows of $\Omega^{[1]}$, the local group means $\mathcal{U}_U(y,x)$ defined for $y \in \Omega^{(1)}$). For all $(b,b') \in \Omega^* \times \Omega^*$,*

$$(\mathsf{H}_U^\Omega)_{bb'} = f_{bb'}(U|_{\mathcal{S}_{bb'}})\,, \tag{B.9}$$

the entry function (3.18) *evaluated on the background restricted to the entry stencil $\mathcal{S}_{bb'}$ of Lemma B.3, and $f_{bb'} = 0$ when $|z(b) - z(b')|_1 > 2$. In particular, on every branch, $(H_\xi)_{bb'} = f_{bb'}(U_1|_{\mathcal{S}_{bb'}})$ for all $b, b' \in \Omega_1^*$, and for $b, b' \in I$ the stencil $\mathcal{S}_{bb'}$ lies inside Ω_1 (within $3L < R_{\mathrm{c}}$ of I, Lemma 3.11(i)).*

Proof. This is the compression identity (3.19) of Section 3.2, whose proof is recalled: the regional form is the sum of the terms of (3.17) with index in Ω (plaquettes meeting Ω, centres in Ω, rows in $\Omega^{[1]}$), its entry at (b,b') is the sum of the polarised entries of those of these terms whose free-input set contains b and b', and every term whose free-input set contains a bond b of Ω^* has its index in Ω: a plaquette term $\mathfrak{a}_p$ with $b \in \mathcal{F}_r$, $r \subset \partial p$, has $r = b$ or $r = b_0(c)$ with $b \in \mathcal{E}_c$, and in either case p contains a bond with an endpoint in the block of the endpoint of b that lies in Ω (b has such an endpoint, being a bond of Ω;

the bonds of $\mathcal{E}_c$ have both endpoints in the two blocks of c, and $b_0(c)$ has one endpoint in each), and that block lies in Ω, a union of blocks; a gauge term $\mathfrak{g}_y$ with $b \in \mathcal{E}_y$ has $y \in \Omega$; a row term $\mathfrak{t}_c$ with $b \in \mathcal{E}_c \cap E_*$ has an endpoint block of c containing an endpoint of b, so $c \in \Omega^{[1]}$. Hence the set of terms contributing to the regional entry is exactly $\mathcal{T}_{bb'}$, and the entry is $f_{bb'}(U|_{\mathcal{S}_{bb'}})$, a finite formula in the background on the stencil in which the rowwise elimination (3.12) uses only the rows of the terms in $\mathcal{T}_{bb'}$, which are rows of $\Omega^{[1]}$, where $\alpha_c(U)$ is invertible: the rows of $\mathfrak{g}_y$ and $\mathfrak{t}_c$ are rows with a centre in Ω by the above, and a row c' of $\mathfrak{a}_p$ has $b_0(c') \subset \partial p$, whose endpoints are the two sites of the segment between the centres of c' next to the common face of its blocks; the other two vertices of p are their translates by one unit in a direction perpendicular to c', which stay in the same two blocks as $L \geqslant 3$; so all four vertices of p lie in the two blocks of c', one of which meets Ω since p does, hence lies in Ω, and $c' \in \Omega^{[1]}$. The vanishing for $|z(b) - z(b')|_1 > 2$ is Lemma B.3. For the last claim, by Lemma B.3 every site of $\mathcal{S}_{bb'}$ is within $3L$ of $b_- \in I$, hence within $3R_{\mathrm{c}}$ of I and in $\Omega_1^{\sim -1}$ by Lemma 3.11(i). *q.e.d.*

Proof of Lemma 3.4. Extend $B \in \mathfrak{g}^I$ by zero. By Lemma B.6,

$$\langle B, H_\xi B\rangle = \sum_{b,b' \in I} \langle B_b, f_{bb'}(U_1|_{\mathcal{S}_{bb'}}) B_{b'}\rangle$$

with the local entries, whose stencils lie inside Ω_1; the vanishing $H_{\xi,II}(b,b') = 0$ for $d(b,b') > \rho_L$ is Lemma B.3. The quantities m_L, c_L of (3.22) bound the row sums $\sum_{b'}$ of $\|f_{bb'}(A)\|$ and $\|Df_{bb'}(A)\|$ over $\|A\|_\infty \leqslant \tau_L$; since $f_{b'b}(A) = f_{bb'}(A)^*$ (the form is symmetric) and hence $Df_{b'b}(A)[X] = (Df_{bb'}(A)[X])^*$, the column sums obey the same bounds.

Partition of unity. Let $\mathcal{Z}_T$ be the coarse grid and $\psi(u) = \prod_{\mu=1}^4 (1 - |u_\mu|/R_{\mathrm{p}})_+$ on the lifted coarse ball of radius R_{p}, extended by zero and periodised (injectively, by the period condition). For $w \in \mathcal{Z}_T$ put

$$\chi_w(b) = \frac{\psi(z(b) - w)}{S^{1/2}}, \qquad S = \sum_u \psi(u)^2 = \Big(\sum_{k \in \mathbb{Z}} (1 - |k|/R_{\mathrm{p}})_+^2\Big)^4 \geqslant \Big(\frac{R_{\mathrm{p}}}{4}\Big)^4 , \tag{B.10}$$

the last inequality because at least R_{p} integers k satisfy $|k| \leqslant R_{\mathrm{p}}/2$ and each contributes at least $1/4$. Then $\sum_w \chi_w(b)^2 = 1$ for every b. A one-step difference $|\psi(u) - \psi(u + e_\nu)|$ is at most $1/R_{\mathrm{p}}$, since only one factor changes and the others are at most one, and it vanishes unless u or $u + e_\nu$ lies in the support, a set of at most $(2R_{\mathrm{p}} + 2)^4 \leqslant (5R_{\mathrm{p}})^4$ points. Telescoping along a coarse path of $n = |z(b) - z(b')|_1$ steps and using $(\sum_{i=1}^n a_i)^2 \leqslant n \sum a_i^2$,

$$\begin{aligned} \sum_w (\chi_w(b) - \chi_w(b'))^2 &\leqslant \frac{1}{S}\, n \sum_{i=1}^{n} \sum_w (\text{one-step difference})^2 \\ &\leqslant \frac{256}{R_{\mathrm{p}}^4}\, n^2\, \frac{(5R_{\mathrm{p}})^4}{R_{\mathrm{p}}^2} = \frac{160000}{R_{\mathrm{p}}^2}\, |z(b) - z(b')|_1^2 . \end{aligned} \tag{B.11}$$

Since $\sum_w \chi_w(b)\chi_w(b') - 1 = -\frac{1}{2}\sum_w (\chi_w(b) - \chi_w(b'))^2$, Lemma B.3 ($n \leqslant 2$ on nonzero entries) and the Schur bound $\sum_{b,b'} \|f_{bb'}\| h(B_b) h(B_{b'}) \leqslant m_L \|B\|^2$, valid because row and column sums of $\|f_{bb'}\|$ are at most m_L (shown below), give

$$\Big|\sum_w \langle \chi_w B, H_\xi \chi_w B\rangle - \langle B, H_\xi B\rangle\Big| \leqslant \frac{1}{2} \cdot \frac{160000 \cdot 4}{R_{\mathrm{p}}^2}\, m_L \|B\|^2 = \frac{c_{1,L}}{R_{\mathrm{p}}^2} \|B\|^2 \leqslant \frac{\gamma_L}{4} \|B\|^2 . \tag{B.12}$$

Local comparison. Fix w with $\chi_w B \neq 0$; since B is supported on I, some $b \in I$ has $\chi_w(b) \neq 0$, so the patch $\{b \,|\, |z(b) - w|_\infty \leqslant R_{\mathrm{p}}\}$ meets I. Let $\mathfrak{P}_w$ be the lifted rectangle of fine sites in

blocks with $|z-w|_\infty \leqslant R_{\mathrm{p}}+2$, of side at most $L(2R_{\mathrm{p}}+5)+1 = s_{\mathfrak{P}}$; it lies within fine distance $s_{\mathfrak{P}}+1 \leqslant 2R_{\mathrm{c}}-1$ of I, so Lemma B.4 applies to all its plaquettes and Lemma B.5 gives a gauge u_w on $\mathfrak{P}_w$ with phases $h(A_e) \leqslant \widehat{\mu}$. Extend u_w by the identity. By Lemma B.3 the stencils of all entries with both labels in the patch lie in $\mathfrak{P}_w$. With $T_w = R_{u_w,*}$ the orthogonal action of Lemma 3.3,

$$\begin{aligned}\langle \chi_w B, H_\xi \chi_w B\rangle &= \sum_{b,b'} \langle (T_w\chi_w B)_b, f_{bb'}(A)(T_w\chi_w B)_{b'}\rangle \\ &= \langle T_w\chi_w B, \mathsf{H}_{\mathbf{1}} T_w\chi_w B\rangle + \sum_{b,b'} \langle (T_w\chi_w B)_b, (f_{bb'}(A)-f_{bb'}(0))(T_w\chi_w B)_{b'}\rangle\ ,\end{aligned}$$

where $f_{bb'}(A)$ is the entry at the gauge-transformed background $U_1^{u_w} = e^{iA}$ on the stencil; the first equality is the entrywise gauge covariance (3.21), $f_{bb'}(U^u) = \mathrm{Ad}_{u(b_-)}\, f_{bb'}(U)\, \mathrm{Ad}^{-1}_{u(b'_-)}$, together with $(T_w\chi_w B)_b = \mathrm{Ad}_{u_w(b_-)}\, \chi_w(b) B_b$ and the invariance of the inner product. The map $A \mapsto f_{bb'}(A)$ is analytic on $\|A\|_\infty \leqslant \tau_L$ (Lemma 3.3) and $\|A\|_\infty \leqslant \widehat{\mu} < \tau_L/2$, so $\|f_{bb'}(A) - f_{bb'}(0)\| \leqslant \sup_{t\in[0,1]} \|Df_{bb'}(tA)\|\, \widehat{\mu}$, and the row and column sums of these bounds are at most $c_L\widehat{\mu}$ by (3.22); the Schur test gives

$$\Big| \langle \chi_w B, H_\xi \chi_w B\rangle - \langle T_w\chi_w B, \mathsf{H}_{\mathbf{1}} T_w\chi_w B\rangle \Big| \leqslant c_L \widehat{\mu}\, \|\chi_w B\|^2\ . \tag{B.13}$$

The operator $\mathsf{H}_{\mathbf{1}}$ is the flat Hessian H of (4.7): at the identity $J_{\mathbf{1}} = 0$, every relative mean $\mathcal{U}_{\mathbf{1}}(y,x)$ is the identity, so $\mathscr{G}_{\mathbf{1}}$ is the gauge functional of (4.5), and $\mathsf{C}_{\mathbf{1}}$ is (4.3). By (A.25) on the branch's torus, $\langle T_w\chi_w B, \mathsf{H}_{\mathbf{1}} T_w\chi_w B\rangle \geqslant \gamma_L \|\chi_w B\|^2$. Summing over w and using $\sum_w \|\chi_w B\|^2 = \|B\|^2$ and (B.8),

$$\langle B, H_\xi B\rangle \geqslant \Big(\gamma_L - \frac{c_{1,L}}{R_{\mathrm{p}}^2} - c_L\widehat{\mu}\Big)\|B\|^2 \geqslant \frac{\gamma_L}{2}\|B\|^2\ . \tag{B.14}$$

Upper bound. For a row $b \in I$ all entries $f_{bb'}$, b' any free bond, have stencils in $\mathfrak{P}_{z(b)}$ (the patch of $w = z(b)$ meets I at b; a nonzero entry has its stencil within ℓ^∞ distance $3L$ of b_- by Lemma B.3, hence in blocks z with $|z - z(b)|_\infty \leqslant 3 \leqslant R_{\mathrm{p}}+2$); in the gauge $u_{z(b)}$ of that patch their phases are at most $\widehat{\mu} \leqslant \tau_L$, and $\|f_{bb'}\|$ is unchanged by the fibre rotations of (3.21), which are orthogonal. Hence, by Lemma B.6 on $\Omega_1^* \times \Omega_1^*$ and (3.22), $\sum_{b'\in\Omega_1^*} \|(H_\xi)_{bb'}\| = \sum_{b'\in\Omega_1^*} \|f_{bb'}(U_1|_{\mathcal{S}_{bb'}})\| \leqslant m_L$, which is the row-sum statement of the lemma; in particular $\sum_{b'\in I} \|f_{bb'}(U_1)\| \leqslant m_L$, and by the symmetry noted at the start the column sums of $H_{\xi,II}$ obey the same bound. The Schur test gives $\|H_{\xi,II}\| \leqslant m_L$. Both bounds pass to principal restrictions: the lower bound by compression, the upper by the Schur test on fewer entries. This proves (3.23) and the range statement. *q.e.d.*

B.3 Uniform Charts and Primitive Bounds

Primitives are evaluated on the input $A = g_0\mathsf{C}_U B$ with the free vector $B = a + tA_{\mathrm{e}}$, $t \in [0,1]$ (the vacuum potentials of Section 3.6 have $t = 0$). For a coarse row c the relevant input is $A|_{\mathcal{E}_c}$, with the Hilbert norm on $\mathfrak{g}^{\mathcal{E}_c}$ (written $\|A|_{\mathcal{E}_c}\|$ or $\|A\|$). Lemma 3.6 asks, for every row c within $2R_{\mathrm{c}}+2L$ of I (every bond of $\mathcal{E}_c$ within $2R_{\mathrm{c}}+2L$ of I), for the existence and uniqueness of the small solution ($h(d_c) < \rho_{\mathrm{d}} = 2K_*\vartheta_*^2$) of the row equation for every real phase with $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$, given by a function φ_c analytic on the complex ball of radius ϑ_* and real on real phases, the positivity of the determinant, and the analyticity in $B|_{\mathcal{E}_c\cap E_*}$ on the row polydisc $\mathcal{U}_c = \{B \,|\, |B(b)| < 2 + 4r\ \forall b \in \mathcal{E}_c \cap E_*\}$ (which $g_0\mathsf{C}_U$ maps into that phase ball under threshold item 3) of the row solution, the phase, the two logarithms and the

plaquette terms of the plaquettes with a bond in $\mathcal{E}_c$; for every centre y within $2R_{\mathrm{c}} + 2L$ of I, for the existence of the relative means and the analyticity of the gauge term on the block polydisc $\mathcal{U}_y = \{|B(b)| < 2 + 4r \ \forall b \in \mathcal{E}_y\}$; in each case as compositions of phase functions with derivatives up to order five bounded on the closed phase ball $\|A\| \leqslant \vartheta_*/2$ by constants depending on L and the group only, with the primitive sizes (B.18); Lemmas B.7 and B.9 supply this.

Lemma B.7 (Uniform Implicit Chart)**.** *There are constants $\varrho_*, \vartheta_* > 0$, $H_*, K_* \geqslant 1$, $C_{\mathsf{C}} \geqslant \max(1, (4L^3(L-1))^{1/2})$, C_{D}, $r_M, r'_M > 0$ with $r'_M \leqslant \varrho_G$, and derivative constants C_k^{D} for $0 \leqslant k \leqslant 6$ and C_k^{a} for $0 \leqslant k \leqslant 5$, depending on L and the group only, with $2\vartheta_* < \varrho_*$, $2\vartheta_* \leqslant \varrho_G$, $\vartheta_* \leqslant \tau_L/2$ and $4K_*H_*\vartheta_* < 1$, such that for every background U admitting a site gauge u in which the phases ϕ, $U^u = e^{i\phi}$, satisfy $\|\phi\|_\infty \leqslant \tau_L/2$ on $\mathcal{E}_c$, the following statements hold. The operator and derivative bounds in (i)–(iii) transfer under the orthogonal fibre gauge action. In (iv), the near-identity assertions for open paths and the mean itself are made in the chosen gauge u; only the relative-mean and gauge-functional conclusions are transferred to U. All formulas depending on ϑ_*, including ρ_{d} and C_k^{D}, use its final reduced value. Norms of phases on $\mathcal{E}_c$ are the Hilbert norms of $\mathfrak{g}^{\mathcal{E}_c}$ and of its complexification (so $\|d\| = h(d)$ for a single $d \in \mathfrak{g}$).*

1. *$\|h_U(c)\| \leqslant H_*$; $\|\mathsf{C}_U B\|_{\mathcal{E}_c} \leqslant C_{\mathsf{C}} \max_{b \in \mathcal{E}_c \cap E_*} |B_b|$ and $\|\mathsf{C}_U B\|_{\mathcal{E}_c} \leqslant C_{\mathsf{C}} \|B|_{\mathcal{E}_c \cap E_*}\|$ (Hilbert norms); and $\mathcal{N}_{U,c}$ is analytic on the open complex ball $\|A\| < 2\varrho_*$ of $\mathfrak{g}_{\mathbb{C}}^{\mathcal{E}_c}$ with $\mathcal{N}_{U,c}(0) = 0$, $D\mathcal{N}_{U,c}(0) = 0$ and $\|D^2\mathcal{N}_{U,c}\| \leqslant K_*$ on the closed ball $\|A\| \leqslant \varrho_*$; $\mathcal{N}_{U,c}$ is real on real A.*

2. *On the open complex ball $\|A\| < \vartheta_*$ there is an analytic function $\varphi_c = \widetilde{D}_{U,c}$, real on real A, with $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$, such that for every A in that ball $\varphi_c(A)$ is the unique solution d of the row equation (3.24) with $\|d\| \leqslant 2K_*\vartheta_*^2 = \rho_{\mathrm{d}}$ (in particular, for real A the unique small solution in the sense of Section 3.3); it agrees with the implicit-function germ of Lemma 3.5 (constructed there from Lemma 3.2, $\mathcal{N}_{U,c}(0) = 0$, $D\mathcal{N}_{U,c}(0) = 0$ and the analyticity of $\mathcal{N}_{U,c}$ near zero, without the present lemma), and satisfies*

$$\|\widetilde{D}_{U,c}(A)\| \leqslant 2K_*\|A\|^2, \qquad \|D_A\widetilde{D}_{U,c}(A)\| \leqslant C_{\mathsf{D}}\|A\|\,, \qquad C_{\mathsf{D}} = 4K_*\,. \tag{B.15}$$

3. *On the closed real ball $\|A\| \leqslant \vartheta_*/2$ the determinant $\det_{\mathfrak{g}}(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c}(A))$ is positive. On a complex neighbourhood of the closed complex ball $\|A\| \leqslant \vartheta_*/2$ the functions $\widetilde{D}_{U,c}$, the phase $q_U(A)|_{\mathcal{E}_c} = A - h_U\widetilde{D}_U(A)$, the normalised Haar logarithm $\log\big(\sigma(q_U(A)(b))/\sigma(0)\big)$, $b \in \mathcal{E}_c$, and the logarithm of that determinant are analytic (the two logarithms being the real logarithms on real A and their analytic continuations on the complex neighbourhood), and on the closed ball their derivatives of order $k \leqslant 5$ are bounded in operator norm by C_k^{a}, uniformly in the background; for $\widetilde{D}_{U,c}$ this holds for $k \leqslant 6$ with $C_k^{\mathsf{D}} = e^k k!\, 2K_*\vartheta_*^2(2/\vartheta_*)^k$. The two logarithms vanish at $A = 0$, and the first derivative of the Haar logarithm vanishes there ($\sigma(0) > 0$ is the value at the identity of the Haar density σ, which (3.30) subtracts); no statement is made about the first derivative of the determinant logarithm at $A = 0$.*

4. *For a block centre y whose block $\mathcal{B}_y$ lies in the gauge patch of u (phases $\|\phi\|_\infty \leqslant \tau_L/2$ on $\mathcal{E}_y$), put $q^u = R_u q$, where $(R_u q)(b) = \mathrm{Ad}_{u(b_-)}\, q(b)$, so $\|q^u|_{\mathcal{E}_y}\| = \|q|_{\mathcal{E}_y}\|$. For every real q with $\|q|_{\mathcal{E}_y}\| \leqslant \vartheta_*$, the path transports $(e^{iq^u}U^u)(\gamma)$, $\gamma \in \mathcal{G}(y,x)$, $x \in \mathcal{B}_y$, are within r_G of $\mathbf{1}$ in operator norm, their logarithms $\log_G$ have $h < r_M$, and the local group mean $\mathcal{U}_{e^{iq^u}U^u}(y,x)$ of (4.5) ([4, (0.10)–(0.11)]), the unique solution near the diagonal value,*

exists, of the form e^{iX} with $h(X) < r'_M \leqslant \varrho_G$; the relative mean $\mathcal{U}_{e^{iq}U}(y,x)\mathcal{U}_U(y,x)^{-1}$ lies within r_G of $\mathbf{1}$ in operator norm (the domain of $\log_G$), and it and the gauge term $\mathscr{G}_{y,U}(q)$ are restrictions to real q of functions analytic on the open complex ball $\|q|_{\mathcal{E}_y}\| < 2\vartheta_*$, with derivatives up to order five bounded by C_k^{a} on the closed complex ball of radius ϑ_*, uniformly in the background, and $\mathscr{G}_{y,U}(0) = 0$, $D\mathscr{G}_{y,U}(0) = 0$.

Proof. There are finitely many row types modulo $L\mathbb{Z}^4$. For each type the maps $U \mapsto h_U(c)$, C_U and $(U, A) \mapsto \mathcal{N}_{U,c}(A)$ are compositions of exponentials, relative block averages, the fibre inverse $\alpha_c(U)^{-1}$ and the chart logarithm, all analytic for phases $\|\phi\|_\infty \leqslant \tau_L$ and A near zero (Lemma 3.2 gives the singular-value bound $L^{-4}/2$ on α_c), and real for real arguments. Under the simultaneous gauge transformation of Lemma 3.3, Q_U, h_U, C_U and $\mathcal{N}_U$ intertwine with the orthogonal fibre action $R_{u,*}$ (proof of that lemma), which preserves the Hilbert norms and the maximum norm $\max_b |B_b|$ ($|\bullet|$ is Ad-invariant); so the bounds in (i)–(iii) may be proved in the gauge u and transferred to U. Part (iv) distinguishes open path transports in that gauge from the relative means and the gauge functional. The set of phase configurations with $\|\phi\|_\infty \leqslant \tau_L/2$ on $\mathcal{E}_c$ is compact, so a common complex radius $2\varrho_*$ of analyticity and bounds H_*, K_*, C_{C} exist (the two bounds on C_U are two operator norms of the same linear map, each bounded on the compact set; C_{C} is the larger, increased if necessary to $\max(1, (4L^3(L-1))^{1/2})$, which only weakens the inequalities); $\mathcal{N}_{U,c}(0) = 0$ because the zero fluctuation reproduces the background averages, and $D\mathcal{N}_{U,c}(0) = 0$ as in the proof of Lemma 3.5.

(ii) Fix A in the complex ball with $\|A\| \leqslant \vartheta \leqslant \vartheta_*$ and consider $\Phi(d) = \mathcal{N}_{U,c}(A - h_U d)$ on the closed complex ball $\|d\| \leqslant 2K_*\vartheta^2$. Then $\|A - h_U d\| \leqslant \vartheta + 2H_*K_*\vartheta^2 \leqslant 2\vartheta \leqslant \varrho_*$; Taylor's formula with $\mathcal{N}(0) = 0$, $D\mathcal{N}(0) = 0$ gives $\|\Phi(d)\| \leqslant \frac{1}{2}K_*(2\vartheta)^2 = 2K_*\vartheta^2$, and $\|D\Phi(d)\| \leqslant \|D\mathcal{N}(A - h_U d)\|H_* \leqslant K_* \cdot 2\vartheta H_* < \frac{1}{2}$. So Φ is a contraction of the closed ball into itself, with a unique fixed point; taking $\vartheta = \vartheta_*$ gives the uniqueness in the ball $\|d\| \leqslant \rho_{\mathrm{d}}$ for every A with $\|A\| < \vartheta_*$, and taking $\vartheta = \|A\|$ gives the first bound of (B.15); the fixed point $\varphi_c(A)$ is analytic in A on the open ball as the uniform limit of the analytic iterates $\Phi^n(0)$ (each iterate is analytic in A, and the convergence is geometric, uniformly in A), and it is real for real A because the iterates are. Differentiating the fixed-point equation, $D_A\widetilde{D} = (\mathbf{1} + D\mathcal{N}(x)h_U)^{-1}D\mathcal{N}(x)$ with $x = A - h_U\widetilde{D}$, $\|D\mathcal{N}(x)\| \leqslant K_*\|x\| \leqslant 2K_*\|A\|$ and $\|(\mathbf{1} + D\mathcal{N}(x)h_U)^{-1}\| \leqslant 2$; this is the second bound, and $D\varphi_c(0) = 0$. The germ of Lemma 3.5 is the solution near zero, and uniqueness in the ball identifies the two.

(iii) Cauchy's estimate on the ball of radius ϑ_*, where $\|\widetilde{D}_{U,c}\| \leqslant 2K_*\vartheta_*^2$, bounds the k-th derivative at a point of the closed ball of radius $\vartheta_*/2$ in a fixed direction v, $\|v\| \leqslant 1$, by $k!\,2K_*\vartheta_*^2(\vartheta_*/2)^{-k}$ for every k, and the polarisation identity for symmetric k-linear maps bounds the operator norm of $D^k\widetilde{D}_{U,c}$ by e^k times the diagonal bound; this gives C_k^{D} for $0 \leqslant k \leqslant 6$. The constants C_k^{a} bounding the other functions are chosen separately after (iv). The phase $q_U(A)|_{\mathcal{E}_c} = A - h_U\widetilde{D}_U(A)$ is a composition of $\widetilde{D}_{U,c}$ with the fixed linear map $h_U(c)$ of norm at most H_*, so it is analytic on the ball of radius ϑ_* and its derivatives are bounded by the chain rule. The Haar density: $\sigma(q)\,\mathrm{d}q$ is the pullback of Haar measure under $q \mapsto e^{iq}$ on the chart ball $h(q) < \varrho_G$, so $\sigma(q) = \sigma(0)|\det D(q \mapsto e^{iq})(q)|$, the determinant of the differential taken with respect to left-invariant frames; the determinant is analytic in q and nonvanishing on the chart ball (the chart has invertible differential, conventions of Section 3), so it has constant sign on the connected ball, positive since it equals one at $q = 0$; thus σ is analytic and positive there, and even, $\sigma(-q) = \sigma(q)$, because Haar measure is invariant under $g \mapsto g^{-1}$ and $e^{-iq} = (e^{iq})^{-1}$. Hence $\log(\sigma/\sigma(0))$, the real logarithm, extends analytically to a complex neighbourhood of the real chart ball, vanishes at 0 with vanishing first derivative (evenness), and its composition with $q_U(A)(b)$ (a phase with $h(q_U(A)(b)) \leqslant$

$\|A\| + H_* 2K_* \|A\|^2 \leqslant \vartheta_*$ on the ball) is analytic on a complex neighbourhood of the closed ball $\|A\| \leqslant \vartheta_*/2$ after reducing ϑ_* so that the compositions stay in the domain of analyticity, with derivatives bounded on the closed ball by the chain rule, and its first derivative at $A = 0$ is $D \log \sigma(0) \circ Dq_U(0) = 0$. The determinant argument $\mathbf{1} - h_U(c) \partial_{A(b_0(c))} \widetilde{D}_{U,c}(A)$ equals $\mathbf{1}$ at $A = 0$ since $D\widetilde{D}(0) = 0$, and is within $H_* C_{\mathrm{D}} \vartheta_*/2 < 1/2$ of $\mathbf{1}$ in operator norm on the ball; for real A it is a real endomorphism of $\mathfrak{g}$ within $1/2$ of $\mathbf{1}$, whose eigenvalues have positive real part, so its determinant is positive, and on a complex neighbourhood of the closed ball $\|A\| \leqslant \vartheta_*/2$ the determinant stays in the disc of radius $1/2$ about 1 (after reducing ϑ_* once more if needed, by continuity and compactness), where log is analytic; so the real logarithm of the determinant extends analytically to that neighbourhood and vanishes at $A = 0$, where the determinant is one. Its derivatives of order $k \leqslant 5$ are bounded on the closed ball by the chain rule, since the argument contains $\partial_{A(b_0(c))} \widetilde{D}_{U,c}$ and so involves the derivatives of $\widetilde{D}_{U,c}$ up to order six, bounded above. Its first derivative at $A = 0$, $-\operatorname{tr}_{\mathfrak{g}}\big(h_U(c) \partial_{A(b_0(c))} D\widetilde{D}_{U,c}(0)[\bullet]\big)$, involves the second derivative $D^2 \widetilde{D}_{U,c}(0) = D^2 \mathcal{N}_{U,c}(0)$, which does not vanish in general ([4, p. 267]: the expansion of $\widetilde{D}$ begins with quadratic terms), and nothing is claimed about it. The bounds are uniform in the background by the compactness of the set of phase configurations. All reductions of ϑ_* are by constants depending on L and the group only, over the finitely many row types.

(iv) Set $q^u = R_u q$; the fibre action is orthogonal on real phases and unitary on their complexifications, so it preserves the phase balls and operator norms of multilinear derivatives. In the gauge u the links of $\mathcal{B}_y$ are $e^{i\phi}$ with $\|\phi\|_\infty \leqslant \tau_L/2$, and $h(q_b) \leqslant \|q|_{\mathcal{E}_y}\| \leqslant \vartheta_* \leqslant \tau_L/2$ for every $b \in \mathcal{E}_y$. A path of $\mathcal{G}(y,x)$ has at most $2(L-1)$ bonds (four coordinate runs of length at most $(L-1)/2$), and for unitaries $\|U_1 \cdots U_n - \mathbf{1}\|_{\mathrm{op}} \leqslant \sum_j \|U_j - \mathbf{1}\|_{\mathrm{op}}$, so every path transport $g_\gamma = (e^{iq^u} U^u)(\gamma)$ lies within $2(L-1)c(\tau_L/2 + \vartheta_*) \leqslant 2(L-1)c\,\tau_L \leqslant r_G$ of $\mathbf{1}$ in operator norm, by (B.6); hence $A_\gamma = \log_G g_\gamma$ is defined with $h(A_\gamma) \leqslant c_G \cdot 2(L-1)c\,\tau_L \leqslant r_M/2$, again by (B.6). The local group mean of a family $(g_\gamma)_{\gamma \in \mathcal{G}(y,x)}$ is the solution $\mathcal{U}$ near the diagonal value of $\sum_\gamma \log_G(g_\gamma \mathcal{U}^{-1}) = 0$; write $g_\gamma = e^{iA_\gamma}, \mathcal{U} = e^{iX}$ and $F(X, (A_\gamma)) = \sum_\gamma \log_G(e^{iA_\gamma} e^{-iX})$, an analytic map on a neighbourhood of $(0,0)$ in $\mathfrak{g} \times \mathfrak{g}^{\mathcal{G}(y,x)}$ with $F(0,0) = 0$ and $D_X F(0,0) = -n\,\mathbf{1}$, $n = |\mathcal{G}(y,x)|$ (as $D \log_G(\mathbf{1}) = \mathbf{1}$). Choose the product neighbourhood in the complexified Lie-algebra variables so that every $e^{iA_\gamma} e^{-iX}$ lies in the analytic logarithm domain there. The analytic implicit function theorem on that neighbourhood gives radii $r_M, r'_M > 0$, with $r'_M \leqslant \varrho_G$ (any smaller solution radius is admissible together with a correspondingly smaller r_M), and an analytic map $X = X((A_\gamma))$ on $\max_\gamma h(A_\gamma) < r_M$, with $h(X) < r'_M$, such that $X((A_\gamma))$ is the unique solution of $F = 0$ with $h(X) < r'_M$; there are finitely many values of n (one per $x \in \mathcal{B}_y$ modulo translations), so r_M, r'_M and the derivative bounds of X are constants of the group and L. This solution is identified with the fixed mean used in Stage 1 as follows. [4, (0.8)] places that specified analytic mean near the identity for input families near the identity, and (0.10) says it solves this equation. Reduce r_M, before the final choice of τ_L, so that all families $\{e^{iA_\gamma}\}$ with $\max_\gamma h(A_\gamma) < r_M$ lie in the small-diameter domain of the specified mean, that this mean has logarithm of norm below r'_M, and that all relative logarithms use the common chart. Use the operator-norm diameter $\max_{\gamma,\gamma'} \|g_\gamma - g_{\gamma'}\|_{\mathrm{op}}$; it is unchanged by the two-sided multiplication $g_\gamma \mapsto u(y) g_\gamma u(x)^{-1}$. Thus the original-gauge families lie in the same small-diameter domain. Their relative products with the mean transform by conjugation at y, which preserves the Ad-invariant logarithm chart, so the original-gauge relative logarithms are defined as well. Local uniqueness from the implicit function theorem then identifies it with e^{iX}. Its left-right covariance is the stated property [4, (0.6)], not a consequence of uniqueness on an unspecified chart domain. Thus for any admissible gauge u its value in the original gauge is $u(y)^{-1} \mathcal{U}_{U^u}(y,x) u(x)$; the value is independent of the choice of admissible gauge because both expressions are values of the

same specified mean. Since $h(A_\gamma) \leqslant r_M/2 < r_M$ for the transport families above, the mean $\mathcal{U} = e^{iX((A_\gamma))}$ exists, lies in the chart domain, and is the unique solution near the diagonal value in the sense of (4.5). Write $A_\gamma(q^u) = \log_G((e^{iq^u}U^u)(\gamma))$ and $X(q^u) = X((A_\gamma(q^u))_\gamma)$. Define $M_x(q) = \mathcal{U}_{e^{iq}U}(y,x)\mathcal{U}_U(y,x)^{-1}$ and $M_x^u(q^u) = \mathcal{U}_{e^{iq^u}U^u}(y,x)\mathcal{U}_{U^u}(y,x)^{-1}$. Gauge covariance gives $M_x^u(q^u) = u(y)M_x(q)u(y)^{-1} = e^{iX(q^u)}e^{-iX(0)}$, with inverse $M_x^u(q^u)^{-1} = e^{iX(0)}e^{-iX(q^u)}$; both are compositions of the analytic map X with $\log_G$, the exponentials of the phases and the fixed links, so they extend analytically to a complex ball $\|q|_{\mathcal{E}_y}\| < 2\vartheta_*$ (after reducing ϑ_* by a constant of the group and L). For real q, $M_x(q)$ is unitary and $\|M_x(q) - \mathbf{1}\|_{\mathrm{op}} = \|e^{iX(q^u)} - e^{iX(0)}\|_{\mathrm{op}} \leqslant C\|q|_{\mathcal{E}_y}\|$ with a constant C of L and the group (the derivatives of X on the ball $\max_\gamma h(A_\gamma) \leqslant r_M/2$, of $\log_G$ on the r_G-ball and of the products of exponentials are bounded), so after reducing ϑ_* once more $M_x(q)$ lies within r_G of $\mathbf{1}$ for $\|q|_{\mathcal{E}_y}\| \leqslant \vartheta_*$. The gauge term is $\mathscr{G}_{y,U}(q) = \sum_x(1 - \operatorname{Re}\operatorname{tr}_N M_x(q))$; for real q, $\operatorname{Re}\operatorname{tr}_N M_x(q) = \frac{1}{2}(\operatorname{tr}_N M_x(q) + \operatorname{tr}_N M_x(q)^{-1})$ because $M_x(q)$ is unitary, and the right side is analytic in q on the complex ball: this is the analytic continuation meant. The derivatives up to order five of M_x and $\mathscr{G}_{y,U}$ are bounded on the closed ball of radius ϑ_* by constants C_k^{a} uniform in the background, because they are analytic jointly in q and in the phases ϕ of the links of $\mathcal{B}_y$ in the gauge u, and the set $\|\phi\|_\infty \leqslant \tau_L/2$ is compact. $\mathscr{G}_{y,U}(0) = 0$ because $M_x(0) = \mathbf{1}$, and $D\mathscr{G}_{y,U}(0) = 0$ because each summand is nonnegative on real q and vanishes at $q = 0$. The gauge u multiplies every transport from y to x on the left by $u(y)$ and on the right by $u(x)^{-1}$, and the mean equation is covariant under this ($\mathcal{U} \mapsto u(y)\mathcal{U}u(x)^{-1}$, so $M_x \mapsto u(y)M_xu(y)^{-1}$, which preserves $\|M_x - \mathbf{1}\|_{\mathrm{op}}$ and $\operatorname{tr}_N M_x$), so the relative-mean and gauge-functional bounds hold for U itself. The near-identity path and unnormalised-mean assertions remain in the gauge u. For each $0 \leqslant k \leqslant 5$, choose C_k^{a} to be at least one and the maximum of C_k^{D} and the uniform derivative bounds for the phase, the two logarithms, the relative means and the gauge functional obtained in (iii)–(iv). Their finiteness and uniformity follow from the compact phase sets, the finitely many templates and the stated analytic neighbourhoods. No explicit value of C_k^{D} is asserted for these other functions. *q.e.d.*

Let c be a row within $2R_{\mathrm{c}} + 2L$ of I (a near row); by Section 3.3 every row of a near primitive (a primitive whose free-input set meets I) is within $2R_{\mathrm{c}}$ of I, since its bonds are within $4L + 3 \leqslant 2R_{\mathrm{c}}$ of a free-input bond in I, and every row of a plaquette with a bond in $\mathcal{E}_c$ for such a row c is within $2R_{\mathrm{c}} + 2L$ of I (its bonds are within $2L - 1 + 1$ of a bond of $\mathcal{E}_c$). The two-block set $\mathcal{E}_c$ lies inside a lifted rectangle of side $2L+1 \leqslant s_{\mathfrak{P}}$ all of whose bonds are within $2R_{\mathrm{c}}+2L+2L+1 \leqslant 3R_{\mathrm{c}}$ of I (as $R_{\mathrm{c}} \geqslant 2R_{\mathrm{loc}}+1 = 4L+5$); Lemmas B.4 and B.5 give a gauge on it with phases at most $\widehat{\mu} < \tau_L/2$ (the bound of Lemma B.5 is monotone in the side), and Lemma 3.11(iii) (range $3R_{\mathrm{c}}$) bounds the retained inputs $A_{\mathrm{e}}(b)$, $b \in \mathcal{E}_c \cap J$, by $r_{\mathrm{e},\xi}$. Likewise a block $\mathcal{B}_y$ with centre y within $2R_{\mathrm{c}} + 2L$ of I is a lifted rectangle of side $L \leqslant s_{\mathfrak{P}}$ whose bonds are within $3R_{\mathrm{c}}$ of I, so Lemmas B.4 and B.5 apply to it directly. For real $B \in \mathcal{U}_y$, the block input satisfies $\|g_0B|_{\mathcal{E}_y}\| \leqslant 2C_{\mathsf{C}}g_0(1+2r) \leqslant \vartheta_*/4$ by threshold item 3 and $C_{\mathsf{C}} \geqslant |\mathcal{E}_y|^{1/2}$. Hence the real relative-mean conclusions of (iv) apply on the caller's entire block polydisc; its analytic continuation is supplied by the same lemma. So Lemma B.7 applies to every row and every centre within $2R_{\mathrm{c}} + 2L$ of I, in particular to every row of every near primitive and of every plaquette with a bond in the two-block set of such a row. Put $s_\xi = 1+r+r_{\mathrm{e},\xi} \leqslant 1+2r$. The next lemma shows that Lemma B.7(i)–(iii) apply at every row of $\Omega_1^{[1]}$, near or far, and that the arguments of the whole fibre $D^{(11)}(V, V_0)$ lie in its balls.

Lemma B.8 (Fibre Smallness)**.** *Let $n_{\mathcal{E}}$ be the largest number of bonds of a two-block set $\mathcal{E}_c$ of* (3.10) *(a constant of L), let $O(L^2)$ be the constant of [3, p. 247] in (B1)(ii), and let g_0 satisfy threshold items 5 and 6 of Section B.8; item 6 is*

$$3c_Gc_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_0 \leqslant \tfrac{1}{2}\tau_L, \qquad n_{\mathcal{E}}^{1/2}c_Gc_{\mathrm{h}}\,2\,O(L^2)\,\varepsilon_0 \leqslant \tfrac{1}{4}\vartheta_*\,. \tag{B.16}$$

The first condition gives $3c_{\rm h}s_{\mathfrak{P}}\varepsilon_0 \leqslant \tau_L/(2c_G) \leqslant r_G$, *since* $\tau_L \leqslant 2c_G r_G$ *by* (B.6). *Let* (Ω_1, V, V_0) *be an admissible pair, with the background* $U_1 = U_1(V, V_0)$ *of (B1) on the bonds of* Ω_1 *and* V_0 *on the other bonds. Phases* $A \in \mathfrak{g}^{\Omega_1}$ *are extended by zero, and* $\varphi(A) \coloneqq (\varphi_c(A|_{\mathcal{E}_c}))_{c\in\Omega_1^{[1]}}$ *wherever every component is defined.*

1. *For every row* $c \in \Omega_1^{[1]}$, U_1 *has a site gauge on* $\mathfrak{P}_c = \mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, *the union of the two blocks of* c, *with phases* $\|\phi\|_\infty \leqslant \tau_L/2$ *on* $\mathcal{E}_c$. *Hence Lemma B.7(i)–(iii) hold at* c *for* $U = U_1$: *the function* $\varphi_c = \widetilde{D}_{U_1,c}$ *of Lemma B.7(ii) is analytic on the ball* $\|A|_{\mathcal{E}_c}\| < \vartheta_*$, *real on real phases, with* $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$, *and for every* A *in that ball* $\varphi_c(A|_{\mathcal{E}_c})$ *is the unique solution* d_c *of the row equation of* c *with* $h(d_c) \leqslant \rho_{\rm d}$.

2. *Let* $U \in D^{(11)}(V, V_0)$, $A' = \frac{1}{i}\log(UU_1^{-1})$ *on the bonds of* Ω_1 *(defined by (B1)),* $d = \mathcal{N}_{U_1}(A')$ *and* $A = A' + h_{U_1}d$, *so that* $A = g_0\mathsf{C}_{U_1}B$ *with* $B = g_0^{-1}A'|_{\Omega_1^*}$ *(the decomposition step of the converse argument of Section 3.1 after* (3.2), *up to the definition of* B, *which uses (B1), the constraint,* $LQ_{U_1}h_{U_1} = \mathbf{1}$ *and the invertibility of* $\alpha_c(U_1)$ *in the elimination maps* (3.12), *given by (i) and Lemma 3.2 as in the Remark under (B2), and neither the cutoff (1.25) nor (iii)). Then for every row* $c \in \Omega_1^{[1]}$

$$\|A'|_{\mathcal{E}_c}\| \leqslant \tfrac{1}{4}\vartheta_*, \qquad h(d_c) \leqslant \tfrac{1}{32}K_*\vartheta_*^2 < \rho_{\rm d}, \qquad \|A|_{\mathcal{E}_c}\| < \tfrac{1}{2}\vartheta_*, \qquad d_c = \varphi_c(A|_{\mathcal{E}_c})\,.$$

3. *For every* $B \in \mathfrak{g}^{\Omega_1^*}$, *every family* d *of chart class at* $A = g_0\mathsf{C}_{U_1}B$ *(Section 3.1) with* $U_d(B) \in D^{(11)}(V, V_0)$ *is* $d = \varphi(A)$.

4. *Let* $\mathcal{N}_\Theta = \{A \in \mathfrak{g}^{\Omega_1} : \|A|_{\mathcal{E}_c}\| < \vartheta_*/2$ *for every* $c \in \Omega_1^{[1]}\}$; *it is open, and by (ii) it contains the argument* A *of every* $U \in D^{(11)}(V, V_0)$. *On* $\mathcal{N}_\Theta$ *the family* φ *is analytic, real on real phases, and solves the row equations* (3.24); *the phase* $\Theta(A) = A - h_{U_1}\varphi(A)$ *has* $\|\Theta(A)|_{\mathcal{E}_c}\| < \vartheta_*$ *at every row and is chart-valued,* $h(\Theta(A)(b)) < \vartheta_* < \varrho_G$ *on every bond* b *of* Ω_1; *for every row,* $\|h_{U_1}(c)\partial_{A(b_0(c))}\varphi_c(A|_{\mathcal{E}_c})\| \leqslant 4H_*K_*\|A|_{\mathcal{E}_c}\| < 1$; *and*

$$\det\Big(\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\varphi(A)\Big) = \prod_{c\in\Omega_1^{[1]}} \det{}_{\mathfrak{g}}\big(\mathbf{1} - h_{U_1}(c)\partial_{A(b_0(c))}\varphi_c(A|_{\mathcal{E}_c})\big) > 0\,. \tag{B.17}$$

Proof. (i) The set $\mathfrak{P}_c$ is a lifted rectangle of side at most $2L+1 \leqslant s_{\mathfrak{P}}$, injective modulo the periods (every fine period is at least $8M > 2L+1$), and it contains every bond of $\mathcal{E}_c$, both endpoints of such a bond lying in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$ by (3.10). The two blocks of c lie in $\Omega_1^\sim$ ((B1)(iv)) and $U_1 \in D^{(11)}(V, V_0)$ ((B1)(iii)), so (B1)(i) gives $|U_1(\partial p) - 1| < \varepsilon_0$, hence $\|U_1(\partial p) - \mathbf{1}\|_{\rm op} \leqslant c_{\rm h}\varepsilon_0 = 2c_{\rm h}(\varepsilon_0/2)$ by the norm comparison of the preamble, for every plaquette $p \subset \mathfrak{P}_c$. By the consequence $3c_{\rm h}s_{\mathfrak{P}}\varepsilon_0 \leqslant r_G$ of (B.16) noted after it, $6c_{\rm h}s_{\mathfrak{P}}(\varepsilon_0/2) \leqslant r_G$, so Lemma B.5 with $\varepsilon_0/2$ in place of ε_1 (the remark after its proof) gives a site gauge on $\mathfrak{P}_c$ with phases $h(A_b) \leqslant 3c_G c_{\rm h}s_{\mathfrak{P}}\varepsilon_0 \leqslant \tau_L/2$, by the first condition. This is the hypothesis of Lemma B.7 at c. The phases of $\mathfrak{g}^{\Omega_1}$, extended by zero, form the subspace of $\mathfrak{g}^{\mathcal{E}_c}$ of the phases vanishing on the bonds of $\mathcal{E}_c$ off Ω_1; for A in it and $d \in \mathfrak{g}$ the phase $A - h_{U_1}(c)d$ lies in it, since $b_0(c)$ is a bond of Ω_1, and Lemma B.7(i)–(iii) hold on it by restriction.

(ii) By (B1)(ii)–(iii), $|U'(b) - 1| < 2\,O(L^2)\varepsilon_0$ for $U' = UU_1^{-1}$ on every bond b of Ω_1: every such bond, the designated bonds included, is a bond of $\square'^{\sim 2}$ for the $L^2M_2R_1$-cube $\square' \subset \Omega_1$ that contains an endpoint of b in Ω_1 (Ω_1 is a union of LMR_1-cubes, p.248, each a union of $L^2M_2R_1$-cubes: the partitions of [3, p. 245] are compatible, that is nested, (H4), and all sides are powers of L, the paragraph following (3.1)). By item 5, $2\,O(L^2)\varepsilon_0 < r_G/\sqrt{N}$, so $\|U'(b) - \mathbf{1}\|_{\rm op} \leqslant c_{\rm h}|U'(b) - 1| < r_G$ and $h(A'(b)) \leqslant c_G\|U'(b) - \mathbf{1}\|_{\rm op} \leqslant c_G c_{\rm h}\,2\,O(L^2)\varepsilon_0$

(preamble); $\mathcal{E}_c$ has at most $n_{\mathcal{E}}$ bonds, and A' vanishes on those off Ω_1, so the second condition of (B.16) gives $\|A'|_{\mathcal{E}_c}\| \leqslant \vartheta_*/4 < \varrho_*$. The row component $d_c = \mathcal{N}_{U_1,c}(A')$ depends on $A'|_{\mathcal{E}_c}$ only (Lemma 3.2), and Taylor's formula with Lemma B.7(i), which applies by (i), gives $h(d_c) \leqslant \frac{1}{2}K_*(\vartheta_*/4)^2 = \frac{1}{32}K_*\vartheta_*^2 < 2K_*\vartheta_*^2 = \rho_{\mathrm{d}}$. The only designated bond in $\mathcal{E}_c$ is $b_0(c)$ (Lemma 3.2) and h_{U_1} has its outputs at the designated bonds, so $A|_{\mathcal{E}_c} = A'|_{\mathcal{E}_c} + h_{U_1}(c)d_c$, and $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/4 + H_*K_*\vartheta_*^2/32 < \vartheta_*/2$ by $4K_*H_*\vartheta_* < 1$. Finally $A'|_{\mathcal{E}_c} = A|_{\mathcal{E}_c} - h_{U_1}(c)d_c$, so $d_c = \mathcal{N}_{U_1,c}(A|_{\mathcal{E}_c} - h_{U_1}(c)d_c)$ solves the row equation of c at A, with $h(d_c) \leqslant \rho_{\mathrm{d}}$ and $\|A|_{\mathcal{E}_c}\| < \vartheta_*$; by the uniqueness in (i), $d_c = \varphi_c(A|_{\mathcal{E}_c})$.

(iii) Since d is of chart class, $q_d(B) = A - h_{U_1}d$ is the chart logarithm of $U_d(B)U_1^{-1}$ on the bonds of Ω_1, so it is the phase A' of (ii) for $U = U_d(B)$. The row equations give $d_c = \mathcal{N}_{U_1,c}(A - h_{U_1}d) = \mathcal{N}_{U_1,c}(A')$, that is $d = \mathcal{N}_{U_1}(A')$, and $A = A' + h_{U_1}d$. So A', d and A are the quantities of (ii) for $U = U_d(B)$, and (ii) gives $d_c = \varphi_c(A|_{\mathcal{E}_c})$ for every row.

(iv) $\mathcal{N}_\Theta$ is a finite intersection of preimages of open balls under restriction maps, hence open, and it contains the arguments by (ii). By (i) each component $\varphi_c(A|_{\mathcal{E}_c})$ is analytic and real on $\mathcal{N}_\Theta$ and solves the row equation of c at A; the row equations decouple ($(A - h_{U_1}\varphi(A))|_{\mathcal{E}_c} = A|_{\mathcal{E}_c} - h_{U_1}(c)\varphi_c(A|_{\mathcal{E}_c})$, Lemma 3.2), so φ solves the system (3.24). By (B.15) and Lemma B.7(i), $\|\Theta(A)|_{\mathcal{E}_c}\| \leqslant \|A|_{\mathcal{E}_c}\| + 2H_*K_*\|A|_{\mathcal{E}_c}\|^2 < \vartheta_*/2 + H_*K_*\vartheta_*^2/2 < \vartheta_*$. Every bond b of Ω_1 lies in $\mathcal{E}_c$ for a row $c \in \Omega_1^{[1]}$: b has an endpoint in Ω_1, whose block $\mathcal{B}_y$ lies in Ω_1 (Ω_1 is a union of blocks); if both endpoints of b lie in $\mathcal{B}_y$, any coarse bond c with $c_- = y$ serves, and otherwise the coarse bond joining y and the centre of the block of the other endpoint does; in both cases a block centre of c lies in Ω_1, so $c \in \Omega_1^{[1]}$, and $b \in \mathcal{E}_c$ by (3.10). Hence $h(\Theta(A)(b)) \leqslant \|\Theta(A)|_{\mathcal{E}_c}\| < \vartheta_* \leqslant \varrho_G/2$. By (B.15) and Lemma B.7(i), $\|h_{U_1}(c)\partial_{A(b_0(c))}\varphi_c\| \leqslant H_*C_{\mathsf{D}}\|A|_{\mathcal{E}_c}\| = 4H_*K_*\|A|_{\mathcal{E}_c}\| < 2H_*K_*\vartheta_* < 1$. The matrix $\frac{\delta}{\delta A}\varphi$ has rows indexed by $\Omega_1^{[1]}$ and columns by the bonds of Ω_1, h_{U_1} maps row c to the bond $b_0(c)$, and $\partial_{A(b_0(c'))}\varphi_c = 0$ for $c' \neq c$, since φ_c depends on $A|_{\mathcal{E}_c}$, which contains no designated bond other than $b_0(c)$ (Lemma 3.2). So, in the ordering (free bonds, designated bonds), $\mathbf{1} - h_{U_1}\frac{\delta}{\delta A}\varphi$ is block triangular, with identity block on the free bonds and designated block $\mathrm{diag}_c(\mathbf{1} - h_{U_1}(c)\partial_{A(b_0(c))}\varphi_c)$; this gives the product in (B.17). Each factor is positive by Lemma B.7(iii), which applies by (i), since $\|A|_{\mathcal{E}_c}\| < \vartheta_*/2$. *q.e.d.*

Lemma B.9 (Primitive Bounds)**.** *Let g_0 satisfy threshold item 3 of Section B.8, $4C_{\mathsf{C}}g_0(1 + 2r) \leqslant \vartheta_*/2$ (hence $4C_{\mathsf{C}}g_0 s_\xi \leqslant \vartheta_*/2$, and $g_0 s_\xi \leqslant \vartheta_*/(8C_{\mathsf{C}}) \leqslant 1$, since $C_{\mathsf{C}} \geqslant 1$ and $\vartheta_* \leqslant \tau_L/2 \leqslant 1/2$ by (B.6)). Then on the enlarged box $\{|a_b| \leqslant 2r\ \forall b \in I\}$ and for $t \in [0,1]$, every primitive p_α with free-input set S_α meeting I is, at the input $a + tA_{\mathrm{e}}$, the restriction to real arguments of a function analytic in (a,t) on a complex neighbourhood of the enlarged box times $[0,1]$, and with a constant C_{p} depending on L and the group only,*

$$|p_{\xi,\alpha}(a+tA_{\mathrm{e}})| \leqslant C_{\mathrm{p}}g_0 s_\xi^3, \qquad |p^{\mathrm{H}}_{\xi,b}(a+tA_{\mathrm{e}})| \leqslant C_{\mathrm{p}}g_0^2 s_\xi^2, \qquad |p^{\mathrm{J}}_{\xi,c}(a+tA_{\mathrm{e}})| \leqslant C_{\mathrm{p}}g_0 s_\xi\ , \tag{B.18}$$

and for each of the three kinds, in the Hilbert norms on $\mathfrak{g}^{S_\alpha}$,

$$\|D_a^2 p_\alpha\| \leqslant C_{\mathrm{p}}g_0 s_\xi, \qquad \|D_a p_\alpha\| \leqslant C_{\mathrm{p}}g_0 s_\xi^2, \qquad |p_\alpha(a + tA_{\mathrm{e}}) - p_\alpha(tA_{\mathrm{e}})| \leqslant C_{\mathrm{p}}g_0 s_\xi^3\ , \tag{B.19}$$

and $\|D_a^3 p_\alpha\| \leqslant C_{\mathrm{p}}g_0$, $\|D_a^4 p_\alpha\| \leqslant C_{\mathrm{p}}g_0^2$; the same four derivative bounds hold for the derivatives $D_B^k p_\alpha$, $k = 1,\dots,4$, in the full input $B = a + tA_{\mathrm{e}}$ on $\mathfrak{g}^{S_\alpha}$, retained inputs included (the input of the primitive is linear in B with the same norm bound, see the proof). Moreover, for every row c within $2R_{\mathrm{c}} + 2L$ of I and every real phase A with $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$, the row equation at A has exactly one small solution $\widetilde{D}_{U,c}(A) = \varphi_c(A|_{\mathcal{E}_c})$ ($h(d_c) < \rho_{\mathrm{d}} = 2K_*\vartheta_*^2$), with φ_c analytic on the complex ball of radius ϑ_*, real on real phases, $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$; $g_0\mathsf{C}_U\mathcal{U}_c$ lies in that phase ball; the determinant $\det_{\mathfrak{g}}(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c}(A))$ is positive there; and

the row solution, the phase $q_U|_{\mathcal{E}_c}$, the normalised Haar logarithm and the real determinant logarithm are analytic in $B|_{\mathcal{E}_c\cap E_*}$ on $\mathcal{U}_c$, as are, when c is within $2R_{\mathrm{c}}$ of I, the plaquette terms $s_{p,U}(q_U(\bullet))$ of the plaquettes p with a bond in $\mathcal{E}_c$ (on the intersection of the polydiscs of the rows of p, all of which are rows within $2R_{\mathrm{c}}+2L$ of I); for every centre y within $2R_{\mathrm{c}}+2L$ of I and every $B\in\mathcal{U}_y$ the relative means exist, lie within r_G of $\mathbf{1}$ in operator norm, and $\mathscr{G}_{y,U}(g_0B|_{\mathcal{E}_y})$ is analytic in $B|_{\mathcal{E}_y}$; all are compositions, with the linear map $B\mapsto g_0\mathsf{C}_UB$ (of norm at most C_{C} for the Hilbert norms, by Lemma B.7(i)), of functions of the phase analytic on a complex neighbourhood of the closed ball $\|A\|\leqslant\vartheta_*/2$ (on the complex ball of radius ϑ_* for the row solution and the phase) with derivatives up to order five bounded on that closed ball by constants depending on L and the group only, so their derivatives in B up to order five are bounded on the polydiscs by such constants. Where $(A_{\mathrm{e}},a)\in\mathcal{D}_\xi$ these objects are Bałaban's [3], by (B4)(i) and the uniqueness of the small solution: for every row within $2R_{\mathrm{c}}+2L$ of I the phase $g_0\mathsf{C}_U\Psi_\xi(a+A_{\mathrm{e}})$ lies in the ball $\|A|_{\mathcal{E}_c}\|\leqslant\vartheta_*/4$, because a is in the open box and $\Psi_\xi(a+A_{\mathrm{e}})=a+A_{\mathrm{e}}$ with $|A_{\mathrm{e}}(b)|<r$ on the two-block set (Lemma 3.11(i),(iii),(iv)), so $\Psi_\xi(a+A_{\mathrm{e}})\in\mathcal{U}_c$ and the bound on $g_0\mathsf{C}_U\mathcal{U}_c$ applies. This is Lemma 3.6.

Proof. The input of a primitive. On the enlarged box, $|a_b+tA_{\mathrm{e}}(b)|\leqslant 2r+r_{\mathrm{e},\xi}\leqslant 2s_\xi$ for every bond b within $3R_{\mathrm{c}}$ of I (Lemma 3.11(iii) bounds the retained inputs by $r_{\mathrm{e},\xi}$), in particular for the free bonds of the two-block sets of the near rows and for the free bonds of the plaquettes and blocks of the near primitives. The rows of a near primitive are within $2R_{\mathrm{c}}$ of I (applicability paragraph). The *input* $\mathfrak{u}$ of a primitive is the vector $g_0\mathsf{C}_UB$ restricted to the following bond set S: for a plaquette primitive $E_{\xi,p}$, $p_{\xi,p}$ the union of the sets $\mathcal{E}_c$ over the rows c of p (at most four) together with the free bonds of ∂p not in those sets; for the Haar primitive $p^{\mathrm{H}}_{\xi,b}$ the set $\mathcal{E}_c$ when $b=b_0(c)$ and $\{b\}$ when b is free; for the Jacobian primitive $p^{\mathrm{J}}_{\xi,c}$ the set $\mathcal{E}_c$; for a gauge primitive $E_{\xi,y}$, $p_{\xi,y}$ the set $\mathcal{E}_y$ (on which C_U is the identity, all its bonds being free). The primitive is a function of the phase $q_U(B)$ on its bonds, and that phase is a function of the input: on $\mathcal{E}_c$ it is $\mathfrak{u}|_{\mathcal{E}_c}-h_U\widetilde{D}_{U,c}(\mathfrak{u}|_{\mathcal{E}_c})$ (Lemma B.7(ii)), and on a free bond outside the row sets it is $\mathfrak{u}_b=g_0B(b)$. With $B=a+tA_{\mathrm{e}}$, by Lemma B.7(i) each row component satisfies $\|\mathfrak{u}|_{\mathcal{E}_c}\|\leqslant C_{\mathsf{C}}g_0\cdot 2s_\xi\leqslant\vartheta_*/4$ and each free-bond component outside the row sets $h(\mathfrak{u}_b)\leqslant|\mathfrak{u}_b|\leqslant 2g_0s_\xi\leqslant\vartheta_*/4$ (threshold item 3 and $C_{\mathsf{C}}\geqslant 1$), so for the plaquette, Haar and Jacobian primitives (at most four rows and four further free bonds)

$$\|\mathfrak{u}\|\leqslant 16C_{\mathsf{C}}g_0s_\xi,\qquad \|D_a\mathfrak{u}\|\leqslant 5C_{\mathsf{C}}g_0,\qquad \|\partial_t\mathfrak{u}\|\leqslant 16C_{\mathsf{C}}g_0s_\xi\ ,$$

the second in the operator norm from the Hilbert norm of $\mathfrak{g}^{S_\alpha}$ (the Hilbert bound of Lemma B.7(i) on each of at most four rows, plus the identity on the free bonds); for the gauge primitive, whose input $g_0B|_{\mathcal{E}_y}$ has $|\mathcal{E}_y|=4L^3(L-1)$ free bonds, $\|\mathfrak{u}\|\leqslant g_0|\mathcal{E}_y|^{1/2}\cdot 2s_\xi\leqslant 2C_{\mathsf{C}}g_0s_\xi\leqslant\vartheta_*/4$ (as $C_{\mathsf{C}}\geqslant(4L^3(L-1))^{1/2}$ and $h\leqslant|\bullet|$), $\|D_a\mathfrak{u}\|\leqslant g_0$ and $\|\partial_t\mathfrak{u}\|\leqslant 2C_{\mathsf{C}}g_0s_\xi$, so the three displayed bounds hold for all four kinds. The input is linear in B, hence affine in (a,t) and linear in a at fixed t; $D_a\mathfrak{u}$ is the restriction of $D_B\mathfrak{u}=g_0\mathsf{C}_U|_S$ to the young directions, and the same count gives $\|D_B\mathfrak{u}\|\leqslant 5C_{\mathsf{C}}g_0$ (respectively $\leqslant g_0$ for the gauge primitive), so every bound below on a derivative in a holds verbatim for the derivative of the same order in B on $\mathfrak{g}^{S_\alpha}$, retained inputs included. Each primitive is $\varphi(\mathfrak{u})$ for a function φ of the input analytic on a complex neighbourhood of the closed set $\mathcal{P}=\{\mathfrak{u}\in\mathfrak{g}^S_{\mathbb{C}}:\|\mathfrak{u}|_{\mathcal{E}_c}\|\leqslant\vartheta_*/2$ for each row c of the primitive, $h(\mathfrak{u}_b)\leqslant\vartheta_*/2$ for each free bond b of S outside the row sets$\}$ (for the gauge primitive, the closed ball $\|\mathfrak{u}\|\leqslant\vartheta_*$ of $\mathfrak{g}^{\mathcal{E}_y}_{\mathbb{C}}$), with derivatives up to order five bounded on $\mathcal{P}$ by constants of L and the group: Lemma B.7(iii) for the row solution, the phase and the two logarithms, each a function of one row component (the Haar logarithm at a free bond outside the row sets is the function of (iii) at a free bond of a two-block set, where $q_U(A)(b)=A(b)$, a function of the single component $\mathfrak{u}_b$); for the plaquette term, with $W(q)=(e^{iq}U)(\partial p)$,

$s_{p,U}(q) = 1 - \operatorname{Re}\operatorname{tr}_N W(q)$ equals $1 - \frac{1}{2}(\operatorname{tr}_N W(q) + \operatorname{tr}_N W(q)^{-1})$ for real q, $W(q)$ being unitary, and $W(q)$ and $W(q)^{-1}$ are ordered products of the fixed links and of $e^{\pm iq(b)}$, entire functions of the phases $q(b)$, which are analytic functions of one row component (with $h(q_U(A)(b)) \leqslant \|A\| + H_* 2K_* \|A\|^2 \leqslant \vartheta_*$ on the ball) or are free-bond components of $\mathfrak{u}$; this is the analytic function meant by $s_{p,U}(q_U(\bullet))$, with derivatives bounded on $\mathcal{P}$ by the chain rule; for the gauge term, Lemma B.7(iv). The segment from 0 to $\mathfrak{u}$ lies in $\mathcal{P}$, each component of $\mathfrak{u}$ being within $\vartheta_*/4$ of zero. Hence every primitive at the input $a + tA_{\mathrm{e}}$ is the restriction to real (a,t) of a function analytic in (a,t) on a complex neighbourhood of the enlarged box times $[0,1]$ (the composition of φ with the affine map), and the bounds below are uniform there.

Energy residual. $E_{\xi,\alpha}(a+tA_{\mathrm{e}}) = g_0^{-2}\varphi(\mathfrak{u})$ with $\varphi(\mathfrak{u}) = -[s_{\alpha,U}(q_U(\mathfrak{u})) - s_{\alpha,U}(0)]$ for a plaquette and $\varphi(\mathfrak{u}) = -\mathscr{G}_{y,U}(\mathfrak{u})$ for a centre. Since $\mathfrak{u}$ is linear in B, the Taylor polynomial of degree two in B at $B = 0$ subtracted in (3.29) is that of φ at 0 composed with the linear map, so $p_{\xi,\alpha} = g_0^{-2}[\varphi(\mathfrak{u}) - D\varphi(0)\mathfrak{u} - \frac{1}{2}D^2\varphi(0)[\mathfrak{u},\mathfrak{u}]]$, the second-order Taylor residual at the input, bounded by $g_0^{-2}\frac{1}{6}\|D^3\varphi\|\|\mathfrak{u}\|^3 \leqslant Cg_0 s_\xi^3$ (the segment from 0 to $\mathfrak{u}$ lies in $\mathcal{P}$). Its first derivative in a is $g_0^{-2}(D\varphi(\mathfrak{u}) - D\varphi(0) - D^2\varphi(0)\mathfrak{u})D_a\mathfrak{u}$, bounded by $g_0^{-2}\frac{1}{2}\|D^3\varphi\|\|\mathfrak{u}\|^2 \cdot 5C_{\mathsf{C}}g_0 \leqslant Cg_0 s_\xi^2$; its second derivative is $g_0^{-2}(D^2\varphi(\mathfrak{u}) - D^2\varphi(0))[D_a\mathfrak{u}, D_a\mathfrak{u}]$, bounded by $g_0^{-2}\|D^3\varphi\|\|\mathfrak{u}\|(5C_{\mathsf{C}}g_0)^2 \leqslant Cg_0 s_\xi$; its third and fourth derivatives are $g_0^{-2}D^3\varphi(\mathfrak{u})[D_a\mathfrak{u}]^3$ and $g_0^{-2}D^4\varphi(\mathfrak{u})[D_a\mathfrak{u}]^4$, bounded by Cg_0 and Cg_0^2. The Taylor polynomial subtracted is that of the germ (Section 3.3), which is the Taylor polynomial of φ at 0 by Lemma B.7(ii).

Haar logarithm. $\psi(\mathfrak{u}) = \log\sigma(q_U(\mathfrak{u})(b)) - \log\sigma(0)$, the primitive (3.30), has $\psi(0) = 0$ and $D\psi(0) = 0$ (Lemma B.7(iii): σ is even and $Dq_U(0)$ is linear). Hence $|\psi(\mathfrak{u})| \leqslant \frac{1}{2}\|D^2\psi\|\|\mathfrak{u}\|^2 \leqslant Cg_0^2 s_\xi^2$, $\|D_a\psi\| \leqslant \|D^2\psi\|\|\mathfrak{u}\| \cdot 5C_{\mathsf{C}}g_0 \leqslant Cg_0^2 s_\xi \leqslant Cg_0 s_\xi^2$, $\|D_a^2\psi\| \leqslant \|D^2\psi\|(5C_{\mathsf{C}}g_0)^2 \leqslant Cg_0 s_\xi$, and the third and fourth derivatives are bounded by $Cg_0^3 \leqslant Cg_0$ and $Cg_0^4 \leqslant Cg_0^2$.

Constraint logarithm. $\lambda(\mathfrak{u}) = \log|\det_{\mathfrak{g}}(\mathbf{1} - h_U(c)\partial_{A(b_0(c))}\widetilde{D}_{U,c}(\mathfrak{u}))| = \log\det_{\mathfrak{g}}(\cdots)$ (the determinant is positive, Lemma B.7(iii)) has $\lambda(0) = 0$ and derivatives up to order five bounded on $\mathcal{P}$; its first derivative at 0 need not vanish. Hence $|\lambda(\mathfrak{u})| \leqslant \|D\lambda\|\|\mathfrak{u}\| \leqslant Cg_0 s_\xi$, $\|D_a\lambda\| \leqslant \|D\lambda\| \cdot 5C_{\mathsf{C}}g_0 \leqslant Cg_0 \leqslant Cg_0 s_\xi^2$, $\|D_a^2\lambda\| \leqslant \|D^2\lambda\|(5C_{\mathsf{C}}g_0)^2 \leqslant Cg_0^2 \leqslant Cg_0 s_\xi$ (as $g_0 \leqslant 1$ and $s_\xi \geqslant 1$), and the third and fourth derivatives are bounded by $Cg_0^3 \leqslant Cg_0$ and $Cg_0^4 \leqslant Cg_0^2$. Let C'_{p} be the maximum over the finitely many templates of the constants C above and put $C_{\mathrm{p}} = 2C'_{\mathrm{p}}$. Then (B.18), the first two bounds of (B.19) and the third- and fourth-order bounds hold with room to spare, and the last bound of (B.19) is the difference of two values each bounded by $C'_{\mathrm{p}}g_0 s_\xi^3$ (for all three kinds, since $s_\xi \geqslant 1$ and $g_0 \leqslant 1$), hence by $2C'_{\mathrm{p}}g_0 s_\xi^3 = C_{\mathrm{p}}g_0 s_\xi^3$.

The chart statements. Let c be a row within $2R_{\mathrm{c}} + 2L$ of I; by the applicability paragraph Lemma B.7 applies to it, and (ii)–(iii) give, for every real A with $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/2$, the unique small solution $\varphi_c(A|_{\mathcal{E}_c})$ with φ_c analytic on the complex ball of radius ϑ_* and real on real phases, $\varphi_c(0) = 0$, $D\varphi_c(0) = 0$, the positivity of the determinant, and the analyticity of the row solution, the phase and the two logarithms on a complex neighbourhood of the closed ball of radius $\vartheta_*/2$, with derivatives up to order five bounded on that ball. For $B \in \mathcal{U}_c$, $\max_{b\in\mathcal{E}_c\cap E_*}|B(b)| < 2 + 4r = 2(1+2r)$ and $\|g_0\mathsf{C}_U B\|_{\mathcal{E}_c} \leqslant 2C_{\mathsf{C}}g_0(1+2r) \leqslant \vartheta_*/4$ by threshold item 3, so $g_0\mathsf{C}_U\mathcal{U}_c$ lies in the phase ball; $B \mapsto g_0\mathsf{C}_U B$ is linear with norm at most $g_0 C_{\mathsf{C}} \leqslant C_{\mathsf{C}}$ for the Hilbert norms (Lemma B.7(i)), so the derivatives in B are bounded by the same constants times C_{C}^k. For c within $2R_{\mathrm{c}}$ of I, the plaquette term of a plaquette p with a bond in $\mathcal{E}_c$ is analytic in B on the intersection of the polydiscs of the rows of p, as above, and these rows are within $2R_{\mathrm{c}} + 2L$ of I (applicability paragraph), so the chart applies to each of them. For a centre y within $2R_{\mathrm{c}} + 2L$ of I, Lemma B.5 applied to the block $\mathcal{B}_y$ (applicability

paragraph) gives a gauge with phases at most $\widehat{\mu} < \tau_L/2$ on $\mathcal{E}_y$; every bond of $\mathcal{E}_y$ is free, so $q_U(B) = g_0 B$ there and $\|g_0 B|_{\mathcal{E}_y}\| \leqslant g_0 |\mathcal{E}_y|^{1/2} \max_b |B(b)| \leqslant g_0 (4L^3(L-1))^{1/2} \cdot 2(1+2r) \leqslant 2C_{\mathsf{C}} g_0 (1+2r) \leqslant \vartheta_*/4$ under threshold item 3 (using $h \leqslant |\bullet|$, $|\mathcal{E}_y| = 4L^3(L-1)$ and $C_{\mathsf{C}} \geqslant (4L^3(L-1))^{1/2}$); Lemma B.7(iv) then gives the existence of the relative means within r_G of $\mathbf{1}$ and the analyticity and derivative bounds of $\mathscr{G}_{y,U}(g_0 B|_{\mathcal{E}_y})$ on $\mathcal{U}_y$. The values at $q = 0$ of the plaquette and gauge terms are subtracted in the primitives. Finally, where $(A_{\mathrm{e}}, a) \in \mathcal{D}_\xi$ and c is within $2R_{\mathrm{c}} + 2L$ of I, the phase $A = g_0 \mathsf{C}_U \Psi_\xi(a + A_{\mathrm{e}})$ has $\|A|_{\mathcal{E}_c}\| \leqslant \vartheta_*/4$ (statement of the lemma), $\varphi_c(A|_{\mathcal{E}_c})$ is the row solution $\widetilde{D}_c(A)$ of (B4)(i), which is that of [3] by (B4)(i) and Lemma B.8(iii), and the phase, the Haar density, the Jacobian and the plaquette and gauge terms of [3, (1.14)–(1.15)] are the same functions of it. *q.e.d.*

Lemma B.10 (Host Multiplicity and Potential Size)**.** *Let $\mathfrak{T}$ be the set of primitive templates (kind, label and free-input shape, including the shapes truncated by Ω_1) modulo $L\mathbb{Z}^4$, anchor each template at one of its input bonds, and let D_{h} be one plus the maximum ℓ^∞ diameter of a lifted host* (3.36) *and B_{h} the maximum number of M-cubes in one, both over $\mathfrak{T}$. With*

$$N_{\mathrm{h}} = |\mathfrak{T}|(2\lceil D_{\mathrm{h}} \rceil + 3)^4, \qquad C_0 = N_{\mathrm{h}} C_{\mathrm{p}} e^{\kappa(B_{\mathrm{h}}+4)}, \qquad \omega(g_0) = C_0 g_0 (1+r)^3 \,, \tag{B.20}$$

at most N_{h} primitives are assigned to any host Y, $d_0(Y) \leqslant B_{\mathrm{h}} + 4$, and on the enlarged box, for every host,

$$\sup |\mathbf{V}_\xi(Y, a)| \leqslant \omega(g_0) e^{-\kappa d_0(Y)} \,. \tag{B.21}$$

Proof. Fix a vertex of Y. Every primitive assigned to Y has its stencil, hence its anchor, in the lifted host, within ℓ^∞ distance D_{h} of that vertex; there are at most $(2\lceil D_{\mathrm{h}} \rceil + 1)^4$ such anchor positions, and a template with an anchor position determines the primitive. Projection to the torus only identifies positions. By Lemma B.1(i), $d_0(Y) \leqslant N_0(Y) - 1 \leqslant B_{\mathrm{h}}$. At $t = 0$, s_ξ may be replaced by $1 + r$ in Lemma B.9, since no retained input enters; each of the three kinds is then bounded by $C_{\mathrm{p}} g_0 (1+r)^3$, and $|\mathbf{V}_\xi(Y, a)| \leqslant N_{\mathrm{h}} C_{\mathrm{p}} g_0 (1+r)^3 \leqslant N_{\mathrm{h}} C_{\mathrm{p}} g_0 (1+r)^3 e^{\kappa(B_{\mathrm{h}}+4)} e^{-\kappa d_0(Y)}$. *q.e.d.*

The definition gives $C_0 \geqslant C_{\mathrm{p}}$, since $N_{\mathrm{h}} \geqslant 1$ and the exponential factor is at least one. The constant C_0 depends on L, the group, M and κ (through B_{h}, D_{h}), not on the volume, the branch or g_0. The hypothesis of Lemma 3.12 holds for $f = f_\xi$ on the enlarged box, which is a neighbourhood of the closed cutoff box: the exponent $\mathsf{Q}_{\xi,t}$ is a quadratic polynomial plus $P_{\xi,t}$, a finite sum of near primitives, each analytic there by Lemma B.9; and $f_\xi(a) = W_\chi(C; U_\xi(a))$ is a character of a product of the reconstructed links (3.26) over the contour bonds, each of which depends on a through the rows of a free-input set meeting I (Lemma 3.9(i) puts $\Psi_\xi(a + A_{\mathrm{e}}) = a + A_{\mathrm{e}}$ on those rows, so the phase is (3.47)), where q_U is analytic in a by Lemma B.7(ii)–(iii); contour bonds whose rows do not meet I do not depend on a. Both factors are therefore analytic on the enlarged box, uniformly in $t \in [0, 1]$.

B.4 Proof of the Gaussian Payment

Proof of Lemma 3.16. Tail of the walk series. Throughout, E_0, C_{E_0}, $\Gamma_{E_0}(\sigma)$ are as in (3.64) and the sentence preceding it (their block decomposition is (3.67)), and $S_{Z_0}(1) = \mathsf{G}^{1/2}$. Let N be the set of bonds $b' \in I$ with $d_{\infty,T}(b'_-, Z_0) \leqslant R+1$. Since $\iota^*_{E_0} H$ has nonzero entries $H(b, b')$ only for $d(b_-, b'_-) \leqslant R$ and $b \in E_0$ meets Z_0, $\Gamma_{E_0}(\sigma) - \Gamma_{E_0}(1) = \iota^*_{E_0} H \Pi_{I \setminus E_0} \Pi_{\mathsf{N}} (S_{Z_0}(\sigma) - \mathsf{G}^{1/2})$. A walk $\omega = (r; b_0, \dots, b_{m+1})$ with $b_0 \in \mathsf{N}$ has $|w_\omega(\sigma) - 1| \neq 0$ only if $D(\omega)$ meets a cube $\Delta \not\subseteq Z'_0$. The open M-neighbourhood of Z_0 is $\bigcup_{Q \subseteq Z_0} \operatorname{int} Q^\sim$, and by definition every LM-cube meeting it lies in Z'_0 (Section 3.5). Hence $D(\omega)$ contains a point at distance at least

M from Z_0, and at least $M-R-1$ from $b_{0,-}$. Each block of A_H^r ($r<n_*$) or P_H vanishes beyond distance R_*, and D_b lies within ρ_{w} of b_-, so $D(\omega)$ lies within $(m+1)R_*+\rho_{\mathrm{w}}$ of $b_{0,-}$. By (3.59), $M/4 \geqslant \rho_{\mathrm{w}}+R+R_*+2$, so $(m+1)R_* \geqslant M-R-1-\rho_{\mathrm{w}} \geqslant \frac{3}{4}M+R_*+1$ and $m > M/(2R_*) \geqslant N_M$. The absolute block row sum of the terms with $m \geqslant N_M$ is, by the bounds in the proof of Lemma 3.13 and $|w_\omega - 1| \leqslant 2a_*^{m+1}$,

$$s_H^{-1/2} \sum_{r<n_*} \sum_{m \geqslant N_M} a_{n_* m+r}\, 2a_*^{m+1} V_* \beta^m \leqslant 2 s_H^{-1/2} \frac{n_* V_* a_*}{1-a_*\beta} 4^{-N_M} = \delta_M \;, \tag{B.22}$$

using $a_*\beta \leqslant 1/4$. The absolute block column sums of these terms obey the same bound, because every walk in the tail has $m \geqslant N_M$ whatever its first bond. Schur's test gives $\|\Pi_{\mathsf{N}}(S_{Z_0}(\sigma) - \mathsf{G}^{1/2})\| \leqslant \delta_M$. With $\|H\| \leqslant h_+$ this is the first bound of (i); with $\|C_{E_0}\| \leqslant \gamma^{-1}$, since $\iota_{E_0}^* H \iota_{E_0} \geqslant \gamma$, the second follows.

Reserve identity. Write $\Gamma = \Gamma_{E_0}(\sigma) = \Gamma_1 + i\Gamma_2$ with real Γ_1, Γ_2; $\Gamma_{E_0}(1)$ is real, so $\Gamma_2 = \mathrm{Im}(\Gamma_{E_0}(\sigma) - \Gamma_{E_0}(1))$. The quadratic form in (3.66) is bilinear, so for real X, $\mathrm{Re}\langle \Gamma X, C_{E_0}\Gamma X\rangle = \langle \Gamma_1 X, C_{E_0}\Gamma_1 X\rangle - \langle \Gamma_2 X, C_{E_0}\Gamma_2 X\rangle$ and $|e^{-\langle B, \Gamma X\rangle}| = e^{-\langle B, \Gamma_1 X\rangle}$ for real B. For $0 \leqslant \alpha_G < \gamma$ put $C_{E_0,\alpha_G} = (\iota_{E_0}^* H \iota_{E_0} - \alpha_G)^{-1}$. The Gaussian integral over B with density $(2\pi)^{-d_G|E_0|/2} \det(C_{E_0})^{-1/2} e^{-\frac{1}{2}\langle B, (C_{E_0}^{-1}-\alpha_G)B\rangle - \langle B, \Gamma_1 X\rangle}$ equals $\det_{E_0}(\mathbf{1} - \alpha_G C_{E_0})^{-1/2} e^{\frac{1}{2}\langle \Gamma_1 X, C_{E_0,\alpha_G}\Gamma_1 X\rangle}$. Collecting the X-exponent,

$$\begin{aligned} &\int \Big| e^{-\frac{1}{2}\langle \Gamma X, C_{E_0}\Gamma X\rangle} e^{-\langle B, \Gamma X\rangle} \Big| e^{\frac{\alpha_G}{2}\|B\|^2} \mathrm{d}\mu_{C_{E_0}}(B) \\ &\qquad = \det_{E_0}(\mathbf{1} - \alpha_G C_{E_0})^{-1/2} e^{\frac{1}{2}\langle X, T_{\alpha_G} X\rangle}, \\ &T_{\alpha_G} = \Gamma_1^{\mathsf{T}}(C_{E_0,\alpha_G} - C_{E_0})\Gamma_1 + \Gamma_2^{\mathsf{T}} C_{E_0}\Gamma_2 \;, \end{aligned} \tag{B.23}$$

a real symmetric positive semidefinite matrix on $\mathfrak{g}^I$ ($C_{E_0,\alpha_G} \geqslant C_{E_0}$ for $\alpha_G \geqslant 0$). By Lemma 3.15 the columns of Γ outside J_Z vanish, so T_{α_G} is supported on J_Z, and $\int e^{\frac{1}{2}\langle X, T_{\alpha_G} X\rangle} \mathrm{d}\mu_0(X) = \det_{J_Z}(\mathbf{1} - T_{\alpha_G})^{-1/2}$ when $\|T_{\alpha_G}\| < 1$. Integrating (B.23) against μ_0 therefore gives the integral of (ii) in closed form, $\det_{E_0}(\mathbf{1} - \alpha_G C_{E_0})^{-1/2} \det_{J_Z}(\mathbf{1} - T_{\alpha_G})^{-1/2}$, once $\|T_{\alpha_G}\| < 1$ is shown.

Norm of T_{α_G}. The resolvent identity gives $C_{E_0,\alpha_G} - C_{E_0} = \alpha_G C_{E_0,\alpha_G} C_{E_0}$, of norm at most $\alpha_G/(\gamma(\gamma-\alpha_G))$. By (i) and $\delta_M \leqslant 1$, $\|\Gamma_1\| \leqslant \|\Gamma_{E_0}(1)\| + h_+\delta_M \leqslant h_+\gamma^{-1/2} + h_+ = A_*$, since $\|\Gamma_{E_0}(1)\| \leqslant \|H\|\|\mathsf{G}^{1/2}\| \leqslant h_+\gamma^{-1/2}$ ($\mathsf{G} = H^{-1}$ and $H \geqslant \gamma$); and $\|\Gamma_2\| \leqslant h_+\delta_M$. Hence, for $\alpha_G \leqslant \min(\gamma/2, \gamma^2/(16A_*^2))$,

$$\|T_{\alpha_G}\| \leqslant \frac{\alpha_G A_*^2}{\gamma(\gamma-\alpha_G)} + \frac{h_+^2\delta_M^2}{\gamma} \leqslant \frac{2\alpha_G A_*^2}{\gamma^2} + \frac{1}{8} \leqslant \frac{1}{4} \;. \tag{B.24}$$

Determinant bounds. The eigenvalues of $\alpha_G C_{E_0}$ lie in $[0, 1/2]$ and $-\frac{1}{2}\log(1-t) \leqslant t$ for $t \leqslant 1/2$, so $\log\det_{E_0}(\mathbf{1} - \alpha_G C_{E_0})^{-1/2} \leqslant \alpha_G \,\mathrm{tr}\, C_{E_0} \leqslant (\alpha_G/\gamma) d_G |E_0|$. The eigenvalues of T_{α_G} lie in $[0, 1/4]$ and $-\frac{1}{2}\log(1-t) \leqslant \frac{2}{3}t$ there, so $\log\det_{J_Z}(\mathbf{1} - T_{\alpha_G})^{-1/2} \leqslant \frac{2}{3}\,\mathrm{tr}\, T_{\alpha_G} \leqslant \frac{2}{3} d_G |J_Z| \|T_{\alpha_G}\|$. With $E_0 \subseteq J_Z$ (Lemma 3.15),

$$\begin{aligned} \log\big[\det_{E_0}(\mathbf{1} - \alpha_G C_{E_0})^{-1/2} \det_{J_Z}(\mathbf{1} - T_{\alpha_G})^{-1/2}\big] &\leqslant d_G|J_Z|\Big(\frac{\alpha_G}{\gamma} + \frac{4\alpha_G A_*^2}{3\gamma^2} + \frac{2h_+^2\delta_M^2}{3\gamma}\Big) \\ &\leqslant C_{\mathrm{d}}(\alpha_G + \delta_M^2)|J_Z|, \\ C_{\mathrm{d}} = d_G \max\Big(\frac{1}{\gamma} + \frac{4A_*^2}{3\gamma^2}, \frac{2h_+^2}{3\gamma}\Big) \;. \end{aligned} \tag{B.25}$$

A bond of J_Z has its initial site in a cube of Z, so $|J_Z| \leqslant 4(LM+1)^4 N_1(Z)$. Since n_*, R_*, C_{w} are fixed before M and $\delta_M = 2C_{\mathrm{w}} 4^{-\lfloor M/(2R_*)\rfloor}$ decays exponentially in M, there is M with $4C_{\mathrm{d}}\delta_M^2 (LM+1)^4 \leqslant 1/2$; then $\alpha_G \leqslant (8C_{\mathrm{d}}(LM+1)^4)^{-1}$ gives the bound $e^{N_1(Z)}$. Both C_{E_0} and Γ_{E_0} are block diagonal over the components D of Z by (3.67), hence so are T_{α_G} and both determinants, which proves the factorisation.

Bad-bond reserve. On $\{|B_b| \geqslant r\}$, $h(B_b)^2 \geqslant c_{\mathrm{n}}|B_b|^2 \geqslant c_{\mathrm{n}} r^2$, so $\mathbf{1}_{\{|B_b|\geqslant r\}} \leqslant e^{\frac{\gamma_2}{2}(h(B_b)^2 - c_{\mathrm{n}} r^2)}$; the product over P is (iii). *q.e.d.*

B.5 The Mayer Constants and the Activity Bounds

Fix, in this order and depending on L and the group only: $C_{\mathrm{g}} = 162$ and $\delta = 1/40$; a decay rate $\kappa \geqslant 3$ with

$$\delta\kappa \geqslant 2C_{\mathrm{g}} \log 128, \qquad \lambda' = (1-5\delta)\frac{L\kappa}{2049}, \qquad \lambda_* = \lambda' - 2C_{\mathrm{g}} \geqslant 2C_{\mathrm{g}} \log 128\,; \tag{B.26}$$

then

$$\begin{aligned} \alpha_6 &= \min\Big(\frac{1}{8}, \frac{\delta\kappa}{2C_{\mathrm{g}} B(\delta\kappa/2)}\Big), \qquad \kappa_1 - 1 \geqslant 5\lambda' + 1, \\ \theta &= \frac{\lambda_*}{16 C_{\mathrm{g}}}, \qquad C_{\mathrm{s}} = \frac{2049 L^3 C_{\mathrm{g}} B(\delta\kappa) e^{L\kappa/2}}{\delta\kappa}\,, \end{aligned} \tag{B.27}$$

and a number $\omega_* > 0$ with

$$\eta_* = 2e^{1/2+5\kappa}\omega_* \leqslant \frac{\alpha_6}{4}, \qquad \varepsilon_* = \frac{C_{\mathrm{s}}\eta_*}{\alpha_6} \leqslant \min\Big(\frac{1}{4}, \frac{1}{72 B(\lambda_*/2)}\Big)\,; \tag{B.28}$$

these conditions involve only κ, α_6, C_{s} and λ_*, so ω_*, η_* and $\varepsilon_* = C_{\mathrm{s}} t_0$ are fixed before M_1, M and the Gaussian reserve α_G, as Lemma 3.19 states. Put $t_0 = \eta_*/\alpha_6 \leqslant 1/4$ and, once α_G of Lemma 3.16 is fixed after M, $\gamma_2 = \alpha_G/2$, so $a_{\mathrm{bad}} = c_{\mathrm{n}}\alpha_G r^2/4$. The coupling is then taken so small that $\omega(g_0) \leqslant \omega_*$, the chart condition of Lemma B.9 holds, and

$$a_{\mathrm{bad}} \geqslant \max\Big(160\big(\kappa + \log(8M^4+8) + 1\big),\ 64 \log\frac{1}{\eta_*}\Big)\,. \tag{B.29}$$

These are compatible because $r = A_1(\log g_0^{-2})^{p_0}$ (the radius of the literal cutoff (4.12)) tends to infinity while $g_0(1+r)^3 \to 0$. Note $B(\delta\kappa/2) \leqslant 2e^{-\delta\kappa/(2C_{\mathrm{g}})}$, $B(\delta\kappa) \leqslant 2e^{-\delta\kappa/C_{\mathrm{g}}}$ and $B(\lambda_*/2) \leqslant 2e^{-\lambda_*/(2C_{\mathrm{g}})}$ by Lemma B.1(iv) and (B.26).

Lemma B.11 (Cover Sum)**.** *For every nonempty union Y_0 of M-cubes,*

$$\sum_{\mathcal{D}\subseteq\mathbf{D}_0 \,|\, \bigcup_{Y\in\mathcal{D}} Y = Y_0} \prod_{Y\in\mathcal{D}} \alpha_6 e^{-\delta\kappa\ell_0(Y)} \leqslant 1\,. \tag{B.30}$$

Proof. The members of $\mathbf{D}_0$ are localisation domains of π_0 and $\mathcal{D}$ is a set of distinct domains. For a cover, $\sum_{Y\in\mathcal{D}} \ell_0(Y) \geqslant \sum_Y N_0(Y)/C_{\mathrm{g}} \geqslant N_0(Y_0)/C_{\mathrm{g}}$ by Lemma B.1(i), so $\prod_Y e^{-\delta\kappa\ell_0(Y)/2} \leqslant e^{-\delta\kappa N_0(Y_0)/(2C_{\mathrm{g}})}$. Dropping the cover constraint, the remaining sum over sets of domains contained in Y_0 is at most $\prod_{Y\subseteq Y_0}(1 + \alpha_6 e^{-\delta\kappa\ell_0(Y)/2}) \leqslant \exp\big(\alpha_6 \sum_{Q\subseteq Y_0}\sum_{Y\ni Q} e^{-\delta\kappa\ell_0(Y)/2}\big) \leqslant \exp(\alpha_6 B(\delta\kappa/2) N_0(Y_0))$ by (B.2). The product of the two bounds is at most one by the choice of α_6. *q.e.d.*

Lemma B.12 (Selected Majorant)**.** *On every branch, with $\mathbf{W} = \mathbf{V}_\xi$, $\omega(g_0) \leqslant \omega_*$ and the chart condition, every selected integral of Definition 3.17 with $F = 1$ satisfies*

$$|\mathcal{A}^{\mathrm{f}}_{(\mathcal{D},\mathsf{P},Z)}[1]| \leqslant e^{-(\kappa_1-1)N_1(Z\setminus Z_0') + N_1(Z) - a_{\mathrm{bad}}|\mathsf{P}|} \prod_{Y\in\mathcal{D}} \omega_* e^{1/2} e^{-\kappa d_0(Y)}\,. \tag{B.31}$$

Proof. The weakening in (3.68) is, for each cube Δ of $Z \setminus Z'_0$, the operation

$$G \mapsto \int_0^1 \mathrm{d}s \oint_{|\sigma_\Delta|=e^{\kappa_1}} \frac{G\,\mathrm{d}\sigma_\Delta}{2\pi i(\sigma_\Delta - s)^2}$$

applied to the inner integral (the t_Y-, X- and B-integrals) as a function of σ_Δ on the circle; it needs only the measurability of the inner integral in σ on the distinguished boundary (the integrand is continuous in σ, since $\Gamma_{E_0}(\sigma)$ is the walk series of Lemma 3.13, and Fubini applies to the iterated integral because the integral of the modulus of the inner integrand is bounded uniformly on the distinguished boundary, by the bounds below) and the bound $\sup_{s\in[0,1],|\sigma|=e^{\kappa_1}} |\sigma - s|^{-2} \cdot e^{\kappa_1} \leqslant e^{\kappa_1}(e^{\kappa_1}-1)^{-2} \leqslant e^{-(\kappa_1-1)}$, the last step because $e^{\kappa_1} \geqslant 3$. Applying it for each cube of $Z \setminus Z'_0$ bounds (3.68) by $e^{-(\kappa_1-1)N_1(Z\setminus Z'_0)}$ times the supremum over the distinguished boundary $\{|\sigma_\Delta| = e^{\kappa_1}\ \forall \Delta \in \pi_1(Z) \setminus \pi_1(Z'_0)\}$ (with $\sigma_\Delta = 0$ off Z, $\sigma_\Delta = 1$ on Z'_0) of the integral of the inner integrand's modulus, which lies in the closed polydisc of Lemma 3.16. There, the Mayer integrals $\int_0^1 \mathrm{d}t_Y$ have mass one; $\chi_b \leqslant 1$; by Lemma 3.16(iii) with $\gamma_2 \leqslant \alpha_G$, $\prod_{b\in\mathsf{P}}(1-\chi_b) \leqslant e^{-a_{\mathrm{bad}}|\mathsf{P}|} e^{\frac{\alpha_G}{2}\|B\|^2}$; on the box $|\mathbf{V}_\xi(Y,B)e^{t_Y \mathbf{V}_\xi(Y,B)}| \leqslant \omega(g_0)e^{-\kappa d_0(Y)}e^{\omega(g_0)} \leqslant \omega_* e^{1/2} e^{-\kappa d_0(Y)}$ by Lemma B.10 and $\omega_* \leqslant 1/2$ (which follows from $\eta_* \leqslant 1/32$); and the integral of the absolute density against $e^{\alpha_G\|B\|^2/2}$ is at most $e^{N_1(Z)}$ by Lemma 3.16(ii). *q.e.d.*

Proof of Lemma 3.19. Fix a connected final domain Z. Each selection $(\mathcal{D}, \mathsf{P})$ in (3.69) determines $Y_0 = \bigcup \mathcal{D}$, the seed Z_0 with wall components $Z_{0,1}, \dots, Z_{0,k'}$, the components Y_i of Y_0 (each inside one $Z_{0,j}$), $n_j = N_0(Z_{0,j} \setminus Y_0)$ and $n = \sum_j n_j$. By Lemma B.12 and $\omega_* e^{1/2} e^{-\kappa d_0(Y)} \leqslant \eta_* e^{-\kappa \ell_0(Y)}$, the sum over selections is organised as $\sum_{Y'\subseteq Z} \sum_{Z_0 \,|\, Z'_0 = Y'} \sum_{Y_0 \subseteq Z_0} \sum_{\mathcal{D} \,|\, \bigcup \mathcal{D} = Y_0} \sum_{\mathsf{P}}$ over unions of cubes Y' of π_1 and Y_0 of π_0, and each inner sum is bounded in turn.

Potential labels. Split $\eta_* e^{-\kappa\ell_0(Y)} = t_0 e^{-(1-\delta)\kappa\ell_0(Y)} \cdot \alpha_6 e^{-\delta\kappa\ell_0(Y)}$. Each $Y \in \mathcal{D}$ lies in one component Y_i, and $\sum_{Y\subseteq Y_i} \ell_0(Y) \geqslant \ell_0(Y_i)$ by Lemma B.1(ii), so with $t_0 \leqslant 1$ and at least one label per component, $\prod_{Y\in\mathcal{D}} t_0 e^{-(1-\delta)\kappa\ell_0(Y)} \leqslant \prod_i t_0 e^{-(1-\delta)\kappa\ell_0(Y_i)}$. Lemma B.11 sums the second factors to at most one.

Bad-bond labels. Given Y_0 and Z_0, every $b \in \mathsf{P}$ has $b \cap Y_0 = \varnothing$ and both endpoints in cubes of Z_0, hence in $Z_0 \setminus Y_0$; there are at most $4M^4 n$ such bonds. Every cube of $Z_0 \setminus Y_0$ meets a bond of P and a bond meets at most two cubes, so $|\mathsf{P}| \geqslant n/2$. Hence

$$\sum_{\mathsf{P}} e^{-a_{\mathrm{bad}}|\mathsf{P}|/2} \leqslant e^{-a_{\mathrm{bad}}n/8} \sum_{\mathsf{P}} e^{-a_{\mathrm{bad}}|\mathsf{P}|/4} \leqslant \exp\Big[-\frac{a_{\mathrm{bad}}}{8}n + 4M^4 e^{-a_{\mathrm{bad}}/4} n\Big] \leqslant e^{-a_{\mathrm{bad}}n/16}\,, \tag{B.32}$$

since $4M^4 e^{-a_{\mathrm{bad}}/4} \leqslant a_{\mathrm{bad}}/16$: by (B.29), $a_{\mathrm{bad}} \geqslant 160(\kappa+1) > 1$ and $64M^4 e^{-a_{\mathrm{bad}}/4} \leqslant 1$ (as $a_{\mathrm{bad}}/4 \geqslant 40\log(8M^4+8) \geqslant \log(64M^4)$). The unused half $e^{-a_{\mathrm{bad}}|\mathsf{P}|/2} \leqslant e^{-a_{\mathrm{bad}}n/4}$ leaves in total $e^{-5a_{\mathrm{bad}}n/16} = \prod_j e^{-5a_{\mathrm{bad}}n_j/16}$, of which $e^{-a_{\mathrm{bad}}n_j/32}$, $e^{-a_{\mathrm{bad}}n_j/64}$ and $e^{-a_{\mathrm{bad}}n_j/64}$ are used below.

Component reserve. For a component $Z_{0,j}$ containing $k_j \geqslant 1$ components Y_i of Y_0, Lemma B.1(iii) gives $\ell_0(Z_{0,j}) \leqslant \sum_i \ell_0(Y_i) + 5n_j$, so with $a_{\mathrm{bad}}/32 \geqslant 5\kappa$,

$$t_0^{k_j} e^{-(1-\delta)\kappa\sum_i \ell_0(Y_i)} e^{-a_{\mathrm{bad}}n_j/32} \leqslant t_0 e^{-(1-\delta)\kappa\ell_0(Z_{0,j})}\,. \tag{B.33}$$

If $k_j = 0$ then $n_j \geqslant 1$, $\ell_0(Z_{0,j}) \leqslant 5n_j$, and $e^{-a_{\mathrm{bad}}n_j/32} e^{-a_{\mathrm{bad}}n_j/64} \leqslant e^{-5\kappa n_j}\eta_* \leqslant t_0 e^{-(1-\delta)\kappa\ell_0(Z_{0,j})}$, using $e^{-a_{\mathrm{bad}}/64} \leqslant \eta_* \leqslant t_0$.

Choice of Y_0. With the reserve $e^{-a_{\rm bad}n/64}$, $\sum_{Y_0\subseteq Z_0} e^{-a_{\rm bad}N_0(Z_0\setminus Y_0)/64} = (1+e^{-a_{\rm bad}/64})^{N_0(Z_0)} \leqslant \exp(e^{-a_{\rm bad}/64}C_{\rm g}\sum_j \ell_0(Z_{0,j})) \leqslant \prod_j e^{\delta\kappa\ell_0(Z_{0,j})}$, since $C_{\rm g}e^{-a_{\rm bad}/64} \leqslant 162e^{-5(\kappa+1)/2} \leqslant \kappa/40$ for $\kappa \geqslant 3$. Altogether, writing $\mathfrak{M}(Z_0, Z)$ for the sum of (B.31) over the selections with seed Z_0,

$$\mathfrak{M}(Z_0, Z) \leqslant e^{-(\kappa_1-1)N_1(Z\setminus Z_0')+N_1(Z)} \prod_j t_0 e^{-(1-2\delta)\kappa\ell_0(Z_{0,j})} \,. \tag{B.34}$$

Scale change. By Lemma B.2, the enlargement Z_0' of Section 3.5 (the LM-cubes meeting $\operatorname{int} Q^\sim$ for some cube Q of Z_0) equals $\bigcup_j Z_{0,j}'$, where $Z_{0,j}'$ is the enlargement of the component $Z_{0,j}$ in the sense of that lemma, so (B.3) applies to each $Z_{0,j}$. Each $Z_{0,j}'$ is connected, so every component Y_i' of $Y' = Z_0'$ is the union of the $Z_{0,j}'$ it contains. Split $e^{-(1-2\delta)\kappa\ell_0(Z_{0,j})} = e^{-\delta\kappa\ell_0(Z_{0,j})}e^{-(1-3\delta)\kappa\ell_0(Z_{0,j})}$ and apply (B.3) in the form $\ell_0(Z_{0,j}) \geqslant \frac{L}{2049}\ell_1(Z_{0,j}') - \frac{5L+1024}{2049}$, with $(5L+1024)/2049 \leqslant L/2$ for $L \geqslant 3$:

$$\prod_{j\,|\,Z_{0,j}'\subseteq Y_i'} e^{-(1-3\delta)\kappa\ell_0(Z_{0,j})} \leqslant e^{L\kappa/2\cdot k_i'} \prod_j e^{-(1-3\delta)\frac{L\kappa}{2049}\ell_1(Z_{0,j}')} \leqslant e^{L\kappa k_i'/2} e^{-(1-3\delta)\frac{L\kappa}{2049}\ell_1(Y_i')} \,,$$

by Lemma B.1(ii), where k_i' counts the fine components in Y_i'. The remaining factors $t_0 e^{L\kappa/2}e^{-\delta\kappa\ell_0(Z_{0,j})}$ pay for the sum over the set of fine components inside Y_i': each contains an M-cube of Y_i', of which there are $L^4N_1(Y_i')$, so by (B.2) the sum over single components is $x_i \leqslant L^4N_1(Y_i')t_0e^{L\kappa/2}B(\delta\kappa) \leqslant \varepsilon_*\frac{\delta L\kappa}{2049}\ell_1(Y_i')$, using $N_1 \leqslant C_{\rm g}\ell_1$ and the definitions of $C_{\rm s}$ and $\varepsilon_* = C_{\rm s}t_0$, and the sum over nonempty sets is at most $e^{x_i} - 1 \leqslant x_i e^{x_i}$. Write $\Lambda_i = \frac{L\kappa}{2049}\ell_1(Y_i')$ and $y = \delta\Lambda_i$, so $x_i \leqslant \varepsilon_* y \leqslant y$. Then $x_i e^{x_i}e^{-(1-3\delta)\Lambda_i} \leqslant \varepsilon_* y e^y e^{-(1-3\delta)\Lambda_i} = \varepsilon_* y e^{-(1-4\delta)\Lambda_i} = \varepsilon_*(ye^{-y})e^{-(1-5\delta)\Lambda_i} \leqslant \varepsilon_* e^{-\lambda'\ell_1(Y_i')}$, using $ye^{-y} \leqslant 1$, so

$$\sum_{Z_0\,|\,Z_0'=Y'} \mathfrak{M}(Z_0, Z) \leqslant e^{-(\kappa_1-1)N_1(Z\setminus Y')+N_1(Z)} \prod_{Y_i'\in{\rm cc}(Y')} \varepsilon_* e^{-\lambda'\ell_1(Y_i')} \,. \tag{B.35}$$

Choice of Y'. For nonempty $Y' \subseteq Z$ with $n' = N_1(Z\setminus Y')$ cubes outside, Lemma B.1(iii) gives $\ell_1(Z) \leqslant \sum_i \ell_1(Y_i') + 5n'$, so $\prod_i e^{-\lambda'\ell_1(Y_i')} \leqslant e^{-\lambda'\ell_1(Z)}e^{5\lambda' n'}$, and $\varepsilon_*^{\#\,{\rm cc}(Y')} \leqslant \varepsilon_*$. With $\kappa_1 - 1 \geqslant 5\lambda' + 1$,

$$\begin{aligned} |k_\xi^{\rm f}(Z)| &\leqslant \varepsilon_* e^{N_1(Z)}e^{-\lambda'\ell_1(Z)} \sum_{\varnothing\neq Y'\subseteq Z} e^{-N_1(Z\setminus Y')} \\ &\leqslant \varepsilon_* e^{(1+e^{-1})N_1(Z)}e^{-\lambda'\ell_1(Z)} \leqslant \varepsilon_* e^{-(\lambda'-2C_{\rm g})\ell_1(Z)} \,, \end{aligned}$$

which is the first inequality of (3.72).

Criterion. Let Z be any union of LM-cubes, connected or not (the lemma states the case of a connected final domain; the proof of Lemma 3.20 uses the general case). A domain Z' incompatible with Z contains one of the at most $9N_1(Z)$ cubes of Z or wall adjacent to Z. Since $(1+6\theta)N_1(Z') \leqslant (1+6\theta)C_{\rm g}\ell_1(Z') \leqslant \frac{\lambda_*}{2}\ell_1(Z')$ by $C_{\rm g} \leqslant \lambda_*/8$ and $6\theta C_{\rm g} = 3\lambda_*/8$, (B.2) gives

$$\sum_{Z'\not\sim Z} |k_\xi^{\rm f}(Z')|e^{(1+6\theta)N_1(Z')} \leqslant 9N_1(Z)\varepsilon_* B(\lambda_*/2) \leqslant \frac{1}{8}N_1(Z) \,. \tag{B.36}$$

The reference data are treated in Section B.7. *q.e.d.*

B.6 Proof of the Vacuum-Cluster Bounds

Proof of Lemma 3.20. Species are the connected final domains D, with weights $\widetilde{z}_D = |k_\xi^{\rm f}(D)|e^{2\theta N_1(D)}$ and sizes $a(D) = N_1(D)/8$; two species are incompatible when they share a

cube or a wall, and each with itself. By (B.36) and $2\theta + \frac{1}{8} \leqslant 1 + 6\theta$,

$$\sum_{D' \not\sim D} \widetilde{z}_{D'} e^{a(D')} \leqslant \sum_{D' \not\sim D} |k_\xi^{\mathrm{f}}(D')| e^{(1+6\theta)N_1(D')} \leqslant a(D) \; .$$

Tuples of species are written $\mathfrak{c} = (D_1, \dots, D_m)$ in this proof (γ is the spectral bound $\gamma_L/2$). The argument of Section 5 following (5.29) applies verbatim to these species, weights and sizes: the absolute Ursell coefficient $|\varphi^T(\mathfrak{c})|$ is bounded by the number of spanning trees of the incompatibility graph of $\mathfrak{c}$, the rooted-tree generating sum with root D is the monotone limit of the recursion (5.30) with $\widetilde{z}$ in place of $|\widetilde{z}_p|$, the displayed criterion gives $T_D^{(k)} \leqslant \widetilde{z}_D e^{a(D)}$ inductively, and the factor $1/m!$ in (3.73) together with the multiplicity $m_D(\mathfrak{c})$ of D in the tuple reproduces the rooted sum. Hence, for every connected final domain D,

$$\sum_{\mathfrak{c}} m_D(\mathfrak{c}) |\Phi_\xi^{\mathrm{v}}(\mathfrak{c})| e^{2\theta \sum_j N_1(D_j)} \leqslant |k_\xi^{\mathrm{f}}(D)| e^{2\theta N_1(D) + N_1(D)/8} \; . \tag{B.37}$$

Here $\mathcal{A}$ is any finite set of unions of LM-cubes, connected or not; the criterion (B.36) was proved for every such union. A tuple hitting $\mathcal{A}$ contains a member D' incompatible with some $D \in \mathcal{A}$, so

$$\begin{aligned} \sum_{\mathfrak{c} \text{ hits } \mathcal{A}} |\Phi_\xi^{\mathrm{v}}(\mathfrak{c})| e^{2\theta \sum_j N_1(D_j)} &\leqslant \sum_{D \in \mathcal{A}} \sum_{D' \not\sim D} \sum_{\mathfrak{c}} m_{D'}(\mathfrak{c}) |\Phi_\xi^{\mathrm{v}}(\mathfrak{c})| e^{2\theta \sum_j N_1(D_j)} \\ &\leqslant \sum_{D \in \mathcal{A}} \sum_{D' \not\sim D} |k_\xi^{\mathrm{f}}(D')| e^{(2\theta + \frac{1}{8}) N_1(D')} \leqslant \frac{1}{8} \sum_{D \in \mathcal{A}} N_1(D) \; . \end{aligned}$$

For (3.74), at fixed volume there are finitely many species. Let $\mathcal{S}_+$ be the set of species with $k_\xi^{\mathrm{f}}(D) \neq 0$; species with $k_\xi^{\mathrm{f}}(D) = 0$ contribute to neither side (a tuple containing one has $\Phi_\xi^{\mathrm{v}} = 0$, and the gas does not see them), so they are discarded. For complex activities $z = (z_D)_{D \in \mathcal{S}_+}$, the hard-core gas $\Xi(z) = \sum_{\mathcal{P} \text{ compatible}} \prod_{D \in \mathcal{P}} z_D$ is a polynomial with $\Xi(0) = 1$: the polymer partition function (5.3) of [13, Chapter 5] for the polymers $\mathcal{S}_+$ with the hard-core interaction $\zeta(D, D') = 0$ on incompatible and $\zeta(D, D') = 1$ on compatible pairs (which satisfies (5.1)–(5.2) there). Let $S(z) = \sum_{\mathfrak{c}} \frac{\varphi^T(\mathfrak{c})}{m!} \prod_j z_{D_j}$, the sum over all ordered tuples $\mathfrak{c} = (D_1, \dots, D_m)$, $m \geqslant 1$, of members of $\mathcal{S}_+$; this is the cluster series of that gas, since the Ursell function of [13, (5.6)] is $\frac{1}{m!} \sum_G \prod_{\{i,j\} \in G} (\zeta(D_i, D_j) - 1) = \frac{\varphi^T(\mathfrak{c})}{m!}$, the sum over the connected graphs G on $\{1, \dots, m\}$, a factor $\zeta - 1$ being -1 on an incompatible pair and 0 otherwise. [13, Proposition 5.3] is the identity $\Xi(z) = \exp(S(z))$ between formal power series in the variables z_D (the book states it as a formal identity; its proof, (5.4)–(5.8) there, rearranges finitely many terms for each monomial). By (B.37) with z_D in place of $k_\xi^{\mathrm{f}}(D)$ (the bound is monotone in the weights and only uses $|z_D| \leqslant |k_\xi^{\mathrm{f}}(D)|$ through the criterion; equivalently [13, Theorem 5.4], whose hypothesis (5.10) is the first display of this proof and whose conclusion (5.9) is this absolute convergence), S converges absolutely at $|z_D| = |k_\xi^{\mathrm{f}}(D)|$, hence absolutely and uniformly on the closed polydisc $\overline{P} = \{|z_D| \leqslant |k_\xi^{\mathrm{f}}(D)| \ \forall D \in \mathcal{S}_+\}$ (Weierstrass M-test), so S is continuous on $\overline{P}$, analytic on the open polydisc P, and its Taylor series at 0 is the formal series S grouped by monomials. The analytic functions Ξ and $\exp \circ S$ on the connected open set $P \ni 0$ have the same Taylor series at 0, by the formal identity, so they coincide on P (identity theorem) and, by continuity, on $\overline{P}$; in particular at $z_D = k_\xi^{\mathrm{f}}(D)$, where $\Xi(k_\xi^{\mathrm{f}}) = \Xi_\xi^{\mathrm{f}}$ and $S(k_\xi^{\mathrm{f}}) = \sum_{\mathfrak{c}} \Phi_\xi^{\mathrm{v}}(\mathfrak{c})$ by (3.73). The restricted gas $(\Xi_\xi^{\mathrm{f}})^{[\cup \mathcal{A}]}$ is the same gas with $z_D = 0$ for the D incompatible with $\bigcup \mathcal{A}$, which are exactly the D incompatible with some member of $\mathcal{A}$; its expansion is the sum over the tuples none of whose members hits $\mathcal{A}$. The quotient is therefore $\exp(-\sum_{\mathfrak{c} \text{ hits } \mathcal{A}} \Phi_\xi^{\mathrm{v}}(\mathfrak{c}))$. *q.e.d.*

B.7 The Reference Data

The reference data of Section 3.8 consist of the identity background $U_1^\circ = \mathbf{1}$ on every bond (the constrained minimiser for identity averages, with zero action and zero gauge functional), $V_0^\circ = \mathbf{1}$, zero retained input $A_{\mathrm{e}}^\circ = 0$ with $\Psi_\xi^\circ = \mathbf{1}$, and the same I, Ω_1, partitions, hosts, cutoff radius r and constants as the underlying branch. They are not a branch, so the conclusions of the lemmas are checked directly. The reference Hessian is the regional Hessian (3.16) at the identity background on the branch's Ω_1, $H_\xi^\circ = \mathsf{H}_{\mathbf{1}}^{\Omega_1}$, and H° denotes its restriction to I. Since I, Ω_1 and R_{c} are those of the branch, the geometric facts used in Lemma B.6 (for $b \in I$ every site of a stencil $\mathcal{S}_{bb'}$ lies within $3L$ of b_-, hence within $3R_{\mathrm{c}}$ of I and in $\Omega_1^{\sim -1}$, by Lemmas B.3 and 3.11(i)) hold for the reference data as well; Lemma B.6 is stated for every background on which the regional form is defined, so it applies to the identity background without change.

Lemma B.13 (Reference Data)**.** *The reference precision H° satisfies* (3.23)*, as do its principal restrictions, and has the range and gauge law of Lemma 3.3; Lemma 3.6 holds for the reference data (at $U = \mathbf{1}$, $A_{\mathrm{e}} = 0$); Lemma 3.16 holds for H° with the same $\gamma, h_+, R, M, \alpha_G$; and* (3.72) *and Lemma 3.20 hold for the reference activities $k_\xi^{\mathrm{f},\circ}$ with the same constants. The reference row solutions, phases and primitives are real, and the reference normalisations $K_{\xi;0}^\circ$, $I_{\xi;0}^\circ$ of Section 3.8 are positive.*

Proof. By Lemma B.6 with the identity background, the entries of H° with a label in I are $f_{bb'}(0)$, the entries of the flat Hessian H of (4.7) on the branch's torus. The compression of $\mathsf{H} \geqslant \gamma_L I$ (A.25) to I, and further to any subset of I, gives the lower bound $\gamma_L \geqslant \gamma_L/2$ for H° and for every principal restriction, and the Schur bound with row sums $\sum_{b'} \|f_{bb'}(0)\| \leqslant m_L$ gives the upper bound for H° and every principal restriction (a principal restriction has smaller row sums); the range is ρ_L by Lemma B.3. The gauge law of Lemma 3.3 is the entrywise covariance (3.21), proved there for every background, hence for $\mathbf{1}$: $f_{bb'}(\mathbf{1}^u) = \mathrm{Ad}_{u(b_-)} f_{bb'}(\mathbf{1}) \mathrm{Ad}_{u(b'_-)}^{-1}$. Lemma 3.16 and Lemmas 3.13, 3.14, 3.15 use only (3.56), the range and the partitions, so they hold for H°. Lemma B.7 applies to the identity background (its hypothesis, a site gauge with phases at most $\tau_L/2$ on the two-block sets and blocks concerned, is satisfied by $u = \mathbf{1}$, $\phi = 0$; the geometric facts of the applicability paragraph depend only on I, Ω_1, R_{c}), so Lemma 3.6 holds at $U = \mathbf{1}$, and Lemma B.9 holds with $r_{\mathrm{e}}^\circ = 0$ and $s^\circ = 1 + r \leqslant s_\xi$; the hosts and the host multiplicity are those of Lemma B.10, so (B.21) holds for $\mathbf{V}_\xi^\circ$. The proofs of Lemmas B.12, 3.19 and 3.20 use nothing else about the branch. Reality: the row solutions are the limits of the contraction iterates of Lemma B.7(ii), which are real for real phases, so for real inputs the phases, the Haar logarithms, the determinant logarithms (positive determinants) and the plaquette and gauge terms are real, and so are the primitives and $\mathbf{V}_\xi^\circ$. Positivity: $K_{\xi;0}^\circ = \mathbf{1}_{\mathcal{D}_I} \exp(-\frac{1}{2}\langle a, H^\circ a\rangle + P_{\xi,0}^\circ(a))$, the function (3.45) for the reference data at $t = 0$, is a real exponential on the open box, the reference primitives being real, hence positive there; and $I_{\xi;0}^\circ = \int K_{\xi;0}^\circ \, \mathrm{d}a$ is positive and finite ($H^\circ \geqslant \gamma$ and the exponent is bounded on the box by Lemma 3.6). *q.e.d.*

B.8 The Thresholds

Before the coupling, in this order: $L > 11$ odd ([4, p. 251], the range in which Bałaban's results are stated); the group constants $c_{\mathrm{h}} = \sqrt{N}$, $c_{\mathrm{n}} = 1/N$, $\varrho_G = \pi/\sqrt{N}$, c, r_G, c_G, and r_M, r'_M (Lemma B.7(iv); these two depend on L and the group); τ_L by (B.6), with m_L, c_L (Lemma 3.3) and γ_L (Appendix A); $R_{\mathrm{loc}} = 2L+2$, R_{p}, $s_{\mathfrak{P}}$, $R_{\mathrm{c}} \geqslant \max(2R_{\mathrm{loc}}+1, s_{\mathfrak{P}}+2, 4L+5)$ and $C_{L,R_{\mathrm{c}}}$; the two-block size $n_{\mathcal{E}}$ of Lemma B.8 (a constant of L); the ranges R_* and R_H

of Section 5 (the perimeter range bounding the primitive-label diameters and the source-to-stencil distances, distinct from the walk range $R_* = n_*R$ below, and the range of the flat Hessian, whose entries couple the bonds of a common plaquette or block; both are constants of L), hence $R_0 = 4(1 + R_* + R_H + L)$; $\vartheta_* \leqslant \tau_L/2$, $\rho_{\mathrm{d}} = 2K_*\vartheta_*^2$, C_{C}, C_{D}, C_k^{a} $(0 \leqslant k \leqslant 5)$ and C_k^{D} $(0 \leqslant k \leqslant 6)$ of Lemma B.7 and C_{p} of Lemma B.9; $\gamma = \gamma_L/2$, $h_+ = m_L$, R; C_{g}, δ, κ, λ', λ_*, α_6, κ_1, θ, C_{s} by (B.26)–(B.27), then ω_*, η_*, ε_*, t_0 by (B.28), with $\omega_* \leqslant 1/2$ (the standing choice of the collar estimates with a Wilson insertion, Section 9) (these involve only κ, α_6, C_{s}, λ_*, so they are fixed before M_1 and M, as Lemma 3.19 states); n_*, R_*, C_{w} (Section 3.5), A_*, C_{d} (Lemma 3.16); M_1 as the M_1 of [3, p. 245], M_2 and M odd powers of L with $M_1 < M_2 < M$, M/M_1 integral, $M \geqslant 8LM_2$, the requirement of [3, p. 245] that M be large, $\rho_{\mathrm{w}} = R_* + R + 1 + 2M_1$, (3.59), $\delta_M \leqslant 1$, $h_+^2\delta_M^2/\gamma \leqslant 1/8$ and $4C_{\mathrm{d}}\delta_M^2(LM+1)^4 \leqslant 1/2$; α_G with $0 < \alpha_G \leqslant \min(\gamma/2, \gamma^2/(16A_*^2), (8C_{\mathrm{d}}(LM+1)^4)^{-1})$, $\gamma_2 = \alpha_G/2$; then N_{h} and C_0 of (B.20), whose hosts depend on the chosen M; then the constants of the collar estimates with a Wilson insertion (Section 9, where they are introduced), in this order: the finite-difference constant $C'_\chi = 2N_{\mathrm{tr}}C_{\mathrm{link}}$ and the character constant K_χ, the dummy weight u_* of the host comparison, the support reserve η of the positive group summation, its pin constant C_{pin} and then the pin radius ρ_{pin} (defined through C_{pin} and η), the host count $N_{\mathrm{host}} = 4^4 \cdot 20^4$ and the probe-host bond count $C_M = 8(4M+1)^4$, all fixed before the coupling, each depending on L, M, κ, ω_*, the group, the character and the constants earlier in this order only. Then the constant β_0 of [3, (2.6)] (p.255; the coupling is then taken small enough for (2.6) to hold with it), the mean radius ρ_{av} of Section D.8.4, the constants $C_{L,G}$, $c_{L,G}$ of Proposition D.17, and C_{H}, c_{H} of Lemma D.18, which depend only on L, the group, M_1, M_2, and the constants of [3] and [7]. The coupling g_0 is then taken below a threshold, and the torus is chosen last: every fine period a multiple of LMR_1 and at least $8M$ ((3.1); the partitions of Section 3.5 need a multiple of LM, the hosts of Section 3.3 need $8M$, and the partitions of [3] into LMR_1-cubes need the multiple of LMR_1; $R_1 = R_1(g_0)$, so the period condition is imposed after the coupling) and every coarse period exceeding $\max(2, 2R_{\mathrm{p}}+8)$. The statements about a square C_n of side nL (Proposition 7.3, Appendices E and F, Sections 9.4–9.8) assume in addition the square/torus geometry of Section 5.3: every coarse period $K_\mu \geqslant 3(n+2)$, so that the square and its fixed neighbourhood have unique lifts and the loop is contractible; and Appendix D (Section D.7) and Sections 9.4–9.8 assume compatible even fine periods, exceeding 200ℓ with $\ell = LMR_1$ where stated (Section 9.4.1). These are lower bounds on the periods in terms of the fixed loop and the coupling; for a given loop and coupling they hold on every sufficiently large compatible torus, and nothing is let tend to infinity. In the threshold conditions, $\varepsilon_1 = g_1 p_0(g_1)$ with $g_1 = g_1(g_0)$ by [3, (1.27)]; R_1 is that of [3], the smallest power of L with $R_1 \geqslant (\log g_1^{-2})^{\mathfrak{r}}$ (p.246, with Bałaban's exponent written $\mathfrak{r}$; $\mathfrak{r} \geqslant 2$ at p.245, $p_0 \geqslant 5\mathfrak{r}$ at p.246, and [3, (2.5)] for the general step); and the cutoff radius is $r = A_1(\log g_0^{-2})^{p_0}$, the radius of the literal cutoff (4.12); it is δ_0/g_0 for the $\delta_k = g_k A_1/A_0\, p_0(g_k)$ of [3] as printed at p.265 (of the equality chain printed for δ_0 at p.247 only the first member differs from this, and it is corrected in (H4); its last member $(A_1/A_0)\varepsilon_0$ agrees; $r = \delta_0/g_0$ identifies Θ_ξ with the test (1.18) and is used throughout, while below only $r \to \infty$ and $g_0(1+r)^3 \to 0$ enter, which hold also for the first member as printed, except item 4(vi), which is stated for the explicit form $r = A_1(\log g_0^{-2})^{p_0}$). The threshold conditions are:

1. $LMR_1 > 4R_{\mathrm{c}} + 4L^2M_2R_1 + 8$ (Lemma 3.11), $LM_2R_1 \geqslant M$ (so that the LM-cubes nest in the $L^2M_2R_1$-cubes, (3.1)) and $4\,O(1)B_3L^2e^{-\delta_{\mathrm{c}}LM_2R_1} \leqslant 1$ (Lemma B.4); both depend on the coupling through R_1;

2. (B.8), the hypotheses of Lemma 3.4, which (3.1) lists as well;

3. $4C_{\mathsf{C}}g_0(1+2r) \leqslant \vartheta_*/2$ (Lemma B.9);

4. the collar thresholds, with $a_{\mathrm{bad}} = c_{\mathrm{n}}\alpha_G r^2/4$, $u_t \coloneqq C'_\chi g_0(1+2r)$, $\rho_t \coloneqq u_t/u_*$ and $\omega_t \coloneqq C_0 g_0(1+2r)^3$: (i) (B.29) for $(9/128)a_{\mathrm{bad}}$ in the role of a_{bad}, that is $(9/128)a_{\mathrm{bad}} \geqslant \max(160(\kappa+\log(8M^4+8)+1), 64\log(1/\eta_*))$, which implies the budget for a_{bad} and for $(9/16)a_{\mathrm{bad}}$; (ii) $\rho_t + u_t/2 + 2\omega_t/\omega_* + 2C_M e^{-(9/32)a_{\mathrm{bad}}} \leqslant 1$, which implies $u_t \leqslant u_*$, $\omega_t \leqslant \omega_*/2$ and the former condition $C_0 g_0(1+r)^3 \leqslant \omega_*$; (iii) $C_0 g_0(1+3r)^3 \leqslant \min(1, \omega_* e^{-\kappa(B_{\mathrm{h}}+4)}/(e^{9/8}N_{\mathrm{host}}))$; (iv) $\rho_t \leqslant \rho_{\mathrm{pin}}$; (v) $2K_\chi C_{\mathrm{pin}}\rho_t \leqslant 1$; (vi) $2C_M \leqslant (1+2r)^3$, $\log g_0^{-2} \geqslant 1$ and $(9c_{\mathrm{n}}\alpha_G/64)A_1^2 \log g_0^{-2} \geqslant 1$, under which the second-order family-sum rate of the collar estimates takes its explicit form (these are the conditions (4.T), (4.T′), (5.T)(i)–(v) and (R.T)(i)–(iii) of the collar estimates, Section 9, in reduced form);

5. Bałaban's own conditions in [3], taken over as they stand: ε_0 small (Theorem 1 of [7], reference [15] of [3], behind (B1); and, the constants $O(L^2)$ of p.247 and $O(1)$ of [8, p. 25] being independent of the coupling ((B1), (H4)), so small that $2\,O(L^2)\varepsilon_0$ and $O(1)L^2\varepsilon_0$ lie below ϱ_G, $r_G/\sqrt{N}$, and the small-diameter radius of the group mean, the domain statement of (B1)); MR_1 large (p.251, the bound on $\mathbf{H}_{1,Ax}$ at the tolerance $\delta_0/40$ of (3.40), behind the change of variables (1.22), Lemma 3.8: by Lemma D.18 it is the condition $\frac{\pi}{2}LC_{\mathrm{H}}(1+\beta_0)(A_0/A_1)e^{-c_{\mathrm{H}}LMR_1} \leqslant \frac{1}{40}$, with C_{H} and c_{H} fixed before g_0, together with the conditions of that lemma's threshold); A_0/A_1 and M_2R_1 large (the two qualitative conditions of (B3), pp.248 and 250; the second requires $O(1)B_3 e^{-\delta_{\mathrm{c}}LM_2R_1}\varepsilon_1$ to be much smaller than δ_0, and $\varepsilon_1/\delta_0 \leqslant (1+\beta_0)A_0/A_1$ stays bounded by [3, (2.8)] and $\delta_0 = (A_1/A_0)\varepsilon_0$, so $R_1 \to \infty$ at fixed L, M_2 suffices, the constants being independent of the coupling by (B3)(iii)). The conditions on MR_1 and M_2R_1 depend on the coupling through R_1 and are met for small g_0 at fixed M, M_2, since $R_1 \to \infty$; the condition on A_0/A_1 is met by the choice before the coupling; Bałaban's conditions on M_1, M_2, M themselves are not listed in [3] (p.245: M_1 the cube size of the previous papers, M_2 not too much larger, M much larger than M_2, "dictated by many conditions involving this constant") and are taken over as the hypothesis that the construction of [3] is valid for the values chosen above, which follow its prescription.

6. the fibre conditions (B.16) of Lemma B.8: $3c_G c_{\mathrm{h}} s_{\mathfrak{P}}\varepsilon_0 \leqslant \tau_L/2$ (which gives $3c_{\mathrm{h}}s_{\mathfrak{P}}\varepsilon_0 \leqslant r_G$, by $\tau_L \leqslant 2c_G r_G$) and $n_{\mathcal{E}}^{1/2} c_G c_{\mathrm{h}}\, 2\,O(L^2)\varepsilon_0 \leqslant \vartheta_*/4$, with $n_{\mathcal{E}}$ the largest number of bonds of a two-block set and $O(L^2)$ the constant of p.247 of item 5.

All hold for small g_0; for item 4(i)–(v) because $a_{\mathrm{bad}} \to \infty$ and $g_0(1+r)^3 \to 0$ (so every term of (ii) tends to zero, the fourth because $a_{\mathrm{bad}} \to \infty$), for item 4(vi) because $r \to \infty$ and $\log g_0^{-2} \to \infty$, and for item 6 because $\varepsilon_0 = g_0 A_0 (\log g_0^{-2})^{p_0} \to 0$ while its other constants are fixed before the coupling, $O(L^2)$ being independent of the coupling and of the branch by [9, Lemma 1, p. 79] ((B1), (H4)). Under (vi), with $c = 9c_{\mathrm{n}}\alpha_G/128$ and $p_0 \geqslant 1$, $e^{-cr^2} \leqslant e^{-cA_1^2(\log g_0^{-2})^2} = g_0^{2cA_1^2 \log g_0^{-2}} \leqslant g_0$, so the fourth term of (ii), $2C_M e^{-(9/32)a_{\mathrm{bad}}} = 2C_M e^{-cr^2}$, is at most $g_0(1+2r)^3$: this is the estimate behind the explicit second-order rate of the collar estimates, not a condition needed for the satisfiability of (ii). The first-step coupling is defined by [3, (1.27)], $g_0^{-2} = g_1^{-2} + \beta_1(g_0)$, and [3, (2.6), p. 255] (with $n = 1$, $m = 0$) states $g_1 \leqslant (1+\beta_0)g_0$ and $g_0 \leqslant (1+\beta_0)g_1$, where $\beta_0 > 0$ is a constant that "can be chosen arbitrarily small, if g is sufficiently small" (p.255); [3] derives (2.6) from the renormalisation group equations (0.20) of [I] and the properties of the β-functions (p.255), and it is taken over as Bałaban's statement, not derived here. Hence $g_1 \to 0$ with g_0, so $R_1 \to \infty$; and [3, (2.8), p. 256] (with $n = 1$, $m = 0$) gives $\varepsilon_1 \leqslant (1+\beta_0)\varepsilon_0$, so $\varepsilon_1 = g_1 A_0 (\log g_1^{-2})^{p_0} \to 0$ with $\varepsilon_0 = g_0 A_0 (\log g_0^{-2})^{p_0}$; while $r \to \infty$ and $g_0(1+r)^3 \to 0$. (The p_0 of [3] denotes both the function $p_0(g) = A_0(\log g^{-2})^{p_0}$ and the integer exponent in it; the notation is Bałaban's and is kept.) The threshold depends on L, the group (through c_{h}, c_{n}, ϱ_G, c, r_G, c_G, r_M, d_G), the

choices τ_L, ϑ_*, R_{c}, κ, α_G, ω_* made above, M_1, M_2, M, the collar constants C'_χ, K_χ, u_*, ρ_{pin}, C_{pin}, η, N_{host}, C_M, and, through ε_1, R_1 and r, on the constants A_0, A_1, p_0, $\mathfrak{r}$ of [3], the map $g_0 \mapsto g_1$ of [3, (1.27)], the constants B_3, δ_{c} and the $O(1)$ of the comparison after (1.19) (p.268; they are Bałaban's constants of (B3)(iii) and (H4), Lemma B.4 uses only their independence from the coupling, and the threshold is stated with whatever dependence on L, the group and M_2 they have), the threshold ε_0 of Theorem 1 of [7] (reference [15] of [3]) behind (B1), the constant $O(L^2)$ of p.247 (items 5 and 6), the constants C_{H} and c_{H} of Lemma D.18 behind (B4)(ii), and the tolerances "much smaller than δ_0" of p.250 behind (B3)(ii).

Further conditions are imposed where the appendices and Sections 9.3–9.9 state them; each is a condition that g_0 lie below a threshold depending on L, the group, the character, M_1, M_2, M, the constants chosen above and those of [3] (and on the parameter ϑ where indicated), holding for all small g_0, or a lower bound on the periods in terms of the loop and the coupling, holding on every sufficiently large compatible torus. They are: $\delta_0 \leqslant \delta_G$ (Lemma 3.8); (D.2) ($r \geqslant 1$ and $C_P g_0(1+2r) \leqslant \gamma/2$, Section D.1); the threshold g_* of Proposition D.15, which contains the threshold of Proposition D.17, the coupling conditions of [3, (2.6)] and the smallness giving $q \leqslant e^{-c_*\pi_*^2}$ (Section D.7); the thresholds of Lemma E.1 (its last inequality), the smallness conditions of Lemma E.9 on the root weight and the smallness of Lemma E.10 (Appendix E, Proposition E.22); the further threshold of (9.53) (Corollary 9.47(c)); the thresholds of Lemmas D.11, D.12 and D.14 (Section D.7; every condition there is an upper bound on the witness threshold θ of (D.26) or a lower bound on R_1); the thresholds of Lemmas 9.5(b) and 9.6, the threshold $g_*(\vartheta)$ of Proposition 9.8 and the smallness required for (9.6) (Section 9.3); (9.15), (9.16), (9.20) and (9.29), which are the collar conditions of item 4 in the forms used there; the period hypothesis of Section 9.4.1 (compatible even fine periods exceeding 200ℓ, $\ell = LMR_1$); and (9.39) (the collar condition with $R_0 = 4(1 + R_* + R_H + L)$ in place of R_{c}, the small-coupling conditions $g_0 C_{\mathsf{C}}(2+4r) < \vartheta_*/2$ and $g_0(2+4r) < a_L$ with the polymer-load inequalities of the proof of Proposition 7.3, and the torus conditions of its item (iii)); the coupling conditions of the flat and regional expansions where they are imposed ($\tau < 1$, the original chart condition, the smallness of $K_1^{s_*}\tau$ and the straight- and bent-path thresholds of Sections 5 and 6, and the threshold of Proposition 7.1); $R_\Upsilon \geqslant 2$, $u \leqslant u_*$, $v \leqslant 1$ and the small-coupling conditions of Sections E.9.1 and E.10.5 (Proposition E.22); the further thresholds of the closing forms of Proposition 9.39 and of Corollary 9.47(a), (c); and the further threshold of Corollary 1.4 ($C\tau^2 L^4 C_K \Phi_c \leqslant 1$ and the bound on $\delta_q^\sharp$), each stated and discussed at the place cited. The list of conditions on the parameters is the enumeration above together with these.

C The Local Interaction at the Identity

Locality of the First-Step Interaction. The trace in the bare action is $\mathrm{tr}_N = N^{-1}\,\mathrm{Tr}$ in its fixed unitary embedding, so $\mathrm{tr}_N(\mathbf{1}) = 1$. Choose a radius a_{ph} for the input A of $\widetilde{D}$, measured in the maximum fine-bond Hilbert–Schmidt norm, on which the local implicit solutions, principal logarithms and composed action and Haar primitives are analytic, with nonzero Jacobian determinants and Haar densities. The radius a_L is a free-input radius: take $a_L \leqslant a_{\mathrm{ph}}/\max(1, \|\mathsf{C}\|_{\infty\to\infty})$, where the operator norm maps the maximum free-bond Hilbert–Schmidt norm to the maximum fine-bond Hilbert–Schmidt norm. This norm is finite and depends on L only, by the fixed row stencils. The Lie-algebra norm is the normalised Hilbert–Schmidt norm, and $|A|_{\mathrm{HS}} \leqslant |A|_{\mathrm{op}}$. Thus $c_{\mathrm{cmp}} = 1$ is a common norm-comparison constant for both cutoff prescriptions (it is not the group constant c_G of Appendix B), and $g_0 r(g_0) < a_L$ suffices. The cutoff sets themselves are unchanged. The componentwise implicit

solutions assemble to a small solution in this maximum norm. Uniqueness of the analytic germ in [4] identifies it with $\widetilde{D}$ on the smaller common chart. The following calculation concerns exactly the identity-background, all-regular integral below. It does not transfer that integral to a nonflat or inherited branch. Let $\mathcal{S}_c$ contain the positive fine bonds with both endpoints in $\mathcal{B}_{c_-} \cup \mathcal{B}_{c_+}$, and put $\Lambda_c = \mathcal{S}_c \cap E_*$. At the present torus sizes, the only designated bond in $\mathcal{S}_c$ is $b_0(c)$. Indeed, an internal path in either block crosses no block interface, and the translated length-L path in [4, (0.4)] crosses only the interface between these two blocks. The reference coarse path crosses the same interface. Thus $\mathcal{Q}_c$ and $\mathcal{N}_c$ depend only on the fine bonds in $\mathcal{S}_c$. In the defining equation for $\widetilde{D}_c(A)$, the only unknown correction entering those bonds is $\widetilde{D}_c(A)$ itself. The analytic implicit-function theorem therefore gives a function of $A|_{\mathcal{S}_c}$ alone, with its radius and constants independent of the number of blocks. Here $h_c = L^3 I$ at the identity, since $Q_0 = L^{-4} I$.

For a free fine bond b, set $\Lambda_b = \{b\}$. For a designated bond $b = b_0(c)$, set $\Lambda_b = \Lambda_c$. Both $(\mathsf{C}B)(b)$ and $q_{g_0}(B)(b)$ depend only on $B|_{\Lambda_b}$. The index α runs over the plaquettes and the pairs (y, x) in (4.5). Define $f_\alpha(q)$ to be the corresponding summand of $\mathscr{A}(e^{iq})$, as a real analytic scalar function near zero in $\mathfrak{g}^E$; use its unique holomorphic extension for complex arguments. Let Λ_α be the union of the sets Λ_b over the bonds in that summand's plaquette or path family. Each Λ_α lies in at most four adjacent blocks. Each free bond belongs to a bounded number of the sets $\Lambda_\alpha, \Lambda_b, \Lambda_c$, with a bound depending on L, not on the torus.

Write $\psi(A) = A - h\widetilde{D}(A)$. Define the following local scalar functions by their real analytic branches at zero:

$$\begin{aligned} v_\alpha(B) &= -\frac{1}{g_0^2}\left(f_\alpha(\psi(g_0 \mathsf{C}B)) - \frac{g_0^2}{2} D^2 f_\alpha(0)[\mathsf{C}B, \mathsf{C}B] \right) , \\ j_b(B) &= \log j_H(q_{g_0}(B)(b)) , \\ d_c(B) &= \log \det_{\mathfrak{g}} \left(I - L^3 \partial_{A(b_0(c))} \widetilde{D}_c(g_0 \mathsf{C}B) \right) . \end{aligned} \tag{C.1}$$

Their supports are respectively $\Lambda_\alpha, \Lambda_b, \Lambda_c$. The full fine Jacobian is block triangular: its nonidentity rows are designated rows, and its designated-to-designated block is diagonal in c. Its determinant is consequently the product of the determinants in d_c. Each real determinant is positive in the chart connected to zero, so this branch agrees with the absolute determinant in (4.14). Summing the action Hessians gives H, and hence

$$\mathcal{P}_{g_0}(B) = \sum_\alpha v_\alpha(B) + \sum_{b \in E} j_b(B) + \sum_{c \in E_1} d_c(B) . \tag{C.2}$$

There is no linear term in either j_b or d_c. Haar density in exponential variables has zero derivative at zero. For d_c, the quadratic part of $\widetilde{D}_c$ is the quadratic part of $\mathcal{N}_c$. Expanding [4, (0.4)] to degree two gives a linear combination of Lie brackets of the bond variables. Differentiating with respect to the designated variable gives a linear combination of adjoint maps. Their traces vanish for a compact Lie algebra. Multiplication by the scalar L^3 does not change this conclusion. Each f_α and its first derivative vanish at zero, and $\psi(A) = A + O(A^2)$, so the subtracted action term begins at degree three.

Use the positive Lie-algebra scalar-product norm for the following estimate. There are constants $a_L, C_L^{\mathrm{loc}} > 0$, depending on L, G and the fixed averaging prescription, such that, on the complex polydisc $g_0 \max_b \|B(b)\| < a_L$, each term v of (C.2), with its indicated support Λ, satisfies

$$|v(B)| \leqslant C_L^{\mathrm{loc}} \left(g_0 \max_{b \in \Lambda} \|B(b)\| + g_0^2 \right) \sum_{b \in \Lambda} \|B(b)\|^2 . \tag{C.3}$$

To obtain the estimate, take a fixed smaller complex chart for the finitely many local templates. Their convergent Taylor series and the vanishing orders just proved give a cubic remainder divided by g_0^2 for v_α and quadratic remainders for j_b, d_c. Equivalence of the finite-dimensional local maximum and Euclidean norms gives the displayed inequality. The bounded overlap then yields a torus-independent quadratic majorant on the cutoff. If c_{cmp} is the fixed norm-equivalence constant from the bond norm of [3] to this norm (the constant of the preceding paragraph), then

$$\sum_\alpha |v_\alpha(B)| + \sum_b |j_b(B)| + \sum_c |d_c(B)| \leqslant C_L^{\mathrm{ov}}(c_{\mathrm{cmp}} g_0 r(g_0) + g_0^2)\|B\|^2 \,. \tag{C.4}$$

The constant C_L^{ov} includes the bounded overlap and is independent of the torus. These estimates concern the flat reference integral and supply the bounded local potentials for its Gaussian-marginal expansion.

The local scalars are evaluated in the identity-background frame and are invariant under simultaneous constant colour conjugation. This follows from equivariance of the logarithm, averages, and implicit correction of [4], cyclicity of the action traces, and conjugation invariance of Haar density and determinants. Their extension to the gauge orbit of this background is unambiguous: if $U_0^u(b) = u(b_-)u(b_+)^{-1}$ and $A^u(b) = u(b_-)A(b)u(b_-)^{-1}$, define the transported scalar at (U_0^u, A^u) to have its identity-frame value at A. Two choices of u representing the same background differ by a constant right factor on the connected torus, and give the same scalar by the stated constant-conjugation invariance. This specifies gauge-invariant scalar contractions on this orbit only; it is not an extension to arbitrary nonflat backgrounds. No open Wilson segment is assigned a separate activity or propagated by this decomposition.

D The Exterior Response and Its History Average

This appendix bounds the exterior response $\mathcal{B}_\xi[f_\xi]$ of (3.51) on every branch (Lemma D.6, Section D.5), and pays its history average, together with the uncovered branches of Lemma 3.28, by one-point witness probabilities on compatible even fine periods (Proposition D.15, Section D.7). Section D.8 proves the deterministic comparison of the two minima used by the witness argument and, by the same argument, the bound on the boundary field of [3, (1.21)] used in Section 3.4 (Lemma D.18). The inputs are the Hessian, Schur, chart and primitive bounds of Appendix B and the collar statements of Section 3.4; Remark D.16 records what is and is not controlled. Section 9 applies Sections D.1–D.5 at the retreated carrier (Lemma 9.17) and replaces the history bound of Proposition D.15 by its labelled version (Proposition 9.39).

D.1 Objects and the Conditional Law

Fix a branch with its data ξ (Section 3.1): the young carrier I, the retained free carrier J, the exterior field $A_{\mathrm{e}} \in \mathfrak{g}^J$, extended by zero on every other free bond, the reconstructed field $U_\xi(a)$ of (3.26), and the cutoff norm $|\bullet|_\sigma$ (the bond norm $|\bullet|$ of (4.12), operator or normalised Hilbert–Schmidt, fixed once). The inner product $\langle \bullet, \bullet \rangle$, the norm h and the product Hilbert norms are those of Section 3; $h \leqslant |\bullet|_\sigma$ for both permitted norms. The cutoff radius is

$$r = \delta_0/g_0 = (A_1/A_0)\, p_0(g_0) = A_1(\log g_0^{-2})^{p_0} \,, \tag{D.1}$$

(the first-step cutoff normalisation (4.13), in which p_0 denotes both the function $p_0(g) = A_0(\log g^{-2})^{p_0}$ and its exponent, as in [3]), so $r \to \infty$ and $g_0 r = \delta_0 \to 0$ as $g_0 \to 0$. Every

derivative, overlap, range and Hessian constant below is independent of g_0 once L, the group, the character and the source constants are fixed; the two conditions

$$r \geqslant 1 \qquad \text{and} \qquad C_P g_0(1+2r) \leqslant \gamma/2 \,, \tag{D.2}$$

with C_P the constant of Lemma D.1 as enlarged in Lemma D.3 (fixed there independently of g_0; (D.2) is imposed with the enlarged constant) and $\gamma = \gamma_L/2$, therefore hold for all g_0 below a threshold (the exponent p_0 is positive, $p_0 \geqslant 5\mathfrak{r}$ by Section B.8, so $r \to \infty$ while $g_0 r \to 0$), and they are assumed throughout this appendix, together with the standing conditions (3.1) and the coupling thresholds of Section B.8 under which Lemmas 3.4, 3.11 and B.9 hold (in particular threshold item 3 there, $4C_{\mathsf{C}} g_0(1+2r) \leqslant \vartheta_*/2$). If $I = \varnothing$ then $F_\xi(1) = F_\xi(0)$ ((3.46)) and the response (3.51) is zero; below $I \neq \varnothing$.

Write

$$H = H_{\xi,II}, \qquad \ell_\xi = H_{\xi,IJ} A_{\mathrm{e}}, \qquad j_\xi(a) = \langle \ell_\xi, a \rangle, \qquad s_\xi = 1 + r + r_{\mathrm{e},\xi} \leqslant 1 + 2r \,, \tag{D.3}$$

with $r_{\mathrm{e},\xi} < r$ the retained radius (3.52), the maximum of $|A_{\mathrm{e}}(b)|_\sigma$ over the retained bonds within $3R_{\mathrm{c}}$ of I (zero if there is none). No bound is imposed on farther retained fields; every retained input entering $H_{\xi,IJ}$ or a near primitive lies in this collar (Lemma 3.9(i)). The collar bound holds on every branch, independently of contour coverage: by Lemma 3.11(i) every retained bond within $3R_{\mathrm{c}}$ of I lies in $\mathcal{G}_\xi \setminus S_1^*$ ($\mathcal{G}_\xi$ the union of the good test carriers, (3.41)), its factor in $\chi_{\xi,\mathrm{e}}$ is $\mathbf{1}\{|\Psi_\xi(A_{\mathrm{e}})(b)| < r\}$ ((3.43)) and Ψ_ξ, the identity off S_1^* ((3.38)), is the identity there, so a nonzero retained weight forces $|A_{\mathrm{e}}(b)| < r$; this is Lemma 3.11(iii), not a condition on the covered branches of Lemma 3.28. On the open box $\mathcal{D}_I = \{a \in \mathfrak{g}^I \,|\, |a_b|_\sigma < r \ \forall b \in I\}$ of (3.45) the interpolating law of that display, which at $t = 1$ is the conditional law of the branch (Lemma 3.9(ii)), is, exactly, for $t \in [0,1]$,

$$\mathrm{d}\nu_{\xi;t}(a) = I_{\xi;t}^{-1} e^{\mathsf{Q}_{\xi,t}(a)} \, \mathrm{d}a \,, \qquad \mathsf{Q}_{\xi,t}(a) = -\tfrac{1}{2}\langle a, Ha \rangle - t\langle \ell_\xi, a \rangle + P_{\xi,t}(a) \,, \tag{D.4}$$

zero off $\mathcal{D}_I$ (the indicator in (3.45)), with $\mathsf{Q}_{\xi,t}$ as in (3.54); $I_{\xi;t}$ is the normaliser of (3.45), not a flat-reference normaliser, and $\mathrm{d}a$ is the product Lebesgue measure of the fixed inner product. Throughout $f_\xi(a) = W_\chi(C; U_\xi(a))$ ((3.46)) with the exterior data fixed at their branch values, defined where $U_\xi(a)$ is (Lemma 3.6), which includes $\mathcal{D}_I$, and extended by zero elsewhere (immaterial below, since $\nu_{\xi;t}$ vanishes off $\mathcal{D}_I$); it is bounded by one, and

$$L_{\xi,t} = -j_\xi + \partial_t P_{\xi,t} \,, \qquad \mathcal{B}_\xi[f] = \int_0^1 \mathrm{Cov}_{\nu_{\xi;t}}(f, L_{\xi,t}) \, \mathrm{d}t \tag{D.5}$$

(Lemma 3.10); $L_{\xi,t} = \partial_t \mathsf{Q}_{\xi,t}$ is called the score below. Derivatives and operator norms refer to the product Hilbert structure, complexified for complex scalar gradients; the fixed matrix-norm equivalence factors of Appendix B are absorbed into the constants before g_0 is chosen. The inputs are Lemmas 3.4 and B.9 under their stated hypotheses (in force by the preceding paragraph): $\gamma\,\mathbf{1} \leqslant H \leqslant m_L\,\mathbf{1}$ with $\gamma = \gamma_L/2$; finite interaction range; and bounded local first and second derivatives of the near primitives on the enlarged box $|a_b|_\sigma \leqslant 2r$ at the inputs $a + tA_{\mathrm{e}}$, $t \in [0,1]$. They are used at the background U_1 of the branch; no evaluation at the identity background is made.

D.2 Local Derivative Bounds

Let S_α be the full free-input stencil, retained inputs included, of a near primitive $\mathfrak{p}_\alpha$ (the p_α of Lemma B.9), let N_{in} bound its size and N_{ov} the number of primitive stencils meeting

any free bond; both are fixed by L and the group. The distance $d(b,b')$ of two bonds is the periodic ℓ^∞ distance of their initial sites, plus one for differing directions (the sum of two pseudometrics; a metric, because a positive bond is determined by its initial site and its direction); $d(b,S)=\inf_{b'\in S} d(b,b')$ for a bond set S, with $d(b,\varnothing)=+\infty$, and this d is the only bond distance used in this appendix. Let $R=8L+8$, a range dominating in this metric every near-primitive stencil diameter (at most $4L+1$, Lemma 3.11, $R_{\mathrm{loc}}=2L+2$) and the Hessian range (at most $\rho_L+1=4L+1$ in this metric, $\rho_L=4L$ being the ℓ^∞ range of initial sites in (I1) of Section 9.1: Lemmas B.3 and 3.3). Put

$$\mathsf{K}_\xi=\{b\in I\,:\,d(b,J)\leqslant R\}\,,\qquad I_C=\{b\in I\,:\,b\in\mathcal{F}_r \text{ for some traversal } r \text{ of } C\}\,, \tag{D.6}$$

with $\mathcal{F}_r$ the free-input stencil (3.13) of the contour bond r (a traversal bond, as in Section 3.3; the letter r denotes a traversal bond only inside (D.6) and in the stencils $\mathcal{F}_r$, and the cutoff radius r of (D.1) everywhere else); the source depends on the young field only through I_C (Lemma 3.5).

Lemma D.1 (Local Bounds)**.** *There are constants C_P, C_S (depending on L and the group only) and $C_{\mathrm{resp},\chi}$ (depending on L, the group, the character and the maximal multiplicity with which C traverses a link only, the latter 1 for the square C_n) such that on the enlarged box $\{|a_b|_\sigma\leqslant 2r\ \forall b\in I\}$, uniformly in $t\in[0,1]$ and in the branch (the fourth bound on every covered branch, Lemma 3.28),*

$$\|D_a^2 P_{\xi,t}\|\leqslant C_P g_0 s_\xi\,,\qquad h(\nabla_b\partial_t P_{\xi,t})\leqslant C_S g_0 s_\xi\, r_{\mathrm{e},\xi}\mathbf{1}_{\mathsf{K}_\xi}(b)\,,\qquad h((\ell_\xi)_b)\leqslant m_L r_{\mathrm{e},\xi}\mathbf{1}_{\mathsf{K}_\xi}(b)\,, \tag{D.7}$$

$$\sup_a h(\nabla_b f_\xi(a))\leqslant C_{\mathrm{resp},\chi} g_0 \mathbf{1}_{I_C}(b)\,. \tag{D.8}$$

Under (D.2) *the function $-\mathsf{Q}_{\xi,t}$ is strongly convex on $\mathcal{D}_I$ with Hessian at least $\rho=\gamma/2$.*

Proof. The first bound of (D.7) is the second-derivative bound of Lemma B.9 on the enlarged box with the finite overlap N_{ov}. For the second, $\partial_t P_{\xi,t}(a)=\sum_\alpha D\mathfrak{p}_\alpha(a+tA_{\mathrm{e}})[A_{\mathrm{e}}]$ (by (3.45), $P_{\xi,t}$ has no other t-dependence), so each summand of $\nabla_b\partial_t P_{\xi,t}$ is $D^2\mathfrak{p}_\alpha$, the full second derivative on $\mathfrak{g}^{S_\alpha}$ (retained inputs included) bounded by Lemma B.9, applied to a young direction at b and to $A_{\mathrm{e}}|_{S_\alpha}$; it vanishes unless S_α contains a retained input, which lies in the collar (stencil diameter at most $4L+1<3R_{\mathrm{c}}$), so $h(A_{\mathrm{e}}(b'))\leqslant|A_{\mathrm{e}}(b')|_\sigma\leqslant r_{\mathrm{e},\xi}$ for each such input ($h\leqslant|\bullet|_\sigma$ for both permitted norms, Section D.1) and $h(A_{\mathrm{e}}|_{S_\alpha})\leqslant\sqrt{N_{\mathrm{in}}}\,r_{\mathrm{e},\xi}$, and summing the at most N_{ov} stencils meeting b gives the bound; the support statement follows from the range R of the stencils. The third bound is the row-sum bound $\sum_{b'\in\Omega_1^*}\|(H_\xi)_{bb'}\|\leqslant m_L$ for the rows $b\in I$, the input (I2) of Section 9.1 (the row-sum statement of Lemma 3.4), applied to A_{e} on the collar: $(H_{\xi,IJ})_{bb'}=0$ unless the initial sites are within ℓ^∞ distance ρ_L ((I1) of Section 9.1), so a nonzero entry has $d(b',I)\leqslant\rho_L+1=4L+1<3R_{\mathrm{c}}$ ($R_{\mathrm{c}}\geqslant 2R_{\mathrm{loc}}+1=4L+5$) and $h(A_{\mathrm{e}}(b'))\leqslant|A_{\mathrm{e}}(b')|_\sigma\leqslant r_{\mathrm{e},\xi}$, as for the second bound, and $\rho_L+1\leqslant R$ gives the support statement. For (D.8), on a covered branch every contour bond r lies in Ω_1 with $\mathcal{F}_r\subseteq I$ (Lemma 3.28), so each contour link is reconstructed by a row within $R_{\mathrm{loc}}=2L+2\leqslant 2R_{\mathrm{c}}+2L$ of I, a row of the charts of Lemma B.9; differentiate the normalised character of the ordered link product: by the rowwise reconstruction (Lemma 3.5) and the uniform charts (Lemma B.7) on the enlarged box with the exterior fixed, whose inputs lie in the phase ball of the charts by Lemma B.9, each affected link has a phase derivative bounded by Cg_0, a fixed input affects a bounded number of links, each link is traversed by C at most as often as its maximal link multiplicity (once for the square C_n, a simple closed contour; the multiplicity enters $C_{\mathrm{resp},\chi}$ as a factor), the derivative of the character is bounded in the invariant tangent norm on the compact group, and the other links remain unitary inside the

closed contraction; no open Wilson segment is estimated separately. The convexity statement is $\gamma\,\mathbf{1} \leqslant H$ (Lemma 3.4) together with $\|D^2 P_{\xi,t}\| \leqslant C_P g_0 s_\xi \leqslant C_P g_0(1+2r) \leqslant \gamma/2$, using $s_\xi \leqslant 1+2r$ (Lemma 3.11(iii)) and (D.2). *q.e.d.*

D.3 An Extension within the Proved Chart

The enlarged-box input matters: a radial extension bounded only in h can leave the operator-norm box, and a radius proportional to $3s_\xi$ can exceed the enlarged box. Let $B_\sigma = \{x \in \mathfrak{g} : |x|_\sigma \leqslant 1\}$, let Π_r be the Hilbert metric projection onto the closed convex set $\frac{3}{2} r B_\sigma$, let ϕ be a fixed nonnegative smooth h-radial mollifier on $\mathfrak{g}$ supported in the unit h-ball with integral one (in this appendix ϕ is only this mollifier, not the phase of Lemma B.7), $\phi_s(y) = s^{-d_G}\phi(y/s)$ for $s>0$ with $d_G = \dim \mathfrak{g}$, and put $\epsilon = 1/(4\sqrt{N})$. Define

$$T_r(x) = \int_{\mathfrak{g}} \Pi_r(x-y)\,\phi_{\epsilon r}(y)\,\mathrm{d}y \qquad (x \in \mathfrak{g})\,. \tag{D.9}$$

Lemma D.2 (The Extension Map)**.** *T_r is smooth and* Ad*-equivariant, and*

$$T_r(x) = x \;\; \text{if } |x|_\sigma \leqslant r, \qquad |T_r(x)|_\sigma \leqslant \tfrac{3}{2} r, \qquad \|DT_r\| \leqslant 1, \qquad \|D^2 T_r\| \leqslant C_\sigma / r\,, \tag{D.10}$$

with $C_\sigma = 4\sqrt{N}\,\|\nabla\phi\|_{L^1}$, which depends on N and the fixed mollifier only, not on r or g_0.

Proof. Both norms are Ad-invariant, so B_σ is Ad-invariant and Ad is h-orthogonal; hence metric projection and the h-radial mollifier commute with the adjoint action, which gives the equivariance. If $h(y) \leqslant \epsilon r$ then $|y|_\sigma \leqslant \sqrt{N}\,h(y) \leqslant r/4$ (Section 9.1: $|\bullet|_\sigma \leqslant \sqrt{N}\,h$), so for $|x|_\sigma \leqslant r$ one has $|x-y|_\sigma \leqslant \frac{5}{4} r < \frac{3}{2} r$ and $\Pi_r(x-y) = x-y$ (indeed for all $|x|_\sigma < \frac{5}{4} r$); the convolution then equals x because the mollifier has integral one and, being h-radial, zero first moment. The range statement is the convexity of $\frac{3}{2} r B_\sigma$: $T_r(x)$ is a probability average of its points. Metric projection is 1-Lipschitz for h and convolution preserves this, giving $\|DT_r\| \leqslant 1$ in the h-operator norm; moving one derivative onto the mollifier of scale ϵr bounds $\|D^2 T_r\|$ by $\|D\Pi_r\|_\infty \|\nabla\phi_{\epsilon r}\|_{L^1} \leqslant \|\nabla\phi\|_{L^1}/(\epsilon r) = C_\sigma/r$. *q.e.d.*

Apply T_r bondwise to the young inputs only and write T for the product map on $\mathfrak{g}^I$. Define

$$\widetilde{P}_t(a) = \sum_{\alpha \text{ near}} \big[\mathfrak{p}_\alpha(Ta + tA_{\mathrm{e}}) - \mathfrak{p}_\alpha(tA_{\mathrm{e}})\big], \qquad \widetilde{f}(a) = f_\xi(Ta), \qquad \widetilde{L}_t(a) = -\langle \ell_\xi, a\rangle + \partial_t \widetilde{P}_t(a)\,. \tag{D.11}$$

Lemma D.3 (The Extended Objects)**.** *$\widetilde{P}_t$, $\widetilde{f}$ and $\widetilde{L}_t$ are defined and smooth on all of $\mathfrak{g}^I$, $|\widetilde{f}| \leqslant 1$, and on $\mathcal{D}_I$: $\widetilde{f} = f_\xi$, $\widetilde{P}_t = P_{\xi,t} - P_{\xi,t}(0)$ (the scalar $P_{\xi,t}(0)$ is removed only in the normalised conditional law; the unnormalised integral keeps it) and $\widetilde{L}_t - L_{\xi,t} = -\partial_t P_{\xi,t}(0)$, a constant, so that $\mathrm{Cov}_{\nu_{\xi;t}}(\widetilde{f}, \widetilde{L}_t) = \mathrm{Cov}_{\nu_{\xi;t}}(f_\xi, L_{\xi,t})$. After enlarging C_P (before g_0 is chosen; the enlarged constant depends on L, the group, C_σ, C_{p}, N_{ov} and N_{in}, all fixed before the coupling, and* (D.2) *is imposed with it), uniformly in $t \in [0,1]$ and $a \in \mathfrak{g}^I$: $\|D^2 \widetilde{P}_t\| \leqslant C_P g_0 s_\xi$, the block Schur row and column sums of $D^2\widetilde{P}_t$ are at most $C_P g_0 s_\xi$ with range R,*

$$h(\nabla_b \partial_t \widetilde{P}_t) \leqslant C_S g_0 s_\xi\, r_{\mathrm{e},\xi} \mathbf{1}_{\mathsf{K}_\xi}(b)\,, \qquad h(\nabla_b \widetilde{f}) \leqslant C_{\mathrm{resp},\chi} g_0 \mathbf{1}_{I_C}(b)\,. \tag{D.12}$$

Proof. The image Ta has young cutoff norm at most $\frac{3}{2} r < 2r$, so it lies in the enlarged box of Lemma B.9 (threshold item 3 of Section B.8, in force by Section D.1), where every near primitive and every reconstruction row within $2R_{\mathrm{c}} + 2L$ of I is analytic, the exterior fields

being fixed; the real reconstruction gives group-valued links, whence $|\widetilde{f}| \leqslant 1$. On $\mathcal{D}_I$, T is the identity (Lemma D.2), which gives the three identities; a constant shift of one operand leaves the covariance unchanged. The Hessian of $\widetilde{P}_t$ is $D^2\mathfrak{p}_\alpha(DT\bullet, DT\bullet) + D\mathfrak{p}_\alpha\, D^2T$ summed over the near primitives: the first term is $O(g_0 s_\xi)$ in operator norm by Lemma D.1 and $\|DT\| \leqslant 1$, and per stencil by the second-derivative bound of Lemma B.9 (the per-stencil form is what the Schur sums below use), the second is $O(g_0 s_\xi^2/r) = O(g_0 s_\xi)$ by the first-derivative bound of Lemma B.9, $\|D^2T\| \leqslant C_\sigma/r$, $r \geqslant 1$ and $s_\xi \leqslant 1 + 2r \leqslant 3r$. Each input belongs to at most N_{ov} stencils and each stencil has at most N_{in} blocks, so summing at most $N_{\mathrm{ov}}N_{\mathrm{in}}$ blocks with this chain-rule bound gives the operator and Schur bounds; the supports and the range are those of the original stencils. The mixed derivative $\nabla_b\partial_t\widetilde{P}_t$ contains no second derivative of T, since T does not depend on t and is bondwise: $\nabla_b\partial_t\widetilde{P}_t(a) = DT_r(a_b)^*(\nabla_b\partial_t P_{\xi,t})(Ta)$, so (D.7) of Lemma D.1 at Ta and $\|DT\| \leqslant 1$ give the first bound of (D.12); the second is (D.8) composed with DT. *q.e.d.*

D.4 Covariance Decay at the Hard Boundary

For a positive integer n define the barrier and the barrier laws

$$\beta_n(a) = n \sum_{b \in I} \big[\mathrm{dist}_h(\bullet, rB_\sigma)^2 * \phi_{n^{-2}}\big](a_b)\ , \tag{D.13}$$

$$V_{n,t}(a) = \tfrac{1}{2}\langle a, Ha\rangle + t\langle \ell_\xi, a\rangle - \widetilde{P}_t(a) + \beta_n(a)\ , \qquad \mathrm{d}\nu_{n,t} = Z_{n,t}^{-1} e^{-V_{n,t}(a)}\,\mathrm{d}a \quad \text{on } \mathfrak{g}^I\ .$$

The squared distance to a convex set is convex and convolution with a nonnegative mollifier preserves convexity, so the barrier Hessian is positive semidefinite; it is diagonal in the bond index because β_n is a sum of functions of the single bond variables. For fixed n the potential is smooth with bounded Hessian (that of the barrier is bounded by $2n$: the gradient $2(x - \Pi_r(x))$ of the squared distance is 2-Lipschitz, and mollification is an average of translates) and Gaussian confinement, its Hessian is at least ρI with $\rho = \gamma/2$ as in Section D.2 ($\gamma\,\mathbf{1} \leqslant H$ by Lemma 3.4, and $\|D^2\widetilde{P}_t\| \leqslant C_P g_0 s_\xi \leqslant C_P g_0(1 + 2r) \leqslant \gamma/2$ by Lemma D.3 and (D.2)–(D.3)), and its off-diagonal block row and column sums are at most

$$M_* = m_L + \gamma/2\ , \tag{D.14}$$

with range at most R in the bond distance d of Section D.2, independently of n, t, the volume and the branch (the barrier is diagonal; the row sums of H are at most m_L by (I2) of Section 9.1, its column sums equal its row sums since $H_{bb'} = H_{b'b}^{\mathsf{T}}$ have equal norms, and its range is $\rho_L = 4L \leqslant R$ by (I1) there; the Schur sums of $D^2\widetilde{P}_t$ are at most $C_P g_0 s_\xi \leqslant \gamma/2$ with range R by Lemma D.3).

Lemma D.4 (Helffer–Sjöstrand Identity and Decay)**.** *Let $\mathscr{A}_{n,t} = \mathscr{L}_{n,t} \otimes I + D^2V_{n,t}$ with $\mathscr{L}_{n,t} = -\Delta + \nabla V_{n,t} \cdot \nabla$, on the Hilbert direct sum over $b \in I$ of $L^2(\nu_{n,t}; \mathfrak{g})$. Then $\mathscr{A}_{n,t} \geqslant \rho I$ ($\mathscr{L}_{n,t} = \nabla^*\nabla \geqslant 0$ on $L^2(\nu_{n,t})$ and $D^2V_{n,t} \geqslant \rho I$), and for real u, v in the weighted space $H^1(\nu_{n,t})$*

$$\mathrm{Cov}_{\nu_{n,t}}(u, v) = \langle \nabla u, \mathscr{A}_{n,t}^{-1}\nabla v\rangle_{L^2(\nu_{n,t})}\ . \tag{D.15}$$

With $\mu = R^{-1}\log(1 + \gamma/(4M_*))$ and Π_b the projection onto the b-component,

$$\|\Pi_b \mathscr{A}_{n,t}^{-1}\Pi_c\| \leqslant \frac{4}{\gamma} e^{-\mu d(b,c)}\ , \qquad |\,\mathrm{Cov}_{\nu_{n,t}}(u, v)| \leqslant \frac{4}{\gamma}\sum_{b,c\in I} e^{-\mu d(b,c)} \|\nabla_b u\|_{L^2(\nu_{n,t})} \|\nabla_c v\|_{L^2(\nu_{n,t})}\ . \tag{D.16}$$

For complex u and real v (the case used: $u = \widetilde{f}$, the Wilson character, and $v = \widetilde{L}_t$), applying (D.16) *to the real and imaginary parts of u gives the same bound with the gradient norm of*

u read as the sum of the gradient norms of its two parts; each is at most the gradient norm of u, a factor at most two that Section D.5 absorbs into $C_{\mathrm{resp},\chi}$.

Proof. The identity (D.15) is [12, Proposition 4.5] (Theorem 1.3 there at $k = 0$) for the real-valued C^2 potential $V_{n,t} + \log Z_{n,t}$ (normalised; the additive constant leaves the derivatives unchanged; $V_{n,t}$ is real because its quadratic, linear and barrier terms are real and $\widetilde{P}_t$ is real, the primitives being real on real inputs with a positive determinant, Lemma B.7(iii) and Lemma B.9). Its conditions are a real C^2 normalised potential and the closed range of the operator A_0 there; the latter follows from $A_1 \geqslant \rho I$, which is $\mathscr{A}_{n,t} \geqslant \rho I$ here, by [12, Theorem 1.3], as explained there after (4.39). The theorem gives the projection onto the orthogonal complement of the constants as $d_0^*(A_1|_{\overline{\mathrm{ran}}\, d_0})^{-1} d_0$, and at fixed n the bounded Hessian identifies A_1 with the displayed operator $\mathscr{A}_{n,t}$: A_1 is the Friedrichs extension of [12, Lemma 4.2] with the action (4.22) there, $\mathscr{L}_{n,t} \otimes I + D^2 V_{n,t}$ on gradient fields, and since $D^2 V_{n,t}$ is a bounded multiplication operator at fixed n its form domain is that of $\mathscr{L}_{n,t} \otimes I$ and A_1 is the operator sum on the domain of $\mathscr{L}_{n,t} \otimes I$, a domain the bounded field-independent weights W_c below preserve. The operands are the source $\widetilde{f}$, bounded with bounded gradient (Lemma D.3), and the score $\widetilde{L}_t = -\langle \ell_\xi, \bullet \rangle + \partial_t \widetilde{P}_t$, which grows at most linearly and has bounded gradient by (D.12); both lie in $H^1(\nu_{n,t})$. For a fixed $c \in I$ let W_c multiply the component b by $e^{\mu d(b,c)}$; it is constant in field space, hence commutes with the scalar generator $\mathscr{L}_{n,t} \otimes I$ and with every diagonal Hessian block, the barrier included, and the off-diagonal Schur bounds give $\|W_c \mathscr{A}_{n,t} W_c^{-1} - \mathscr{A}_{n,t}\| \leqslant M_*(e^{\mu R} - 1) \leqslant \rho/2$ by the choice of μ. W_c and its inverse are bounded at finite volume, so a resolvent Neumann series gives $\|W_c \mathscr{A}_{n,t}^{-1} W_c^{-1}\| \leqslant (\rho - \rho/2)^{-1} = 4/\gamma$; compressing by Π_b on the left and Π_c on the right, where W_c acts as $e^{\mu d(b,c)}$ and as $e^{\mu d(c,c)} = 1$, gives the first bound of (D.16); the second follows from (D.15) and the Cauchy–Schwarz inequality. The large diagonal barrier never enters the decay constant. *q.e.d.*

Lemma D.5 (The Barrier Limit)**.** *For u, v bounded or of linear growth with bounded gradients,* $\mathrm{Cov}_{\nu_{n,t}}(u, v) \to \mathrm{Cov}_{\nu_{\xi;t}}(u, v)$ *as $n \to \infty$. Consequently the second bound of* (D.16) *holds for $\nu_{\xi;t}$ on the hard box $\mathcal{D}_I$, with the gradient norms taken as global suprema, $\|\nabla_b u\|_\infty \|\nabla_c v\|_\infty$ in place of the L^2 norms.*

Proof. At finite volume $\beta_n \to 0$ on the closed cutoff body $\{|a_b|_\sigma \leqslant r \ \forall b\}$ and $\beta_n \to \infty$ outside it, the boundary of the body has Lebesgue measure zero, and $\beta_n \geqslant 0$ together with the Gaussian confinement supplies an integrable majorant for u, v and uv (bounded or of linear growth); dominated convergence gives the convergence of the unnormalised integrals, and of the normaliser to $\int_{\mathcal{D}_I} e^{-V_t^\circ}\, \mathrm{d}a > 0$ (the box has nonempty interior), where $V_t^\circ = V_{n,t} - \beta_n$ is the potential without the barrier, independent of n, hence of the covariances, to those of the law with density $e^{-V_t^\circ}$ on $\mathcal{D}_I$, which is $\nu_{\xi;t}$ by Lemma D.3 ($\widetilde{P}_t = P_{\xi,t} - P_{\xi,t}(0)$ on $\mathcal{D}_I$, and (D.4)). On the right of (D.16) the $L^2(\nu_{n,t})$ norms depend on n, but they are bounded by the global suprema $\|\nabla_b u\|_\infty$, $\|\nabla_c v\|_\infty$, which do not, and μ and $4/\gamma$ are independent of n, so the bound passes to the limit. No translated cutoff and no new exterior law is introduced. *q.e.d.*

D.5 The Response

Put

$$D_\xi = \sum_{b \in I_C} e^{-\mu d(b,J)/2}\,, \tag{D.17}$$

with μ of Lemma D.4 and the bond distance d of Section D.2; $D_\xi = 0$ when $J = \varnothing$ (by $d(b, \varnothing) = +\infty$; then also $A_{\mathrm{e}} = 0$ and the score vanishes).

Lemma D.6 (The Exterior Response)**.** *Under* (D.2)*, on every covered branch (Lemma 3.28; the bound is used on covered branches only, Section D.6 and Lemma 9.17) and for every* $t \in [0, 1]$*,*

$$|\operatorname{Cov}_{\nu_{\xi;t}}(f_\xi, L_{\xi,t})| \leqslant C_{\mathrm{resp},\chi}\, g_0 r \sum_{b \in I_C} e^{-\mu d(b,J)/2}\,, \qquad |\mathcal{B}_\xi[f_\xi]| \leqslant C_{\mathrm{resp},\chi}\, g_0 r\, D_\xi\,, \tag{D.18}$$

after enlarging the constant $C_{\mathrm{resp},\chi}$ *of Lemma D.1 (the enlarged constant keeps the name); it depends on* L*,* γ*,* m_L*,* M_**,* R*,* N_{in}*,* N_{ov}*,* C_P*,* C_S*,* C_σ*, the character, the group and the fixed source parameters, and is independent of* g_0*, the branch, the volume,* t *and the barrier index.*

Proof. By Lemma D.3, $\operatorname{Cov}_{\nu_{\xi;t}}(f_\xi, L_{\xi,t}) = \operatorname{Cov}_{\nu_{\xi;t}}(\widetilde{f}, \widetilde{L}_t)$, and Lemma D.5 bounds it by $(8/\gamma)\sum_{b,c\in I} e^{-\mu d(b,c)}\|\nabla_b \widetilde{f}\|_\infty \|\nabla_c \widetilde{L}_t\|_\infty$: the bound (D.16) is applied to the real and imaginary parts of the complex $\widetilde{f}$ and the real $\widetilde{L}_t$ (Lemma D.4), the sum of the two gradient norms being at most $2\|\nabla_b \widetilde{f}\|_\infty$; this is the factor two that Lemma D.4 leaves to this section. By (D.12) and (D.7), $\|\nabla_c \widetilde{L}_t\|_\infty \leqslant (m_L + C_S g_0 s_\xi) r_{\mathrm{e},\xi} \mathbf{1}_{\mathsf{K}_\xi}(c) \leqslant (m_L + C_S\gamma/(2C_P))\, r\, \mathbf{1}_{\mathsf{K}_\xi}(c)$ (as $r_{\mathrm{e},\xi} < r$ and $C_S g_0 s_\xi \leqslant C_S g_0(1 + 2r) \leqslant C_S\gamma/(2C_P)$ by (D.2); the coefficient is a constant fixed before g_0) and $\|\nabla_b \widetilde{f}\|_\infty \leqslant C_{\mathrm{resp},\chi} g_0 \mathbf{1}_{I_C}(b)$ with the constant of Lemma D.1. For the shell step, $d(c, J) \leqslant R$ on K_ξ, so $d(b, c) \geqslant d(b, J) - R$ and

$$e^{-\mu d(b,c)} = e^{-\mu d(b,c)/2} e^{-\mu d(b,c)/2} \leqslant e^{\mu R/2} e^{-\mu d(b,J)/2}\, e^{-\mu d(b,c)/2}\,;$$

the sum of $e^{-\mu d(b,c)/2}$ over all periodic bonds c is bounded by the infinite-lattice shell sum $S_{\mu/2} = \sup_b \sum_c e^{-\mu d(b,c)/2}$ (the nearest-lift map from torus bonds to lattice bonds is injective and the terms are nonnegative), uniformly in the periods. Collecting the constants, $(8/\gamma)(m_L + C_S\gamma/(2C_P)) e^{\mu R/2} S_{\mu/2}$ times the constant of Lemma D.1, which depends on μ through $e^{\mu R/2}$ and $S_{\mu/2}$ and hence on M_*, gives the first bound of (D.18) with the enlarged $C_{\mathrm{resp},\chi}$; integrating over $t \in [0, 1]$ in (D.5) gives the second. *q.e.d.*

Remark D.7 (The Surface Terms Are Retained). In the notation (3.53)–(3.54), Lemma 3.12 (its hypothesis holds: f_ξ is C^1 on the enlarged box $\{|a_b|_\sigma \leqslant 2r\}$, a neighbourhood of $\overline{\mathcal{D}_I}$, by the charts of Lemma B.9 (Lemma 3.6) and Lemma B.7, as in the proof of Lemma D.1) writes the integrand of the response as

$$-\mathbb{E}_{\nu_{\xi;t}}[D_{m_\xi} f_\xi] + \operatorname{Cov}_{\nu_{\xi;t}}(f_\xi, \partial_t P_{\xi,t} - D_{m_\xi} P_{\xi,t}) + \mathcal{T}_{\xi,t}[f_\xi] - \mathbb{E}_{\nu_{\xi;t}}[f_\xi]\, \mathcal{T}_{\xi,t}[1]\,, \tag{D.19}$$

with the hard-cutoff flux $\mathcal{T}_{\xi,t}[u] = I_{\xi;t}^{-1} \int_{\partial\mathcal{D}_I} u\, e^{\mathsf{Q}_{\xi,t}} \langle m_\xi, n\rangle \,\mathrm{d}S$. Lemma D.6 bounds this whole expression: neither flux is set to zero or estimated separately. The extension T_r of Lemma D.2 is the identity on the neighbourhood $\{|a_b|_\sigma < \frac{5}{4} r\}$ of $\overline{\mathcal{D}_I}$ and the barrier limit of Lemma D.5 returns the original measure, so the boundary traces and the normalised covariance in (D.19) are those of Section 3.4.

D.6 The Physical Substitution

Let $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1} = \mathcal{J}_\xi^{\mathrm{a}}[1]$ be the branch weight (3.49) and $\mathcal{O}$ the outer operation of (3.4), so that $\mathcal{Z}_0 = \int \mathcal{O}[\mathsf{w}_\xi](V)\,\mathrm{d}V$ ((3.7)). Define the history expectation and probability

$$\mathbb{E}_{\mathrm{h}}[X] = \mathcal{Z}_0^{-1} \int \mathcal{O}[\mathsf{w}_\xi X](V)\,\mathrm{d}V\,, \qquad \mathbb{P}_{\mathrm{h}}(A) = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_A]\,, \tag{D.20}$$

for nonnegative or bounded Borel functions X of the branch data; $\mathcal{O}$ has a nonnegative kernel (its Haar and constraint factors are measures and its cutoffs and exponentials are nonnegative, (3.4)) and $0 < \mathcal{Z}_0 < \infty$ (the measurability paragraph of Section 3.1), so $\mathbb{E}_{\mathrm{h}}$ is positive, $|\mathbb{E}_{\mathrm{h}}[X]| \leqslant \sup|X|$ and $\mathbb{E}_{\mathrm{h}}[1] = 1$. Let A_{cov} be the set of covered branches of Lemma 3.28: every contour bond is a bond of Ω_1 and $\mathcal{F}_r \subseteq I$ for every contour bond r. The event A_{cov} is determined by Ω_1, R_1 and the fixed contour templates ($I = \Lambda_1^*$, the free bonds with an endpoint in Λ_1, Section 3.1, with Λ_1 derived from Ω_1 and R_1 by [3, (1.20)] and its filling; the stencils $\mathcal{F}_r$ are fixed); it and D_ξ are measurable functions of the outer branch data, constant in the young integration, and $0 \leqslant D_\xi \leqslant |I_C|$, a bound by the number of free inputs of the contour. The response bound is applied on covered branches only. The combined exterior-response and uncovered contribution to $\mathbb{E}_\varepsilon[W_\chi(C)]$ of (3.7) is

$$E_{\mathrm{bdry}} = \mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}}\mathcal{B}_\xi[f_\xi] + \mathbf{1}_{A_{\mathrm{cov}}^c}F_\xi(1)\big] \,, \tag{D.21}$$

the exterior response on the covered branches and the whole conditional expectation on the uncovered ones ((3.51), (3.97)); the integrand is a bounded Borel function of the branch data ($F_\xi(t) = \int f_\xi K_{\xi;t}\,\mathrm{d}a/I_{\xi;t}$ is Borel in the branch data for each t, by the composition argument of Section 3.1 applied to the interpolated kernel of (3.46) and Fubini's theorem, and $|\mathcal{B}_\xi[f_\xi]| \leqslant C_{\mathrm{resp},\chi} g_0 r D_\xi$ is bounded), so (D.20) applies to it. No bound on $\mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}D_\xi]$ or on $\mathbb{P}_{\mathrm{h}}(A_{\mathrm{cov}}^c)$ is used in this subsection; both are bounded in Section D.7 (Proposition D.15).

Corollary D.8. *Under* (D.2)*,*

$$\Big|\mathcal{Z}_0^{-1}\int \mathcal{O}\big[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}}}\mathcal{B}_\xi[f_\xi]\big](V)\,\mathrm{d}V\Big| \leqslant C_{\mathrm{resp},\chi}\, g_0 r\, \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}D_\xi] \,, \tag{D.22}$$

$$|E_{\mathrm{bdry}}| \leqslant C_{\mathrm{resp},\chi}\, g_0 r\, \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}D_\xi] + \mathbb{P}_{\mathrm{h}}(A_{\mathrm{cov}}^c) \,.$$

Proof. $\mathsf{w}_\xi \geqslant 0$ and $\mathcal{O}$ is positive, so Lemma D.6 gives the first bound pointwise on the branches; the triangle inequality, (D.22), positivity and $|F_\xi(1)| \leqslant 1$ give the second. No conditional or flat-reference normaliser replaces $\mathcal{Z}_0$, and c_ξ stays inside w_ξ. *q.e.d.*

D.7 The History Average by One-Point Witnesses

This section assumes compatible even fine periods: every fine period is a multiple of $\ell = LMR_1$ and at least $8M$ ((3.1)) and even. This is a restriction relative to the compatible periods of Section 3; it includes the torus with periods $2L^m$ when the partitions fit. Lemma D.6 needs no parity restriction, and no odd-period history bound is asserted.

Write $\ell = LMR_1$ for the macro side (R_1 the scale integer), $B_\xi = \Lambda_1^{\mathrm{c}}$ for the filled bad region, $E(X)$ for the set of positive fine bonds with at least one endpoint in a set X of sites, and n_C for the number of contour traversals. Let Γ_C be the deterministic set of the contour bonds together with the free-input templates of their reconstruction rows, projected periodically from the torus templates before any history is selected (so $\mathcal{F}_r \subseteq \Gamma_C$ for every contour bond r, with $\mathcal{F}_r$ the free-input set of (3.13), the set appearing in the covered event of Lemma 3.28); there is a constant C_L, depending on L only (the templates are fixed lattice stencils), with $|\Gamma_C| \leqslant C_L n_C$, and $I_C \subseteq \Gamma_C$ on every branch (the young designated variables are eliminated, so their free-input stencils, not the designated bonds, are the members of Γ_C). For a positive bond z define the marked probability

$$p_z = \mathcal{Z}_0^{-1}\int \mathcal{O}\big[\mathsf{w}_\xi \mathbf{1}\{z \in E(B_\xi)\}\big](V)\,\mathrm{d}V = \mathbb{P}_{\mathrm{h}}(z \in E(B_\xi)) \,; \tag{D.23}$$

the mark depends on the regions and on the deterministic filling, not on the young variable.

Lemma D.9 (Reduction to One-Point Marks)**.** *Let $c = \mu/2$ and $S_c = \sup_b \sum_z e^{-cd(b,z)}$, the periodic bond shell sum, bounded by its infinite-lattice counterpart uniformly in the periods. Pointwise on every branch*

$$D_\xi \leqslant \sum_{b\in\Gamma_C}\sum_{z} e^{-cd(b,z)}\mathbf{1}\{z \in E(B_\xi)\}\,, \qquad \mathbf{1}_{A^c_{\mathrm{cov}}} \leqslant \sum_{b\in\Gamma_C}\mathbf{1}\{b \in E(B_\xi)\}\,. \tag{D.24}$$

Hence, if $p_z \leqslant q$ for every positive bond z,

$$\mathbb{E}_{\mathrm{h}}[D_\xi] \leqslant C_L S_c n_C q\,, \qquad \mathbb{P}_{\mathrm{h}}(A^c_{\mathrm{cov}}) \leqslant C_L n_C q\,. \tag{D.25}$$

Proof. $J \subseteq E(B_\xi)$: a free bond not in I has no endpoint in Λ_1. So each $b \in I_C \subseteq \Gamma_C$ contributes $e^{-cd(b,J)} \leqslant \sum_{z\in J} e^{-cd(b,z)} \leqslant \sum_z e^{-cd(b,z)}\mathbf{1}\{z \in E(B_\xi)\}$. If coverage fails, either a contour bond is not a bond of Ω_1, and then it meets B_ξ since $\Lambda_1 \subseteq \Omega_1$, or a free input of the contour is not in I (some $b \in \mathcal{F}_r \setminus I$ for a contour bond r, and $\mathcal{F}_r \subseteq \Gamma_C$), and then it is retained in J or lies outside Ω_1 and meets B_ξ; in both cases some $b \in \Gamma_C$ lies in $E(B_\xi)$. Taking $\mathbb{E}_{\mathrm{h}}$ of (D.24) (Tonelli; all integrands nonnegative) and using $p_z \leqslant q$ and $|\Gamma_C| \leqslant C_L n_C$ gives (D.25). *q.e.d.*

D.7.1 Cutoff Failures Give Local Witnesses

The witness argument operates on the original unresummed cutoffs [3, (1.1), (1.4), (1.8), (1.18)], before the boundary substitution; it is proved in Lemmas D.10, D.11 and D.12 below. Its conclusion: a bad eligible test forces a fine plaquette with $|U(\partial p) - \mathbf{1}|_\sigma \geqslant \theta$ in its neighbourhood, or an active mean-transport variable with $|w - \mathbf{1}|_\sigma \geqslant \theta$ there (the P_0 and P_1 alternatives need no gauge-variable witness), with the threshold

$$\theta = \min\Big(\varepsilon_0,\ \frac{\varepsilon_1}{4B_3N^{3/2}L^2},\ \frac{\delta_0}{2C_Q}\Big)\,, \qquad C_Q = (9K_L + 2)(1 + 2B_3N^{3/2}L^2)\,, \tag{D.26}$$

where K_L is the plaquette-filling constant of the loops of [4, (0.4), (0.11)] and of the bond loops of one block or of two adjacent blocks, defined in the paragraph "Constants of the block comparison" below. The norm is the cutoff norm $|\bullet|_\sigma$, and the mean and comparison constants are chosen for it. The seed is used on the original earlier-cutoff and averaging-constraint support only, not for arbitrary frozen exterior data. Its common local width w is LM_2R_1 or $L^2M_2R_1$, with one convention throughout, and its macro width is ℓ; these are the conventions of the branch hypotheses of Section 3.1. For a cube Q and $k \geqslant 1$, $Q^{[k]}_w$ denotes Q enlarged by k layers of width w (the $\square'^{\sim k}$ of [3] when $w = L^2M_2R_1$), so that $\mathrm{side}(Q^{[k]}_w) = \mathrm{side}(Q) + 2kw$ and $Q^{[2]}_w$ is at ℓ^∞ distance $2w$ from the complement of $Q^{[4]}_w$; no mixed-width version is asserted. An eligible cube is a cube $\square'$ of the partition of [3] into $L^2M_2R_1$-cubes carrying the test (1.18), which Bałaban performs for $\square' \subset \Omega_1^{\sim -1}$ (p.250), so every eligible cube satisfies the hypothesis $Q \subseteq \Omega_1^{\sim -1}$ of Proposition D.17.

Constants of the block comparison. For a block centre y and a site x of its block let $\mathbf{G}(y,x)$ be Bałaban's finite family of shortest paths from y to x ([4, p. 252], the definition preceding (0.4), used in (0.11)): for each permutation π of the axes the path moves first along $e_{\pi(1)}$ by the full coordinate difference, then along $e_{\pi(2)}$, and so on; each coordinate is monotone along the path, so the path stays in the block $B(y)$. Let $\Gamma_x \in \mathbf{G}(y,x)$ be a fixed representative (Γ_y the empty path, with transport $\mathbf{1}$). Let K_L bound the number of plaquettes needed to fill each closed loop of the following finite family, each filling chosen inside the union of the block or blocks containing the loop (a lattice box, in which every closed lattice loop bounds

a plaquette filling): the difference of two paths of one $\mathbf{G}(y,x)$; the loop $\Gamma_x \cup b \cup (-\Gamma_{x'})$ of a block-internal bond $b=(x,x')$; the loop $\Gamma_x \cup b \cup (-\Gamma_{x'}) \cup (-c)$ of an interface bond $b=(x,x')$ of the coarse bond c, $x \in B(c_-)$, $x' \in B(c_+)$, $\Gamma_x \in \mathbf{G}(c_-,x)$, $\Gamma_{x'} \in \mathbf{G}(c_+,x')$; and the loops $\Gamma \cup [x,x'] \cup (-\Gamma') \cup (-c)$ of [4, (0.4)], for a site $x \in B(c_-)$, with $[x,x']$ the parallel transport of the coarse bond c to x ([4, p. 252], the sentence preceding the definition of the averages: the straight contour of L fine bonds in the direction of c from x to $x'=x+(c_+-c_-) \in B(c_+)$) and paths $\Gamma \in \mathbf{G}(c_-,x)$, $\Gamma' \in \mathbf{G}(c_+,x')$. Each loop lies in one block or in the union $B(c_-) \cup B(c_+)$ of two adjacent blocks (the contour $[x,x']$ runs inside this union, a box of sides $2L,L,L,L$), so K_L depends on L only. The transports of a unitary field F along two paths with the same endpoints differ in $|\bullet|_\sigma$ by at most the sum of $|F(\partial p)-\mathbf{1}|_\sigma$ over the plaquettes filling their loop (unitary invariance of the norm and telescoping), hence by at most $K_L\eta$ when $|F(\partial p)-\mathbf{1}|_\sigma \leqslant \eta$ for every plaquette p with its four vertices in $B(c_-) \cup B(c_+)$ ("the plaquettes of the two blocks"; this includes the plaquettes straddling the interface). For the group mean M of [4, (0.5)–(0.10)] let $\mathcal{N}_L$ be the finite set of cardinalities $|\mathbf{G}(y,x)|$ and, for $n \in \mathcal{N}_L$, $\mathfrak{m}_n(A_1,\dots,A_n)=(1/i)\log M(e^{iA_1},\dots,e^{iA_n})$; it is analytic near zero with $\mathfrak{m}_n(0)=0$ and derivative $A \mapsto n^{-1}\sum_j A_j$ at zero ([4, (0.8)]). Let $r_{\mathrm{M}}>0$ be a radius such that every $\mathfrak{m}_n$, $n \in \mathcal{N}_L$, is analytic on the closed maximum-norm ball of n-tuples of radius r_{M} (the minimum over the finite set $\mathcal{N}_L$ of such radii), and C_{M} half the maximum over $n \in \mathcal{N}_L$ of the second-derivative norms on these balls; Taylor's theorem gives $|\mathfrak{m}_n(A)|_\sigma \leqslant \max_j |A_j|_\sigma + C_{\mathrm{M}}(\max_j |A_j|_\sigma)^2$ for $\max_j |A_j|_\sigma \leqslant r_{\mathrm{M}}$. On the principal unitary chart $|(1/i)\log W|_\sigma \leqslant |W-\mathbf{1}|_\sigma + C_{\log}|W-\mathbf{1}|_\sigma^2$ with a fixed $C_{\log}$: eigenvalue by eigenvalue, $2\arcsin(t/2) \leqslant t+Ct^2$ for $0 \leqslant t \leqslant 2$, $t_j=|e^{i\varphi_j}-1|$, which gives the operator-norm case directly and the normalised Hilbert–Schmidt case with a constant depending on N ($\max_j t_j \leqslant \sqrt{N}|W-\mathbf{1}|_\sigma$). Put $C'_{\mathrm{M}}=C_{\log}+4C_{\mathrm{M}}$. If $|W_jW_k^{-1}-\mathbf{1}|_\sigma \leqslant \rho$ for all j, with $\rho \leqslant 1/\max(1,C_{\log})$ and $2\rho \leqslant r_{\mathrm{M}}$, then $|M(W_1,\dots,W_n)-W_k|_\sigma \leqslant \rho+C'_{\mathrm{M}}\rho^2$: translate by W_k^{-1} ([4, (0.6)]), so that the inputs $A_j=(1/i)\log(W_jW_k^{-1})$ have norm at most $\rho+C_{\log}\rho^2 \leqslant 2\rho$; the phase $H=\mathfrak{m}_n(A)$ is Hermitian ([4, (0.9)], the W_j being unitary), so $|e^{iH}-\mathbf{1}|_\sigma \leqslant |H|_\sigma \leqslant \rho+C_{\log}\rho^2+4C_{\mathrm{M}}\rho^2$. Fix $\eta_0>0$ with $C'_{\mathrm{M}}K_L\eta_0 \leqslant 1$, $2K_L\eta_0 \leqslant r_{\mathrm{M}}$, $K_L\eta_0 \leqslant 1$, $C_{\log}K_L\eta_0 \leqslant 1$ (so $\rho=K_L\eta_0 \leqslant 1/\max(1,C_{\log})$) and $2K_L\eta_0 \leqslant r_G/\sqrt{N}$, so that every $W \in G$ with $|W-\mathbf{1}|_\sigma \leqslant 2K_L\eta_0$ lies in the chart domain: $\|W-\mathbf{1}\|_{\mathrm{op}} \leqslant \sqrt{N}|W-\mathbf{1}|_\sigma \leqslant r_G$, and r_G is the radius of Appendix B with this property (the eigenvalue bound alone does not place W in the chart domain of a general compact G, the preamble) (every logarithm above is taken at such a W: the loops have $|W-\mathbf{1}|_\sigma \leqslant K_L\eta_0$, the translated means $|W_jW_k^{-1}-\mathbf{1}|_\sigma \leqslant \rho \leqslant K_L\eta_0$); for $\eta \leqslant \eta_0$ and $\rho=K_L\eta$ this gives $\rho+C'_{\mathrm{M}}\rho^2 \leqslant 2K_L\eta$. The constants K_L, r_{M}, C_{M}, $C_{\log}$, η_0 depend on L, the group and the norm only.

Lemma D.10 (Block Comparison)**.** *Let F be a unitary field on the bonds of two blocks $B(c_-)$, $B(c_+)$ joined by the coarse bond c (or of one block), with $|F(\partial p)-\mathbf{1}|_\sigma \leqslant \eta \leqslant \eta_0$ for every plaquette with its four vertices in $B(c_-) \cup B(c_+)$ and $|F(y,x)-\mathbf{1}|_\sigma \leqslant \eta$ for its averaged contour variables $F(y,x)=M(\{F(\Gamma)\}_{\Gamma \in \mathbf{G}(y,x)})$ ([4, (0.11)]), $y \in \{c_-,c_+\}$. Then, with $C_{\mathrm{G}}=9K_L+2$ and $M(F,c)$ the block average of [4, (0.4)],*

$$\begin{aligned} |F(b)-\mathbf{1}|_\sigma &\leqslant C_{\mathrm{G}}\eta \quad \text{for a block-internal bond } b, \\ |F(b)-M(F,c)|_\sigma &\leqslant C_{\mathrm{G}}\eta \quad \text{for an interface bond } b \in B(c)\,. \end{aligned} \tag{D.27}$$

Proof. Two paths of $\mathbf{G}(y,x)$ have transports within $K_L\eta$ of each other, so the group-mean estimate with $\rho=K_L\eta$ puts each $F(\Gamma)$, $\Gamma \in \mathbf{G}(y,x)$, within $2K_L\eta$ of $F(y,x)$, hence within $(2K_L+1)\eta$ of the identity. For an internal bond $b=(x,x')$ the loop $\Gamma_x \cup b \cup (-\Gamma_{x'})$ gives $|F(\Gamma_x)F(b)F(\Gamma_{x'})^{-1}-\mathbf{1}|_\sigma \leqslant K_L\eta$, and replacing $F(\Gamma_x)$, $F(\Gamma_{x'})$ by the identity costs $2(2K_L+1)\eta$, so $|F(b)-\mathbf{1}|_\sigma \leqslant (5K_L+2)\eta$. For an interface bond $b=(x,x')$ the loop $\Gamma_x \cup b \cup (-\Gamma_{x'}) \cup (-c)$ gives in the same way $|F(b)-F(c)|_\sigma \leqslant (5K_L+2)\eta$, $F(c)$ the transport

along the straight centre-to-centre path. By [4, (0.4)], $M(F,c) = e^{iH}F(c)$ with H the average, with weights summing to one (the weight L^{-4} of each site $x \in B(c_-)$ and the path weights of (0.4)), of the logarithms $(1/i)\log F(\Gamma \cup [x,x'] \cup (-\Gamma') \cup (-c))$ over the sites $x \in B(c_-)$ and the path pairs $\Gamma \in \mathbf{G}(c_-,x)$, $\Gamma' \in \mathbf{G}(c_+,x')$, $[x,x']$ the transported coarse bond; each of these loops belongs to the family defining K_L, so each logarithm has norm at most $K_L\eta + C_{\log}(K_L\eta)^2 \leqslant 2K_L\eta$ and is Hermitian, so $|M(F,c) - F(c)|_\sigma \leqslant |H|_\sigma \leqslant 2K_L\eta \leqslant 4K_L\eta$. Together $|F(b) - M(F,c)|_\sigma \leqslant (9K_L + 2)\eta$. *q.e.d.*

Lemma D.11 (Witnesses for Bad Tests)**.** *Let $k_N = \sqrt{N}$ (so $|\bullet|_\sigma$ and the norms of [8] and [7] are within factors k_N of each other: each of these is the operator norm or the Hilbert–Schmidt norm in one of its two normalisations, and $h \leqslant |\bullet|_\sigma \leqslant \|\bullet\|_{\mathrm{op}} \leqslant \sqrt{N}\,h$ with the unnormalised Hilbert–Schmidt norm equal to $\sqrt{N}\,h$, the conventions of Section 3, so any two of the three norms are within a factor $\sqrt{N}$ of each other), let $\alpha = 2k_N^2L^2\theta$ with θ of* (D.26)*, and let g_0 be small enough that $LM_2R_1 > 4LM_1 + 4L + 8$, $\theta \leqslant \eta_0$, $2B_3k_N^3L^2\theta \leqslant \eta_0$, $k_N\theta \leqslant c_2'$ and $C_0L^2k_N\theta \leqslant 1$ (the constants of [8, Proposition 1, (51)]) and $\alpha \leqslant a_1$ (the constant of [7, Theorem 1]; then $B_3\alpha \leqslant B_3a_1 \leqslant a_0$); every condition is an upper bound on θ or a lower bound on R_1, and all hold for small g_0 since $\theta \leqslant \varepsilon_0 \to 0$ and $R_1 \to \infty$ ([3, (2.4)–(2.8)], the definition of R_1). For a cube Q let $N(Q)$ be the set of fine sites within distance 6ℓ of Q. On the support of the averaging constraint $\delta(\bar{U}V^{-1})$ of [3, (1.6), (1.11)], for the gauge-fixed field $\widetilde{U}$ of that integrand: (a) a P_0 test is good whenever every plaquette of its defining enlargement has $|U(\partial p) - \mathbf{1}|_\sigma < \theta$; (b) if every plaquette $p \subseteq N(Q)$ has $|U(\partial p) - \mathbf{1}|_\sigma < \theta$, where Q is any cube of the partition of [3] into $L^2M_2R_1$-cubes (in particular any cube carrying the test (1.4); the position of Q is not used), the P_1 test of Q is good, and $|U_{1,Q}(\partial p) - \mathbf{1}|_\sigma \leqslant 2B_3k_N^3\theta \leqslant \varepsilon_1L^{-2}/2$ on $Q_w^{[1]}$ and on $Q_w^{[2]}$, its incident bonds and their block stencils (for a bond b, its block stencil is the block or the pair of adjacent blocks containing its endpoints, with the plaquettes having their four vertices in their union and the averaged contour variables of their centres; these are the sets to which Lemma D.10 is applied in (c)); (c) if moreover every active averaged contour variable $w_{yx} = \widetilde{U}(y,x)$ with $x \in N(Q)$ has $|w_{yx} - \mathbf{1}|_\sigma < \theta$, then, for a Q_1- or R_1-eligible cube Q,*

$$\sup_{b \in (Q_w^{[2]})^*} |\widetilde{U}(b)U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_Q\theta \leqslant \frac{\delta_0}{2} < 2\delta_0\,, \tag{D.28}$$

so the Q_1 test of Q is good. The R_1 test is treated in the proof of Lemma D.12. The hypotheses enter only through the stated upper bounds, so the lemma holds verbatim with θ replaced by any $\theta' \leqslant \theta$ and $C_Q\theta$ by $C_Q\theta'$ in (D.28)*.*

Proof. (a) is $\theta \leqslant \varepsilon_0$. Collar payments: $w \leqslant a = L^2M_2R_1$ and $M \geqslant 8LM_2$ give $4w + 2L + 4 \leqslant 4a + 2L + 4 < \ell$, and $w \geqslant LM_2R_1 > 4LM_1 + 4L + 8$ gives $2w - 2LM_1 > 2L + 4$. The $\sim$ operations on P_0', P_1' and $B(P_1^1)^c$ are by LMR_1-cubes ([3, pp. 246 and 248]), so the first inequality places $Q_w^{[4]}$ and the blocks meeting it inside $B(P_1^1)$ for a Q_1-eligible cube $Q \subseteq B(P_1^1)^{\sim -1}$ ([3, (1.8)]) and, by [3, (1.10)] ($\Omega_1 \subseteq B(P_1^1)^{\sim -3}$), for an R_1-eligible cube $Q \subseteq \Omega_1^{\sim -1}$ as well; the second places $Q_w^{[2]}$, its incident bonds and their block stencils at distance more than $2L + 4$ inside $Q_w^{[4]}$ eroded by $2LM_1$. All these sets lie in $N(Q)$, since they are within $4w + 2L + 4 \leqslant 4a + 2L + 4 < \ell < 6\ell$ of Q. The placement inside $B(P_1^1)$ is used in (c) only; (b) uses the hypothesis on $N(Q)$ alone.

(b) The block average is gauge covariant, $M(U^v,c) = v(c_-)M(U,c)v(c_+)^{-1}$ for a gauge transformation v: in [4, (0.4)] every loop is based at c_-, so its holonomy is conjugated by $v(c_-)$, the chart logarithm is conjugation covariant, and $U(c)$ is a transport from c_- to c_+. So the coarse plaquette deviations $|\bar{U}(\partial p') - \mathbf{1}|_\sigma$ of $\bar{U} = M(U,\bullet)$ coincide for U and its gauge transform $\widetilde{U}$, as do the fine plaquette deviations, and the plaquette hypothesis on $N(Q)$ may

be read for either field. [8, Proposition 1, (51)], in the averaging norm (at most $k_N|\bullet|_\sigma$) with $\alpha_0 = k_N\theta \leqslant c_2'$ gives $|\bar U(\partial p') - \mathbf{1}| \leqslant L^2 k_N\theta + C_0(L^2k_N\theta)^2 \leqslant 2L^2k_N\theta$ for every coarse plaquette p' whose blocks lie in $N(Q)$; a further factor k_N converts this to the norm of [7], so on the support of $\delta(\bar U V^{-1})$ the coarse data V on $Q_w^{[4]}$ satisfy the data bound of [7] with $\alpha = 2k_N^2L^2\theta$. The determining data of $U_{1,Q}$ are those of [3, (1.2)]: the minimal determining set $\mathbf{B}_1(Q_w^{[4]})$ with the data $M^\bullet(Q_1^{s*}V)$, no exterior data. The extension (1.3) has identity block-internal links and interface links $V(c)$; its fine plaquettes are the identity (two internal and two interface links across one coarse bond) or coarse plaquette holonomies (four interface links at a corner of four blocks), and its block averages are V (the loops of [4, (0.4)] are the identity on it). So the ordinary and mixed determining plaquettes of [7, (7)] obey the bound $\alpha \leqslant a_1$, and [7, Theorem 1] gives a minimal critical orbit in the class (8) with bound $B_3\alpha$. [3, p. 256, (2.12)] defines $U_{1,Q}$ as an element of the unique minimal orbit of the local action over the regular configurations U with $M_{\mathbf{B}}(U) = V$, and refers for "regular" and for the variational theory to [7]; "regular" is membership in the class [7, (6)], the space [7, (2)] with a radius ε satisfying $B_3\alpha \leqslant \varepsilon \leqslant a_0$, the setting of [7, Theorem 1] (such ε exist, as $B_3\alpha \leqslant B_3a_1 \leqslant a_0$). Theorem 1 states that the minimal orbit of the class (8) is the unique critical orbit of the class (6), for every such ε, so the choice of the radius is immaterial; a minimal configuration is critical, so the minimal orbit over the regular configurations is this orbit, $U_{1,Q}$ satisfies the bounds (8) with the parameter of [7] (written ε_1 there; not the $\varepsilon_1 = g_1p_0(g_1)$ of [3]) equal to α, and the axial gauge condition ([3, p. 247]) fixes the representative. The construction of $\mathbf{B}_1$ in [3, pp. 256–257] (after (2.12); $Q_w^{[4]}$ is a union of w-cubes, $w \in \{LM_2R_1, L^2M_2R_1\}$ a multiple of LM_1, hence a union of LM_1-cubes and of M_1-cubes, $M_1 < M_2$ being powers of L) gives that the coarse determining interior contains $(Q_w^{[4]})^{\sim -2}$ with the erosion in M_1-cubes of $T_{L^{-1}}$, of side LM_1 in fine units, so the loss is at most $2LM_1$, and by the collar payment $Q_w^{[2]}$, its incident bonds and their block stencils lie inside it. There [7, (8)] bounds the fine plaquettes of the minimum by $B_3\alpha L^{-2}$ in the norm of [7], that is by $k_NB_3\alpha L^{-2} = 2B_3k_N^3\theta \leqslant \varepsilon_1L^{-2}/2$ in $|\bullet|_\sigma$ by (D.26); on $Q_w^{[1]}$ this is the good P_1 test of [3, (1.4)].

(c) The blocks meeting $(Q_w^{[2]})^*$ are active by the collar payments. Apply Lemma D.10 to $\widetilde U$ with $\eta = \theta$ (plaquettes and averaged contour variables below θ) and to $U_{1,Q}$ with $\eta = 2B_3k_N^3L^2\theta \leqslant \eta_0$ (its plaquette deviations are at most $2B_3k_N^3\theta$ by (b), and its averaged contour variables are the identity in the axial gauge). On the support of the averaging constraint $M(\widetilde U, c) = V(c)$, and $M(U_{1,Q}, c) = V(c)$ on the coarse bonds of the determining interior (the constraint of the local problem, data (1.3)). For an internal bond compare both fields with the identity, for an interface bond both with $V(c)$: $|\widetilde U(b)U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma = |\widetilde U(b) - U_{1,Q}(b)|_\sigma \leqslant C_{\mathrm{G}}\theta(1 + 2B_3k_N^3L^2) = C_Q\theta \leqslant \delta_0/2$ by (D.26), the good Q_1 test of [3, (1.8)]. The final sentence of the statement holds because every step used θ only through the hypotheses and through the premise on the witnesses. *q.e.d.*

Lemma D.12 (A Bad Test Forces a Witness)**.** *For g_0 below a threshold depending on L, the group, M_1, M_2, M and the source constants, on the original cutoff support: every bad P_0, P_1, Q_1 or R_1 test forces a plaquette or gauge-variable witness at threshold θ within distance $K\ell$ of every bond $z \in E(B_\xi)$ whose membership it causes, with the fixed $K = 256$; and the number of plaquette and gauge-variable labels in such a neighbourhood is at most*

$$n_W = 7(2K\ell + 7)^4 \,. \tag{D.29}$$

Proof. The P_0, P_1 and Q_1 alternatives are Lemma D.11(a)–(c). For the R_1 test, Lemma D.11(c) is combined with the comparison of the two minima on the twice enlarged

carrier of every eligible cube Q, uniformly in its label, proved in Proposition D.17 below:

$$\sup_{b\in(Q_w^{[2]})^*} |U_1(b)U_{1,Q}(b)^{-1}-\mathbf{1}|_\sigma \leqslant C_{L,G}\varepsilon_1 e^{-c_{L,G}LM_2R_1} < \delta_0/10\,. \tag{D.30}$$

Of the characteristic factors of [3, (1.11)] it uses only their nonvanishing at U_1 itself, the membership $U_1 \in D^{(11)}(V,V_0)$ of (B1) ([3, p. 248], in the sense recorded there and in (H4)); it does not assume that U_1 lies in the support of the fluctuation cutoffs (1.18) or (1.25) at zero fluctuation. On the native principal chart a bad R_1 label of the cube Q forces $|g_0A(b)|_\sigma \geqslant \delta_0$ at a bond $b \in (Q_w^{[2]})^*$, the carrier of the test in [3, (1.18)]; the eigenvalue inequality $|e^{iX}-\mathbf{1}|_\sigma \geqslant (2/\pi)|X|_\sigma$, valid when every eigenvalue of X lies in $[-\pi,\pi]$ (as on the chart) in both permitted norms, and (D.30) give, for the gauge-fixed field $\widetilde{U} = e^{ig_0A}U_1$ at the free bond b ([3, (1.13)]: $U' = UU_1^{-1} = e^{iA'}$ with $A' = g_0A$ the fluctuation field; the variable tested in (1.18) is the free variable $A(b)$ of (1.15), and the fluctuation at a free bond is exactly $e^{ig_0A(b)}$, because the substitution (1.14), $A' = B - h_{U_1}\widetilde{D}(B)$ with $B = g_0\mathsf{C}_{U_1}A$, changes the phase only at the designated bonds: h_{U_1} is supported on them and $\mathsf{C}_{U_1}A$ agrees with A on the free bonds, (4.3) at the background, Section 3.2),

$$|\widetilde{U}(b)U_{1,Q}(b)^{-1}-\mathbf{1}|_\sigma \geqslant (2/\pi - 1/10)\delta_0 > \delta_0/2\,. \tag{D.31}$$

If every witness in $N(Q)$ were below θ, (D.28) (Lemma D.11(c) for the R_1-eligible cube Q; the earlier cutoffs and Ω_1 are fixed term by term in the sum (1.11), which resums P_0, P_1, Q_1 at fixed Ω_1, and $\Omega_1 \subseteq B(P_1^1)^{\sim-3}$ holds for each term; the comparison (D.30) holds whatever the R_1 label of Q) would give $|\widetilde{U}(b)U_{1,Q}(b)^{-1}-\mathbf{1}|_\sigma \leqslant C_Q\theta \leqslant \delta_0/2$, a contradiction; so the label is excluded. [3, (1.1), (1.4), (1.8), (1.18)] are sharp characteristic partitions at their printed thresholds, so these inequalities make the corresponding bad factor vanish; off-chart configurations are not added to (1.18), which is introduced after the logarithmic reconstruction (1.13)–(1.15). For the distances: before filling, [3, (1.10), (1.20)] give the bad region as $Q_1'^{\sim4} \cup (B(P_1^1)^c)^{\sim5} \cup R_1'^{\sim1}$, with $B(P_1^1)^c \subseteq P_0'^{\sim2} \cup P_1'^{\sim1}$ (the definition of P_1^1 following [3, (1.4)], as printed, gives $B(P_1^1)^c = P_0'^{\sim2}$; the reading consistent with the active-region construction of [3, (3.2)–(3.3)] gives $P_0'^{\sim2} \cup P_1'^{\sim}$; both are contained in $P_0'^{\sim2} \cup P_1'^{\sim1}$), so a point of this region is within 10ℓ of a bad test. The modification after (1.20) fills only components contained in cubes of side less than 100ℓ, so every added point is within 100ℓ of an original unfilled bad point; this also holds if small merged components are filled again, since by induction every current component contains an original bad point and each fill is performed inside a cube of side less than 100ℓ containing that component (every point of the rectangular hull of such a component is within 100ℓ of a chosen original bad point of it, and overlapping hulls or later merges preserve the presence of an original point), so there is no cumulative distance loss. [3, (2.9)] gives, for its scale R_0 (the range of its expansions; not this paper's fixed range constant R_0 of (7.28)), $R_0 \leqslant (L+1)R_1$, hence $R_0 \leqslant LR_1$ as both are powers of L (the conventions of Section 3). Including each test's witness neighbourhood and incident bonds, every $z \in E(B_\xi)$ forces a witness within $K\ell$ of z with $K = 256$: a point of the filled bad region is within 100ℓ of an original bad point, which is within 10ℓ of a bad test cube Q of side at most $a \leqslant \ell/8$, whose witness lies in $N(Q)$, within 6ℓ of Q; an incident bond adds one; in all $100\ell + 10\ell + a + 6\ell + 1 \leqslant 117\ell < 256\ell$. The count (D.29) of the plaquette and gauge-variable labels in that neighbourhood is an overcount on a small periodic torus, which is harmless. *q.e.d.*

D.7.2 The Marked Probability Retains the Physical Normaliser

Lemma D.13 (The Marked Density and Its Reversal)**.** *Let $A_{\mathrm{e}}^{\Psi} = \Psi_\xi(A_{\mathrm{e}})$ and let $\mathscr{E}_\xi(B)$ be the pre-boundary-change exponent of [3, (1.15)], its outer prefactor excluded ((3.34)). With*

c_ξ *of* (3.48)*, pointwise on* $\mathfrak{g}^I$*,*

$$e^{c_\xi} K_{\xi;1}(a) = \mathbf{1}_{\mathcal{D}_I}(a)\,|\det D\Psi_\xi(A_{\mathrm{e}})|\,\exp \mathscr{E}_\xi\big(\Psi_\xi(a+A_{\mathrm{e}})\big)\,, \tag{D.32}$$

and the composed retained cutoff $\chi_{\xi,\mathrm{e}}$ *and* ζ_ξ *multiply both sides; integrating over* a*,*

$$\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1} = \int e^{c_\xi} K_{\xi;1}(a)\,\mathrm{d}a\,, \tag{D.33}$$

the branch constant c_ξ *and every far primitive present (*(3.49)*). The reversal of the changes of variables carried out in the proof below is a change of variables for the integral of the density times any nonnegative measurable function of* U*, of the regions and of the labels of a term, performed term by term before the characteristic partitions are summed; it is stated here for the mark* $\mathbf{1}\{z \in E(B_\xi)\}$ *and is used in that general form in Proposition 9.8 and Lemma 9.45. Together with Lemma D.12, on compatible even fine periods, for every positive bond* z*,*

$$p_z \leqslant \big(\mu_0 \otimes q_g^{\otimes \mathcal{I}} \otimes \mathrm{Haar}^{\otimes T}\big)(W_z)\,, \qquad \mu_0(\mathrm{d}U) = \mathcal{Z}_0^{-1} e^{-\mathsf{E}} e^{-S(U)/g_0^2}\,\mathrm{d}U\,, \tag{D.34}$$

where W_z *is the event that some plaquette or some auxiliary gauge variable in the deterministic* $K\ell$*-neighbourhood of* z *is a witness at threshold* θ*,* $\mathcal{I}$ *is the set of pairs* (y,x) *of a block centre* y *and a noncentral site* x *of its block,* T *the set of centres and unused sites, and*

$$q_g(\mathrm{d}w) = z_g^{-1}\mathbf{1}\{|w-\mathbf{1}|_\sigma < \varepsilon_0\} e^{-s_G(w)/g_0^2}\,\mathrm{d}w\,, \qquad s_G(w) = 1 - \mathrm{Re}\,\mathrm{tr}_N\, w\,, \tag{D.35}$$

with z_g *the integral of the displayed density, the gauge normaliser of [3].*

Proof. Ψ_ξ is the identity in every near stencil and its support S_1^* is disjoint from I, so $\Psi_\xi(a+A_{\mathrm{e}}) = a + A_{\mathrm{e}}^\Psi$ ((3.27)). Expanding the quadratic form of $\mathscr{E}_\xi$ into its II, IJ, JJ blocks, leaving every far primitive at A_{e}^Ψ and retaining the boundary determinant gives (D.32), which is the computation of Lemma 3.9(ii); the retained cutoff and ζ_ξ are factors of the integrand outside the a-integral. Together with the outer prefactor κ_ξ ((3.5)) the right side recombines into the transformed density of [3, (1.15)], boundary Jacobian included. Now reverse the changes of variables of Section 3.1 with the mark $\mathbf{1}\{z \in E(B_\xi)\}$ inserted. The reversal is performed in each term of the history sum separately, where the regions, the filling and the labels are fixed, and each change of variables is an identity between integrals of a density times a nonnegative measurable integrand (the term's partitions and cutoffs multiply the integrand and are carried along), so it holds for the mark, and equally for any nonnegative measurable function of U, the regions and the labels in its place; the partitions are summed only after the domination by a label-free event below. For the present mark this is also seen directly: it commutes with the earlier resummation because the final regions and the filling are fixed in each term. Undo Ψ_ξ, the free-variable constraint elimination, the scaling, the nonlinear constraint linearisation of (1.14), the logarithmic chart of (1.13) with its Haar density, and the translation about U_1, in this order; the retained cutoff stays composed with Ψ_ξ and ζ remains in that composition, boundary-fluctuation dependence included. The marked integral is then [3, (1.6)] with its four original partitions and the site-gauge integration, no determinant or normalisation discarded; the unmarked chain is $\mathcal{Z}_0 = e^{-\mathsf{E}} \int e^{-S(U)/g_0^2}\,\mathrm{d}U$. For each active block centre y and noncentral x let w_{yx} be its mean transport after the gauge change; [4, (0.6)] and [3, (1.5)] give the Haar substitution $v(x) = w_{yx}^{-1} h_y U(y,x)$, $v(y) = h_y$, with Jacobian one, and the law of w is (D.35). Add independent q_g variables of integral one for the inactive indices and independent Haar variables for the centres and unused sites. The earlier P_0, P_1 choices depend on U and its coarse gauge orbit, not on these auxiliary variables. By Lemma D.12 the mark is dominated by $\mathbf{1}_{W_z}$, and W_z does not depend on the later Q_1, R_1 labels; only after this

enlargement are the characteristic partitions summed to one. This gives (D.34). The product measure is a representation of the integral before the history conditions, not an independence statement about them; plaquette witness norms and the conjugation-invariant gauge-variable norms are gauge-invariant scalars under the simultaneous gauge transformations, and no open Wilson contraction is estimated. *q.e.d.*

D.7.3 One-Point Tails and Summation

Lemma D.14 (Single-Witness Tails)**.** *Let $d_G = \dim G$, $b_G = 1/(2N)$, so that $s_G(w) \geqslant b_G|w-\mathbf{1}|_\sigma^2$ ($s_G(w) = 1 - \operatorname{Re}\operatorname{tr}_N w$ with the normalised trace $\operatorname{tr}_N = N^{-1}\operatorname{Tr}$,* (D.35)*; the action is $S(U) = \sum_p s_G(U(\partial p))$ over the unoriented plaquettes,* (1.2)*, so that $\mathcal{Z}_0 = \int \rho_0 \,\mathrm{d}U = e^{-\mathsf{E}} \int e^{-S/g_0^2}\,\mathrm{d}U$ by* (1.7) *with $\mathsf{E} = E_\varepsilon$ the vacuum scalar of* (1.2)*; $\mathrm{d}U$, $\mathrm{d}w$ are normalised Haar measures, and μ_0 and q_g are probability measures by their normalisations $\mathcal{Z}_0$ and z_g* (D.35)*). There are constants c_H, $g_H > 0$ depending on the group only such that the Haar measure of the operator ball of radius g about $\mathbf{1}$ is at least $c_H g^{d_G}$ for $0 < g \leqslant g_H$: in the chart, $\mathrm{d}w = \sigma(A)\,\mathrm{d}A$ with the continuous Haar density σ, $\sigma(0) = \sigma_0 > 0$ (the steps listed after* (3.7)*); choose $g_H < \varrho_G$ with $\sigma \geqslant \sigma_0/2$ on $\{\|A\|_{\mathrm{op}} \leqslant g_H\}$, a subset of the chart ball as $h \leqslant \|\bullet\|_{\mathrm{op}}$; for $g \leqslant g_H$ the map $A \mapsto e^{iA}$ sends $\{\|A\|_{\mathrm{op}} < g\}$ injectively into the operator ball of radius g ($\|e^{iA}-\mathbf{1}\|_{\mathrm{op}} \leqslant \|A\|_{\mathrm{op}}$), so that ball has Haar measure at least $(\sigma_0/2)\operatorname{Leb}\{\|A\|_{\mathrm{op}} < 1\}\, g^{d_G} =: c_H g^{d_G}$. Let $g_0 \leqslant g_H$ be small enough that the operator ball of radius g_0 lies in $\{|w-\mathbf{1}|_\sigma < \varepsilon_0\}$; this holds for small g_0, since the two norms are comparable with fixed constants and $\varepsilon_0/g_0 = A_0(\log g_0^{-2})^{p_0} \to \infty$, $A_0 > 0$ ((4.13), [3, p. 246]: $\varepsilon_0 = g_0 A_0(\log g_0^{-2})^{p_0}$). With V_T the number of fine sites,*

$$\mathcal{Z}_0 \geqslant e^{-\mathsf{E}}(c_H g_0^{d_G})^{4V_T} e^{-48V_T}\,. \tag{D.36}$$

On compatible even fine periods, for every threshold $\theta' \in (0, \varepsilon_0]$, every plaquette p and every auxiliary gauge variable w,

$$\mu_0\big(|U(\partial p)-\mathbf{1}|_\sigma \geqslant \theta'\big) \leqslant z_P(\theta') \coloneqq \exp\Big[-\frac{b_G\theta'^2}{4g_0^2} + 4d_G\log g_0^{-1} + 4\log c_H^{-1} + 48\Big]\,, \tag{D.37}$$

$$q_g\big(|w-\mathbf{1}|_\sigma \geqslant \theta'\big) \leqslant z_G(\theta') \coloneqq e^{1/2}(c_H g_0^{d_G})^{-1} e^{-b_G\theta'^2/g_0^2}\,, \tag{D.38}$$

and therefore, with $z_P \coloneqq z_P(\theta)$ and $z_G \coloneqq z_G(\theta)$ at the threshold $\theta \leqslant \varepsilon_0$ of (D.26) *and for g_0 below the threshold of Lemma D.12 (which* (D.34) *inherits through Lemma D.13), $p_z \leqslant q \coloneqq n_W \max(z_P, z_G)$ for every positive bond z.*

Proof. On the event that every fine link lies in the operator ball of radius g_0 every plaquette has $\|U(\partial p)-\mathbf{1}\|_{\mathrm{op}} \leqslant 4g_0$ (telescoping over its four links) and contributes $s_G(U(\partial p)) = \frac{1}{2N}\sum_j |e^{i\varphi_j}-1|^2 \leqslant \frac{1}{2}\|U(\partial p)-\mathbf{1}\|_{\mathrm{op}}^2 \leqslant 8g_0^2$ to S ($e^{i\varphi_j}$ the eigenvalues of $U(\partial p)$), and there are at most $6V_T$ plaquettes and $4V_T$ links, which gives (D.36). Fix $\theta' \in (0, \varepsilon_0]$. For (D.37) use reflection positivity and the chessboard estimate of μ_0 on the even fine periods N_μ. For a coordinate reflection ϑ through the planes $x_\mu = 0$, $x_\mu = N_\mu/2$ put $(\vartheta U)(x,x') = U(\vartheta x, \vartheta x')$ (inverse values on reversed bonds; $s_G(W)$ and $|W-\mathbf{1}|_\sigma$ are invariant under $W \mapsto W^{-1}$ and under conjugation, so the plaquette terms of S and the plaquette indicators below do not depend on the orientation or the base point of a plaquette) and $\Theta F = F \circ \vartheta$. Every plaquette lies in one of the two closed half-tori; write U_∂ for the links in the two planes, U_+ for the remaining links of one half, and S_+ for the sum of $s_G(U(\partial p))$ over the plaquettes of that closed half, the plaquettes lying in the planes (functions of U_∂) counted with weight $1/2$, so

that $S = S_+ + \Theta S_+$. For a bounded F of the links (U_+, U_∂), since Θ fixes U_∂ and carries the Haar measure of U_+ to that of the other half,

$$\int \overline{F}\,\Theta F\,\mathrm{d}\mu_0 = \frac{e^{-\mathsf{E}}}{\mathcal{Z}_0}\int \Big|\int F e^{-S_+/g_0^2}\,\mathrm{d}U_+\Big|^2\,\mathrm{d}U_\partial \geqslant 0\,. \tag{D.39}$$

Let $C_x = x + [0,1]^4$, x a fine site, be the closed unit cubes and, for a bounded function f of the links of C_0, let $f^{[x]}$ be its transport to C_x by successive reflections (the four coordinate reflections commute and the periods are even, so $f^{[x]}$ depends neither on the order of the directions nor on the winding of the path of reflections; and the reflection through any site plane $x_\mu = m$ carries the transport to the slab at position k in direction μ to the transport to position $2m-1-k$, since every composition of integer-plane reflections carrying position 0 to position k has orientation $(-1)^k$ in that direction and so coincides with the defining transport). For nonnegative bounded functions f_x of the links of C_0, indexed by the fine sites x,

$$\int \prod_x f_x^{[x]}\,\mathrm{d}\mu_0 \leqslant \prod_x \Big(\int \prod_y f_x^{[y]}\,\mathrm{d}\mu_0\Big)^{1/V_T}\,, \tag{D.40}$$

the chessboard estimate. It is proved here, by the argument of [13, Theorem 10.11, p. 445] with (D.39) as the reflection positivity; that theorem is stated for a spin system on the sites with reflections through planes between sites, and is not applied as stated to the link variables with the site reflections used here. In one direction, with $N_\mu = 2h$: by translation invariance of μ_0, (D.39) holds for the reflection through any pair of antipodal site planes $x_\mu = k$, $x_\mu = k+h$; let $\Phi(f_0,\dots,f_{2h-1})$ be the μ_0-expectation of the reflected product of functions of the $2h$ closed unit slabs (slab j carrying $f_j^{[j]}$, the transport of f_j to slab j); adjacent slabs share only plane links; the two planes bounding h consecutive slabs are fixed by the reflection through them and the plane links interior to these slabs belong to U_+, so the product over any h consecutive slabs is a function of (U_+, U_∂) for the reflection through the planes bounding it, and (D.39) with the Cauchy–Schwarz inequality for the sesquilinear form $(F,F') \mapsto \int \overline{F}\,\Theta F'\mathrm{d}\mu_0$ bounds Φ^2 by the product of the two values of Φ obtained by reflecting either half of the tuple onto the other. Add a positive constant to each function and divide it by the $2h$-th root of its fully disseminated expectation (the normalised functions $\widetilde{f}_j$ have $\Phi(\widetilde{f}_j,\dots,\widetilde{f}_j) = 1$); among the finitely many tuples with entries from $\{\widetilde{f}_0,\dots,\widetilde{f}_{2h-1}\}$ take a maximiser of Φ with a longest constant run (runs counted cyclically), of length r; both tuples obtained by reflecting either half onto the other are maximisers as well (the maximum is positive, every function being bounded below by its added constant, and the Cauchy–Schwarz bound, the square of the maximum being at most the product of the two reflected values, forces both to equal the maximum). If $r \leqslant h$, reflecting the half that ends with the run onto the other half gives a maximiser with a constant run of length $2r > r$, contradicting the choice of r; so $r > h$, a half lying inside the run is reflected, and the resulting maximiser is constant: the maximum is $\Phi(\widetilde{f}_j,\dots,\widetilde{f}_j) = 1$ for some j, which gives the one-dimensional inequality for the normalised functions; letting the added constants decrease to zero (dominated convergence) gives it for the f_j. Applying it successively in the four directions (the reflections commute, and the slab functions produced after disseminating in the earlier directions are again nonnegative and bounded) gives the exponent $1/V_T$. Every plaquette is a translate of a plaquette of C_0 and μ_0 is translation invariant, so it suffices to apply (D.40) with $f_0 = \mathbf{1}\{|U(\partial p) - \mathbf{1}|_\sigma \geqslant \theta'\}$ for a plaquette p of C_0 and $f_x = 1$ for $x \neq 0$: the dissemination $\{f_0^{[y]}\}_y$ contains $V_T/4$ distinct plaquettes of the plane orientation $(\mu\nu)$ of p (the two transverse coordinates of a copy range over one parity class; repeated copies of an indicator are harmless), each with $s_G \geqslant b_G\theta'^2$ on the event, and $S \geqslant 0$ termwise, so the disseminated expectation is at most $e^{-\mathsf{E}}e^{-b_G\theta'^2 V_T/(4g_0^2)}/\mathcal{Z}_0$; dividing by (D.36) and taking the

$1/V_T$-th root gives $z_P(\theta')$, a normalised probability estimate. For (D.38) restrict one gauge variable to its operator ball of radius g_0, where its action is at most $g_0^2/2$ and its cutoff factor is one (the ball lies in $\{|w-\mathbf{1}|_\sigma < \varepsilon_0\}$ by the hypothesis on g_0), which bounds z_g from below by $c_H g_0^{d_G} e^{-1/2}$; on the event $|w-\mathbf{1}|_\sigma \geqslant \theta'$ the action is at least $b_G\theta'^2$, which gives $z_G(\theta')$. Neither step uses anything about θ' beyond $0 < \theta' \leqslant \varepsilon_0$ (q_g is carried by the ball $|w-\mathbf{1}|_\sigma < \varepsilon_0$, so for larger thresholds the event in (D.38) is empty). The union bound over the at most n_W witness labels of Lemma D.12, without any independence between witnesses, and (D.34) at the threshold θ give $p_z \leqslant q$. *q.e.d.*

Proposition D.15 (The Exterior-Response and Uncovered Term)**.** *Let*

$$\pi_* = \min(p_0(g_0), p_0(g_1)) \,.$$

There are c_θ, c_*, $g_* > 0$, fixed after L, the group, the character, M_1, M_2, M and the source constants, such that $\theta \geqslant c_\theta g_0 \pi_*$ and $q \leqslant e^{-c_*\pi_*^2}$ for $0 < g_0 < g_*$, and on compatible even fine periods, for $0 < g_0 < g_*$ and under (D.2)*,*

$$|E_{\mathrm{bdry}}| \leqslant C_{\mathrm{resp},\chi}\, n_C (1 + g_0 r)\, e^{-c_*\pi_*^2} \,, \tag{D.41}$$

after replacing $C_{\mathrm{resp},\chi}$ by $C_L \max(C_{\mathrm{resp},\chi} S_c, 1)$; g_* is at most the thresholds of Lemma D.12, the g_H and the smallness condition of Lemma D.14, the threshold of Proposition D.17 and the coupling below which the comparison of [3, (2.6)] used in the proof holds. The constants are uniform in the compatible even periods, the contour location and the first-step histories; the factor n_C is retained.

Proof. [3, (2.6)–(2.9)] and $\delta_0 = (A_1/A_0)\varepsilon_0$ give $\theta \geqslant c_\theta g_0 \pi_*$ for small coupling: each of the three terms of (D.26) is at least a fixed constant times $g_0 p_0(g_0)$ or $g_1 p_0(g_1)$, and g_1/g_0 is bounded above and below by [3, (2.6)]. Since $\ell = LMR_1$ grows as a fixed power of $\log g_1^{-2}$ ((9.9)), $\log n_W = O(\log\log g_0^{-2})$; with the exponent of [3], $p_0 \geqslant 5\mathfrak{r} \geqslant 10$, $\pi_*^2 \geqslant A_0^2(\frac{1}{2}\log g_0^{-2})^{2p_0}$ dominates $4d_G \log g_0^{-1}$ and $\log n_W$ for small g_0, so $q = n_W \max(z_P, z_G) \leqslant e^{-c_*\pi_*^2}$ with $c_* = b_G c_\theta^2/8$, for $g_0 < g_*$. Then Corollary D.8, Lemma D.9 and Lemma D.14 give $|E_{\mathrm{bdry}}| \leqslant C_{\mathrm{resp},\chi} g_0 r\, C_L S_c n_C q + C_L n_C q \leqslant C_L \max(C_{\mathrm{resp},\chi} S_c, 1)\, n_C (1 + g_0 r)\, q$, and $q \leqslant e^{-c_*\pi_*^2}$ gives (D.41). *q.e.d.*

Remark D.16 (Scope). Lemma D.6 is proved from the Hessian, Schur, enlarged-chart and primitive derivative bounds of Appendix B and the collar cutoff of Section 3.4, with both hard-cutoff surface terms inside the bounded covariance; its physical caller is restricted to A_{cov}. Proposition D.15 holds on compatible even-period tori, from the single-witness tails of Lemma D.14 alone, without a component probability theorem, and uses the reconstruction and measure identities of Section 3; the marked density (D.32) is displayed without equating any older two-parameter normaliser with $I_{\xi;1}$, and the filling proximity of Lemma D.12 holds for a simultaneous or an iterated filling of small components. The R_1-label step of Lemma D.12 rests on Proposition D.17, which uses local data of [7], with the local hypotheses of the existence and uniqueness theorem [7, Theorem 1] verified where this paper's inputs allow and otherwise taken over as the application of [7] to the determining sets of [3, p. 256], uses the derivative-decay results of [7] and the identification [3, (3.17)–(3.19)] as stated, and accepts Bałaban's gauge-conversion results at their stated regular-chart hypotheses through the extension statements of [4]; of the characteristic factors of [3, (1.11)] it uses their nonvanishing at the minimum U_1 itself, the membership $U_1 \in D^{(11)}(V, V_0)$ of (B1) ([3, p. 248], in the sense recorded in (B1)(iii) and (H4)), and nothing about the fluctuation cutoffs (1.18), (1.25) at zero fluctuation. This appendix controls the exterior-response and uncovered term only:

it does not control the joint defects $R_\xi(\mathfrak{B})$ of (3.95) and does not identify the regional scalar $S^\circ_{\xi,C}$. Section 9 replaces Proposition D.15 on the labelled route only (at the original carrier the proposition stands and bounds E_{bdry} of (D.21) in Theorem 1.2(ii)): its history weight is the labelled weight of the second collar test with the retreat of Λ_1, its boundary bound is Proposition 9.39 with the label accounting (9.33)–(9.34), in which the one-point cost q becomes $q + N_{\mathrm{ret}} q_\vartheta$, the constant $C_{\mathrm{resp},\chi}$ enters its C_1, a prefactor r^{d_G} and a family-sum factor appear, and the leading coefficient is identified at the retreated carrier $D(\Lambda)$ (Section 9.9); Sections D.1–D.6, the tails of Section D.7 and Proposition D.17 are used there as stated, with the covered event A_{cov} replaced by the labelled event $A^{\mathrm{fin}}_{\mathrm{cov}}$ (Section 9.4.2(a)).

D.8 The Minimum Comparison from Local Data

This section proves the deterministic comparison (D.30) of the two minima from local data and, in Section D.8.6, the bound on the boundary field of [3, (1.21)] used in Section 3.4 and (B4) (Lemma D.18). Of the branch hypothesis (B1) of Section 3.1 it uses the membership $U_1 \in D^{(11)}(V, V_0)$ of the minimum in the domain of integration of [3, (1.11), p. 248] (by [7, Theorem 1]) with the smallness statement (i) derived from it in (B1), the averaging constraint and the source axial mean gauge, which place U_1 in the regularity class of [7] near Ω_1 and so identify it with the regular critical orbit constructed below; it does not use, and does not claim, that U_1 lies in the support of the fluctuation cutoffs (1.18) or (1.25) at zero fluctuation, which is false on R_1 (the large-fluctuation tests there exclude the configurations near U_1, Section 3.1). The sources are [7], [3], [4], and the averaging paper [8]; Bałaban's existence and uniqueness result ([7, Theorem 1]) is used at local hypotheses verified below where this paper's inputs allow, and otherwise as the application of [7] to the determining sets of [3, p. 256] (where Bałaban states that all the results of [7] are used for the averaging); his derivative-decay results ([7, Proposition 9, (172), and (190)]), the gauge transformation of [7, Proposition 2] and the identification [3, (3.17)–(3.19)] are used as stated, and his gauge-conversion results are accepted at their stated regular-chart hypotheses through the extension statements of [4]; no claim to have re-proved the averaging paper is made.

Proposition D.17 (Comparison of the Global and Local Minima)**.** *There are constants $C_{L,G}$, $c_{L,G} > 0$, depending on L, G, M_1, M_2 and the source parameters and fixed before g_0, independent of the fluctuation, its R_1 label, the exterior history and the torus, such that, on compatible even fine periods (every fine period is then at least 2ℓ, which the lifts below need), for every admissible (Ω_1, V, V_0) and every eligible cube $Q \subseteq \Omega_1^{\sim -1}$, on its twice enlarged free-bond carrier,*

$$\sup_{b \in (Q_w^{[2]})^*} |U_1(b) U_{1,Q}(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_{L,G} \varepsilon_1 e^{-c_{L,G} L M_2 R_1} < \delta_0/10 \,, \tag{D.42}$$

for g_0 below a threshold depending on L, G, M_1, M_2 and the source constants (the conditions $\alpha \leqslant a_1$, $B_3 \alpha \leqslant a_0$, the smallness of α at [7, (172) and (187)], $CL^2\varepsilon_0 \leqslant 1$, $\ell > \max(2L + 4, RM_1)$, $LM_2R_1 \geqslant 2LM_1 + C_L$ and the chart reductions of the proof; this threshold is part of g_* of Proposition D.15), the last inequality by Section D.8.5.

The fine periods are multiples of ℓ (compatibility, Section 3) and, by the hypothesis of the proposition, even (the restriction of Section D.7), and $\ell = LMR_1$ is odd (L is odd and M, R_1 are powers of L, (3.1)), so every compatible fine period is at least 2ℓ; since $w \leqslant L^2 M_2 R_1$ and $M \geqslant 8LM_2$ ((3.1)), $\mathrm{side}(Q_w^{[4]}) \leqslant 9L^2 M_2 R_1 \leqslant 9\ell/8 < 2\ell$, so this determining box and its fixed-range stencils have lifted copies before the next periodic image for small coupling. If Ω_1 is the full torus there is no exterior collar to freeze and the same argument uses the

periodic problem of [7]. The quantity in (D.42) concerns a simultaneously gauge-covariant pair of links: their relative product transforms by conjugation at the initial endpoint, so its norm is gauge invariant; it is not an estimate on an open Wilson source. The proof occupies the rest of this section.

D.8.1 Source Inputs and the Local-Domain Verification

U_1 is the critical minimum of [3, (1.12)], the unique critical point in the domain of integration in (1.11), supplied by [7, Theorem 1] with the prescribed V, V_0 and the source axial mean gauge ([3, p. 248], recorded in (B1) as $U_1 \in D^{(11)}(V, V_0)$). This membership gives, by (B1)(i), $|U_1(\partial p) - \mathbf{1}| < \varepsilon_0$ for every plaquette $p \subset \Omega_1^{\sim}$ and the axial condition $|U_1(y, x) - \mathbf{1}| < \varepsilon_0$ at the block centres of $\Omega_1^{\sim(1)}$, so that U_1 lies in the regularity class [7, (6)] with radius $C_L\varepsilon_0$ on the outer domain Ω_0 of [7] below, by the same conjugation and counting argument as for $\widehat{U}$ in the local-domain verification ([7] counts as plaquettes of a region those with a corner in it, so Ω_0 is taken to be the enlarged domain Ω^+ shrunk by $\ell/2$, a union of M_1-blocks every plaquette with a corner in which lies in $\Omega_1^{\sim}$; the remaining separation $\ell/2$ from Ω_1 keeps the spacing condition [7, (1)]); the localisation below verifies the hypotheses of [7, Theorem 1] for the localised problem near Ω_1 and does not involve a distant large-field exterior. The plaquette variables of U_1 on $\Omega_1^{\sim}$ are thus within $\varepsilon_0 \leqslant (1 + \beta_0)^2\varepsilon_1$ of the identity ([3, (2.8)]). The local minimum $U_{1,Q}$ is [3, (1.2)] with determining set $\mathfrak{B}_Q = \mathbf{B}_1(Q_w^{[4]})$ and reference data $D_Q^0 = \dot{M}(Q_1^{s*}V)|_{\mathfrak{B}_Q}$; it has the corresponding mixed-scale regularity of [7], with the scale-zero boundary bound described below, and the same coarse averages V on the interior determining region (the interior coarse labels of $Q_w^{[4]}$ lie in Ω_1, where the constraint $\mathsf{M}(U_1) = V$ holds: $Q \subseteq \Omega_1^{\sim -1}$ is at ℓ^∞ distance at least ℓ from Ω_1^c, and $4w \leqslant 4L^2M_2R_1 \leqslant \ell/2$ by $M \geqslant 8LM_2$, (3.1); [3, p. 250] states this containment). These are the hypotheses of the derivation [3, (3.17)–(3.19)], whose only field is the minimum, not the integrated fluctuation; at $k = 0$ the fine spacing is $\eta = 1$ and $V^{(0)} = U_1$, $V_Q^{(0)} = U_{1,Q}$, and no preceding effective interaction is needed.

The local-domain verification. Work before the P_0, P_1, Q_1 resummation, at any supported history contributing to the fixed Ω_1, V, V_0, with fine field $\widehat{U}$. Take the enlarged fine domain $\Omega^+ = \Omega_1^{\sim 1}$ (the macro enlargement of [3, (1.10)]) and fix $\widehat{U} = V_0$ on the collar $\Omega^+ \setminus \Omega_1$. Outside Ω^+ retain the original field but omit its action terms from the variational problem: once $\ell > 2L + 4$, every plaquette or block-gauge term containing an unfrozen variable lies in Ω^+, so the omitted terms are constants, and the constrained first variation is exactly that of [3, (1.12)], with no change of boundary condition or integration law (the gauge functional G of (1.12) has zero first variation at U_1, each of its nonnegative summands vanishing there since $U_1(y, x) = \mathbf{1}$ in the source axial mean gauge, [3, p. 248], so the first variations of $A+G$ and of A agree on the constraint tangent space at U_1). [3, (1.10)] and the following sentence separate Ω_1 from the earlier bad regions by at least two macro layers, so every plaquette and averaging stencil in Ω^+ lies in the P_0-good region, where at this unresummed stage $|\widehat{U}(\partial p) - \mathbf{1}|_\sigma < \varepsilon_0$. If the canonical outer operation has already omitted these factors, they equal one on the support of the retained original-field factors ((B3)(i), the form in which the removal statement of [3, p. 248] before (1.11) is recorded), so restoring them changes no integral; since the history summands are nonnegative, a point of nonzero resummed density has at least one supported unresummed history. This restoration concerns $\widehat{U}$, not the minimum U_1, and controls the prescribed collar V_0; the distant exterior is never required to be regular. Conjugation and the bounded number of incident plaquettes give $|D^*_{\widehat{U}}\partial\widehat{U}| \leqslant C_{\mathrm{inc}}\varepsilon_0$ on the collar and the variable region (C_{inc} the incidence count, not the Gaussian constant C_{d}), and multiplying by the

largest of the fixed scale weights L^2, L^3 puts $\widehat{U}$ in the class [7, (6)] with radius $C_L\varepsilon_0$. Its fixed data are exactly V_0 on the fine collar and V on the interior coarse labels, since the original averaging constraint is retained.

Use the positive-bond carriers $E(X)$. The fixed fine collar carrier is $E(\Omega^+) \setminus E(\Omega_1)$, so no crossing bond that is a variable of [3, (1.12)] is fixed. The hierarchy for [7] has outer fine domain Ω^+, inner block domain Ω_1 and determining set this bond-set difference at scale zero together with Ω_1 at scale one, with the mixed-bond convention of [7, (7)]; its spacing condition (1) holds because the separating macro layer has width ℓ, exceeding the fixed RM_1 of [7] for small coupling. For a mixed determining plaquette p', replace every coarse bond of p' outside the scale-one determining set by the block average of the scale-zero data, as prescribed in [7, (7)]: the original constraint makes the resulting loop an ordinary fine plaquette of $\widehat{U}$ or a coarse plaquette of its block average, and [8, Proposition 1, (51)], bounds the latter, in the operator norm of [8], by $L^2\alpha_0 + C(L^2\alpha_0)^2 \leqslant 2L^2\alpha_0$ with $\alpha_0 = \sqrt{N}\varepsilon_0$ for g_0 small ($CL^2\alpha_0 \leqslant 1$; the conversion $\|\bullet\|_{\mathrm{op}} \leqslant \sqrt{N}h$ of the conventions covers the case of the normalised Hilbert–Schmidt cutoff norm), which is also the explicit bound of [3] before (1.2); the same $O_L(\varepsilon_0)$ bound follows from the elementary two-block comparison of Lemma D.10. The constraint [7, (3)] and [8, Proposition 1] are stated for Bałaban's earlier averaging; their use with the averaging [4, (0.4)] of [3, (1.12)] is Bałaban's own transfer in [3, p. 256] (all the results of [7] used for its averaging) and the extension statement of [4] after (0.12), accepted as such. So the entire determining set of the localised problem has data regularity in the sense of [7] at most $C_L\varepsilon_0$ (C_L including $\sqrt{N}$), and no regularity of the rest of Ω_1^c is involved.

[7, Theorem 1] therefore constructs the regular critical orbit for these data, at radius $B_3C_L\varepsilon_0$, unique in the regularity class (6) of radius $\varepsilon_0^{\mathrm{V}} := \max(C_L\varepsilon_0, B_3C_L\varepsilon_0) \leqslant a_0$ for small g_0, a class containing U_1, with both the plaquette and the divergence bounds (2). The minimum U_1 of [3, (1.12)] lies on this orbit: it is critical for the localised problem, since the localisation changes the action by constants only, for every admissible variation, and it lies in the regularity class of [7] on Ω^+ by its membership $U_1 \in D^{(11)}(V, V_0)$ (the first paragraph of this subsection); the uniqueness in the class identifies the two. Restricted to $Q_w^{[4]}$ the same bound gives membership in the local uniqueness class, with fixed scale and norm conversions (that the results of [7] hold for the determining set $\mathbf{B}_1(Q_w^{[4]})$ of [3, (1.2)], whose fine layer has width between LM_1 and $2LM_1$, p.256, in place of the spacing condition [7, (1)] for that layer, is Bałaban's application of [7] to the determining sets of [3], p.256, taken over as such); since $\varepsilon_0 \leqslant (1+\beta_0)^2\varepsilon_1$ by [3, (2.8), p. 256] (with $n = 1$, $m = 0$), the radius is $C_{L,G}\varepsilon_1$, below the a_0 of [7] for small coupling. The fine boundary layer of $U_{1,Q}$ has the scale-zero bound of [7], not an $\varepsilon_1 L^{-2}$ bound: the plaquettes of its straight-extension data are trivial or equal to coarse plaquette holonomies of V, bounded by $2L^2\varepsilon_0$, so its mixed-scale data are regular with the same parameter $C_{L,G}\varepsilon_1$. What is used of the characteristic factors of [3, (1.11)] at U_1 is their nonvanishing at U_1 itself, the statement $U_1 \in D^{(11)}(V, V_0)$ of [3, p. 248] ((B1)); nothing is assumed about the fluctuation cutoffs (1.18), (1.25) at zero fluctuation, and no branch at $A = 0$ is introduced.

D.8.2 The Boundary Data and Their Size

Let $D_Q = \dot{M}(U_1)|_{\mathfrak{B}_Q}$ and define on the determining set

$$B_Q(e) = \tfrac{1}{i}\log\big(D_Q(e)D_Q^0(e)^{-1}\big)\ . \tag{D.43}$$

Both data are defined by the source averaging operation on this regular configuration; the determining interior carries scale-one coarse bonds and its boundary layer the finer deter-

mining variables, as defined after [3, (2.12)]; the mixed determining plaquettes and the local averages are those of [3, (1.2), (2.12)–(2.16)]. For the straight extension of [3, (1.3)] every intrablock contour is the identity, every centre-to-centre path crossing a specified interface has transport $V(c)$, and every relative closed holonomy of [4, (0.4)] is the identity, so the logarithmic correction vanishes and $\mathsf{M}(Q_1^{s*}V)(c) = V(c)$; on the interior both data equal V, hence $B_Q = 0$ there, and $\operatorname{supp} B_Q$ lies in the boundary layer of width $2LM_1$ of $Q_w^{[4]}$, the support statement preceding [3, (3.18)]. The principal logarithm is legitimate because the reasoning after [3, (3.17)] bounds the argument of the logarithm by $C_{L,G}\varepsilon_1$ (the printed coefficient $44d^2B_3$ of [3] belongs to its own regularity coefficient; the enlarged local coefficient is kept inside $C_{L,G}$). At one step this can also be read from the two-block comparison: in the axial mean gauge a regular minimum is within $C_{L,G}\varepsilon_1$ of the identity on internal bonds and of $V(c)$ on interface bonds, which are exactly the values of $Q_1^{s*}V$; the bound follows from the regularity of the minimum and the equality of the coarse averages, not from a fluctuation cutoff. Thus

$$\|B_Q\|_\infty \leqslant C_{L,G}\varepsilon_1 \,. \tag{D.44}$$

To meet [7, (172)] with its absolute constant C_1, take its small parameter $\alpha = K_\alpha\varepsilon_0$ with K_α dominating all determining-data, uniqueness-class and B_Q bounds (for the last, $\varepsilon_1 \leqslant (1+\beta_0)\varepsilon_0$ by [3, (2.8)]); then $|e^{itB_Q} - \mathbf{1}| \leqslant |tB_Q| \leqslant \alpha$ and $C_1 = 1$ suffices. [7, Proposition 9 and (190)] are applied with this α, not with the identically named ε_1 of [3]; for small g_0, $\alpha \leqslant a_1$, $B_3\alpha \leqslant a_0$, and α lies below the thresholds of [7] "for ε_1 sufficiently small" at (172) and (187) (pp.305 and 308); and $\varepsilon_0 \leqslant (1+\beta_0)^2\varepsilon_1$ ([3, (2.8)]) converts the resulting $C\alpha$ bounds into $C'\varepsilon_1$; this uses the printed absolute-C_1 statement of [7] without extending it. No bound is imposed on the exterior integration variables away from this local regular domain.

D.8.3 Integrating the Local Derivative Estimate

Let $\mathscr{H}_Q(B)$ be the relative Landau-gauge logarithm of the local minimal-configuration map of [7, Proposition 9] on $\mathfrak{B}_Q$ with reference $U_{1,Q}$, normalised by $\mathscr{H}_Q(0) = 0$; this is the map of [3, (3.17)], with its L^{-1} scale factor at $k = 0$. [3, (3.17)] identifies the map at B_Q with the restriction of U_1 up to the source gauge restoration: both lie on the unique regular critical orbit with determining data D_Q, the former satisfying the constraint up to the gauge transformation of [7, Proposition 2]. For the criticality, a local admissible variation fixes the determining fine layer and the averages on the interior coarse rows; extended by zero outside $Q_w^{[4]}$ it preserves every global average and exterior value in [3, (1.12)], because the fixed layer is wider than the plaquette and averaging range. The first variation of the global action vanishes and its local part is the local action variation; the gauge functional of the source axial mean gauge, in which the variational problem [3, (1.12)] is posed (as recalled in the local-domain verification of Section D.8.1), has zero first variation at an axial mean configuration, where each of its nonnegative summands attains zero. So the restriction of U_1 is critical for the local action, it lies in the uniqueness class of Section D.8.1 by its regularity there (from $U_1 \in D^{(11)}(V, V_0)$), and the uniqueness identifies its orbit with the local analytic solution. The identity concerns the minimum U_1 and its regularity; no membership of the integrated fine field in a cutoff event is involved.

[7, (190), Proposition 9], gives the functional derivative with the mixed-scale weights; at one step the scale labels are 0 and 1, so the powers of L in those weights are fixed constants, and a determining bond e of the outer layer and an interior bond b satisfy the bound $C_{L,G}e^{-c_{L,G}d(b,e)}$, the unit-density measure at the finer level and the scale-one determining measure contributing fixed L-dependent factors. Here $d(b,e)$ is the ℓ^∞ distance between the initial vertices of b and e on the fine lattice. The distance in [7, (190)] is that of [6, (2.46)] ([7, p. 288]), which

charges one unit for each fine bond of a contour in the scale-zero region and one unit for each coarse bond, of fine length L, in the scale-one region, and is evaluated at the block points of the two endpoints; so it is at least $L^{-1}(d(b,e)-L)$, and the rate $\delta_{\mathrm{P}}/8$ of (190), with δ_{P} the decay rate of [10, p. 423] ((H4)), gives $c_{L,G}=\delta_{\mathrm{P}}/(8L)$, the factor $e^{\delta_{\mathrm{P}}/8}$ entering $C_{L,G}$. Integrating along tB_Q, $t\in[0,1]$, and summing the labels,

$$|\mathscr{H}_Q(B_Q;b)|_\sigma \leqslant C_{L,G}\varepsilon_1 \sum_{e\in\operatorname{supp} B_Q} e^{-c_{L,G}d(b,e)} \leqslant C'_{L,G}\varepsilon_1 e^{-c'_{L,G}d(b,\operatorname{supp} B_Q)}\,, \tag{D.45}$$

where for the second inequality each exponential is split into two halves, $e^{-c_{L,G}d(b,e)} \leqslant e^{-c_{L,G}d(b,\operatorname{supp} B_Q)/2}e^{-c_{L,G}d(b,e)/2}$, and the remaining sum over e is bounded by the infinite-lattice sum $\sum_e e^{-c_{L,G}d(b,e)/2}$ over the bonds of $\mathbb{Z}^4$ (four per site) through injective nearest lifts, a constant depending on $c_{L,G}$ only; so $c'_{L,G}=c_{L,G}/2$ and no volume-dependent constant enters. For a bond $b\in(Q_w^{[2]})^*$ (its initial vertex is within one unit of $Q_w^{[2]}$, a difference absorbed in C_L),

$$d(b,\operatorname{supp} B_Q) \geqslant 2w - 2LM_1 - C_L \geqslant LM_2R_1 \tag{D.46}$$

once $w\geqslant LM_2R_1$ (the collar-width convention of Section D.7) and $LM_2R_1\geqslant 2LM_1+C_L$, which holds for small g_0 since $R_1\to\infty$ (Section B.8); the first inequality is the distance from $Q_w^{[2]}$ to the complement of $Q_w^{[4]}$ less the width $2LM_1$ of the boundary layer carrying $\operatorname{supp} B_Q$ and the bond offsets. This covers either candidate common width w and never uses a good/bad R_1 label.

D.8.4 Restoring the Source Gauge

The transformation of [7, Proposition 2] is not centre-fixing: its condition is $(\overline{R_0u})^j=\mathbf{1}$ on Λ_j, in particular $u=\mathbf{1}$ on the fine determining layer ([7, p. 281]), and at one step its interior condition $\overline{R_0u}(y)=\mathbf{1}$ does not imply $u(y)=\mathbf{1}$. Keep the restoring transformation u_Q of [3, (3.17)], defined by this source normalisation: the configuration $e^{iL^{-1}\mathscr{H}_Q(B_Q)}U_{1,Q}$ of Proposition 9 is in the relative gauge of Proposition 2 and satisfies the constraint with data D_Q up to that gauge, so it lies on the orbit of the restriction of U_1, and u_Q is the transformation carrying it to U_1, whose values at the block centres need not be the identity and are retained. At the first step, with the $\eta=1$ of [3] and its convention (1.17), on the required bonds

$$U_1(b) = u_Q(b_-)^{-1}e^{iL^{-1}\mathscr{H}_Q(B_Q;b)}U_{1,Q}(b)\,u_Q(b_+)\,. \tag{D.47}$$

The local inputs of the transferred source estimates are the uniqueness and regularity class of Section D.8.1, the small supported B_Q of Section D.8.2 and the response bound (D.45) on both endpoint blocks and the crossing bond. Fix an analytic mean neighbourhood radius $\rho_{\mathrm{av}}>0$ for the finite source contour families, with derivative bounds on its closed half-radius neighbourhood, and choose g_0 after it so that the reference path-family diameter and its perturbation are both less than $\rho_{\mathrm{av}}/4$; this is the regular-chart reduction. The block-local gauge inequality (D.48) below is the transferred source result; the following path description explains its one-step support and its constants, and does not assert the single-contour formula of [8] unchanged for the modified mean. In [8] the equations determining u_Q are (77)–(87), with its expression in the relative fields given in (106)–(108): at one step the product in (106) has a single $j=0$ factor, a reference-unitary conjugation of the inverse block average followed by a local relative path transport, and the centre factor (87) is retained. Unitary conjugations preserve the norm, each path has length at most $(2d+1)L$, so the telescoping product identity bounds the relative deviation by C_L times the largest relative-link deviation in the block, and

the analytic averaging bound of (161)–(163) bounds the block-average deviation by the same quantity after reducing the chart. The transferred result reads

$$\sup_{x\in\mathcal{B}_y}|u_Q(x)-\mathbf{1}|_\sigma \leqslant C_{L,G}\sup_{b\in E(\mathcal{B}_y)}|\mathscr{H}_Q(B_Q;b)|_\sigma\,, \tag{D.48}$$

the factor L^{-1} of [3, (3.17)] absorbed into the constant. The restoring map is relative to the reference pair: under a simultaneous gauge change h of field and reference its representative is $u_Q^h(x)=h(x)u_Q(x)h(x)^{-1}$ (conjugate the relative action, [7, (16)–(17)], and its reference-dependent gauge condition), so $|u_Q(x)-\mathbf{1}|_\sigma$ is a conjugation-invariant scalar of the relative construction, used only inside this gauge restoration; the centre contribution $u_Q(y)$ is retained. [8] uses its older averaging definition; the application here rests on Bałaban's extension statements, [4] after (0.10) (the considerations and results of [4], and those of the earlier papers, hold for every group mean with the properties (0.5)–(0.9)) and [4] after (0.12) (the average (0.12) has properties similar to those of (0.4), all results of [8] hold for it, with proofs mostly unchanged and otherwise with minor modifications, and both definitions are equally good for the purposes of [4]; the coverage of (0.4) rests on this last remark). These are blanket transfer statements, none of which displays a restoring bound, so (D.48) is an accepted transfer of the results of [8] to the averaging and mean gauge of [4], used as that source result and not by asserting that the older individual-contour identities stay unchanged; [3, (3.17)–(3.19)] uses the transferred results in the same way for the later scheme, as does [4] for its gauge conversions ((3.26)–(3.27) with [8, (77)], and (3.39) with [8, (97)], retaining the centre-block gauge factors; reference [12] of [4] is [8]). No change of the physical averaging operation is made.

For the bond b of (D.42) apply (D.45) on both endpoint blocks, on b itself and on the paths entering the gauge factors, which enlarges the input set by at most $2L+2$, already included in the C_L of (D.46). The relative exponential and both endpoint factors of (D.47) then give $|U_1(b)U_{1,Q}(b)^{-1}-\mathbf{1}|_\sigma \leqslant C_{L,G}\varepsilon_1 e^{-c_{L,G}LM_2R_1}$; both endpoint gauge factors of [3, (3.18)] are retained, and the quantity is the simultaneous-gauge-invariant norm of the relative pair of links, not a comparison modulo an unspecified gauge transformation.

D.8.5 The Caller and the Denominator

[3, (2.6), (2.8)] bound ε_1/δ_0 by fixed source constants and R_1 grows as a power of $\log g_1^{-2}$, so the bound is $<\delta_0/10$ for small g_0, which completes the proof of Proposition D.17. In Lemma D.12 the same norm $|\bullet|_\sigma$ is used for the test of [3, (1.18)], the comparison and the no-witness bound (the norm convention fixed before that lemma), and the eigenvalue inequality $|e^{iX}-\mathbf{1}|_\sigma \geqslant (2/\pi)|X|_\sigma$, valid when every eigenvalue of X lies in $[-\pi,\pi]$, as on the native chart (the eigenvalues of a chart element lie in $(-\pi,\pi)$, the conventions of Section 3), holds in both permitted norms without a norm conversion; the replay ($|g_0A(b)|_\sigma \geqslant \delta_0$ on a bad R_1 test, (D.31), and the contradiction with the no-witness two-block bound $C_Q\theta \leqslant \delta_0/2$) is carried out in the proof of that lemma. The physical integral is never altered: for each original history the comparison is used inside its cutoff-support implication, the numerator of (D.23) remains $\int \mathcal{O}[\mathsf{w}_\xi \mathbf{1}\{z\in E(B_\xi)\}]\,\mathrm{d}V$ with $\mathsf{w}_\xi = e^{c_\xi}I_{\xi;1}$ and its denominator $\mathcal{Z}_0$; no ζ factor is removed, evaluated at U_1 or divided out, and the covariance estimate of Section D.5 is unchanged. Lemmas D.13 and D.14 and Proposition D.15 then apply with their existing hypotheses and even-period scope.

D.8.6 The Boundary Field of Bałaban's Equation (1.21)

The same argument bounds the field $\mathbf{H}_{1,Ax}$ of [3, (1.21)] on the bonds of S_1^*, where it enters the change of variables (1.22) through Lemma 3.8 and (3.40).

Lemma D.18 (The Boundary Field on S_1)**.** *There are constants C_{H}, $c_{\mathrm{H}} > 0$, depending on L, G, M_1, M_2, and the source constants and fixed before g_0, independent of the fluctuation, its R_1 label, the exterior history, and the torus, such that, for g_0 below a threshold depending on the same quantities (the conditions of Sections D.8.1, D.8.2, and D.8.4 and those named in the proof), for every admissible (Ω_1, V, V_0), every large-fluctuation region R_1 of the history sum, and every $b \in S_1^*$, with $U_1' := U_1(\Omega_1 \cap \Lambda_1^c, V)$ the representative fixed in (B1),*

$$|U_1(b)U_1'(b)^{-1} - \mathbf{1}|_\sigma \leqslant C_{\mathrm{H}}\varepsilon_1 e^{-c_{\mathrm{H}}LMR_1} \,. \tag{D.49}$$

Consequently $U_1(b)U_1'(b)^{-1}$ lies in the chart domain, $\mathbf{H}_{1,Ax}(b) = L\log\big(U_1(b)U_1'(b)^{-1}\big)$ satisfies $|\mathbf{H}_{1,Ax}(b)|_\sigma \leqslant \frac{\pi}{2}LC_{\mathrm{H}}\varepsilon_1 e^{-c_{\mathrm{H}}LMR_1}$, and (3.40) holds once $\frac{\pi}{2}LC_{\mathrm{H}}(1+\beta_0)(A_0/A_1)e^{-c_{\mathrm{H}}LMR_1} \leqslant \frac{1}{40}$ (item 5 of Section B.8).

Proof. This is the proof of Proposition D.17, Sections D.8.1–D.8.4, for the pair (U_1, U_1') in place of $(U_1, U_{1,Q})$, without the cube localisation: the reference problem is that of [3], so no lift and no condition on the periods enters. The norm $|\bullet|_\sigma$ is the cutoff norm of (3.40).

The problem and the argument. U_1' is the minimal configuration for the determining data named at p.251 ((B1)): the conditions of (1.12) on the rows of $(\Lambda_1^\sim)^c$, and on $\Lambda_1^\sim \setminus \Lambda_1$ the determining set $\mathbf{B}_1(\Lambda_1^c)$ with the data $D^0 = M^\bullet(Q_1^{s*}V)$. The configuration, and with it the first equality of (1.21), lives on $\Lambda_1^c = (\Lambda_1^\sim)^c \cup (\Lambda_1^\sim \setminus \Lambda_1)$, the union of the two domains on which p.251 determines it, and Λ_1 carries no variable of the problem; every plaquette with a corner in Λ_1 has its bonds in Λ_1 or in the fixed layer, which is at least LM_1 wide, so these terms are constants of the problem and are omitted, as are those off Ω^+. The hierarchy for [7] is that of Section D.8.1 with Λ_1 removed from its domains: V_0 is fixed on the collar $\Omega^+ \setminus \Omega_1$, $\Omega^+ = \Omega_1^{\sim 1}$, and the action terms off Ω^+ are omitted as constants; the scale-zero determining set is the collar together with the fine layer of $\mathbf{B}_1(\Lambda_1^c)$, and the scale-one determining set is the set of coarse bonds of $\Omega_1 \cap \Lambda_1^c$ off that layer. The data are regular as there: the collar lies in the P_0-good region, the plaquettes of $Q_1^{s*}V$ on the layer are trivial or coarse plaquette holonomies of V, bounded by $2L^2\varepsilon_0$, and the mixed determining plaquettes are bounded by [8, Proposition 1, (51)]. That the results of [7] hold for this determining set, whose fine layer has width between LM_1 and $2LM_1$ (p.256), in place of the spacing condition [7, (1)] for the layer, is Bałaban's application of [7] to the determining sets of [3] (p.256, and p.251 for this problem), taken over as such, as for $\mathbf{B}_1(Q_w^{[4]})$ in Section D.8.1; no condition relating L to the R of [7] is imposed. By the first equality of (1.21), $U_1 = U_1(\Omega_1 \cap \Lambda_1^c, M^\bullet(U_1))$, U_1 lies on the minimal orbit of the same problem with the data $D = M^\bullet(U_1)$ (the criticality and uniqueness argument of Section D.8.3 gives the same identification). [3, p. 251] applies [7, Proposition 9] to this problem ("The dependence is analytic, and the function $\mathbf{H}_{1,Ax}$ has the exponential decay property. It was formulated in (190) Sect. G of [15] for the Landau gauge"); let $\mathscr{H}$ be its relative Landau-gauge function with reference U_1', normalised by $\mathscr{H}(0) = 0$, and let B be the chart logarithm of $D(D^0)^{-1}$ (the $\frac{1}{i}\log$ of [3]), the argument of $\mathbf{H}_{1,Ax}$ in (1.21).

Support and size of B. On the rows of $(\Lambda_1^\sim)^c$ the two data coincide, the conditions there being those of (1.12) for both. On the coarse bonds of $\mathbf{B}_1(\Lambda_1^c)$ in $\Lambda_1^\sim \setminus \Lambda_1$, the only ones entering (p.251), both data equal V: these bonds lie in Ω_1, where $\bar{U}_1 = V$, and $M(Q_1^{s*}V) = V$ for the straight extension (Section D.8.2). So B vanishes off the fine layer of $\mathbf{B}_1(\Lambda_1^c)$, on which D and D^0 are the bond variables of U_1 and of $Q_1^{s*}V$ ([3, (2.11)] at the first step). That

layer lies within ℓ^∞ distance $2LM_1$ of Λ_1 ([3, p. 256] for the domain Λ_1^c: the coarse part of $\mathbf{B}_1(\Omega)$ contains $\Omega^{\sim -2}$, the enlargement taken for M_1-cubes, of side LM_1 in fine units); its width is between one and two M_1-cubes (p.256), and [3, p. 251] describes it as "one layer of M_1-cubes touching $\partial\Lambda_1$ and contained in Λ_1^c". Since Ω_1 and R_1' are unions of LMR_1-cubes and $\Lambda_1 \subseteq \Omega_1^{\sim -2} \cap (R_1'^{\sim})^c$, every site within ℓ^∞ distance $LMR_1 - 1$ of Λ_1 lies in $\Omega_1^{\sim -1} \setminus R_1'$; for small g_0 ($2LM_1 + 1 < LMR_1 - 1$) the layer therefore lies in $\Omega_1^{\sim -1}$, where U_1 is regular and in the source axial mean gauge ((B1)(i)), and the two-block comparison of Section D.8.2 gives $\|B\|_\infty \leqslant C_{L,G}\varepsilon_1$. [7, (172)] is met as there, with $\alpha = K_\alpha \varepsilon_0$, K_α dominating also this bound, and $C_1 = 1$.

Distance. Every site of $S_1 = (\Omega_1 \setminus \Omega_1^{\sim -1}) \cup R_1$ is thus at ℓ^∞ distance at least LMR_1 from Λ_1 ($R_1 \subseteq R_1'$), and the filling modification after (1.20) only removes cubes from Λ_1. For $b \in S_1^*$ (it has an endpoint in S_1) and b' equal to b or a bond of one of its two endpoint blocks, the initial vertex of b' is within $2L+1$ of a site of S_1, and the initial vertex of a bond of the layer is within $2LM_1+1$ of Λ_1; so, with d the ℓ^∞ distance between initial vertices of Section D.8.3,

$$d(b', \operatorname{supp} B) \geqslant LMR_1 - 2LM_1 - 2L - 2 \geqslant \tfrac{1}{2} LMR_1 \tag{D.50}$$

once $LMR_1 \geqslant 4LM_1 + 4L + 4$, which holds for small g_0 since $R_1 \to \infty$.

The Landau-gauge function. [7, (190)], integrated along tB, $t \in [0,1]$, and summed over $\operatorname{supp} B$ as in (D.45), gives $|\mathscr{H}(B;b')|_\sigma \leqslant C'_{L,G}\varepsilon_1 e^{-c'_{L,G} d(b', \operatorname{supp} B)}$ for every bond b' of the problem, with the constants of (D.45); the rate $c'_{L,G} = \delta_{\mathrm{P}}/(16L)$ is fixed before g_0, coming from the rate $\delta_{\mathrm{P}}/8$ of [7, (190)] in the distance [6, (2.46)], converted to the fine ℓ^∞ distance in Section D.8.3, with δ_{P} the decay rate defined in [10, p. 423] after Theorem 3.12, a constant of d and L. [7] states (190), and the representation (189) behind it, without the calculation (p.308, "We do not perform these calculations here"); (190) is Bałaban's statement, taken over as stated.

Restoring the axial gauge. As in Section D.8.4, the configuration $e^{iL^{-1}\mathscr{H}(B)}U_1'$ of [7, Proposition 9] lies on the orbit of U_1, and the restoring transformation u carrying it to U_1, with its centre values retained, gives (D.47) with $(U_1', \mathscr{H})$ in place of $(U_{1,Q}, \mathscr{H}_Q)$ and satisfies the block-local bound (D.48), $\sup_{x \in \mathcal{B}_y} |u(x) - \mathbf{1}|_\sigma \leqslant C_{L,G} \sup_{b' \in E(\mathcal{B}_y)} |\mathscr{H}(B;b')|_\sigma$: at one step the transformation on a block is fixed by the relative transports along the contours inside that block and by its centre factor ([8, (77)–(87) and (106)–(108)], as recorded there), which is the locality of the axial gauge on the unit lattice; (D.48) is the accepted transfer of Section D.8.4, applied to the function of (1.21) as [3] applies (3.17)–(3.18) to the corresponding function at the general step, "with similar bounds" (p.269). The target is U_1 itself, not another axial element of its orbit: the minimal configurations with the data D form one orbit, in the class of Section D.8.1, of the gauge transformations equal to the identity on the set $\mathfrak{B}_k$ of [7] ([3, p. 256]; [7, (4) and Theorem 1, pp. 278–279]), in which the axial gauge determines a unique element ([4, (0.6) and (0.11)], as recorded in (B1)), and U_1 is in the axial gauge ([3, p. 248]), so it is that element. At an endpoint off Ω_1, a point of the fixed exterior field V_0 (scale-zero data, as in the hierarchy of Section D.8.1), $u = \mathbf{1}$ ([7, p. 281] for the transformation of [7, Proposition 2], and [7, (4)] for the orbit element, the collar being part of $\mathfrak{B}_k$). For $b \in S_1^*$, unitary invariance and conjugation give

$$\begin{aligned} |U_1(b)U_1'(b)^{-1} - \mathbf{1}|_\sigma &\leqslant |u(b_-) - \mathbf{1}|_\sigma + L^{-1}|\mathscr{H}(B;b)|_\sigma + |u(b_+) - \mathbf{1}|_\sigma \\ &\leqslant (2C_{L,G} + 1) C'_{L,G}\varepsilon_1 e^{-c'_{L,G} LMR_1/2} \end{aligned}$$

by (D.50), which is (D.49) with $C_{\mathrm{H}} = (2C_{L,G}+1)C'_{L,G}$ and $c_{\mathrm{H}} = c'_{L,G}/2$.

The consequences. For small g_0 the right side of (D.49) is below $r_G/\sqrt{N}$, so $U_1(b)U_1'(b)^{-1}$ lies in the chart domain (the norm facts at the start of the proof of Lemma 3.8), and its chart

logarithm $X = L^{-1}\mathbf{H}_{1,Ax}(b)$ satisfies $|X|_\sigma \leqslant \frac{\pi}{2}|e^{iX} - \mathbf{1}|_\sigma$, which is the bound on $\mathbf{H}_{1,Ax}(b)$. With $\varepsilon_1/\delta_0 \leqslant (1+\beta_0)A_0/A_1$ ([3, (2.8)] and $\delta_0 = (A_1/A_0)\varepsilon_0$), it is at most $\delta_0/40$ under the stated condition, which holds for small g_0 at fixed L, M, and M_1 since $R_1 \to \infty$. *q.e.d.*

E The Signed Integrated Wilson Defects

This appendix bounds the defect term of the branch identity (3.97) at the carrier I of Section 3, under the original history law: the joint defects $R_\xi(\mathfrak{B})$ of (3.95) are summed with their restricted reference sums before absolute values are taken, each closed group is compared with its flat value by one background derivative along a local coefficient path, and the comparison is paid by the Gaussian reserve, the positive host comparison and the positive group summation of Sections E.4, E.6 and E.7. The result is Proposition E.22: on compatible even fine periods and for the square geometry of Section 5.3,

$$|E_{\mathrm{def}}| \leqslant C'_L n_C e^{-(c_*/2)\pi_*^2} + e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1 \,,$$

with $\rho = O(g_0^2(1+r)^4)$, $\varepsilon_1 = g_1 p_0(g_1)$, $b = \lfloor LMR_1/16 \rfloor$, $\pi_* = \min(p_0(g_0), p_0(g_1))$, $C'_L = 8C_L$ with the contour constant C_L of Section D.7 and the tail exponent c_* of Proposition D.15, and C_χ, c fixed before the coupling. The regional scalar $S^\circ_{\xi,C}$ is retained inside the expectation and is not identified with the perimeter/corner coefficient here; the exponential is not replaced by its linear term unless its argument is bounded. Section 9 keeps the history law and splits its weight w_ξ into labelled terms ($\sum_{\mathrm{lab}} \mathsf{w}_\xi^{\mathrm{lab}} = \mathsf{w}_\xi$, (9.31)) on the branches where a second collar test fails, retreating Λ_1 and the carrier I to a labelled carrier $D(\Lambda)$; the reading of this appendix at $D(\Lambda)$ with the labelled weight $\mathbf{1}_{\mathrm{lab}} e^{c_\xi^\Lambda} I_1^\Lambda$ (with $I \to D(\Lambda)$, $S^\circ_{\xi,C} \to S^\circ_\Lambda$, $R_\xi \to R^0_\Lambda$), which that section needs for its labelled defect $E^{\mathrm{lab}}_{\mathrm{def}}$, is Proposition 9.40: it lists the inputs of Sections E.2–E.11 and where each is supplied at $D(\Lambda)$. Sections E.2–E.11 refer to the original law. The appendix shares the names ϖ, η, C_{pin}, K_χ and $M(W)$ with Section 9.6, which uses the same positive summation at $D(\Lambda)$; the objects here are those at I.

E.1 Operand and Retained Normalisation

Use exactly the groups W of Definition 3.24, the compatible families $\mathfrak{B}$, the closed pairings $A_\xi(\mathcal{S})$ of (3.90), the scalar references a_W of (3.92), the restricted reference sums $S^{\circ,[\mathfrak{B}]}_{\xi,C}$ and the joint defects $R_\xi(\mathfrak{B})$ of Proposition 3.27. Every source term, the Wilson contraction f_ξ and each group's closed contraction of Definition 3.24, is a gauge-invariant closed contraction (Section E.10.2). Put

$$\mathscr{R}_\xi = \sum_{\substack{\mathfrak{B}\neq\varnothing \\ \mathfrak{B}\ \text{compatible}}} S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B}), \qquad E_{\mathrm{def}} = \frac{1}{\mathcal{Z}_0}\int \mathcal{O}\big[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}}} \mathscr{R}_\xi\big](V)\,\mathrm{d}V \,. \tag{E.1}$$

The weight is the branch weight $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$ of (3.49), the outer operation $\mathcal{O}$ of (3.4) contains the original cutoffs, ζ_ξ with its boundary-fluctuation dependence, the gauge and Haar factors, the Jacobians, the vacuum prefactor and the deferred integrations, $\mathcal{Z}_0 = \int \mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V$, and $\mathbb{E}_{\mathrm{h}}$, $\mathbb{P}_{\mathrm{h}}$ are the history expectation and probability (D.20) for this positive weight; A_{cov} is the covered event of Lemma 3.28 (every contour bond is a bond of Ω_1 and every free input of the contour lies in I). The weight is positive: $K_{\xi;1}$ is the exponential of a real function times indicators (Section 3.4) and c_ξ is real, the logarithm of the positive a-independent factor collected in the proof of (3.49). At fixed finite volume the signed sum (E.1) converges

absolutely: the family sum (3.91) converges absolutely, and the proof of Proposition 3.27 obtains (E.1) from it by Boolean inversion over finitely many root families, each with an absolutely convergent cluster-list sum; no uniform absolute summability is inferred from that fact.

E.2 Histories near the Bad Region

Let $B_\xi = \Lambda_1^c$ be the filled bad region and $E(B_\xi)$ the set of positive fine bonds with at least one endpoint in B_ξ. Let Γ_C be the deterministic set of contour bonds and free-input templates of Section D.7, with $|\Gamma_C| \leqslant C_L n_C$, n_C the number of contour traversals; distances are those of the periodic fine-bond metric d_T of (5.2), $d_T(b,c) = d_{\infty,T}(x,y) + \mathbf{1}\{i \neq j\}$ for $b = (x,i)$, $c = (y,j)$. For an integer $b \geqslant 0$ define

$$\mathsf{N}_b = \{\xi \mid \operatorname{dist}(\Gamma_C, E(B_\xi)) \leqslant b\}\,, \tag{E.2}$$

with distance to the empty set infinite. This is a measurable history event (Λ_1 is a function of the history labels); it changes neither the covered event nor the law. Write $E_{\mathrm{def}}^{\mathrm{near}} = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}\mathbf{1}_{\mathsf{N}_b}\mathscr{R}_\xi]$ and $E_{\mathrm{def}}^{\mathrm{far}} = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}}\mathbf{1}_{\mathsf{N}_b^c}\mathscr{R}_\xi]$, so that $E_{\mathrm{def}} = E_{\mathrm{def}}^{\mathrm{near}} + E_{\mathrm{def}}^{\mathrm{far}}$ (both integrable, by the bound $|\mathscr{R}_\xi| \leqslant 2$ of Lemma E.1).

Lemma E.1 (Near Histories)**.** *Let $b \geqslant 0$ be an integer and let q_{pt} be the one-point bound of Lemma D.14 ($\mathbb{P}_{\mathrm{h}}(z \in E(B_\xi)) = p_z \leqslant q_{\mathrm{pt}}$ for every bond z, on compatible even fine periods and for $g_0 \leqslant g_H$), with $q_{\mathrm{pt}} \leqslant e^{-c_*\pi_*^2}$ for $0 < g_0 < g_*$ by Proposition D.15. Then $|\mathscr{R}_\xi| \leqslant 2$ pointwise on every branch, and*

$$\mathbb{P}_{\mathrm{h}}(\mathsf{N}_b) \leqslant 4C_L n_C (2b+3)^4 q_{\mathrm{pt}}, \qquad |E_{\mathrm{def}}^{\mathrm{near}}| \leqslant 8C_L n_C (2b+3)^4 q_{\mathrm{pt}} \leqslant C_L' n_C e^{-(c_*/2)\pi_*^2}\,, \tag{E.3}$$

the last inequality for $b = \lfloor LMR_1/16 \rfloor$, $C_L' \coloneqq 8C_L$ and g_0 below a threshold depending on L, M_1, M_2, M, the group, the character and the source constants.

Proof. By (3.96), $\mathscr{R}_\xi = F_\xi(0) - S_{\xi,C}^{\circ} f_\xi(0)$. Both $F_\xi(0) = \int f_\xi \,\mathrm{d}\nu_{\xi;0}$ ((3.46)) and $S_{\xi,C}^{\circ}$ ((3.94)) are normalised integrals of a character of modulus at most one against the real positive kernels $K_{\xi;0}$ (the exponential of a real function times indicators, Section 3.4) and $K_{\xi;0}^{\circ}$ (Lemma B.13), and $|f_\xi(0)| \leqslant 1$; hence $|\mathscr{R}_\xi| \leqslant 2$. This bound is used only after the joint defects have been summed; it is not a bound on any $R_\xi(\mathfrak{B})$, group or restricted reference factor. The members of Γ_C are bonds (contour bonds and the bonds of the free-input templates), and there are at most $4(2b+3)^4$ positive fine-bond labels within d_T-distance b of a fixed fine bond (an overcount on short periods), so a union bound over Γ_C and the one-point estimate $p_z \leqslant q_{\mathrm{pt}}$ of Lemma D.14 (p_z of (D.23)) give the first bound; positivity of w_ξ and $\mathcal{O}$, used only after the signed sum has been formed, and $|\mathscr{R}_\xi| \leqslant 2$ give the second. For the last: $b = \lfloor LMR_1/16 \rfloor$ exceeds every fixed source stencil for small g_0, and $\log(2b+3) = O(\log\log g_0^{-2})$ since $\ell = LMR_1 \leqslant L^2 M (2\log g_0^{-2})^{\mathfrak{r}}$ ((9.9)), while $\pi_* = \min(p_0(g_0), p_0(g_1)) \geqslant A_0(\tfrac{1}{2}\log g_0^{-2})^{p_0}$ with $p_0 > 0$, by $p_0(g) = A_0(\log g^{-2})^{p_0}$ ((D.1)) and $\log g_1^{-2} \geqslant \tfrac{1}{2}\log g_0^{-2}$ (as at (9.9)); so $4\log(2b+3) \leqslant (c_*/2)\pi_*^2$, that is $8C_L(2b+3)^4 e^{-c_*\pi_*^2} \leqslant C_L' e^{-(c_*/2)\pi_*^2}$ with $C_L' = 8C_L$, for small g_0 at fixed source constants. The constants are uniform in the compatible even-period tori and the histories; no lower bound on a physical denominator is substituted in this proof (the normaliser bound (D.36) enters only through the cited one-point estimate), no conditional scalar is cancelled across histories, and no pointwise smallness of a Wilson-loop random variable is claimed. *q.e.d.*

E.3 The Interior Operand

The remaining operand is $E^{\mathrm{far}}_{\mathrm{def}}$, with the full signed group-family sum retained under its original weight. A group's random-field support is not its background support: its Wilson contraction carries the background transports along its contour interval $\mathcal{H}(W)$ as well. The flat comparisons of Sections E.4 and E.5 compare each closed group with its flat value along the group's own interval transport closed back to its base (Remark E.6, Lemma E.19), so these transports, in their order, are part of the compared word: no group, in particular none meeting separated sides of the square, is flattened on its random support alone while the intervening transports are changed. The analytic step is to insert the background derivative of the closed source-containing selected integrals and of their joint vacuum ratio before taking absolute values or summing group families (Section E.4), and to sum the groups afterwards (Sections E.5, E.7 and E.8). The derivative bounds used are those of Sections E.6 and E.9.

E.4 The Joint Signed Selected-Integral Estimate

Fix a compatible root family $\mathcal{A} = \{D_1, \dots, D_m\}$. For each D_j use the selected measures $\Lambda^{\mathrm{f}}_{\xi;\ell_j}$ of (3.81) with $\ell_j = (\mathcal{D}_j, \mathsf{J}_j, \mathsf{P}_j, Z_j)$, $Z_j = D_j$ and $\mathsf{J}_j \neq \varnothing$; put $q_j = |\mathsf{J}_j|$ and $q = \sum_j q_j$. The centred measure $\widehat{\nu}^{\mathrm{f}}_{\xi,D} = \nu^{\mathrm{f}}_{\xi,D} - k^{\mathrm{f}}_{\xi}(D)\delta_{0,\mathsf{L}_D}$ of (3.83) is exactly the sum of the nonempty-probe selections of (3.81): the empty-probe selections give the mass times δ_0 and disappear under centring. This is the measure identity of Lemma 3.21; no new expansion is introduced. Define the finite-difference operator on a bounded function f of the young variable by

$$\Delta_{\mathsf{J}} f(B) = \sum_{K \subseteq \mathsf{J}} (-1)^{|\mathsf{J}\setminus K|} f(\mathbf{1}_K B) \,, \tag{E.4}$$

every unselected young label being zero; for several roots compose these operators on their disjoint label sets, writing $\Delta_{\mathbf{J}} f$. The integrand is a sum of closed Wilson contractions with the ordering and background transports retained; the source may carry a fixed complementary closed holonomy $h \in G$, transforming by conjugation at the base, and all bounds are uniform in h. With (3.84),

$$\mathfrak{Q}_{\xi}(\mathcal{A}; f) = \sum_{(\ell_j)_{j\leqslant m} \,|\, Z_j = D_j,\ \mathsf{J}_j \neq \varnothing} \int \Delta_{\mathbf{J}} f(B)\, \mathrm{d} \bigotimes_{j=1}^{m} \Lambda^{\mathrm{f}}_{\xi;\ell_j}(B_j) \,, \tag{E.5}$$

the integral of each Dirac tensor $\bigotimes_{b\in\mathsf{J}}(\delta_{B_b} - \delta_0)$ against f being the finite difference (E.4).

E.4.1 The Powers Supplied by the Signed Source

Set $u = C_\chi g_0(1 + r)$, increasing the fixed row/character constant C_χ once (Section E.9.4). On the real protected box the row reconstruction depends on $g_0 B$ and has analytic derivative bounds on a larger fixed phase chart (Lemma 3.6, Lemma B.7). Each selected free label affects a bounded number of reconstructed contour links, each link has a bounded input set, and for the square the traversal multiplicity is bounded; applying the product rule inside the closed character therefore gives C_χ^q derivative assignments, not n_C^q or $q!$. Derivative orders on any one row are bounded by twice its fixed input count plus one, including the remainders below (Section E.9.4); the higher row derivatives are obtained from the bounded analytic phase map on its larger fixed ball by Cauchy estimates (Lemma B.7(i)–(ii): $\widetilde{D}_{U,c}$ is analytic on the ball of radius ϑ_* with $\|\widetilde{D}_{U,c}(A)\| \leqslant 2K_*\|A\|^2$ and $\|h_U(c)\| \leqslant H_*$, so Cauchy's estimates

on the ball of radius $\vartheta_*/2$ bound every order; the finite list of derivative constants of item (iii) is not relied on beyond order five), and the maximum order used is fixed by L. All untouched transports are unitary matrices inside the same closed contraction. For the closed character f of the source (Lemma E.3 below) the mixed integral remainder formula (E.59) gives

$$\sup |\Delta_{\mathsf{J}} f| \leqslant u^q \,, \tag{E.6}$$

the supremum over the real protected box, which contains the support of every selected measure in the selected variables (the measure (3.81) carries the cutoffs χ_b over $E(Y_0)$ in the variables it integrates; its bad cutoffs $1-\chi_b$ act on the labels of P, which are integrated out, not on the selected probes J).

For a root with odd q_j take the homogeneous term of degree q_j in its selected variables B_{D_j}, the other root variables fixed; it is odd under $B_{D_j} \to -B_{D_j}$, and the difference from it carries an additional factor u ((E.61); the choice of the extra increment is bounded by $q_j \leqslant 2^{q_j}$, absorbed in C_χ). The same holds simultaneously for any set of odd roots, the product-rule allocation count remaining exponential in q. Every term is a closed contraction; no open interval is estimated separately. Each selected Gaussian kernel (3.66) with its cutoffs is invariant under $(B_{D_j}, X_{D_j}) \to (-B_{D_j}, -X_{D_j})$ once its ordinary Mayer factors are removed: the determinant, the quadratic cross term, the compensator, the real cutoffs χ_b, $1-\chi_b$ and the spatial Cauchy integrations are even. The tensor product in (E.5) gives separate integration variables and a separate reversal for each root. Thus the leading odd part integrates to zero in the signed selected integral, also when several roots are present.

Let $\omega(g_0) = C_0^{\partial} g_0 (1+r)^3$ with the host derivative constant $C_0^{\partial} \geqslant C_0$ of Section E.6.1, let ω_* be the fixed host budget of (B.28), and put $R_\Upsilon = \omega_*/(2\omega(g_0))$; choose g_0 so that $R_\Upsilon \geqslant 2$. In root j multiply the selected ordinary potentials $\mathbf{V}_\xi(Y)$, $Y \in \mathcal{D}_j$, by an independent complex parameter ζ_j, $|\zeta_j| \leqslant R_\Upsilon$, leaving the Gaussian kernel, cutoffs and probes unchanged. If $\mathcal{D}_j \neq \varnothing$ the selected integral vanishes at $\zeta_j = 0$ (each Mayer factor $\int_0^1 \zeta \mathbf{V} e^{t\zeta\mathbf{V}} \mathrm{d}t = e^{\zeta\mathbf{V}} - 1$ does), so the Cauchy difference between $\zeta_j = 0$ and $\zeta_j = 1$ of the analytic function of ζ_j costs at most $2/(R_\Upsilon - 1)$ times its supremum on the disc; if $\mathcal{D}_j = \varnothing$ the density is the even Gaussian/cutoff kernel and the reversal applies. Independent parameters on disjoint roots fit the same local potential budget simultaneously (Section E.6.1). Therefore an odd q_j costs

$$v = u + \frac{2}{R_\Upsilon - 1} \,. \tag{E.7}$$

No separate cancellation of a group coefficient is assumed; the cancellations occur inside the original signed selected integrals. Both u and v tend to zero as $g_0 \to 0$, since $g_0(1+r)^3 \to 0$ by (D.1). (ω as a background direction, with $\|\omega\|_\infty$ and $\mathscr{D}_\omega$, and $\omega(g_0)$, ω_* as host bounds are distinguished by their arguments.)

E.4.2 Numerical Majorants Paid with the Same Gaussian Reserve

For a selected root ℓ define

$$M_\ell = \sup_{|\zeta| \leqslant R_\Upsilon} \|\Lambda^{\mathrm{f}}_{\xi;\ell,\zeta}\|_{\mathrm{TV}} \,, \tag{E.8}$$

where $\Lambda^{\mathrm{f}}_{\xi;\ell,\zeta}$ is the measure (3.81) with each selected ordinary potential replaced by ζ times that potential; the cutoffs, Gaussian density and normalisers, compensator, integration carriers, weakening contours and seed data are unchanged. The total variation is a numerical majorant used after the signed source cancellation; it is not a bound on the centred Dirac tensors.

Lemma E.2 (Dummy Comparison)**.** *There are at most $P_M = 4(4M+3)^4$ probe labels with a given host (a label b lies in the interior of its host Y_b, a rectangle of at most four M-cubes per side, (3.75)), and each probe host has $N_0(Y_b) \leqslant 4^4 \leqslant s_{\mathrm{p}} = 5^4$ cubes ((3.75)). Let $R_\Upsilon \geqslant 2$ and let g_0 be below the thresholds of Lemmas 3.19 and B.12. Choose once, after M and before g_0,*

$$u_* = \min\Big(\frac{1}{4},\ \frac{\omega_* e^{-\kappa s_{\mathrm{p}}}}{8P_M}\Big)\,. \tag{E.9}$$

Then for every connected final domain D, with $m(D) = \varepsilon_* e^{-\lambda_* \ell_1(D)}$ the majorant of Lemma 3.19,

$$\sum_{\ell \,|\, Z(\ell)=D,\ \mathsf{J}(\ell)\neq\varnothing} u_*^{q(\ell)} M_\ell \leqslant m(D)\,. \tag{E.10}$$

Proof. The number u_* is a fixed comparison weight, not a measure. For every probe label b insert, in the numerical comparison only, the dummy potential $p_b^+ = u_*$ on its protected host box and zero outside; its selected Mayer factor $e^{u_*} - 1$ is at least u_*. The summed dummy potential at a host is at most $P_M u_* \leqslant \omega_* e^{-\kappa s_{\mathrm{p}}}/8 \leqslant \omega_* e^{-\kappa d_0(Y)}/8$, the input B_Y of (E.27) (for a probe host $d_0(Y) \leqslant N_0(Y) - 1 < s_{\mathrm{p}}$: its cubes form a rectangular array, and a path through the centres of wall-adjacent cubes visiting each once meets every cube and has length $N_0(Y) - 1$ in (B.1)), while the complex ordinary potentials use at most half the budget $\omega_* e^{-\kappa d_0(Y)}$ on $|\zeta| \leqslant R_\Upsilon$ (Lemma B.10 and $R_\Upsilon \omega(g_0) = \omega_*/2$). Split the labels with a common host exactly as in the proof of Lemma 3.21; the seeds, conditioning carrier and Gaussian kernel depend on the active hosts only and are unchanged. Sum all split selections with the same active-host set first: Lemma E.13 proves that their many-to-one positive sum is dominated hostwise by the undifferentiated majorant (B.31). That majorant pays its determinant and compensator with the Gaussian reserve α_G (Lemma 3.16(ii)), and the cover and seed sums (B.34), (B.35) with the final Y'-sum in the proof of Lemma 3.19 (Appendix B) sum the seeds and final domains to $m(D)$. All parameters in $m(D)$ are the existing positive-majorant parameters, and (E.10) holds with the exact λ_*. (The polynomial factors inserted in Sections E.6 and E.7 are paid there by a fixed decrease of the remaining decay of $m(D)$.) Nothing is changed in the physical potential or in a normalisation of E_{def}. *q.e.d.*

Lemma E.3 (Joint Selected-Integral Bound)**.** *Let f be a normalised character of an ordered closed reconstructed word, optionally closed with its inverse background interval and a fixed complementary h (Section E.6.2). For g_0 small enough that $u \leqslant u_*$ and $v \leqslant 1$, put*

$$\rho = \Big(\frac{u}{u_*}\Big)^2 + \frac{u}{u_*}\,v\,. \tag{E.11}$$

Then for every compatible root family $\mathcal{A}$ and every choice of selections (ℓ_j) with $\mathsf{J}_j \neq \varnothing$,

$$\Big|\int \Delta_{\mathsf{J}} f(B)\,\mathrm{d}\bigotimes_{j=1}^{m} \Lambda^{\mathrm{f}}_{\xi;\ell_j}(B_j)\Big| \leqslant \prod_{j=1}^{m} u^{q_j} v^{q_j \bmod 2} M_{\ell_j}, \qquad |\mathfrak{Q}_\xi(\mathcal{A}; f)| \leqslant \prod_{D\in\mathcal{A}} \rho\, m(D)\,. \tag{E.12}$$

Proof. The first bound combines the reversal, the Cauchy differences and (E.6)–(E.7) as written out in Section E.6.2: the roots with odd q_j and empty ordinary selection lose their leading odd part to the even kernel and keep the remainder u^{q_j+1}, the roots with odd q_j and nonempty ordinary selection pay $u^{q_j} \cdot 2/(R_\Upsilon - 1)$ by the Cauchy difference in ζ_j (the supremum on the disc being $u^{q_j} M_{\ell_j}$ by (E.8)), the roots with even q_j pay $u^{q_j} M_{\ell_j}$. The bound retains the full joint source through each cancellation. For every nonempty probe set $(u/u_*)^{q_j} v^{q_j \bmod 2} \leqslant \rho$ ($q_j = 1$ gives $(u/u_*)v$, $q_j \geqslant 2$ gives $(u/u_*)^2$). Summing over the selections of every root with (E.5) and (E.10) gives the second bound. *q.e.d.*

The gain is $\rho = O(g_0^2(1+r)^4)$ with constants fixed before the coupling; it is obtained after the signed integration, not by replacing a probe tensor by its total variation $2^{|\mathsf{J}|}$.

E.4.3 Background Derivative and the Joint Vacuum Ratio

For the flat subtraction take a real background direction ω along one of the globally small analytic extensions of Section E.8.1; the geometry, carriers, cutoff norm, means, Gaussian reserve and walk recipe are unchanged. Let L_f be the number of occurrences of contour background links in the closed word whose local coefficient arguments vary; when a group's relative interval is compared this includes the interval and its inverse closing transport. Complementary closed holonomies are held fixed, and no outer history variable or physical weight is differentiated. Write $\mathscr{D}_\omega$ for this derivative.

Lemma E.4 (One Background Derivative)**.** *On the extensions of Section E.8.1, with ρ, $m(D)$ of Lemma E.3 and a constant C fixed before the coupling,*

$$|\mathscr{D}_\omega \mathfrak{Q}_\xi(\mathcal{A}; f)| \leqslant C\|\omega\|_\infty \Big(L_f + \sum_{D\in\mathcal{A}} N_1(D)\Big) \prod_{D\in\mathcal{A}} \rho\, m(D)\ . \tag{E.13}$$

Proof. The derivative of the selected Gaussian factors (density, normaliser, compensator) inserts a scalar polynomial of degree at most two in (B, X), including the determinant trace; its moment is paid by the joint coercive Gaussian bound of Section E.9.3 at cost $CN_1(D)\|\omega\|_\infty$. Differentiated walk coefficients sum in the convergent differentiated series of Section E.9.2. The derivative of a host Mayer t-integrand is $\zeta(\mathscr{D}_\omega \mathbf{V})e^{t\zeta\mathbf{V}}(1+t\zeta\mathbf{V})$, whose t-integral $\zeta(\mathscr{D}_\omega\mathbf{V})e^{\zeta\mathbf{V}}$ vanishes at $\zeta = 0$; its size is bounded by the differentiated host comparison of Lemma E.13. A Gaussian derivative is even under each component reversal and the leading source derivative has the parity of the undifferentiated term ((E.62)), so the signed powers of Section E.4.1 survive one derivative, with the additional factor $C\|\omega\|_\infty(L_f + \sum_D N_1(D))$: the source derivative contributes L_f plus the number of selected labels, the latter at most $4(LM+1)^4 \sum_D N_1(D)$ (Section E.6.1), and the host, Gaussian and walk derivatives contribute one factor $N_1(D)$ each by Lemma E.13. No division by a potential, activity or character is used. The factor $N_1(D)$ is kept explicitly; it is paid by $m(D)$ in Section E.6. *q.e.d.*

The vacuum ratio is kept as the joint ratio of the whole root family, not a product of one-root ratios: by (3.84) and (3.74),

$$q_\xi(\mathcal{A}) = \exp\Big[-\sum_{\mathfrak{c}\text{ hits }\mathcal{A}} \Phi^{\mathrm{v}}_\xi(\mathfrak{c})\Big]\ , \tag{E.14}$$

a cluster hitting several roots being counted once, and its derivative is the derivative of this exponential.

Lemma E.5 (The Joint Ratio and Its Derivative)**.** *With the constant C_M of Lemma E.13,*

$$|q_\xi(\mathcal{A})| \leqslant e^{\sum_{D\in\mathcal{A}} N_1(D)/8}, \qquad |\mathscr{D}_\omega q_\xi(\mathcal{A})| \leqslant \frac{C_M}{2}\|\omega\|_\infty \sum_{D\in\mathcal{A}} N_1(D)\, e^{\sum_{D\in\mathcal{A}} N_1(D)/8}\ , \tag{E.15}$$

and consequently

$$\big|\mathscr{D}_\omega[q_\xi(\mathcal{A})\mathfrak{Q}_\xi(\mathcal{A}; f)]\big| \leqslant C\|\omega\|_\infty \Big(L_f + \sum_{D\in\mathcal{A}} N_1(D)\Big) \prod_{D\in\mathcal{A}} \big[\rho\, m(D) e^{N_1(D)/8}\big]\ . \tag{E.16}$$

Proof. The first bound is Lemma 3.20. The second is Lemma E.16, which differentiates one vacuum activity in its connected-cluster tree majorant, keeping every occurrence and symmetry factor and marking the single derivative in the fixed positive cluster majorant, without shrinking a previously chosen parameter or dividing by an activity. The third follows from the product rule with Lemmas E.3 and E.4 and the two bounds of (E.15), with the constant C of (E.13) increased by $C_M/2$. *q.e.d.*

The actual joint ratio is differentiated before estimating. The normaliser $I_{\xi;0}$ encoded by this ratio is not cancelled against the physical history factor $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$.

E.5 The Closed Group Subtraction and Its Signed Caller

Remark E.6 (What the Subtraction Must Pay). Section E.4 supplies the derivative estimate when the flat subtraction of each closed group is made inside $\mathscr{R}_\xi$. The group-family summation (this section and Sections E.7, E.8) must pay the overlap of contour intervals and the spatial decorations, and must control the restricted reference sums $S^{\circ,[\mathfrak{B}]}_{\xi,C}$ with their regional flat kernel; no product of one-component ratios and no uncontrolled exponential in the loop length is admissible. The group contraction for this caller uses the interval transport relative to its background interval, closed back to its base, with an arbitrary complementary closed holonomy h; its flat value is $a_W \chi(h)$ by Lemma 3.26. The product bound for such closed contractions is the finite-dimensional norm equivalence of Lemma E.7; no independent open matrix norm is an inductive carrier, and the multi-group factorisation is checked against (3.90), not presumed from spatial independence of the physical law.

Geometry. The square/torus geometry of Section 5.3 is in force: C is a coarse square of side $s = nL$ fine bonds on fine periods LK_i with $K_i \geqslant 3(n+2)$, and $n_C = 4s$. For a group W, $\mathcal{H}(W)$ is its interval (Definition 3.24: a cyclic interval of fewer than $n_C/2$ contour traversals, or the whole contour read from the base point) and $|\mathcal{H}(W)|$ the number of traversals in it. No loop separation hypothesis is added. Contour hull length is paid by connected spatial extent as in (5.24): contacts of diameter $< s/2$ occupy one side or two adjacent sides and fit in an arc of length at most $4\,\mathrm{diam}+1$; otherwise $n_C \leqslant 8\,\mathrm{diam}$. Periodic distances between square contacts equal their lifted distances.

For a group W let $n(W)$ be the sum of N_1 over its roots and over every constituent domain of every occurrence in its cluster list, with multiplicities, and define

$$\mathfrak{s}(W) = n(W) + \frac{|\mathcal{H}(W)|}{LM} \,. \tag{E.17}$$

The footprint $F(W)$ is the set of LM-cubes meeting a constituent domain, its fixed coefficient stencil halo, or the contour/free-input stencil of a traversal of $\mathcal{H}(W)$ ("meeting" in the site sense: a cube containing a site or a bond endpoint of the set; the ranges are made explicit in Lemma E.8; the comparison box Q_W of Section E.7.4 carries its own fixed enlargement, absorbed in C_{box}, covering the Hessian stencils and reconstruction rows consulted by the coefficient, and is not part of the footprint); the source geometry gives $|F(W)| \leqslant C\mathfrak{s}(W)$ (Lemma E.8(a)). For a component of the roots/cluster-occurrence graph the union of its domains is wall connected, so its diameter is at most CLM times its cube count, and its contact hull is bounded by the square estimate above; overlapping hulls have a connected interval union, which pays the hull of a group by the sum of its component counts. If all components are merged because some hull is long (Section 3.8), the sum of the component sizes in that class is at least $cn_C/(LM)$. Hence $|\mathcal{H}(W)|/(LM) \leqslant Cn(W)$. Coefficients whose local rectangle fails to lift are kept in the large-domain class of Section E.7.4.

E.5.1 The Closed Relative Interval Functional

For W use its root variables only, every other young variable zero, in the reconstructed links along $\mathcal{H}(W)$. Let $T_W(a;U_1)$ be the ordered product of these links (the interval transport) and $P_W(U_1)$ the same product at zero young variable, both with their source orientations and with the same two endpoint vertices. The relative product $T_W(a;U_1)P_W(U_1)^{-1}$ is based and closed at the initial vertex. Define the scalar closed contraction

$$c_W(U_1;h) = w_\xi(W) \int \chi\big(T_W(a;U_1)P_W(U_1)^{-1}h\big)\,\mathrm{d}\Big(\bigotimes_{D\in\mathcal{A}(W)} \widehat{\nu}^{\mathrm{f}}_{\xi,D}\Big)(a)\,, \tag{E.18}$$

$h \in G$ a complementary closed holonomy at that base, transforming by simultaneous conjugation; the product $T_W P_W^{-1}$ and h transform at one base, so the displayed scalar is gauge invariant and no open interval functional is estimated. Put

$$d_W(U_1;h) = c_W(U_1;h) - a_W\chi(h), \qquad \mathfrak{d}_W(U_1) = \sup_{h\in G} |d_W(U_1;h)|\,. \tag{E.19}$$

At the reference data, where the same formula is taken with the reference weights and measures, $c_W(\mathbf{1};h) = a_W\chi(h)$ by Lemma 3.26 (there the interval transport at the root variables is the $P^\circ_W(a)$ of that lemma and the background transport is the identity). The weights and measures in c_W are those of the joint extraction, including every cluster occurrence of the group; they are not a newly normalised conditional law.

Lemma E.7 (Factorisation and Product Inequality)**.** *Let π be the irreducible unitary representation with $\chi = \dim(\pi)^{-1}\operatorname{tr}\pi$. (a) There is a constant K_χ of the character such that $\|\Upsilon\|_{\mathrm{op}} \leqslant K_\chi \sup_{h\in G} |\dim(\pi)^{-1}\operatorname{tr}(\Upsilon\pi(h))|$ for every $\Upsilon \in \mathrm{End}(V_\pi)$. (b) For a compatible family $\mathfrak{B} = \{W_1,\dots,W_m\}$ on a covered branch, $R_\xi(\mathfrak{B})$ is the normalised trace of the ordered product of the matrices $\Upsilon_{W_j}\pi(G_j)$, where $\Upsilon_W = w_\xi(W)\int \pi(T_W P_W^{-1})\,\mathrm{d}\bigotimes \widehat{\nu}^{\mathrm{f}}_\xi - a_W\mathbf{1}$ and G_j is the background transport of the gap after $\mathcal{H}(W_j)$ composed with P_{W_j}. (c) Consequently*

$$|R_\xi(\mathfrak{B})| \leqslant \prod_{W\in\mathfrak{B}} K_\chi \mathfrak{d}_W(U_1)\,. \tag{E.20}$$

Proof. (a) $\Upsilon \mapsto (h \mapsto \operatorname{tr}(\Upsilon\pi(h)))$ is linear and injective on $\mathrm{End}(V_\pi)$, since $\pi(G)$ spans $\mathrm{End}(V_\pi)$ for an irreducible representation (Burnside); so the supremum over h is a norm on the finite-dimensional space $\mathrm{End}(V_\pi)$, equivalent to the operator norm, and K_χ is the equivalence constant. (b) In a compatible family a traversal of $\mathcal{H}(W)$ has no contact with a root of a different group (that contact hull would meet $\mathcal{H}(W)$), so its reconstructed link uses the root variables of W only (Lemma 3.5); a traversal outside every interval has its background link. Number the groups in cyclic order and start the ordered link product at the first traversal of $\mathcal{H}(W_1)$ (cyclicity of the trace). Expanding the ordered product $\prod_j (X_j - a_{W_j}\mathbf{1})\pi(G_j)$ with $X_j = w_\xi(W_j)\int \pi(T_{W_j}P_{W_j}^{-1})\mathrm{d}\bigotimes\widehat{\nu}^{\mathrm{f}}$ over subsets $\mathcal{S}$ of slots, a slot of the first kind contributes the links of $\mathcal{H}(W_j)$ at the root variables followed by the gap transport, a slot of the second kind contributes $-a_{W_j}$ times the links of $\mathcal{H}(W_j)$ at zero young variable followed by the gap; by multilinearity of the trace and Fubini over the product measure (as in the proof of Lemma 3.26) the normalised trace of the term of $\mathcal{S}$ is $\big(\prod_{W\notin\mathcal{S}}(-a_W)\big)\big(\prod_{W\in\mathcal{S}} w_\xi(W)\big)\,\mathfrak{Q}_\xi(\mathcal{A}(\mathcal{S});f_\xi) = \big(\prod_{W\notin\mathcal{S}}(-a_W)\big)A_\xi(\mathcal{S})$ by (3.90), since on a covered branch the links outside the intervals are the background links at U_1 (Section E.7.1; Lemma 3.28 gives $f_\xi(0) = W_\chi(C;U_1)$). Summing over $\mathcal{S}$ reproduces (3.95). This is an algebraic factorisation of the existing polymer integrals, not independence of physical field regions. (c) $|\operatorname{tr}(Y_1\cdots Y_{2m})| \leqslant \dim(\pi)\prod\|Y_i\|_{\mathrm{op}}$, the gap transports are unitary, and

$\|\Upsilon_W\|_{\mathrm{op}} \leqslant K_\chi \mathfrak{d}_W(U_1)$ by (a), since $\dim(\pi)^{-1}\operatorname{tr}(\Upsilon_W \pi(h)) = d_W(U_1; h)$. Matrices are used to prove the finite-dimensional multilinear identity; no separately estimated endpoint-covariant matrix or open Wilson carrier is introduced. *q.e.d.*

E.5.2 Positive Graph Sums for the Reference Factors

This is the summation with the estimates of Sections E.4 and E.6; it concerns the source-containing groups, not the vacuum gas alone. A cluster occurrence $\mathfrak{c}$ has cube cost $n(\mathfrak{c}) = \sum_{D \in \mathfrak{c}} N_1(D)$; its positive weight $M_v(\mathfrak{c})$ is the tree majorant of its Ursell coefficient: for the ordered tuple defining $\mathfrak{c}$, divide by the tuple factorial and sum over the incompatibility spanning trees, multiplying the majorants $m(D)$ of the activities (the positive tree sum of the proof of Lemma 3.20, Section B.6). Source roots have the weights $\rho\, m(D)$ of Lemma E.3. For each W the product of its root weights, its cluster weights $M_v(\mathfrak{c}_j)$ and $1/k(W)!$ is denoted $M(W)$; it bounds $|a_W|$, the pairings $|w_\xi(W)\mathfrak{Q}_\xi(\mathcal{A}(W); f)|$ for the words of Lemma E.3, and the undifferentiated closed source functional $|c_W(U_1; h)|$, the physical and reference data having the same numerical majorant (Lemma E.3 applies at the reference data too, whose hypotheses Lemma B.13 verifies).

Lemma E.8 (Footprints and Envelopes)**.** *The cubes are those of the partition π_1 (side LM, Section B.1): $\Delta_z = LMz + [-LM/2, LM/2]^4$ reduced modulo the fine periods, which are multiples of LM by* (3.1)*; every site lies in exactly one cube, and the index distance of two cubes is the periodic maximum-norm distance of their indices z. For a set S of sites, $\pi_1(S)$ is the set of cubes containing a site of S; throughout, a cube meets a set of sites or bonds (as in the footprint definition following* (E.17)*) when it contains a site or a bond endpoint of the set, the site reading; and $\pi_1(\mathcal{H}(W))$ abbreviates π_1 of the vertex set of the traversals of $\mathcal{H}(W)$. For a set X of cubes, $\mathfrak{N}_R(X)$ is the set of cubes at index distance at most R from a cube of X, so $|\mathfrak{N}_R(X)| \leqslant (2R+1)^4|X|$. The three ingredients of the footprint $F(W)$ have the following fixed ranges, in fine ℓ^∞ units. The free-input stencil $\mathcal{F}_r$ of a bond r* (3.13) *is $\{r\}$ or $\mathcal{E}_c \cap E_*$, bonds with both endpoints in two adjacent blocks of side L* (3.10)*, or empty (for a contour bond that is not a bond of Ω_1, the convention of Section 3.8), so every endpoint of a bond of $\mathcal{F}_r \cup \{r\}$ is within $2L$ of the initial vertex of r; this is the contour/free-input stencil of a traversal. The coefficient stencil halo of a domain D is the union of the hosts* (3.36)*,* (3.75) *whose base lies within R_{h} of D, where R_{h} is the maximal base-to-host offset of Section E.6.1, fixed by the host constructions: the maximal ℓ^∞ distance from the base of a host to a point of the host, so that each host lies within R_{h} of its base (the active hosts of a selection with final domain D are among these hosts, their bases lying in the R_{h}-neighbourhood of D); the halo therefore lies within $2R_{\mathrm{h}}$ of D. A root contact $n \in \mathsf{T}_D$* (3.87) *has a bond $b \in \mathcal{F}_{r_n} \cap \mathsf{L}_D$, whose endpoints lie in D ($Y_b \subseteq D$ by* (3.76)*, and Y_b contains b by its definition* (3.75)*, a rectangle containing the neighbourhood of the lift of b), so the initial vertex of r_n is within $2L$ of D. Put $R_{\mathrm{e}} = \lceil 2\max(R_{\mathrm{h}}, L)/(LM) \rceil$, a constant of L and M fixed before the coupling, $A_F = A_E = 9(2R_{\mathrm{e}}+1)^4$, $X(W) = \bigcup_D \pi_1(D)$ over the constituent domains D of W (roots and cluster constituents; a set, so $|X(W)| \leqslant n(W)$, which counts with multiplicity) and $\mathfrak{E}(W) = \mathfrak{N}_{R_{\mathrm{e}}}(X(W) \cup \pi_1(\mathcal{H}(W)))$, the envelope. Then: (a) $|\pi_1(\mathcal{H}(W))| \leqslant 8 + |\mathcal{H}(W)|/(LM)$, $|X(W) \cup \pi_1(\mathcal{H}(W))| \leqslant 9\mathfrak{s}(W)$, $F(W) \subseteq \mathfrak{E}(W)$ and $|F(W)| \leqslant |\mathfrak{E}(W)| \leqslant A_E \mathfrak{s}(W)$; (b) $\mathfrak{E}(W)$ is connected under maximum-norm adjacency of cubes (two distinct cubes are adjacent when their index distance is one); (c) $\mathcal{H}(W) \neq \varnothing$, and if $\Delta \in F(W)$ then $\operatorname{dist}(\Delta, C) \leqslant A_E LM\, \mathfrak{s}(W)$, that is $\mathfrak{s}(W) \geqslant \operatorname{dist}(\Delta, C)/(A_E LM)$, where $\operatorname{dist}(\Delta, C)$ is the ℓ^∞ distance in fine units from the nearest site of Δ to the vertex set of C (the proof bounds the distance from every site of Δ); (d) two incompatible groups have footprints in contact (sharing a cube, or adjacent across a face, edge or corner, that is at*

index distance at most one) or overlapping intervals, so an incompatible W' has a footprint cube in $\mathfrak{N}_1(F(W))$, a set of at most $3^4 A_F \mathfrak{s}(W)$ cubes, or a traversal in $\mathcal{H}(W)$.

Proof. Index conversion: if sites $x \in \Delta_z$ and $y \in \Delta_{z'}$ are at fine distance at most ρ, then, for lifts of x, y realising that distance and the indices z, z' of their lifted cubes, $|LM(z-z')|_\infty \leqslant \rho + 2 \cdot \frac{LM-1}{2} < \rho + LM$, so the index distance of Δ_z and $\Delta_{z'}$ is at most $\lceil \rho/(LM) \rceil$; in particular the cube of a site within $2L$ of a cube Δ is at index distance at most $\lceil 2L/(LM) \rceil = 1$ from Δ ($M \geqslant 8$), and a cube meeting the halo of D is at index distance at most $\lceil 2R_\mathrm{h}/(LM) \rceil \leqslant R_\mathrm{e}$ from a cube of D. Every root D of W is marked, that is $\mathsf{T}_D \neq \varnothing$ ((3.87) and the clause following it; the construction preceding Definition 3.24: every component has a marked root, and the groups are formed from the components), and $\mathsf{T}_D \subseteq \mathcal{H}(W)$, because $\mathcal{H}(W)$ is the hull of the union of the contacts of the group's components, which include the contact sets of its roots (the same construction, from "Every component has a marked root" to the definition of $\mathcal{H}(K)$). In particular $\mathcal{H}(W) \neq \varnothing$. (a) $\mathcal{H}(W)$ is a cyclic interval of fewer than $n_C/2$ traversals of the square C, or the whole contour (Definition 3.24), so the vertices of its traversals on each side form one run of consecutive collinear sites (the whole side, for the whole contour), and C does not wrap around the torus ($LK_i \geqslant 3L(n+2) > nL$). A run of t traversals along a side has $t+1$ consecutive collinear vertices, which lie in at most $\lfloor t/(LM) \rfloor + 2$ cubes, since the cubes cut the side into runs of LM consecutive sites; summing over the four sides, $|\pi_1(\mathcal{H}(W))| \leqslant 8 + |\mathcal{H}(W)|/(LM)$. Hence $|X(W) \cup \pi_1(\mathcal{H}(W))| \leqslant n(W) + 8 + |\mathcal{H}(W)|/(LM) = \mathfrak{s}(W) + 8 \leqslant 9\mathfrak{s}(W)$ by (E.17) and $\mathfrak{s}(W) \geqslant n(W) \geqslant 1$ (a group has a nonempty root family, Definition 3.24, and each root is a connected final domain, a nonempty union of cubes, Section 3.5). A cube of $F(W)$ (its definition follows (E.17)) is a cube of a constituent domain, in $X(W)$; or meets the halo of a constituent domain, at index distance at most R_e from $X(W)$; or contains an endpoint of a bond of the stencil of a traversal of $\mathcal{H}(W)$, within $2L$ of the initial vertex of that traversal, hence at index distance at most $1 \leqslant R_\mathrm{e}$ from $\pi_1(\mathcal{H}(W))$. So $F(W) \subseteq \mathfrak{E}(W)$ and $|\mathfrak{E}(W)| \leqslant (2R_\mathrm{e}+1)^4 \cdot 9\mathfrak{s}(W) = A_E \mathfrak{s}(W)$. (b) The set $X(W) \cup \pi_1(\mathcal{H}(W))$ is connected under adjacency: the two vertices of a traversal are at fine distance one and consecutive traversals of $\mathcal{H}(W)$ share a vertex, so the vertices lie in the same or in adjacent cubes, and $\pi_1(\mathcal{H}(W))$ is connected; each constituent domain is wall connected (Section 3.5), so its cubes are connected; within a cluster occurrence the constituent domains are linked by incompatibilities (the incompatibility graph of a cluster occurrence is connected: the Ursell coefficient of Lemma 3.20 is a sum over the connected spanning subgraphs of that graph, so a tuple with a disconnected graph has $\varphi^T = 0$ and is dropped from (3.88), as stated after that display), a cluster occurrence of W hits a root of W (Definition 3.24: some constituent is incompatible with a root), and incompatible domains share a cube or a wall (Section 3.6), that is contain cubes at index distance at most one; finally every root D has a contact $n \in \mathsf{T}_D \subseteq \mathcal{H}(W)$ (shown above), the initial vertex of r_n being within $2L$ of D, so a cube of $\pi_1(\mathcal{H}(W))$ is at index distance at most one from a cube of D. Thus every constituent domain is joined to $\pi_1(\mathcal{H}(W))$ through adjacent cubes, and $\mathfrak{E}(W)$, the union of the connected boxes $\mathfrak{N}_{R_\mathrm{e}}(\{\Delta\})$ over the cubes Δ of a connected set, each containing its centre Δ, is connected. (c) $\mathcal{H}(W) \neq \varnothing$ was shown above. A connected set of k cubes contains a path of at most k cubes between any two of them, so the cubes of any two of its sites are at index distance at most $k-1$ and the sites are at fine distance less than kLM; with $k \leqslant A_E \mathfrak{s}(W)$, the ℓ^∞ diameter of $\mathfrak{E}(W)$ is at most $A_E LM\, \mathfrak{s}(W)$. $\mathfrak{E}(W)$ contains $\Delta \in F(W)$ and the nonempty set $\pi_1(\mathcal{H}(W))$, whose cubes contain vertices of C. (d) Two groups are compatible when all their constituent domains are mutually compatible and their intervals are disjoint, and a family containing a whole-contour group is a singleton (Definition 3.24). So if W and W' are incompatible, either one of them is a whole-contour group, and then the intervals overlap, as $\mathcal{H}(W)$ and $\mathcal{H}(W')$ are nonempty by (c); or the intervals overlap, so

W' has a traversal in $\mathcal{H}(W)$; or some constituent domains D of W and D' of W' share a cube or a wall, so a cube of $D' \subseteq F(W')$ lies at index distance at most one from a cube of $D \subseteq F(W)$, that is in $\mathfrak{N}_1(F(W))$, and $|\mathfrak{N}_1(F(W))| \leqslant 3^4|F(W)| \leqslant 3^4 A_F \mathfrak{s}(W)$. *q.e.d.*

Lemma E.9 (Positive Group Sums)**.** *Choose $\eta > 0$ after the fixed geometry and forest decay constants, as in Section E.7.2 ($1000\eta A_{\mathrm{s}} < \varpi/8$ there). Under the smallness conditions $C\rho LM \leqslant \eta/8$ and the root-incidence bound of Section E.7.2, which hold for g_0 small with constants fixed before the coupling and independent of the loop length,*

$$\sup_{n \in \mathcal{I}_C} \sum_{W \,|\, n \in \mathcal{H}(W)} M(W) e^{4\eta \mathfrak{s}(W)} + \sup_{\Delta \in \pi_1} \sum_{W \,|\, \Delta \in F(W)} M(W) e^{4\eta \mathfrak{s}(W)} \leqslant C\rho \,, \tag{E.21}$$

with a constant C fixed before the coupling (the constants of Lemma E.17(b) and of Section E.7.3), where $\mathcal{I}_C$ is the set of contour traversals; the bound holds in the form "$\leqslant C\rho$ for every ρ below the threshold", the conditions being monotone in ρ. For footprint contact, wall adjacency and corner contact are included (Section E.11.4).

Proof. This is the combinatorial payment of Sections E.7.2 and E.7.3; its structure is as follows. First form the connected components of the roots/cluster-occurrence graph before hull merging. The factorials are the exponential-generating factorials of these objects: roots form an unordered distinct family and cluster occurrences carry $1/k!$. Dropping the compatibility constraints is an upper bound, and a connected graph indicator is at most its sum of incompatibility spanning trees. For a root node D the incident vacuum nodes have weighted mass at most $N_1(D)/8$ by the hit bound of Lemma 3.20 (its rooted form is (B.37)), in the exponentially weighted form (E.34) with weight $e^{n(\mathfrak{c})/4}$ that Lemma E.17 uses (Section E.11.4); for a vacuum node $\mathfrak{c}$ the same estimate summed over its constituents gives $n(\mathfrak{c})/8$; the added source-root mass is at most $C\rho$ times the same cost by the animal sum (B.2) and Lemma E.3, hence at most that cost/8 for small g_0. The two-species positive tree recursion is thus bounded with node cost one quarter of the cube cost: if t_v is a node weight and a_v its cost, the iterates $T_v^{(k+1)} = t_v \exp(\sum_{v' \sim v} T_{v'}^{(k)})$ satisfy $T_v^{(k)} \leqslant t_v e^{a_v}$, because $\sum_{v' \sim v} t_{v'} e^{a_{v'}} \leqslant a_v$, and their monotone limit sums the rooted trees. This pays the vacuum decorations and leaves a factor ρ for a component with a marked root. The same recursion pays the polynomial loss when a component is pinned at a point of its hull rather than at a contact: the hull length is bounded by the spatial extent, so a root lies within CLM times the cube cost of that point, and the root locations and the polynomial in the cube count are absorbed by the unused exponential decay; a spatial footprint pin is handled by a tree path from the marked root to the pinned cube, and the square has bounded local traversal density, so the sum over source contacts weighted by that distance is uniform in n_C. Then merge the components whose hull intervals overlap, summing the connected-overlap indicators by interval spanning trees with the same recursion: a hull $\mathcal{H}$ meets components of total weighted mass at most $C\rho|\mathcal{H}|$, its node cost contains $\eta|\mathcal{H}|/(LM)$, and $C\rho LM \leqslant \eta/8$ pays the branching; the weighted union size is at most the sum of the component sizes, and the exponential-generating factorials agree with the interleavings of the cluster lists exactly as in Lemma 3.25. The exceptional merge (a long hull merges every remaining component into the whole-contour group) is paid in Lemma E.18: a long class costs at least $cn_C/(LM)$ in the sum of its component sizes, the arbitrary remaining components cost at most $e^{C\rho n_C}$, and at small g_0 this is absorbed into a fraction of the long-class decay, not left as a loop-length factor. Lemmas E.17 and E.18 carry out these steps with the explicit weights. *q.e.d.*

Lemma E.10 (The Reference Group Gas)**.** *Put $\mathfrak{a}_W = \eta \mathfrak{s}(W)$. Under Lemma E.9, for every*

group W,

$$\sum_{W' \not\sim W} |a_{W'}| e^{\mathfrak{a}_{W'}} \leqslant C\rho\big(|F(W)| + |\mathcal{H}(W)|\big) \leqslant \frac{1}{8}\mathfrak{a}_W \tag{E.22}$$

after choosing g_0 small, every constant fixed before the coupling and independently of the loop length. Hence the reference group gas (3.94) *has a convergent connected-cluster logarithm, $S^{\circ}_{\xi,C} \neq 0$, and for every compatible family $\mathfrak{B}$*

$$\Big|\frac{S^{\circ,[\mathfrak{B}]}_{\xi,C}}{S^{\circ}_{\xi,C}}\Big| \leqslant \exp\Big(\frac{1}{8}\sum_{W\in\mathfrak{B}} \mathfrak{a}_W\Big)\,. \tag{E.23}$$

Proof. Incompatibility of W and W' implies contact of their footprints or overlap of their intervals, so the two pin sums (E.21) (with $|a_{W'}| \leqslant M(W')$ and $e^{\mathfrak{a}_{W'}} \leqslant e^{4\eta\mathfrak{s}(W')}$) give the first inequality of (E.22), and $|F(W)| + |\mathcal{H}(W)| \leqslant C_1 LM\mathfrak{s}(W)$ with $C\rho LM \leqslant \eta/8$, the constant C there taken to include C_1, gives the second. This is the Kotecký–Preiss criterion for the species W with activities a_W and costs $\mathfrak{a}_W$; the pinned-tree argument of Section B.6 gives the convergent logarithm and the exclusion identity, whose countable-species form is Lemma E.20. The restricted ratio is the single joint exclusion ratio of that gas, and the pinned cluster estimate counts each cluster hitting $\mathfrak{B}$ once, giving (E.23). There is no product-of-ratios assertion; this is the step that prevents a factor $e^{C\rho n_C}$ from replacing the flat scalar in E_{def}. *q.e.d.*

E.5.3 Subtraction and Its Curvature Payment

For a group W its local comparison box Q_W contains every constituent domain's coefficient halo and every background transport along $\mathcal{H}(W)$, including the inverse interval used to close the source; its lifted side is at most $CLMn(W)$ (Section E.7.4). For the good interior boxes of Section E.7.4, contained in Λ_1 and disjoint from S_1, root a tree gauge at the base of the complementary holonomy. The ordered plaquette filling of Lemma E.21 gives a phase bound $C\,\mathrm{diam}(Q_W)^2\delta_{Q_W}(U_1)$, where $\delta_Q(U)$ is the maximum plaquette deviation $|U(\partial p) - \mathbf{1}|_\sigma$ over the plaquettes of Q. The source is a closed contraction throughout. The derivative bound of Section E.7.4 ((E.44) with Lemmas E.16 and E.4), applied to this coefficient with its cluster list and the whole relative interval retained, costs the factor $n(W) + |\mathcal{H}(W)|$ and the factor ρ for each marked root; integrating the real phase path from the flat projection to U_1 (Lemma E.19) gives $\mathfrak{d}_W \leqslant C\delta_{Q_W}\,\mathrm{diam}(Q_W)^2$ times that polynomial times $M(W)$. If the phase bound exceeds the fixed interpolation chart, the two undifferentiated endpoint bounds give the same inequality with a larger constant (Section E.11.2). No g_0-dependent restriction on the group size is added.

The regional carrier I is retained in the reference coefficient. The comparison is used only for groups whose box, with every coefficient and source halo, lies in Λ_1 and misses S_1; boxes crossing that integration boundary are kept in the large-domain class and are not interpolated, which leaves the target unchanged because their full contribution is paid by the tail below. On an interior box the centred kernel depends on the branch fields only through U_1 there: the retained field and the boundary change are absent from its source rows and primitive arguments, and V_0 enters only through that background (Section E.7.1). On far histories $\mathrm{dist}(\Gamma_C, E(\Lambda_1^c)) > b$, and Section E.7.4 uses the good comparison only when $n(W) < b/(4C_{\mathrm{box}}LM)$, which places every required stencil in an interior lifted box; all other groups, which have $n(W) \geqslant b/(4C_{\mathrm{box}}LM)$, keep both endpoint coefficients and the undifferentiated estimate, no assertion about their background continuation being made, and the reserved exponential pays this tail.

Lemma E.11 (The Group Pin Sum)**.** *On covered far histories, with* $\varepsilon_1 = g_1 p_0(g_1)$*,* $b = \lfloor LMR_1/16 \rfloor$ *and constants* C_χ*,* c *fixed before the coupling,*

$$\sum_W K_\chi \mathfrak{d}_W(U_1) e^{\mathfrak{a}_W/8} \leqslant C_\chi n_C \rho\big(\varepsilon_1 + e^{-cb/(LM)}\big) \,. \tag{E.24}$$

The proof is Section E.11.2, from the local estimate of Section E.7.4 and the pins of Lemma E.9. The polynomial factors $n(W)^k$, the squared box size and the interval length are paid by the reserved domain decay, not placed in a loop-dependent constant.

E.5.4 The Signed Integrated Bound

Lemma E.12 (The Family Sum)**.** *On covered far histories,*

$$|\mathscr{R}_\xi| \leqslant |S^\circ_{\xi,C}| \Big[\exp\Big(C_\chi n_C \rho\big(\varepsilon_1 + e^{-cb/(LM)}\big)\Big) - 1 \Big] \,. \tag{E.25}$$

Proof. Use the sum (E.1): absolute values are taken only after the flat subtraction has been made in each $R_\xi(\mathfrak{B})$ and the joint reference ratio (E.23) has been retained. By (E.20) and (E.23), $|S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B})| \leqslant |S^\circ_{\xi,C}| \prod_{W\in\mathfrak{B}} K_\chi \mathfrak{d}_W(U_1) e^{\mathfrak{a}_W/8}$, and the sum over nonempty compatible families is bounded by the exponential series of the one-group sum (E.24) minus its zeroth term. The restricted reference sums, not a new vacuum normalisation, supply the factor $|S^\circ_{\xi,C}|$; at no stage is w_ξ cancelled across histories. The countable-group limit is Lemma E.20. *q.e.d.*

Together with Lemma E.1 this gives the first form of the defect bound, (E.78) below, with ρ of (E.11) and all physical history weights and $\mathcal{Z}_0$ retained. It leaves the regional scalar $S^\circ_{\xi,C}$ inside the expectation; the identification of that scalar with the perimeter/corner coefficient is Appendix F at the carrier I, and, for the labelled scalar S°_Λ at the retreated carrier, Section 9.9 (the defect bound at $D(\Lambda)$ is Proposition 9.40); neither is part of this appendix. The reconstructed fine background loop is not replaced by the coarse loop.

E.6 The Joint Kernel Payment

This section supplies the quantitative steps used in Section E.4; it adds no assumption on E_{def}.

E.6.1 Host Budget and Fixed Parameter Order

C_{p} bounds one primitive (Lemma B.9) and $C_0 = N_{\mathrm{h}} C_{\mathrm{p}} e^{\kappa(B_{\mathrm{h}}+4)}$ bounds their sum at a host (Lemma B.10). Let C^∂_{p} be the maximum of the finite local coefficient-derivative bounds of Section E.9.1: the energy primitive is g_0 times the third-order Taylor remainder in the protected variables, and its background derivative has the same factor $g_0(1+r)^3$ because its local analytic coefficients and their background derivatives are bounded on the larger fixed row chart; the Haar primitive and its background derivative start at order $g_0^2(1+r)^2$; the constraint-Jacobian primitive and its derivative are bounded by $Cg_0(1+r)$. Set

$$C^\partial_0 = N_{\mathrm{h}} \max(C_{\mathrm{p}}, C^\partial_{\mathrm{p}}) e^{\kappa(B_{\mathrm{h}}+4)}, \qquad \omega(g_0) = C^\partial_0 g_0 (1+r)^3, \qquad R_\Upsilon = \frac{\omega_*}{2\omega(g_0)} \,, \tag{E.26}$$

C_0^∂ chosen after M and before g_0, as C_0 is ($\omega(g_0)$ and R_Υ then depend on g_0 through $g_0(1+r)^3$); neither ω_* nor ε_* nor any earlier choice is shrunk after M. Summing the same finite host assignments proves $\sup|\mathscr{D}_\omega \mathbf{V}_\xi(Y)| \leqslant \omega(g_0)e^{-\kappa d_0(Y)}\|\omega\|_\infty$ on the background neighbourhood of Section E.9.1 (phases within τ_{b} of the branch background, Lemma E.15), where the primitive bounds hold with the same constants, so that $\omega(g_0) \geqslant C_0 g_0(1+r)^3$ also bounds $\sup|\mathbf{V}_\xi(Y)|e^{\kappa d_0(Y)}$ there ((B.21)); in Sections E.4 and E.5 this $\omega(g_0)$ replaces the $\omega(g_0)$ of (B.20), which it dominates. The coupling threshold is $\omega(g_0) \leqslant \omega_*/4$, that is $R_\Upsilon \geqslant 2$ (Lemma E.2). Let N_{type} be the number of tagged ordinary primitive and probe types based in one fine translation cell of the fixed L-periodic template and R_{h} the maximal base-to-host offset, both fixed by the host constructions (3.36), (3.75). Distinct active hosts of a selection with final domain D have a labelled base in the fixed R_{h}-neighbourhood of D, so their number is at most $H_M N_1(D)$ with $H_M = N_{\mathrm{type}}(LM + 2R_{\mathrm{h}} + 3)^4$, and the selected probe count is at most the number of fine bonds in D, $q \leqslant 4(LM+1)^4 \sum_D N_1(D)$. The quantities $\omega(g_0)$ and R_Υ of (E.26) are used throughout Sections E.4 and E.5; only the coupling threshold changes.

Lemma E.13 (Positive Host Comparison)**.** *For a host Y let $n_Y \leqslant P_M$ be the number of probe labels with host Y and put*

$$A_Y = \frac{\omega_*}{2}e^{-\kappa d_0(Y)}, \qquad B_Y = n_Y u_* \leqslant \frac{\omega_*}{8}e^{-\kappa d_0(Y)} \tag{E.27}$$

(the inequality by $n_Y \leqslant P_M$, $u_* \leqslant \omega_* e^{-\kappa s_{\mathrm{p}}}/(8P_M)$ from (E.9)*, and $d_0(Y) \leqslant N_0(Y) - 1 < s_{\mathrm{p}}$ for a probe host,* (B.1) *and Lemma E.2). (a) With $v_{\mathrm{ordinary}} = e^{A_Y} - 1$ and $v_{\mathrm{probe}} = u_*$, the sum over the nonempty label subsets at Y satisfies*

$$\begin{aligned}\sum_{\varnothing \neq S \subseteq \{\mathrm{ordinary,probes}\}} \prod_{\ell' \in S} v_{\ell'} &\leqslant e^{A_Y}(1+u_*)^{n_Y} - 1 \leqslant e^{A_Y + B_Y} - 1 \\ &\leqslant (A_Y + B_Y)e^{A_Y+B_Y} \leqslant \omega_* e^{1/2} e^{-\kappa d_0(Y)} .\end{aligned} \tag{E.28}$$

(b) For a differentiated host, the summed weight of the label subsets at Y is bounded by

$$R_\Upsilon \sup|\mathscr{D}_\omega \mathbf{V}_\xi(Y)|\, e^{A_Y}(1+u_*)^{n_Y} \leqslant A_Y e^{A_Y+B_Y}\|\omega\|_\infty \leqslant \omega_* e^{1/2} e^{-\kappa d_0(Y)}\|\omega\|_\infty . \tag{E.29}$$

(c) There is a constant C_M, fixed after M and before g_0, such that for every connected final domain D

$$\sum_{\ell \,|\, Z(\ell) = D} u_*^{q(\ell)} \sup_{|\zeta| \leqslant R_\Upsilon} \|\mathscr{D}_\omega \Lambda^{\mathrm{f}}_{\xi;\ell,\zeta}\|_{\mathrm{TV}} \leqslant C_M N_1(D)\|\omega\|_\infty \varepsilon_* e^{-\lambda_* \ell_1(D)} , \tag{E.30}$$

the derivative including every Gaussian normaliser and the compensator; and $|\mathscr{D}_\omega k^{\mathrm{f}}_\xi(D)| \leqslant C_M N_1(D)\|\omega\|_\infty m(D)$ for the vacuum activities, without shrinking ε_*.

Proof. The bound $n_Y \leqslant P_M$ holds because a probe label b lies in its host, whose fine side is at most $4M$, and there are four positive directions per site. After taking absolute values inside the real Mayer t-integral the ordinary factor is at most $e^{A_Y} - 1$ on $|\zeta| \leqslant R_\Upsilon$ (as $R_\Upsilon \omega(g_0) = \omega_*/2$). The last inequality of (E.28) uses $\omega_* \leqslant 1/2$ (the standing choice of Section B.8) and $A_Y + B_Y \leqslant \frac{5}{8}\omega_* e^{-\kappa d_0(Y)}$. This is a positive numerical comparison, not the Mayer factor of a complex combined potential. All split selections with the same active-host set have the same Y_0, seed, conditioning carrier and Gaussian kernel and, at fixed bad-bond choice P and final domain Z, the same final-domain weakening (proof of Lemma 3.21), so their many-to-one sum is dominated at each host by (E.28) before the cover and seed sums of Appendix B are applied; this is the domination used in Lemma E.2, the final domain sum being the chain (B.34), (B.35) and the final Y'-sum in the proof of Lemma 3.19. (b) is

the same comparison with the differentiated host bound of Section E.6.1. (c) There are at most $H_M N_1(D)$ active hosts; the Gaussian and walk derivatives have the same factor after their polynomial moment is paid (Sections E.9.2, E.9.3): the uniform Gaussian coercivity gives polynomial moments at most $C_M N_1(D)$ times the same exponential volume majorant $e^{N_1(D)}$, so no fresh exponential reserve is spent. Executing the differentiated sum with the positive host comparison (b) gives (E.30), and the same sum over the selections with $\mathsf{J} = \varnothing$ gives the vacuum-activity bound. *q.e.d.*

E.6.2 The Multi-root Cancellation

The input f is a single normalised character of the ordered closed reconstructed word, optionally closed with its inverse background interval and a fixed complementary h; for a linear combination the bound is multiplied by the sum of the absolute coefficients, and an arbitrary unbounded sum is not an input. On covered branches its contour input labels lie in I, so the retained A_{e} and the boundary map do not enter the source rows (Lemma 3.28); the centred kernel $K_{\xi;0}$ likewise uses zero retained input, and the reference group construction uses the same row and mean conventions.

Lemma E.14 (Commuting Projections)**.** *Let O_0 be the set of roots with odd q_j and empty ordinary selection, O_1 the set of roots with odd q_j and nonempty ordinary selection. Then*

$$\Big|\int \Delta_{\mathsf{J}} f \,\mathrm{d} \bigotimes_j \Lambda^{\mathrm{f}}_{\xi;\ell_j}\Big| \leqslant u^q \prod_{j\in O_0\cup O_1} \Big(u + \frac{2}{R_\Upsilon - 1}\Big) \prod_j M_{\ell_j} \,, \tag{E.31}$$

with the exponential-in-q constants absorbed in C_χ; after one background derivative the same holds with the factor M_{ℓ_j} of the differentiated root replaced by $\sup_{|\zeta|\leqslant R_\Upsilon} \|\mathscr{D}_\omega \Lambda^{\mathrm{f}}_{\xi;\ell_j,\zeta}\|_{\mathrm{TV}}$, which (E.30) *sums, and with the additional factor of Lemma E.4. Moreover $\Delta_{\mathsf{J}} f = 0$ if some selected label does not influence the closed source word, so every surviving root has a contact with Γ_C.*

Proof. For $j \in O_0$ let P_j take the homogeneous part of degree q_j of $\Delta_{\mathsf{J}} f$ in the selected variables of root j, the other root variables fixed; these projections commute. Expand $\prod_{j\in O_0}(P_j + (1 - P_j))$. A term containing P_j is odd under the reversal of exactly root j and integrates to zero against that root's even source-free density; only $\prod_{j\in O_0}(1 - P_j)$ remains, with one extra u per such root by the mixed Taylor remainder (E.61) and the fixed row-overlap count. For $j \in O_1$ the independent ζ_j-integral vanishes at $\zeta_j = 0$, so the multivariable Cauchy difference supplies $\prod_{j\in O_1} 2/(R_\Upsilon - 1)$ times the supremum $\prod M_{\ell_j}$ of (E.8). Together this is (E.31), and for even q_j only $u^{q_j} M_{\ell_j}$ is paid. After one background derivative the Gaussian derivative is even, the leading source derivative has the same parity in the selected variables (a background derivative hits a row coefficient or a background occurrence of the closed word, not the selected root variables, Section E.9.4; the resulting bound is (E.62)) and a differentiated Mayer factor still vanishes at $\zeta_j = 0$, so the argument repeats. The last statement is the pairwise cancellation of K with $K \cup \{b\}$ in the K-sum (E.4) for a label b of a root that the word does not see; it is the pin used in the $\mathcal{A}$- and W-sums, and no sum over unmarked roots in the volume is claimed to be uniform. *q.e.d.*

E.6.3 The Common Hessian Bounds on the Extension

Lemma E.15 (Hessian Bounds along the Coefficient Path)**.** *Choose $\tau_{\mathrm{b}} > 0$ once, after τ_L, c_L, γ_L and before g_0, with $\tau_{\mathrm{b}} \leqslant \tau_L/4$, $c_L\tau_{\mathrm{b}} \leqslant \gamma_L/4$ and the chart conditions of Section E.9.1 (the matrix-logarithm neighbourhood and the entry-chart norm, each a smallness condition*

on τ_{b} in terms of constants fixed before g_0); this one τ_{b} is used throughout Sections E.8–E.11. For every $t \in [0,1]$ and every globally small extension $U = e^{itB}U^{\circ}$ with $\|B\|_{\infty} \leqslant \tau_{\mathrm{b}}$ of the flat background on a comparison box, every compression of the regional Hessian to the same I satisfies

$$\frac{3}{4}\gamma_L \, \mathbf{1} \leqslant H_{e^{itB}} \leqslant m_L \, \mathbf{1} \, . \tag{E.32}$$

Proof. On physical branches Lemma 3.4 gives $\gamma_L/2 \leqslant H_{\xi,II} \leqslant m_L$ with the force–constraint and soft-gauge terms. The local entry formula $f_{bb'}(A)$ of Lemma B.6 has row sums at most m_L on the phase box τ_L by the definition of m_L in (3.22), column sums equal to its row sums since $f_{bb'} = f_{b'b}^{\mathsf{T}}$ have equal norms, entries vanishing beyond the range of that lemma, and Lipschitz bound c_L in the phase; so the extension has operator difference from $H^{\circ} = \mathsf{H}_{\mathbf{1}}^{\Omega_1}$ at most $c_L\tau_{\mathrm{b}} \leqslant \gamma_L/4$ by the Schur test, uniformly in t. The flat lower bound γ_L ((A.25)) passes to every compression to I by the compression argument in the proof of Lemma B.13, and with the difference bound this gives (E.32). The recipe $\gamma = \gamma_L/2$, $h_+ = m_L$ of Section 3.5 therefore works at the physical endpoint and throughout the extension; no new expansion instantiation or coefficientwise identification of different recipes is used, and nothing is asserted about the global closeness of the physical background to the identity. *q.e.d.*

E.6.4 Differentiating the Joint Vacuum Ratio

$\Phi^{\mathrm{v}}_{\xi}(\mathfrak{c})$ is the Ursell expression $\varphi^{T}(\mathfrak{c}) \prod_j k^{\mathrm{f}}_{\xi}(D_j)/m!$ of (3.73) with the fixed-kernel vacuum activities; that formula is used directly.

Lemma E.16 (Derivative of the Cluster Sum)**.** *Let $M_v(\mathfrak{c})$ be the positive incompatibility-tree majorant of $\Phi^{\mathrm{v}}_{\xi}(\mathfrak{c})$ built from $m(D)$ (Section E.5.2) and $n(\mathfrak{c}) = \sum_{D\in\mathfrak{c}} N_1(D)$. Then*

$$|\mathscr{D}_{\omega}\Phi^{\mathrm{v}}_{\xi}(\mathfrak{c})| \leqslant C_M \|\omega\|_{\infty} \, n(\mathfrak{c}) M_v(\mathfrak{c}) \, , \tag{E.33}$$

and for every finite set $\mathcal{A}$ of connected final domains

$$\sum_{\mathfrak{c} \text{ hits } \mathcal{A}} M_v(\mathfrak{c}) e^{n(\mathfrak{c})/2} \leqslant \frac{1}{8} \sum_{D\in\mathcal{A}} N_1(D), \qquad \sum_{\mathfrak{c} \text{ hits } \mathcal{A}} |\mathscr{D}_{\omega}\Phi^{\mathrm{v}}_{\xi}(\mathfrak{c})| \leqslant \frac{C_M}{2}\|\omega\|_{\infty} \sum_{D\in\mathcal{A}} N_1(D) \, . \tag{E.34}$$

Proof. (E.33) is the product rule over the constituents with the activity derivative bound of Lemma E.13(c): a sum over the possible differentiated vertices, not a division by an activity. The numerical criterion of Lemma 3.19, (B.36), is stated for $|k^{\mathrm{f}}_{\xi}|$; its proof uses only $|k^{\mathrm{f}}_{\xi}(D)| \leqslant m(D) = \varepsilon_* e^{-\lambda_* \ell_1(D)}$, so it gives $\sum_{D' \not\sim D} m(D') e^{(1+6\theta)N_1(D')} \leqslant N_1(D)/8$ as well. Apply the rooted-tree proof of Lemma 3.20 (Appendix B.6) to the activity $m(D)e^{N_1(D)/2}$ with node cost $N_1(D)/8$, the combined exponent $5/8$ being below $1 + 6\theta$ (Section E.11.4 records this $n/2$ form beside the $n/4$ form). This gives the first bound of (E.34), the family version following by summing the single-root bound over $D \in \mathcal{A}$ (clusters hitting several roots are overcounted in this majorant only, not in the exact ratio). Since $n \leqslant 4e^{n/4} \leqslant 4e^{n/2}$, inserting (E.33) gives the second bound. Inserting the second bound into the derivative of the exponential (E.14) proves (E.15): C_M is an outside prefactor for a single derivative and is not put into every cluster weight, so no M-dependent shrinking of ω_* or ε_* is needed after those constants have been fixed. The same $m(D)$ controls the undifferentiated activity along the whole extension ($|k^{\mathrm{f}}_{\xi}(D)| \leqslant m(D)$ there by Lemma E.15 and the host bounds of Section E.9) and its one-derivative polynomial majorant, the polynomial being paid by the displayed exponential moment. *q.e.d.*

E.7 Quantitative Group and Caller Checks

E.7.1 Carrier, Source and Physical Weights

All group comparisons and flat-subtraction identities are made on A_{cov}. There the source rows have their free inputs in I, so Lemma 3.28 gives $U_\xi(0)(r) = U_1(r)$ for every contour bond r, and the free vector $B = a + \Psi_\xi(A_{\mathrm{e}})$ equals a on I ($\Psi_\xi(A_{\mathrm{e}})$ vanishes on I, proof of that lemma), so neither A_{e} nor Ψ_ξ enters the fluctuating source rows; Lemma 3.9 and (3.49) give $K_{\xi;1}$, and $K_{\xi;0}$ is the $s = 0$ member of the interpolation (E.66), with $H = H_{\xi,II}$, $P_{\xi,0}(a) = \sum_{\mathfrak{p}\ \mathrm{near}} \mathfrak{p}(a)$ and the chosen r-box, every primitive argument having zero retained input. The boundary Jacobian, the far primitives and the retained quadratic term are in c_ξ, not in the centred coefficients, and every fixed-kernel operation, probe measure and vacuum activity is built from those centred objects. Hence the local background data of a coefficient are U_1, the unchanged carrier geometry and the fixed coupling/cutoff/mean prescription; V_0 is not replaced in the physical integral, and its influence on these local coefficients is through U_1.

The local-data comparison of Section E.10 applies on a box containing all source and coefficient stencils and lying in Λ_1 outside S_1. The fixed-kernel support statements (Section E.9.2) place every surviving walk coefficient, seed inverse and primitive stencil there, and the regional I and Ω_1 are kept on both sides of the interpolation. At the flat endpoint these local formulas are the reference formulas for a_W, since their arguments contain no retained field, boundary map or additional V_0: an equality of the selected coefficient, not of physical densities. The force–constraint and relative soft-gauge terms are those of the regional Hessian (3.16) and are included throughout.

Under a simultaneous site gauge each Hessian entry, restricted inverse and fixed walk coefficient is conjugated by the free-field adjoint action (Lemma 3.3, Lemma 3.13), the protected cutoffs are invariant, and the row solution is covariant; orthogonal changes of the integration variables leave the closed selected integrals unchanged, including their spatial Cauchy integrals, and their Laplace transforms give the covariance of each probe measure; the vacuum activities and Ursell products are scalar invariants; finally $T_W P_W^{-1}$ and h are simultaneously conjugated at the interval base. This is the gauge law of the group closed contraction factor by factor (Section E.10.2); no straight-side assumption enters it.

$\mathcal{O}$ is the positive outer operation (3.4): the Haar and constraint factors are measures and the displayed cutoffs and exponentials are nonnegative, and $0 < \mathcal{Z}_0 < \infty$ (finite by the measurability paragraph of Section 3.1; $\mathcal{Z}_0 = \int \mathcal{O}[\mathcal{J}_\xi^{\mathrm{a}}[1]]\,\mathrm{d}V = \int \mathcal{O}[\mathsf{w}_\xi]\,\mathrm{d}V$ is the normaliser of (3.7), the integral of a positive function, since $\nu_{\xi;1}$ is a probability measure and so $\mathcal{J}_\xi^{\mathrm{a}}[1] = \mathsf{w}_\xi$ by (3.49)), so $\mathbb{E}_{\mathrm{h}}$ is a positive normalised functional. Multiply (3.97) by $\mathbf{1}_{A_{\mathrm{cov}}}$ and write $\mathcal{J}_\xi^{\mathrm{a}}[f_\xi] = \mathsf{w}_\xi F_\xi(1)$ on A_{cov}^c; every term is integrable against the finite positive measure $\mathcal{O}[\mathsf{w}_\xi \bullet]\,\mathrm{d}V$, because $|S^\circ_{\xi,C} f_\xi(0)| = |F_\xi(0) - \mathscr{R}_\xi| \leqslant 3$ by (3.96), $|F_\xi(0)| \leqslant 1$ and Lemma E.1, $|F_\xi(1)| \leqslant 1$, and $|\mathcal{B}_\xi[f_\xi]| \leqslant C_{\mathrm{resp},\chi} g_0 r |I_C|$ on A_{cov} (Section D.6). With E_{bdry} of (D.21), the full identity is

$$\mathbb{E}_\varepsilon[W_\chi(C)] = \mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}} S^\circ_{\xi,C} f_\xi(0)\big] + E_{\mathrm{def}} + E_{\mathrm{bdry}}\,, \tag{E.35}$$

by (3.97) and (3.7). The far uncovered event is empty: if coverage fails, either a contour bond r is not a bond of $\Omega_1 \supseteq \Lambda_1$, so that r has an endpoint in $\Lambda_1^c = B_\xi$, or a bond of a stencil $\mathcal{F}_r \not\subseteq I$ is a free bond with no endpoint in Λ_1 ($I = \Lambda_1^*$, Section 3.1); in both cases an element of Γ_C (a contour bond or a bond of a free-input stencil) lies in $E(B_\xi)$, so $\mathrm{dist}(\Gamma_C, E(B_\xi)) = 0$ and $A_{\mathrm{cov}}^c \subseteq \mathsf{N}_0 \subseteq \mathsf{N}_b$ ((E.2); the events N_b increase with b). The exterior response is not dropped: it is the separately controlled E_{bdry} of (D.21) and Appendix D, not a part of E_{def}.

E.7.2 Two Quantitative Pins

A component γ consists of a nonempty unordered root family $\mathcal{A}(\gamma)$ and an ordered list $(\mathfrak{c}_1, \dots, \mathfrak{c}_k)$ of cluster occurrences with connected incidence graph as in Section 3.8; every root is a marked domain, so γ has at least one marked root and $\mathsf{T}_\gamma \neq \varnothing$ (only families of marked domains contribute, Section 3.8). Here n_γ counts every constituent cube with its occurrence multiplicity, T_γ is the union of the contact sets of its roots, H_γ the hull of T_γ (a cyclic interval of fewer than $n_C/2$ traversals, or the whole contour, Section 3.8), X_γ the union of its domains (roots and cluster constituents), and $F(\gamma)$ its footprint: the LM-cubes meeting a domain of X_γ, its fixed coefficient stencil halo, or the contour/free-input stencil of a traversal of H_γ, as $F(W)$ in Section E.5, with the contact conventions of Section E.11.4 (wall adjacency and corner contact) used in (E.21). A class of components (defined below) is long when the union of the hulls of its members has at least $n_C/2$ positions, in which case all components merge into the whole-contour group (Section 3.8). Define

$$t_\gamma = \frac{\rho^{|\mathcal{A}(\gamma)|}}{k!} \prod_{D \in \mathcal{A}(\gamma)} m(D) \prod_{j=1}^{k} M_v(\mathfrak{c}_j) \,. \tag{E.36}$$

Summing unordered distinct roots is overcounted by ordered root tuples divided by their factorial. A class K is an unordered family of components with connected hull-overlap graph, with weight $t_K = \prod_{\gamma \in K} t_\gamma$; for the positive recursion the sum over classes is overcounted by ordered tuples divided by the tuple factorial, dropping the disjointness restrictions, which only enlarges the sum and fixes every multiplicity convention. Fix

$$0 < \varpi \leqslant \min\Big(\frac{1}{16}, \frac{\lambda_*}{64 C_{\mathrm{g}}}\Big) \,, \tag{E.37}$$

with C_{g} of Lemma B.1.

Lemma E.17 (Component Pins)**.** *There are constants C_{pin}, A_{h}, A_{x} fixed before the coupling such that, for ρ below a threshold depending on them: (a) for every root D_0,*

$$\sum_{\gamma \,|\, D_0 \in \mathcal{A}(\gamma)} t_\gamma e^{\varpi n_\gamma} \leqslant \rho\, m(D_0) e^{(\varpi + 1/4) N_1(D_0)}, \qquad \sup_{a \in \mathcal{I}_C} \sum_{\gamma \,|\, a \in \mathsf{T}_\gamma} t_\gamma e^{\varpi n_\gamma} \leqslant C_{\mathrm{pin}} \rho \,; \tag{E.38}$$

(b) $|H_\gamma| \leqslant A_{\mathrm{h}} LM n_\gamma$, and the distance from a source contact of γ to any cube of X_γ, halos included, is at most $A_{\mathrm{x}} LM n_\gamma$; with $\theta_{\mathrm{h}} = \varpi/(4A_{\mathrm{h}}LM)$, $\theta_{\mathrm{x}} = \varpi/(4A_{\mathrm{x}}LM)$ and d_C the contour distance,

$$\sum_{\gamma \,|\, a \in H_\gamma} t_\gamma e^{\varpi n_\gamma/4} \leqslant \sum_{a' \in \mathcal{I}_C} e^{-\theta_{\mathrm{h}} d_C(a,a')} \sum_{\gamma \,|\, a' \in \mathsf{T}_\gamma} t_\gamma e^{\varpi n_\gamma/2} \leqslant C_{\mathrm{pin}} \rho \Big(1 + \frac{2}{e^{\theta_{\mathrm{h}}} - 1}\Big) \,, \tag{E.39}$$

$$\sum_{\gamma \,|\, \Delta \subset X_\gamma} t_\gamma e^{\varpi n_\gamma/4} \leqslant C_{\mathrm{pin}} \rho \sum_{a' \in \mathcal{I}_C} e^{-\theta_{\mathrm{x}} d_T(a', \Delta)} \leqslant C\rho \,, \tag{E.40}$$

with $d_T(a', \Delta)$ the periodic fine distance (Section E.2) from the bond of the traversal a' to the cube Δ, and the footprint pin $\sum_{\gamma \,|\, \Delta \in F(\gamma)} t_\gamma e^{\varpi n_\gamma/4} \leqslant C\rho$ with the constant C enlarged by the stencil factor CLM; (c) with $A_{\mathrm{s}} = 1 + A_{\mathrm{h}}$ and $\eta > 0$ chosen after these constants with $1000\eta A_{\mathrm{s}} < \varpi/8$, the three pins of (b) hold with the weight $e^{100\eta \mathfrak{s}(\gamma)}$, $\mathfrak{s}(\gamma) = n_\gamma + |H_\gamma|/(LM)$, leaving a fixed exponential margin.

Proof. (a) The sum is a two-species positive tree recursion, the recursion of the proof of Lemma 3.20 (Appendix B.6) with two kinds of nodes: the roots D, with weight

$\rho\, m(D) e^{\varpi N_1(D)}$ and cost $N_1(D)/4$, and the cluster occurrences $\mathfrak{c}$, with weight $M_v(\mathfrak{c}) e^{\varpi n(\mathfrak{c})}$ and cost $n(\mathfrak{c})/4$; a connected incidence graph is bounded by the sum of its spanning trees, the factorials are the exponential-generating factorials of the unordered roots and the ordered occurrence list, and the iterates $T_v^{(k+1)} = t_v \exp(\sum_{v' \sim v} T_{v'}^{(k)})$ satisfy $T_v^{(k)} \leqslant t_v e^{a_v}$ once $\sum_{v' \sim v} t_{v'} e^{a_{v'}} \leqslant a_v$ for every node v (weights t_v, costs a_v). The criterion holds. The occurrences hitting a node have weighted mass at most one half of its cost, by (E.34) applied to the node's constituents, since $e^{\varpi n} e^{n/4} \leqslant e^{n/2}$ ($\varpi + 1/4 < 1/2$). A root hitting a node is incompatible with one of its constituents, so it contains one of the at most $9N_1$ cubes meeting a constituent cube or sharing a wall with it (N_1 the node's cube count); for a fixed cube Q,

$$\sum_{D \ni Q} m(D) e^{(\varpi + 1/4) N_1(D)} \leqslant \varepsilon_* \sum_{D \ni Q} e^{-(\lambda_* - (\varpi + 1/4) C_{\mathrm{g}}) \ell_1(D)} \leqslant \varepsilon_* B(\lambda_*/2) =: C_{\mathrm{a}} \,,$$

by $N_1(D) \leqslant C_{\mathrm{g}} \ell_1(D)$ (Lemma B.1(1); the final domains are localisation domains) and the animal bound (B.2), applicable because $\lambda_* - (\varpi + 1/4) C_{\mathrm{g}} \geqslant \lambda_* - \frac{5}{16} C_{\mathrm{g}} \geqslant \lambda_*/2 > C_{\mathrm{g}} \log 64$ by $\varpi \leqslant 1/16$ and $\lambda_* \geqslant 2 C_{\mathrm{g}} \log 128$ ((B.26)). So the roots hitting a node have weighted mass at most $9\rho C_{\mathrm{a}} N_1 \leqslant N_1/32$, one eighth of the cost, once $9\rho C_{\mathrm{a}} \leqslant 1/32$, a condition on ρ alone, that is on g_0. The rooted sum at D_0 is therefore at most $\rho\, m(D_0) e^{\varpi N_1(D_0)} e^{N_1(D_0)/4}$, the first bound of (E.38). A root D_0 with source contact a has a bond of L_{D_0} in the stencil $\mathcal{F}_{r_a}$ ((3.87)), so it contains one of the at most C_{st} cubes meeting that stencil (C_{st} a constant of L and M); summing the first bound over these roots by the display gives the second bound with $C_{\mathrm{pin}} = C_{\mathrm{st}} C_{\mathrm{a}}$. (b) The two geometric bounds follow from the square geometry and the wall connectivity of a component's domains (the geometry paragraph of Section E.5). If a traversal a lies in H_γ and a' is a contact of γ, then $d_C(a, a') \leqslant A_{\mathrm{h}} LM n_\gamma$, so $e^{\varpi n_\gamma/4} \leqslant e^{\varpi n_\gamma/2} e^{-\theta_{\mathrm{h}} d_C(a,a')}$, and the contour sum $\sum_{a'} e^{-\theta_{\mathrm{h}} d_C(a,a')} \leqslant 1 + 2/(e^{\theta_{\mathrm{h}}} - 1)$ is uniform in n_C. For a cube $\Delta \subset X_\gamma$ the analogous inequality holds with θ_{x} and the torus distance; for each of the four straight sides the sum over a' is at most $1 + 2/(e^{\theta_{\mathrm{x}}} - 1)$ up to the fixed cube/halo offset, bounded by the full one-dimensional periodic lattice sum independently of n and the periods. This retains ρ even when Δ lies in a cluster domain. A cube $\Delta \in F(\gamma)$ either meets a domain of X_γ or its stencil halo (a fixed offset, absorbed as above) or meets the stencil of a traversal of H_γ, and at most CLM traversals have stencils meeting a given cube; so the footprint pin is at most (E.40) plus CLM times the hull pin (E.39), which is the stated bound with the enlarged constant. (c) $100\eta\mathfrak{s}(\gamma) \leqslant 100\eta A_{\mathrm{s}} n_\gamma < \varpi n_\gamma/8$, so the enhanced weights are dominated by the $e^{\varpi n_\gamma/4}$ of (b) with a margin $e^{\varpi n_\gamma/8}$. All constants are fixed before g_0; the only condition on ρ in this lemma is $9\rho C_{\mathrm{a}} \leqslant 1/32$ from (a), and Lemma E.18 adds $C\rho LM \leqslant \eta/8$ with the constant C of the pins (b). *q.e.d.*

E.7.3 Short Classes and the Exceptional Whole-Contour Merge

Lemma E.18 (Class Pins)**.** *Let C be the constant of the pins (b)–(c) of Lemma E.17 and assume $C\rho LM \leqslant \eta/8$. Put $\sigma_K = \sum_{\gamma \in K} \mathfrak{s}(\gamma)$ for a class K. Then: (a) for every overlap-connected class K, before the promotion of a long class to the whole contour, the hull and footprint pin sums $\sup_{a \in \mathcal{I}_C} \sum_{K \,|\, a \in \bigcup_{\gamma \in K} H_\gamma} t_K e^{64\eta\sigma_K}$ and $\sup_{\Delta} \sum_{K \,|\, \Delta \in \bigcup_{\gamma \in K} F(\gamma)} t_K e^{64\eta\sigma_K}$ are at most $C\rho$; (b) a long class satisfies*

$$\sigma_K \geqslant \frac{1}{LM} \sum_{\gamma \in K} |H_\gamma| \geqslant \frac{n_C}{2LM}, \qquad \sum_{K \text{ long}} t_K e^{4\eta\sigma_K} \leqslant C\rho n_C e^{-30\eta n_C/(LM)} \,; \tag{E.41}$$

(c) the whole-contour groups created by the exceptional merge have total weighted mass $\sum_{W \text{ whole}} M(W) e^{4\eta\mathfrak{s}(W)}$ *at most*

$$C\rho n_C e^{-30\eta n_C/(LM)} e^{4\eta n_C/(LM)} e^{C\rho n_C} \leqslant C\rho n_C e^{-25\eta n_C/(LM)} \leqslant C'\rho\,, \tag{E.42}$$

uniformly in n_C*. A group* W *with a short interval determines a class* K *(Section 3.8), and the groups with the same class* K *are interleavings of the cluster lists of its components: their number is at most* $k(W)!/\prod_{\gamma\in K} k_\gamma!$ *with* $k(W) = \sum_{\gamma\in K} k_\gamma$ *(the interleaving count of Lemma 3.25), and each has*

$$M(W) = \prod_{\gamma\in K} \big(\rho^{|\mathcal{A}(\gamma)|} \prod_{D\in\mathcal{A}(\gamma)} m(D) \prod_j M_v(\mathfrak{c}_j)\big)/k(W)!$$

($M(W)$ *is the product of the root weights* $\rho\, m(D)$*, the cluster weights* $M_v(\mathfrak{c}_j)$ *and* $1/k(W)!$*, Section E.5.2), so the sum of* $M(W)$ *over the groups with class* K *is at most* $\prod_{\gamma\in K}\big(\rho^{|\mathcal{A}(\gamma)|}\prod_D m(D)\prod_j M_v(\mathfrak{c}_j)/k_\gamma!\big) = t_K$ *(the per-group inequality* $M(W) \leqslant t_K$ *alone would not account for this many-to-one correspondence); moreover every such* W *has* $\mathfrak{s}(W) \leqslant \sigma_K$ *(*$n(W) = \sum_{\gamma\in K} n_\gamma$*, and* $\mathcal{H}(W)$*, the union of the overlapping hulls* H_γ*, has at most* $\sum_{\gamma\in K} |H_\gamma|$ *traversals); its footprint* $F(W)$ *lies in* $\bigcup_{\gamma\in K} F(\gamma)$*. Hence the two sums of* (E.21) *over short groups are bounded by the pin sums of (a) (*$e^{4\eta\sigma_K} \leqslant e^{64\eta\sigma_K}$*), and the sums over whole-contour groups by (c): together, (a) and (c) prove the two weighted group pins* (E.21)*.*

Proof. (a) Run the positive overlap-tree recursion on the decorated components with the weights enhanced by $e^{64\eta\mathfrak{s}(\gamma)}$. The pin of Lemma E.17(c) controls the overlap sum of a hull: the components whose hulls meet H_γ have weighted mass at most $C\rho|H_\gamma| \leqslant \eta\mathfrak{s}(\gamma)/8$ by the hypothesis $C\rho LM \leqslant \eta/8$, so the rooted recursion over components (node weight $t_\gamma e^{64\eta\mathfrak{s}(\gamma)}$, node cost $\eta\mathfrak{s}(\gamma)/8$; the pins of Lemma E.17(c) carry the weight $e^{100\eta\mathfrak{s}(\gamma)} \geqslant e^{64\eta\mathfrak{s}(\gamma)} e^{\eta\mathfrak{s}(\gamma)/8}$, which covers the node cost) gives the hull and footprint pins for every class before promotion: the sum over the classes containing a component pinned by the hull pin at a, or by the footprint pin at Δ, is the rooted sum with that component as root. No property of the union interval enters, so the class pins hold for long classes as well; for a short class the interval $\mathcal{H}(K)$ is the union of its hull intervals (Section 3.8), so its size is bounded by the additive class cost σ_K. (b) The first inequality is the definition of $\mathfrak{s}$; the second is the definition of a long class (the union of its hulls has at least $n_C/2$ positions, Section 3.8). Anchor the class at a traversal of its union interval; there are at most n_C anchors, and the class pin (a) with the weight $e^{64\eta\sigma_K}$ gives the second bound of (E.41), the remaining exponent $60\eta\sigma_K$ being at least $30\eta n_C/(LM)$. (c) Once a long class occurs, every other component is merged into the whole-contour group; the other components with their 4η weights sum to at most $e^{C\rho n_C}$ (each has a contact on the contour and is pinned there by Lemma E.17(c), so the sum over the families of further components is at most $\prod_{a\in\mathcal{I}_C} \exp(C\rho)$), and the whole-contour interval weight costs $e^{4\eta n_C/(LM)}$. The middle inequality of (E.42) uses $C\rho LM \leqslant \eta$, implied by the hypothesis, and the final maximum is finite uniformly in n_C because LM and η are fixed. If a spatial pin lies in an additional component, root that component by its footprint pin at a further cost $C\rho$, the anchored long-class factor and remaining exponential being paid by (E.42); if it lies in the long class, use the footprint pin in place of the hull pin; pins on the whole contour are covered by the total whole-group bound. This proves the exceptional-merge pin without assigning its cost to a single original component. Polynomial moments are allowed by using a smaller part of the same fixed enhanced-weight margin. *q.e.d.*

E.7.4 Interior Boxes, Endpoints and the Differentiated Group Weight

Let C_{box} be a fixed geometric constant for the group footprint and the row/contour halos such that a liftable enclosing comparison rectangle has diameter at most $C_{\mathrm{box}}LMn(W)$, increased so that a group with

$$n(W) < \frac{b}{4C_{\mathrm{box}}LM} \tag{E.43}$$

has its entire rectangle and stencils within $b/2$ of a source contact and an injective lift (Section E.10.4). On a far history this box lies in Λ_1, hence in Ω_1, and outside $S_1 = (\Omega_1 \setminus \Omega_1^{\sim -1}) \cup R_1$; its plaquettes satisfy $\delta_{Q_W}(U_1) \leqslant C\varepsilon_1$ by Lemma B.4 and the distance check of Section E.10.5, and, for a group passing the phase test of Lemma E.19 (the groups failing it are treated in Section E.11.2), the Hessian bounds (E.32) of Lemma E.15 hold on the globally small extension used for its comparison. All other groups keep their undifferentiated difference, $|c_W| + |a_W\chi| \leqslant 2M(W)$ (Section E.5.2); their cube count $n(W)$ is at least $b/(4C_{\mathrm{box}}LM)$, and if a rectangle fails to lift the same lower bound holds, since $N_{\min} = \min_i N_i \geqslant 2LMR_1 > 32b$ (Section E.8.2) and a wrapping connected network has length at least a fixed fraction of $N_{\min}$. No interpolation is made across the integration boundary.

For a good box use the coefficient locality and covariance of Sections E.7.1 and E.10: root the tree gauge at the initial interval base, keep h fixed, and compare $c_W(U_1; h)$ with $c_W(\mathbf{1}; h) = a_W\chi(h)$ by the small-phase extension and the fundamental theorem of calculus (Lemma E.19, for a group passing the phase test). It is the same regional reference coefficient, with I, all selection indices and the local geometry retained; no bulk comparison replaces it by a different coefficient. The group weight is differentiated by the product rule, each occurrence retained:

$$\mathscr{D}_\omega w_\xi(W) = \frac{(-1)^{k(W)}}{k(W)!} \sum_{j=1}^{k(W)} \mathscr{D}_\omega \Phi^{\mathrm{v}}_\xi(\mathfrak{c}_j) \prod_{l \neq j} \Phi^{\mathrm{v}}_\xi(\mathfrak{c}_l) \,, \tag{E.44}$$

bounded by Lemma E.16 by $C_M\|\omega\|_\infty \sum_j n(\mathfrak{c}_j)$ times the positive product majorant, without dividing by any Φ. With the joint source derivative of Lemma E.4, whose factor $L_f + \sum_D N_1(D)$ is at most $C(|\mathcal{H}(W)| + n(W))$ because the number L_f of varying background occurrences in the relative word is at most $2|\mathcal{H}(W)|$ times the fixed reconstruction-row overlap constant (Section E.8.1, the sentence defining L_f) and $\sum_D N_1(D) \leqslant n(W)$, this gives $|\mathscr{D}_\omega c_W| \leqslant C\|\omega\|_\infty(n(W) + |\mathcal{H}(W)|)M(W)$ uniformly in h ((E.73)); the curvature phase bound contributes $C\,\mathrm{diam}(Q_W)^2\delta_{Q_W}$, and with $|\mathcal{H}(W)| \leqslant CLMn(W)$ all these losses are a fixed constant times ε_1 times a fixed polynomial in $\mathfrak{s}(W)$, absorbed by the enhanced pin ($x^k e^{-\eta x} \leqslant k!\eta^{-k}$). For a large group the factor $e^{-\eta b/(8C_{\mathrm{box}}LM)}$ is extracted from its reserved weight $e^{\eta\mathfrak{s}(W)/2}$. Combining the two classes and summing over a contour pin gives (E.24); the computation is Section E.11.2. The same matrices and the same cutoff norm are used at the physical and flat endpoints, and every source estimated is the closed functional c_W: no partial interval is replaced by a scalar before its closed subtraction is formed.

E.8 The Coefficient Extension and the Countable-Gas Limit

E.8.1 Background Scope and the Coefficient Path

Sections E.4 and E.6 give the joint source and one-derivative bounds at the physical centred background and on every globally small real extension described here, for each closed character $\chi(T_W P_W^{-1} h)$ uniformly in h; they do not assert a path from the entire physical background to the identity in one global small chart.

Fix a good interior group of Section E.7.4 and its enclosing fine rectangle Q_W, which contains all coefficient, reconstruction, averaging and interval/closing stencils. By the support statements of Section E.9.2, for the fixed selection indices and the fixed spatial Cauchy coefficient defining the group only the background links on Q_W are used: the restricted seed inverse uses entries on E_0, every walk coefficient has its domains in the final component, and the potential and source rows have their finite stencils. The regional I, Ω_1 and every auxiliary parameter are held fixed. Gauge U_1 on Q_W by its rooted tree gauge g_W with $g_W(\mathrm{base}) = \mathbf{1}$ (Lemma E.21). If $C_{\mathrm{tg}} \operatorname{diam}(Q_W)^2 \delta_{Q_W} \leqslant \tau_{\mathrm{b}}/2$ (the phase test; since $\operatorname{diam}(Q_W)$ is at least the largest side R of Q_W, it implies the hypothesis $C_{\mathrm{tg}} R^2 \delta_{Q_W} \leqslant \tau_{\mathrm{b}}/2$ of Lemma E.21), write the principal phases $B_W(b) = \frac{1}{i} \log U_1^{g_W}(b)$ on the rectangle and extend $B_W = 0$ on every other bond, and define the coefficient argument and its direction by

$$U_{W,t}(b) = e^{itB_W(b)}, \qquad \omega_W(b) = B_W(b), \qquad \|\omega_W\|_\infty = \max_b h(B_W(b)), \qquad 0 \leqslant t \leqslant 1 \, ; \tag{E.45}$$

the norm is the invariant norm h, the comparison with the cutoff norm being absorbed in C. Set L_f equal to the number of occurrences of background links in the relative word $T_W P_W^{-1}$ on which this argument varies, $L_f \leqslant 2|\mathcal{H}(W)|$ times the fixed reconstruction-row overlap constant. The complementary h is held fixed at the root base.

At each t form $H_t = \iota_I^* \mathsf{H}^{\Omega_1}_{U_{W,t}} \iota_I$, the row reconstruction $q_{U_{W,t}}$ and the near primitives at $U_{W,t}$, with zero retained input and the same protected r-box. For a generic noncritical $U_{W,t}$ the energy primitive is defined by subtracting its constant, linear and quadratic Taylor terms as in (3.29), the Hessian keeping the force–constraint term; the Haar and constraint-Jacobian primitives use their analytic row formulas. These definitions specify an analytic coefficient extension. This auxiliary kernel is not identified with the physical integral of [3] at $t \in (0,1)$, and the original fixed-V constraint is not imposed along the path; at the physical endpoint each compensated primitive agrees with its Taylor-subtracted formula by locality, criticality being used in the primitive-sum identity (3.33) for the physical exponent and not to assert that $U_{W,1}$ is a global minimum; at the flat endpoint the formulas agree with the reference formulas. Lemma E.15 gives $\gamma_L/2 \leqslant H_t \leqslant m_L$; the designated inverses, reconstruction and means use the fixed smaller row charts since $\|B_W\|_\infty \leqslant \tau_{\mathrm{b}}/2$ and $\tau_{\mathrm{b}} \leqslant \tau_L/4$; the host norms and their derivatives are bounded by $C_0^\partial g_0 (1+r)^3$ on these charts; so the Gaussian payment, the positive forest budget and the source estimates hold uniformly in t with the same recipe. The force term is one of the local entries and is not omitted off the minimum.

Lemma E.19 (The Coefficient Path)**.** *For a small interior group (*(E.43)*, a good box of Section E.7.4) passing the phase test of Section E.8.1, and for each $h \in G$ in the closed character $\chi(T_W P_W^{-1} h)$,*

$$d_W(U_1; h) = \int_0^1 \mathscr{D}_{\omega_W} c_W(U_{W,t}; h) \, \mathrm{d}t \, , \tag{E.46}$$

and $|\mathscr{D}_{\omega_W} c_W(U_{W,t}; h)| \leqslant C \|\omega_W\|_\infty (L_f + n(W)) M(W)$ uniformly in t and h.

Proof. At $t = 1$ the group coefficient agrees with the original one by coefficient locality on Q_W and simultaneous gauge covariance (Section E.10.3): the values outside Q_W can be replaced in the argument because no term of the spatial Cauchy coefficient consults them, and the root gauge $g_W(\mathrm{base}) = \mathbf{1}$ leaves h unchanged. At $t = 0$ the coefficient is exactly the regional reference one, with the same I and the same group index W, so $c_W(U_{W,0}; h) = a_W \chi(h)$. The fundamental theorem of calculus applies: by the dominated-convergence hypotheses stated in the last paragraph of Section E.10.3 and the termwise differentiation statement of Lemma E.20, $t \mapsto c_W(U_{W,t}; h)$ is continuously differentiable on $[0,1]$ with derivative $\mathscr{D}_{\omega_W} c_W(U_{W,t}; h)$. The derivative of c_W differentiates $\prod_j \Phi^{\mathrm{v}}(\mathfrak{c}_j)$, giving the occurrence sum (E.44) with majorant

$C\|\omega\|_\infty(\sum_j n(\mathfrak{c}_j))M(W)$, and the root integral, giving $C\|\omega\|_\infty(L_f + \sum_D N_1(D))M(W)$ by Lemmas E.4 and E.13; the sum is the displayed bound. This is a cluster-list bound for the same group weight, not an identification of a one-root vacuum ratio with it: the joint-ratio estimate of Lemma E.16 controls the original family normalisation, the occurrence estimate controls its regrouping. If the phase test fails, the two endpoint estimates give the curvature bound of Section E.11.2; no continuation is claimed in that regime. The path is the connection between the local physical coefficient and the globally small extension; it does not assume global near-identity physical fields. *q.e.d.*

E.8.2 Parameter and Period Conventions

All results inherit the standing conditions (3.1) and the compatible even fine periods of Section D.7: every fine period N_i is an even integer multiple of $\ell = LMR_1$, and ℓ is odd (L odd, M and R_1 powers of L). If $N_i = q_i\ell$ then q_i is even and positive, so

$$N_{\min} = \min_i N_i \geqslant 2LMR_1 > 32\lfloor LMR_1/16\rfloor = 32b\,, \tag{E.47}$$

the strict middle inequality because ℓ is odd. This is independent of the square-fit condition $K_i \geqslant 3(n+2)$; both are retained, with the source-row and coarse-grid conditions of Section 3. Thus the good comparison box and every fixed stencil have a unique lift under the small-group criterion (E.43), and the periodic failure lies in the large-group class; no condition on the fluctuating field or a denominator is added. Γ_C consists of the individual positive contour bonds and all their fixed input templates, C_L including the number and size of the templates, and the radius-b ball of Lemma E.1 is around each individual bond, so the count $4(2b+3)^4$ needs no further n_C-dependent factor. The input of Appendix D is the one-point probability $p_z \leqslant q \leqslant e^{-c_*\pi_*^2}$ of Lemma D.14 and Proposition D.15 (this q is the q_{pt} of Lemma E.1) with $p_0(g) = A_0(\log g^{-2})^{p_0}$, $p_0 \geqslant 5\mathfrak{r} \geqslant 10$. E_{bdry} of (D.21) is bounded separately by Proposition D.15; E_{def} contains only the covered centred-kernel subtraction, so neither the exterior response nor the uncovered contribution is dropped from (E.35).

E.8.3 Countably Many Group Species and Differentiation

Lemma E.20 (Countable Species)**.** *At a fixed finite torus let $\mathcal{W}_N = \{W \mid n(W) \leqslant N\}$, a finite set, with partition sum S_N and connected-cluster logarithm $\mathscr{L}_N$, $S_N = e^{\mathscr{L}_N}$ (the cluster logarithm is written $\mathscr{L}$, the block scale being L). Under the hypotheses of Lemma E.10 ($C\rho LM \leqslant \eta/8$ and g_0 small),*

$$\sum_W |a_W| e^{\mathfrak{a}_W/8} \leqslant C\rho n_C < \infty, \qquad \mathscr{L}_N \to \mathscr{L}, \qquad S_N \to S^{\circ}_{\xi,C} = e^{\mathscr{L}} \neq 0\,, \tag{E.48}$$

the restricted identity $S^{\circ,[\mathfrak{B}]}_{\xi,C} = e^{\mathscr{L}^{[\mathfrak{B}]}}$ ($\mathscr{L}^{[\mathfrak{B}]}$ the connected-cluster logarithm of the species compatible with $\mathfrak{B}$) and the ratio bound (E.23) *hold for the countable index set, and the background derivatives of Lemmas E.4, E.13 and E.16 may be taken termwise and integrated in t before the countable limit.*

Proof. $\mathcal{W}_N$ is finite: the domain set is finite, and $n(W)$ bounds the number of roots, occurrences and members of every occurrence. Its gas satisfies the same Kotecký–Preiss inequality, deleting species only decreasing the positive sums. Every group has a nonempty contour interval, $|a_W| \leqslant M(W)$ (Section E.5.2) and $e^{\mathfrak{a}_W/8} = e^{\eta\mathfrak{s}(W)/8} \leqslant e^{4\eta\mathfrak{s}(W)}$, so the pin (E.21) summed over the n_C traversals gives the first bound of (E.48); the rooted-tree estimate bounds the

total absolute connected-cluster series by this finite sum up to a fixed constant; each finite cluster tuple is present for all large N, so absolute domination gives $\mathscr{L}_N \to \mathscr{L}$; independently the full gas sum over finite compatible families is absolutely bounded by $\exp(\sum_W |a_W|)$, and dominated convergence gives $S_N \to S^\circ_{\xi,C}$. The same argument after excluding every species incompatible with a fixed compatible family $\mathfrak{B}$ gives the restricted identity; the logarithmic difference is the connected-cluster sum of tuples hitting $\mathfrak{B}$, counted once, with absolute sum at most $\sum_{W\in\mathfrak{B}} \mathfrak{a}_W/8$ by the pinned estimate. For the derivatives, Lemmas E.13 and E.16 give summable positive majorants independent of t on the extension; on each finite truncation the product and integral derivatives are justified by the uniform Gaussian coercivity (Section E.9.3) and the fixed protected box, and the walk derivative series converges on its unchanged contours (Section E.9.2); the cluster majorant $n(\mathfrak{c})M_v(\mathfrak{c})$ is paid by its exponential moment and the group majorant has only the polynomial in $\mathfrak{s}(W)$, paid by the reserved weight. Uniform absolute convergence permits termwise differentiation and t-integration before the countable limit; no conditionally convergent series is differentiated. *q.e.d.*

E.8.4 Integrability at the Physical Caller

The near-history contribution is bounded after forming $\mathscr{R}_\xi$, without exchanging its sum with the outer integral (Lemma E.1). On covered far histories the proof of Lemma E.12 bounds $\sum_{\mathfrak{B}\neq\varnothing} |S^{\circ,[\mathfrak{B}]}_{\xi,C} R_\xi(\mathfrak{B})|$ by $|S^\circ_{\xi,C}|$ times the exponential of the one-group sum (E.25) minus one; since $|S^\circ_{\xi,C}| \leqslant 1$ ($S^\circ_{\xi,C}$ is the normalised integral (3.94) of f°_ξ, a normalised character of modulus at most one, against the positive reference kernel $K^\circ_{\xi;0}$) and the exponential is uniform in the branch fields and retained variables, it is integrable against the normalised positive history law at each fixed loop and cutoff, which justifies the far-history sum/integral exchanges. The physical $\mathcal{Z}_0$ and $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$ remain those of Section E.1.

E.9 Analytic Inputs for the Coefficient Extension

E.9.1 Joint Background/Phase Analyticity and Primitive Derivatives

Fix one of the finitely many row/primitive stencil types in its translated fine-lattice frame and write the background on the stencil as $U_Z(b) = e^{iZ_b}$; background directions use the maximum h-norm on this finite label set. Choose τ_{b} and a phase radius $\theta_0 > 0$ so that $\|Z\| < 2\tau_{\mathrm{b}}$ and $\|A\| < 2\theta_0$ keep every finite path product and relative product entering the bond average and the mean of [4, (0.4), (0.11)] ((4.1)) inside the matrix-logarithm neighbourhood $\|W - \mathbf{1}\|_{\mathrm{op}} < 1/2$; this is possible uniformly over the finite types, every word having bounded length and $e^{iA}e^{iZ}$ being analytic in both variables. The radii τ_{b}, θ_0 used below are those of the final joint domain, after the reductions of this section and of Section E.10.5; in particular $2\tau_{\mathrm{b}} \leqslant \tau_L/2$ (Lemma E.15) and $2\theta_0 \leqslant \vartheta_*/2$ (Section E.10.5), so that Lemma B.7(ii)–(iv) applies at each real Z with $\|Z\| < 2\tau_{\mathrm{b}}$ and each A with $\|A\| < 2\theta_0$.

For the mean of [4, (0.11)] write its defining equation $\sum_j \log(W_j M^{-1}) = 0$. At $W_j = \mathbf{1}$, $M = \mathbf{1}$ its derivative in the Lie-algebra parameter of M is $-n\,\mathbf{1}$, so the parametric analytic implicit function theorem gives a jointly analytic local mean in the background and fluctuating phases; it agrees with the mean of Lemma B.7(iv), both taking the value $\mathbf{1}$ at zero and solving that equation in the same smaller neighbourhood. The joint radii are shrunk so that the mean stays in that neighbourhood; no global uniqueness is inferred from a logarithm chart.

The bond constraint on a row c is the jointly analytic function

$$\mathcal{Q}_Z(A) = \frac{1}{i}\log\bigl(\mathsf{M}(e^{iA}U_Z, c)\mathsf{M}(U_Z, c)^{-1}\bigr), \quad LQ_Z = D_A\mathcal{Q}_Z(0), \quad N_Z(A) = \mathcal{Q}_Z(A) - LQ_Z A\,, \tag{E.49}$$

M the fixed bond average of [4, (0.4)], not the finite-group mean, with $N_Z(0) = D_A N_Z(0) = 0$ for every Z. The coefficient of the designated phase in LQ_Z is uniformly invertible on a smaller background polydisc, by the identity-background designated inverse (Lemma 3.2) and continuity, so the designated inverse h_Z with $LQ_Z h_Z = \mathbf{1}$ and $\mathsf{C}_Z = \iota_{\mathrm{free}} - h_Z LQ_Z \iota_{\mathrm{free}}$ are analytic in Z with uniformly bounded first background derivatives; these formulas fix their maps and normalisations, which are those of Section 3.2, without a new free-variable convention. The row solution is defined by

$$\widetilde{D}_Z(A) = N_Z\bigl(A - h_Z\widetilde{D}_Z(A)\bigr)\,. \tag{E.50}$$

On smaller joint balls the derivative of the right side in $\widetilde{D}$ has norm at most $1/2$ uniformly in Z, since $D_A N_Z(A) = O(\|A\|)$; uniform analytic contraction (the parametric implicit function theorem) gives joint analyticity and $\|\widetilde{D}_Z(A)\| \leqslant C\|A\|^2$ (Lemma B.7(ii) at each real Z). Cauchy estimates on the larger joint balls, followed by the order-two zero in A, give for $l = 0, 1$ and each fixed k needed below

$$\|D_Z^l D_A^k \widetilde{D}_Z(A)\| \leqslant C_{l,k}\|A\|^{\max(2-k,0)}\,. \tag{E.51}$$

All constants are maxima over finitely many types, independent of the volume, the history and g_0; the dimensions of the types depend on L and G only. The protected free input satisfies $g_0\|\mathsf{C}_Z a\| < \theta_0/2$ after reducing the coupling, and the phase $q_Z(a) = g_0\mathsf{C}_Z a - h_Z\widetilde{D}_Z(g_0\mathsf{C}_Z a)$ satisfies

$$\|D_Z q_Z(a)\| \leqslant C g_0(1+r), \qquad \|D_Z^l D_a^k q_Z(a)\| \leqslant C_{l,k} g_0^k \tag{E.52}$$

for $l = 0, 1$ and $1 \leqslant k \leqslant 2K+1$, K the fixed row input count of Section E.9.4. These are joint background/young-field derivative bounds; they are not the assertion that the finite list of phase-only derivative constants of Lemma B.7 contains them.

Analytic traces of X and X^{-1} extend $\operatorname{Re}\operatorname{tr} X$ from unitary fields, where it equals $\operatorname{tr}(X + X^{-1})/2$; the energy and relative gauge expressions are therefore jointly analytic finite word functions. For the local energy function $E_Z(s)$ in the dimensionless free phase s (the action and gauge-fixing functional of (4.5) at the background U_Z), the compensated primitive has the Taylor-remainder form

$$p_Z^{\mathrm{E}}(a) = -\frac{g_0}{2}\int_0^1 (1-t)^2 D_s^3 E_Z(t g_0 a)[a,a,a]\,\mathrm{d}t\,. \tag{E.53}$$

Its first background derivative has the same factor g_0, since $D_Z D_s^3 E_Z$ is uniformly bounded on the smaller joint domain; hence $|p^{\mathrm{E}}| + \|D_Z p^{\mathrm{E}}\| \leqslant C g_0(1+r)^3$, the relative means and the soft-gauge term being included in E_Z. The Haar function $\log(\sigma(q)/\sigma(0))$ is analytic in q with an order-two zero, so composed with q_Z it and its Z-derivative are at most $C g_0^2(1+r)^2$. The constraint-Jacobian logarithm is $\log\det(\mathbf{1} - h_Z D_A\widetilde{D}_Z)$ on a matrix ball of radius $< 1/2$; its value and Z-derivative are at most $C g_0(1+r)$, as $D_A\widetilde{D}_Z = O(g_0(1+r))$ and $D_Z D_A\widetilde{D}_Z$ has the same order, and the real determinant is positive on that ball, so the analytic logarithm restricts to the real logarithm of Lemma B.7(iii). Summing the finite primitive types assigned to one host gives the host derivative bound with $C_0^{\partial} = N_{\mathrm{h}}\max(C_{\mathrm{p}}, C_{\mathrm{p}}^{\partial})e^{\kappa(B_{\mathrm{h}}+4)}$ of Section E.6.1; this constant and the finite source derivative constants are chosen after M and τ_{b}, before g_0, and no earlier Mayer, walk or reserve choice depends on them.

E.9.2 Differentiated Fixed Walks and Exact Locality

Use the series (3.62) with its parameters $s_H = 2h_+$, $\mathsf{A}_H = \mathbf{1} - H/s_H$, $\mathsf{P}_H = \mathsf{A}_H^{n_*}$, $q_H = 1 - \gamma/s_H$, $\beta = V_* q_H^{n_*}$ ($V_* = V_{n_*}$), $a_* = e^{81\kappa_1}$ and n_*, fixed from the common input bounds and not re-evaluated at the varied background. Of the background these constructions use only the input bounds (3.56), $\gamma \leqslant H \leqslant h_+$, and the R-locality of the entries; both hold along $U_{W,t}$, the two-sided bound by Lemma E.15 and the locality by the entry statement of Lemma 3.3, whose radius R does not depend on the background. Include every geometrically allowed tuple of bond indices, including coefficients vanishing at a particular background, so that differentiation does not change the index set. The weights $w_\omega(\sigma)$ and domains $D(\omega)$ depend only on the labels and the fixed geometry. Let c_H bound the Schur row and column norms of $\mathscr{D}_\omega H/\|\omega\|_\infty$, supplied by the finite analytic entries of Section E.9.1. Then $\mathscr{D}_\omega \mathsf{A}_H = -\mathscr{D}_\omega H/s_H$ and $\mathscr{D}_\omega \mathsf{A}_H^j = \sum_{l=0}^{j-1} \mathsf{A}_H^l (\mathscr{D}_\omega \mathsf{A}_H) \mathsf{A}_H^{j-1-l}$, a block sum of which is bounded by $V_j j q_H^{j-1} c_H \|\omega\|_\infty / s_H$. Put $d_* = V_* n_* q_H^{n_*-1} c_H / s_H$ and let d_{ini} be the maximum of the corresponding bounds for $j < n_*$. Differentiating the grouped walk products gives the absolute Schur bound

$$\|\mathscr{D}_\omega S_{Z_0}(\sigma)\|_{\mathrm{Schur}} \leqslant C_{\mathrm{w}}^{\partial} \|\omega\|_\infty, \qquad C_{\mathrm{w}}^{\partial} = s_H^{-1/2} n_* \Big(\frac{a_* d_{\mathrm{ini}}}{1 - a_*\beta} + \frac{a_*^2 V_* d_*}{(1 - a_*\beta)^2} \Big) , \tag{E.54}$$

the denominators positive because $a_*\beta \leqslant 1/4$ ((3.57)); the same bound holds on the spatial Cauchy circles. The derivative of C_{E_0} is $-C_{E_0}(\mathscr{D}_\omega H_{E_0}) C_{E_0}$, with $H_{E_0} = \iota_{E_0}^* H \iota_{E_0}$, and that of Γ_{E_0} is the product rule on $\iota_{E_0}^* H \Pi_{I \setminus E_0} S_{Z_0}$ ((3.64)); their operator norms are fixed constants times $\|\omega\|_\infty$.

For a selected final domain Z the spatial variables outside Z are zero and the protected seed enlargement Z_0' lies in Z; a walk with a coefficient domain meeting a cube outside Z has a zero spatial factor, which remains zero after background differentiation. Every surviving walk, including every hidden fine step in its grouped powers, has its coefficient stencil inside its declared domains by the construction of D_b (Section 3.5). The restricted inverse C_{E_0} consults only entries on the seed E_0, included in the comparison rectangle; it is not a global inverse restricted afterwards, and the normalised Gaussian determinant is $\det C_{E_0}$, not a determinant of H on all of I. All potential and source stencils and every constituent vacuum domain of c_W lie in Q_W. These facts prove the locality used at $t = 1$ in Lemma E.19, with the fixed regional I: no surviving selected coefficient consults the background outside Q_W, and no field-dependent global spectral quantity appears among the recipe parameters. The common full Gaussian prefactor $\mathfrak{n}_\xi$ (Section 3.5) has cancelled inside the exact conditional ratio defining $F_\xi(0)$ ((3.85)); it is not a missing local activity factor, and that conditional cancellation does not cancel $\mathsf{w}_\xi = e^{c_\xi} I_{\xi;1}$ or the physical $\mathcal{Z}_0$.

E.9.3 The Complex-Gaussian Moment Payment

On a selected component write E for its seed carrier E_D, J_Z for its auxiliary carrier J_D (the block of $\bigsqcup_D J_D$ belonging to this component, Lemma 3.15; in this subsection J_Z denotes the carrier of one component), $H_E = \iota_E^* H \iota_E$, $C_E = H_E^{-1}$ and $\Gamma = \Gamma_1 + i\Gamma_2$ for the block of $\Gamma_{E_0}(\sigma)$, the pairings complex bilinear. Before cutoffs and Mayer factors the joint density with respect to real Lebesgue measure $\mathrm{d}B\,\mathrm{d}X$ is the two centred Gaussian normalisations times

$$\exp\Big[-\frac{1}{2}\langle B, H_E B\rangle - \langle B, \Gamma X\rangle - \frac{1}{2}\langle \Gamma X, C_E \Gamma X\rangle - \frac{1}{2}\|X\|^2 \Big] , \tag{E.55}$$

call it G_σ. The columns of Γ outside J_Z, and those of its derivative, vanish identically by Lemma 3.15, so the auxiliary standard Gaussians integrate to one and no moment cost

proportional to $|I|$ arises. With $\alpha = \alpha_G$ and $C_\alpha = (H_E - \alpha\mathbf{1})^{-1}$, the absolute density times $e^{\alpha\|B\|^2/2}$ is a positive real Gaussian with the original determinant prefactor; in the variables $y = B + C_\alpha\Gamma_1 X$ its exponent is

$$-\frac{1}{2}\langle y, (H_E - \alpha)y\rangle - \frac{1}{2}\langle X, (\mathbf{1} - T_\alpha)X\rangle \,, \qquad T_\alpha = \Gamma_1^{\mathsf{T}}(C_\alpha - C_E)\Gamma_1 + \Gamma_2^{\mathsf{T}} C_E \Gamma_2 \,, \tag{E.56}$$

the reserve matrix of Appendix B.4. With $H_E - \alpha \geqslant \gamma/2$ (from $H \geqslant \gamma$, (3.56), and $\alpha_G \leqslant \gamma/2$, Lemma 3.16(ii)), $\|T_\alpha\| \leqslant 1/4$ ((B.24)) and $\|\Gamma_1\| \leqslant A_*$ (the norm paragraph of Appendix B.4) and

$$c_{\mathrm{G}} = \frac{\min(\gamma/2, 3/4)}{2(1 + 2A_*/\gamma)^2} > 0 \,, \tag{E.57}$$

the real quadratic form is at least $c_{\mathrm{G}}(\|B\|^2 + \|X\|^2)$. Its total unnormalised mass is the determinant product bounded by $e^{N_1(Z)}$ in Lemma 3.16(ii), and its normalised covariance is at most $c_{\mathrm{G}}^{-1}\mathbf{1}$. With $d = d_G(|E| + |J_Z|)$,

$$\int (1 + \|B\|^2 + \|X\|^2) e^{\alpha_G\|B\|^2/2} |G_\sigma(B, X)| \,\mathrm{d}B \,\mathrm{d}X \leqslant (1 + d/c_{\mathrm{G}}) e^{N_1(Z)} \,, \tag{E.58}$$

and $d \leqslant 8d_G(LM + 1)^4 N_1(Z)$, so the loss is linear in the cube count $N_1(Z)$ with a fixed M-dependent constant; it uses the positive $3/4$ margin in $\mathbf{1} - T_\alpha$, not a new exponential reserve. The logarithmic derivative of the Gaussian normalisation contributes $\mathrm{Tr}(C_E \mathscr{D}_\omega H_E)/2$; the exponent derivative contributes $-\langle B, (\mathscr{D}_\omega H_E)B\rangle/2$, $-\langle B, (\mathscr{D}_\omega \Gamma)X\rangle$ and $-\mathscr{D}_\omega\langle \Gamma X, C_E \Gamma X\rangle/2$ with $\mathscr{D}_\omega C_E = -C_E(\mathscr{D}_\omega H_E)C_E$; each is bounded by $C\|\omega\|_\infty(d + \|B\|^2 + \|X\|^2)$ by Section E.9.2. Hence (E.58) proves the differentiated Gaussian payment $C_M N_1(Z)\|\omega\|_\infty e^{N_1(Z)}$ on the same Cauchy circles and carriers, all normalisers and the negative compensator $-\frac{1}{2}\langle \Gamma X, C_E \Gamma X\rangle$ differentiated. The protected cutoff domain is fixed throughout; no cutoff translation or surface term is suppressed in this derivative.

E.9.4 Source Remainder Bounds

Let $\mathsf{J} = (b_1, \ldots, b_q)$ be distinct selected labels and let a' denote all other selected root variables. The mixed fundamental theorem of calculus gives the exact identity

$$\Delta_{\mathsf{J}} f(B, a') = \int_{[0,1]^q} D_{b_1} \cdots D_{b_q} f\Big(a' + \sum_{j=1}^{q} s_j B_{b_j}\Big)[B_{b_1}, \ldots, B_{b_q}] \,\mathrm{d}\mathbf{s} \,, \tag{E.59}$$

the sum in the argument meaning the canonical embedding into the free carrier. Let K be a fixed bound for the number of source rows affected by one free label and for the number of inputs of one row. Only the source $f_W(a; h) = \chi(T_W(a)P_W^{-1}h)$, or the original full closed Wilson character, is used (a normalised character, so $|f| \leqslant 1$); untouched factors are unitary in its representation. The derivative product rule has at most K^q row assignments, and the derivatives within a row have order at most $2K + 1$, whose constants (E.52) are fixed. Consequently, with a fixed C_χ enlarged once,

$$\|D_{b_1} \cdots D_{b_q} f\| \leqslant (C_\chi g_0)^q, \qquad \sup|\Delta_{\mathsf{J}} f| \leqslant (C_\chi g_0 r)^q \,. \tag{E.60}$$

For root j let P_j be its homogeneous degree-q_j projection after taking Δ_{J_j}. Subtracting the integrand of (E.59) at zero root input and integrating its first derivative along the segment $t(\mathbf{s}B)$ gives q_j terms, each with one additional derivative and one additional selected B factor. Thus, for any set O of roots,

$$\sup\Big|\prod_{j\in O}(1 - P_j)\Delta_{\mathsf{J}} f\Big| \leqslant u^{q+|O|}, \qquad u = C_\chi g_0(1 + r) \,, \tag{E.61}$$

the choices of the extra increment, at most $\prod_{j\in O} q_j \leqslant 2^q$, being absorbed in C_χ. The row-order bound $2K+1$ covers these remainders simultaneously because the roots have disjoint selected labels and a row has at most K inputs. A single background derivative hits either a row coefficient or one of the L_f background occurrences of the closed word; (E.52) and the product rule give

$$\sup\Big|\mathscr{D}_\omega \prod_{j\in O}(1-P_j)\Delta_{\mathfrak{J}} f\Big| \leqslant C_\chi \|\omega\|_\infty (L_f+q)\, u^{q+|O|}\ . \tag{E.62}$$

These are the inequalities used in the commuting-projection cancellation of Lemma E.14. Roots with an empty ordinary selection retain only these remainders under their even Gaussian integration; roots with an ordinary selection use their independent Cauchy vanishing. This recovers the product $u^{q_j} v^{q_j \bmod 2}$, and its one-background-derivative version, before any total variation majorant is summed; every term remains a closed contraction with the same fixed h and the original word order.

E.9.5 Tree Gauge on a Comparison Rectangle

Lemma E.21 (Tree Gauge)**.** *Let Q be an injectively lifted fine rectangle of largest side $R \geqslant 1$ with a root $o \in Q$, let U be a unitary field, and let $\delta_Q(U) = \max_{p\subset Q} |U(\partial p)-\mathbf{1}|_\sigma$. For $x \in Q$ let P_x run from o to x in coordinate order, each coordinate run oriented by its displacement; these paths form a rooted tree. Put $g(o)=\mathbf{1}$, $g(x)=U(P_x)$, so that $U^g(b) = U(P_x)U(b)U(P_{x'})^{-1}$ for $b=(x,x')$ is the based closed transport at o of the fundamental loop of b. Then*

$$U^g(b) = \prod_{j=1}^{m_b} G_j U(\partial p_j)^{\epsilon_j} G_j^{-1}, \qquad m_b \leqslant 3R, \qquad \epsilon_j \in \{1,-1\}\ , \tag{E.63}$$

with every plaquette p_j in Q, and

$$|U^g(b)-\mathbf{1}|_\sigma \leqslant 3R\,\delta_Q(U)\ . \tag{E.64}$$

There is a constant C_{tg}, depending only on the fixed group chart and τ_{b}, such that $C_{\mathrm{tg}} R^2 \delta_Q(U) \leqslant \tau_{\mathrm{b}}/2$ implies that every $U^g(b)$, $b\subset Q$, lies in the group chart, with principal phase B_b satisfying

$$\max_{b\subset Q} h(B_b) \leqslant C_{\mathrm{tg}} R^2 \delta_Q(U) \leqslant \tau_{\mathrm{b}}/2\ . \tag{E.65}$$

Proof. For a bond of direction μ, move its last μ-step past the later coordinate runs of P_x. Each interchange with a perpendicular fine edge replaces the ordered word by one plaquette transport conjugated by the preceding word; all plaquettes stay in Q, and the number of interchanges is at most $\sum_{\nu>\mu} |x_\nu - o_\nu| \leqslant 3R$. Negative coordinate runs reverse a plaquette orientation without altering this count. Induction on the interchanges gives the exact nonabelian word (E.63), the order being that of the step interchanges and G_j the preceding transport to the plaquette, so every factor is a conjugated closed plaquette; no commutative plaquette replacement is used. Unitary invariance of $|\bullet|_\sigma$ and telescoping give (E.64), with a constant independent of R. Norm conversion to the operator norm and the fixed local group logarithm give $h(\log U^g(b)) \leqslant 3C_G\sqrt{N}\,R\,\delta_Q$ whenever that distance is inside the group chart; since $R\geqslant 1$, requiring $C_{\mathrm{tg}} \geqslant 3C_G\sqrt{N}$ and $C_{\mathrm{tg}} \geqslant 6\sqrt{N}\tau_{\mathrm{b}}/r_G$ makes $C_{\mathrm{tg}} R^2\delta_Q \leqslant \tau_{\mathrm{b}}/2$ sufficient for the chart condition and for (E.65). The constants do not grow with b or the scale R_1. If the sufficient condition fails, the two-endpoint coefficient bound (E.73) is used instead of the interpolation (Section E.11.2), so the lemma covers growing good boxes without imposing their smallness as a hypothesis. The root $g(o)=\mathbf{1}$ leaves the complementary h fixed in every closed contraction. *q.e.d.*

E.10 Identification of the Centred Coefficient and Its Local Extension

This section concerns c_W of Section E.5.1, with W fixed on a covered branch; its first caller is the interpolation identity of Lemma E.19. The conclusions retain the hypotheses of Section 3 (its Section 3.8 included) and Appendix B, with their fixed charts, carriers, cutoffs and parameters, and do not identify the regional scalar $S^{\circ}_{\xi,C}$ with the perimeter/corner scalar.

E.10.1 Which Exterior Dependence Is Removed

Write s for the exterior-response parameter and t for the background-path parameter. Section 3 defines

$$K_{\xi;s}(a) = \Theta_\xi(a) \exp\Big(-\frac{1}{2}\langle a, H_{\xi,II} a\rangle - s\langle a, H_{\xi,IJ} A_{\mathrm{e}}\rangle + \sum_{\mathfrak{p} \text{ near}} \mathfrak{p}(a + sA_{\mathrm{e}})\Big) , \tag{E.66}$$

with U_1 held fixed in the s-interpolation. The collar identity and the boundary-response identity give, with $F_\xi(s)$ defined with the same Wilson function f_ξ,

$$\mathcal{J}^{\mathrm{a}}_\xi[1] = e^{c_\xi} I_{\xi;1}, \qquad F_\xi(1) = F_\xi(0) + \mathcal{B}_\xi[f_\xi] . \tag{E.67}$$

On a covered branch every contour row has its entire free-input set in I. Since Ψ_ξ is the identity on I and A_{e} vanishes there, row locality implies, for every a in the integration box, that every contour link in $f_\xi(a)$ is its reconstruction from a alone at background U_1: this is the argument of the proof of Lemma 3.28 with a retained instead of set to zero, using Lemma 3.5, and it uses no estimate. Thus both the source and $K_{\xi;0}$ have zero retained free input, while the dependence of U_1 itself on the exterior determining data remains. Every selected measure in c_W is constructed from $K_{\xi;0}$: its potentials are the near primitives at a, its Hessian is $H_{\xi,II}$ and its cutoff is Θ_ξ. The factors involving $\Psi_\xi(A_{\mathrm{e}})$, the far primitives and the boundary-map Jacobian are in c_ξ, and the interaction between a and A_{e} is accounted for by $\mathcal{B}_\xi$; they are not factors of c_W. Consequently the physical comparison keeps

$$\frac{1}{\mathcal{Z}_0}\int \mathcal{O}\big[e^{c_\xi} I_{\xi;1} F_\xi(0)\big]\mathrm{d}V + \frac{1}{\mathcal{Z}_0}\int \mathcal{O}\big[e^{c_\xi} I_{\xi;1}\mathcal{B}_\xi[f_\xi]\big]\mathrm{d}V , \tag{E.68}$$

and does not replace $I_{\xi;1}$ by $I_{\xi;0}$. This is the role of the boundary separation in [3, (1.26)–(1.30)], not a claim that the exterior input of the original conditional integral vanishes.

E.10.2 Covariance of the Selected Coefficient

For a site gauge g let $R_g a(b) = \mathrm{Ad}_{g(b_-)}\, a(b)$ on free bonds; R_g is orthogonal for h, preserves each free bond subspace and commutes with geometric restrictions. The intertwinings in the proof of Lemma 3.3, $Q_{U^g} R_g = R_g^{(1)} Q_U$, $\mathcal{N}_{U^g} \circ R_g = R_g^{(1)} \mathcal{N}_U$, $h_{U^g} = R_g h_U (R_g^{(1)})^{-1}$ and $\mathsf{C}_{U^g} = R_g \mathsf{C}_U R_{g,*}^{-1}$ with $(R_g^{(1)} v)(c) = \mathrm{Ad}_{g(c_-)}\, v(c)$ (the row covariance of Section 3.2, used in the same way in the proof of Lemma 3.26), give $q_{U^g}(R_g a) = R_g q_U(a)$, branch included: $R_g q_U(a)$ solves the transformed row equation and lies in the same invariant small ball, and the uniqueness of the small row solution there (Lemma 3.5; the uniqueness ball of Lemma B.7(ii)) identifies it with the selected solution. The joint fixed point of Section E.9.1 is identified with this solution by choosing its phase radius so that its $O(\|A\|^2)$ bound lies inside the uniqueness ball of Lemma B.7 (Section E.10.5); the inverse and mean branches are those fixed in the row construction.

The action of each plaquette is invariant and each relative soft-gauge mean is conjugated at its base, so the constant, linear and quadratic Taylor coefficients of the energy primitives transform with their arguments, the force–constraint term included; subtracting them preserves the covariance of the compensated primitives even at noncritical backgrounds. The Haar density is conjugation invariant. The constraint-Jacobian endomorphism transforms by conjugation between the same free and designated spaces, so its determinant and its selected logarithm are unchanged, the logarithm branch being fixed by continuation from zero within the nonvanishing chart. Hence each primitive and each host sum $\mathbf{V}_{\xi}(Y)$ is invariant under the simultaneous transformation of background and free variables.

The Hessian, each restricted seed inverse and every fixed walk transform by R_g on their indices, by the entry identities of Lemma 3.3 and the geometric walk formula; all geometric choices, Cauchy variables and Mayer parameters are unchanged. In the selected integral change variables $B \mapsto R_g B$, $X \mapsto R_g X$: Lebesgue and standard Gaussian measures, the two allowed cutoff norms ((5.4)), the conditional Gaussian determinant and the cross and compensator terms are preserved, so each selected measure is pushed forward by R_g, and in the probe formula $\delta_B - \delta_0$ is pushed to $\delta_{R_g B} - \delta_0$. Summing selections,

$$\widehat{\nu}^{\mathrm{f}}_{U^g,D} = (R_g)_* \widehat{\nu}^{\mathrm{f}}_{U,D} \, . \tag{E.69}$$

The vacuum activities, their Ursell products and $w_{\xi}(W)$ are scalar invariants, the incompatibility graph and every numerical combinatorial factor being field independent. Both $T_W P_W^{-1}$ and the complementary h transform at the initial interval vertex o, so

$$c_W(U^g; g(o) h g(o)^{-1}) = c_W(U; h) \, . \tag{E.70}$$

The rooted comparison gauge has $g(o) = \mathbf{1}$. Open transports occur only in this equality inside the closed character; no norm of an open transport is used.

E.10.3 Domain and Endpoint

Section E.7.4 defines the small interior groups by (E.43). The support of such a group includes every root domain, each vacuum-cluster domain occurrence, their coefficient halos and the whole contour interval $\mathcal{H}(W)$ with its closing transport; the hull-extent bound is needed here, since a box enclosing only the random inputs would not suffice. Under (E.43) the enclosing rectangle has diameter at most $C_{\mathrm{box}} LMn(W)$, lies within $b/2$ of a source contact and is injectively liftable under the period convention of Section E.8.2; on the far-history event the box and its halos lie in the regular region. All other groups, including winding or boundary-reaching groups, stay in the large-domain tail of Section E.7.4. This is the small-local-domain versus large/distant-domain separation of [4, (3.5)–(3.9)] and of the boundary separation of [3]; no interpolation is asserted for a group in the tail.

On a small interior group first use its rooted tree gauge. If $C_{\mathrm{tg}} \operatorname{diam}(Q_W)^2 \delta_{Q_W}(U_1) \leqslant \tau_{\mathrm{b}}/2$, define B_W and $U_{W,t}$ as in (E.45). With $\tau_{\mathrm{b}} \leqslant \tau_L/4$ and $c_L \tau_{\mathrm{b}} \leqslant \gamma_L/4$ (Lemma E.15) the entry Lipschitz estimate and the flat compression gap supply the two-sided Hessian bounds throughout the path. That the path need not consist of constrained minima is allowed only because its compensated primitive formulas are defined on the joint chart; physical criticality identifies those formulas at the endpoint, not at intermediate t.

For each selected coefficient Section E.9.2 fixes all walk parameters and restricts every surviving walk and coefficient stencil to its final domain; the seed inverse is the inverse on the seed, not the restriction of a global inverse. Thus, on the supports included in Q_W, the $t = 1$ recipe

equals the gauge-transformed physical centred recipe, its values on bonds outside Q_W cannot enter the coefficient, and (E.70) returns the coefficient to the original gauge. At $t = 0$ the recipe is the regional reference recipe with exactly the same I, Ω_1, W, cutoffs and constants. This yields the endpoint identities

$$c_W(U_{W,1}; h) = c_W(U_1; h), \qquad c_W(U_{W,0}; h) = a_W \chi(h) \ , \tag{E.71}$$

equalities of the selected coefficient, not of the outer history density. If the tree-gauge phase condition fails, the two-endpoint majorant of Section E.5.3 is used; if the box condition fails, the whole group is in the tail. Neither case is assigned an endpoint identity it does not satisfy.

For differentiation under the selected integrals, the norm of X in Section E.9.3 is the norm of X restricted to J_Z; all other auxiliary columns integrate out with mass one and have zero derivative. The common strictly positive c_{G} of (E.57) supplies a background-independent Gaussian majorant for that finite-dimensional integral and its first derivative; the selected sums use the summable majorants of Lemma E.13 and Section E.6.4, and the vacuum occurrence sums those of Lemma E.16 and Section E.7.4. These are the dominated-convergence hypotheses used in the proof of Lemma E.19; the group derivative includes the derivative of $w_\xi(W)$. [4, (3.21), (4.35)–(4.36)] estimate changes of reconstruction and domain rather than declaring them zero; no such change is made in the identities above, the regional carrier and each local recipe being fixed. A failure of the stated finite-support or covariance identity would invalidate the endpoint step and would require a retained comparison error, which the tree-gauge estimate alone could not absorb.

E.10.4 Endpoint Conventions and the Lifting Claim

The endpoint statement is restricted to covered far histories and the small interior groups of Section E.7.4. Extend the rooted site gauge from Q_W to the whole torus by the identity on sites outside Q_W. Every background bond consulted by the coefficient has both endpoints inside Q_W: in defining Q_W the enclosing rectangle is enlarged to contain both endpoints of each Hessian stencil, row set, block-mean path, plaquette and interval or closing-path bond, an enlargement that is part of the fixed halo included in C_{box}. Boundary bonds of the site-gauge extension are not consulted by the coefficient.

Only the finite formula $w_\xi(W) = (-1)^{k(W)}/k(W)!$ times the product of the specified vacuum cluster occurrences is varied along $U_{W,t}$; neither the global partition function nor the global exclusion ratio $q_\xi(\mathcal{A})$ is an operand of the interpolation. Lemma E.16 supplies the sum of the occurrence derivative majorants and (E.44) applies the finite product rule; the exclusion ratio was expanded at the original background before the group construction. Thus no identity or positivity statement for a global partition function at $U_{W,t}$ is needed to identify either endpoint. The regional scalar a_W is the one defined in Section 3.8 from $w^\circ_\xi(W)$ and the same root measures at the identity background ((3.92)); it is not an infinite-volume coefficient, and its closed Schur identity (3.93) is the $t = 0$ identity used here.

For the lifting claim, represent each constituent connected domain by a tree of its constituent coarse cubes and adjoining faces and root the union at a source contact. Each root or cluster-domain tree has at most its number of cubes minus one edges; incompatibility joins have bounded range, and the finite reconstruction halos add a fixed range at their contacts. The contour-hull extent bound bounds the additional contour edges joining the components by $CLMn(W)$; the network contains the whole interval $\mathcal{H}(W)$ and its closing path, not only their contacts. The resulting connected bond network has a spanning traversal of length at most $CLMn(W)$ with C fixed. If this is less than $N_{\min}/4$, lift the traversal from its root:

every lifted vertex lies within that distance of the root, and two lifts of the same vertex could differ only by a nonzero period vector, of length at least $N_{\min}$, while the two paths giving them have total length less than $N_{\min}/2$, a contradiction. Every point of the enclosing coordinate rectangle and of its stipulated halos is within $\operatorname{diam}(Q_W) \leqslant C_{\mathrm{box}} LMn(W)$ of the root, and two lifts of such a point would differ by a nonzero period vector, of length at least $N_{\min}$, while being joined through the root by two segments of total length at most $2C_{\mathrm{box}} LMn(W) < b < N_{\min}$ ((E.43), (E.47)); so the rectangle and its halos project injectively. C_{box} is chosen to include these constants; with (E.43) and (E.47) the small-group criterion implies this sufficient lifting condition, and failure to lift cannot occur in the small-group class. No curvature estimate for a winding loop is used. In applying the entry estimates, τ_{b} is taken small enough for both the h-maximum and the stated entry-chart norm, using $h \leqslant |\bullet|_\sigma \leqslant \sqrt{N} h$, before g_0 is chosen. Geometrically allowed zero walks re-index the same series and do not change its value or support domains.

E.10.5 Source and Radius Specifications

The imported mean covariance is [4, (0.6)], $\mathsf{M}(\{uU_j v\}) = u\mathsf{M}(\{U_j\})v$ for the specified mean on its small-diameter domain, which [4, (0.10)] asserts for the selected mean; it is used for unitary site gauges with no smallness restriction on the gauge values, the relative distances and the real chart domains being preserved. The cutoff invariance is (5.4).

The curvature input of Section E.7.4 is Lemma B.4. For a plaquette of Q_W choose one of its vertices x; $x \in \Lambda_1$ since $Q_W \subset \Lambda_1$. Within the L-block containing x choose a neighbouring vertex still in that block ($L \geqslant 3$). That internal fine bond is free, since designated bonds join distinct blocks, and belongs to I because it has the endpoint $x \in \Lambda_1$; so the plaquette has a bond within two fine-bond metric units of I, within the $3R_{\mathrm{c}}$ the lemma requires. This yields the bound $\delta_{Q_W}(U_1) \leqslant C\varepsilon_1$, with the fixed conversion between the lemma's operator norm and the norm $|\bullet|_\sigma$ of Lemma E.21.

For the branch identification of Section E.10.2 let $C_{\mathrm{D,joint}}$ be the constant in $\|\widetilde{D}_Z(A)\| \leqslant C_{\mathrm{D,joint}}\|A\|^2$ on the joint chart of Section E.9.1. After choosing that chart reduce θ_0 so that $2\theta_0 \leqslant \vartheta_*/2$ and $4C_{\mathrm{D,joint}}\theta_0^2 < \rho_{\mathrm{d}}$; the norms of Z and A are those of Lemma B.7, in which ϑ_*, ρ_{d} and τ_L are measured, and the background radius satisfies $2\tau_{\mathrm{b}} \leqslant \tau_L/2$ (Lemma E.15), so every real background with $\|Z\| < 2\tau_{\mathrm{b}}$ lies in the chart of that lemma. At real backgrounds the joint solution for $\|A\| < 2\theta_0$ then lies in the uniqueness ball of Lemma B.7 and is its selected solution. The coupling threshold is chosen after these radii so that the phase inputs $g_0 \mathsf{C}_U B$ of the enlarged free box remain in $\|A\| < \theta_0$; no expansion parameter is enlarged after this choice.

E.11 Quantitative Assembly at the Signed Physical Caller

This section proves the first-step estimate for E_{def} of (E.1). It uses the hypotheses of Section 3 (its Section 3.8 included) and Appendix B, the compatible even-period and square-fit conventions of Section E.8.2, and the same irreducible normalised character. Constants may depend on the fixed group, the character, L, M and the fixed expansion and chart choices, but not on the coupling, the square length, the compatible torus size or the history; the coupling is chosen last. The regional scalar $S^{\circ}_{\xi,C}$ is retained; no identification with a perimeter/corner scalar is asserted.

E.11.1 The Joint Derivative Supplied to the Group Coefficient

The quantitative derivative is obtained from Lemmas E.13 and E.14 and Section E.9. For each selected root the source is $\Delta_\jmath f$, a closed character difference; (E.61) bounds its mixed Taylor remainder by u^q and supplies an extra u for each odd leading term removed by reversal. The root's Gaussian derivative is even in its joint (B, X) variables, including its determinant and compensator derivatives, and a derivative of a selected ordinary potential carries a factor ζ and vanishes at $\zeta = 0$; so the independent-root parity/Cauchy argument of Lemma E.14 applies after one background derivative. Lemma E.13(b),(c) ((E.29), (E.30)) bounds a differentiated potential without dividing by that potential; (E.54) sums the derivatives of all geometrically indexed walks on the same spatial contours; (E.58) pays each Gaussian derivative by $C_M N_1(D)$ times the same $e^{N_1(D)}$ determinant majorant, with X restricted to J_Z, all other auxiliary variables integrating to one. Host and Gaussian derivatives therefore insert one factor $N_1(D)$, not one new constant per selected label. The source derivative inserts the number L_f of varying contour/closing links plus the number of selected labels, which Section E.6.1 bounds by $C_M \sum_D N_1(D)$. The resulting joint estimate is

$$|\mathscr{D}_\omega \mathfrak{Q}_\xi(\mathcal{A}; f)| \leqslant C_\chi \|\omega\|_\infty \Big(L_f + \sum_{D \in \mathcal{A}} N_1(D) \Big) \prod_{D \in \mathcal{A}} \rho\, m(D) \,, \tag{E.72}$$

with ρ of (E.11) and $m(D) = \varepsilon_* e^{-\lambda_* \ell_1(D)}$ of Lemma E.2; the pairing $\mathfrak{Q}_\xi(\mathcal{A}; f)$ and its source are evaluated at the background $U_{W,t}$ of the fixed coefficient path of Section E.10 in place of U_1 (the subscript ξ is kept for brevity), and the estimate differentiates neither an outer history weight nor a global exclusion ratio. For a vacuum occurrence $\mathfrak{c}$ the selected-integral derivative gives (E.33); differentiating the finite product $w_\xi(W)$ by (E.44), its occurrences $\Phi^{\mathrm{v}}(\mathfrak{c}_j)$ likewise evaluated at $U_{W,t}$, and adding the root-integral derivative yields, uniformly for $h \in G$,

$$|c_W(U; h)| \leqslant M(W), \qquad |\mathscr{D}_\omega c_W(U; h)| \leqslant C_\chi \|\omega\|_\infty \big(n(W) + |\mathcal{H}(W)| \big) M(W) \,. \tag{E.73}$$

The undifferentiated estimate holds at the original physical background and at the regional flat reference. $M(W)$ contains ρ for every marked root and the original factorials; polynomial factors have not been summed or hidden in $M(W)$, and the factor from the relative interval's inverse word, at most a fixed multiple of $|\mathcal{H}(W)|$, is included in C_χ.

E.11.2 Payment of the Local and Tail Losses

Proof of Lemma E.11. Write $s = \mathfrak{s}(W) \geqslant 1$, so that $\mathfrak{a}_W = \eta s$. The fixed hull and box bounds give $n(W) + |\mathcal{H}(W)| \leqslant CLMs$ and $\operatorname{diam}(Q_W) \leqslant C_{\mathrm{box}} LMn(W) \leqslant C_{\mathrm{box}} LMs$ (Section E.7.4). The endpoint identities (E.71), the path integral (E.46) with the derivative bound (E.73) along it, and the plaquette estimate $\delta_{Q_W}(U_1) \leqslant C\varepsilon_1$ of Section E.10.5 with the phase bound (E.65) therefore give, for a small interior group passing the phase test,

$$\mathfrak{d}_W(U_1) \leqslant C_\chi \varepsilon_1 s^3 M(W) \,. \tag{E.74}$$

If the phase test of such a group fails, then $C_{\mathrm{tg}} \operatorname{diam}(Q_W)^2 \delta_{Q_W} > \tau_{\mathrm{b}}/2$; with $\delta_{Q_W} \leqslant C\varepsilon_1$ and $\operatorname{diam}(Q_W) \leqslant CLMs$ this gives $1 \leqslant C\varepsilon_1 s^2$ with constants fixed before the coupling, so the two undifferentiated endpoint bounds (E.73) give (E.74) again, as $s \geqslant 1$. No curvature comparison is used for a group outside the small interior class; for such a group

$$\mathfrak{d}_W(U_1) \leqslant 2M(W), \qquad s \geqslant T_b \coloneqq \frac{b}{4C_{\mathrm{box}} LM} \,. \tag{E.75}$$

For the local polynomial payment let

$$C_{\eta,3} = \sup_{s\geqslant 1} s^3 e^{-(31\eta/8)s} < \infty \,, \tag{E.76}$$

so that $s^3 e^{\eta s/8} \leqslant C_{\eta,3} e^{4\eta s}$; for the tail, $s \geqslant T_b$ gives $e^{\eta s/8} \leqslant e^{-2\eta T_b} e^{4\eta s}$. Hence the two weighted pin estimates of Lemmas E.17 and E.18 imply

$$\sum_W K_\chi \mathfrak{d}_W(U_1) e^{\mathfrak{a}_W/8} \leqslant C_\chi\big(\varepsilon_1 + e^{-2\eta T_b}\big) \sum_W M(W) e^{4\eta \mathfrak{s}(W)} \leqslant C_\chi n_C \rho\big(\varepsilon_1 + e^{-cb/(LM)}\big),$$
$$c = \frac{\eta}{2C_{\mathrm{box}}} \,. \tag{E.77}$$

The last summation uses that every W has a nonempty contour interval: summing the contour-pin estimate (E.21) over the n_C traversals counts every group at least once. The cubic moment and the tail use the existing 4η weight, not an additional size-dependent hypothesis or a loss in the scalar $S^\circ_{\xi,C}$. The exceptional whole-contour merge is the one paid in Lemma E.18, where the long class pays both the arbitrary extra components and the whole-contour weight. This is (E.24). *q.e.d.*

E.11.3 The Signed Family Sum and the Physical Expectation

Proposition E.22 (The Signed Defect Bound)**.** *Under the hypotheses stated at the start of this section, for g_0 below the thresholds of Lemmas E.1, E.9 and E.10, with $R_\Upsilon \geqslant 2$ (Section E.6.1, Lemma E.2), $u \leqslant u_*$ and $v \leqslant 1$ (Lemma E.3) and the small-coupling conditions of Sections E.9.1 and E.10.5, with ρ of* (E.11)*, $\rho = O(g_0^2(1+r)^4)$, $\varepsilon_1 = g_1 p_0(g_1)$, $b = \lfloor LMR_1/16 \rfloor$, $\pi_* = \min(p_0(g_0), p_0(g_1))$ and the constants C'_L, c_* of* (E.3) *and C_χ, c of* (E.24)*,*

$$|E_{\mathrm{def}}| \leqslant C'_L n_C e^{-(c_*/2)\pi_*^2} + \frac{1}{\mathcal{Z}_0} \int \mathcal{O}\Big[\mathsf{w}_\xi \mathbf{1}_{A_{\mathrm{cov}} \cap \mathsf{N}_b^c} |S^\circ_{\xi,C}| \Big(e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1\Big)\Big](V)\, \mathrm{d}V \,, \tag{E.78}$$

and consequently

$$|E_{\mathrm{def}}| \leqslant C'_L n_C e^{-(c_*/2)\pi_*^2} + e^{C_\chi n_C \rho(\varepsilon_1 + e^{-cb/(LM)})} - 1 \,. \tag{E.79}$$

Proof. Lemma E.10 supplies the single joint ratio (E.23). Combined with the closed-contraction product bound (E.20), formed after each d_W, the nonempty compatible-family sum is bounded by the exponential series of the summable one-group weight (E.24), which on covered far histories is (E.25) (Lemma E.12); the countable-group limit is the dominated finite-species limit of Lemma E.20, with the same positive majorant and the nonzero reference partition function supplied by its convergent logarithm, so no quotient is formed at a zero partition function. Combining the near-history estimate (E.3) with the positivity of the original history measure gives (E.78); all history weights, deferred integrations and the physical normalisation are those of (E.1), and the exact source on the left was summed before absolute values were taken. The regional reference scalar $S^\circ_{\xi,C} = \int f^\circ_\xi K^\circ_{\xi;0}\, \mathrm{d}a / I^\circ_{\xi;0}$ ((3.94)) is a normalised integral of f°_ξ, a normalised character of modulus at most one, against the positive reference kernel $K^\circ_{\xi;0}$, so $|S^\circ_{\xi,C}| \leqslant 1$, and the same physical normalisation yields (E.79). *q.e.d.*

The exponential in (E.79) is not replaced by its linear term unless its argument is bounded; in particular no assertion of UV summability for a fixed physical loop is made. The conclusion has exactly the hypotheses and quantitative scope stated at the beginning of this section.

E.11.4 Fixed Reserve Conventions

The exponential vacuum estimate used in Lemma E.17 holds with $n(\mathfrak{c})/2$ as well as with $n(\mathfrak{c})/4$: the rooted-tree proof of Lemma 3.19 applied to the numerical activity $m(D)e^{N_1(D)/2}$ with node cost $N_1(D)/8$ has combined exponent $5/8$, below the $1+6\theta$ of the activity payment (3.72), which gives the first bound of (E.34). No reserve of Appendix B is re-spent. Adjacency allowance: in the footprint-contact sums of Lemma E.9 ((E.21)) and Lemma E.10, footprint contact includes wall adjacency of two LM-cubes (a shared face, edge or corner), not only intersection of the two cube sets; counting every cube touching a given one at a face, an edge or a corner gives at most $3^4 - 1$ neighbours, so the spatial pin is summed over the neighbouring cubes as well, which inserts a fixed factor in the constant C multiplying ρ. The smallness condition $C\rho LM \leqslant \eta/8$ of Lemma E.9 is imposed after incorporating this factor; the allowance does not alter ρ, the reserve exponent or the loop-length dependence. The tail rate is $c = \eta/(2C_{\mathrm{box}})$ as in (E.77); earlier weaker tail displays remain consequences of it with a reduced positive c.

F The Regional Scalar of the Covered Branch

This appendix identifies the scalar $S^{\circ}_{\xi,C}$ of the branch identity (3.97), defined in Section 3.8 as the reference-gas sum (3.94), with the regional square scalar (7.31) at the carrier I, and thereby with the perimeter/corner coefficient κ_C of (8.1) up to the error of Proposition 7.3. It is the statement used in Section 8 at (8.15); its labelled counterpart at the retreated carrier $D(\Lambda)$ is Lemma 9.42 with Corollary 9.43. The quantitative comparison of the regional square scalar with the bulk square is the proof of Proposition 7.3, in particular (7.44) and (7.46); nothing is added to it here.

F.1 Conventions

Fix one of the two cutoff prescriptions σ, the bare coupling g_0, the irreducible normalised character χ, the averaging and mean prescription of [4] and the designated-bond convention, as in the rest of the paper, with the standing conditions (3.1) and the collar conditions (7.28). The fine periods are $N_\mu = LK_\mu$, and $C = C_n$ is a coarse translate of the counterclockwise square with vertices 0, nLe_1, $nL(e_1+e_2)$, nLe_2, with n coarse segments of length L on each side and $K_\mu \geqslant 3(n+2)$ for every coarse period; its physical perimeter is $P(C_n) = 4nL\varepsilon$. Let A_{cov} be the set of covered branches of Lemma 3.28 and let $\xi \in A_{\mathrm{cov}}$. The young carrier is $D = I = \Lambda_1^*$ (Section 3), the set of free bonds with an endpoint in Λ_1, which is the carrier D_+ of Section 7 and contains its D_-; D^c is its complement in the torus free carrier $E_{*,T}$; distances are those of the periodic fine-bond metric d_T of (5.2). For a coarse segment j of the cyclic index set $J_C = \{0, \dots, 4n-1\}$ let K_j be the union of the fixed free-input stencils of its fine traversals, and with the fixed range $R_0 = 4(1 + R_* + R_H + L)$ of Section 7 (the paragraph following (7.28); the R_0 of (7.32)) put

$$d_j = \max\bigl(0, d_T(K_j, I^c) - R_0\bigr), \qquad \iota_T = \max\bigl(0, N_{\min}/4 - R_0\bigr), \qquad N_{\min} = \min_{\mu} N_\mu\ , \quad \text{(F.1)}$$

the distance to the empty set being infinite with $e^{-\infty} = 0$; there is no distance to a side endpoint in d_j. The constants below depend on the character, the group, L and the forest parameters only, not on n, the compatible periods or the shape of I; g_0 is chosen after them, below the common flat and regional thresholds. τ is the marked-forest parameter of (5.18);

enlarging a fixed majorant constant multiplies τ by a fixed factor, absorbed in the constants and the coupling threshold.

F.2 Identification with the Regional Square Integral

Lemma F.1 (The Reference-Gas Scalar at the Carrier I)**.** *On A_{cov}, under the conventions of Section F.1 and the thresholds displayed in the proof of Proposition 7.3 (also* (9.39)*(ii)),*

$$S^{\circ}_{\xi,C_n} = S^{\circ}_{I,C_n,T} = \frac{\displaystyle\int \Theta^{\sigma}_{I}(B) e^{P_{I,T}(B)} \chi\big(\mathrm{Hol}_{U_{g_0,T}(i_{I,T}B)}\, C_n\big)\, \mathrm{d}\mu_{C_{I,T}}(B)}{\displaystyle\int \Theta^{\sigma}_{I}(B) e^{P_{I,T}(B)}\, \mathrm{d}\mu_{C_{I,T}}(B)}\,, \tag{F.2}$$

the right side being (7.31) *at $D = I$, with every original primitive label retained, including the labels crossing the boundary of I, and with the exterior link value the identity.*

Proof. By (3.94), $S^{\circ}_{\xi,C_n} = \int f^{\circ}_{\xi} K^{\circ}_{\xi;0}\, \mathrm{d}a / I^{\circ}_{\xi;0}$, the normalised expectation of the full reference Wilson loop under the centred kernel at the identity background, whose precision is $H^{\circ}_{\xi} = \iota_I^* \mathsf{H}^{\Omega_1}_{\mathbf{1}} \iota_I$, whose potentials are the near primitives evaluated at zero retained input and whose cutoff is the bond cutoff on I. Both sides of (F.2) are ratios of two integrals over $\mathfrak{g}^I$, so it suffices to identify the factors of the integrands. Precision: by Lemma B.6 at the identity background the entries of H°_{ξ} are $f_{bb'}(0)$, the entries of the flat Hessian (4.7) on the branch's torus (as in the proof of Lemma B.13), and the collar puts every precision and primitive stencil touching I inside the regular region (the identification paragraph of Section 7 preceding (7.29)), so $H^{\circ}_{\xi} = i^*_{I,T} H_T i_{I,T}$, the diagonal contributions of the plaquettes crossing the boundary of I included; the Gaussian normaliser of $\mu_{C_{I,T}}$ is common to numerator and denominator and cancels within the ratio, not against the outer physical weight. This is the full-square form of (7.29): the kernel identification uses only that the source is a closed contraction with the given input stencils, not the straightness of the segment. Reconstruction: the row solutions $\widetilde{D}_{U,c}$ of Section 3.3 (Lemma 3.6) at the identity background and those of Section 7 agree as analytic germs at zero by their defining equation and uniqueness; the Hilbert-ball domain of Lemma 3.6 (reference version) and the maximum-norm domain of the flat expansion have a convex common phase domain containing the closed cutoff box at the stated thresholds, and analytic continuation on that domain identifies the two solutions throughout the box. Primitives: Section 3.3 groups the gauge primitives by their centre y while (7.20) indexes them by the pairs (y, x); the sums coincide, the degree-two Taylor residual being linear in the local term, and every primitive whose free support misses I vanishes at $i_{I,T}B$ while every primitive meeting I is retained on both sides. Cutoff: at zero exterior free field the conditional cutoff is Θ^{σ}_{I} at the threshold $r = \delta_0/g_0$ in the norm $|\bullet|_{\sigma}$, as in Section 7. Wilson factor: on a covered branch every contour traversal has its free-input set in I, so the proof of Lemma 3.28 applies verbatim with a retained instead of set to zero and gives that each contour link of $f^{\circ}_{\xi}(a)$ is its reconstruction from a alone, that is, $U_{g_0,T}(i_{I,T}B)$ at $B = a$; the boundary map is the identity by the reference-data convention $\Psi^{\circ} = \mathbf{1}$ of Section 3.8. Numerator and denominator agree factor by factor, so the ratios agree. The equality is one of integral ratios; no identification of a separately chosen logarithm of the reference group gas of Section 3.8 with the connected source logarithm of the proof of Proposition 7.3 is used. *q.e.d.*

F.3 The Scalar of the Branch Identity

Proposition F.2 (The Regional Scalar of the Covered Branch)**.** *Under the conventions of Section F.1 and the hypotheses of Proposition 7.3, on A_{cov},*

$$S^{\circ}_{\xi,C_n} = e^{-\delta m^{\sigma}(g_0,\varepsilon)P(C_n)}\big(z^{\sigma}_{\mathrm{c}}(g_0,\pi/2)\big)^4 e^{E^{\circ}_{\xi}} = \kappa_{C_n} e^{E^{\circ}_{\xi}} , \tag{F.3}$$
$$|E^{\circ}_{\xi}| \leqslant C\tau^2\Big[\sum_{j\in J_C} e^{-cd_j} + 4ne^{-c\iota_T} + (n+1)e^{-cn}\Big] ,$$

with $\kappa_{C_n} \coloneqq e^{-\delta m^{\sigma}(g_0,\varepsilon)P(C_n)}(z^{\sigma}_{\mathrm{c}}(g_0,\pi/2))^4$ the coefficient of (8.1)*, built from the coefficients of* (5.39) *and* (6.12)*, and*

$$E^{\circ}_{\xi} = E^{\circ}_{I,C_n,T} = (F_{I,C,T} - F_{C,T}) + (F_{C,T} - F_{C,\infty}(n)) + E^{\sigma}_{n} , \tag{F.4}$$

the error of Proposition 7.3 at $D = I$: as in its proof, $F_{I,C,T} = \log S^{\circ}_{I,C_n,T}$ on the connected source branch, $F_{C,T} = \log(N^{\sigma}_{C_n,T}/D^{\sigma}_{T})$, $F_{C,\infty}(n) = F_{\square,\infty}(n)$ is the bulk logarithm (6.9)*, and $E^{\sigma}_{n} \coloneqq F_{\square,\infty}(n) - n\sum_{v\in\mathscr{V}_{\square}}\kappa_v - \sum_{(v,w)\in\mathscr{E}_{\square}}\zeta_{v,w}$ is the corner-extraction error, $|E^{\sigma}_{n}| \leqslant C\tau^2(n+1)e^{-cn}$ by* (6.11)*, with $n\sum_v \kappa_v = -\delta m^{\sigma}P(C_n)$ (*(5.39)*) and $\sum_{(v,w)}\zeta_{v,w} = 4\log z^{\sigma}_{\mathrm{c}}$ (*(6.12)*), so that $F_{I,C,T} = \log\kappa_{C_n} + E^{\circ}_{\xi}$. The scalar and its nonzero logarithm are those of the regional conditional reference integral; no physical history weight or large-field indicator is evaluated at flat fields, and the bound does not assume the loop to be far from the integration boundary: $d_j = 0$ retains a boundary contribution of size $C\tau^2$.*

Proof. Lemma F.1 and Proposition 7.3 at $D = I$, which is covered on A_{cov} and is the carrier D_+ itself (Section F.1), so $D_- \subseteq D \subseteq D_+$ holds with the collar; the three terms of (F.4) are (7.44), (7.46) and the corner extraction (6.11), the first two retaining the changes of kernel and of summation range as errors before the translation-invariant coefficients are identified, as in [4, (4.35)–(4.37)], while the exterior operation and its boundary terms are kept as they stand in this paper ([3, (1.26)–(1.30)], which perform Bałaban's own separation, are not used, except (1.27) for g_1; Section 8). *q.e.d.*

Corollary F.3 (The Physical Identity with the Scalar Substituted)**.** *Under the hypotheses of Proposition F.2, with E_{def} of* (E.1) *and E_{bdry} of* (D.21)*,*

$$\mathbb{E}_{\varepsilon}[W_{\chi}(C_n)] = \kappa_{C_n}\,\mathbb{E}_{\mathrm{h}}\big[\mathbf{1}_{A_{\mathrm{cov}}} e^{E^{\circ}_{\xi}} W_{\chi}(C_n;U_1)\big] + E_{\mathrm{def}} + E_{\mathrm{bdry}} , \tag{F.5}$$

where $\mathbb{E}_{\mathrm{h}}$ uses the original outer operation, the weight $\mathsf{w}_{\xi} = e^{c_{\xi}} I_{\xi;1}$ and the physical $\mathcal{Z}_0$. Every term is integrable against the normalised history weight, and the reconstructed fine loop $W_{\chi}(C_n;U_1)$ is retained.

Proof. By (E.35), $\mathbb{E}_{\varepsilon}[W_{\chi}(C_n)] = \mathbb{E}_{\mathrm{h}}[\mathbf{1}_{A_{\mathrm{cov}}} S^{\circ}_{\xi,C_n} f_{\xi}(0)] + E_{\mathrm{def}} + E_{\mathrm{bdry}}$, with $f_{\xi}(0) = W_{\chi}(C_n;U_1)$ on A_{cov} (Lemma 3.28); substitute (F.3). E°_{ξ} is a function of the finite geometry at fixed torus and reference parameters, hence of the history labels through I alone, and is independent of the field values at fixed labels, so the first term is measurable; its bound has at most $4n$ boundary terms, $|E^{\circ}_{\xi}| \leqslant C\tau^2(9n+1)$, so the factor $e^{E^{\circ}_{\xi}}$ is bounded and the term integrable. For the other terms, $|S^{\circ}_{\xi,C_n}| \leqslant 1$ because it is the normalised integral of a character of modulus at most one against the positive reference kernel $K^{\circ}_{\xi;0}$ (Lemma B.13); $|F_{\xi}(s)| \leqslant 1$ since $F_{\xi}(s) = \int f_{\xi}\,\mathrm{d}\nu_{\xi;s}$ ((3.46)) with $\nu_{\xi;s}$ a probability measure, and $|f_{\xi}(0)| \leqslant 1$; hence $|\mathcal{B}_{\xi}[f_{\xi}]| = |F_{\xi}(1) - F_{\xi}(0)| \leqslant 2$ ((3.51)) and the summed signed defect $|F_{\xi}(0) - S^{\circ}_{\xi,C_n} f_{\xi}(0)| \leqslant 2$, and both are integrable against the unchanged probability history weight. After the substitution the first integrand is still exactly $S^{\circ}_{\xi,C_n} f_{\xi}(0)$, so no independent bound on $e^{E^{\circ}_{\xi}}$ is needed to justify the identity. No new denominator estimate or cancellation is invoked. *q.e.d.*

Remark F.4 (Scope). The callers of (F.3) at $D = I$ are the unlabelled branch identity at (8.15) and Corollary F.3, through which (i) and the first bound of (ii) of the main theorem use it; Appendix E carries $S^{\circ}_{\xi,C}$ inside the expectation and does not use (F.3). Section 9 carries out the same identification at the retreated carrier $D(\Lambda) \subseteq I$ of its labelled terms (Lemma 9.42: factorwise agreement of precision, cutoff, primitives and Wilson factor; Corollary 9.43: $S^{\circ}_{\Lambda} = \kappa_C e^{E^{\circ}_{\Lambda}}$ with the error at $D(\Lambda)$; Lemma 9.44: the label average of the boundary sum paid by one-point witness probabilities). The ingredients are the same in both cases: the primitives of (7.4), the regional stencil identity and the collar. The identity (F.5) compares the physical expectation with the reconstructed fine loop at the background U_1; a comparison with the bare coarse loop, or convergence over all scales, is not asserted.

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